

Fractal Kernel SVR report

April 29, 2025

1 Time Series Support Vector Regression with Fractal Kernel

Applications of fractal-type kernels in Gaussian process regression and support vector machine regression

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Paper Date: 18 October 2024

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1.1 Imports

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.svm import SVR
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_squared_error, r2_score
from rkhs import RKHS
```

1.2 Data Preprocessing

```
[2]: df = pd.read_csv('./data/Crude Oil WTI Futures Historical Data.csv',
    ↳ parse_dates=['Date'])
df.sort_values(by='Date', ascending=True, inplace=True)
df.reset_index(drop=True, inplace=True)
df
```

```
[2]:
```

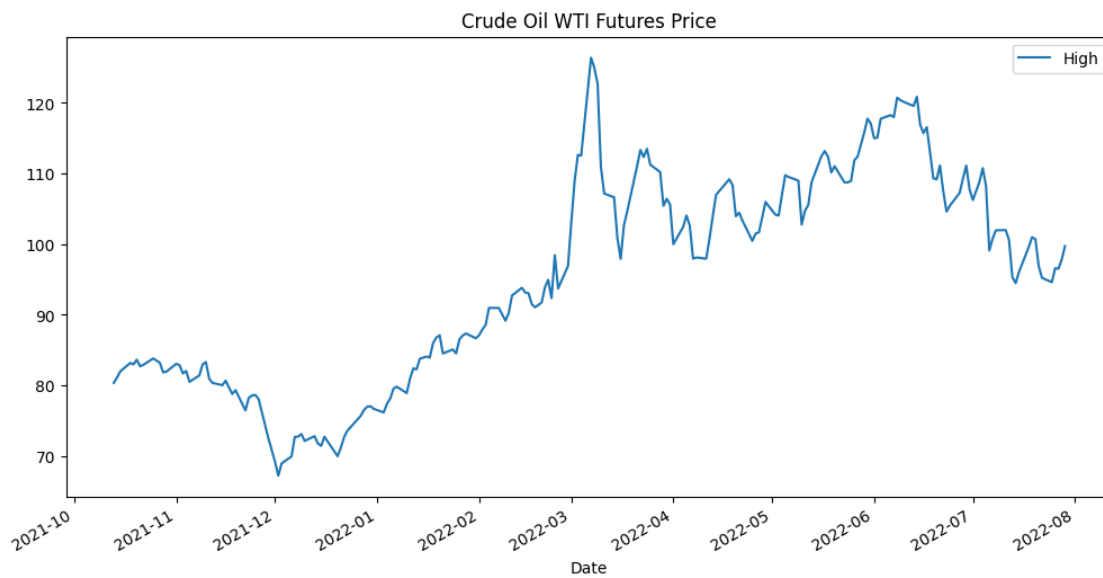
	Date	Price	Open	High	Low	Vol.	Change %
0	2021-10-13	79.82	79.83	80.36	78.78	304.96K	-0.13%
1	2021-10-14	80.77	79.94	81.14	79.79	300.57K	1.19%
2	2021-10-15	81.73	80.93	82.00	80.86	284.73K	1.19%
3	2021-10-18	81.69	82.00	83.18	81.21	501.07K	-0.05%
4	2021-10-19	82.44	81.58	82.99	81.08	531.47K	0.92%
..	...	...	...	...	...	...	...

206	2022-07-25	94.42	92.83	94.62	90.79	92.70K	2.15%
207	2022-07-26	92.88	93.95	96.60	92.64	89.07K	-1.63%
208	2022-07-27	95.29	93.51	96.53	92.30	105.40K	2.59%
209	2022-07-28	94.65	96.09	97.82	94.28	96.14K	-0.67%
210	2022-07-29	96.75	95.42	99.75	94.59	116.64K	2.22%

[211 rows x 7 columns]

```
[3]: df.plot(x='Date', y='High', figsize=(12, 6), title='Crude Oil WTI Futures_
Price')
```

```
[3]: <Axes: title={'center': 'Crude Oil WTI Futures Price'}, xlabel='Date'>
```



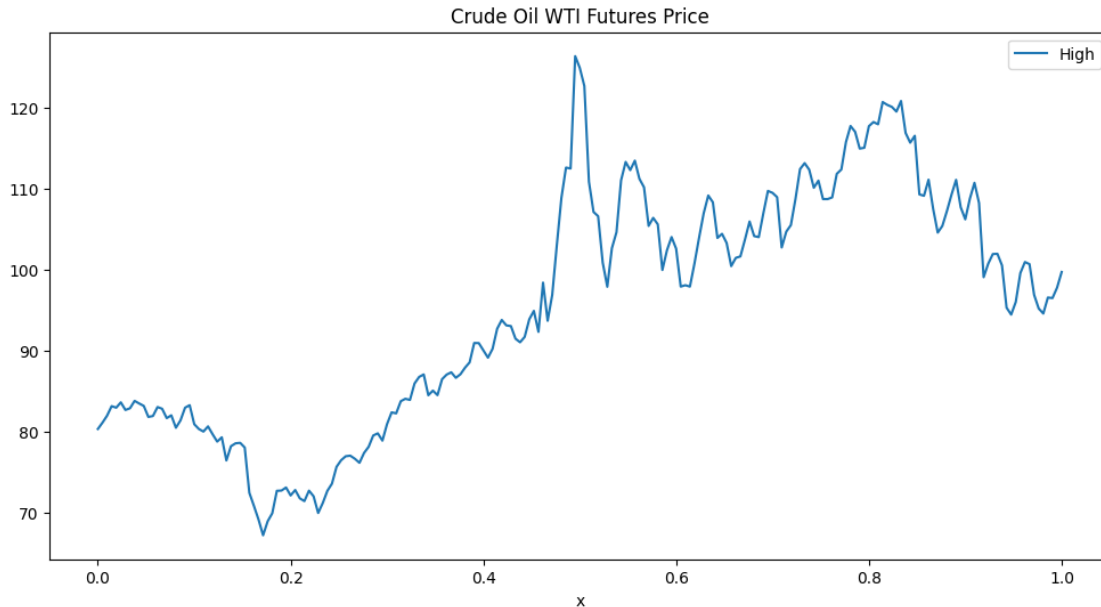
```
[4]: df['x'] = df.index/210
df
```

	Date	Price	Open	High	Low	Vol.	Change %	x
0	2021-10-13	79.82	79.83	80.36	78.78	304.96K	-0.13%	0.000000
1	2021-10-14	80.77	79.94	81.14	79.79	300.57K	1.19%	0.004762
2	2021-10-15	81.73	80.93	82.00	80.86	284.73K	1.19%	0.009524
3	2021-10-18	81.69	82.00	83.18	81.21	501.07K	-0.05%	0.014286
4	2021-10-19	82.44	81.58	82.99	81.08	531.47K	0.92%	0.019048
...	...	...	...	...	...	...	...	...
206	2022-07-25	94.42	92.83	94.62	90.79	92.70K	2.15%	0.980952
207	2022-07-26	92.88	93.95	96.60	92.64	89.07K	-1.63%	0.985714
208	2022-07-27	95.29	93.51	96.53	92.30	105.40K	2.59%	0.990476
209	2022-07-28	94.65	96.09	97.82	94.28	96.14K	-0.67%	0.995238
210	2022-07-29	96.75	95.42	99.75	94.59	116.64K	2.22%	1.000000

[211 rows x 8 columns]

```
[5]: df.plot(x='x', y='High', figsize=(12, 6), title='Crude Oil WTI Futures Price')
```

```
[5]: <Axes: title={'center': 'Crude Oil WTI Futures Price'}, xlabel='x'>
```



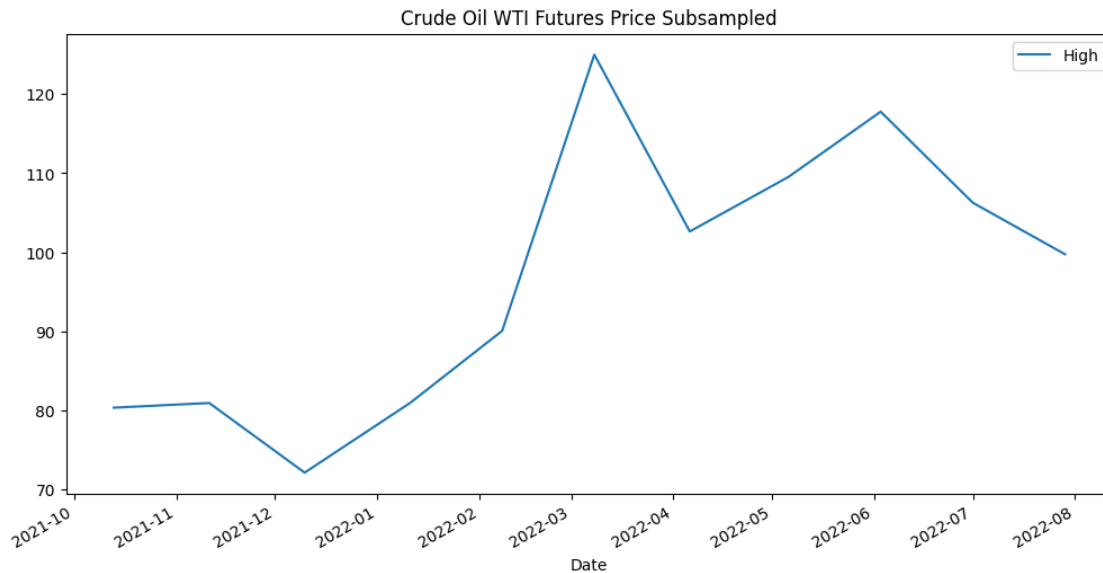
```
[6]: # 2021/10/13, 11/11, 12/10, 2022/1/11, 2/8, 3/8, 4/6, 5/6, 6/3, 7/1, 7/29.
date_strings = ['2021-10-13', '2021-11-11', '2021-12-10', '2022-01-11',
               ↪ '2022-02-08', '2022-03-08', '2022-04-06', '2022-05-06', '2022-06-03',
               ↪ '2022-07-01', '2022-07-29']
target_dates = pd.to_datetime(date_strings)
df_subsample = df[df['Date'].isin(target_dates)]
df_subsample
```

```
[6]:
```

	Date	Price	Open	High	Low	Vol.	Change %	x
0	2021-10-13	79.82	79.83	80.36	78.78	304.96K	-0.13%	0.0
21	2021-11-11	80.42	80.10	80.95	78.98	209.70K	0.45%	0.1
42	2021-12-10	71.48	70.52	72.15	70.19	185.50K	0.97%	0.2
63	2022-01-11	80.67	77.87	80.95	77.83	270.56K	3.82%	0.3
84	2022-02-08	87.90	89.98	90.07	87.07	188.54K	-2.08%	0.4
105	2022-03-08	119.65	116.98	124.98	113.66	200.16K	3.32%	0.5
126	2022-04-06	95.41	99.95	102.63	94.93	159.99K	-5.23%	0.6
147	2022-05-06	108.33	107.07	109.53	105.72	119.33K	1.49%	0.7
168	2022-06-03	116.23	114.90	117.77	112.73	74.48K	1.69%	0.8
189	2022-07-01	105.39	103.33	106.26	101.94	99.69K	2.22%	0.9
210	2022-07-29	96.75	95.42	99.75	94.59	116.64K	2.22%	1.0

```
[8]: df_subsample.plot(x='Date', y='High', figsize=(12, 6), title='Crude Oil WTI_
↳Futures Price Subsampled')
```

```
[8]: <Axes: title={'center': 'Crude Oil WTI Futures Price Subsampled'},
xlabel='Date'>
```



```
[9]: data_series = df_subsample[['x', 'High']].to_numpy()
data_series
```

```
[9]: array([[0.0000e+00, 8.0360e+01],
[1.0000e-01, 8.0950e+01],
[2.0000e-01, 7.2150e+01],
[3.0000e-01, 8.0950e+01],
[4.0000e-01, 9.0070e+01],
[5.0000e-01, 1.2498e+02],
[6.0000e-01, 1.0263e+02],
[7.0000e-01, 1.0953e+02],
[8.0000e-01, 1.1777e+02],
[9.0000e-01, 1.0626e+02],
[1.0000e+00, 9.9750e+01]])
```

```
[10]: fractal_depth = 2
rkhs = RKHS(data_series=data_series, m=11, fractal_depth=fractal_depth)
```

Vertical scaling factors s_values were not provided, so they were initialised randomly.

Building the Basis functions for the RKHS:

Built $F_{[u_1]}$.

```

Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[11]: print("Matrix A (inner products between fractal basis functions):")
print(np.round(rkhs.A, decimals=2))
print("\nMatrix B (inverse of A):")
print(np.round(rkhs.B, decimals=2))

```

Matrix A (inner products between fractal basis functions):

```

[[ 1.    0.    0.16 -0.    0.15 -0.    0.06 0.    -0.05 -0.    0.15]
 [ 0.    0.51 -0.    0.03 0.04 -0.01 0.06 0.    0.07 -0.03 -0.03]
 [ 0.16 -0.    1.03 -0.11 0.14 -0.17 -0.03 -0.15 -0.27 0.03 0.18]
 [-0.    0.03 -0.11 1.06 0.06 0.    0.09 0.13 -0.24 0.04 -0. ]
 [ 0.15 0.04 0.14 0.06 1.08 -0.09 -0.01 -0.04 0.06 0.14 0.17]
 [-0.   -0.01 -0.17 0.   -0.09 1.1   0.1   0.03 -0.08 0.03 -0.04]
 [ 0.06 0.06 -0.03 0.09 -0.01 0.1   0.95 0.01 -0.25 0.09 0.02]
 [ 0.    0.   -0.15 0.13 -0.04 0.03 0.01 0.79 -0.07 0.1   0.14]
 [-0.05 0.07 -0.27 -0.24 0.06 -0.08 -0.25 -0.07 1.13 -0.02 -0.06]
 [-0.   -0.03 0.03 0.04 0.14 0.03 0.09 0.1   -0.02 0.88 0.01]
 [ 0.15 -0.03 0.18 -0.    0.17 -0.04 0.02 0.14 -0.06 0.01 1.1 ]]

```

Matrix B (inverse of A):

```

[[ 1.06 0.01 -0.14 -0.   -0.13 -0.03 -0.07 -0.02 -0.    0.04 -0.09]
 [ 0.01 2.04 -0.08 -0.08 -0.09 0.   -0.18 -0.06 -0.19 0.11 0.09]
 [-0.14 -0.08 1.23 0.18 -0.11 0.18 0.13 0.27 0.39 -0.08 -0.18]
 [-0.   -0.08 0.18 1.05 -0.1 0.04 -0.02 -0.12 0.27 -0.02 0.02]
 [-0.13 -0.09 -0.11 -0.1 1.03 0.06 0.02 0.08 -0.1 -0.17 -0.14]
 [-0.03 0.   0.18 0.04 0.06 0.95 -0.07 0.01 0.1 -0.04 0. ]
 [-0.07 -0.18 0.13 -0.02 0.02 -0.07 1.17 0.07 0.28 -0.13 -0.04]
 [-0.02 -0.06 0.27 -0.12 0.08 0.01 0.07 1.42 0.13 -0.18 -0.23]
 [-0.   -0.19 0.39 0.27 -0.1 0.1   0.28 0.13 1.13 -0.04 -0.01]
 [ 0.04 0.11 -0.08 -0.02 -0.17 -0.04 -0.13 -0.18 -0.04 1.2   0.05]
 [-0.09 0.09 -0.18 0.02 -0.14 0.   -0.04 -0.23 -0.01 0.05 1.  ]]

```

```

[12]: np.linalg.eigvals(rkhs.A)

```

```
[12]: array([1.66082579, 1.49667447, 0.42737145, 0.55203237, 0.62499452,
          1.20079553, 1.10594822, 0.7922898 , 0.85235997, 0.91272238,
          1.01521956])
```

```
[13]: K = rkhs.kernel_array(data_series[:, 0][:, None], data_series[:, 0][None, :])
      K[:3,:3]
```

```
[13]: array([[12.44367008,  0.51448384,  0.89951005],
          [ 0.51448384,  6.76738538,  0.68684319],
          [ 0.89951005,  0.68684319,  8.37846012]])
```

```
[14]: total_data_series = df[['x', 'High']].to_numpy()
      total_data_series.shape
```

```
[14]: (211, 2)
```

1.3 Modelling

```
[15]: X = total_data_series[:, 0][:, None]
      y = total_data_series[:, 1]
```

```
[16]: def kernel_builder(X, Y):
      return rkhs.kernel_array(X, Y.T)
```

```
[17]: svr_model = SVR(kernel=kernel_builder, C = 10)

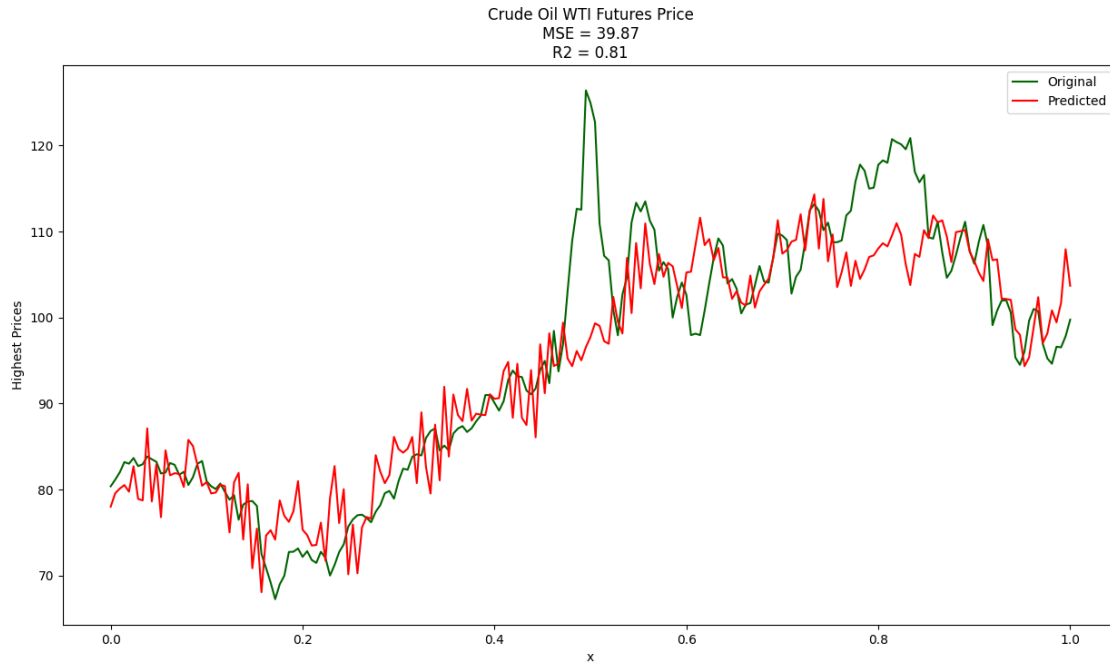
      svr_model.fit(X, y)

      y_pred = svr_model.predict(X)

      y_pred[:5], y[:5]
      mean_squared_error(y, y_pred)
```

```
[17]: 39.86688208491785
```

```
[18]: plt.figure(figsize=(15, 8))
      plt.plot(X[:,0], y, label='Original', color='darkgreen')
      plt.plot(X[:,0], y_pred, label='Predicted', color='red')
      plt.xlabel('x')
      plt.ylabel('Highest Prices')
      plt.title('Crude Oil WTI Futures Price\n'
                f'MSE = {mean_squared_error(y, y_pred):.2f}\n'
                f'R2 = {r2_score(y, y_pred):.2f}')
      plt.legend()
      plt.show()
```



1.4 Hyperparameter Tuning

```
[19]: import optuna
      from sklearn.model_selection import cross_val_score, TimeSeriesSplit
```

```
[20]: def fractal_kernel(rkhs: RKHS):
      def k(X, Y):
          return rkhs.kernel_array(X, Y.T)

      return k
```

```
[23]: # Optuna objective
def objective(trial):
    # 1) Suggest each element of the matrix B
    B = np.zeros((11, 10), dtype=float)
    for i in range(11):
        for j in range(10):
            B[i, j] = trial.suggest_float(f"B_{i}_{j}", -0.99, 0.99)

    rkhs = RKHS(data_series=data_series, m=11, fractal_depth=2, s_values=B)

    kernel = fractal_kernel(rkhs)

    svr = SVR(kernel=kernel, C = 10)
```

```

loocv = TimeSeriesSplit(n_splits=5)

scores = cross_val_score(svr, X, y,
                        cv=loocv,
                        scoring="neg_mean_squared_error",
                        n_jobs=-1)

# maximize mean squared error
# since cross_val_score returns negative mean squared error
return - scores.mean()

```

```

[43]: sampler = optuna.samplers.TPESampler(n_startup_trials=100, seed=42)
study = optuna.create_study(direction="minimize", sampler=sampler)

study.optimize(objective, n_trials=200, show_progress_bar=True)

print("Best vertical scaling factors:")
for k, v in study.best_params.items():
    print(f" {k} = {v:.4f}")
print("Best MSE:", study.best_value)

```

[I 2025-04-29 13:43:47,124] A new study created in memory with name: no-name-57e04187-09dc-459f-8db4-b4ccd860d77b

0%| | 0/200 [00:00<?, ?it/s]

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 13:44:08,098] Trial 0 finished with value: 185.20794658897242 and parameters: {'B_0_0': -0.24841056468222222, 'B_0_1': 0.892414326691634, 'B_0_2': 0.4593480047865821, 'B_0_3': 0.19534379871013252, 'B_0_4': -0.6810830919239756, 'B_0_5': -0.6811308497343187, 'B_0_6': -0.874994447906965, 'B_0_7': 0.7250287686343717, 'B_0_8': 0.20020772325155334, 'B_0_9': 0.41198370403617, 'B_1_0': -0.9492427012943111, 'B_1_1': 0.9304215072807487, 'B_1_2': 0.658236428784835, 'B_1_3': -0.5695685608570131, 'B_1_4': -0.6299865649299408, 'B_1_5': -0.6268590704902011, 'B_1_6': -0.38760035894011535, 'B_1_7':

0.04901773463183101, 'B_1_8': -0.13474886308861078, 'B_1_9': -0.413366302407877,
'B_2_0': 0.22146873155031144, 'B_2_1': -0.7138021559089571, 'B_2_2':
-0.4115535959002681, 'B_2_3': -0.2646035502784905, 'B_2_4':
-0.08698143125026891, 'B_2_5': 0.5646484035581669, 'B_2_6': -0.5946459113264477,
'B_2_7': 0.028184188058951065, 'B_2_8': 0.18298084634684408, 'B_2_9':
-0.8980281828144046, 'B_3_0': 0.2129388067648479, 'B_3_1': -0.6523622350991627,
'B_3_2': -0.8611978458891465, 'B_3_3': 0.8887933637615999, 'B_3_4':
0.9219514254876275, 'B_3_5': 0.610626749270593, 'B_3_6': -0.3868647370367261,
'B_3_7': -0.79660921426736, 'B_3_8': 0.3647813924940706, 'B_3_9':
-0.11849806239558947, 'B_4_0': -0.7483642950073379, 'B_4_1':
-0.009549717979685068, 'B_4_2': -0.9219107281918676, 'B_4_3':
0.8104543961159885, 'B_4_4': -0.4776156364319665, 'B_4_5': 0.3217941230208843,
'B_4_6': -0.3728120693429663, 'B_4_7': 0.03973468193206542, 'B_4_8':
0.09248635309969377, 'B_4_9': -0.6239881780594565, 'B_5_0': 0.9297775629738259,
'B_5_1': 0.5447629902550069, 'B_5_2': 0.8702079042970945, 'B_5_3':
0.7817581538467446, 'B_5_4': 0.1938419580459485, 'B_5_5': 0.8353109853457714,
'B_5_6': -0.8147848459371994, 'B_5_7': -0.6019539324100924, 'B_5_8':
-0.9004499679571346, 'B_5_9': -0.34584594508873656, 'B_6_0':
-0.2204189664148256, 'B_6_1': -0.45272891708768614, 'B_6_2': 0.65090026812082,
'B_6_3': -0.2836284131466932, 'B_6_4': -0.4337496708189861, 'B_6_5':
0.08453824465333204, 'B_6_6': -0.7109700345499699, 'B_6_7': 0.5983500218929985,
'B_6_8': -0.8423897255140538, 'B_6_9': 0.9640361344690243, 'B_7_0':
0.5390446432073817, 'B_7_1': -0.5965429505623386, 'B_7_2': -0.9790662080952672,
'B_7_3': 0.6246136283405717, 'B_7_4': 0.4095775408182818, 'B_7_5':
0.4534341927211549, 'B_7_6': 0.5371152864381725, 'B_7_7': -0.8433915895665011,
'B_7_8': -0.2802378574823402, 'B_7_9': -0.7605792621402432, 'B_8_0':
0.7189447832336753, 'B_8_1': 0.24413029111856477, 'B_8_2': -0.3348219107917546,
'B_8_3': -0.8641544664336732, 'B_8_4': -0.37425500300298886, 'B_8_5':
-0.3461370223870408, 'B_8_6': 0.45462023310936694, 'B_8_7': 0.27236379328332205,
'B_8_8': 0.7666812303011266, 'B_8_9': -0.055014448179340425, 'B_9_0':
-0.7532033930421627, 'B_9_1': 0.42222467870153, 'B_9_2': 0.516354396261457,
'B_9_3': 0.1213288511876025, 'B_9_4': 0.5365150163100307, 'B_9_5':
-0.012284719198506378, 'B_9_6': 0.045011002176348125, 'B_9_7':
-0.14346878365007176, 'B_9_8': -0.9396701290466916, 'B_9_9':
-0.7763749745532572, 'B_10_0': -0.9277702123402661, 'B_10_1':
0.27009261430228526, 'B_10_2': -0.36757515746887315, 'B_10_3':
0.016969968506111544, 'B_10_4': 0.8069816183736642, 'B_10_5':
-0.4964013862852276, 'B_10_6': -0.1774418123894531, 'B_10_7':
0.5059912543152363, 'B_10_8': -0.5369796323265875, 'B_10_9':
-0.8375797785389899}. Best is trial 0 with value: 185.20794658897242.

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
```

```

    Built F_[u_8].
    Built F_[u_9].
    Built F_[u_10].
    Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 13:44:25,461] Trial 1 finished with value: 230.68588330650027 and
parameters: {'B_0_0': -0.4162921232307393, 'B_0_1': -0.6707818512370712,
'B_0_2': 0.8508013516382946, 'B_0_3': 0.6100783515375456, 'B_0_4':
0.26413943789063854, 'B_0_5': 0.735491968571681, 'B_0_6': 0.6012707122602465,
'B_0_7': -0.6205912834056491, 'B_0_8': 0.777266817010156, 'B_0_9':
0.0778976389929884, 'B_1_0': 0.6087315072248438, 'B_1_1': 0.7842607738485166,
'B_1_2': -0.3603531195557095, 'B_1_3': -0.7720971894352, 'B_1_4':
-0.5386883781669555, 'B_1_5': -0.14432657852001252, 'B_1_6': 0.6296692365265364,
'B_1_7': 0.7142465548475601, 'B_1_8': -0.9762347815482424, 'B_1_9':
0.021279659103580206, 'B_2_0': -0.16352621376541765, 'B_2_1':
-0.5502265352679541, 'B_2_2': -0.752666572679308, 'B_2_3': -0.3215219606208166,
'B_2_4': 0.8769612137467879, 'B_2_5': -0.35005819459890464, 'B_2_6':
0.0372054310518648, 'B_2_7': 0.4019775386124522, 'B_2_8': -0.27001338728899793,
'B_2_9': 0.9341285237875021, 'B_3_0': 0.9156456439853802, 'B_3_1':
-0.491471054265779, 'B_3_2': -0.005447958333076763, 'B_3_3':
-0.3942609465627961, 'B_3_4': -0.42601582113261416, 'B_3_5': -0.916963844238025,
'B_3_6': 0.2169373812801958, 'B_3_7': 0.005304465993145713, 'B_3_8':
-0.8880720725250211, 'B_3_9': -0.4382800008115093, 'B_4_0': 0.8083664542139744,
'B_4_1': -0.5156674564793946, 'B_4_2': -0.7031081532593783, 'B_4_3':
-0.02088353465042525, 'B_4_4': 0.9615878991389895, 'B_4_5': -0.5107305624072291,
'B_4_6': 0.34082838386363945, 'B_4_7': 0.5180068383508607, 'B_4_8':
-0.5194776628950486, 'B_4_9': 0.45186837025148185, 'B_5_0':
-0.26178939721587857, 'B_5_1': 0.2619655445752873, 'B_5_2': 0.26438882730657154,
'B_5_3': 0.0708338744680217, 'B_5_4': -0.8112262552922715, 'B_5_5':
0.6638989412666911, 'B_5_6': -0.35485547135596307, 'B_5_7': -0.6206933494082887,
'B_5_8': -0.9092652197215675, 'B_5_9': 0.1799680275127189, 'B_6_0':
0.3515774364477191, 'B_6_1': -0.9571560987228448, 'B_6_2': 0.02394425543257639,
'B_6_3': -0.5415383651080828, 'B_6_4': 0.28744212501071065, 'B_6_5':
-0.6447544705701169, 'B_6_6': 0.3780567214428825, 'B_6_7': -0.22426401432493592,
'B_6_8': 0.8647253776987343, 'B_6_9': -0.7177085305909334, 'B_7_0':
-0.3146886249204881, 'B_7_1': -0.7653224279436337, 'B_7_2': 0.8408933641915544,
'B_7_3': 0.7471319196943425, 'B_7_4': -0.47927557712399194, 'B_7_5':
0.3167684111476745, 'B_7_6': 0.6280999563984073, 'B_7_7': 0.10929760696693536,
'B_7_8': 0.058708145144892754, 'B_7_9': -0.5111324640171057, 'B_8_0':
-0.8056565197443195, 'B_8_1': 0.7864872007475869, 'B_8_2': 0.7928277531833943,
'B_8_3': 0.26354088540107057, 'B_8_4': -0.31872101372357264, 'B_8_5':
-0.2985650422669315, 'B_8_6': 0.447392244163074, 'B_8_7': 0.7862783147061025,
'B_8_8': 0.7664311200449323, 'B_8_9': 0.5541535807980953, 'B_9_0':
0.28122265938548985, 'B_9_1': -0.8234028693098033, 'B_9_2': -0.6699751460926647,
'B_9_3': 0.7891372932836169, 'B_9_4': 0.21072953812598816, 'B_9_5':
-0.9717898377990732, 'B_9_6': -0.7890863451252564, 'B_9_7': 0.3237335028339505,

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'B_9_8': -0.979978063984487, 'B_9_9': -0.6716000581933527, 'B_10_0': 0.09649290294584056, 'B_10_1': 0.37995249143153265, 'B_10_2': 0.3008832938151491, 'B_10_3': -0.5459467672680915, 'B_10_4': 0.4201148582681209, 'B_10_5': -0.5202468067563358, 'B_10_6': -0.3457085976446499, 'B_10_7': 0.48805298213368786, 'B_10_8': 0.29627314011348505, 'B_10_9': 0.6914623527784722}. Best is trial 0 with value: 185.20794658897242.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:44:42,529] Trial 2 finished with value: 172.61952341356672 and parameters: {'B_0_0': 0.3120735267546799, 'B_0_1': 0.1352510346042337, 'B_0_2': -0.8045239597003768, 'B_0_3': -0.2619227099423216, 'B_0_4': -0.46489931199018364, 'B_0_5': -0.5069005061094145, 'B_0_6': 0.9365608984098421, 'B_0_7': -0.21166650515981444, 'B_0_8': 0.7762521792506842, 'B_0_9': 0.25965447947458054, 'B_1_0': 0.5837263810124638, 'B_1_1': 0.005221444348280424, 'B_1_2': 0.1522696915601911, 'B_1_3': -0.01481496623864953, 'B_1_4': -0.6034188841598719, 'B_1_5': 0.4404551882177805, 'B_1_6': -0.43407072236710553, 'B_1_7': -0.9418543864657214, 'B_1_8': 0.2880351458961923, 'B_1_9': -0.6393208547740431, 'B_2_0': 0.8721079970187704, 'B_2_1': 0.8987785824651231, 'B_2_2': 0.8214314926364881, 'B_2_3': -0.2570857734942201, 'B_2_4': -0.9593958992728425, 'B_2_5': 0.8480707539236962, 'B_2_6': -0.14219538633171758, 'B_2_7': 0.9239765417064658, 'B_2_8': 0.9179675546367205, 'B_2_9': 0.698958721825373, 'B_3_0': -0.40699119370222037, 'B_3_1': -0.22750649736818795, 'B_3_2': 0.6952506096033766, 'B_3_3': -0.3624944297905702, 'B_3_4': -0.6544043615615369, 'B_3_5': 0.1124664996675333, 'B_3_6': 0.8635864528383463, 'B_3_7': 0.38813899741644664, 'B_3_8': 0.13872111677694265, 'B_3_9': -0.7975905423338783, 'B_4_0': 0.22771430886435606, 'B_4_1': 0.9703066232064412, 'B_4_2': -0.7126336498316824, 'B_4_3': 0.036292711680198764, 'B_4_4': 0.7471986824173518, 'B_4_5': 0.47672186315332477, 'B_4_6': 0.39009116717063064, 'B_4_7': 0.40091848629447635, 'B_4_8': -0.27820752058488474, 'B_4_9': -0.4086881483563032, 'B_5_0': 0.6125350878474569, 'B_5_1': 0.6140245214647779, 'B_5_2': 0.7268031907886052, 'B_5_3': 0.8182162940618132, 'B_5_4': 0.02245794974465687, 'B_5_5': 0.003002263480655243, 'B_5_6': 0.5906244543542147, 'B_5_7': 0.29692858293997504, 'B_5_8': 0.3998944169702525, 'B_5_9': 0.5856694854834801, 'B_6_0': 0.7722105767987812, 'B_6_1': -0.32076958943395906, 'B_6_2': -0.2463457537729109, 'B_6_3': -0.8039157591150794, 'B_6_4':

```

0.15499467917242438, 'B_6_5': -0.9188342978824506, 'B_6_6':
-0.06811592409772893, 'B_6_7': 0.08443637672100168, 'B_6_8':
-0.4226483207859969, 'B_6_9': 0.17984985592664127, 'B_7_0': -0.9296095051206821,
'B_7_1': -0.9160505862765554, 'B_7_2': 0.6387491101061233, 'B_7_3':
-0.27682253000569945, 'B_7_4': -0.7384201849492682, 'B_7_5':
0.04404165490851275, 'B_7_6': 0.5345872351352494, 'B_7_7': -0.5626743655562505,
'B_7_8': 0.24332314212162043, 'B_7_9': -0.8210120193123394, 'B_8_0':
-0.8876701920861567, 'B_8_1': 0.06208217050493303, 'B_8_2': 0.08045754078801082,
'B_8_3': 0.2721112049664489, 'B_8_4': 0.44766084077086976, 'B_8_5':
0.9421871173358185, 'B_8_6': 0.032274689636366816, 'B_8_7':
-0.35054618357633294, 'B_8_8': 0.5844686656420333, 'B_8_9': -0.4537521425010931,
'B_9_0': -0.1208365870028405, 'B_9_1': -0.8346563649423134, 'B_9_2':
-0.9398055280373941, 'B_9_3': 0.9160438610622916, 'B_9_4': 0.6652406386141676,
'B_9_5': 0.388028928065522, 'B_9_6': -0.18027317005974564, 'B_9_7':
-0.6468772462597254, 'B_9_8': -0.6802546555112496, 'B_9_9':
-0.49451906163410125, 'B_10_0': 0.09746879611811843, 'B_10_1':
0.42489992694612333, 'B_10_2': 0.31719080590110793, 'B_10_3':
-0.4357308840470332, 'B_10_4': 0.9006332557131242, 'B_10_5':
0.47103589505762167, 'B_10_6': 0.10762102397257345, 'B_10_7':
0.22120707754401736, 'B_10_8': -0.15919187639297594, 'B_10_9':
-0.4994926407877082}. Best is trial 2 with value: 172.61952341356672.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 13:44:58,276] Trial 3 finished with value: 146.62841443584088 and
parameters: {'B_0_0': -0.28517409627050205, 'B_0_1': 0.5105352987194509,
'B_0_2': -0.9615008925130833, 'B_0_3': -0.7601761717963058, 'B_0_4':
-0.8989147687969296, 'B_0_5': -0.9093569714084391, 'B_0_6': 0.7038119563417944,
'B_0_7': 0.40324256157244687, 'B_0_8': -0.051135818407096134, 'B_0_9':
-0.796288361911017, 'B_1_0': -0.016600567268671984, 'B_1_1':
-0.05252589385447992, 'B_1_2': -0.6470602975781699, 'B_1_3':
-0.13097373450881333, 'B_1_4': -0.20096062589320063, 'B_1_5':
0.2293831941433886, 'B_1_6': 0.2674854287179347, 'B_1_7': -0.9002980606513519,
'B_1_8': -0.24826702303958703, 'B_1_9': 0.2492026331141881, 'B_2_0':
0.006209791988573632, 'B_2_1': 0.7058498855528781, 'B_2_2': 0.3142133906055111,
'B_2_3': -0.6673898343787692, 'B_2_4': -0.8502738801471489, 'B_2_5':

```

0.28199017084850486, 'B_2_6': -0.9375076051275888, 'B_2_7': 0.16983565092145736,
 'B_2_8': 0.8716558780214161, 'B_2_9': 0.14943887219424035, 'B_3_0':
 -0.2214235461110866, 'B_3_1': 0.28371067251585935, 'B_3_2':
 -0.08265927682679708, 'B_3_3': 0.0903212428455511, 'B_3_4': 0.8741003213775198,
 'B_3_5': -0.22551677715446694, 'B_3_6': 0.9131573163713502, 'B_3_7':
 0.8025942710730063, 'B_3_8': -0.602333553117193, 'B_3_9': -0.8526646242671724,
 'B_4_0': -0.7904595572726952, 'B_4_1': -0.9539207852099315, 'B_4_2':
 -0.8030029377032618, 'B_4_3': 0.3623534113643865, 'B_4_4': -0.8490464760487466,
 'B_4_5': -0.3584282520183526, 'B_4_6': 0.6828531157195201, 'B_4_7':
 -0.9439215672430648, 'B_4_8': 0.6226475955260928, 'B_4_9': -0.4319275459486681,
 'B_5_0': -0.7560336413091207, 'B_5_1': 0.38953958742101835, 'B_5_2':
 0.2553068366241702, 'B_5_3': 0.7473945867835647, 'B_5_4': 0.46544066673169393,
 'B_5_5': 0.6008922421620002, 'B_5_6': -0.4315715463088131, 'B_5_7':
 -0.6386697033161488, 'B_5_8': 0.4962172082488996, 'B_5_9': 0.6075327837491828,
 'B_6_0': 0.9712001811613331, 'B_6_1': -0.17301699971537554, 'B_6_2':
 -0.25340419013028936, 'B_6_3': 0.5472976622691537, 'B_6_4': -0.3152089902990247,
 'B_6_5': 0.852899504695058, 'B_6_6': 0.7096572486491635, 'B_6_7':
 -0.14059182579746365, 'B_6_8': 0.4967247142271647, 'B_6_9': 0.503994890687671,
 'B_7_0': -0.7858147397048534, 'B_7_1': 0.797054755225542, 'B_7_2':
 0.010399697446757061, 'B_7_3': 0.6463857828933284, 'B_7_4':
 -0.35630178995938877, 'B_7_5': 0.783135992422477, 'B_7_6': -0.21938067610635703,
 'B_7_7': -0.9685414500690093, 'B_7_8': 0.8026563133101421, 'B_7_9':
 -0.8092523799634556, 'B_8_0': -0.3577589975709785, 'B_8_1': 0.8911226947605937,
 'B_8_2': 0.8922021509363611, 'B_8_3': 0.14540701848410653, 'B_8_4':
 0.26103768009620265, 'B_8_5': -0.10207786648292683, 'B_8_6':
 -0.40944267203783224, 'B_8_7': -0.3392442001675664, 'B_8_8': 0.341586543032536,
 'B_8_9': 0.4997015682866064, 'B_9_0': 0.57732650657718, 'B_9_1':
 0.5734439227332166, 'B_9_2': -0.8094119159635931, 'B_9_3':
 -0.011047796688888667, 'B_9_4': -0.8760336551670443, 'B_9_5':
 0.09806718700099637, 'B_9_6': -0.1157696072807135, 'B_9_7': 0.7676542818614336,
 'B_9_8': -0.29518827514688417, 'B_9_9': -0.7582073074733404, 'B_10_0':
 -0.7068764695353851, 'B_10_1': 0.517791050800595, 'B_10_2': 0.23407176536619678,
 'B_10_3': -0.7897771012768753, 'B_10_4': -0.8234685238923051, 'B_10_5':
 0.39791888028905764, 'B_10_6': -0.8459292473988969, 'B_10_7':
 0.6372829173949053, 'B_10_8': 0.4083596097698625, 'B_10_9':
 -0.8289294143290384}. Best is trial 3 with value: 146.62841443584088.

Building the Basis functions for the RKHS:

Built F__[u₁].
 Built F__[u₂].
 Built F__[u₃].
 Built F__[u₄].
 Built F__[u₅].
 Built F__[u₆].
 Built F__[u₇].
 Built F__[u₈].
 Built F__[u₉].
 Built F__[u₁₀].
 Built F__[u₁₁].

Established the refined x grid with 1001 points.
 Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].
 Built matrix B = inverse(A).
 [I 2025-04-29 13:45:13,972] Trial 4 finished with value: 202.92332611563316 and
 parameters: {'B_0_0': -0.8220213261113201, 'B_0_1': 0.9635463654323273, 'B_0_2':
 -0.24894382440288176, 'B_0_3': -0.256128548807556, 'B_0_4': 0.6193431431698551,
 'B_0_5': 0.8855521832200401, 'B_0_6': 0.9622821063692844, 'B_0_7':
 0.5016888068127043, 'B_0_8': -0.24500602064878674, 'B_0_9': -0.8246685809366359,
 'B_1_0': 0.5487508935363248, 'B_1_1': 0.11564041447689388, 'B_1_2':
 -0.15004042169098697, 'B_1_3': 0.8045816824875773, 'B_1_4': -0.7698289850338204,
 'B_1_5': -0.014602293504098918, 'B_1_6': -0.9675197833605103, 'B_1_7':
 -0.06205192885163002, 'B_1_8': -0.878519514149962, 'B_1_9': -0.7547405257892176,
 'B_2_0': -0.7572980313813323, 'B_2_1': 0.2954363981898058, 'B_2_2':
 0.487168860945538, 'B_2_3': 0.1650701548923761, 'B_2_4': 0.9151016459795931,
 'B_2_5': -0.247756252543066, 'B_2_6': -0.4242900691619157, 'B_2_7':
 0.7298262738151313, 'B_2_8': -0.5472802397314838, 'B_2_9': 0.9171806280924104,
 'B_3_0': -0.9659341401141637, 'B_3_1': 0.9303600768811253, 'B_3_2':
 -0.9045433743378593, 'B_3_3': 0.7744633651221808, 'B_3_4': 0.05484819599087376,
 'B_3_5': 0.9760702963162147, 'B_3_6': -0.8438828018239102, 'B_3_7':
 0.10663148311461512, 'B_3_8': 0.929219020525816, 'B_3_9': 0.04573373145689463,
 'B_4_0': 0.2562093035078199, 'B_4_1': 0.38758240418954193, 'B_4_2':
 -0.09000869175980897, 'B_4_3': 0.25256499856644554, 'B_4_4':
 0.16694233760773858, 'B_4_5': 0.7942928607721582, 'B_4_6': -0.9000161669239134,
 'B_4_7': -0.433692884607384, 'B_4_8': 0.8918147384715862, 'B_4_9':
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 'B_5_2': -0.4407852576973573, 'B_5_3': -0.6175201037469527, 'B_5_4':
 -0.07187715821883534, 'B_5_5': -0.29036258850841545, 'B_5_6':
 0.1656391014647267, 'B_5_7': -0.8360854188093301, 'B_5_8': 0.9393017191790096,
 'B_5_9': 0.9626972740696138, 'B_6_0': 0.3923601937590955, 'B_6_1':
 0.07147080536135841, 'B_6_2': -0.377135319753071, 'B_6_3': 0.6213141390197583,
 'B_6_4': 0.36576772165668103, 'B_6_5': -0.6680184600971153, 'B_6_6':
 0.813635825297808, 'B_6_7': 0.6386237409878746, 'B_6_8': 0.8906038283180095,
 'B_6_9': 0.4469246266089528, 'B_7_0': 0.2245620879528638, 'B_7_1':
 -0.16187878814457468, 'B_7_2': 0.8568023970409462, 'B_7_3': 0.7248065012108085,
 'B_7_4': -0.900467033189745, 'B_7_5': -0.937793390495441, 'B_7_6':
 -0.24460253358146178, 'B_7_7': 0.614895594948029, 'B_7_8': 0.9648067360435901,
 'B_7_9': -0.6921745556150142, 'B_8_0': 0.18637881639722753, 'B_8_1':
 -0.2358361038705774, 'B_8_2': 0.9304305076729142, 'B_8_3': 0.6773954678087033,
 'B_8_4': 0.669890835328053, 'B_8_5': -0.06198754360595882, 'B_8_6':
 -0.16865738537142294, 'B_8_7': -0.4486539975772017, 'B_8_8':
 -0.8783765166311643, 'B_8_9': 0.7221503049850053, 'B_9_0': 0.6195439980775537,
 'B_9_1': 0.9894409931065387, 'B_9_2': 0.9833409374063327, 'B_9_3':
 0.10975477709320236, 'B_9_4': 0.5325950820574108, 'B_9_5': 0.8806361451672076,
 'B_9_6': 0.6923018335412747, 'B_9_7': -0.5002507585484686, 'B_9_8':
 -0.09792261208601483, 'B_9_9': -0.7342643580000399, 'B_10_0':
 0.8990210339722704, 'B_10_1': 0.2102257762127424, 'B_10_2': -0.5372872451031439,
 'B_10_3': 0.3399673551235962, 'B_10_4': 0.2338939161066338, 'B_10_5':
 -0.2808378182949758, 'B_10_6': -0.7651559674447346, 'B_10_7':

0.3397149272737432, 'B_10_8': 0.04020924778951063, 'B_10_9':
0.5391904156365659}. Best is trial 3 with value: 146.62841443584088.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:45:32,102] Trial 5 finished with value: 234.0815751548317 and
parameters: {'B_0_0': 0.03992373220174694, 'B_0_1': 0.6973193706307095, 'B_0_2':
0.10277554077348139, 'B_0_3': 0.12065718364006472, 'B_0_4': 0.745774133263523,
'B_0_5': -0.19110392489945383, 'B_0_6': -0.7246498476677313, 'B_0_7':
-0.9330103008995888, 'B_0_8': 0.5051717662337656, 'B_0_9': 0.23821291167985992,
'B_1_0': 0.40407794083646276, 'B_1_1': -0.5683309602123567, 'B_1_2':
-0.7199844783381959, 'B_1_3': -0.9612015619775938, 'B_1_4': -0.295836633562938,
'B_1_5': 0.17803701997217347, 'B_1_6': -0.21335679070253, 'B_1_7':
-0.12379965439301643, 'B_1_8': 0.8002342150976218, 'B_1_9': -0.3004541752938653,
'B_2_0': 0.02769918853642528, 'B_2_1': 0.5616329652274632, 'B_2_2':
-0.20484529100388504, 'B_2_3': 0.2417316664511895, 'B_2_4': 0.7174801433185554,
'B_2_5': 0.8900508348421314, 'B_2_6': -0.6987945077605049, 'B_2_7':
0.844643497819759, 'B_2_8': -0.015609739702514336, 'B_2_9': -0.4786761111680624,
'B_3_0': -0.08091120264824259, 'B_3_1': 0.9504644990652444, 'B_3_2':
-0.014616173894118178, 'B_3_3': -0.33907181163073374, 'B_3_4':
0.2641336915471171, 'B_3_5': -0.5145116748191777, 'B_3_6': -0.8397906103448454,
'B_3_7': -0.7348181506169145, 'B_3_8': -0.7364692388636105, 'B_3_9':
-0.6892326668456572, 'B_4_0': -0.715122198154168, 'B_4_1': 0.278931994710365,
'B_4_2': -0.6298774328896932, 'B_4_3': -0.30557877901875086, 'B_4_4':
0.7856410516139034, 'B_4_5': -0.05155595227951271, 'B_4_6': 0.33176432227163377,
'B_4_7': -0.648806655020773, 'B_4_8': -0.6092677427588319, 'B_4_9':
-0.9090801397923718, 'B_5_0': -0.6555085751171141, 'B_5_1': -0.4383911287167219,
'B_5_2': -0.6395192411320413, 'B_5_3': -0.8143689831610299, 'B_5_4':
-0.7511409752208104, 'B_5_5': -0.07765803929520299, 'B_5_6':
-0.5814592375565308, 'B_5_7': -0.2687456751248106, 'B_5_8':
0.006766196292616655, 'B_5_9': 0.3769817606861432, 'B_6_0': -0.912161963114624,
'B_6_1': 0.5928325898399045, 'B_6_2': 0.25324277119199734, 'B_6_3':
-0.8281171167412336, 'B_6_4': 0.739685675731419, 'B_6_5': 0.8333273530529901,
'B_6_6': -0.8690656394873698, 'B_6_7': -0.44178225666853665, 'B_6_8':
0.6062785339902614, 'B_6_9': 0.4915541869596436, 'B_7_0': -0.6246483816743729,
'B_7_1': -0.5754883397933136, 'B_7_2': -0.25646523647306363, 'B_7_3':

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-0.030644489321777768, 'B_7_4': 0.2341444476299861, 'B_7_5':
-0.25955099365185064, 'B_7_6': -0.07418126205636721, 'B_7_7': 0.489992457504838,
'B_7_8': -0.9173672582766161, 'B_7_9': -0.4901748501988389, 'B_8_0':
0.4224321800514139, 'B_8_1': 0.7825095386206546, 'B_8_2': 0.02312133538901895,
'B_8_3': 0.06358470082532497, 'B_8_4': -0.7777994175472434, 'B_8_5':
-0.10412351368955985, 'B_8_6': 0.0645821875809458, 'B_8_7': -0.5099084028032352,
'B_8_8': -0.45689840272022564, 'B_8_9': -0.24297735705284718, 'B_9_0':
-0.9502590284001018, 'B_9_1': -0.352283252145307, 'B_9_2': -0.5713329461468416,
'B_9_3': -0.34155524268772897, 'B_9_4': -0.7528709789978826, 'B_9_5':
0.773244015864992, 'B_9_6': 0.18531305803701636, 'B_9_7': 0.35462259190608947,
'B_9_8': 0.5725590524425299, 'B_9_9': -0.003084446120466544, 'B_10_0':
-0.817897829586901, 'B_10_1': 0.07347095280072469, 'B_10_2':
0.17194541368134075, 'B_10_3': 0.4859701588849734, 'B_10_4':
-0.1353140984652348, 'B_10_5': -0.7373910004647837, 'B_10_6':
-0.4281237065185255, 'B_10_7': -0.27109705313070254, 'B_10_8':
0.28891613783657033, 'B_10_9': 0.14014104324444543}. Best is trial 3 with value:
146.62841443584088.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 13:45:46,751] Trial 6 finished with value: 204.06071087777502 and
parameters: {'B_0_0': -0.28492848272226445, 'B_0_1': 0.9633001926100997,
'B_0_2': 0.2094341423266366, 'B_0_3': -0.5202909523627309, 'B_0_4':
-0.7884707042116006, 'B_0_5': -0.6873389044150227, 'B_0_6': -0.5030036977986738,
'B_0_7': -0.6718508809460797, 'B_0_8': -0.6205972923784147, 'B_0_9':
-0.4255115659861828, 'B_1_0': -0.6467202813163855, 'B_1_1': 0.7855955407603219,
'B_1_2': -0.8311371835899484, 'B_1_3': 0.04853255134910417, 'B_1_4':
-0.1774142825604702, 'B_1_5': 0.9551096614790406, 'B_1_6': -0.7681629737072564,
'B_1_7': -0.20224591388943158, 'B_1_8': 0.9295514578852304, 'B_1_9':
0.7237041092700809, 'B_2_0': 0.6278027004795743, 'B_2_1': -0.4793524024510192,
'B_2_2': -0.6516425769676697, 'B_2_3': 0.33391357545037326, 'B_2_4':
0.8501644584726198, 'B_2_5': 0.112390528167581, 'B_2_6': 0.14179312515040166,
'B_2_7': -0.4356413945526373, 'B_2_8': 0.533596007720035, 'B_2_9':
-0.6196533778561037, 'B_3_0': -0.34911511191959754, 'B_3_1':
-0.14763585153949477, 'B_3_2': 0.015068549795220987, 'B_3_3':
-0.5100287298181412, 'B_3_4': -0.762623087016377, 'B_3_5': 0.2190276840344325,

```


'B_3_6': -0.41851150458429354, 'B_3_7': 0.1608516784167724, 'B_3_8': -0.6843618237570794, 'B_3_9': -0.03734259832746134, 'B_4_0': 0.06452707645213995, 'B_4_1': -0.8873893970915947, 'B_4_2': -0.3235235291760372, 'B_4_3': -0.723858939660831, 'B_4_4': -0.8645175584639199, 'B_4_5': 0.9701212601320917, 'B_4_6': -0.35173938695004847, 'B_4_7': 0.6135514027921771, 'B_4_8': -0.4858115035677476, 'B_4_9': 0.35937539000338004, 'B_5_0': 0.5152511625815794, 'B_5_1': 0.18936470640353176, 'B_5_2': -0.05627914667068645, 'B_5_3': -0.17455498998840835, 'B_5_4': -0.2992408322448693, 'B_5_5': 0.8504677056106951, 'B_5_6': 0.6546264274197038, 'B_5_7': 0.920753283119695, 'B_5_8': -0.7438914974986215, 'B_5_9': 0.4571176009032156, 'B_6_0': 0.8679141045056549, 'B_6_1': -0.6311585289919929, 'B_6_2': -0.8583373906137806, 'B_6_3': 0.4774188855943169, 'B_6_4': 0.14745676409622566, 'B_6_5': 0.6768209779813787, 'B_6_6': -0.7132506942799468, 'B_6_7': 0.5846292774825825, 'B_6_8': -0.5907779063054659, 'B_6_9': -0.665961233125905, 'B_7_0': -0.664753720096634, 'B_7_1': 0.6228579460581365, 'B_7_2': 0.32709049697847625, 'B_7_3': 0.045669541042856254, 'B_7_4': -0.27951564143546503, 'B_7_5': 0.7468570708099544, 'B_7_6': -0.2129586873031819, 'B_7_7': 0.6268668901537227, 'B_7_8': -0.12051288103096758, 'B_7_9': -0.243650029738683, 'B_8_0': -0.07389402437417925, 'B_8_1': -0.3932718091549987, 'B_8_2': 0.4902665727489772, 'B_8_3': 0.005386372383108684, 'B_8_4': -0.5302188636093017, 'B_8_5': 0.7911576550836457, 'B_8_6': -0.2298953816810414, 'B_8_7': 0.08623466500569732, 'B_8_8': 0.8048147797098029, 'B_8_9': 0.2459912319097044, 'B_9_0': -0.7585418793974391, 'B_9_1': 0.8708676047546808, 'B_9_2': 0.25286194508140736, 'B_9_3': -0.3268868829789695, 'B_9_4': -0.7142412961264932, 'B_9_5': 0.582169874755186, 'B_9_6': 0.23774405673845678, 'B_9_7': 0.06625296211311671, 'B_9_8': 0.7799073144408961, 'B_9_9': 0.5714224782245709, 'B_10_0': -0.6896837381291527, 'B_10_1': -0.37279030576481453, 'B_10_2': -0.49799150316735785, 'B_10_3': 0.4830136592939005, 'B_10_4': -0.9236057792231568, 'B_10_5': 0.13838157604520673, 'B_10_6': 0.5196681977665671, 'B_10_7': 0.745995960788264, 'B_10_8': -0.3126781375425032, 'B_10_9': 0.6360894632505854}. Best is trial 3 with value: 146.62841443584088.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$\langle F_{[u_i]}, F_{[u_j]} \rangle$].

Built matrix B = inverse(A).

[I 2025-04-29 13:46:03,868] Trial 7 finished with value: 174.1567247424878 and parameters: {'B_0_0': -0.7709491608286897, 'B_0_1': 0.6859755376343459, 'B_0_2':

-0.7375724485826748, 'B_0_3': -0.20337116469047278, 'B_0_4': 0.5886448242435163,
 'B_0_5': -0.693163493849428, 'B_0_6': -0.5360822374393699, 'B_0_7':
 0.44006008541827124, 'B_0_8': 0.4356723423612272, 'B_0_9': 0.27947231311288867,
 'B_1_0': 0.38401792004485813, 'B_1_1': 0.08459439782824041, 'B_1_2':
 -0.49143786336423356, 'B_1_3': -0.3055219328622395, 'B_1_4':
 -0.6304365207331772, 'B_1_5': 0.8087321114405841, 'B_1_6': 0.16511575363691855,
 'B_1_7': -0.19631419480799306, 'B_1_8': -0.07522850878461729, 'B_1_9':
 0.8856210124313943, 'B_2_0': -0.6863642218301612, 'B_2_1': 0.1707350673932586,
 'B_2_2': 0.011659584191242578, 'B_2_3': 0.22067938616060268, 'B_2_4':
 -0.9541418360347358, 'B_2_5': 0.7368053397094199, 'B_2_6': 0.8555941993175524,
 'B_2_7': 0.12896370350663378, 'B_2_8': 0.38936863127624677, 'B_2_9':
 0.8365487747310454, 'B_3_0': 0.41033249594052923, 'B_3_1': -0.6879726950297625,
 'B_3_2': 0.15105095313029016, 'B_3_3': 0.2112957918380547, 'B_3_4':
 -0.15022127082127568, 'B_3_5': 0.4681595865369512, 'B_3_6': 0.8600466892426495,
 'B_3_7': 0.842625655555417, 'B_3_8': -0.09733804461981843, 'B_3_9':
 -0.7657886692353046, 'B_4_0': 0.9599855739454224, 'B_4_1': 0.6710182111629495,
 'B_4_2': -0.7431678912175317, 'B_4_3': 0.8332669275823972, 'B_4_4':
 0.732394796883014, 'B_4_5': 0.037299353109622846, 'B_4_6': 0.18072536277495987,
 'B_4_7': -0.19997464633714213, 'B_4_8': -0.881571955132378, 'B_4_9':
 -0.3263094615411162, 'B_5_0': 0.5996498282240625, 'B_5_1': -0.9808285944508863,
 'B_5_2': -0.3296716400515345, 'B_5_3': -0.20162598668993226, 'B_5_4':
 0.07404329381708741, 'B_5_5': 0.8313141204972658, 'B_5_6': -0.3042349311553967,
 'B_5_7': -0.30303266024546915, 'B_5_8': 0.47025247125730174, 'B_5_9':
 -0.09460847703818187, 'B_6_0': -0.5452824505788357, 'B_6_1':
 -0.09416975805726702, 'B_6_2': -0.7111030996479963, 'B_6_3':
 -0.6407537667176779, 'B_6_4': -0.003231809975830058, 'B_6_5':
 -0.16052760998099513, 'B_6_6': 0.8213948841148382, 'B_6_7': -0.2724600797490664,
 'B_6_8': 0.15956493355052603, 'B_6_9': 0.26188329008067024, 'B_7_0':
 -0.9640729759550994, 'B_7_1': 0.323803996593087, 'B_7_2': -0.6374887855978921,
 'B_7_3': 0.9129192285895211, 'B_7_4': -0.6956477990488363, 'B_7_5':
 -0.16904423502049304, 'B_7_6': -0.8210076572042851, 'B_7_7': 0.9838110186549758,
 'B_7_8': 0.004346120455860358, 'B_7_9': 0.1888623342936866, 'B_8_0':
 -0.8571885747709136, 'B_8_1': 0.494921731390372, 'B_8_2': -0.5743869256707401,
 'B_8_3': 0.7881474930926131, 'B_8_4': -0.5838235118456259, 'B_8_5':
 -0.6124383130859402, 'B_8_6': -0.9176316576607721, 'B_8_7': -0.0553074486822015,
 'B_8_8': 0.12838544385998052, 'B_8_9': -0.8598968939318623, 'B_9_0':
 0.5455446810561209, 'B_9_1': -0.09248810719890532, 'B_9_2': 0.0482927332686085,
 'B_9_3': -0.11728976106230826, 'B_9_4': -0.19648913946685453, 'B_9_5':
 0.11808785599027138, 'B_9_6': -0.6826243130571892, 'B_9_7': -0.6297823016193631,
 'B_9_8': 0.7163355296067642, 'B_9_9': 0.8833086150245928, 'B_10_0':
 -0.25084755376608914, 'B_10_1': -0.4539255471757635, 'B_10_2':
 0.2851190956132512, 'B_10_3': -0.18070634122576912, 'B_10_4':
 -0.9397350157925172, 'B_10_5': -0.680817857214943, 'B_10_6': 0.4276250131178472,
 'B_10_7': 0.3146694049820997, 'B_10_8': -0.9363499348431027, 'B_10_9':
 -0.5504951193727601}. Best is trial 3 with value: 146.62841443584088.

Building the Basis functions for the RKHS:

Built $F_{[u_1]}$.

Built $F_{[u_2]}$.

```

Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:46:19,082] Trial 8 finished with value: 119.8836634080008 and parameters: {'B_0_0': -0.5324719027556186, 'B_0_1': 0.34034763232548215, 'B_0_2': -0.950973135246359, 'B_0_3': -0.7838650076705438, 'B_0_4': 0.593833849038915, 'B_0_5': -0.6364815691324195, 'B_0_6': 0.3024372935467119, 'B_0_7': -0.5183980935274815, 'B_0_8': -0.7931060423364966, 'B_0_9': -0.5085190618210809, 'B_1_0': 0.44008852507420504, 'B_1_1': 0.7042790068504459, 'B_1_2': 0.6538353318426433, 'B_1_3': -0.20357661135545946, 'B_1_4': 0.3328085704098793, 'B_1_5': -0.5841310950766745, 'B_1_6': -0.4095674940831937, 'B_1_7': 0.7847449206718171, 'B_1_8': -0.9642561914487426, 'B_1_9': -0.8206931089081532, 'B_2_0': -0.5783852148108659, 'B_2_1': -0.9374662363298369, 'B_2_2': -0.6307578385222012, 'B_2_3': 0.1644222907199906, 'B_2_4': -0.15557938982685404, 'B_2_5': 0.7774899879324102, 'B_2_6': 0.628538252242113, 'B_2_7': -0.3132016436382056, 'B_2_8': -0.4763416018060409, 'B_2_9': -0.23820903182881148, 'B_3_0': 0.17878398617931923, 'B_3_1': -0.45923399117070496, 'B_3_2': 0.2458148375412854, 'B_3_3': -0.1793649286613439, 'B_3_4': 0.10305341808692092, 'B_3_5': -0.12646947231207295, 'B_3_6': -0.406957796107003, 'B_3_7': 0.8879375477850702, 'B_3_8': 0.5219394724363264, 'B_3_9': -0.712575911982424, 'B_4_0': 0.7295665922778674, 'B_4_1': -0.02488622746596303, 'B_4_2': 0.7812134092503009, 'B_4_3': 0.5937134067756842, 'B_4_4': -0.1480772611509178, 'B_4_5': -0.9455107695261675, 'B_4_6': -0.4580188284178067, 'B_4_7': 0.08243574502851647, 'B_4_8': 0.26428687525577166, 'B_4_9': -0.4793823828422594, 'B_5_0': -0.7140749733358083, 'B_5_1': 0.6631618688626122, 'B_5_2': 0.9591163177930333, 'B_5_3': 0.050866560959317964, 'B_5_4': -0.650075014020359, 'B_5_5': -0.4508314934916494, 'B_5_6': -0.9535864604360158, 'B_5_7': 0.820311636989788, 'B_5_8': -0.7568528558775205, 'B_5_9': 0.15150262151822202, 'B_6_0': -0.44737066303933215, 'B_6_1': 0.10727244498127098, 'B_6_2': 0.29981236893669116, 'B_6_3': 0.6528887713402591, 'B_6_4': -0.5812858819140057, 'B_6_5': -0.9682282592562086, 'B_6_6': -0.7189664524637616, 'B_6_7': 0.792036910859248, 'B_6_8': 0.7403023535737803, 'B_6_9': 0.19287794229721045, 'B_7_0': 0.1990233836586337, 'B_7_1': 0.32677261560158577, 'B_7_2': -0.6427648683255698, 'B_7_3': 0.8205356529314203, 'B_7_4': -0.16083436071369528, 'B_7_5': -0.23138571406599506, 'B_7_6': 0.03745705646001829, 'B_7_7': -0.897007385785391, 'B_7_8': -0.6607589298629628, 'B_7_9': 0.4713065605242135, 'B_8_0': -0.8260586375082496, 'B_8_1': 0.20424117674344866, 'B_8_2': -0.5042087628309845, 'B_8_3': -0.21919468419688648, 'B_8_4': -0.41838640119401427, 'B_8_5':

```
-0.2857680213994007, 'B_8_6': 0.4337108922648061, 'B_8_7': -0.4016990030661135,
'B_8_8': 0.13148118778785634, 'B_8_9': -0.0474202036457827, 'B_9_0':
0.3240689074180434, 'B_9_1': 0.8649228838630207, 'B_9_2': 0.4604927524762956,
'B_9_3': -0.5644180503903016, 'B_9_4': -0.9282573925786564, 'B_9_5':
-0.47071719228634634, 'B_9_6': 0.18825430278650446, 'B_9_7':
-0.8881768893011843, 'B_9_8': -0.007194830541552011, 'B_9_9':
0.19174884085544064, 'B_10_0': -0.32819709618241755, 'B_10_1':
0.5364061634167274, 'B_10_2': -0.7789354587951318, 'B_10_3':
-0.8412271921630986, 'B_10_4': 0.45181373728313456, 'B_10_5':
-0.008927193911727027, 'B_10_6': 0.3730367449269174, 'B_10_7':
-0.12904186956458252, 'B_10_8': -0.5021239741865685, 'B_10_9':
0.6318225889949154}. Best is trial 8 with value: 119.8836634080008.
```

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```
[I 2025-04-29 13:46:34,222] Trial 9 finished with value: 175.8682874985775 and
parameters: {'B_0_0': 0.5928434403585789, 'B_0_1': 0.385499012291765, 'B_0_2':
-0.45115262828467384, 'B_0_3': 0.1786567204007925, 'B_0_4': -0.2752716840587469,
'B_0_5': -0.8086674948132644, 'B_0_6': 0.8262808794152408, 'B_0_7':
-0.7190991107804564, 'B_0_8': 0.8914699605651888, 'B_0_9': -0.10690856954324768,
'B_1_0': -0.623436800899533, 'B_1_1': 0.0829638758091491, 'B_1_2':
0.7384327550352885, 'B_1_3': 0.45980527509093116, 'B_1_4': 0.6069910727656704,
'B_1_5': 0.31439106608722045, 'B_1_6': 0.3807075977453478, 'B_1_7':
0.6914073900993323, 'B_1_8': -0.4956573424588118, 'B_1_9': -0.02093857198658189,
'B_2_0': -0.5520053051971876, 'B_2_1': 0.9655826558333611, 'B_2_2':
0.8792374925794944, 'B_2_3': -0.9119349134903583, 'B_2_4': 0.40703884158106307,
'B_2_5': 0.8419916684830184, 'B_2_6': -0.6324608166478796, 'B_2_7':
0.13453155649420645, 'B_2_8': 0.8226668292243229, 'B_2_9': -0.9227869624001183,
'B_3_0': 0.39089212914874305, 'B_3_1': -0.40124896540234944, 'B_3_2':
0.8403044668455302, 'B_3_3': 0.9326953254274282, 'B_3_4': 0.8796476484445992,
'B_3_5': -0.05105585101822174, 'B_3_6': 0.7168444489588406, 'B_3_7':
0.6822078090994395, 'B_3_8': -0.35818106297835384, 'B_3_9': 0.6512526388183411,
'B_4_0': -0.9167248832633246, 'B_4_1': 0.19061435939446492, 'B_4_2':
-0.534582502170348, 'B_4_3': -0.751277566160988, 'B_4_4': -0.8376326607741659,
'B_4_5': 0.3886517762387167, 'B_4_6': -0.31704757173922904, 'B_4_7':
0.4450382076269477, 'B_4_8': -0.8605944452180941, 'B_4_9': -0.3657251310953915,
```

```

'B_5_0': 0.07819275890316768, 'B_5_1': 0.5756318663811477, 'B_5_2':
-0.3588700441945015, 'B_5_3': 0.2492649253452781, 'B_5_4': 0.7642359415076505,
'B_5_5': 0.22940911260096297, 'B_5_6': -0.5287402399878052, 'B_5_7':
-0.9416864525180547, 'B_5_8': 0.7327957703238412, 'B_5_9': -0.9478865665162335,
'B_6_0': 0.7419093119147149, 'B_6_1': 0.05729552537387983, 'B_6_2':
0.8693540430555344, 'B_6_3': 0.5915908068318574, 'B_6_4': 0.9859095388560082,
'B_6_5': -0.2955906054056139, 'B_6_6': 0.5290328120836312, 'B_6_7':
-0.19417679105370056, 'B_6_8': -0.0398462717982605, 'B_6_9': 0.2524608171723728,
'B_7_0': 0.7398806860889506, 'B_7_1': 0.9584852690146044, 'B_7_2':
0.531181359451746, 'B_7_3': -0.16282177130867914, 'B_7_4': -0.1557131354914011,
'B_7_5': 0.4704129571460052, 'B_7_6': -0.5172212513787613, 'B_7_7':
-0.7712612559998386, 'B_7_8': -0.28784812787126246, 'B_7_9':
-0.42126679652491816, 'B_8_0': -0.4033099214971396, 'B_8_1':
-0.5274566529211784, 'B_8_2': -0.9066554845200034, 'B_8_3': -0.9546096092278407,
'B_8_4': 0.9656903316773424, 'B_8_5': -0.1430091952029926, 'B_8_6':
-0.2290332386238303, 'B_8_7': 0.3557016197322782, 'B_8_8': -0.5578573020271729,
'B_8_9': 0.8909231442214463, 'B_9_0': 0.566963128542793, 'B_9_1':
-0.8129662154217261, 'B_9_2': -0.1631900639458419, 'B_9_3': 0.7506542489730861,
'B_9_4': 0.8805694041369925, 'B_9_5': -0.06454500772525795, 'B_9_6':
0.2245545506372013, 'B_9_7': -0.6592727867376911, 'B_9_8': 0.9725138797512136,
'B_9_9': -0.531290031260782, 'B_10_0': 0.8766089127875487, 'B_10_1':
0.29630036500488965, 'B_10_2': 0.21331885386014116, 'B_10_3':
0.02512325181268693, 'B_10_4': -0.5332737727988378, 'B_10_5':
-0.6404744966290936, 'B_10_6': -0.5534373060410528, 'B_10_7':
-0.620852240954376, 'B_10_8': 0.5535772576621716, 'B_10_9':
-0.29675198684987436}. Best is trial 8 with value: 119.8836634080008.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 13:46:49,233] Trial 10 finished with value: 146.0430568798616 and
parameters: {'B_0_0': -0.8754715004044752, 'B_0_1': 0.9288232076788063, 'B_0_2':
0.7598960522275828, 'B_0_3': 0.8469495207265216, 'B_0_4': 0.9799174888399418,
'B_0_5': -0.6456874065454152, 'B_0_6': -0.20544080257273567, 'B_0_7':
0.5113121818941009, 'B_0_8': 0.3881208237465086, 'B_0_9': -0.6852861054470869,
'B_1_0': 0.6253495874814232, 'B_1_1': -0.5456076677634119, 'B_1_2':
-0.5468411226512063, 'B_1_3': 0.07320935732895895, 'B_1_4': 0.18402107098657217,

```

'B_1_5': 0.15857069151886605, 'B_1_6': -0.8088560619524453, 'B_1_7':
 0.747372508008153, 'B_1_8': -0.46411191567436194, 'B_1_9': -0.7335604558600477,
 'B_2_0': 0.7697211981405749, 'B_2_1': 0.9021899664949116, 'B_2_2':
 0.7170126821855922, 'B_2_3': 0.6128418279552585, 'B_2_4': 0.30737912166526304,
 'B_2_5': 0.10069759380609589, 'B_2_6': -0.817766215375408, 'B_2_7':
 -0.18126263812001653, 'B_2_8': -0.25207673631561833, 'B_2_9':
 -0.4756875081545541, 'B_3_0': 0.4423718251033981, 'B_3_1':
 -0.008166044544033801, 'B_3_2': -0.8295284925028571, 'B_3_3':
 -0.5540372601393735, 'B_3_4': 0.36285235204599986, 'B_3_5': -0.8392608982092056,
 'B_3_6': 0.6953896898165619, 'B_3_7': -0.009609876512330806, 'B_3_8':
 -0.03843857689323693, 'B_3_9': 0.1829674136258448, 'B_4_0': 0.6428683125317953,
 'B_4_1': -0.3013377683536578, 'B_4_2': 0.35247198206694574, 'B_4_3':
 0.1301492887116662, 'B_4_4': -0.4612840250645458, 'B_4_5': 0.7496873729832128,
 'B_4_6': 0.5889035227817176, 'B_4_7': 0.3137346326236825, 'B_4_8':
 0.6941518236065989, 'B_4_9': 0.7272425179004087, 'B_5_0': 0.4125586938957688,
 'B_5_1': 0.6672863901600705, 'B_5_2': 0.3909934941051816, 'B_5_3':
 0.3566787280853949, 'B_5_4': 0.23485052886606494, 'B_5_5': 0.5003789463241297,
 'B_5_6': -0.6759618915197141, 'B_5_7': 0.7541241032138783, 'B_5_8':
 0.7362501849355594, 'B_5_9': -0.9320903795915729, 'B_6_0': 0.64511716611823,
 'B_6_1': -0.7348376624025785, 'B_6_2': -0.32646466856689904, 'B_6_3':
 0.4821463474573835, 'B_6_4': -0.6716954058243497, 'B_6_5': 0.6295747077557432,
 'B_6_6': 0.657625672356329, 'B_6_7': 0.014786112846455701, 'B_6_8':
 -0.9773559740006695, 'B_6_9': -0.42166449631367264, 'B_7_0':
 0.23151529838396967, 'B_7_1': 0.9527486324942984, 'B_7_2': 0.2609907834930034,
 'B_7_3': -0.4755889094929636, 'B_7_4': 0.2653312921372304, 'B_7_5':
 0.07917105183739381, 'B_7_6': 0.5540938823992643, 'B_7_7': -0.7781783351226224,
 'B_7_8': 0.5168352469540154, 'B_7_9': 0.08170782577869828, 'B_8_0':
 0.9167241676408093, 'B_8_1': -0.31309311124340156, 'B_8_2': 0.26259134840521736,
 'B_8_3': 0.8554156489098346, 'B_8_4': -0.7870307385784551, 'B_8_5':
 0.8657124047281732, 'B_8_6': 0.37201373015561034, 'B_8_7': -0.8556826229717596,
 'B_8_8': -0.3940921374453038, 'B_8_9': 0.41218073551765744, 'B_9_0':
 -0.8566458090918321, 'B_9_1': 0.16269751114902098, 'B_9_2':
 -0.30515154723314564, 'B_9_3': 0.23941272517960477, 'B_9_4':
 -0.8994307730512245, 'B_9_5': 0.7356428761817051, 'B_9_6': 0.937508158971246,
 'B_9_7': 0.9283781534656692, 'B_9_8': 0.4943106268509909, 'B_9_9':
 -0.7324292445413487, 'B_10_0': 0.5113611279394648, 'B_10_1': -0.941317905411573,
 'B_10_2': -0.9461953679725854, 'B_10_3': -0.3492517660839083, 'B_10_4':
 -0.02248648299876177, 'B_10_5': 0.5354066872594305, 'B_10_6':
 0.36292484568088024, 'B_10_7': -0.10711264137403831, 'B_10_8':
 -0.4482192007623016, 'B_10_9': 0.9843065103122681}. Best is trial 8 with value:
 119.8836634080008.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].

```

Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 13:47:06,911] Trial 11 finished with value: 189.85779242389356 and
parameters: {'B_0_0': -0.14616102157277355, 'B_0_1': -0.09625369182724253,
'B_0_2': -0.6660248340404951, 'B_0_3': 0.5837229065248601, 'B_0_4':
0.3834908070473475, 'B_0_5': -0.5528761666782545, 'B_0_6': -0.8268855296806314,
'B_0_7': 0.357388618107932, 'B_0_8': 0.3059322042766477, 'B_0_9':
-0.44894613654354554, 'B_1_0': 0.8927098532558124, 'B_1_1': -0.6909053742738176,
'B_1_2': -0.13397709393564816, 'B_1_3': 0.8783595219318459, 'B_1_4':
-0.1589399124862778, 'B_1_5': 0.2742813763748895, 'B_1_6': -0.20276309203366738,
'B_1_7': -0.44705389934963335, 'B_1_8': 0.9582757429604598, 'B_1_9':
-0.1795186674962148, 'B_2_0': 0.7803164232846862, 'B_2_1': -0.5346898803356493,
'B_2_2': -0.568052686030335, 'B_2_3': -0.9283545158886675, 'B_2_4':
0.3003003142442524, 'B_2_5': -0.26031783942695297, 'B_2_6': 0.7214293346824814,
'B_2_7': -0.05304438475071682, 'B_2_8': 0.9270229872711491, 'B_2_9':
-0.6226594789078201, 'B_3_0': 0.7298738725521094, 'B_3_1': 0.5476617685276486,
'B_3_2': 0.5364252523952229, 'B_3_3': 0.6826707916714376, 'B_3_4':
0.5168275020671378, 'B_3_5': 0.24991623683020237, 'B_3_6': -0.7301351421888727,
'B_3_7': -0.9255981646073212, 'B_3_8': 0.83327873867506, 'B_3_9':
0.2309676227510613, 'B_4_0': 0.5871438361328292, 'B_4_1': -0.0365857440051508,
'B_4_2': -0.7577297858544589, 'B_4_3': -0.7421321314389502, 'B_4_4':
0.3674192687133633, 'B_4_5': -0.13799432809906387, 'B_4_6': -0.5929610411333333,
'B_4_7': -0.01664279745195496, 'B_4_8': -0.8628663045912345, 'B_4_9':
0.16230337579050969, 'B_5_0': -0.4573930592185066, 'B_5_1': 0.5891670192616325,
'B_5_2': -0.3754833213313239, 'B_5_3': -0.08866410481803366, 'B_5_4':
-0.9669913309819608, 'B_5_5': -0.8465551621667734, 'B_5_6': -0.2128627583147641,
'B_5_7': -0.03972110726009548, 'B_5_8': 0.19804068527622132, 'B_5_9':
-0.41250809415557954, 'B_6_0': 0.3860641345305542, 'B_6_1': 0.7130423464346252,
'B_6_2': 0.5541049580088326, 'B_6_3': -0.9115547258102529, 'B_6_4':
-0.038596244429498716, 'B_6_5': -0.7822382467320137, 'B_6_6':
-0.5107508685921955, 'B_6_7': 0.9635919346689499, 'B_6_8': -0.7078588250550037,
'B_6_9': -0.0022014561663149523, 'B_7_0': 0.2339483539498759, 'B_7_1':
0.40088064167732584, 'B_7_2': 0.11810439330408995, 'B_7_3': -0.9706537221100163,
'B_7_4': -0.34360660967689605, 'B_7_5': 0.03506905390172865, 'B_7_6':
-0.81602433169323, 'B_7_7': -0.2957586762058221, 'B_7_8': -0.9242578445930942,
'B_7_9': -0.834414575633059, 'B_8_0': -0.20409191312083974, 'B_8_1':
-0.7272228069947405, 'B_8_2': 0.1337308795578851, 'B_8_3': 0.37514063889178617,
'B_8_4': 0.595161664235984, 'B_8_5': -0.593702516395276, 'B_8_6':
-0.6583844871270419, 'B_8_7': -0.7829556761378875, 'B_8_8': 0.2701318940964006,
'B_8_9': 0.40882193844406434, 'B_9_0': -0.9274594332452287, 'B_9_1':
0.8637002475625057, 'B_9_2': -0.8870968583700847, 'B_9_3': 0.0817667438961196,
'B_9_4': 0.4139398285128144, 'B_9_5': 0.7345188650172494, 'B_9_6':

```

0.423892125622207, 'B_9_7': 0.5974216044781877, 'B_9_8': -0.317888618765244,
 'B_9_9': 0.6233537252180943, 'B_10_0': -0.8313726041583432, 'B_10_1':
 0.7817369789998447, 'B_10_2': 0.09423290478439772, 'B_10_3': 0.6282495845257374,
 'B_10_4': -0.0944097966537657, 'B_10_5': 0.28428383648911937, 'B_10_6':
 0.05227726865350424, 'B_10_7': 0.4585472530755572, 'B_10_8':
 -0.8283726355794326, 'B_10_9': -0.8705028736986837}. Best is trial 8 with value:
 119.8836634080008.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 13:47:25,597] Trial 12 finished with value: 253.8239212822955 and
 parameters: {'B_0_0': -0.5007355966599101, 'B_0_1': -0.6741015333758871,
 'B_0_2': 0.7361314618525594, 'B_0_3': -0.5559563050310723, 'B_0_4':
 0.94221320652188, 'B_0_5': -0.32294633229320846, 'B_0_6': -0.6294065269363753,
 'B_0_7': 0.5736030441421087, 'B_0_8': 0.3142413954917347, 'B_0_9':
 -0.003572481422784768, 'B_1_0': 0.10961983085650995, 'B_1_1':
 0.4340195209790825, 'B_1_2': -0.5376596121640285, 'B_1_3': 0.9827411537923489,
 'B_1_4': 0.9400904610506307, 'B_1_5': 0.29764485896693493, 'B_1_6':
 -0.5949059471602927, 'B_1_7': 0.3568519200139568, 'B_1_8': -0.8470471502212318,
 'B_1_9': -0.929308045632504, 'B_2_0': -0.4797878807477969, 'B_2_1':
 -0.07400654565615372, 'B_2_2': 0.7291795607085934, 'B_2_3': 0.44979475813728986,
 'B_2_4': 0.48055891197596257, 'B_2_5': -0.14752319779281076, 'B_2_6':
 -0.30504871475701345, 'B_2_7': -0.2553432492904867, 'B_2_8': 0.9655461361973945,
 'B_2_9': -0.9105838010032847, 'B_3_0': 0.7267223623224481, 'B_3_1':
 0.15577730897333897, 'B_3_2': -0.12154147000461091, 'B_3_3': 0.4460101676219703,
 'B_3_4': -0.0263954959790883, 'B_3_5': 0.7393780114016879, 'B_3_6':
 0.7933896907422919, 'B_3_7': -0.15499256479055823, 'B_3_8': -0.4418809614745979,
 'B_3_9': 0.18285365061485725, 'B_4_0': 0.816479424321048, 'B_4_1':
 -0.5728888657687802, 'B_4_2': 0.243473835455696, 'B_4_3': 0.2604892357831907,
 'B_4_4': 0.4615637843822564, 'B_4_5': -0.7294959834480256, 'B_4_6':
 0.42733343007053537, 'B_4_7': 0.8098843909179692, 'B_4_8': -0.6342274444369209,
 'B_4_9': -0.5196642166507223, 'B_5_0': 0.9333622862024464, 'B_5_1':
 -0.6316656336352102, 'B_5_2': 0.7016824848717667, 'B_5_3':
 -0.015289844232891148, 'B_5_4': -0.5004824726817094, 'B_5_5':
 0.7340848045195725, 'B_5_6': -0.10829559509472231, 'B_5_7':
 0.029338360780786665, 'B_5_8': -0.2787179285884639, 'B_5_9':

0.18404268584115635, 'B_6_0': -0.666222732281655, 'B_6_1': -0.2156585574295138,
 'B_6_2': 0.9294363982238694, 'B_6_3': -0.4788958032517675, 'B_6_4':
 0.3103385957917586, 'B_6_5': -0.34612367283510415, 'B_6_6': 0.5414767888594694,
 'B_6_7': -0.7308701517776673, 'B_6_8': 0.9302456692555185, 'B_6_9':
 -0.09149670806039434, 'B_7_0': -0.5226200825740014, 'B_7_1': -0.844476440286575,
 'B_7_2': -0.6538793479266694, 'B_7_3': 0.03915241814091508, 'B_7_4':
 -0.32273371066605316, 'B_7_5': 0.6511890644475666, 'B_7_6': -0.1368427031496242,
 'B_7_7': -0.4975457402764928, 'B_7_8': 0.23194707347597943, 'B_7_9':
 0.4094188894331827, 'B_8_0': -0.6592570223561193, 'B_8_1': -0.658113951749131,
 'B_8_2': -0.9173905746715849, 'B_8_3': 0.46807598982996823, 'B_8_4':
 0.3243329646911741, 'B_8_5': -0.050230866015368525, 'B_8_6': 0.6814574889590104,
 'B_8_7': 0.6052269028411053, 'B_8_8': 0.1690016415055693, 'B_8_9':
 0.7291771354162522, 'B_9_0': -0.5824344041271989, 'B_9_1': -0.7683991535924988,
 'B_9_2': -0.45589576919636055, 'B_9_3': -0.8769680249431566, 'B_9_4':
 0.06171566544204343, 'B_9_5': 0.8644792707440236, 'B_9_6': -0.912099789476351,
 'B_9_7': -0.7482223702601469, 'B_9_8': -0.09464592399879812, 'B_9_9':
 0.8590725347052517, 'B_10_0': -0.364010912146034, 'B_10_1':
 0.014324921203016983, 'B_10_2': -0.9076857390802237, 'B_10_3':
 -0.696280462095064, 'B_10_4': 0.9635276434585169, 'B_10_5': 0.92093501894337,
 'B_10_6': -0.980218837749869, 'B_10_7': 0.8945873351380131, 'B_10_8':
 0.2754574768746907, 'B_10_9': 0.7284782231496438}. Best is trial 8 with value:
 119.8836634080008.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:47:42,153] Trial 13 finished with value: 233.64473830083656 and
 parameters: {'B_0_0': -0.08961508584495903, 'B_0_1': 0.030880136586915574,
 'B_0_2': -0.022083771091298, 'B_0_3': 0.33039122994128656, 'B_0_4':
 -0.7134905155825004, 'B_0_5': -0.9306522920520977, 'B_0_6':
 -0.38029871564944195, 'B_0_7': 0.40526791021856057, 'B_0_8':
 -0.5903301647965622, 'B_0_9': 0.3433962179833363, 'B_1_0': 0.9304258512923955,
 'B_1_1': -0.8040765827369516, 'B_1_2': 0.34175219408574997, 'B_1_3':
 -0.11137456577694915, 'B_1_4': 0.7289216636674607, 'B_1_5': -0.6392434168941759,
 'B_1_6': 0.3813993854078057, 'B_1_7': 0.6694682735032582, 'B_1_8':
 0.880336154558566, 'B_1_9': 0.3628310960134986, 'B_2_0': -0.005593967263313959,
 'B_2_1': 0.2333375355984033, 'B_2_2': 0.730431869267862, 'B_2_3':

0.1398072983448828, 'B_2_4': -0.929833621800263, 'B_2_5': 0.8532784171038159,
 'B_2_6': 0.37526296704249096, 'B_2_7': 0.34949650383894815, 'B_2_8':
 -0.5629631982525201, 'B_2_9': 0.31459323106066117, 'B_3_0':
 -0.21014847680476623, 'B_3_1': 0.2994412945421543, 'B_3_2': -0.7789457999901783,
 'B_3_3': 0.3125337031140114, 'B_3_4': 0.9888391770259197, 'B_3_5':
 -0.8945401630560991, 'B_3_6': 0.9448048847582082, 'B_3_7': -0.18432223776871615,
 'B_3_8': 0.7340918316677731, 'B_3_9': 0.5591232584353814, 'B_4_0':
 0.13269219675360233, 'B_4_1': 0.47212943431762655, 'B_4_2': 0.7494608113766688,
 'B_4_3': -0.18980216313095466, 'B_4_4': -0.3424743401047874, 'B_4_5':
 0.33183490341589783, 'B_4_6': 0.6095349650436921, 'B_4_7': 0.5193245668576991,
 'B_4_8': 0.5896710247400963, 'B_4_9': -0.12754503667039263, 'B_5_0':
 0.6293117480265429, 'B_5_1': -0.7519860703661952, 'B_5_2': 0.08808841396248202,
 'B_5_3': -0.9785978522137193, 'B_5_4': -0.3473200567974648, 'B_5_5':
 -0.2644061609546322, 'B_5_6': -0.20557807061367428, 'B_5_7':
 0.38702506924592717, 'B_5_8': -0.22065495948471547, 'B_5_9':
 -0.10158662720738931, 'B_6_0': -0.5196626208688316, 'B_6_1':
 -0.2509614525739067, 'B_6_2': -0.54000613801896, 'B_6_3': -0.8450720709384816,
 'B_6_4': 0.20482821489512815, 'B_6_5': 0.33306134113893227, 'B_6_6':
 0.2365908850990368, 'B_6_7': -0.07228179327486672, 'B_6_8': -0.238024155245125,
 'B_6_9': 0.7194006261522139, 'B_7_0': 0.037781934569711684, 'B_7_1':
 -0.04121988229313267, 'B_7_2': -0.9392287097032573, 'B_7_3':
 -0.3143293013002104, 'B_7_4': -0.237212674804003, 'B_7_5': -0.20033089382413705,
 'B_7_6': 0.15874129107011803, 'B_7_7': 0.06653304251921943, 'B_7_8':
 0.2136520837310203, 'B_7_9': 0.524468857853414, 'B_8_0': 0.619711762657865,
 'B_8_1': 0.4318836909284589, 'B_8_2': 0.9019369179907706, 'B_8_3':
 -0.953899486300804, 'B_8_4': -0.6023595883465787, 'B_8_5': -0.9750255075432785,
 'B_8_6': 0.2919999340261139, 'B_8_7': 0.788100486920974, 'B_8_8':
 -0.5079050527789275, 'B_8_9': 0.8455283986966871, 'B_9_0': -0.8706705672266695,
 'B_9_1': 0.8601833330749182, 'B_9_2': -0.29378707926552383, 'B_9_3':
 -0.7891867596705513, 'B_9_4': -0.027973916811075195, 'B_9_5':
 -0.48158239591339425, 'B_9_6': -0.4259516541153928, 'B_9_7':
 -0.3815658694459526, 'B_9_8': 0.5999912777086736, 'B_9_9': 0.07753932781700379,
 'B_10_0': -0.3736107556711128, 'B_10_1': 0.21846089095561827, 'B_10_2':
 0.4279783349619233, 'B_10_3': -0.45020447854760315, 'B_10_4':
 -0.17117277897980399, 'B_10_5': -0.7486655351928438, 'B_10_6':
 -0.631324287921333, 'B_10_7': 0.3586133508506659, 'B_10_8': -0.6307520715626587,
 'B_10_9': 0.04982349966077915}. Best is trial 8 with value: 119.8836634080008.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].

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Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 13:47:57,210] Trial 14 finished with value: 158.23005663679118 and
parameters: {'B_0_0': 0.4139115982104167, 'B_0_1': -0.7783836922724072, 'B_0_2':
0.13327819486522907, 'B_0_3': -0.48200568827129975, 'B_0_4': 0.916595212970408,
'B_0_5': -0.03257961949454902, 'B_0_6': 0.605865248554813, 'B_0_7':
0.0994485534693983, 'B_0_8': -0.9040431849474581, 'B_0_9': 0.2636397236761643,
'B_1_0': 0.8937786175691809, 'B_1_1': 0.20119140388973578, 'B_1_2':
0.6319939416650446, 'B_1_3': 0.7607287974298473, 'B_1_4': -0.5384020515411136,
'B_1_5': -0.5701512169817112, 'B_1_6': 0.21974235792496355, 'B_1_7':
-0.1761636297827247, 'B_1_8': 0.6729253796761063, 'B_1_9': 0.7920457843280622,
'B_2_0': -0.2902256689806182, 'B_2_1': -0.5209962932305255, 'B_2_2':
0.5554405201828465, 'B_2_3': -0.4458840496146905, 'B_2_4': 0.6387763506570832,
'B_2_5': -0.15099825777544962, 'B_2_6': 0.3317488005971114, 'B_2_7':
-0.8008400786073571, 'B_2_8': 0.24524146285961446, 'B_2_9':
-0.09549999610028637, 'B_3_0': 0.1714847570874387, 'B_3_1': -0.6573318686651703,
'B_3_2': 0.4690100149894807, 'B_3_3': 0.7183382135355925, 'B_3_4':
-0.5608551958202642, 'B_3_5': -0.8004851800700797, 'B_3_6': -0.943195599628254,
'B_3_7': 0.2811035708776668, 'B_3_8': 0.21204619145578585, 'B_3_9':
0.09246087717385865, 'B_4_0': -0.5307447496903687, 'B_4_1':
-0.21600612810708097, 'B_4_2': 0.18706314366862253, 'B_4_3':
-0.006401619458844654, 'B_4_4': 0.9658153300254768, 'B_4_5':
-0.7198492899376684, 'B_4_6': 0.3863862170142356, 'B_4_7': -0.18944883915350164,
'B_4_8': -0.14216476244728282, 'B_4_9': 0.430843576003189, 'B_5_0':
0.38102357962225186, 'B_5_1': 0.9726868675303328, 'B_5_2': -0.735779306880832,
'B_5_3': -0.7838628942193506, 'B_5_4': 0.4441908579149074, 'B_5_5':
0.15520609567268995, 'B_5_6': -0.44716188021158443, 'B_5_7':
-0.8327496494018065, 'B_5_8': -0.8203966657990676, 'B_5_9': 0.7804979318349543,
'B_6_0': -0.610102697056889, 'B_6_1': -0.3497243070112689, 'B_6_2':
-0.5412203235905637, 'B_6_3': -0.28710731445276416, 'B_6_4':
-0.8525407913627351, 'B_6_5': 0.03773838599837531, 'B_6_6': -0.8561271240874144,
'B_6_7': 0.5947058854882759, 'B_6_8': -0.5272500778173426, 'B_6_9':
0.07922359091693099, 'B_7_0': 0.7525565933716873, 'B_7_1': 0.2987371907134444,
'B_7_2': 0.0652564173716439, 'B_7_3': -0.3478191688315563, 'B_7_4':
-0.33065621214476815, 'B_7_5': 0.33558417129241636, 'B_7_6': 0.9783959352179117,
'B_7_7': 0.3204415959017697, 'B_7_8': 0.11441116639264926, 'B_7_9':
0.4566880102737072, 'B_8_0': -0.06889288767823809, 'B_8_1': -0.8709181616504452,
'B_8_2': 0.12334769543826152, 'B_8_3': 0.9060980633206528, 'B_8_4':
-0.6429001741318339, 'B_8_5': 0.37620968382491826, 'B_8_6': -0.5921512964550681,
'B_8_7': 0.07093881525670098, 'B_8_8': -0.7985806296994952, 'B_8_9':
-0.09826554620879346, 'B_9_0': 0.5072033912125873, 'B_9_1': -0.3018083820068155,
'B_9_2': 0.32652521448469174, 'B_9_3': 0.5849909223636638, 'B_9_4':
0.8458120825693163, 'B_9_5': -0.5254086779262019, 'B_9_6': -0.19935448669756983,
'B_9_7': -0.6882162936057734, 'B_9_8': 0.9751173342735828, 'B_9_9':
0.8454619164901047, 'B_10_0': 0.07911511795550807, 'B_10_1': 0.6772259228653141,
'B_10_2': 0.04149679499864356, 'B_10_3': 0.24469966224546424, 'B_10_4':

```

-0.8135336244102871, 'B_10_5': 0.5054354325836796, 'B_10_6':
-0.7371273024022351, 'B_10_7': 0.6456139079000063, 'B_10_8': 0.5584156135873641,
'B_10_9': 0.4133145015310933}. Best is trial 8 with value: 119.8836634080008.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 13:48:11,805] Trial 15 finished with value: 119.98613874768425 and
parameters: {'B_0_0': -0.918402446560971, 'B_0_1': -0.3898058552282494, 'B_0_2':
-0.46903711157653116, 'B_0_3': -0.27692991320639215, 'B_0_4':
-0.8164673614075997, 'B_0_5': 0.8651764896015044, 'B_0_6': 0.10652843661050504,
'B_0_7': -0.38506186489711336, 'B_0_8': -0.20397659619291886, 'B_0_9':
-0.10453896980042543, 'B_1_0': 0.19917678005402228, 'B_1_1': 0.0310452668390655,
'B_1_2': 0.8303961069645462, 'B_1_3': -0.006012303846456124, 'B_1_4':
0.974472869350159, 'B_1_5': 0.6958214163352896, 'B_1_6': -0.5771491815724751,
'B_1_7': 0.8525785249949542, 'B_1_8': -0.7595945321867573, 'B_1_9':
0.6285504228125594, 'B_2_0': -0.2363658789163714, 'B_2_1': 0.7483891544812376,
'B_2_2': 0.7287522464127516, 'B_2_3': 0.6057322922328898, 'B_2_4':
0.5742602638879815, 'B_2_5': -0.38673530458587924, 'B_2_6': -0.8297798195585206,
'B_2_7': -0.1920992460449803, 'B_2_8': -0.6464214603267129, 'B_2_9':
0.3860031560268742, 'B_3_0': -0.3047225414551846, 'B_3_1': 0.9417081977026316,
'B_3_2': 0.2791247131437573, 'B_3_3': 0.6385115159960346, 'B_3_4':
-0.7276011479711402, 'B_3_5': 0.7167886748997927, 'B_3_6': 0.8370592372165209,
'B_3_7': -0.025617400738506935, 'B_3_8': 0.21038081674693543, 'B_3_9':
0.5243234057879309, 'B_4_0': -0.6438195180243831, 'B_4_1': 0.005080831397615948,
'B_4_2': -0.20064720501082, 'B_4_3': -0.7001794955187468, 'B_4_4':
-0.26228184386276776, 'B_4_5': -0.8550188274167227, 'B_4_6':
-0.9388924242400157, 'B_4_7': -0.7223707474734509, 'B_4_8': 0.9169679209465642,
'B_4_9': 0.0980684810721335, 'B_5_0': 0.9223278790833755, 'B_5_1':
-0.13365420090633873, 'B_5_2': -0.372604056491186, 'B_5_3':
0.012160888423683236, 'B_5_4': -0.11976685650683794, 'B_5_5':
-0.780783925161747, 'B_5_6': 0.27883610308660844, 'B_5_7': -0.5622443986309928,
'B_5_8': 0.23678415069770264, 'B_5_9': 0.29739816294834376, 'B_6_0':
-0.6889907908052768, 'B_6_1': -0.868527738320877, 'B_6_2': 0.5559079404155107,
'B_6_3': -0.07959516093674601, 'B_6_4': -0.874835684899158, 'B_6_5':
0.979835307996139, 'B_6_6': -0.8755944892271941, 'B_6_7': 0.3861697531399353,
'B_6_8': 0.957684272021559, 'B_6_9': -0.5164233936524673, 'B_7_0':

```

-0.7083462499993516, 'B_7_1': -0.7496578199435908, 'B_7_2': -0.3895152078161608,
'B_7_3': -0.7899292941989196, 'B_7_4': 0.3804794565543812, 'B_7_5':
-0.8666622397024759, 'B_7_6': 0.018655813736583537, 'B_7_7': 0.9834597706288941,
'B_7_8': 0.6216611336219173, 'B_7_9': 0.22813448769740163, 'B_8_0':
-0.3836178308425089, 'B_8_1': 0.24531375585744408, 'B_8_2':
0.053542098545070305, 'B_8_3': -0.14635489989706452, 'B_8_4':
-0.7311934540544051, 'B_8_5': 0.765476346176124, 'B_8_6': -0.09942639533378139,
'B_8_7': -0.6046474265948851, 'B_8_8': -0.26183648526296655, 'B_8_9':
-0.17002308999272453, 'B_9_0': 0.6485250332224111, 'B_9_1': 0.46255647351157614,
'B_9_2': 0.5332236894098528, 'B_9_3': -0.9681580964312785, 'B_9_4':
-0.16601508360407224, 'B_9_5': -0.036938579762059276, 'B_9_6':
-0.9519992918903477, 'B_9_7': -0.47556984668483226, 'B_9_8': 0.5153738497818272,
'B_9_9': -0.7185227826565819, 'B_10_0': 0.06991406027613145, 'B_10_1':
-0.5639002934058697, 'B_10_2': -0.9660008662050913, 'B_10_3':
-0.5124211139529078, 'B_10_4': 0.9422300347219066, 'B_10_5': 0.5970434815283194,
'B_10_6': 0.9099617557556461, 'B_10_7': -0.024048993025802323, 'B_10_8':
-0.772722326618543, 'B_10_9': 0.09495978816791206}. Best is trial 8 with value:
119.8836634080008.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$\langle F_{[u_i]}, F_{[u_j]} \rangle$].

Built matrix B = inverse(A).

```

[I 2025-04-29 13:48:28,385] Trial 16 finished with value: 191.637720813924 and
parameters: {'B_0_0': -0.0903328775712775, 'B_0_1': 0.6818270238244395, 'B_0_2':
-0.7957964900471957, 'B_0_3': -0.02328254869687918, 'B_0_4':
-0.6929036439059111, 'B_0_5': -0.34714166475768127, 'B_0_6': 0.4699670132278433,
'B_0_7': -0.04748410384984225, 'B_0_8': -0.24574119541137673, 'B_0_9':
-0.20893696843133014, 'B_1_0': -0.08029539855614642, 'B_1_1':
0.5643327567973726, 'B_1_2': 0.7763276841016642, 'B_1_3': 0.901562679277778,
'B_1_4': 0.5680687047854842, 'B_1_5': -0.36549434641893475, 'B_1_6':
0.3725067180513535, 'B_1_7': -0.12354581437032397, 'B_1_8': -0.4857521683488186,
'B_1_9': 0.6749257272633267, 'B_2_0': -0.9139158290545965, 'B_2_1':
0.7954887444281387, 'B_2_2': -0.07627462004302632, 'B_2_3': 0.27165892408039527,
'B_2_4': 0.3155207659674497, 'B_2_5': 0.7823331265989419, 'B_2_6':
0.2706059706910704, 'B_2_7': 0.2255884978249223, 'B_2_8': -0.8580289596898338,
'B_2_9': 0.03644788345150407, 'B_3_0': -0.6926653783822575, 'B_3_1':
0.4701188620685415, 'B_3_2': 0.024199403209095527, 'B_3_3': 0.35685100283253957,

```

'B_3_4': -0.9074876568072792, 'B_3_5': -0.8221118082293096, 'B_3_6': 0.42832030836411095, 'B_3_7': -0.8472730147226267, 'B_3_8': -0.8489116790247879, 'B_3_9': -0.9660252190349229, 'B_4_0': 0.9038727686329706, 'B_4_1': 0.4702665517288338, 'B_4_2': -0.29056221116591296, 'B_4_3': -0.40285954905044263, 'B_4_4': -0.2975876051032179, 'B_4_5': 0.5438139917387401, 'B_4_6': 0.3195138099194206, 'B_4_7': -0.6233127756224135, 'B_4_8': -0.645263516038513, 'B_4_9': -0.7951766135674734, 'B_5_0': 0.3173993850086496, 'B_5_1': 0.5234578711990372, 'B_5_2': -0.46520807191649205, 'B_5_3': -0.9485289771587926, 'B_5_4': -0.8273000983966738, 'B_5_5': 0.9263628689931602, 'B_5_6': -0.4050193405947079, 'B_5_7': 0.5330618327195045, 'B_5_8': 0.24683386633144222, 'B_5_9': -0.2337595134872661, 'B_6_0': -0.5827392208528008, 'B_6_1': -0.7496548920443753, 'B_6_2': 0.22772567671695598, 'B_6_3': 0.5437748856628202, 'B_6_4': 0.2849304220114388, 'B_6_5': 0.05999822383333342, 'B_6_6': -0.9069365776053602, 'B_6_7': 0.9276077796368798, 'B_6_8': 0.5914540994760578, 'B_6_9': -0.41021236733208977, 'B_7_0': 0.9503412521305243, 'B_7_1': 0.20172554747991778, 'B_7_2': 0.163196863250348, 'B_7_3': 0.4911848952704736, 'B_7_4': 0.6173041804184725, 'B_7_5': 0.3098276425163753, 'B_7_6': -0.7363704221949239, 'B_7_7': -0.3202303381308932, 'B_7_8': 0.8476056078455785, 'B_7_9': -0.5452610885491731, 'B_8_0': -0.2531092928017237, 'B_8_1': -0.13448777177049198, 'B_8_2': -0.11997812040498468, 'B_8_3': 0.22362037333916063, 'B_8_4': 0.877290158734521, 'B_8_5': -0.5134284305716401, 'B_8_6': -0.7494272749241823, 'B_8_7': -0.5990084300773242, 'B_8_8': 0.7661113024143535, 'B_8_9': 0.2887054090573078, 'B_9_0': -0.4239045573064629, 'B_9_1': 0.6255749391578622, 'B_9_2': 0.7155125927996033, 'B_9_3': 0.6860983906067506, 'B_9_4': 0.8294745366063179, 'B_9_5': -0.49056278005674403, 'B_9_6': 0.504983018917631, 'B_9_7': -0.07813179539720283, 'B_9_8': 0.6771571347223218, 'B_9_9': 0.45241154077999424, 'B_10_0': 0.5473659440694547, 'B_10_1': 0.30920043860852164, 'B_10_2': -0.6386910364499804, 'B_10_3': 0.08915329941593786, 'B_10_4': 0.9596460842585262, 'B_10_5': 0.8660283716642954, 'B_10_6': -0.9045160031219169, 'B_10_7': -0.6636666648557145, 'B_10_8': -0.7291770285599675, 'B_10_9': 0.44744022177088505}. Best is trial 8 with value: 119.8836634080008.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F_[u_i], _F_[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 13:48:43,461] Trial 17 finished with value: 187.77263425804722 and

```

parameters: {'B_0_0': 0.6292149531167512, 'B_0_1': -0.5672474947048696, 'B_0_2':
0.011588332044621508, 'B_0_3': 0.674591995462176, 'B_0_4': 0.4609470586371762,
'B_0_5': 0.08362966687385964, 'B_0_6': 0.17888842710837038, 'B_0_7':
0.01655388289423443, 'B_0_8': -0.4008540663887352, 'B_0_9': 0.1287435458608519,
'B_1_0': 0.37399289805917957, 'B_1_1': 0.7391793734043992, 'B_1_2':
0.26985688087891124, 'B_1_3': 0.5170206433454527, 'B_1_4': -0.6730581627899455,
'B_1_5': -0.07611620064881597, 'B_1_6': -0.9715233927423552, 'B_1_7':
-0.5015758482471864, 'B_1_8': 0.4483941959237496, 'B_1_9': 0.9737837036717396,
'B_2_0': -0.7936273628173125, 'B_2_1': -0.19504125459464927, 'B_2_2':
0.5941405180243358, 'B_2_3': -0.5860094434347802, 'B_2_4': 0.10906819961852676,
'B_2_5': 0.4614811661387561, 'B_2_6': 0.22965119151515445, 'B_2_7':
-0.617711025461476, 'B_2_8': -0.28633855400223596, 'B_2_9': 0.5619077332810865,
'B_3_0': 0.10736849798139292, 'B_3_1': -0.9796453651850328, 'B_3_2':
0.5167617048556259, 'B_3_3': -0.9200835172183266, 'B_3_4': 0.48655288980783973,
'B_3_5': -0.58908849087283, 'B_3_6': 0.9069854906363728, 'B_3_7':
-0.2614773124218728, 'B_3_8': -0.34267539965060334, 'B_3_9':
-0.6952016603417595, 'B_4_0': -0.3849036533293745, 'B_4_1': 0.7457685084198371,
'B_4_2': 0.9827419885019497, 'B_4_3': -0.2607471295964403, 'B_4_4':
-0.10175095071000573, 'B_4_5': 0.4397004576414194, 'B_4_6': 0.7646676453208476,
'B_4_7': 0.18422778168983256, 'B_4_8': -0.21477911760789792, 'B_4_9':
-0.17300875511972502, 'B_5_0': 0.3873239300636775, 'B_5_1': -0.9836278380635282,
'B_5_2': 0.23678688817048466, 'B_5_3': -0.2861238393261124, 'B_5_4':
0.5825107159286269, 'B_5_5': -0.8058785331501983, 'B_5_6': 0.17464048855781322,
'B_5_7': -0.037673678025474944, 'B_5_8': 0.2818045412923502, 'B_5_9':
-0.8615898829237226, 'B_6_0': 0.1583678991077, 'B_6_1': 0.12173948415985691,
'B_6_2': 0.12010698694401833, 'B_6_3': 0.20490557986098046, 'B_6_4':
0.34940651570819714, 'B_6_5': 0.6038782193679242, 'B_6_6': -0.45575497414613564,
'B_6_7': 0.6435978321565228, 'B_6_8': -0.0034537491183618263, 'B_6_9':
-0.8374246073140243, 'B_7_0': -0.8740691600969211, 'B_7_1':
-0.32820812898179963, 'B_7_2': 0.5640960146807057, 'B_7_3': 0.4112082507870132,
'B_7_4': 0.5714576300513647, 'B_7_5': 0.03419273175438664, 'B_7_6':
-0.11840598404812874, 'B_7_7': -0.6980439971215577, 'B_7_8':
-0.34017834785356316, 'B_7_9': -0.130641654975727, 'B_8_0': -0.8145711492343579,
'B_8_1': -0.5531883336318018, 'B_8_2': 0.1944860822462029, 'B_8_3':
0.4666129661935958, 'B_8_4': 0.9867280725580403, 'B_8_5': 0.8575643960602441,
'B_8_6': 0.2822790950796792, 'B_8_7': -0.1559288544777374, 'B_8_8':
0.26963118212230075, 'B_8_9': 0.5655902037988594, 'B_9_0': -0.755694341971693,
'B_9_1': -0.17838831840134928, 'B_9_2': 0.6728085255585221, 'B_9_3':
-0.23001075517257286, 'B_9_4': 0.14230709702982924, 'B_9_5':
0.17378333450715244, 'B_9_6': -0.624737018795394, 'B_9_7': -0.27277382552846896,
'B_9_8': -0.3276676482909744, 'B_9_9': -0.9381305171065117, 'B_10_0':
-0.9421003074951166, 'B_10_1': 0.6567601464780315, 'B_10_2':
-0.44931979625383145, 'B_10_3': 0.03579595715285455, 'B_10_4':
-0.3985233642942293, 'B_10_5': 0.8725449053283743, 'B_10_6':
-0.47659241744576797, 'B_10_7': -0.1392795107769086, 'B_10_8':
0.7380058959595117, 'B_10_9': 0.6770284622477811}. Best is trial 8 with value:
119.8836634080008.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```

[I 2025-04-29 13:49:00,365] Trial 18 finished with value: 121.53323691307283 and
parameters: {'B_0_0': -0.6215191926643708, 'B_0_1': 0.5992337533955066, 'B_0_2':
-0.08278996425897256, 'B_0_3': -0.03372163327099664, 'B_0_4':
-0.7257096546164313, 'B_0_5': -0.8304090027608932, 'B_0_6': 0.451319827808055,
'B_0_7': -0.0070069183365064625, 'B_0_8': -0.12503560882194453, 'B_0_9':
0.4544262927317455, 'B_1_0': 0.5257155400023974, 'B_1_1': -0.6753618280146799,
'B_1_2': 0.2182457959657298, 'B_1_3': -0.7219989170901826, 'B_1_4':
0.4977226703375104, 'B_1_5': 0.3107712094089363, 'B_1_6': 0.9040969497452467,
'B_1_7': -0.8534631276142861, 'B_1_8': -0.8770316521205165, 'B_1_9':
-0.43126959210746374, 'B_2_0': -0.4718227462029062, 'B_2_1':
-0.5009819778374404, 'B_2_2': 0.8043840694316526, 'B_2_3': -0.4958985242979779,
'B_2_4': -0.45153954226531057, 'B_2_5': 0.5136085595875681, 'B_2_6':
-0.09951511194793083, 'B_2_7': 0.5478869027712587, 'B_2_8': -0.8605750080225172,
'B_2_9': -0.024609036526699102, 'B_3_0': -0.9234450716371003, 'B_3_1':
-0.865946657158398, 'B_3_2': 0.8047461576219337, 'B_3_3': -0.7142941651871615,
'B_3_4': 0.06419295090492994, 'B_3_5': -0.17603070685060107, 'B_3_6':
-0.3022602140074915, 'B_3_7': 0.7916700244607995, 'B_3_8': -0.9467896743845308,
'B_3_9': 0.32430357862766646, 'B_4_0': 0.9175209797428299, 'B_4_1':
0.11913300325444709, 'B_4_2': 0.8649084748255218, 'B_4_3': -0.8865293990136729,
'B_4_4': -0.16078922823316044, 'B_4_5': -0.4748875738951387, 'B_4_6':
0.45702551066184505, 'B_4_7': 0.9529682391845484, 'B_4_8': -0.48207047636038936,
'B_4_9': 0.3052657109186674, 'B_5_0': -0.5977666871357146, 'B_5_1':
0.1293539040517877, 'B_5_2': -0.07141367635709428, 'B_5_3': 0.9345705526704433,
'B_5_4': 0.21488399993900376, 'B_5_5': -0.29797738326809875, 'B_5_6':
-0.764090345921892, 'B_5_7': -0.6905312791048296, 'B_5_8': -0.5438724786965545,
'B_5_9': -0.49308601054621, 'B_6_0': 0.6942197945254529, 'B_6_1':
0.1212211207993994, 'B_6_2': 0.046314004521245566, 'B_6_3': -0.7627576336989494,
'B_6_4': 0.7130765928318812, 'B_6_5': 0.44117232035279774, 'B_6_6':
-0.8560166434379969, 'B_6_7': 0.41151349251494196, 'B_6_8': 0.08620567033843929,
'B_6_9': -0.8281838154213791, 'B_7_0': -0.08256472977326024, 'B_7_1':
-0.03030135148250701, 'B_7_2': -0.661766410476714, 'B_7_3': 0.882482331060559,
'B_7_4': 0.6929512355317107, 'B_7_5': 0.3346642279116345, 'B_7_6':
-0.07465478921879776, 'B_7_7': -0.17470422766267313, 'B_7_8':
0.29892746495888933, 'B_7_9': 0.08995510142567276, 'B_8_0': -0.8666992503538664,

```


'B_8_1': 0.024755244565256795, 'B_8_2': 0.6066791621210221, 'B_8_3':
-0.080705042358023, 'B_8_4': -0.8871259764425485, 'B_8_5': 0.5668308347803854,
'B_8_6': -0.5912997113707708, 'B_8_7': -0.47793074692168325, 'B_8_8':
-0.6638814202040542, 'B_8_9': -0.33617417155443485, 'B_9_0': 0.5083679812634974,
'B_9_1': 0.03838400181779589, 'B_9_2': -0.5843350771815892, 'B_9_3':
0.7481035547410408, 'B_9_4': 0.7515720728040278, 'B_9_5': 0.7337452815910936,
'B_9_6': -0.5171835014645523, 'B_9_7': -0.09654609852164353, 'B_9_8':
0.9602795191589326, 'B_9_9': 0.5385847030937809, 'B_10_0': -0.9362085059733916,
'B_10_1': -0.8608949093042257, 'B_10_2': -0.07141587391158111, 'B_10_3':
0.8102560076058891, 'B_10_4': 0.07662955994637421, 'B_10_5':
-0.0043312340921053405, 'B_10_6': -0.7811620740858715, 'B_10_7':
0.31042460735410216, 'B_10_8': 0.6377642579884464, 'B_10_9':
-0.23676838513502663}. Best is trial 8 with value: 119.8836634080008.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [<F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 13:49:17,870] Trial 19 finished with value: 176.40126406844252 and
parameters: {'B_0_0': 0.5457114648321484, 'B_0_1': 0.9196637747236069, 'B_0_2':
-0.5865424109271755, 'B_0_3': 0.04619266976558767, 'B_0_4':
-0.42146684299254045, 'B_0_5': 0.5798517496870019, 'B_0_6': 0.15363486444469188,
'B_0_7': 0.2664731867496015, 'B_0_8': 0.5898700360394702, 'B_0_9':
-0.2059784641283393, 'B_1_0': 0.8218783354459456, 'B_1_1': 0.06539715714735705,
'B_1_2': -0.677249450786011, 'B_1_3': 0.38788025397422876, 'B_1_4':
0.5806574738677437, 'B_1_5': -0.3628118794062972, 'B_1_6': 0.7072149288575309,
'B_1_7': 0.8041636444362175, 'B_1_8': -0.44172911499193834, 'B_1_9':
0.9573725159761397, 'B_2_0': -0.7113911744634583, 'B_2_1': -0.5900089765630698,
'B_2_2': -0.6252348183087082, 'B_2_3': 0.7800996258155835, 'B_2_4':
0.30549925360342445, 'B_2_5': -0.6888335192198998, 'B_2_6': -0.118159631630669,
'B_2_7': 0.22829008963820807, 'B_2_8': -0.824741111721021, 'B_2_9':
0.7571844094465308, 'B_3_0': 0.6011350001742397, 'B_3_1': 0.010309411681743708,
'B_3_2': 0.9249106982131574, 'B_3_3': -0.1628332610353026, 'B_3_4':
0.9585384316922656, 'B_3_5': 0.3324815976311817, 'B_3_6': 0.26664912104280414,
'B_3_7': -0.6614093073143457, 'B_3_8': 0.7562169614431951, 'B_3_9':
-0.14357041791718206, 'B_4_0': -0.6687778565506909, 'B_4_1':
-0.9650371104479971, 'B_4_2': 0.11831625360399745, 'B_4_3':
0.054251251802899025, 'B_4_4': 0.434320172290934, 'B_4_5': 0.7727109449752483,

'B_4_6': -0.8327749624068015, 'B_4_7': 0.4583630508739174, 'B_4_8':
-0.6189243506867872, 'B_4_9': 0.7091905652511847, 'B_5_0': 0.631745757446079,
'B_5_1': 0.08077163487137784, 'B_5_2': 0.4162807412741567, 'B_5_3':
-0.36758677589617816, 'B_5_4': -0.05708756011302707, 'B_5_5':
0.6368410574667338, 'B_5_6': -0.08065490959703647, 'B_5_7': -0.2815594346556253,
'B_5_8': -0.011459354444406267, 'B_5_9': 0.6499335198256924, 'B_6_0':
-0.32628806620567197, 'B_6_1': -0.6459520455912273, 'B_6_2': 0.4197877602549458,
'B_6_3': 0.645436861162002, 'B_6_4': -0.790737792523404, 'B_6_5':
-0.5150492567432294, 'B_6_6': -0.708895460273232, 'B_6_7': -0.30107600506731,
'B_6_8': -0.09830506176728293, 'B_6_9': 0.49267614718734687, 'B_7_0':
0.2992722180657148, 'B_7_1': 0.23943856360729376, 'B_7_2': -0.2922976057128843,
'B_7_3': 0.6760661551078246, 'B_7_4': -0.056851408551634175, 'B_7_5':
0.948627777270006, 'B_7_6': 0.265601433825559, 'B_7_7': -0.7399958117022846,
'B_7_8': 0.3488315729093594, 'B_7_9': -0.346294049445236, 'B_8_0':
0.3689278494384729, 'B_8_1': -0.8521105266384366, 'B_8_2': -0.6437347543916003,
'B_8_3': 0.704359772633457, 'B_8_4': -0.5401844770934405, 'B_8_5':
0.6673414279810213, 'B_8_6': -0.43703334486026735, 'B_8_7': 0.28290618191624106,
'B_8_8': 0.38441813554946536, 'B_8_9': 0.0250564112272047, 'B_9_0':
-0.385484860140472, 'B_9_1': -0.5689641347852297, 'B_9_2': -0.924285077966396,
'B_9_3': -0.38818612554336684, 'B_9_4': 0.3032623252519404, 'B_9_5':
0.8678435405279956, 'B_9_6': 0.734984874880843, 'B_9_7': 0.5268082575398991,
'B_9_8': 0.5711257396415315, 'B_9_9': 0.3266701088824706, 'B_10_0':
-0.4746320220193141, 'B_10_1': 0.8062461607975431, 'B_10_2': 0.3380499775735515,
'B_10_3': 0.11967244934127108, 'B_10_4': -0.7702408509573074, 'B_10_5':
-0.10483002326327762, 'B_10_6': -0.07849062302018595, 'B_10_7':
0.7218359468548865, 'B_10_8': 0.09236347597182126, 'B_10_9':
-0.23680691303858892}. Best is trial 8 with value: 119.8836634080008.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].
Built F__[u_10].
Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 13:49:33,173] Trial 20 finished with value: 197.57990607554808 and
parameters: {'B_0_0': 0.9440646504511054, 'B_0_1': -0.770773225213895, 'B_0_2':
-0.1533541119423497, 'B_0_3': -0.9067911431606175, 'B_0_4': 0.47501015911852207,
'B_0_5': 0.8277915733543182, 'B_0_6': -0.43552608749003097, 'B_0_7':
0.7095165200622593, 'B_0_8': -0.41140790368544244, 'B_0_9': 0.8133249200549522,
'B_1_0': 0.5028445691216321, 'B_1_1': 0.6037154181460485, 'B_1_2':

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-0.9543470199111, 'B_1_3': 0.9163645070065713, 'B_1_4': 0.4488085642992401,
'B_1_5': -0.38659283113545606, 'B_1_6': 0.6522014364205169, 'B_1_7':
-0.4325880271384387, 'B_1_8': 0.7380526023409333, 'B_1_9': -0.7670908269843647,
'B_2_0': 0.4032982353723755, 'B_2_1': 0.08058298375692252, 'B_2_2':
-0.7988617490130272, 'B_2_3': -0.5110502200474988, 'B_2_4': -0.9654400081709167,
'B_2_5': -0.06183942838027845, 'B_2_6': -0.3934947836951993, 'B_2_7':
0.1947478189095624, 'B_2_8': -0.40146914073693796, 'B_2_9': -0.3961601592161703,
'B_3_0': 0.48152253348857466, 'B_3_1': -0.8946771847628602, 'B_3_2':
0.7977320904801348, 'B_3_3': 0.6974824712945393, 'B_3_4': 0.332253272788285,
'B_3_5': 0.18457870881262095, 'B_3_6': 0.7767589463165114, 'B_3_7':
-0.6230462445331454, 'B_3_8': -0.8336410588621593, 'B_3_9': -0.5157698635791146,
'B_4_0': 0.5832649918263768, 'B_4_1': -0.921352871881563, 'B_4_2':
0.1639575334653094, 'B_4_3': 0.9809662820587937, 'B_4_4': 0.704278265798328,
'B_4_5': 0.04246246430630718, 'B_4_6': -0.8639909996124482, 'B_4_7':
0.6561195584837054, 'B_4_8': 0.19597744619087698, 'B_4_9': -0.7624326658220577,
'B_5_0': -0.8041625760497547, 'B_5_1': 0.811061102484453, 'B_5_2':
0.3350165206173954, 'B_5_3': 0.651987862647176, 'B_5_4': 0.7503782225588893,
'B_5_5': 0.1421092570556779, 'B_5_6': 0.034543776474981636, 'B_5_7':
-0.13775372756443938, 'B_5_8': -0.36244575617358377, 'B_5_9':
-0.12949999749975072, 'B_6_0': 0.5422817183457096, 'B_6_1': 0.2018083934099606,
'B_6_2': 0.7771961141105952, 'B_6_3': -0.11210757152920825, 'B_6_4':
0.21203736586313293, 'B_6_5': 0.2599889551221377, 'B_6_6': 0.18156038136652453,
'B_6_7': 0.4012148707093224, 'B_6_8': -0.5198817326178835, 'B_6_9':
0.02448027954066556, 'B_7_0': -0.7836348904394717, 'B_7_1': -0.2286675712632712,
'B_7_2': -0.02441920412989651, 'B_7_3': 0.30140403390003545, 'B_7_4':
0.892051484120683, 'B_7_5': 0.1992891117657447, 'B_7_6': 0.48231595402958827,
'B_7_7': 0.012406797800501357, 'B_7_8': 0.2655259821251359, 'B_7_9':
-0.8495542120551286, 'B_8_0': -0.4863046699331429, 'B_8_1': -0.2735308754538254,
'B_8_2': -0.05446315708710314, 'B_8_3': -0.8996154839052802, 'B_8_4':
-0.7127522868444309, 'B_8_5': -0.44190767057321456, 'B_8_6': 0.9336347279840802,
'B_8_7': -0.33393292593595947, 'B_8_8': -0.03555870493807656, 'B_8_9':
-0.601726528777982, 'B_9_0': 0.2193445407300212, 'B_9_1': -0.4342472332783709,
'B_9_2': -0.5801546946919149, 'B_9_3': 0.0328138043230386, 'B_9_4':
-0.9790985176623668, 'B_9_5': -0.9748252153545338, 'B_9_6': -0.5562437579783216,
'B_9_7': -0.9172917022483741, 'B_9_8': -0.7761090068212311, 'B_9_9':
-0.3190559143092718, 'B_10_0': 0.5991196455680294, 'B_10_1':
0.14265631387121558, 'B_10_2': 0.025082097393704483, 'B_10_3':
-0.40889213004791913, 'B_10_4': 0.8548723644769765, 'B_10_5':
-0.2039102183503605, 'B_10_6': -0.8175563217443724, 'B_10_7':
0.2317917019132465, 'B_10_8': -0.7645999809163292, 'B_10_9':
-0.3064581772526497}. Best is trial 8 with value: 119.8836634080008.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].

```

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Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 13:49:48,872] Trial 21 finished with value: 167.2951254938717 and
parameters: {'B_0_0': 0.014675638661261647, 'B_0_1': 0.7409605988766481,
'B_0_2': -0.012777770926565069, 'B_0_3': 0.4004723567426953, 'B_0_4':
0.9757773412181594, 'B_0_5': -0.7296515426019964, 'B_0_6': -0.4460315794713513,
'B_0_7': -0.20873937031666012, 'B_0_8': -0.15478020649311386, 'B_0_9':
-0.17617827097421934, 'B_1_0': 0.8070697850619788, 'B_1_1': 0.4237846360443758,
'B_1_2': 0.21365232389959332, 'B_1_3': -0.37744201863169147, 'B_1_4':
0.6411049607581636, 'B_1_5': 0.901019812740278, 'B_1_6': 0.6359725994494485,
'B_1_7': -0.9869010917208837, 'B_1_8': 0.270074712963146, 'B_1_9':
-0.8887519447446406, 'B_2_0': -0.4799377037392307, 'B_2_1': -0.8721410933114121,
'B_2_2': 0.2055149968397878, 'B_2_3': 0.3694480287153614, 'B_2_4':
-0.7633144766295079, 'B_2_5': -0.2299990115709406, 'B_2_6':
-0.08665171226802681, 'B_2_7': -0.25927501691338517, 'B_2_8':
-0.7503704990393177, 'B_2_9': -0.16048337542879865, 'B_3_0':
0.49733253466066785, 'B_3_1': -0.8494048242241542, 'B_3_2': -0.8312435721178312,
'B_3_3': -0.2875528421647513, 'B_3_4': 0.8746169090054716, 'B_3_5':
0.33377384051136993, 'B_3_6': 0.3537665142116784, 'B_3_7': -0.2733984181410274,
'B_3_8': 0.1854483665929061, 'B_3_9': -0.9699402556250681, 'B_4_0':
0.2694701718908936, 'B_4_1': 0.8183081562009487, 'B_4_2': 0.2228954673227419,
'B_4_3': 0.7399232150633632, 'B_4_4': 0.44346664982286566, 'B_4_5':
-0.7512943355206934, 'B_4_6': 0.7968575484259666, 'B_4_7': -0.8584411881755657,
'B_4_8': 0.06725559564457217, 'B_4_9': -0.7085684948889538, 'B_5_0':
-0.96682091075281, 'B_5_1': -0.1543721649472748, 'B_5_2': -0.4058193988447706,
'B_5_3': -0.027719022280746075, 'B_5_4': 0.15285577573441067, 'B_5_5':
-0.9033966368282842, 'B_5_6': -0.7464534756549837, 'B_5_7': 0.11611194497184218,
'B_5_8': -0.3105286407181367, 'B_5_9': 0.453763505347907, 'B_6_0':
0.30153746421204053, 'B_6_1': 0.6842966387200156, 'B_6_2': 0.38113579680506615,
'B_6_3': -0.13873646676750673, 'B_6_4': 0.3424739123409761, 'B_6_5':
-0.4447452749742713, 'B_6_6': -0.38349927241913895, 'B_6_7': 0.5721906166431707,
'B_6_8': -0.10608935752486726, 'B_6_9': 0.5907930742263379, 'B_7_0':
0.6383962998631914, 'B_7_1': 0.7079734285693418, 'B_7_2': 0.8249377669655904,
'B_7_3': -0.13664926338367345, 'B_7_4': -0.3586436997241451, 'B_7_5':
0.16275351825634554, 'B_7_6': -0.2550786015934974, 'B_7_7': 0.2001268502644613,
'B_7_8': 0.4070600080347728, 'B_7_9': 0.3730419366144311, 'B_8_0':
-0.24838528370015978, 'B_8_1': -0.6596168006364694, 'B_8_2':
-0.13755190984482912, 'B_8_3': -0.7076635947178207, 'B_8_4': 0.7723921296639404,
'B_8_5': -0.30516262773904235, 'B_8_6': -0.6841697004978189, 'B_8_7':
-0.9396134353866132, 'B_8_8': 0.28872832484723876, 'B_8_9': 0.27106658856039156,
'B_9_0': -0.3156037501690425, 'B_9_1': -0.848010460081651, 'B_9_2':
-0.17893478810744057, 'B_9_3': -0.3737893597744807, 'B_9_4': 0.3506988666753186,

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'B_9_5': 0.20944291391632697, 'B_9_6': -0.26810441816887076, 'B_9_7':
-0.5585727425751196, 'B_9_8': 0.9663112129880718, 'B_9_9': -0.09107679003169256,
'B_10_0': 0.3727829868448693, 'B_10_1': -0.7117051241948457, 'B_10_2':
-0.028532394674392392, 'B_10_3': -0.9354772946584583, 'B_10_4':
0.01080113346269096, 'B_10_5': 0.9187544660930507, 'B_10_6':
-0.22927070828811558, 'B_10_7': -0.912787895101675, 'B_10_8':
-0.9287075339796999, 'B_10_9': -0.22178621530519838}. Best is trial 8 with
value: 119.8836634080008.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 13:50:04,246] Trial 22 finished with value: 188.11046988374972 and
parameters: {'B_0_0': -0.6731506258028758, 'B_0_1': -0.9437628492701919,
'B_0_2': 0.5073029556213502, 'B_0_3': -0.08212444241385641, 'B_0_4':
-0.417295351494144, 'B_0_5': 0.7921647023726317, 'B_0_6': -0.7600448588029594,
'B_0_7': 0.9027192813315321, 'B_0_8': -0.36834589959284136, 'B_0_9':
0.7690392711711889, 'B_1_0': 0.20382689542933385, 'B_1_1': 0.6468805722881452,
'B_1_2': 0.9583460631567193, 'B_1_3': -0.41895723311888344, 'B_1_4':
0.9130482611070341, 'B_1_5': -0.2188258480932077, 'B_1_6': -0.22668642862089383,
'B_1_7': -0.3160335335760788, 'B_1_8': 0.08198706254821864, 'B_1_9':
-0.6847928145065956, 'B_2_0': 0.10634874100886216, 'B_2_1': 0.08257552857180794,
'B_2_2': 0.5185655889576426, 'B_2_3': 0.6606269246364194, 'B_2_4':
-0.11792612638140043, 'B_2_5': -0.3913801662840424, 'B_2_6':
-0.4765169033121529, 'B_2_7': -0.6046448033243536, 'B_2_8': -0.8758577872459308,
'B_2_9': -0.3119530019072676, 'B_3_0': -0.4549095695810491, 'B_3_1':
0.9235064240154867, 'B_3_2': 0.1142634759750869, 'B_3_3': -0.30239146271959505,
'B_3_4': 0.15932584996751809, 'B_3_5': -0.7143905313355229, 'B_3_6':
-0.11079337657963828, 'B_3_7': 0.24994395641370648, 'B_3_8':
-0.021982293050946233, 'B_3_9': -0.19447797470443795, 'B_4_0':
0.9778198100214024, 'B_4_1': 0.7530232290548755, 'B_4_2': 0.24434379715569143,
'B_4_3': 0.13736717514091978, 'B_4_4': 0.2389077500526764, 'B_4_5':
-0.5914020265115821, 'B_4_6': -0.20762726833664913, 'B_4_7':
-0.9118701248760194, 'B_4_8': -0.04841884577608324, 'B_4_9': 0.0851886762361731,
'B_5_0': -0.539055522515645, 'B_5_1': 0.9187771173877173, 'B_5_2':
0.8107201296557391, 'B_5_3': 0.4398436062026936, 'B_5_4': 0.06610552495541788,
'B_5_5': 0.7325389361024162, 'B_5_6': -0.7313083592295434, 'B_5_7':

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0.5752093926642723, 'B_5_8': -0.7428547883456403, 'B_5_9': 0.5825371118478226,
'B_6_0': -0.4439802148623132, 'B_6_1': 0.746639350416014, 'B_6_2':
0.8792033457742876, 'B_6_3': -0.695383980577871, 'B_6_4': -0.07390230197052083,
'B_6_5': 0.9523550087621164, 'B_6_6': -0.03285327677316974, 'B_6_7':
0.7198240945787995, 'B_6_8': 0.1756884138200132, 'B_6_9': -0.24684678231918833,
'B_7_0': -0.424148560058777, 'B_7_1': -0.5876182796495473, 'B_7_2':
0.5183604189278406, 'B_7_3': -0.22464979649040484, 'B_7_4':
0.022325415157487205, 'B_7_5': -0.015195549597659785, 'B_7_6':
0.15301247788159977, 'B_7_7': 0.7238427483366134, 'B_7_8': 0.9518639002951903,
'B_7_9': -0.1829832670238425, 'B_8_0': 0.64848746577048, 'B_8_1':
0.5237650334163408, 'B_8_2': 0.1455873238820351, 'B_8_3': 0.9029733443516275,
'B_8_4': -0.5930604592537625, 'B_8_5': -0.7736568635869496, 'B_8_6':
0.7008446945717359, 'B_8_7': -0.12048336306517837, 'B_8_8': 0.6869983313299797,
'B_8_9': 0.7783186062869216, 'B_9_0': -0.8663339646342385, 'B_9_1':
0.7592602307124001, 'B_9_2': -0.10232827036854408, 'B_9_3':
0.020656328228379817, 'B_9_4': 0.25065317196455617, 'B_9_5': 0.8442375613255717,
'B_9_6': -0.9521501330290673, 'B_9_7': -0.04584993273072213, 'B_9_8':
0.37168964401251015, 'B_9_9': 0.4409597762424924, 'B_10_0': 0.3812142144434738,
'B_10_1': -0.7238184437719187, 'B_10_2': -0.39719673502407116, 'B_10_3':
-0.27974325967431335, 'B_10_4': 0.6027856712054198, 'B_10_5':
-0.438055452619185, 'B_10_6': -0.5728041715429377, 'B_10_7': 0.9057470776954304,
'B_10_8': -0.9724506954753316, 'B_10_9': 0.9856852942503169}. Best is trial 8
with value: 119.8836634080008.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 13:50:18,258] Trial 23 finished with value: 169.19681286181148 and
parameters: {'B_0_0': 0.35010617173764325, 'B_0_1': 0.6503692129092518, 'B_0_2':
-0.4066535952345375, 'B_0_3': -0.9616557502192212, 'B_0_4': 0.4709816467518295,
'B_0_5': 0.6616063020613381, 'B_0_6': 0.47614274126796796, 'B_0_7':
-0.7071564311443396, 'B_0_8': 0.5017872860687989, 'B_0_9': 0.5324671085731292,
'B_1_0': 0.3139102210267086, 'B_1_1': 0.5269095469352607, 'B_1_2':
0.6849244505619096, 'B_1_3': 0.22493614619016777, 'B_1_4': -0.8145626663400355,
'B_1_5': -0.024496936858922513, 'B_1_6': -0.8362543173502219, 'B_1_7':
-0.18306479855578695, 'B_1_8': -0.18392915153125733, 'B_1_9':
-0.8593005086023217, 'B_2_0': -0.29933534274043094, 'B_2_1':

```

```

-0.7702237643931966, 'B_2_2': 0.6103057158182599, 'B_2_3': 0.886422291409652,
'B_2_4': -0.8468133789892816, 'B_2_5': 0.9011281104375688, 'B_2_6':
0.04470166668675546, 'B_2_7': -0.39685995765132964, 'B_2_8':
-0.8378130909531962, 'B_2_9': 0.0012360507794272335, 'B_3_0':
0.5831407780916824, 'B_3_1': 0.41003122498806244, 'B_3_2': -0.8905524973013319,
'B_3_3': -0.8456543659715511, 'B_3_4': -0.19231090038737852, 'B_3_5':
-0.40532482178507234, 'B_3_6': -0.5298790403490252, 'B_3_7':
-0.4336110921607471, 'B_3_8': 0.6008958317278315, 'B_3_9': 0.8498715461910706,
'B_4_0': -0.18789666483541845, 'B_4_1': 0.8040997842223443, 'B_4_2':
-0.3534385115291937, 'B_4_3': -0.04665476703929261, 'B_4_4':
-0.5424625843243205, 'B_4_5': 0.2781425635081711, 'B_4_6': 0.9483826080461215,
'B_4_7': 0.20491633016588962, 'B_4_8': -0.28152811345517326, 'B_4_9':
0.2926785430612462, 'B_5_0': -0.7466170566385326, 'B_5_1': 0.7695449789254893,
'B_5_2': 0.006106222544076001, 'B_5_3': -0.10028751076150866, 'B_5_4':
0.17001228113796807, 'B_5_5': 0.24707204832301688, 'B_5_6': -0.8478839036715042,
'B_5_7': 0.36158209802233676, 'B_5_8': -0.5109752729752515, 'B_5_9':
0.42362621236304787, 'B_6_0': 0.6386188928807095, 'B_6_1': 0.6018378449539903,
'B_6_2': 0.1039519154676205, 'B_6_3': 0.039936386775287325, 'B_6_4':
-0.7071055965479648, 'B_6_5': 0.5451853777982085, 'B_6_6': -0.4526094267155055,
'B_6_7': -0.006543062820465795, 'B_6_8': -0.4271372960035169, 'B_6_9':
-0.7250198412606076, 'B_7_0': 0.25652423984563844, 'B_7_1': -0.8824225710506504,
'B_7_2': 0.492317563569876, 'B_7_3': -0.361178145382989, 'B_7_4':
-0.989733307851119, 'B_7_5': 0.022035695689714974, 'B_7_6': -0.8972332211153458,
'B_7_7': -0.44318427551461903, 'B_7_8': 0.4098134451623181, 'B_7_9':
-0.8658745491887055, 'B_8_0': 0.6718902148648846, 'B_8_1': -0.9824365531360054,
'B_8_2': -0.5012887389802336, 'B_8_3': 0.4769900311811015, 'B_8_4':
-0.3637851035359726, 'B_8_5': -0.7882530716292933, 'B_8_6': -0.2767368407751528,
'B_8_7': -0.4546212929586394, 'B_8_8': 0.6785695662778988, 'B_8_9':
-0.3695711231814166, 'B_9_0': 0.5720862941164944, 'B_9_1': 0.7759068467252548,
'B_9_2': -0.13104150923847158, 'B_9_3': 0.8116874248316961, 'B_9_4':
-0.24291025315287817, 'B_9_5': 0.9188752646504141, 'B_9_6': -0.8132067414393656,
'B_9_7': 0.37029814767642066, 'B_9_8': -0.012248199895281253, 'B_9_9':
-0.22245536519781883, 'B_10_0': 0.26276951992222974, 'B_10_1':
0.4036370414359758, 'B_10_2': -0.981360726092388, 'B_10_3': -0.6594328842805466,
'B_10_4': 0.4218311802559511, 'B_10_5': 0.3294431188848488, 'B_10_6':
0.9227740601547809, 'B_10_7': 0.5168548246965503, 'B_10_8': 0.8925299613080493,
'B_10_9': 0.40103019303891774}. Best is trial 8 with value: 119.8836634080008.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].

```

```

Built F[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F[u_i], F[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 13:50:33,066] Trial 24 finished with value: 157.005740072509 and
parameters: {'B_0_0': -0.39985662689390256, 'B_0_1': -0.7813774539433042,
'B_0_2': 0.5580112494229725, 'B_0_3': 0.2854154585223134, 'B_0_4':
-0.8945916954612735, 'B_0_5': -0.2771000775160629, 'B_0_6': 0.9044636346588866,
'B_0_7': 0.0007942685061984589, 'B_0_8': -0.1331963106809848, 'B_0_9':
-0.08375478178459794, 'B_1_0': -0.5764122064697759, 'B_1_1':
-0.2599598016877055, 'B_1_2': -0.2577697531453903, 'B_1_3': -0.8863370600288046,
'B_1_4': 0.5297839044654782, 'B_1_5': -0.16532122483031964, 'B_1_6':
0.6379161758965182, 'B_1_7': 0.6936890422173929, 'B_1_8': -0.5702517716966571,
'B_1_9': 0.3115598946968545, 'B_2_0': -0.054882027757296226, 'B_2_1':
0.7527103410845404, 'B_2_2': -0.5628334552306209, 'B_2_3': 0.35207013661388764,
'B_2_4': 0.21334920206039065, 'B_2_5': -0.40530307440777114, 'B_2_6':
-0.7195302107706782, 'B_2_7': 0.30024672574500766, 'B_2_8': 0.4724229671020148,
'B_2_9': -0.36505785934856705, 'B_3_0': 0.28676855596418127, 'B_3_1':
-0.2076408029292235, 'B_3_2': 0.4221496288995832, 'B_3_3': -0.5955537557097749,
'B_3_4': 0.7726257052196253, 'B_3_5': -0.42092899899815217, 'B_3_6':
-0.26178374424281725, 'B_3_7': -0.8749779348809075, 'B_3_8':
-0.7692058196038727, 'B_3_9': 0.03140656737040959, 'B_4_0': -0.4601665000786165,
'B_4_1': 0.664250846969545, 'B_4_2': -0.9609070904631913, 'B_4_3':
-0.23939378362956631, 'B_4_4': -0.32205581739719646, 'B_4_5':
-0.9517312907831499, 'B_4_6': -0.7437505699789897, 'B_4_7':
-0.17098750366110216, 'B_4_8': -0.01412812279405562, 'B_4_9':
-0.1895063004353592, 'B_5_0': 0.06125660325594584, 'B_5_1': 0.18836071631324947,
'B_5_2': -0.9703508341668854, 'B_5_3': -0.0710915842820623, 'B_5_4':
0.9177284969574249, 'B_5_5': 0.037683914656506445, 'B_5_6': 0.3515336895687273,
'B_5_7': -0.37250956132457125, 'B_5_8': 0.5425029287424699, 'B_5_9':
0.5403845341147078, 'B_6_0': 0.042166174355195096, 'B_6_1': 0.942507370092843,
'B_6_2': -0.7414105747699463, 'B_6_3': -0.9564436235186297, 'B_6_4':
0.53491325927905, 'B_6_5': 0.6081679639824675, 'B_6_6': -0.7519896863412591,
'B_6_7': -0.46414599074195995, 'B_6_8': -0.9552463205340748, 'B_6_9':
-0.40924895065448186, 'B_7_0': 0.5408209103589696, 'B_7_1': 0.0355618448709063,
'B_7_2': -0.30077072552678963, 'B_7_3': -0.25395619608546605, 'B_7_4':
-0.987319821083373, 'B_7_5': -0.3963949554208329, 'B_7_6': 0.2899883084431023,
'B_7_7': 0.9389049987563871, 'B_7_8': 0.687180437428121, 'B_7_9':
-0.9432819700018523, 'B_8_0': 0.789149997256182, 'B_8_1': 0.5607328583312794,
'B_8_2': 0.5551447246705412, 'B_8_3': -0.08316839717292845, 'B_8_4':
-0.20194951995008437, 'B_8_5': -0.3902268569102786, 'B_8_6':
-0.8599241792703558, 'B_8_7': -0.5380939802824758, 'B_8_8': -0.5017248027435113,
'B_8_9': -0.031527401973800884, 'B_9_0': 0.489939925061454, 'B_9_1':
-0.051856539772065724, 'B_9_2': -0.875467990094622, 'B_9_3': 0.9064238059566436,
'B_9_4': 0.8765941518063423, 'B_9_5': 0.5646752954338485, 'B_9_6':
0.9728292450971567, 'B_9_7': 0.08807501894452052, 'B_9_8': 0.916281333501348,
'B_9_9': -0.8403020195788334, 'B_10_0': -0.2659521184424848, 'B_10_1':
-0.5437450893466036, 'B_10_2': -0.6022504934141453, 'B_10_3':

```


-0.7112101611925014, 'B_10_4': 0.24237901030067333, 'B_10_5': 0.557017274379664, 'B_10_6': 0.15503083085231406, 'B_10_7': -0.6990138350153449, 'B_10_8': 0.6160250644682723, 'B_10_9': 0.2691903201841823}. Best is trial 8 with value: 119.8836634080008.

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:50:48,113] Trial 25 finished with value: 171.73810464266518 and parameters: {'B_0_0': -0.22143190186854889, 'B_0_1': 0.34477798516140457,

'B_0_2': -0.4753881410991909, 'B_0_3': -0.3065194271814632, 'B_0_4': 0.8266219694013406, 'B_0_5': -0.4148010787705876, 'B_0_6': -0.059423229047285586, 'B_0_7': 0.772479452044109, 'B_0_8': 0.4112882735297043, 'B_0_9': -0.8668376121938886, 'B_1_0': -0.6983399723276387, 'B_1_1': -0.9744736502716047, 'B_1_2': 0.25922514465374347, 'B_1_3': -0.10360661768737167, 'B_1_4': -0.7241357952134944, 'B_1_5': 0.9067095596924055, 'B_1_6': 0.0587262534123274, 'B_1_7': -0.5110512776564794, 'B_1_8': 0.0011950708863209814, 'B_1_9': 0.35564825881551343, 'B_2_0': -0.8390463484123423, 'B_2_1': -0.4460864462051739, 'B_2_2': 0.6077916719503074, 'B_2_3': -0.0798635566027307, 'B_2_4': 0.0922227412239438, 'B_2_5': -0.13302431562525086, 'B_2_6': -0.9030781135668499, 'B_2_7': -0.6618154359560913, 'B_2_8': -0.10782755529105603, 'B_2_9': -0.5758127784923028, 'B_3_0': -0.8910415232538361, 'B_3_1': 0.680426334822611, 'B_3_2': 0.9527612077449492, 'B_3_3': 0.580422890513544, 'B_3_4': 0.7004941449752407, 'B_3_5': -0.5108852819722501, 'B_3_6': 0.9120414115161355, 'B_3_7': -0.6000871051822119, 'B_3_8': 0.8938310245164278, 'B_3_9': 0.979742202052583, 'B_4_0': 0.41921115537717646, 'B_4_1': 0.9526645889478074, 'B_4_2': 0.1376887635428652, 'B_4_3': -0.4761072693113412, 'B_4_4': -0.12474814638207066, 'B_4_5': 0.18525065204999613, 'B_4_6': -0.8452985071865103, 'B_4_7': 0.24223966394214735, 'B_4_8': 0.9527321167893432, 'B_4_9': -0.6135868534282193, 'B_5_0': 0.5793383302509989, 'B_5_1': 0.8076396116836464, 'B_5_2': 0.8785291537840372, 'B_5_3': 0.9110684967227856, 'B_5_4': 0.04249008466851434, 'B_5_5': 0.9450696578231772, 'B_5_6': 0.5094742101261889, 'B_5_7': -0.6698905562508155, 'B_5_8': -0.04573778919341942, 'B_5_9': 0.43229572723656795, 'B_6_0': -0.500266174678215, 'B_6_1': 0.278418063426797, 'B_6_2': 0.3298689485123061, 'B_6_3': -0.6678545072590396, 'B_6_4': 0.12886891865645733, 'B_6_5': 0.5378212837376433, 'B_6_6':

```

-0.002186584218834997, 'B_6_7': -0.9660358287833073, 'B_6_8':
-0.9721038312230015, 'B_6_9': -0.28308189893937585, 'B_7_0': 0.8438642554949283,
'B_7_1': -0.5372199763848546, 'B_7_2': 0.2660401039659406, 'B_7_3':
-0.5502897760272252, 'B_7_4': -0.35309342047409376, 'B_7_5': 0.6891233374109382,
'B_7_6': 0.45314565569449483, 'B_7_7': -0.801109430919843, 'B_7_8':
-0.1411699984468614, 'B_7_9': -0.9322438679823764, 'B_8_0':
-0.03783750972604216, 'B_8_1': 0.3216193431681147, 'B_8_2': -0.7553648052756599,
'B_8_3': -0.418007820959852, 'B_8_4': -0.20228348150572928, 'B_8_5':
0.8306714288189785, 'B_8_6': 0.9766449305374687, 'B_8_7': -0.9010756279886806,
'B_8_8': 0.5167956885208762, 'B_8_9': -0.25398634852336044, 'B_9_0':
-0.21293279126562958, 'B_9_1': 0.503444647227105, 'B_9_2': 0.828484335647909,
'B_9_3': 0.8928386171150458, 'B_9_4': 0.15270272035587507, 'B_9_5':
-0.2829001045850045, 'B_9_6': 0.5693471900003828, 'B_9_7': -0.4930172611406455,
'B_9_8': 0.1268665770903652, 'B_9_9': -0.2800146461807511, 'B_10_0':
0.3101295259415946, 'B_10_1': -0.5140101524708564, 'B_10_2': -0.610645967445045,
'B_10_3': 0.8281133916173653, 'B_10_4': -0.7884287077599169, 'B_10_5':
0.01179665731516466, 'B_10_6': -0.5527114897332424, 'B_10_7':
-0.9129186726151785, 'B_10_8': -0.9186827397829758, 'B_10_9':
-0.6430608912304618}. Best is trial 8 with value: 119.8836634080008.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 13:51:04,000] Trial 26 finished with value: 143.9728279050659 and
parameters: {'B_0_0': 0.7262053019218806, 'B_0_1': -0.4306920869254872, 'B_0_2':
0.8919091577390854, 'B_0_3': 0.16161214011860325, 'B_0_4': -0.1255039042578432,
'B_0_5': 0.15857589264552008, 'B_0_6': 0.03306280274801021, 'B_0_7':
0.5123769824395039, 'B_0_8': -0.43065778937375254, 'B_0_9':
-0.29096022463140125, 'B_1_0': 0.7803067190158364, 'B_1_1': 0.8839838960378028,
'B_1_2': 0.7772652904232196, 'B_1_3': -0.15949297196451906, 'B_1_4':
0.5551236959536257, 'B_1_5': -0.04685524806382435, 'B_1_6':
-0.0048710927507176205, 'B_1_7': -0.5847333840665778, 'B_1_8':
0.18043800045186265, 'B_1_9': -0.6214509224483711, 'B_2_0':
-0.33350189584695633, 'B_2_1': 0.7029597992074716, 'B_2_2': -0.5799898977625066,
'B_2_3': -0.849106595356603, 'B_2_4': -0.8533649666266236, 'B_2_5':
0.8727533505961536, 'B_2_6': 0.013702467164432575, 'B_2_7':
-0.17936409780230367, 'B_2_8': 0.6155395903956784, 'B_2_9': 0.6649353372628217,

```

```

'B_3_0': -0.33226175443110284, 'B_3_1': 0.3833637537981145, 'B_3_2':
0.5368231753867354, 'B_3_3': 0.30620835999458995, 'B_3_4': -0.6898417518958032,
'B_3_5': 0.7442485053637728, 'B_3_6': 0.07740721441729725, 'B_3_7':
-0.43070459578739273, 'B_3_8': -0.148048192833366, 'B_3_9': -0.9156092410101634,
'B_4_0': -0.7368237335796133, 'B_4_1': 0.5257825660214044, 'B_4_2':
-0.989976963184375, 'B_4_3': -0.16519965960506144, 'B_4_4': 0.04457015854946511,
'B_4_5': -0.8818238483980748, 'B_4_6': 0.9366947626852191, 'B_4_7':
-0.5422718441256495, 'B_4_8': -0.38768653512016393, 'B_4_9':
-0.3881938257350459, 'B_5_0': -0.5337750090076472, 'B_5_1': -0.9870818322691034,
'B_5_2': 0.45410269229423283, 'B_5_3': 0.924354089602585, 'B_5_4':
-0.5458989030956107, 'B_5_5': 0.3228334397707764, 'B_5_6': 0.47895472737825595,
'B_5_7': 0.689882250599239, 'B_5_8': -0.1531941899130539, 'B_5_9':
-0.3901968068319501, 'B_6_0': -0.34591563463439834, 'B_6_1': 0.4209902395350553,
'B_6_2': 0.6272233436811594, 'B_6_3': -0.6304043438419267, 'B_6_4':
-0.2555368038476249, 'B_6_5': 0.7958415895633171, 'B_6_6': 0.6072531689955838,
'B_6_7': 0.9600196374697021, 'B_6_8': 0.50341155417052, 'B_6_9':
-0.21147350196471393, 'B_7_0': 0.17946291130938286, 'B_7_1': 0.3188082234096943,
'B_7_2': -0.8346574836920946, 'B_7_3': 0.08810397902565281, 'B_7_4':
0.41445524037576686, 'B_7_5': -0.6587038180779005, 'B_7_6': 0.5556508005026031,
'B_7_7': 0.165870105417941, 'B_7_8': 0.8953991356812374, 'B_7_9':
-0.9060038495503544, 'B_8_0': -0.4646542476401754, 'B_8_1': 0.20107668101365173,
'B_8_2': -0.4028113738322572, 'B_8_3': 0.42455984486105947, 'B_8_4':
0.5128304593934234, 'B_8_5': -0.7870183911483806, 'B_8_6': 0.02743183912940017,
'B_8_7': 0.01760330007029065, 'B_8_8': -0.2588240749732287, 'B_8_9':
0.8571912278879508, 'B_9_0': 0.6484625133489781, 'B_9_1': 0.39047457618108083,
'B_9_2': 0.4243667387552952, 'B_9_3': -0.07580203525002416, 'B_9_4':
0.8335691517712203, 'B_9_5': 0.38529894596150505, 'B_9_6': 0.4533825020785165,
'B_9_7': 0.7161479894989435, 'B_9_8': -0.4473382531024618, 'B_9_9':
0.608000389656072, 'B_10_0': -0.6034236061461458, 'B_10_1':
-0.30622293455512206, 'B_10_2': -0.3254913005457114, 'B_10_3':
0.9474804274104982, 'B_10_4': 0.705943703020895, 'B_10_5': 0.39831632594871436,
'B_10_6': 0.44957238799847254, 'B_10_7': 0.12290413291709079, 'B_10_8':
0.8852395208748678, 'B_10_9': -0.007407669292924135}. Best is trial 8 with
value: 119.8836634080008.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix $B = \text{inverse}(A)$.

[I 2025-04-29 13:51:19,075] Trial 27 finished with value: 159.01948332559027 and parameters: {'B_0_0': -0.2365749344576088, 'B_0_1': -0.6671900328754499, 'B_0_2': 0.5666871914768947, 'B_0_3': 0.4641989855096711, 'B_0_4': -0.22897702372221296, 'B_0_5': -0.9401170403607259, 'B_0_6': 0.6712147088816611, 'B_0_7': -0.9673923927609342, 'B_0_8': 0.40332556366552086, 'B_0_9': 0.931109077257593, 'B_1_0': -0.12343057026655713, 'B_1_1': -0.5247526581080135, 'B_1_2': 0.40564459988589574, 'B_1_3': 0.6279137211908585, 'B_1_4': 0.09193202583413762, 'B_1_5': 0.9247298450071695, 'B_1_6': -0.887695931117237, 'B_1_7': 0.009495284642965673, 'B_1_8': 0.43253881569864916, 'B_1_9': 0.7180281328753744, 'B_2_0': -0.6350738884225757, 'B_2_1': 0.5940068938920893, 'B_2_2': 0.10436000992271954, 'B_2_3': -0.20482370965987284, 'B_2_4': -0.7292042434133947, 'B_2_5': 0.7232856026396466, 'B_2_6': -0.6785990478041636, 'B_2_7': -0.3766200387659835, 'B_2_8': -0.41570984671437494, 'B_2_9': 0.7353997876978915, 'B_3_0': 0.3419519285336181, 'B_3_1': 0.5874291664389173, 'B_3_2': -0.49407356039085204, 'B_3_3': 0.24725071722185787, 'B_3_4': 0.14205704662462248, 'B_3_5': 0.6590041460407767, 'B_3_6': 0.8040523797528958, 'B_3_7': -0.9659295924747742, 'B_3_8': 0.34455943973640957, 'B_3_9': -0.8873651177429365, 'B_4_0': 0.09674015685316473, 'B_4_1': -0.4204871964935172, 'B_4_2': -0.3825823329461604, 'B_4_3': -0.29114216316433394, 'B_4_4': 0.24015904911822794, 'B_4_5': -0.32858106797583697, 'B_4_6': 0.4607441201687552, 'B_4_7': -0.18903577656373804, 'B_4_8': -0.8546606633862819, 'B_4_9': 0.5618444881185323, 'B_5_0': -0.4241985173064421, 'B_5_1': -0.13312157839345995, 'B_5_2': 0.36717888779365, 'B_5_3': -0.33173680884555024, 'B_5_4': -0.877960400829503, 'B_5_5': -0.24963632691462911, 'B_5_6': 0.8800081937347208, 'B_5_7': 0.2806341066511453, 'B_5_8': 0.3395287143610697, 'B_5_9': 0.26191866420252685, 'B_6_0': -0.5959955648603151, 'B_6_1': -0.1616989490774854, 'B_6_2': 0.4968608007593702, 'B_6_3': -0.7892816628447449, 'B_6_4': -0.43985150334236967, 'B_6_5': -0.4428881652376099, 'B_6_6': -0.13460252336287815, 'B_6_7': 0.9511301070655265, 'B_6_8': -0.8563449575660326, 'B_6_9': 0.037027968873578176, 'B_7_0': -0.6348579296158233, 'B_7_1': 0.9319395599112934, 'B_7_2': -0.7656590606396209, 'B_7_3': -0.19087001532404935, 'B_7_4': 0.4710122699725203, 'B_7_5': 0.4050177674943416, 'B_7_6': -0.15299729895001413, 'B_7_7': -0.30388214745361786, 'B_7_8': -0.20272749046531624, 'B_7_9': -0.46673299894589426, 'B_8_0': -0.5834407813605365, 'B_8_1': -0.033581398833397835, 'B_8_2': -0.4583031629861375, 'B_8_3': -0.42082590350344473, 'B_8_4': 0.3103770056796229, 'B_8_5': 0.9277039167657866, 'B_8_6': 0.20520165710756277, 'B_8_7': -0.8375806552113667, 'B_8_8': -0.8403444445689474, 'B_8_9': 0.8938179772742205, 'B_9_0': -0.40136423519147946, 'B_9_1': -0.8077073675566986, 'B_9_2': 0.1961082825577556, 'B_9_3': 0.24482458101472693, 'B_9_4': 0.2940395438350025, 'B_9_5': -0.46054398484206827, 'B_9_6': -0.9600808452553394, 'B_9_7': 0.9207304315198543, 'B_9_8': -0.49323176953161707, 'B_9_9': 0.3485320503440441, 'B_10_0': 0.40912714235984504, 'B_10_1': 0.21781468581738173, 'B_10_2': -0.3704433594569765, 'B_10_3': -0.45322941754714974, 'B_10_4': 0.1933831997513522, 'B_10_5': 0.7248693545511491, 'B_10_6': 0.8845327832683232, 'B_10_7': -0.78030650728056, 'B_10_8': -0.6834393470565148, 'B_10_9': 0.8805777923904976}. Best is trial 8 with value: 119.8836634080008.

Building the Basis functions for the RKHS:

```
Built F_[u_1].  
Built F_[u_2].  
Built F_[u_3].  
Built F_[u_4].  
Built F_[u_5].  
Built F_[u_6].  
Built F_[u_7].  
Built F_[u_8].  
Built F_[u_9].  
Built F_[u_10].  
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:51:33,544] Trial 28 finished with value: 184.49127708504838 and parameters: {'B_0_0': 0.46833975210288936, 'B_0_1': 0.7583277731629507, 'B_0_2': -0.5887873844120206, 'B_0_3': 0.17342002479661578, 'B_0_4': 0.39825639694659776, 'B_0_5': 0.3566215331850058, 'B_0_6': -0.18185965208393773, 'B_0_7': -0.9595181144512092, 'B_0_8': 0.16419359247244092, 'B_0_9': -0.4888589524243746, 'B_1_0': -0.09849659263707711, 'B_1_1': 0.9060104098659583, 'B_1_2': -0.1999100710679793, 'B_1_3': 0.6728071811955649, 'B_1_4': -0.6166896006164828, 'B_1_5': 0.3414717860895571, 'B_1_6': 0.9444737697891081, 'B_1_7': -0.7882516602715851, 'B_1_8': -0.9735265143901598, 'B_1_9': -0.13150607058742358, 'B_2_0': -0.8066015435582929, 'B_2_1': 0.49180101235893803, 'B_2_2': 0.8208063544201467, 'B_2_3': -0.13063847403665751, 'B_2_4': -0.4777509485322269, 'B_2_5': -0.12988142836691263, 'B_2_6': 0.44242667934300206, 'B_2_7': -0.9720727600708654, 'B_2_8': 0.17711889917970036, 'B_2_9': 0.22431753607701044, 'B_3_0': 0.2726227288539522, 'B_3_1': -0.5107962816378631, 'B_3_2': 0.42382439228869107, 'B_3_3': -0.8090448570630827, 'B_3_4': -0.5954447337957344, 'B_3_5': 0.7473906923758742, 'B_3_6': 0.4726718384386013, 'B_3_7': -0.9627838030575364, 'B_3_8': -0.4982150044677956, 'B_3_9': -0.5654751636735342, 'B_4_0': -0.45380347048679426, 'B_4_1': -0.4998316548458249, 'B_4_2': -0.8661979344291655, 'B_4_3': -0.08129777713829578, 'B_4_4': 0.460812280076607, 'B_4_5': 0.21132881494052413, 'B_4_6': 0.3422859038241979, 'B_4_7': -0.8293246264982465, 'B_4_8': 0.8939516061500388, 'B_4_9': 0.6702138310091892, 'B_5_0': 0.6040788306214122, 'B_5_1': 0.6395079339826997, 'B_5_2': 0.8567668061028675, 'B_5_3': 0.0876228529051215, 'B_5_4': -0.5934416046221007, 'B_5_5': 0.23123154354803144, 'B_5_6': 0.4809053574625193, 'B_5_7': 0.4710764388409703, 'B_5_8': 0.042469067850422704, 'B_5_9': -0.8544517357195154, 'B_6_0': -0.2551963819083469, 'B_6_1': 0.8331178847574474, 'B_6_2': 0.16720884131970348, 'B_6_3': 0.07589724764631267, 'B_6_4': -0.45767004152325685, 'B_6_5': -0.2604711385651738, 'B_6_6': 0.7827852565232751, 'B_6_7': 0.3295968341031348, 'B_6_8': 0.5690263129696191, 'B_6_9': -0.0903960623260428, 'B_7_0': 0.25788949452319265, 'B_7_1': -0.4982004546884491, 'B_7_2': 0.4068124439613028, 'B_7_3': -0.14334935926739212, 'B_7_4': -0.11375988066680376, 'B_7_5': 0.29565820266222564, 'B_7_6': 0.8638356305084747, 'B_7_7': -0.8632674337149305,

```

'B_7_8': 0.6429916099470752, 'B_7_9': -0.4110814413838576, 'B_8_0':
-0.11103948007737197, 'B_8_1': -0.9466116090254822, 'B_8_2':
-0.39392876588806025, 'B_8_3': 0.005209860407658362, 'B_8_4':
-0.8787713668520707, 'B_8_5': -0.017630455821333046, 'B_8_6':
0.8456790451396861, 'B_8_7': -0.7813214281465457, 'B_8_8': 0.5235926415894985,
'B_8_9': -0.17885131431553458, 'B_9_0': 0.30724395475818933, 'B_9_1':
-0.4747315359892459, 'B_9_2': -0.6742053201681106, 'B_9_3': -0.6722840237429717,
'B_9_4': -0.8504259476265736, 'B_9_5': -0.6224195242955388, 'B_9_6':
0.3251543822781571, 'B_9_7': 0.7557474114501459, 'B_9_8': 0.6219688574705005,
'B_9_9': 0.3665659335514786, 'B_10_0': -0.7713449104460167, 'B_10_1':
-0.41740885633401936, 'B_10_2': -0.37658238686151146, 'B_10_3':
-0.49508524613271, 'B_10_4': 0.029711206225251807, 'B_10_5':
0.07039655346175167, 'B_10_6': -0.2833657691612018, 'B_10_7':
-0.28949746570185664, 'B_10_8': 0.6505400838540412, 'B_10_9':
0.5727998396729774}. Best is trial 8 with value: 119.8836634080008.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$\langle F_{[u_i]}, F_{[u_j]} \rangle$].

Built matrix B = inverse(A).

```

[I 2025-04-29 13:51:49,635] Trial 29 finished with value: 103.51132933229894 and
parameters: {'B_0_0': -0.3805642532721236, 'B_0_1': 0.8193011188690533, 'B_0_2':
0.896573733664962, 'B_0_3': -0.3430497667805191, 'B_0_4': -0.2882386013897513,
'B_0_5': 0.011155623775478407, 'B_0_6': 0.8734192088166737, 'B_0_7':
0.7451124776217768, 'B_0_8': -0.7869154340844922, 'B_0_9': -0.21239312832259594,
'B_1_0': 0.10567495570454821, 'B_1_1': 0.006202295652049217, 'B_1_2':
-0.6061511254849525, 'B_1_3': 0.7104572069129462, 'B_1_4': 0.35034275033373796,
'B_1_5': 0.6690585822065711, 'B_1_6': 0.7103532731119548, 'B_1_7':
0.49174459025816475, 'B_1_8': -0.12036300952555967, 'B_1_9':
0.21888257493054053, 'B_2_0': -0.6724764958829086, 'B_2_1': 0.3438179266003172,
'B_2_2': -0.6350785536362241, 'B_2_3': 0.3840201892101105, 'B_2_4':
-0.535395641707142, 'B_2_5': -0.7572484257025645, 'B_2_6': -0.6627367095169443,
'B_2_7': -0.9860555722662409, 'B_2_8': 0.43310444334999, 'B_2_9':
0.46015407713559053, 'B_3_0': 0.02956803282825371, 'B_3_1': -0.6705016539103403,
'B_3_2': -0.8245624117288343, 'B_3_3': -0.9521246464541717, 'B_3_4':
-0.661818609608863, 'B_3_5': 0.7738939592666305, 'B_3_6': -0.5117570534306701,
'B_3_7': -0.28854277210991297, 'B_3_8': -0.7814159407269603, 'B_3_9':
-0.5497246165986228, 'B_4_0': 0.03806465014480609, 'B_4_1': 0.21335031252939207,

```

'B_4_2': -0.5042024179509021, 'B_4_3': -0.8756903247342679, 'B_4_4':
-0.2157549191277507, 'B_4_5': -0.5262773255658584, 'B_4_6': -0.5552832320049921,
'B_4_7': 0.9105283493726415, 'B_4_8': 0.23027496213293364, 'B_4_9':
0.11261931748074994, 'B_5_0': -0.16676529502382265, 'B_5_1':
-0.1406902198083143, 'B_5_2': 0.08099190759940011, 'B_5_3': 0.38893101793149243,
'B_5_4': 0.40024927044002223, 'B_5_5': -0.6500460627432099, 'B_5_6':
0.00022324911172144635, 'B_5_7': -0.17432297183247814, 'B_5_8':
0.7336433932133808, 'B_5_9': 0.2602094640123853, 'B_6_0': 0.06435805150364393,
'B_6_1': -0.7615186084857436, 'B_6_2': 0.20916302086035476, 'B_6_3':
-0.7579362667531379, 'B_6_4': -0.3223309803798109, 'B_6_5': -0.7071729096676229,
'B_6_6': 0.3803058211082504, 'B_6_7': -0.5816208146044178, 'B_6_8':
-0.2141184410877085, 'B_6_9': 0.7834360596851817, 'B_7_0': -0.5854535608700389,
'B_7_1': 0.01545402663838269, 'B_7_2': -0.15984211089127642, 'B_7_3':
-0.9541143987294551, 'B_7_4': 0.5796610648741694, 'B_7_5': -0.8532891054970926,
'B_7_6': -0.050975058465631884, 'B_7_7': 0.1201430239791208, 'B_7_8':
0.2542843840968825, 'B_7_9': 0.3734906260295936, 'B_8_0': -0.4885206806250242,
'B_8_1': -0.9702420222162702, 'B_8_2': 0.4424184514627256, 'B_8_3':
0.07060006670820562, 'B_8_4': 0.6655130523503525, 'B_8_5': 0.6363489585433746,
'B_8_6': 0.6797858122200087, 'B_8_7': -0.029512573240510487, 'B_8_8':
-0.329415776243006, 'B_8_9': 0.5772852411309126, 'B_9_0': -0.09643765620207523,
'B_9_1': -0.6267844511211997, 'B_9_2': 0.7028485587317193, 'B_9_3':
0.7574685560092476, 'B_9_4': -0.06670668284246317, 'B_9_5': -0.8401145399266208,
'B_9_6': -0.22210434196447615, 'B_9_7': 0.6010045657359546, 'B_9_8':
0.7955125765215327, 'B_9_9': -0.5871239698547602, 'B_10_0': -0.8573922481899624,
'B_10_1': 0.7471538972030205, 'B_10_2': -0.21897822082182694, 'B_10_3':
0.08268670924977517, 'B_10_4': 0.9267703078605027, 'B_10_5':
-0.8582018387298387, 'B_10_6': 0.29366965475514917, 'B_10_7':
-0.8432712633017632, 'B_10_8': -0.24657152163632923, 'B_10_9':
0.6015527994830245}. Best is trial 29 with value: 103.51132933229894.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]]>].

Built matrix B = inverse(A).

[I 2025-04-29 13:52:05,497] Trial 30 finished with value: 214.51161135292918 and
parameters: {'B_0_0': -0.13172497914384385, 'B_0_1': 0.9844210607278419,
'B_0_2': 0.11693738419427224, 'B_0_3': -0.35409893778081214, 'B_0_4':
-0.5541795747085888, 'B_0_5': -0.29596934211339965, 'B_0_6':

-0.2523841179560894, 'B_0_7': -0.8539469181065573, 'B_0_8':
 -0.25824308513498917, 'B_0_9': -0.07080773091633463, 'B_1_0':
 0.4410207494328333, 'B_1_1': 0.3103243797487838, 'B_1_2': 0.4133559973537253,
 'B_1_3': -0.9734399098838229, 'B_1_4': -0.5590185488305386, 'B_1_5':
 0.3201348910317159, 'B_1_6': -0.0317010510041007, 'B_1_7': -0.979479318176675,
 'B_1_8': 0.6028993920150083, 'B_1_9': 0.5401629551148681, 'B_2_0':
 0.09586994272725402, 'B_2_1': -0.8584026149629812, 'B_2_2': 0.5276997980745175,
 'B_2_3': 0.1657529037081007, 'B_2_4': 0.5582807547727591, 'B_2_5':
 0.49845904494376403, 'B_2_6': 0.5999441024765175, 'B_2_7': 0.03565597148115707,
 'B_2_8': -0.7121574539768403, 'B_2_9': 0.33889179729178354, 'B_3_0':
 0.23853809440290186, 'B_3_1': 0.4796613106541927, 'B_3_2': -0.653635608019544,
 'B_3_3': -0.6039491041974274, 'B_3_4': 0.7730108440942705, 'B_3_5':
 0.49495604093510015, 'B_3_6': 0.8082924076266393, 'B_3_7': 0.5122563359045631,
 'B_3_8': 0.19236750586079765, 'B_3_9': 0.30518281931518976, 'B_4_0':
 0.7704732894012196, 'B_4_1': 0.15603737393364225, 'B_4_2': 0.2621348221702071,
 'B_4_3': -0.6801761023128708, 'B_4_4': -0.05163598417011661, 'B_4_5':
 0.42846628167703593, 'B_4_6': -0.4535334573335965, 'B_4_7': -0.5895367225614209,
 'B_4_8': -0.3686229197156572, 'B_4_9': -0.5118287565741897, 'B_5_0':
 -0.5644729786679362, 'B_5_1': -0.1487953311211878, 'B_5_2': 0.8075202574893976,
 'B_5_3': 0.013978100542662242, 'B_5_4': -0.6179176459856603, 'B_5_5':
 -0.8375960946258694, 'B_5_6': 0.3883891673730919, 'B_5_7': -0.23205844641595208,
 'B_5_8': 0.6372252742390114, 'B_5_9': 0.315831321908725, 'B_6_0':
 0.5865580720719097, 'B_6_1': -0.45154684195147143, 'B_6_2': 0.38087066647289514,
 'B_6_3': -0.46715727030643683, 'B_6_4': 0.8693555893115774, 'B_6_5':
 0.2700210774360814, 'B_6_6': -0.34746663565282265, 'B_6_7': -0.4563665126613001,
 'B_6_8': -0.6119653615581458, 'B_6_9': 0.3853890986095809, 'B_7_0':
 -0.5569430862641039, 'B_7_1': 0.18870046653642825, 'B_7_2': -0.4660226466877925,
 'B_7_3': 0.32069851069185984, 'B_7_4': 0.6235806625102676, 'B_7_5':
 0.5504899814264133, 'B_7_6': 0.5164856352429479, 'B_7_7': -0.6183096367293409,
 'B_7_8': -0.814997804970653, 'B_7_9': 0.3939186804050234, 'B_8_0':
 -0.2606725166871844, 'B_8_1': -0.13395447172137231, 'B_8_2':
 -0.9283247962690857, 'B_8_3': -0.4760386304354026, 'B_8_4': -0.9233207203285613,
 'B_8_5': 0.7507878297504704, 'B_8_6': -0.5080745672273097, 'B_8_7':
 0.11352759575455584, 'B_8_8': -0.9128209450909618, 'B_8_9': 0.330357834785203,
 'B_9_0': -0.3504058554068942, 'B_9_1': 0.7878799901761784, 'B_9_2':
 0.7684740140893909, 'B_9_3': -0.34592457564373935, 'B_9_4': 0.7939025734671263,
 'B_9_5': 0.9823921935275821, 'B_9_6': 0.6443230149654171, 'B_9_7':
 0.6828452685553534, 'B_9_8': -0.496962702504002, 'B_9_9': 0.15187074973835757,
 'B_10_0': -0.8568332903446116, 'B_10_1': -0.802062485093094, 'B_10_2':
 0.9878073718259155, 'B_10_3': -0.3432542861366963, 'B_10_4':
 0.49140414585671577, 'B_10_5': 0.6071968049930965, 'B_10_6': 0.7088860765943139,
 'B_10_7': 0.98530402885627, 'B_10_8': -0.5118369591914016, 'B_10_9':
 -0.9101053628177945}. Best is trial 29 with value: 103.51132933229894.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].


```
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```
[I 2025-04-29 13:52:21,355] Trial 31 finished with value: 129.05093738191803 and
parameters: {'B_0_0': -0.17584079437071976, 'B_0_1': -0.732440273738685,
```

```
'B_0_2': -0.9455967394591308, 'B_0_3': -0.2763548822598575, 'B_0_4':
0.5618017371289681, 'B_0_5': 0.13103776862564032, 'B_0_6': -0.370699581363499,
'B_0_7': 0.3055971708755849, 'B_0_8': -0.5306040029885548, 'B_0_9':
-0.9615247367772717, 'B_1_0': 0.5234197051441263, 'B_1_1': 0.2450113080307379,
'B_1_2': 0.5193591965119007, 'B_1_3': -0.9129027269584221, 'B_1_4':
0.6674968631046914, 'B_1_5': 0.2366618760718806, 'B_1_6': 0.12552279441102754,
'B_1_7': 0.24673581526677246, 'B_1_8': 0.7215778287993608, 'B_1_9':
0.1721342829866359, 'B_2_0': 0.16028175217859508, 'B_2_1': 0.9715447309902163,
'B_2_2': 0.5083257307957647, 'B_2_3': -0.1142582652067522, 'B_2_4':
0.4106460600651969, 'B_2_5': -0.21932366605025877, 'B_2_6': -0.536826584282672,
'B_2_7': 0.19176671623776365, 'B_2_8': 0.847800554265624, 'B_2_9':
0.8496962496306162, 'B_3_0': -0.3130275493511292, 'B_3_1': 0.05480138311895599,
'B_3_2': -0.5709609208316748, 'B_3_3': 0.9813495968077892, 'B_3_4':
0.9526134575785128, 'B_3_5': 0.2960276808207458, 'B_3_6': 0.6027398381613973,
'B_3_7': 0.42589831934172606, 'B_3_8': 0.184453273698147, 'B_3_9':
-0.8843709659619825, 'B_4_0': -0.08956660174573106, 'B_4_1':
0.34607687829147293, 'B_4_2': 0.3515125954896956, 'B_4_3': -0.2512172155236805,
'B_4_4': 0.8745501335829005, 'B_4_5': -0.6586330761711736, 'B_4_6':
0.0007117035068100463, 'B_4_7': 0.37795027357062905, 'B_4_8':
0.3903288956657822, 'B_4_9': 0.2943051344022547, 'B_5_0': -0.4449293373653169,
'B_5_1': -0.6807109940004885, 'B_5_2': 0.2691618471168875, 'B_5_3':
0.19661818995608682, 'B_5_4': -0.6348750996273029, 'B_5_5': 0.4061755082297154,
'B_5_6': -0.08875945636736027, 'B_5_7': 0.33216597137424286, 'B_5_8':
0.6679868179448984, 'B_5_9': -0.6539377114176921, 'B_6_0': -0.9520931613662633,
'B_6_1': 0.5526231210636579, 'B_6_2': 0.21720090180217966, 'B_6_3':
0.39528634785180095, 'B_6_4': 0.6695468504775122, 'B_6_5': 0.5995656509525109,
'B_6_6': 0.9126218909190729, 'B_6_7': 0.07115145612719398, 'B_6_8':
-0.022778647333387414, 'B_6_9': -0.1942633442163545, 'B_7_0':
-0.685691735168475, 'B_7_1': 0.1437623211064043, 'B_7_2': -0.44151520183456205,
'B_7_3': 0.8341888741816914, 'B_7_4': 0.16471779246865315, 'B_7_5':
0.18382355740605805, 'B_7_6': -0.28724917146352014, 'B_7_7':
-0.8870610830841392, 'B_7_8': -0.9271473445614825, 'B_7_9':
-0.15179641457666893, 'B_8_0': -0.8225605513674691, 'B_8_1':
0.21155697196947543, 'B_8_2': 0.7543265511665216, 'B_8_3': 0.7581713367070699,
'B_8_4': 0.3149099120618202, 'B_8_5': -0.5717609736835966, 'B_8_6':
0.7187337260905666, 'B_8_7': 0.7644650019201522, 'B_8_8': -0.6006259017931417,
```

'B_8_9': 0.470103649825941, 'B_9_0': -0.42269129312429654, 'B_9_1': 0.5992278184588007, 'B_9_2': 0.9845310418091333, 'B_9_3': -0.9305485094478022, 'B_9_4': 0.7867844142660452, 'B_9_5': 0.242808456525343, 'B_9_6': 0.9372975859463459, 'B_9_7': -0.06928115529767331, 'B_9_8': 0.68782810253132, 'B_9_9': -0.866459744175411, 'B_10_0': -0.32614748084497447, 'B_10_1': -0.8573072552526042, 'B_10_2': 0.9411794042885597, 'B_10_3': 0.6276083363982368, 'B_10_4': 0.6980438940670215, 'B_10_5': 0.8671711582391106, 'B_10_6': -0.821495097482825, 'B_10_7': -0.2265167668235999, 'B_10_8': -0.8493509196928758, 'B_10_9': -0.5726296589309433}. Best is trial 29 with value: 103.51132933229894.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 13:52:37,420] Trial 32 finished with value: 111.39609192370645 and parameters: {'B_0_0': -0.5358557614654087, 'B_0_1': -0.061167722125794244, 'B_0_2': -0.4582469084417877, 'B_0_3': -0.790063348500833, 'B_0_4': -0.6583676435936667, 'B_0_5': -0.69839438416686, 'B_0_6': 0.9366920598663506, 'B_0_7': 0.5133991636985777, 'B_0_8': 0.9261677012337326, 'B_0_9': -0.11931383671434115, 'B_1_0': -0.4388304063079399, 'B_1_1': 0.5906205304741896, 'B_1_2': -0.34401724227852937, 'B_1_3': -0.39722092006341037, 'B_1_4': -0.5296381354284942, 'B_1_5': -0.733235623116014, 'B_1_6': -0.48220313918680335, 'B_1_7': -0.28646689802170233, 'B_1_8': 0.34450853212854615, 'B_1_9': -0.8661216124054859, 'B_2_0': -0.573196259462621, 'B_2_1': 0.6118668077775193, 'B_2_2': -0.6990945913041339, 'B_2_3': -0.3115626098854074, 'B_2_4': 0.7220246652791174, 'B_2_5': -0.6832478839359835, 'B_2_6': -0.826181065475984, 'B_2_7': -0.030614769741578507, 'B_2_8': -0.3912093676684979, 'B_2_9': 0.1255485616780312, 'B_3_0': 0.6015336831572808, 'B_3_1': -0.7184459934575685, 'B_3_2': 0.15978418741748435, 'B_3_3': 0.010907345571930627, 'B_3_4': -0.7047240036938649, 'B_3_5': 0.24551700134738508, 'B_3_6': -0.4481919726356721, 'B_3_7': -0.02408804928285524, 'B_3_8': -0.8271007979314174, 'B_3_9': -0.08002356671702637, 'B_4_0': -0.3832023332752158, 'B_4_1': 0.6383147981470016, 'B_4_2': -0.8771821205452108, 'B_4_3': -0.16115759566170673, 'B_4_4': -0.08230578021975277, 'B_4_5': 0.4455120310722551, 'B_4_6': 0.1470893986786177, 'B_4_7': 0.33108042764447854, 'B_4_8': 0.5485575855904772, 'B_4_9': 0.7171618878432138, 'B_5_0': -0.3685265164895084, 'B_5_1': 0.07528495662052781, 'B_5_2': 0.6734139953096157, 'B_5_3': 0.9691132393108117, 'B_5_4':

```

0.771248235866651, 'B_5_5': -0.2544948200037339, 'B_5_6': -0.6035702959070721,
'B_5_7': -0.021002775934954365, 'B_5_8': 0.4784500746353637, 'B_5_9':
-0.014713239908470732, 'B_6_0': -0.033301890638936404, 'B_6_1':
0.6696962337419652, 'B_6_2': -0.27442051744697105, 'B_6_3': 0.7123934561270584,
'B_6_4': -0.18452649411558741, 'B_6_5': -0.34009845075688794, 'B_6_6':
-0.09109997582459595, 'B_6_7': 0.5196449781762396, 'B_6_8': -0.7405159848292913,
'B_6_9': -0.6010899525734056, 'B_7_0': 0.893860754812922, 'B_7_1':
-0.6425265777849124, 'B_7_2': 0.13410008950348806, 'B_7_3': 0.15704264399063694,
'B_7_4': -0.020543565140515363, 'B_7_5': 0.28659559218210084, 'B_7_6':
-0.5349430829207156, 'B_7_7': 0.10427129019796144, 'B_7_8': -0.2530969546111437,
'B_7_9': 0.3201011740657942, 'B_8_0': -0.710377384121164, 'B_8_1':
0.14033728534216294, 'B_8_2': -0.6231646963410191, 'B_8_3': -0.4382867350322279,
'B_8_4': -0.5569545953593051, 'B_8_5': -0.6279811958597421, 'B_8_6':
0.6449076840901207, 'B_8_7': -0.42429901603143716, 'B_8_8': 0.8457050880916097,
'B_8_9': 0.930454520183478, 'B_9_0': 0.14109573047545276, 'B_9_1':
-0.7068577771218617, 'B_9_2': -0.24839447810842907, 'B_9_3': 0.5895098814294637,
'B_9_4': -0.2625663571798005, 'B_9_5': -0.8177059077604782, 'B_9_6':
0.11243412363781213, 'B_9_7': 0.6833392859621488, 'B_9_8': 0.5859342465441346,
'B_9_9': -0.6432561825419005, 'B_10_0': 0.34207393653668405, 'B_10_1':
-0.5534015583974188, 'B_10_2': -0.557949081586699, 'B_10_3': 0.7406621111442666,
'B_10_4': -0.4956554735743443, 'B_10_5': -0.4687702538943379, 'B_10_6':
-0.9887062862926269, 'B_10_7': 0.733924687568525, 'B_10_8': 0.5793377301540867,
'B_10_9': 0.2521112914795425}. Best is trial 29 with value: 103.51132933229894.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 13:52:52,348] Trial 33 finished with value: 154.09264920900836 and
parameters: {'B_0_0': 0.4955262125989077, 'B_0_1': -0.6889438364995062, 'B_0_2':
-0.0825192827166007, 'B_0_3': -0.2945856130494139, 'B_0_4': -0.8043224780111073,
'B_0_5': -0.027461639274450333, 'B_0_6': 0.833064507678418, 'B_0_7':
-0.9106829442796249, 'B_0_8': -0.41290739128228815, 'B_0_9':
-0.5790689463127071, 'B_1_0': -0.5184696257419212, 'B_1_1': 0.808700412721163,
'B_1_2': -0.0632232726161055, 'B_1_3': -0.06672716237915877, 'B_1_4':
0.5159397720614589, 'B_1_5': -0.6841054360702469, 'B_1_6':
-0.025214598881899475, 'B_1_7': -0.13862927604030173, 'B_1_8':
0.19188316728237909, 'B_1_9': 0.9891242524358683, 'B_2_0': 0.5335088102824532,

```

'B_2_1': -0.20222590802549412, 'B_2_2': 0.6485550435026131, 'B_2_3':
 -0.6519974406412892, 'B_2_4': -0.929681901108259, 'B_2_5': -0.5851922225207393,
 'B_2_6': -0.3159851188937659, 'B_2_7': 0.02094989976992867, 'B_2_8':
 0.22795569827157558, 'B_2_9': 0.8129993200845551, 'B_3_0': 0.019510938909906184,
 'B_3_1': 0.0025257763589441318, 'B_3_2': -0.8905188405504953, 'B_3_3':
 -0.9208751257406769, 'B_3_4': 0.10129672459441674, 'B_3_5':
 -0.12240634576827791, 'B_3_6': 0.6715771820609913, 'B_3_7': -0.6718544222705465,
 'B_3_8': -0.9405561090082418, 'B_3_9': -0.10090323949835078, 'B_4_0':
 -0.5197924576305302, 'B_4_1': -0.8917635407891018, 'B_4_2': 0.44470460795830524,
 'B_4_3': -0.76904586008465, 'B_4_4': 0.21447261001538798, 'B_4_5':
 -0.43352149466955947, 'B_4_6': -0.6464778202970564, 'B_4_7':
 -0.23809198461975145, 'B_4_8': 0.5968626369214483, 'B_4_9':
 -0.21365471772894606, 'B_5_0': 0.49652390543789027, 'B_5_1':
 -0.7415371323376816, 'B_5_2': 0.5400874109368567, 'B_5_3': -0.5208720166848357,
 'B_5_4': 0.3512383971799682, 'B_5_5': 0.13024747858517283, 'B_5_6':
 0.849706540079141, 'B_5_7': -0.2231014144088782, 'B_5_8': -0.858992926107808,
 'B_5_9': -0.9520721386085192, 'B_6_0': 0.6471799507260512, 'B_6_1':
 0.04897656256963567, 'B_6_2': 0.5449832226331996, 'B_6_3': -0.5258373119772008,
 'B_6_4': -0.30673418388379725, 'B_6_5': -0.9303943330215885, 'B_6_6':
 0.9137548026567421, 'B_6_7': 0.33237845526761056, 'B_6_8': 0.8577460737634264,
 'B_6_9': -0.4643171132904368, 'B_7_0': 0.2206979502268085, 'B_7_1':
 0.35342993721942295, 'B_7_2': -0.36125159348997493, 'B_7_3': 0.6892562853309476,
 'B_7_4': 0.8853796965424907, 'B_7_5': 0.7620131330380886, 'B_7_6':
 0.4733526123599576, 'B_7_7': -0.440619934144591, 'B_7_8': -0.43240023663084914,
 'B_7_9': 0.9174534344831551, 'B_8_0': -0.9696366980473298, 'B_8_1':
 0.42777894701944597, 'B_8_2': 0.4083674018594192, 'B_8_3': 0.24292529108059902,
 'B_8_4': 0.9704155152519942, 'B_8_5': -0.372478717187824, 'B_8_6':
 -0.3168557096893686, 'B_8_7': -0.83369998216582, 'B_8_8': -0.11325968964339983,
 'B_8_9': -0.47312391805500076, 'B_9_0': -0.31009432968826445, 'B_9_1':
 0.6624521447061427, 'B_9_2': 0.8622593367844416, 'B_9_3': -0.6222485174297325,
 'B_9_4': -0.2524323155241376, 'B_9_5': 0.8486885129884802, 'B_9_6':
 -0.8676192478353392, 'B_9_7': -0.8071261543394705, 'B_9_8': -0.6676738413716659,
 'B_9_9': 0.18852141157121194, 'B_10_0': -0.692064903408125, 'B_10_1':
 0.9267912352724523, 'B_10_2': -0.10550523602812967, 'B_10_3':
 0.003005561493329423, 'B_10_4': -0.5016159407052951, 'B_10_5':
 -0.057498700179227646, 'B_10_6': 0.3205889926212828, 'B_10_7':
 0.4973307311690047, 'B_10_8': 0.5026588037656277, 'B_10_9':
 0.15422440313429497}. Best is trial 29 with value: 103.51132933229894.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].

```

    Built F_[u_10].
    Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 13:53:06,576] Trial 34 finished with value: 168.25267355341393 and
parameters: {'B_0_0': 0.8006323544878373, 'B_0_1': 0.6284675376660966, 'B_0_2':
0.5089552671032784, 'B_0_3': -0.879901067072688, 'B_0_4': -0.9763816348050943,
'B_0_5': -0.5693543911529635, 'B_0_6': 0.32377671594863555, 'B_0_7':
-0.17582897201255931, 'B_0_8': -0.19323560658736272, 'B_0_9':
0.7614820386946353, 'B_1_0': 0.7832144393010407, 'B_1_1': 0.8101670228260998,
'B_1_2': -0.36852140850024595, 'B_1_3': 0.3782905092237383, 'B_1_4':
-0.45216630107674816, 'B_1_5': -0.6125925305125894, 'B_1_6':
-0.6244277344522089, 'B_1_7': -0.31262691069008386, 'B_1_8':
-0.13868391283595294, 'B_1_9': 0.6552793586890353, 'B_2_0': -0.75261563492403,
'B_2_1': 0.4651531778418905, 'B_2_2': 0.06228073009482449, 'B_2_3':
-0.4198682215015349, 'B_2_4': -0.014217222625007353, 'B_2_5':
-0.3961147018414951, 'B_2_6': 0.1902422713718459, 'B_2_7': -0.13098903274661644,
'B_2_8': -0.6644770524238202, 'B_2_9': -0.7591655123797205, 'B_3_0':
0.09245745050281773, 'B_3_1': 0.7967778059064847, 'B_3_2': -0.30896378061578256,
'B_3_3': 0.4613402123007939, 'B_3_4': 0.3141494827807385, 'B_3_5':
0.8544787303327723, 'B_3_6': 0.635542414252299, 'B_3_7': 0.13187841263281674,
'B_3_8': 0.3109203865035324, 'B_3_9': 0.7888052038363891, 'B_4_0':
-0.19895280651786906, 'B_4_1': -0.34294648240160086, 'B_4_2':
-0.9685940544023113, 'B_4_3': 0.6465711268990992, 'B_4_4': 0.5959417867986625,
'B_4_5': -0.7831800541617002, 'B_4_6': 0.15217761464274648, 'B_4_7':
-0.07131623762956296, 'B_4_8': -0.7549559908919593, 'B_4_9': 0.9515250895802325,
'B_5_0': -0.5649021428225562, 'B_5_1': -0.8568763234206889, 'B_5_2':
0.18779781141378105, 'B_5_3': 0.4729560207234047, 'B_5_4': -0.9257619768265698,
'B_5_5': 0.31426092467950384, 'B_5_6': 0.06393791438387986, 'B_5_7':
-0.7855352824274041, 'B_5_8': -0.6484027236462031, 'B_5_9': 0.1345891086858395,
'B_6_0': -0.403211925243689, 'B_6_1': 0.8678651427296058, 'B_6_2':
0.6311056543784206, 'B_6_3': 0.9574630688970527, 'B_6_4': -0.47585173887818033,
'B_6_5': 0.9300527929399944, 'B_6_6': -0.13615693781271543, 'B_6_7':
-0.30191278527876586, 'B_6_8': -0.8903079366006805, 'B_6_9':
-0.8855247894379868, 'B_7_0': 0.38018228677437627, 'B_7_1': -0.0833005910206247,
'B_7_2': -0.5414871804485106, 'B_7_3': 0.2256612065239607, 'B_7_4':
-0.48934092586477984, 'B_7_5': 0.1537108736019568, 'B_7_6':
-0.27876668097427315, 'B_7_7': 0.6406334086914016, 'B_7_8': 0.6349783983182231,
'B_7_9': -0.04484621171764536, 'B_8_0': -0.2953781635166691, 'B_8_1':
-0.27213160281299276, 'B_8_2': 0.6053147608230713, 'B_8_3':
-0.34020462406072827, 'B_8_4': -0.5760624400604448, 'B_8_5':
-0.8223170830868924, 'B_8_6': -0.06594762064887139, 'B_8_7':
-0.03537844413871305, 'B_8_8': 0.6749992465977561, 'B_8_9': -0.5519473810367933,
'B_9_0': -0.2356372781365682, 'B_9_1': 0.6106433630095081, 'B_9_2':
0.6407747492829676, 'B_9_3': -0.22847006956203897, 'B_9_4':
-0.08093443389237187, 'B_9_5': -0.38949393611841976, 'B_9_6':
0.8564586592702379, 'B_9_7': -0.7550853438316709, 'B_9_8': 0.8588238575429106,

```

'B_9_9': 0.36405557463397953, 'B_10_0': 0.06251286231972775, 'B_10_1':
-0.9253350596466982, 'B_10_2': -0.04831954019242357, 'B_10_3':
-0.18265875209459004, 'B_10_4': -0.6707867522892573, 'B_10_5':
0.3096910149313006, 'B_10_6': 0.9328787965892058, 'B_10_7': 0.12313384590775978,
'B_10_8': 0.4252114873590127, 'B_10_9': -0.8544681039192867}. Best is trial 29
with value: 103.51132933229894.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:53:21,057] Trial 35 finished with value: 141.68478355775937 and
parameters: {'B_0_0': -0.16301934828235043, 'B_0_1': -0.7596101993742982,
'B_0_2': 0.2233357543622907, 'B_0_3': 0.868194673264026, 'B_0_4':
0.3203199102938443, 'B_0_5': -0.8379299643374454, 'B_0_6': -0.28776091412953597,
'B_0_7': 0.10031349129288603, 'B_0_8': -0.19164864285616057, 'B_0_9':
0.6606151407408412, 'B_1_0': 0.6230485893257787, 'B_1_1': 0.22142541980437835,
'B_1_2': -0.2505532686372436, 'B_1_3': -0.48443397994877657, 'B_1_4':
-0.7803688499380039, 'B_1_5': -0.287302586696623, 'B_1_6': -0.17191841592871537,
'B_1_7': 0.3481293917060657, 'B_1_8': 0.3131564521091954, 'B_1_9':
-0.8511858835103112, 'B_2_0': -0.20738655544630757, 'B_2_1':
-0.6300778429096234, 'B_2_2': -0.6795108434727247, 'B_2_3': 0.648058256479747,
'B_2_4': -0.9070078709018101, 'B_2_5': -0.1600179057137998, 'B_2_6':
-0.656956761582294, 'B_2_7': 0.7510738270433786, 'B_2_8': 0.11294021628456963,
'B_2_9': -0.5318663399738409, 'B_3_0': 0.004806028589425249, 'B_3_1':
0.45834299221216934, 'B_3_2': 0.9070736576604788, 'B_3_3': -0.5535169115940706,
'B_3_4': 0.7646399088161033, 'B_3_5': 0.8599221460516306, 'B_3_6':
0.824512678173491, 'B_3_7': 0.26812042658658153, 'B_3_8': 0.2570448957807212,
'B_3_9': -0.19164294629603096, 'B_4_0': 0.4997478069615, 'B_4_1':
0.06196215633484514, 'B_4_2': 0.351327381032023, 'B_4_3': -0.14165335037938498,
'B_4_4': 0.4583629988054154, 'B_4_5': 0.6424536048570071, 'B_4_6':
-0.7000562119264241, 'B_4_7': 0.6592008368369715, 'B_4_8': 0.07978713294480833,
'B_4_9': 0.6824384999986441, 'B_5_0': -0.13602262144479105, 'B_5_1':
-0.23949983894159066, 'B_5_2': 0.8218202503841003, 'B_5_3': -0.4926848502541654,
'B_5_4': 0.6815404040665634, 'B_5_5': -0.03163926643137982, 'B_5_6':
0.027908976251494133, 'B_5_7': -0.3792328026508216, 'B_5_8':
0.14462355722565112, 'B_5_9': -0.34608183137310156, 'B_6_0':
-0.9122465598015284, 'B_6_1': -0.4514427614181751, 'B_6_2': -0.9773694940549504,

'B_6_3': 0.9471424783725761, 'B_6_4': 0.9221794754431001, 'B_6_5':
-0.207592511004836, 'B_6_6': 0.4521742915924991, 'B_6_7': -0.3053039251473585,
'B_6_8': 0.33876445103023256, 'B_6_9': 0.6047514852310385, 'B_7_0':
0.8845773644906367, 'B_7_1': -0.1975796938176494, 'B_7_2': 0.5609300062529745,
'B_7_3': -0.4634616677484048, 'B_7_4': 0.9708579615114237, 'B_7_5':
-0.9389761944465848, 'B_7_6': 0.20498703687221953, 'B_7_7': 0.3158218436696276,
'B_7_8': 0.3723302030067752, 'B_7_9': -0.7516325939976037, 'B_8_0':
0.8682889712767983, 'B_8_1': -0.631257107480836, 'B_8_2': 0.24305173841488537,
'B_8_3': -0.5491013633362918, 'B_8_4': -0.38171149429788365, 'B_8_5':
0.09198791262168626, 'B_8_6': -0.16395703124213856, 'B_8_7':
-0.6727533100665175, 'B_8_8': -0.6515459232067204, 'B_8_9':
-0.16207331943765668, 'B_9_0': 0.5094202969854915, 'B_9_1': 0.7879700007586212,
'B_9_2': -0.8234272287264219, 'B_9_3': -0.21165426303238588, 'B_9_4':
-0.7915137838287043, 'B_9_5': -0.9571468883474562, 'B_9_6': 0.3202029916350342,
'B_9_7': 0.20284989509384377, 'B_9_8': -0.6672221546465253, 'B_9_9':
-0.527050590475733, 'B_10_0': -0.9430586477838536, 'B_10_1': 0.6631954672749194,
'B_10_2': 0.9398873801761318, 'B_10_3': -0.7217901171465122, 'B_10_4':
-0.5316790767447097, 'B_10_5': 0.7294508574149294, 'B_10_6': 0.8443983199532459,
'B_10_7': -0.1591377820425155, 'B_10_8': -0.8897675049072332, 'B_10_9':
-0.9120041692882176}. Best is trial 29 with value: 103.51132933229894.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:53:35,354] Trial 36 finished with value: 95.85134184163296 and
parameters: {'B_0_0': 0.14642073481664064, 'B_0_1': -0.21206768439150558,
'B_0_2': -0.933268193396244, 'B_0_3': 0.16483973677886232, 'B_0_4':
-0.9676209775537356, 'B_0_5': 0.5689964771355605, 'B_0_6': -0.3832630110473243,
'B_0_7': -0.9102891540417204, 'B_0_8': 0.1749373975466435, 'B_0_9':
-0.20258485942255167, 'B_1_0': 0.938124818447156, 'B_1_1': 0.08751335175733033,
'B_1_2': -0.44502642154672356, 'B_1_3': 0.4146889519752701, 'B_1_4':
-0.45244583496543156, 'B_1_5': 0.8004017169869788, 'B_1_6':
-0.24805049839539461, 'B_1_7': 0.09873975429688642, 'B_1_8':
-0.8898652544280269, 'B_1_9': -0.14627994885458306, 'B_2_0': 0.65807980852877,
'B_2_1': 0.605184634619417, 'B_2_2': -0.545849253470018, 'B_2_3':
-0.5423634393990859, 'B_2_4': 0.6282645181822089, 'B_2_5': 0.8521888719352644,
'B_2_6': -0.8011853758794529, 'B_2_7': -0.098883893936074, 'B_2_8':

```

-0.3218405276508266, 'B_2_9': 0.73396314180573, 'B_3_0': -0.8246206689335629,
'B_3_1': -0.5719706337350859, 'B_3_2': 0.4998873084212587, 'B_3_3':
-0.8884381790171931, 'B_3_4': -0.014779946185707216, 'B_3_5':
-0.11462807133415442, 'B_3_6': -0.3278856656022776, 'B_3_7':
-0.20874681566853093, 'B_3_8': 0.05928236212186122, 'B_3_9':
-0.6704926303910015, 'B_4_0': 0.1425518390015137, 'B_4_1': 0.604756012013973,
'B_4_2': 0.515118640984515, 'B_4_3': -0.6852781887063282, 'B_4_4':
-0.6944860499118817, 'B_4_5': -0.45901475186507157, 'B_4_6':
-0.2750720418447463, 'B_4_7': -0.1812579488112045, 'B_4_8': 0.35580048933002373,
'B_4_9': -0.8777727443116459, 'B_5_0': -0.9213480456323886, 'B_5_1':
-0.21401708537106012, 'B_5_2': 0.3903841104332013, 'B_5_3': -0.6069980209461645,
'B_5_4': 0.2801788728873784, 'B_5_5': -0.47554028687820704, 'B_5_6':
0.7644504954507623, 'B_5_7': 0.7834660896479007, 'B_5_8': -0.40137133552297355,
'B_5_9': -0.5346123671527838, 'B_6_0': -0.17561813530886206, 'B_6_1':
-0.5137474613433055, 'B_6_2': 0.34132000971757326, 'B_6_3': 0.6456080754056053,
'B_6_4': 0.34272241920420976, 'B_6_5': 0.6422138855537118, 'B_6_6':
-0.20395551882920127, 'B_6_7': -0.6804923979964554, 'B_6_8':
0.47114287584723247, 'B_6_9': -0.27626047981972335, 'B_7_0': 0.3391162339536824,
'B_7_1': -0.454124994876104, 'B_7_2': -0.8291653230827881, 'B_7_3':
0.9753119550657714, 'B_7_4': -0.6807210142974134, 'B_7_5': 0.9670735680476925,
'B_7_6': 0.9450142664815657, 'B_7_7': 0.5817598457379185, 'B_7_8':
0.31565746492235514, 'B_7_9': 0.15405795949714762, 'B_8_0': 0.7248810598481126,
'B_8_1': -0.4169097471934139, 'B_8_2': -0.06399119927994346, 'B_8_3':
0.23639213934567382, 'B_8_4': -0.1758428577784077, 'B_8_5':
-0.14357682090126944, 'B_8_6': -0.33603635834587287, 'B_8_7':
0.12717899703538138, 'B_8_8': 0.6941375654699298, 'B_8_9': -0.5909738115475005,
'B_9_0': 0.8601774167318139, 'B_9_1': 0.3743935544214947, 'B_9_2':
0.6400809611064597, 'B_9_3': 0.1112575714074231, 'B_9_4': 0.5534430688196457,
'B_9_5': -0.9579232132678549, 'B_9_6': 0.6303932410634796, 'B_9_7':
-0.9105250590109742, 'B_9_8': 0.7720284851984678, 'B_9_9': 0.9740862395325138,
'B_10_0': -0.4077464248899799, 'B_10_1': -0.5735692585541285, 'B_10_2':
0.5254194604389206, 'B_10_3': -0.4890078449778986, 'B_10_4': 0.7238135186809629,
'B_10_5': -0.7863716675281961, 'B_10_6': -0.7406087111688706, 'B_10_7':
0.9487200790992187, 'B_10_8': 0.34420101857460916, 'B_10_9':
0.6868367043693266}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.


```

Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 13:53:52,990] Trial 37 finished with value: 133.26065906838838 and
parameters: {'B_0_0': -0.3478278328439943, 'B_0_1': 0.3494378240507967, 'B_0_2':
0.18659881807102252, 'B_0_3': 0.20423723784343495, 'B_0_4': 0.3614142698846887,
'B_0_5': 0.1492111627956152, 'B_0_6': -0.14045650878072147, 'B_0_7':
-0.4436730651265173, 'B_0_8': 0.5317910352081401, 'B_0_9': -0.5419805885936912,
'B_1_0': 0.38086152618574, 'B_1_1': -0.5280010111543332, 'B_1_2':
0.24815373650981876, 'B_1_3': 0.4892149689154772, 'B_1_4': -0.5569464999650524,
'B_1_5': -0.8713122642946096, 'B_1_6': -0.7306719888715354, 'B_1_7':
0.20987399519663041, 'B_1_8': 0.6919038843269563, 'B_1_9': -0.9009062939538598,
'B_2_0': 0.4636194929024595, 'B_2_1': -0.31418908101889675, 'B_2_2':
-0.04251382755934541, 'B_2_3': 0.84914190869422, 'B_2_4': -0.33269642062302596,
'B_2_5': -0.06863287079981761, 'B_2_6': -0.9629088187738514, 'B_2_7':
-0.82843453439042, 'B_2_8': -0.4779886495531276, 'B_2_9': -0.9348525631742727,
'B_3_0': 0.26012314557553684, 'B_3_1': -0.14573125353278737, 'B_3_2':
0.0953275014014836, 'B_3_3': -0.6441985719252674, 'B_3_4': -0.4040545817155001,
'B_3_5': 0.3243724602083957, 'B_3_6': 0.9212993927881916, 'B_3_7':
-0.8901747882676257, 'B_3_8': 0.7729609552934269, 'B_3_9': 0.15222477859619277,
'B_4_0': 0.126560742176135, 'B_4_1': 0.0005514997113654774, 'B_4_2':
-0.8525997099097169, 'B_4_3': -0.8120438335624018, 'B_4_4': 0.19980577120499055,
'B_4_5': -0.3149020285575945, 'B_4_6': 0.8261907718238077, 'B_4_7':
-0.18495073602824497, 'B_4_8': -0.7063027117687706, 'B_4_9': 0.4251432903340313,
'B_5_0': -0.40916022554680054, 'B_5_1': 0.05043120856466854, 'B_5_2':
0.39170891530023666, 'B_5_3': 0.7925229401800993, 'B_5_4': 0.5785385657073188,
'B_5_5': 0.34919491438799377, 'B_5_6': 0.3556737011423843, 'B_5_7':
0.8829307951835312, 'B_5_8': -0.4042682869568587, 'B_5_9': -0.9877980534915489,
'B_6_0': -0.45152997239156656, 'B_6_1': -0.5585750683390469, 'B_6_2':
0.32043247852571644, 'B_6_3': 0.2647489675357475, 'B_6_4': 0.18505353775535305,
'B_6_5': -0.9578379824452816, 'B_6_6': 0.4529805478122819, 'B_6_7':
-0.34940978327872674, 'B_6_8': 0.3268108582331295, 'B_6_9': 0.11202528843426363,
'B_7_0': -0.3113927676527438, 'B_7_1': -0.7235389911338667, 'B_7_2':
-0.8030505826322477, 'B_7_3': 0.6564062995622026, 'B_7_4': 0.8284271333214848,
'B_7_5': 0.2965803325933132, 'B_7_6': -0.7852004525762429, 'B_7_7':
-0.19398611056128334, 'B_7_8': 0.45329961620821124, 'B_7_9': 0.5540453503399347,
'B_8_0': -0.7559271278636446, 'B_8_1': -0.9899391766861428, 'B_8_2':
0.420031350843598, 'B_8_3': -0.28393957154731286, 'B_8_4': -0.4861269633958447,
'B_8_5': -0.9644630098167396, 'B_8_6': 0.07979282538776933, 'B_8_7':
0.6952784561434437, 'B_8_8': 0.9061106596197239, 'B_8_9': 0.13017500221431266,
'B_9_0': 0.028633428924990545, 'B_9_1': -0.8215883567313026, 'B_9_2':
0.09676814705434755, 'B_9_3': -0.2376414236485186, 'B_9_4': 0.21212785614948237,
'B_9_5': -0.2203218312044375, 'B_9_6': -0.5142077355682738, 'B_9_7':
-0.8019687550478145, 'B_9_8': -0.36646147971398646, 'B_9_9':
-0.7986432281102505, 'B_10_0': -0.6397440373685118, 'B_10_1': 0.964925593987501,
'B_10_2': -0.11067495732489507, 'B_10_3': 0.0631060976799327, 'B_10_4':
0.7397646868996863, 'B_10_5': 0.9812264229715619, 'B_10_6': 0.1637746097024695,
'B_10_7': 0.6186981656674599, 'B_10_8': -0.34265677181338294, 'B_10_9':
-0.3846269642543636}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```
Built F_[u_1].  
Built F_[u_2].  
Built F_[u_3].  
Built F_[u_4].  
Built F_[u_5].  
Built F_[u_6].  
Built F_[u_7].  
Built F_[u_8].  
Built F_[u_9].  
Built F_[u_10].  
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:54:06,877] Trial 38 finished with value: 174.76402725245208 and parameters: {'B_0_0': -0.19264663243077085, 'B_0_1': 0.3415643800938253,

'B_0_2': 0.3607395447250956, 'B_0_3': -0.3658259193940874, 'B_0_4': -0.7262443502094752, 'B_0_5': 0.262144587801707, 'B_0_6': -0.7359147928662462, 'B_0_7': 0.15541516629407348, 'B_0_8': 0.3829758135514918, 'B_0_9': 0.3972107628610142, 'B_1_0': 0.5018747791060016, 'B_1_1': 0.7382616547000158, 'B_1_2': -8.119590971888968e-05, 'B_1_3': 0.4558933952685509, 'B_1_4': 0.23596201602435674, 'B_1_5': -0.6204529192005885, 'B_1_6': -0.9391345031585718, 'B_1_7': -0.42660805635967114, 'B_1_8': -0.11279326145151358, 'B_1_9': 0.23186976996873954, 'B_2_0': 0.6929194743963467, 'B_2_1': -0.601433480921792, 'B_2_2': -0.7414114160010153, 'B_2_3': 0.9176902959991684, 'B_2_4': -0.7760717724347139, 'B_2_5': -0.04258768118776546, 'B_2_6': 0.1694161830107377, 'B_2_7': 0.08192135339672779, 'B_2_8': -0.8196563290747725, 'B_2_9': -0.8770867102728337, 'B_3_0': -0.7811428136271493, 'B_3_1': 0.16946361406777277, 'B_3_2': 0.08780287751595584, 'B_3_3': -0.5271932317672463, 'B_3_4': 0.27279280085427127, 'B_3_5': 0.6337336540701857, 'B_3_6': -0.9076736595037443, 'B_3_7': -0.003702711774876022, 'B_3_8': 0.3748346159633267, 'B_3_9': -0.4904223844020196, 'B_4_0': -0.38061360751956475, 'B_4_1': 0.22500304791897996, 'B_4_2': 0.787911881018599, 'B_4_3': 0.6136503521997407, 'B_4_4': 0.1645978338171754, 'B_4_5': 0.4558390011593534, 'B_4_6': -0.2676515089885586, 'B_4_7': 0.2791117456355403, 'B_4_8': -0.06696044956743719, 'B_4_9': -0.6139264498140293, 'B_5_0': 0.39944178227696914, 'B_5_1': 0.11179998195066765, 'B_5_2': -0.2799292677318891, 'B_5_3': 0.8144515963556889, 'B_5_4': -0.9485504467030681, 'B_5_5': -0.36476916052599795, 'B_5_6': -0.8773748324002598, 'B_5_7': 0.5277240861188217, 'B_5_8': 0.40017275914530925, 'B_5_9': -0.3339226069020923, 'B_6_0': 0.3478125463700046, 'B_6_1': -0.20604429213201936, 'B_6_2': 0.50738250701642, 'B_6_3': -0.09071634778132198, 'B_6_4': -0.1736861388933022, 'B_6_5': 0.860765232666836, 'B_6_6': -0.4932593822164135, 'B_6_7': -0.7515142668068004, 'B_6_8': 0.16739536425372714, 'B_6_9': 0.9292571883226741, 'B_7_0': -0.24187835682017045, 'B_7_1': -0.8671672080150126, 'B_7_2': -0.3181755707270565, 'B_7_3': 0.011744675552725736, 'B_7_4': -0.6693924468682169, 'B_7_5': 0.31304239175185433, 'B_7_6': 0.9855437902401971, 'B_7_7': -0.09555583755674124,

'B_7_8': -0.2876374576741789, 'B_7_9': -0.196605749667756, 'B_8_0': -0.7620105667775017, 'B_8_1': 0.7588817308239231, 'B_8_2': -0.16864520206772304, 'B_8_3': -0.22319666537881422, 'B_8_4': 0.3161867404023391, 'B_8_5': -0.11477382655637547, 'B_8_6': 0.2935022682395352, 'B_8_7': -0.8683363455410408, 'B_8_8': 0.6211722205700103, 'B_8_9': 0.8721978267012171, 'B_9_0': 0.2945554645955841, 'B_9_1': 0.9000405391404454, 'B_9_2': -0.6901640175474727, 'B_9_3': -0.044715944828899734, 'B_9_4': 0.20157733321198568, 'B_9_5': 0.16183856189394974, 'B_9_6': -0.19510567740369444, 'B_9_7': -0.3205928694548025, 'B_9_8': -0.7388884353501708, 'B_9_9': -0.21434994788668915, 'B_10_0': -0.6679315860911395, 'B_10_1': 0.4640440394545924, 'B_10_2': -0.5765147477207118, 'B_10_3': -0.874040695536623, 'B_10_4': 0.10368495287812385, 'B_10_5': 0.18855487078441513, 'B_10_6': 0.7363878613354009, 'B_10_7': 0.1524235925870101, 'B_10_8': -0.30594790962757745, 'B_10_9': 0.5999635254221147}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$\langle F_{[u_i]}, F_{[u_j]} \rangle$].

Built matrix B = inverse(A).

[I 2025-04-29 13:54:20,053] Trial 39 finished with value: 220.8114858715504 and parameters: {'B_0_0': 0.07975220851109377, 'B_0_1': -0.8330250364302403,

'B_0_2': 0.13038410942022827, 'B_0_3': 0.8078767199348915, 'B_0_4': -0.20530100151352326, 'B_0_5': -0.40327491223839584, 'B_0_6': -0.7059304751043203, 'B_0_7': -0.6901378665679925, 'B_0_8': -0.13311246100319862, 'B_0_9': 0.18944605183982843, 'B_1_0': -0.8303814977641886, 'B_1_1': 0.8703458062698282, 'B_1_2': 0.505164562541488, 'B_1_3': 0.17151535820165198, 'B_1_4': 0.6503085684592018, 'B_1_5': -0.8312302215120156, 'B_1_6': -0.04585902208841375, 'B_1_7': 0.25758439118609355, 'B_1_8': 0.6507327541142127, 'B_1_9': 0.5612697288975501, 'B_2_0': -0.44140288556746876, 'B_2_1': 0.8729427357272244, 'B_2_2': -0.7437952339668767, 'B_2_3': 0.7427645648698813, 'B_2_4': 0.9324616942171446, 'B_2_5': -0.6392360240192014, 'B_2_6': 0.43969080712349484, 'B_2_7': -0.910234672819563, 'B_2_8': -0.18726280622447322, 'B_2_9': 0.032240932411394585, 'B_3_0': 0.15967754730107542, 'B_3_1': 0.8631557515321477, 'B_3_2': 0.34105636084250546, 'B_3_3': -0.03706965658591077, 'B_3_4': 0.6131565718064609, 'B_3_5': 0.8920549854897297, 'B_3_6': -0.9463090256090642, 'B_3_7': 0.9553921190669565, 'B_3_8': -0.8194583646539129, 'B_3_9':

0.6224858229521475, 'B_4_0': -0.4349371899482033, 'B_4_1': -0.7282842776473339,
'B_4_2': -0.11908583336660694, 'B_4_3': 0.2850340273062748, 'B_4_4':
-0.23571534685351947, 'B_4_5': -0.881960491600256, 'B_4_6': 0.19693305652425885,
'B_4_7': 0.796758486699136, 'B_4_8': -0.23123357542444167, 'B_4_9':
-0.5616611002909624, 'B_5_0': -0.11369685505302551, 'B_5_1':
-0.8942418321234139, 'B_5_2': 0.6345654172709441, 'B_5_3': 0.6461548056813697,
'B_5_4': 0.1742409175906643, 'B_5_5': -0.2908562799019888, 'B_5_6':
0.593571061602651, 'B_5_7': 0.10828030908096697, 'B_5_8': 0.6453459709290321,
'B_5_9': 0.2602884093143809, 'B_6_0': 0.5631228369613386, 'B_6_1':
0.19515472657116484, 'B_6_2': -0.17000068493298093, 'B_6_3': 0.9067328026537524,
'B_6_4': 0.08140537326011255, 'B_6_5': 0.20807724671026606, 'B_6_6':
-0.5534344739587964, 'B_6_7': 0.24878856306600738, 'B_6_8': 0.14178314597622887,
'B_6_9': -0.6240199720648907, 'B_7_0': -0.8720175472407754, 'B_7_1':
0.20663145303524666, 'B_7_2': 0.5221106933136346, 'B_7_3': 0.04633471442557635,
'B_7_4': -0.5409902018671016, 'B_7_5': 0.32971351385220915, 'B_7_6':
-0.8313906613129876, 'B_7_7': -0.11516926449212983, 'B_7_8': -0.666533818371128,
'B_7_9': -0.6247710789014309, 'B_8_0': -0.5894579806667568, 'B_8_1':
-0.22432002705292375, 'B_8_2': -0.8887225928716127, 'B_8_3':
-0.20236898696038563, 'B_8_4': 0.023180745287107563, 'B_8_5':
-0.03307731576667605, 'B_8_6': -0.23208748120197964, 'B_8_7':
0.6715102839800899, 'B_8_8': -0.7022531490074019, 'B_8_9': 0.011279552687120553,
'B_9_0': -0.8671579944260225, 'B_9_1': -0.8491117558217387, 'B_9_2':
0.14625438236065058, 'B_9_3': 0.13071648127788538, 'B_9_4': 0.7485063554320248,
'B_9_5': 0.11458987780694496, 'B_9_6': 0.9103862204016033, 'B_9_7':
-0.8935842001681391, 'B_9_8': -0.7959447841341327, 'B_9_9': -0.9027495940906448,
'B_10_0': -0.6231518363395907, 'B_10_1': 0.08171481966452254, 'B_10_2':
0.28823291922630734, 'B_10_3': -0.8997413872616358, 'B_10_4':
0.8830223618942865, 'B_10_5': 0.6774874457677544, 'B_10_6': -0.4027072853289556,
'B_10_7': -0.8371686901515935, 'B_10_8': -0.6536235950262665, 'B_10_9':
-0.738650060765154}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].
Built F__[u_10].
Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 13:54:41,154] Trial 40 finished with value: 138.94395785560428 and
parameters: {'B_0_0': -0.7458981169880852, 'B_0_1': 0.03675214355815326,
'B_0_2': -0.5032220638630673, 'B_0_3': -0.28088336127229396, 'B_0_4':

0.9692205601009585, 'B_0_5': 0.36494221287218265, 'B_0_6': 0.8889817207530626,
'B_0_7': -0.7077380210344311, 'B_0_8': -0.23336383983151288, 'B_0_9':
0.10836897049906136, 'B_1_0': -0.8380190374602987, 'B_1_1': -0.9817088616166434,
'B_1_2': 0.33745998091789664, 'B_1_3': 0.280802643266842, 'B_1_4':
-0.17663815620076873, 'B_1_5': -0.020504729456575332, 'B_1_6':
-0.1674228588276626, 'B_1_7': -0.94247001120712, 'B_1_8': -0.32596421968942435,
'B_1_9': -0.6393771343133661, 'B_2_0': -0.7951201055792899, 'B_2_1':
0.9043064591009127, 'B_2_2': 0.19673225558348784, 'B_2_3': 0.4527394427884732,
'B_2_4': -0.36557978833308735, 'B_2_5': -0.21156868360184988, 'B_2_6':
-0.5208904920844875, 'B_2_7': -0.798007971488459, 'B_2_8': -0.6355173578656641,
'B_2_9': 0.5890778096870419, 'B_3_0': 0.35154069426387613, 'B_3_1':
0.09291917445124431, 'B_3_2': -0.04932205172379922, 'B_3_3': 0.8364198816391264,
'B_3_4': -0.8459848033982366, 'B_3_5': -0.43441038978892643, 'B_3_6':
-0.29712304992998617, 'B_3_7': 0.55920310818132, 'B_3_8': 0.9751789479207031,
'B_3_9': -0.5134723807808369, 'B_4_0': 0.7411285215914196, 'B_4_1':
0.6546330656549468, 'B_4_2': -0.5451783526693021, 'B_4_3': -0.19932149120553933,
'B_4_4': -0.17884771880081218, 'B_4_5': 0.9468769236825854, 'B_4_6':
-0.6318340352697813, 'B_4_7': 0.5923327827874685, 'B_4_8': -0.32829842593724234,
'B_4_9': 0.45712328463679763, 'B_5_0': -0.15837821149458753, 'B_5_1':
0.1551523309707492, 'B_5_2': 0.6585713111857396, 'B_5_3': 0.6022478186680906,
'B_5_4': 0.7255179034086063, 'B_5_5': -0.8719576958175751, 'B_5_6':
0.38026009501869473, 'B_5_7': -0.7128249940909095, 'B_5_8':
-0.16667923532354623, 'B_5_9': 0.09620506180828325, 'B_6_0':
-0.19257582020127517, 'B_6_1': 0.038831118089954764, 'B_6_2':
0.9831220914212659, 'B_6_3': -0.7222697188485476, 'B_6_4': 0.3474029697661225,
'B_6_5': -0.20591808655648192, 'B_6_6': -0.726640598163489, 'B_6_7':
-0.6757012085188672, 'B_6_8': 0.8895607503663214, 'B_6_9': 0.7516109634892887,
'B_7_0': 0.8059430816217432, 'B_7_1': 0.9732348961382253, 'B_7_2':
-0.5775355277831344, 'B_7_3': -0.2862301120191577, 'B_7_4': 0.33505367925123597,
'B_7_5': -0.032070971075918386, 'B_7_6': -0.16158872391089962, 'B_7_7':
-0.28153372631552576, 'B_7_8': 0.18568565124803538, 'B_7_9': 0.1496749887073623,
'B_8_0': -0.6703134885528046, 'B_8_1': -0.055955121713490596, 'B_8_2':
0.10587411689550286, 'B_8_3': 0.13916813868815314, 'B_8_4': -0.5739877580335434,
'B_8_5': 0.47969735665952773, 'B_8_6': -0.9401057922542821, 'B_8_7':
-0.28743292602453796, 'B_8_8': 0.5548210441005543, 'B_8_9': 0.12650798368969673,
'B_9_0': -0.4728983836284967, 'B_9_1': 0.38600717629923453, 'B_9_2':
0.13343406942902525, 'B_9_3': 0.5865623467236465, 'B_9_4': 0.4653498686558504,
'B_9_5': 0.2180136856317445, 'B_9_6': -0.023408202884588225, 'B_9_7':
-0.7262993911783037, 'B_9_8': -0.47256062458694015, 'B_9_9':
-0.16018304362417113, 'B_10_0': 0.1966125882726495, 'B_10_1':
0.027141553326303525, 'B_10_2': -0.4193946386001316, 'B_10_3':
-0.9772005715763731, 'B_10_4': -0.007446340636388382, 'B_10_5':
-0.4244605426467587, 'B_10_6': 0.4645244354068594, 'B_10_7':
-0.9425863947644427, 'B_10_8': 0.16795906638546598, 'B_10_9':
0.8741189902620095}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built $F_{[u_1]}$.

Built $F_{[u_2]}$.

```

Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```

[I 2025-04-29 13:55:02,519] Trial 41 finished with value: 161.12967996577697 and
parameters: {'B_0_0': -0.6446696511192309, 'B_0_1': -0.056134008313821204,
'B_0_2': -0.8089844562020664, 'B_0_3': 0.24885018627747812, 'B_0_4':
0.10005272758052675, 'B_0_5': -0.18341626205859196, 'B_0_6':
0.042326502518007514, 'B_0_7': 0.7852178237686402, 'B_0_8': -0.6013372814240873,
'B_0_9': -0.9453222668234125, 'B_1_0': 0.71609647170422, 'B_1_1':
0.15269182312805474, 'B_1_2': 0.7735037418144479, 'B_1_3': 0.19195154235808176,
'B_1_4': 0.6141189581202109, 'B_1_5': -0.13924674055680708, 'B_1_6':
0.49472266395077225, 'B_1_7': 0.8178221044419149, 'B_1_8': 0.14280346816996836,
'B_1_9': -0.6320454018864885, 'B_2_0': -0.4577807222402076, 'B_2_1':
-0.5966837377431365, 'B_2_2': -0.5012230853604489, 'B_2_3': -0.383297410284948,
'B_2_4': 0.10916489777131311, 'B_2_5': 0.17422342633838883, 'B_2_6':
-0.14553050046111493, 'B_2_7': 0.22853314404525404, 'B_2_8':
-0.8284816472894955, 'B_2_9': -0.8161792906248433, 'B_3_0': -0.6497776684918363,
'B_3_1': 0.035793446119882555, 'B_3_2': -0.5662905361773392, 'B_3_3':
-0.4290968810846969, 'B_3_4': -0.19738058437747175, 'B_3_5': 0.617494925540552,
'B_3_6': -0.962552970892537, 'B_3_7': 0.29577006406851236, 'B_3_8':
0.334375531460557, 'B_3_9': 0.5913384451061403, 'B_4_0': 0.8568508193394202,
'B_4_1': -0.9501270590804062, 'B_4_2': -0.685508263083832, 'B_4_3':
0.7646191622308662, 'B_4_4': -0.08213324686970958, 'B_4_5': 0.12788946508825783,
'B_4_6': 0.3227938633718319, 'B_4_7': 0.35455187690136203, 'B_4_8':
0.8548459985666863, 'B_4_9': 0.9839598464751058, 'B_5_0': 0.43591177264326175,
'B_5_1': -0.4213020057738014, 'B_5_2': 0.7519097565428003, 'B_5_3':
-0.892542589209292, 'B_5_4': -0.5324599113084976, 'B_5_5': 0.7868237205781554,
'B_5_6': -0.48274080141007725, 'B_5_7': -0.553110901948938, 'B_5_8':
-0.7165119195823104, 'B_5_9': 0.7110412801408947, 'B_6_0':
0.00011535319835165314, 'B_6_1': 0.3462885420472266, 'B_6_2':
-0.5166485595314412, 'B_6_3': 0.5109200336022086, 'B_6_4': 0.5148109790249551,
'B_6_5': -0.37105591068156574, 'B_6_6': -0.17684838663963998, 'B_6_7':
-0.5748132506882023, 'B_6_8': 0.9435908261793504, 'B_6_9': 0.288293102575655,
'B_7_0': 0.8603968838167206, 'B_7_1': -0.3892966236143163, 'B_7_2':
0.5581440077338418, 'B_7_3': 0.5264001148917892, 'B_7_4': 0.3794781906957523,
'B_7_5': 0.9219651710282648, 'B_7_6': -0.21076043450949522, 'B_7_7':
-0.7311796992922805, 'B_7_8': 0.3399835413032404, 'B_7_9': 0.4540334049448682,
'B_8_0': 0.14861993544845298, 'B_8_1': -0.5805005715958753, 'B_8_2':
0.0088227645053236, 'B_8_3': -0.37186432159312377, 'B_8_4': -0.1436162742299556,

```

'B_8_5': 0.28753278017243455, 'B_8_6': 0.48552519052785237, 'B_8_7':
-0.5301951120153405, 'B_8_8': -0.18745578591903234, 'B_8_9':
-0.45706008943657417, 'B_9_0': -0.5967094045534991, 'B_9_1': 0.5611066955917601,
'B_9_2': 0.810320334985245, 'B_9_3': -0.2569402723445381, 'B_9_4':
0.4975322702600584, 'B_9_5': 0.786050672941534, 'B_9_6': 0.6853259368498585,
'B_9_7': -0.6280294906923327, 'B_9_8': 0.9034579253964554, 'B_9_9':
-0.24418064794966432, 'B_10_0': -0.318303490875804, 'B_10_1':
-0.8663859719899949, 'B_10_2': -0.027376164872331632, 'B_10_3':
-0.7741530410309612, 'B_10_4': -0.49789115456369654, 'B_10_5':
-0.3624536089007828, 'B_10_6': 0.6114761205135344, 'B_10_7': 0.7434675277838751,
'B_10_8': -0.39111669142629, 'B_10_9': 0.7102222800642564}. Best is trial 36
with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 13:55:18,691] Trial 42 finished with value: 195.28532573642286 and
parameters: {'B_0_0': 0.7073172559921863, 'B_0_1': 0.5240088817620496, 'B_0_2':
-0.6995436742499406, 'B_0_3': 0.37683923686344345, 'B_0_4': 0.8811476478695262,
'B_0_5': 0.361430147958971, 'B_0_6': -0.3333986387664888, 'B_0_7':
0.001147048513045923, 'B_0_8': 0.2885355858350862, 'B_0_9':
0.020929409282437028, 'B_1_0': -0.7524931753726798, 'B_1_1':
-0.37211319158899914, 'B_1_2': 0.5987441404961924, 'B_1_3': 0.7156077312691245,
'B_1_4': -0.7181459352231709, 'B_1_5': 0.8957257697494747, 'B_1_6':
-0.33893121665975356, 'B_1_7': 0.32138120268682546, 'B_1_8': 0.4986137889942286,
'B_1_9': 0.6176535445830091, 'B_2_0': 0.8825844104073417, 'B_2_1':
-0.37041929490293124, 'B_2_2': 0.7098427800120779, 'B_2_3': -0.72925993119631,
'B_2_4': 0.4059809945801136, 'B_2_5': 0.47911590443769203, 'B_2_6':
0.35189201757861444, 'B_2_7': -0.4904168224523971, 'B_2_8': 0.7386254310901026,
'B_2_9': -0.6556602425266178, 'B_3_0': -0.8838031127887549, 'B_3_1':
0.430580681411586, 'B_3_2': -0.04855808383918725, 'B_3_3': 0.6912140795181787,
'B_3_4': -0.23087366070464732, 'B_3_5': -0.6568083947547286, 'B_3_6':
0.6661555410903421, 'B_3_7': 0.09684107064734193, 'B_3_8': -0.6153923310670973,
'B_3_9': 0.43636031720116253, 'B_4_0': 0.022005004209211032, 'B_4_1':
0.20653970483583328, 'B_4_2': -0.0774212903509679, 'B_4_3': 0.6508437879596527,
'B_4_4': 0.6525341770524387, 'B_4_5': 0.41458986721228475, 'B_4_6':
-0.7813052727550047, 'B_4_7': 0.6709306465700564, 'B_4_8': 0.33860570145516133,

'B_4_9': 0.390107848688513, 'B_5_0': -0.2712939760003341, 'B_5_1': 0.7509735211908732, 'B_5_2': 0.4462357684650178, 'B_5_3': -0.767070876272195, 'B_5_4': -0.18259949187579794, 'B_5_5': 0.7650331166810806, 'B_5_6': -0.8965505306391895, 'B_5_7': 0.913604906891258, 'B_5_8': 0.3547418852727344, 'B_5_9': 0.26012443573388566, 'B_6_0': -0.454292672153777, 'B_6_1': -0.4973922582964294, 'B_6_2': 0.875275274587419, 'B_6_3': -0.39942054061643695, 'B_6_4': 0.5455004875036187, 'B_6_5': -0.21161883954998328, 'B_6_6': -0.714276939846286, 'B_6_7': -0.9687837105631892, 'B_6_8': 0.08762952729019657, 'B_6_9': 0.9420420607268867, 'B_7_0': -0.25911033625777513, 'B_7_1': -0.2827591786937712, 'B_7_2': -0.3086985966581046, 'B_7_3': 0.040676797863619774, 'B_7_4': -0.11748353081989882, 'B_7_5': 0.7522309609194886, 'B_7_6': 0.010931167612510784, 'B_7_7': 0.015360519613366153, 'B_7_8': -0.7160242424532979, 'B_7_9': 0.9558242910025643, 'B_8_0': -0.49505367607971723, 'B_8_1': -0.8447391464299867, 'B_8_2': 0.3230295729709134, 'B_8_3': 0.18882541291060795, 'B_8_4': -0.16801160509436341, 'B_8_5': -0.15405937691625748, 'B_8_6': 0.5644265636719337, 'B_8_7': -0.6024889726026948, 'B_8_8': 0.6341015715108818, 'B_8_9': 0.2945192855121359, 'B_9_0': -0.16043984791654176, 'B_9_1': -0.08310333916394008, 'B_9_2': -0.4818385297715303, 'B_9_3': 0.49918637874001015, 'B_9_4': 0.025491989764430123, 'B_9_5': -0.35470800535530966, 'B_9_6': 0.1901792292066815, 'B_9_7': 0.9815542150616197, 'B_9_8': -0.12818068628002344, 'B_9_9': 0.8329689735416668, 'B_10_0': -0.3068666569493591, 'B_10_1': -0.881629240288528, 'B_10_2': -0.5856006293905108, 'B_10_3': -0.1609836162111853, 'B_10_4': -0.5551228413827534, 'B_10_5': -0.989550464327854, 'B_10_6': 0.8217030447401039, 'B_10_7': 0.6808812342372395, 'B_10_8': 0.4797386183441552, 'B_10_9': -0.6577271006208634}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:55:35,010] Trial 43 finished with value: 191.05829171604 and parameters: {'B_0_0': -0.7516801191042276, 'B_0_1': -0.8561425960446813,

'B_0_2': 0.443577793770384, 'B_0_3': 0.8490397621620129, 'B_0_4': -0.13753907459218673, 'B_0_5': -0.7541014694478246, 'B_0_6': 0.024537785049570937, 'B_0_7': -0.29931282323191233, 'B_0_8': -0.24784664978700455, 'B_0_9': 0.2402346358283005, 'B_1_0': 0.4330618642591606,

'B_1_1': -0.7710048742311415, 'B_1_2': -0.9195313430031767, 'B_1_3':
 0.9787739993177154, 'B_1_4': -0.5291978691472073, 'B_1_5': -0.883415560681251,
 'B_1_6': -0.6140468483900785, 'B_1_7': -0.9128634642300408, 'B_1_8':
 -0.2142144286580001, 'B_1_9': 0.5987555332136221, 'B_2_0': -0.3143359894797132,
 'B_2_1': -0.11051685438083425, 'B_2_2': 0.34766181166352883, 'B_2_3':
 0.01761025873836286, 'B_2_4': 0.715662458322222, 'B_2_5': 0.7247506365905978,
 'B_2_6': -0.9703768229275158, 'B_2_7': 0.5505316868367343, 'B_2_8':
 -0.07774747152148576, 'B_2_9': 0.9679042717812529, 'B_3_0':
 -0.033200489766638364, 'B_3_1': 0.964053971868829, 'B_3_2': 0.5040473351873584,
 'B_3_3': -0.4977623284656535, 'B_3_4': 0.07602205898817616, 'B_3_5':
 0.9744285661675938, 'B_3_6': 0.707910603348809, 'B_3_7': -0.7972135502933823,
 'B_3_8': 0.40861655368379224, 'B_3_9': -0.30575241997579017, 'B_4_0':
 -0.9761751214769108, 'B_4_1': -0.19110768132654077, 'B_4_2':
 -0.5251862449845568, 'B_4_3': 0.6313611556305336, 'B_4_4': -0.3220853944853843,
 'B_4_5': 0.6484969140285579, 'B_4_6': -0.9888160811071053, 'B_4_7':
 -0.38051808294828404, 'B_4_8': -0.5124928756706313, 'B_4_9':
 -0.5275366250841502, 'B_5_0': -0.8549904928643839, 'B_5_1': -0.6194939445872065,
 'B_5_2': 0.10998814837352744, 'B_5_3': -0.4131050757812659, 'B_5_4':
 -0.1635583309882871, 'B_5_5': -0.2438035769125667, 'B_5_6': 0.7442917514997913,
 'B_5_7': 0.8024920180408206, 'B_5_8': -0.013945196121397108, 'B_5_9':
 -0.33629121837510323, 'B_6_0': -0.77853460607532, 'B_6_1': 0.8977377011765482,
 'B_6_2': -0.8707506490974111, 'B_6_3': -0.4006524504148047, 'B_6_4':
 -0.3998289216833918, 'B_6_5': -0.3556054534423745, 'B_6_6': -0.6647120503206296,
 'B_6_7': 0.5606596332289204, 'B_6_8': -0.7474542333073413, 'B_6_9':
 0.26711518512326227, 'B_7_0': -0.7447848671224815, 'B_7_1': -0.850339437565946,
 'B_7_2': 0.8106704172980355, 'B_7_3': -0.6135450426889628, 'B_7_4':
 0.6360753060282145, 'B_7_5': 0.088369124792129, 'B_7_6': -0.8358440911746362,
 'B_7_7': 0.9017874190473736, 'B_7_8': -0.5010168893560528, 'B_7_9':
 0.7120977837981608, 'B_8_0': -0.6557561885465629, 'B_8_1': 0.7008069066408686,
 'B_8_2': -0.9638069504475179, 'B_8_3': 0.023325304007571823, 'B_8_4':
 0.5334963072546826, 'B_8_5': 0.860895124635711, 'B_8_6': 0.8102202296888443,
 'B_8_7': 0.632857241704337, 'B_8_8': 0.7686188723293821, 'B_8_9':
 -0.5955450220425885, 'B_9_0': -0.4273150944907771, 'B_9_1': -0.4178423206563041,
 'B_9_2': -0.2434332166166363, 'B_9_3': -0.21122507263732637, 'B_9_4':
 0.08821954281625755, 'B_9_5': -0.6734380685344461, 'B_9_6': 0.37866839429625876,
 'B_9_7': -0.6582182521770772, 'B_9_8': -0.3755892054329959, 'B_9_9':
 0.006623358600185036, 'B_10_0': 0.586952311816485, 'B_10_1': 0.4741490818126799,
 'B_10_2': 0.288158025692834, 'B_10_3': -0.6815689440512631, 'B_10_4':
 0.0931497843188438, 'B_10_5': -0.8168559264362268, 'B_10_6':
 -0.2428209099426315, 'B_10_7': -0.17123727002588085, 'B_10_8':
 -0.9111733498974074, 'B_10_9': -0.4475035630494558}. Best is trial 36 with
 value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u₁].
 Built F_[u₂].
 Built F_[u₃].
 Built F_[u₄].
 Built F_[u₅].

```

Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 13:55:48,767] Trial 44 finished with value: 133.09453556155984 and
parameters: {'B_0_0': 0.9021859451876488, 'B_0_1': 0.8981925635027852, 'B_0_2':
-0.29239243257997216, 'B_0_3': -0.9046617842823184, 'B_0_4':
-0.6353920201302006, 'B_0_5': -0.21340876900159056, 'B_0_6': 0.886603801979075,
'B_0_7': -0.7104258541963738, 'B_0_8': 0.5217613704630601, 'B_0_9':
-0.7748085720381097, 'B_1_0': 0.9528900032347281, 'B_1_1': -0.2343993317639127,
'B_1_2': 0.5449643870201308, 'B_1_3': -0.311669656033959, 'B_1_4':
0.2527843767744413, 'B_1_5': -0.6098081522147273, 'B_1_6': -0.7008783817920141,
'B_1_7': 0.8673669963449, 'B_1_8': 0.8269416102507783, 'B_1_9':
0.767188501621507, 'B_2_0': 0.24835823132985513, 'B_2_1': 0.6103710582082309,
'B_2_2': -0.8533068649154777, 'B_2_3': -0.3271176970564553, 'B_2_4':
-0.3688678129808307, 'B_2_5': 0.2685777374661742, 'B_2_6': -0.28520360852813176,
'B_2_7': -0.695956572345512, 'B_2_8': 0.19100480931881658, 'B_2_9':
0.6224801629247814, 'B_3_0': -0.2333954809692531, 'B_3_1': -0.7281079441050062,
'B_3_2': -0.6005573693206103, 'B_3_3': -0.5197830557646197, 'B_3_4':
-0.39620276587081826, 'B_3_5': 0.2827342396024921, 'B_3_6': 0.19987205938468366,
'B_3_7': 0.38823812590273743, 'B_3_8': -0.272246070627604, 'B_3_9':
0.641515350722357, 'B_4_0': -0.5859064309115947, 'B_4_1': -0.06129931991405524,
'B_4_2': 0.595858668652032, 'B_4_3': -0.6332858774894139, 'B_4_4':
-0.7724710347932497, 'B_4_5': 0.7596942560222197, 'B_4_6': -0.3109992677367921,
'B_4_7': 0.36406284731988325, 'B_4_8': -0.9398097635574018, 'B_4_9':
0.3762928390507103, 'B_5_0': -0.3641194031369984, 'B_5_1': -0.8796331098968654,
'B_5_2': 0.2174292678652716, 'B_5_3': -0.513208444175117, 'B_5_4':
-0.22412166073727757, 'B_5_5': -0.8304499916727031, 'B_5_6':
-0.6986167628320082, 'B_5_7': -0.40000507371822935, 'B_5_8':
-0.8122761888908889, 'B_5_9': 0.8134360040789042, 'B_6_0': -0.9780268544957227,
'B_6_1': -0.031956766648663426, 'B_6_2': -0.7459700328975482, 'B_6_3':
-0.38079540283366187, 'B_6_4': 0.8435979284401656, 'B_6_5': -0.669844784517506,
'B_6_6': 0.9223479930422986, 'B_6_7': -0.582527133560945, 'B_6_8':
0.4783008947643226, 'B_6_9': 0.09242098449667724, 'B_7_0': 0.7612678806415831,
'B_7_1': 0.10690664589619936, 'B_7_2': 0.06842725392949123, 'B_7_3':
-0.6029931592705682, 'B_7_4': -0.6608658631377158, 'B_7_5': 0.8246485192087181,
'B_7_6': -0.3408803662769325, 'B_7_7': -0.25327713517346606, 'B_7_8':
0.5144508436882089, 'B_7_9': -0.3191295048028414, 'B_8_0': 0.8950836443452255,
'B_8_1': 0.47109698261762456, 'B_8_2': 0.19290990029537158, 'B_8_3':
0.75417241787861, 'B_8_4': -0.8917803014680241, 'B_8_5': -0.4810158694653667,
'B_8_6': -0.06922805824259826, 'B_8_7': -0.5405824918463884, 'B_8_8':
0.48293227085826085, 'B_8_9': 0.8300319014863757, 'B_9_0': -0.15093516253677164,
'B_9_1': -0.312577451778448, 'B_9_2': 0.9138704496323813, 'B_9_3':

```

```

-0.10919218300146916, 'B_9_4': -0.021684167110419206, 'B_9_5':
-0.2937332527630274, 'B_9_6': 0.8109059765640341, 'B_9_7': 0.13653185689806446,
'B_9_8': 0.5191056826196565, 'B_9_9': 0.6173773023347551, 'B_10_0':
-0.3131388970038288, 'B_10_1': -0.13128522398564746, 'B_10_2':
0.689974113324404, 'B_10_3': -0.27075440368189574, 'B_10_4': 0.5819915642346656,
'B_10_5': 0.8731647986973854, 'B_10_6': 0.24395653867465383, 'B_10_7':
0.4784220291547401, 'B_10_8': 0.604092677841447, 'B_10_9': 0.6831785048153522}.
Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 13:56:02,093] Trial 45 finished with value: 161.7047522332583 and
parameters: {'B_0_0': -0.20779439104354402, 'B_0_1': -0.8884983508315236,
'B_0_2': -0.19428498295125574, 'B_0_3': 0.34403733497249767, 'B_0_4':
-0.09830623615403034, 'B_0_5': 0.492947038703589, 'B_0_6': -0.28317478965204557,
'B_0_7': 0.26679339938597857, 'B_0_8': -0.6208887863891304, 'B_0_9':
0.4937514845570148, 'B_1_0': 0.09047801824979107, 'B_1_1': -0.5859525106777308,
'B_1_2': -0.41201587647434157, 'B_1_3': 0.4922452429920281, 'B_1_4':
-0.03444955339018907, 'B_1_5': -0.562257748021717, 'B_1_6': 0.9251969569331893,
'B_1_7': 0.4259475392374381, 'B_1_8': 0.9051269684287964, 'B_1_9':
-0.14433201018652042, 'B_2_0': -0.04930132543039345, 'B_2_1':
0.7879513280463379, 'B_2_2': 0.04959769368586153, 'B_2_3': 0.41041224557322087,
'B_2_4': -0.49069770255885065, 'B_2_5': -0.10548159196877172, 'B_2_6':
-0.7455692656954969, 'B_2_7': -0.5821595289636159, 'B_2_8': -0.9881320377700904,
'B_2_9': -0.9122551977356528, 'B_3_0': 0.94502196989032, 'B_3_1':
-0.5101396836292107, 'B_3_2': 0.3233552048179238, 'B_3_3': 0.6714174036232308,
'B_3_4': 0.10030770928839727, 'B_3_5': -0.6867738898337667, 'B_3_6':
0.4516705459256094, 'B_3_7': 0.19849779199717443, 'B_3_8': 0.45787616690780264,
'B_3_9': 0.5340983132570325, 'B_4_0': 0.9413125280157952, 'B_4_1':
0.14565529445288572, 'B_4_2': -0.31355059687260833, 'B_4_3':
0.29395982266246934, 'B_4_4': -0.8544873956996002, 'B_4_5': 0.7868466902767408,
'B_4_6': -0.7536255975503704, 'B_4_7': -0.3408711809860798, 'B_4_8':
0.6251758536397194, 'B_4_9': 0.19267850909480932, 'B_5_0': -0.21060166985066464,
'B_5_1': -0.05259739443745759, 'B_5_2': 0.7020038385403171, 'B_5_3':
-0.3167913156014136, 'B_5_4': 0.7319063759280993, 'B_5_5': -0.8154938266661509,
'B_5_6': 0.5480609056730887, 'B_5_7': 0.688144313448378, 'B_5_8':

```

```

-0.6300010473820692, 'B_5_9': -0.1379138659030431, 'B_6_0': -0.6623104461659619,
'B_6_1': 0.40907617540855123, 'B_6_2': 0.06998858076208037, 'B_6_3':
0.26794022648247573, 'B_6_4': -0.6009558054267019, 'B_6_5': -0.5707431931362094,
'B_6_6': -0.9078352954903854, 'B_6_7': -0.35230628031245514, 'B_6_8':
0.11829442944067647, 'B_6_9': 0.7087111297147111, 'B_7_0': 0.3305163459681879,
'B_7_1': -0.12791527966849214, 'B_7_2': 0.8971778265003929, 'B_7_3':
0.434105497140147, 'B_7_4': 0.8517271016678489, 'B_7_5': 0.05462418707888794,
'B_7_6': -0.47736953590958364, 'B_7_7': -0.8854031609978504, 'B_7_8':
0.4476480059380239, 'B_7_9': -0.7498200639490424, 'B_8_0': -0.39050524602150494,
'B_8_1': 0.06429239761011685, 'B_8_2': 0.1275448438039004, 'B_8_3':
0.19915473003452933, 'B_8_4': -0.6608144459075928, 'B_8_5': -0.2381199384531355,
'B_8_6': 0.23204456969159404, 'B_8_7': 0.9302413519748938, 'B_8_8':
0.45070965264538354, 'B_8_9': 0.8367554390460366, 'B_9_0': 0.5191798754620705,
'B_9_1': 0.18159959840695294, 'B_9_2': -0.6097937291100378, 'B_9_3':
0.33003759074717576, 'B_9_4': 0.24429154060061609, 'B_9_5': 0.2028889575053876,
'B_9_6': -0.020223981383283296, 'B_9_7': 0.05674007589876573, 'B_9_8':
-0.32780379630615397, 'B_9_9': 0.038239131431760276, 'B_10_0':
-0.5988393067566131, 'B_10_1': 0.604245306773546, 'B_10_2': -0.6221616296965585,
'B_10_3': -0.8217667460188565, 'B_10_4': -0.1262328678128284, 'B_10_5':
0.31344873022200237, 'B_10_6': -0.12332068663158813, 'B_10_7':
-0.4420268187364089, 'B_10_8': 0.11517894410307261, 'B_10_9':
-0.3029353758838441}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 13:56:16,785] Trial 46 finished with value: 130.95624518100448 and
parameters: {'B_0_0': 0.8570780264507334, 'B_0_1': 0.8368373764150727, 'B_0_2':
0.004696203544748978, 'B_0_3': -0.3407896918438481, 'B_0_4':
0.46952217963414533, 'B_0_5': -0.917308371903528, 'B_0_6': -0.04995736803357054,
'B_0_7': -0.3240796793447396, 'B_0_8': 0.8326610525030582, 'B_0_9':
-0.9662340060000886, 'B_1_0': 0.10457345392233108, 'B_1_1': 0.4765138905119195,
'B_1_2': -0.029664384702398294, 'B_1_3': -0.8209020002778111, 'B_1_4':
0.9354735522552464, 'B_1_5': 0.035660654124673785, 'B_1_6': 0.22608876375164244,
'B_1_7': -0.5215126302391355, 'B_1_8': -0.03267379148852845, 'B_1_9':
-0.14028400592252277, 'B_2_0': -0.8417062328360722, 'B_2_1':
-0.7798024999760149, 'B_2_2': 0.6681955888451736, 'B_2_3': -0.5151564946309513,

```

'B_2_4': -0.6039826224327115, 'B_2_5': 0.009724673475275725, 'B_2_6': 0.5321193200977756, 'B_2_7': -0.8673458932136039, 'B_2_8': 0.15207968088374657, 'B_2_9': -0.7539633231756647, 'B_3_0': -0.9178071193652755, 'B_3_1': -0.8855669747188095, 'B_3_2': 0.6616449342527138, 'B_3_3': -0.756414482416438, 'B_3_4': -0.900600298463662, 'B_3_5': -0.12295822345703367, 'B_3_6': 0.681107074445209, 'B_3_7': -0.4704314145511822, 'B_3_8': -0.15500901729868555, 'B_3_9': -0.9109533855237726, 'B_4_0': -0.10082454201294577, 'B_4_1': 0.15367389966152478, 'B_4_2': 0.14039596642924668, 'B_4_3': -0.33179514949475697, 'B_4_4': -0.3652443022869917, 'B_4_5': -0.7788852345055362, 'B_4_6': -0.26373932512038856, 'B_4_7': -0.79300055563153, 'B_4_8': 0.5278599705298621, 'B_4_9': 0.9232231095414081, 'B_5_0': 0.9307474242000797, 'B_5_1': 0.7228407767632798, 'B_5_2': 0.19956845569244575, 'B_5_3': 0.39771408054288226, 'B_5_4': -0.4252925106927802, 'B_5_5': 0.25892543194178885, 'B_5_6': -0.08781389210904267, 'B_5_7': -0.03560566724167369, 'B_5_8': 0.5103975895428128, 'B_5_9': -0.4318016559174417, 'B_6_0': -0.36028156884485063, 'B_6_1': 0.8407907464347006, 'B_6_2': -0.8786758961950749, 'B_6_3': -0.028444089141586026, 'B_6_4': 0.8173455848075661, 'B_6_5': 0.2185187554799508, 'B_6_6': 0.09188600694002158, 'B_6_7': -0.4955065251258767, 'B_6_8': -0.3103145316523924, 'B_6_9': 0.6970558632932919, 'B_7_0': 0.11725190090154713, 'B_7_1': 0.029189543638424187, 'B_7_2': -0.7928321671907428, 'B_7_3': -0.9582642502188562, 'B_7_4': 0.9180966736003573, 'B_7_5': -0.24356738877205353, 'B_7_6': 0.20436186295701853, 'B_7_7': -0.8321641550891757, 'B_7_8': 0.3625662857316321, 'B_7_9': 0.8564820832378026, 'B_8_0': 0.14981720093489104, 'B_8_1': -0.740291661696687, 'B_8_2': 0.1618086243974839, 'B_8_3': 0.5273674819472116, 'B_8_4': -0.185295257583443, 'B_8_5': 0.7520787409241063, 'B_8_6': 0.29439224764758176, 'B_8_7': 0.7905128582126209, 'B_8_8': 0.24454051626972761, 'B_8_9': -0.730049815430448, 'B_9_0': -0.3768104120117255, 'B_9_1': -0.5904460024601352, 'B_9_2': 0.8094764654793873, 'B_9_3': 0.5155921332292157, 'B_9_4': 0.34926877044181204, 'B_9_5': -0.3940866871293891, 'B_9_6': -0.6255913513405772, 'B_9_7': 0.5077731542180455, 'B_9_8': -0.051167267159422214, 'B_9_9': -0.5429366867417543, 'B_10_0': 0.23091151676724464, 'B_10_1': -0.9101389727658651, 'B_10_2': -0.34418289682605807, 'B_10_3': -0.0622571507095534, 'B_10_4': -0.6970424984633764, 'B_10_5': 0.9596158656298623, 'B_10_6': -0.5759872181125638, 'B_10_7': -0.7319008025143342, 'B_10_8': -0.5870009497039451, 'B_10_9': 0.5327410436562541}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].

```

Built F[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [F[u_i], F[u_j]].
Built matrix B = inverse(A).
[I 2025-04-29 13:56:33,354] Trial 47 finished with value: 322.49083994175214 and
parameters: {'B_0_0': -0.1474987946417512, 'B_0_1': 0.12952356071002535,
'B_0_2': -0.9484018691670058, 'B_0_3': -0.9218260090248371, 'B_0_4':
-0.107265411342508, 'B_0_5': 0.6269773815392146, 'B_0_6': 0.7627729798409792,
'B_0_7': -0.8183209961606324, 'B_0_8': 0.07595894216002508, 'B_0_9':
0.8181947788329555, 'B_1_0': -0.22181394629218498, 'B_1_1': 0.6525403462281698,
'B_1_2': -0.7252826042891163, 'B_1_3': 0.8620810895422415, 'B_1_4':
0.49781816871197915, 'B_1_5': 0.8738420977180652, 'B_1_6': 0.3497021224339627,
'B_1_7': -0.27046021464113457, 'B_1_8': 0.8675987215989505, 'B_1_9':
-0.4443777829078266, 'B_2_0': -0.33229840631776963, 'B_2_1': 0.3976006676434458,
'B_2_2': 0.5253262730602648, 'B_2_3': 0.8502188980213246, 'B_2_4':
-0.5844912300687052, 'B_2_5': 0.5900361097340634, 'B_2_6': 0.4723211857259877,
'B_2_7': -0.8636867786178762, 'B_2_8': -0.22329818040379024, 'B_2_9':
-0.43012439502100475, 'B_3_0': -0.3871089597790198, 'B_3_1': 0.9562348215183076,
'B_3_2': 0.2826992786902571, 'B_3_3': 0.43099879243112915, 'B_3_4':
0.9437029218837181, 'B_3_5': -0.24314047073387535, 'B_3_6': 0.5976435335365557,
'B_3_7': -0.1296286926570992, 'B_3_8': 0.7316766701936599, 'B_3_9':
-0.6322980682036197, 'B_4_0': 0.8864186305872737, 'B_4_1': -0.5568320173464583,
'B_4_2': -0.34408339617775563, 'B_4_3': 0.5072606235314254, 'B_4_4':
-0.2103648197004947, 'B_4_5': 0.21425852599756445, 'B_4_6':
-0.10974446788253156, 'B_4_7': 0.47989247543776803, 'B_4_8':
-0.5379041203101504, 'B_4_9': -0.8742531852709835, 'B_5_0': -0.3961397930033288,
'B_5_1': -0.050868843560508026, 'B_5_2': -0.6573202966584965, 'B_5_3':
-0.2875013664803504, 'B_5_4': -0.19803362377873435, 'B_5_5':
-0.8763641494972184, 'B_5_6': 0.16494835140472497, 'B_5_7': 0.7609999873831492,
'B_5_8': -0.6895490323529562, 'B_5_9': 0.19365986653499778, 'B_6_0':
0.3261836257234756, 'B_6_1': -0.15971733276980904, 'B_6_2': 0.39817322298125446,
'B_6_3': -0.17675154388779613, 'B_6_4': 0.00918183019532115, 'B_6_5':
-0.9754409717822345, 'B_6_6': 0.3796058259866839, 'B_6_7': 0.31945850431055445,
'B_6_8': -0.9139662628938378, 'B_6_9': -0.2608647873576365, 'B_7_0':
0.9484830447675014, 'B_7_1': -0.15694160611413877, 'B_7_2':
0.003851902161046583, 'B_7_3': 0.8116617131562427, 'B_7_4': 0.4301785142141976,
'B_7_5': 0.16249885962596267, 'B_7_6': 0.5896532690553236, 'B_7_7':
0.721445207582827, 'B_7_8': -0.08536161796021158, 'B_7_9': -0.04928543743496594,
'B_8_0': -0.22561914740936517, 'B_8_1': 0.924078157236552, 'B_8_2':
0.3903891834948938, 'B_8_3': -0.8252183289080168, 'B_8_4': 0.7186600541597765,
'B_8_5': -0.038491317144189185, 'B_8_6': -0.8538672652834748, 'B_8_7':
0.09844016522335597, 'B_8_8': -0.16401472762810987, 'B_8_9': 0.7479438216279972,
'B_9_0': -0.586090684853413, 'B_9_1': 0.6463999267945222, 'B_9_2':
0.11422360854719638, 'B_9_3': -0.8818715996373313, 'B_9_4': 0.9342536966244597,
'B_9_5': 0.02497818332433588, 'B_9_6': -0.4140638286214666, 'B_9_7':
-0.060574452861268946, 'B_9_8': 0.7085584113646282, 'B_9_9':
-0.38629181371757426, 'B_10_0': -0.6085007349563334, 'B_10_1':
-0.7344389310768547, 'B_10_2': -0.4002383911873817, 'B_10_3':

```

0.5544952123392861, 'B_10_4': -0.05617979691338182, 'B_10_5':
-0.5395864246867909, 'B_10_6': -0.6609537007726967, 'B_10_7':
-0.3311956365094494, 'B_10_8': 0.8710502583137034, 'B_10_9':
-0.3108071090717517}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:56:49,249] Trial 48 finished with value: 124.33716985010842 and
parameters: {'B_0_0': 0.9664856703901781, 'B_0_1': -0.029243697584594774,
'B_0_2': -0.6352554417855012, 'B_0_3': 0.7744573817511486, 'B_0_4':
-0.07849636119304282, 'B_0_5': 0.41514926013895836, 'B_0_6': 0.9577489347541204,
'B_0_7': -0.8906528063664515, 'B_0_8': -0.7985447716206171, 'B_0_9':
0.43650434809625716, 'B_1_0': -0.4319864934749654, 'B_1_1': -0.6250832340388038,
'B_1_2': -0.9120634779569357, 'B_1_3': -0.9467435102673204, 'B_1_4':
0.5434814941771859, 'B_1_5': -0.723821813808291, 'B_1_6': -0.06391318037712124,
'B_1_7': 0.20967569405537634, 'B_1_8': -0.14186343487368447, 'B_1_9':
-0.9363288283742585, 'B_2_0': 0.5613329751240594, 'B_2_1': -0.9691838154075261,
'B_2_2': 0.5185419908674183, 'B_2_3': 0.7787428977004829, 'B_2_4':
-0.13297695630679884, 'B_2_5': 0.07037409673364059, 'B_2_6':
-0.04557288261704551, 'B_2_7': -0.21936693512559113, 'B_2_8':
0.7671005800333133, 'B_2_9': 0.7799883511265813, 'B_3_0': -0.3250835980097616,
'B_3_1': 0.22783811294910206, 'B_3_2': 0.845276973048811, 'B_3_3':
0.914875827903876, 'B_3_4': 0.017738900035654037, 'B_3_5': -0.09340688553703302,
'B_3_6': -0.4381852544615189, 'B_3_7': 0.9085663089170117, 'B_3_8':
-0.7072276771437227, 'B_3_9': 0.15806256057449541, 'B_4_0': -0.430550470313095,
'B_4_1': -0.43756358511136273, 'B_4_2': -0.1881621969951468, 'B_4_3':
0.9876100129149961, 'B_4_4': -0.8811667124571523, 'B_4_5': 0.15551280004299217,
'B_4_6': -0.8265071270901512, 'B_4_7': -0.6840487566187725, 'B_4_8':
-0.11966537528966059, 'B_4_9': 0.02071106006595369, 'B_5_0':
-0.47759103143824244, 'B_5_1': -0.3634148808686559, 'B_5_2':
-0.8178052998932526, 'B_5_3': -0.38962482798838394, 'B_5_4': 0.9237857099734772,
'B_5_5': 0.8805206157115137, 'B_5_6': -0.3269409855287697, 'B_5_7':
0.14029079383194798, 'B_5_8': 0.8109988743996699, 'B_5_9': -0.9777560311710246,
'B_6_0': 0.615153469268432, 'B_6_1': 0.8877717145220596, 'B_6_2':
-0.809175046019557, 'B_6_3': -0.6703335816548164, 'B_6_4': 0.9122776794872838,
'B_6_5': 0.07432948440788567, 'B_6_6': -0.9618906637459871, 'B_6_7':

```

-0.4798886424482123, 'B_6_8': -0.09911152494765796, 'B_6_9': 0.38195964382904,
'B_7_0': 0.4616575487108632, 'B_7_1': 0.9809345955924031, 'B_7_2':
-0.42072782522015717, 'B_7_3': 0.07567580169958088, 'B_7_4':
0.06598062476201783, 'B_7_5': -0.1868223862944698, 'B_7_6': -0.655059450192008,
'B_7_7': 0.9453639951040083, 'B_7_8': 0.6995680295998403, 'B_7_9':
-0.003300603977410388, 'B_8_0': 0.20442593691437905, 'B_8_1':
-0.5400231139518686, 'B_8_2': 0.08669839937565893, 'B_8_3': 0.38604086358052214,
'B_8_4': 0.03003071620544584, 'B_8_5': 0.9707646377581838, 'B_8_6':
0.22045647937049795, 'B_8_7': -0.2554720490675053, 'B_8_8': 0.5388610146793595,
'B_8_9': -0.6524842270683487, 'B_9_0': -0.22865868886159224, 'B_9_1':
-0.30302127825238, 'B_9_2': -0.13502195168660103, 'B_9_3': 0.557924055338878,
'B_9_4': -0.03133739631645327, 'B_9_5': 0.5821486115266061, 'B_9_6':
0.15144802821693437, 'B_9_7': -0.3127025193780404, 'B_9_8': 0.6137579124932755,
'B_9_9': -0.5181864523048787, 'B_10_0': -0.545721699996405, 'B_10_1':
0.008915789628574866, 'B_10_2': -0.473518492995064, 'B_10_3':
-0.9720076707965917, 'B_10_4': 0.293629104051627, 'B_10_5':
0.021965766515348095, 'B_10_6': -0.10787472730354086, 'B_10_7':
0.9245976589442644, 'B_10_8': 0.14097438424656916, 'B_10_9':
0.7195945240229813}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 13:57:07,220] Trial 49 finished with value: 172.59517653189866 and
parameters: {'B_0_0': -0.7138885323137829, 'B_0_1': 0.1370287969962669, 'B_0_2':
0.7285996609271876, 'B_0_3': 0.164549399414198, 'B_0_4': 0.431668072691463,
'B_0_5': -0.2533537357318294, 'B_0_6': -0.8069499562666081, 'B_0_7':
-0.0833574156483986, 'B_0_8': -0.20893080613906212, 'B_0_9': 0.7267791611103158,
'B_1_0': -0.506889301342105, 'B_1_1': 0.9241337705195141, 'B_1_2':
0.9071134772129024, 'B_1_3': -0.49968934671855136, 'B_1_4': -0.3008571281549054,
'B_1_5': 0.7832003669302923, 'B_1_6': 0.9239119973742929, 'B_1_7':
-0.11199057638463283, 'B_1_8': -0.3544258496310223, 'B_1_9':
0.05166614623805321, 'B_2_0': 0.3717830286041548, 'B_2_1': -0.44852045955753994,
'B_2_2': 0.46821688574576403, 'B_2_3': -0.8877184329130108, 'B_2_4':
-0.8923969043264555, 'B_2_5': -0.22515034381182042, 'B_2_6':
-0.29012307971977325, 'B_2_7': -0.36504623357303845, 'B_2_8':
-0.017780869915558162, 'B_2_9': -0.35494750702308897, 'B_3_0':

```



```

-0.609329321433178, 'B_3_1': -0.37725029280457023, 'B_3_2': 0.9403104601642651,
'B_3_3': -0.17599543435019982, 'B_3_4': -0.10946870182049517, 'B_3_5':
0.15155908010325714, 'B_3_6': 0.9570147214601685, 'B_3_7': -0.5116294726571893,
'B_3_8': 0.358938578854219, 'B_3_9': 0.4276496932126137, 'B_4_0':
0.21250234525027056, 'B_4_1': -0.8146148875194448, 'B_4_2': 0.3502071061386014,
'B_4_3': -0.8630924721483726, 'B_4_4': 0.19036244610879582, 'B_4_5':
0.47437073089905035, 'B_4_6': -0.5722796165724982, 'B_4_7': 0.6764115749346926,
'B_4_8': -0.9298994650446516, 'B_4_9': 0.680749470043446, 'B_5_0':
-0.07854847972732515, 'B_5_1': -0.5591758789353439, 'B_5_2':
-0.7251192729119433, 'B_5_3': 0.8975481918130976, 'B_5_4': -0.8461221365893097,
'B_5_5': -0.5957950285268672, 'B_5_6': -0.301311752082569, 'B_5_7':
0.23806500094301852, 'B_5_8': 0.08870614884667472, 'B_5_9': -0.7412258850236183,
'B_6_0': -0.416338592198335, 'B_6_1': -0.5172714932496723, 'B_6_2':
-0.5671851481091241, 'B_6_3': -0.1759342137696256, 'B_6_4': 0.40398227613148063,
'B_6_5': -0.5594431942098728, 'B_6_6': -0.2639758181880968, 'B_6_7':
-0.21598349485449875, 'B_6_8': 0.9433839282821121, 'B_6_9': 0.9002393866205562,
'B_7_0': 0.5942724138491102, 'B_7_1': -0.5115153367780134, 'B_7_2':
-0.855286118098494, 'B_7_3': -0.362529807313163, 'B_7_4': -0.4617467921210149,
'B_7_5': 0.011733084289859974, 'B_7_6': 0.5499012619342303, 'B_7_7':
0.7094794189801388, 'B_7_8': 0.6828308891128068, 'B_7_9': -0.2778423887783258,
'B_8_0': 0.3707240000521521, 'B_8_1': -0.3705882889020443, 'B_8_2':
-0.8694070243943383, 'B_8_3': 0.39909922589382574, 'B_8_4': -0.97054144591182,
'B_8_5': -0.17981847683219265, 'B_8_6': -0.3378290467818901, 'B_8_7':
0.3828875262673277, 'B_8_8': 0.5271660360655501, 'B_8_9': -0.38789262077685804,
'B_9_0': 0.6170214886008101, 'B_9_1': -0.4080448142002312, 'B_9_2':
-0.10627380722660873, 'B_9_3': 0.4494966834879852, 'B_9_4': 0.38198205386508133,
'B_9_5': 0.9649895751781561, 'B_9_6': -0.5441811720524549, 'B_9_7':
0.32138946880837316, 'B_9_8': 0.21717017838603914, 'B_9_9': 0.8218949398902593,
'B_10_0': -0.6804029595431473, 'B_10_1': 0.4985603639021272, 'B_10_2':
0.13517186021216143, 'B_10_3': -0.44616157781994725, 'B_10_4':
-0.05061886998859533, 'B_10_5': -0.32912233780180034, 'B_10_6':
0.15704053752115255, 'B_10_7': 0.7702212094130383, 'B_10_8':
0.13722386913616003, 'B_10_9': 0.2608103882866364}. Best is trial 36 with value:
95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix $B = \text{inverse}(A)$.

[I 2025-04-29 13:57:22,325] Trial 50 finished with value: 186.3284501844343 and parameters: {'B_0_0': 0.03712073728712473, 'B_0_1': -0.8886922972379577, 'B_0_2': -0.5289400573853333, 'B_0_3': -0.015836575724476165, 'B_0_4': -0.3757341381854845, 'B_0_5': -0.6301080764328355, 'B_0_6': 0.008183026998507925, 'B_0_7': -0.3508366089732563, 'B_0_8': -0.47894686651663887, 'B_0_9': -0.07130887346804604, 'B_1_0': -0.7260971689972971, 'B_1_1': 0.8469365514893064, 'B_1_2': 0.7280501538800932, 'B_1_3': 0.41108875976626, 'B_1_4': -0.7796966213748567, 'B_1_5': 0.5462369638499922, 'B_1_6': 0.6097555096558576, 'B_1_7': 0.5301739430743886, 'B_1_8': -0.7643475435039024, 'B_1_9': 0.28493775499417406, 'B_2_0': 0.6619630955838092, 'B_2_1': 0.15315973441494202, 'B_2_2': 0.2898866029516163, 'B_2_3': -0.08244031073591374, 'B_2_4': 0.05684049337042407, 'B_2_5': -0.9219338204635841, 'B_2_6': 0.27514678350710353, 'B_2_7': -0.8287345184755048, 'B_2_8': -0.12867110320635766, 'B_2_9': -0.41054290661853254, 'B_3_0': 0.0899473414191918, 'B_3_1': -0.72583565566405, 'B_3_2': 0.37945393590717, 'B_3_3': 0.5841228431183714, 'B_3_4': -0.6665581895085502, 'B_3_5': -0.7834600870432733, 'B_3_6': -0.7018793440810289, 'B_3_7': 0.7467125725590333, 'B_3_8': 0.055814738277585096, 'B_3_9': -0.06482322790514483, 'B_4_0': -0.7787105981773386, 'B_4_1': 0.29612089337878444, 'B_4_2': 0.7286391919522781, 'B_4_3': 0.6498080212604369, 'B_4_4': 0.24958832378198648, 'B_4_5': -0.8409481410296862, 'B_4_6': -0.7674538861914943, 'B_4_7': -0.8923682055225471, 'B_4_8': -0.48019115134491075, 'B_4_9': -0.721390295022452, 'B_5_0': -0.47510488030018094, 'B_5_1': -0.22758504967645588, 'B_5_2': 0.8771059124608487, 'B_5_3': 0.2185445983677503, 'B_5_4': 0.6632480956522655, 'B_5_5': 0.4164959714104177, 'B_5_6': -0.7672438015572691, 'B_5_7': 0.46712240462041876, 'B_5_8': -0.44530421451293367, 'B_5_9': 0.2754335370495187, 'B_6_0': -0.3148952921868827, 'B_6_1': 0.9853910812640252, 'B_6_2': -0.4681680705963799, 'B_6_3': -0.43732166786694604, 'B_6_4': 0.8527310660135485, 'B_6_5': -0.9412040788200811, 'B_6_6': 0.3894441959889299, 'B_6_7': 0.44614625365703864, 'B_6_8': 0.6660295832725907, 'B_6_9': 0.9393898350641812, 'B_7_0': 0.9416917999301717, 'B_7_1': -0.8607520200184015, 'B_7_2': 0.6372418590507469, 'B_7_3': 0.28969366864161494, 'B_7_4': 0.29173742605872044, 'B_7_5': -0.7358671453029043, 'B_7_6': 0.2973705895623935, 'B_7_7': 0.9553155095904706, 'B_7_8': 0.815400767688105, 'B_7_9': -0.12052129160639591, 'B_8_0': -0.9109030128935331, 'B_8_1': -0.2726588897995944, 'B_8_2': 0.6462528169690993, 'B_8_3': 0.3989682209010288, 'B_8_4': -0.08049956192396635, 'B_8_5': -0.8808989603855144, 'B_8_6': -0.14474843890986078, 'B_8_7': -0.06133061935859063, 'B_8_8': 0.8485399551303892, 'B_8_9': 0.2793504850926758, 'B_9_0': -0.37292722502379516, 'B_9_1': 0.05013016361242828, 'B_9_2': -0.8592104905121231, 'B_9_3': 0.9356851065864789, 'B_9_4': -0.2770813941771625, 'B_9_5': -0.043822637271991094, 'B_9_6': -0.2865377310108933, 'B_9_7': 0.36137322563007634, 'B_9_8': 0.3212196734925943, 'B_9_9': 0.42633817484768866, 'B_10_0': -0.4590764053683968, 'B_10_1': -0.36497640729514935, 'B_10_2': -0.7420025624295082, 'B_10_3': 0.30790974425530293, 'B_10_4': -0.9112016878754079, 'B_10_5': 0.19341102433644886, 'B_10_6': 0.7319978520062349, 'B_10_7': 0.7738476536134744, 'B_10_8': -0.1009052344674759, 'B_10_9': -0.08057869733811396}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

```
Built F_[u_1].  
Built F_[u_2].  
Built F_[u_3].  
Built F_[u_4].  
Built F_[u_5].  
Built F_[u_6].  
Built F_[u_7].  
Built F_[u_8].  
Built F_[u_9].  
Built F_[u_10].  
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:57:38,287] Trial 51 finished with value: 187.7181204139805 and parameters: {'B_0_0': -0.8376010176441256, 'B_0_1': 0.33811475037721084, 'B_0_2': 0.6360936523382341, 'B_0_3': -0.5055829918826682, 'B_0_4': -0.15582425436593794, 'B_0_5': 0.6709018867952352, 'B_0_6': 0.7466985259713304, 'B_0_7': -0.5085036390768327, 'B_0_8': 0.10012985639190108, 'B_0_9': -0.6088635617410625, 'B_1_0': -0.788280047003678, 'B_1_1': 0.8140185927560759, 'B_1_2': 0.634604858379576, 'B_1_3': 0.5148691055702843, 'B_1_4': -0.893442688749982, 'B_1_5': 0.6805830095663097, 'B_1_6': -0.12267272970734022, 'B_1_7': -0.31205765587419787, 'B_1_8': 0.09202319483024546, 'B_1_9': -0.8091205794073595, 'B_2_0': 0.5081601677177496, 'B_2_1': 0.16322145223882578, 'B_2_2': -0.6205516081911283, 'B_2_3': -0.6173002444572897, 'B_2_4': -0.625187669886105, 'B_2_5': -0.8771352726471707, 'B_2_6': 0.1738409297559358, 'B_2_7': 0.673889391625246, 'B_2_8': -0.9553535363049706, 'B_2_9': -0.8929708029568784, 'B_3_0': 0.14408514784015258, 'B_3_1': 0.6326014570643836, 'B_3_2': -0.3430569966741055, 'B_3_3': -0.29776349036071015, 'B_3_4': 0.19076606504361626, 'B_3_5': -0.447058479534189, 'B_3_6': -0.9373154351661854, 'B_3_7': 0.46575246584116026, 'B_3_8': 0.24878682762407722, 'B_3_9': 0.15989687813020503, 'B_4_0': 0.4785345754038097, 'B_4_1': 0.3469570583728343, 'B_4_2': -0.5823536634069242, 'B_4_3': -0.2920875270521648, 'B_4_4': -0.7403019194763225, 'B_4_5': -0.7328831134325955, 'B_4_6': 0.34886757778594757, 'B_4_7': 0.7100282118253465, 'B_4_8': 0.9162105160450553, 'B_4_9': 0.2541272927478999, 'B_5_0': -0.6254793797453052, 'B_5_1': -0.7808123160111172, 'B_5_2': 0.6194449236427804, 'B_5_3': 0.15593782319279859, 'B_5_4': 0.2342043416071653, 'B_5_5': 0.5572949357343242, 'B_5_6': 0.40248891091025407, 'B_5_7': 0.6622273406550714, 'B_5_8': -0.8786964671894406, 'B_5_9': 0.4900908607580796, 'B_6_0': 0.6946884594977489, 'B_6_1': -0.4234526159325376, 'B_6_2': 0.2780856025066407, 'B_6_3': -0.3679591136027952, 'B_6_4': 0.9067827116352589, 'B_6_5': -0.9238213265777026, 'B_6_6': 0.15798983078925088, 'B_6_7': -0.6603229768419601, 'B_6_8': 0.42019600625077413, 'B_6_9': -0.641737238831049, 'B_7_0': -0.523941141441252, 'B_7_1': -0.011145794129398978, 'B_7_2': 0.8213732009938186, 'B_7_3': -0.5708055431470177, 'B_7_4': -0.704404758186155, 'B_7_5': 0.5127741270709807, 'B_7_6': 0.0012266740211833715, 'B_7_7': 0.8153345225159074, 'B_7_8': -0.6997031768235122, 'B_7_9':

0.28893722818561796, 'B_8_0': -0.5288992906194062, 'B_8_1': -0.6796143512697541,
 'B_8_2': 0.3631902420113957, 'B_8_3': 0.18549554741929186, 'B_8_4':
 0.5045583675574159, 'B_8_5': -0.37726409863655896, 'B_8_6':
 -0.10301612036326391, 'B_8_7': -0.18716608086577147, 'B_8_8': 0.958497138653698,
 'B_8_9': -0.411347065656012, 'B_9_0': -0.4062618680734016, 'B_9_1':
 -0.2879945465193574, 'B_9_2': 0.12789628095483652, 'B_9_3': -0.4904009663577844,
 'B_9_4': 0.8222740819509624, 'B_9_5': -0.4639425815647983, 'B_9_6':
 0.7837151484891856, 'B_9_7': -0.7851193739797624, 'B_9_8': 0.30999415075131154,
 'B_9_9': 0.8058313265077226, 'B_10_0': -0.6558761119715055, 'B_10_1':
 0.927287380996525, 'B_10_2': -0.09329094637728685, 'B_10_3': 0.8825159147132284,
 'B_10_4': 0.3501036863874931, 'B_10_5': -0.3928579976503679, 'B_10_6':
 -0.6171610006641293, 'B_10_7': -0.3886046084743032, 'B_10_8':
 -0.8058255296599035, 'B_10_9': -0.7051414266402071}. Best is trial 36 with
 value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:57:54,942] Trial 52 finished with value: 157.9050975220017 and
 parameters: {'B_0_0': -0.7827723281139674, 'B_0_1': -0.15257178339171573,
 'B_0_2': 0.5852451849900211, 'B_0_3': -0.7100605626516464, 'B_0_4':
 -0.731965542318775, 'B_0_5': 0.2839893200443149, 'B_0_6': -0.5995776957032288,
 'B_0_7': -0.7967321245441966, 'B_0_8': -0.6596398151529563, 'B_0_9':
 0.8389881638360166, 'B_1_0': 0.6365130804968577, 'B_1_1': -0.2889521665369137,
 'B_1_2': 0.30852511930823767, 'B_1_3': 0.07380583386271544, 'B_1_4':
 0.3507060416954766, 'B_1_5': 0.3616888818762636, 'B_1_6': 0.254893349818786,
 'B_1_7': -0.46311929029612897, 'B_1_8': -0.91276728688962, 'B_1_9':
 -0.08797777165958454, 'B_2_0': 0.7741282916875936, 'B_2_1': 0.1376045718667802,
 'B_2_2': 0.6581342625697777, 'B_2_3': 0.9734161990632317, 'B_2_4':
 -0.6660399158062129, 'B_2_5': 0.37971972550112487, 'B_2_6': 0.3451012759260488,
 'B_2_7': 0.9503117857693912, 'B_2_8': 0.9614894532265448, 'B_2_9':
 0.36962365121377627, 'B_3_0': -0.5076427118752553, 'B_3_1': 0.5911405594985861,
 'B_3_2': -0.7780823142621446, 'B_3_3': 0.5557846222598248, 'B_3_4':
 0.8121883258064988, 'B_3_5': -0.23198987835849816, 'B_3_6': 0.25865788406649926,
 'B_3_7': -0.13831739044828395, 'B_3_8': -0.1232251911235056, 'B_3_9':
 -0.37546563956592227, 'B_4_0': 0.5190166484916428, 'B_4_1': 0.8636295970496803,
 'B_4_2': 0.872686701284233, 'B_4_3': -0.6439330559476311, 'B_4_4':

0.7419924263200008, 'B_4_5': 0.4701142540257017, 'B_4_6': -0.4893819466332273,
 'B_4_7': -0.8567764664154012, 'B_4_8': -0.4593654115689316, 'B_4_9':
 0.8161723088274206, 'B_5_0': 0.2636937762936291, 'B_5_1': 0.43768822857778633,
 'B_5_2': -0.48083834480399024, 'B_5_3': 0.4343885480381442, 'B_5_4':
 -0.9343194175697109, 'B_5_5': -0.24686373536943274, 'B_5_6':
 0.25097163011381585, 'B_5_7': -0.700728800228186, 'B_5_8': 0.6395470774049308,
 'B_5_9': -0.524638028352401, 'B_6_0': 0.30151660338052366, 'B_6_1':
 -0.18770965770084724, 'B_6_2': 0.19016631904574144, 'B_6_3':
 -0.7772479057947619, 'B_6_4': -0.9565126561552175, 'B_6_5': 0.8109375662129963,
 'B_6_6': -0.4561325826409781, 'B_6_7': 0.6329604655053089, 'B_6_8':
 0.9887135949349564, 'B_6_9': -0.5713661091621559, 'B_7_0': -0.46496193245695594,
 'B_7_1': 0.32246237951663814, 'B_7_2': 0.9215882965092301, 'B_7_3':
 0.8233124094176729, 'B_7_4': 0.005693529729858038, 'B_7_5': 0.5843183594029149,
 'B_7_6': -0.7212232108621959, 'B_7_7': 0.9453276047239443, 'B_7_8':
 0.9194194033912533, 'B_7_9': -0.6658166633992415, 'B_8_0': -0.39714221288984886,
 'B_8_1': -0.7916931049366015, 'B_8_2': -0.4519991541816042, 'B_8_3':
 0.625592463042109, 'B_8_4': -0.24093509781278621, 'B_8_5': -0.8149740050494184,
 'B_8_6': -0.312393460937769, 'B_8_7': 0.9531390017977346, 'B_8_8':
 0.25677351861322184, 'B_8_9': -0.16600132611580765, 'B_9_0':
 -0.6602537432754054, 'B_9_1': 0.9626300586217156, 'B_9_2': 0.635281038340995,
 'B_9_3': -0.43267494114717964, 'B_9_4': -0.8512639824720449, 'B_9_5':
 -0.8778813963924351, 'B_9_6': -0.751651263979043, 'B_9_7': 0.15992179555655484,
 'B_9_8': 0.8489515739868749, 'B_9_9': -0.00644801254815075, 'B_10_0':
 0.6790320552632703, 'B_10_1': -0.10714449267758053, 'B_10_2':
 0.41299542062029393, 'B_10_3': 0.4591080198272717, 'B_10_4':
 -0.6455518623113894, 'B_10_5': -0.27479220322789266, 'B_10_6':
 0.2718582699452874, 'B_10_7': -0.850020715794955, 'B_10_8': -0.2741504282293459,
 'B_10_9': 0.06890294323676116}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 13:58:14,254] Trial 53 finished with value: 170.18018643731858 and
 parameters: {'B_0_0': 0.9825244962283846, 'B_0_1': -0.05042354600756738,
 'B_0_2': 0.6257923790863607, 'B_0_3': -0.13450677556809687, 'B_0_4':
 0.5868536307870236, 'B_0_5': 0.18892130720781464, 'B_0_6': 0.7654589741747486,
 'B_0_7': -0.17633374634678234, 'B_0_8': 0.2065227512465282, 'B_0_9':

0.2574369876252782, 'B_1_0': -0.1635817752350629, 'B_1_1': -0.7042621534987865, 'B_1_2': -0.8047523743135756, 'B_1_3': -0.9569903298127265, 'B_1_4': 0.739597843205525, 'B_1_5': -0.5217796737673848, 'B_1_6': -0.5712166364220992, 'B_1_7': 0.4673500928436096, 'B_1_8': -0.3563680960869643, 'B_1_9': 0.011968490975436508, 'B_2_0': -0.6682280907110365, 'B_2_1': 0.9102965284946991, 'B_2_2': 0.6961311720341299, 'B_2_3': 0.7203034534707065, 'B_2_4': -0.18458797782742964, 'B_2_5': 0.5194752031402021, 'B_2_6': -0.7051252123282352, 'B_2_7': -0.972934374868275, 'B_2_8': -0.9504535934955043, 'B_2_9': 0.17147144006102977, 'B_3_0': -0.08831808760907722, 'B_3_1': 0.03567039083886914, 'B_3_2': -0.9752229819739056, 'B_3_3': -0.15308049307829708, 'B_3_4': 0.4317749591777045, 'B_3_5': -0.11346647010447697, 'B_3_6': 0.45979402849820006, 'B_3_7': -0.5286224504942671, 'B_3_8': 0.6200211688738164, 'B_3_9': 0.17509385609783545, 'B_4_0': -0.5726890631287856, 'B_4_1': -0.5212934578094108, 'B_4_2': 0.2535003198476833, 'B_4_3': 0.6532803945606378, 'B_4_4': -0.16705699158227272, 'B_4_5': -0.47856515032736746, 'B_4_6': 0.7266829816829887, 'B_4_7': 0.3351512176666531, 'B_4_8': -0.10279073517950854, 'B_4_9': -0.02395761334698232, 'B_5_0': -0.5855721934859351, 'B_5_1': 0.2510692075283909, 'B_5_2': -0.27968982912508045, 'B_5_3': 0.9813774498152115, 'B_5_4': -0.03923663978146208, 'B_5_5': -0.027946499744565867, 'B_5_6': 0.13038653907313025, 'B_5_7': 0.8251768861751185, 'B_5_8': 0.5956198377223056, 'B_5_9': -0.09676246180027392, 'B_6_0': 0.433024546626136, 'B_6_1': 0.17074628758590804, 'B_6_2': -0.015113243571527102, 'B_6_3': -0.609214769886986, 'B_6_4': -0.07140619835931594, 'B_6_5': -0.5032791976953022, 'B_6_6': 0.8213874567062762, 'B_6_7': -0.32589450927723307, 'B_6_8': 0.895482748776077, 'B_6_9': 0.06544947438752824, 'B_7_0': 0.26045786977879737, 'B_7_1': 0.08502208008886902, 'B_7_2': -0.7338263784059322, 'B_7_3': -0.24587398636427693, 'B_7_4': -0.7936906976903046, 'B_7_5': 0.24344900748309595, 'B_7_6': -0.10518909350313099, 'B_7_7': 0.550410243710455, 'B_7_8': -0.5534684476212477, 'B_7_9': -0.6641592043099169, 'B_8_0': -0.18546218269173975, 'B_8_1': 0.8446390587723946, 'B_8_2': 0.41663519372457536, 'B_8_3': -0.1605312229100211, 'B_8_4': 0.03419889611353488, 'B_8_5': -0.8232582378693453, 'B_8_6': 0.9594602748247039, 'B_8_7': 0.32948646329593956, 'B_8_8': -0.4248999444424235, 'B_8_9': 0.1699006205461837, 'B_9_0': -0.7732249151833919, 'B_9_1': 0.9223566042390154, 'B_9_2': -0.6439257239909355, 'B_9_3': 0.29310002974932403, 'B_9_4': -0.5040001641729177, 'B_9_5': 0.4307715376441008, 'B_9_6': 0.5481063071357566, 'B_9_7': 0.34337106324463673, 'B_9_8': 0.22670893130528125, 'B_9_9': 0.7770888060249832, 'B_10_0': -0.8343191122402791, 'B_10_1': -0.3347887637552135, 'B_10_2': -0.498390790330925, 'B_10_3': 0.9662676448895346, 'B_10_4': -0.21900823625959054, 'B_10_5': 0.4605731444959895, 'B_10_6': 0.36575582448438904, 'B_10_7': -0.5700791497947921, 'B_10_8': -0.5167845931811283, 'B_10_9': -0.4330565641649925}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u₁].

Built F__[u₂].

Built F__[u₃].

```

Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:58:34,798] Trial 54 finished with value: 157.20130224989185 and parameters: {'B_0_0': -0.17596798904642919, 'B_0_1': -0.5260363700119569,

```

'B_0_2': -0.9510900641612413, 'B_0_3': 0.751198252478247, 'B_0_4':
0.9219074661341504, 'B_0_5': -0.3136906336774955, 'B_0_6': -0.3197563787375637,
'B_0_7': -0.9759582107067455, 'B_0_8': 0.3568963628941406, 'B_0_9':
0.644370993582285, 'B_1_0': -0.6997155652871938, 'B_1_1': -0.6738606282530415,
'B_1_2': 0.5743648349434467, 'B_1_3': 0.23025568174120647, 'B_1_4':
-0.9011760958971203, 'B_1_5': 0.24264304111531776, 'B_1_6': -0.3783429087693432,
'B_1_7': -0.259924192167725, 'B_1_8': 0.5340682082278227, 'B_1_9':
0.615651681491769, 'B_2_0': -0.19729387885713534, 'B_2_1': 0.8853628132925824,
'B_2_2': -0.18520381845489697, 'B_2_3': 0.5382353497556176, 'B_2_4':
-0.03192301616276705, 'B_2_5': 0.7871140065137263, 'B_2_6': 0.8073672624273536,
'B_2_7': -0.8169147204152539, 'B_2_8': 0.8642567959942469, 'B_2_9':
0.6445589862104231, 'B_3_0': 0.5767952143390804, 'B_3_1': -0.5921955474226656,
'B_3_2': 0.6041360333342365, 'B_3_3': 0.6932628146763729, 'B_3_4':
-0.41748233778078114, 'B_3_5': 0.895474955157401, 'B_3_6': -0.8912769101296548,
'B_3_7': -0.6931796785706634, 'B_3_8': 0.07441065334199837, 'B_3_9':
0.1500690928221733, 'B_4_0': 0.2873971649357887, 'B_4_1': -0.9564011239553764,
'B_4_2': 0.9115262137518179, 'B_4_3': -0.9017155763863949, 'B_4_4':
-0.7067373336076632, 'B_4_5': -0.9622512552984072, 'B_4_6': 0.13237969792230753,
'B_4_7': 0.855888430163525, 'B_4_8': 0.32857538883931925, 'B_4_9':
0.6388617875912461, 'B_5_0': -0.9637032171520115, 'B_5_1': 0.08319029098985764,
'B_5_2': -0.08006530453906668, 'B_5_3': -0.00204347044282982, 'B_5_4':
-0.8470178930363279, 'B_5_5': 0.3651820149172287, 'B_5_6': 0.006376421420096201,
'B_5_7': 0.5249947227253027, 'B_5_8': -0.029124544557803556, 'B_5_9':
-0.6942243602716645, 'B_6_0': 0.29354876907408767, 'B_6_1': -0.648675003031771,
'B_6_2': 0.7373412354425335, 'B_6_3': 0.22397015335709813, 'B_6_4':
-0.6787363105572306, 'B_6_5': 0.9154293538127896, 'B_6_6': 0.03636361740393479,
'B_6_7': -0.8456610652941231, 'B_6_8': 0.2511291507418332, 'B_6_9':
-0.4886661641146357, 'B_7_0': 0.6013130347170588, 'B_7_1': 0.6292877056610575,
'B_7_2': 0.9483157645922304, 'B_7_3': 0.0037016714285172947, 'B_7_4':
-0.08918537813908045, 'B_7_5': 0.5018816987840315, 'B_7_6': -0.7277081004429854,
'B_7_7': 0.09289271731038595, 'B_7_8': 0.09157024825908477, 'B_7_9':
-0.8128432168354334, 'B_8_0': -0.16239151069631108, 'B_8_1': 0.6995039011630599,
'B_8_2': 0.7051841715494274, 'B_8_3': -0.7935094234590316, 'B_8_4':
-0.8103726521592949, 'B_8_5': -0.4698629555361681, 'B_8_6': 0.7420368038569072,
'B_8_7': -0.7347728815544721, 'B_8_8': 0.43559366920234366, 'B_8_9':

```

```

-0.7896920731660556, 'B_9_0': 0.24381295042069429, 'B_9_1':
-0.28498674774409327, 'B_9_2': 0.5720496466724168, 'B_9_3': -0.52729291132578,
'B_9_4': 0.27561962219212055, 'B_9_5': -0.5233485815397478, 'B_9_6':
0.4230473605620646, 'B_9_7': 0.7398825084302825, 'B_9_8': -0.7389324377290152,
'B_9_9': 0.7258594634190969, 'B_10_0': 0.1852790037403098, 'B_10_1':
-0.7383212280832925, 'B_10_2': -0.14346712638777026, 'B_10_3':
-0.6714966188995062, 'B_10_4': 0.3966888199558076, 'B_10_5': 0.5125589061475404,
'B_10_6': -0.7805228898598581, 'B_10_7': 0.012156908335395844, 'B_10_8':
0.622957610931316, 'B_10_9': 0.8636028222709731}. Best is trial 36 with value:
95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 13:58:51,383] Trial 55 finished with value: 114.0643137672243 and
parameters: {'B_0_0': 0.026765376148704556, 'B_0_1': 0.891552862750586, 'B_0_2':
0.07066416150293908, 'B_0_3': -0.2095028206921823, 'B_0_4': 0.6894421945347331,
'B_0_5': -0.013894233292983671, 'B_0_6': -0.055633029082244434, 'B_0_7':
-0.019181270968539943, 'B_0_8': 0.6866343037020959, 'B_0_9': 0.6572442525423432,
'B_1_0': -0.1910304149683818, 'B_1_1': -0.3300662774563604, 'B_1_2':
-0.01598251240252402, 'B_1_3': 0.14408956927947592, 'B_1_4': -0.518092322956566,
'B_1_5': 0.5820310677368694, 'B_1_6': -0.026764123849802646, 'B_1_7':
-0.32858981412848, 'B_1_8': -0.7784944339312986, 'B_1_9': -0.5170865824008584,
'B_2_0': 0.8895767200768476, 'B_2_1': -0.38590262009733733, 'B_2_2':
-0.6556586070532701, 'B_2_3': -0.07786787252311345, 'B_2_4':
-0.38196626814670764, 'B_2_5': -0.9203846844630356, 'B_2_6':
-0.35017246978675387, 'B_2_7': 0.08733507540014784, 'B_2_8':
0.10239049659953459, 'B_2_9': -0.2840133360697604, 'B_3_0': -0.1043405426198507,
'B_3_1': 0.9217254781048374, 'B_3_2': -0.24285001508398296, 'B_3_3':
-0.47871090974523367, 'B_3_4': -0.5079352527478149, 'B_3_5':
-0.08362606684581442, 'B_3_6': 0.5138498944595493, 'B_3_7': -0.5821433953068231,
'B_3_8': 0.41440633173872077, 'B_3_9': -0.9481233497681557, 'B_4_0':
-0.16051270359178693, 'B_4_1': 0.546212829101518, 'B_4_2': -0.31930874756662575,
'B_4_3': 0.12619946086189082, 'B_4_4': -0.5841411190046966, 'B_4_5':
0.4791508744298212, 'B_4_6': -0.886487375527856, 'B_4_7': 0.6980344368534908,
'B_4_8': -0.597571154885097, 'B_4_9': 0.7413081027824948, 'B_5_0':
-0.29199210222497574, 'B_5_1': 0.4409134774594834, 'B_5_2': -0.3579339103833469,

```


'B_5_3': 0.8037051090446021, 'B_5_4': -0.7392584914736188, 'B_5_5':
-0.6365308172402546, 'B_5_6': -0.013800352185571518, 'B_5_7':
-0.8277618907044706, 'B_5_8': -0.6019592342905113, 'B_5_9': 0.8786574367797884,
'B_6_0': 0.9434495290728167, 'B_6_1': -0.2450065279257646, 'B_6_2':
0.8381063216075926, 'B_6_3': 0.08875285783339493, 'B_6_4': -0.45590991756994126,
'B_6_5': 0.4007494677395853, 'B_6_6': 0.31462557701821714, 'B_6_7':
0.9684681318352921, 'B_6_8': -0.46520637092961903, 'B_6_9': 0.3406778152702361,
'B_7_0': -0.8869929650416951, 'B_7_1': -0.7649808433804955, 'B_7_2':
-0.9822416175968354, 'B_7_3': 0.9889321484981528, 'B_7_4': 0.5985019203520059,
'B_7_5': 0.5677115691724108, 'B_7_6': 0.019257629070595916, 'B_7_7':
-0.39586717584722897, 'B_7_8': -0.6904967167639017, 'B_7_9':
-0.42634401708400194, 'B_8_0': -0.24557698058030453, 'B_8_1':
0.5074939920209822, 'B_8_2': 0.07853034202824172, 'B_8_3': -0.8365824194201443,
'B_8_4': -0.9543025622613538, 'B_8_5': -0.787382284522555, 'B_8_6':
-0.17721254687918886, 'B_8_7': 0.8577219757570562, 'B_8_8': 0.2720569127942587,
'B_8_9': 0.5976152817535199, 'B_9_0': -0.06657830288099387, 'B_9_1':
-0.2739011552738153, 'B_9_2': -0.775367262021087, 'B_9_3': 0.7443473839286858,
'B_9_4': 0.5606568574752475, 'B_9_5': -0.6571261667565043, 'B_9_6':
0.024275916474995718, 'B_9_7': -0.024997880315935084, 'B_9_8':
0.24768127431932663, 'B_9_9': 0.8834500501026972, 'B_10_0': 0.15794745468970972,
'B_10_1': 0.07963018822166479, 'B_10_2': -0.8903605927893654, 'B_10_3':
-0.8654431974843126, 'B_10_4': -0.6190533531729581, 'B_10_5':
-0.9257078720503384, 'B_10_6': 0.48183873322862447, 'B_10_7':
0.44219400190804703, 'B_10_8': -0.7206086558253469, 'B_10_9':
0.6654362235746563}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].
Built F__[u_10].
Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 13:59:09,109] Trial 56 finished with value: 136.66382574008804 and
parameters: {'B_0_0': -0.6835236510046983, 'B_0_1': -0.3936723377452296,
'B_0_2': -0.4970257396604973, 'B_0_3': -0.2528319448253654, 'B_0_4':
-0.4857657435378755, 'B_0_5': -0.021877165651864927, 'B_0_6':
-0.1833323949177229, 'B_0_7': 0.9634075917430664, 'B_0_8': -0.8792658267461132,
'B_0_9': -0.7084808747530438, 'B_1_0': -0.8854047300609671, 'B_1_1':
0.24916556606510398, 'B_1_2': -0.8820716307208352, 'B_1_3': 0.6661506800047241,
'B_1_4': 0.023638488750217013, 'B_1_5': 0.053475096671436795, 'B_1_6':

0.12195712527014457, 'B_1_7': -0.6517888615137027, 'B_1_8': -0.5985731998333146, 'B_1_9': -0.35874170071145073, 'B_2_0': 0.1851229033272659, 'B_2_1': -0.5739668036383752, 'B_2_2': -0.7596631802761428, 'B_2_3': 0.9308314351084956, 'B_2_4': -0.9464760066769466, 'B_2_5': 0.5993408625653229, 'B_2_6': 0.5703095961585436, 'B_2_7': -0.4015160545867099, 'B_2_8': -0.7214593003055194, 'B_2_9': -0.7016552795232616, 'B_3_0': 0.390914602129641, 'B_3_1': 0.2827570503506238, 'B_3_2': 0.5598364350177458, 'B_3_3': 0.8662127213502273, 'B_3_4': 0.38001617926785203, 'B_3_5': -0.6814139576281419, 'B_3_6': -0.5079836188044868, 'B_3_7': 0.6775435871287667, 'B_3_8': -0.575573969848267, 'B_3_9': -0.8464058124301268, 'B_4_0': 0.40249450700365585, 'B_4_1': -0.19299781612951872, 'B_4_2': 0.8316793040408892, 'B_4_3': 0.2975050407654498, 'B_4_4': -0.8109704239725117, 'B_4_5': 0.8867560031622483, 'B_4_6': -0.47805661799231314, 'B_4_7': -0.3223192689955542, 'B_4_8': -0.19352438090658675, 'B_4_9': -0.8885758931194597, 'B_5_0': -0.006899463983402354, 'B_5_1': -0.7458293876709213, 'B_5_2': -0.19607875315706358, 'B_5_3': -0.29559461977567314, 'B_5_4': 0.5402233651844766, 'B_5_5': 0.6002776130328076, 'B_5_6': -0.6862424928696057, 'B_5_7': -0.2642286489891321, 'B_5_8': -0.2507607411569732, 'B_5_9': -0.6902073210065516, 'B_6_0': 0.5909379400013572, 'B_6_1': 0.41031209010413816, 'B_6_2': -0.6130440631318712, 'B_6_3': -0.26389062098427785, 'B_6_4': -0.7805060507841018, 'B_6_5': 0.2538490622442271, 'B_6_6': -0.749683589393733, 'B_6_7': 0.9621496538773373, 'B_6_8': 0.5557505078819673, 'B_6_9': -0.21732767182368973, 'B_7_0': 0.15555658591312183, 'B_7_1': 0.8144567961084894, 'B_7_2': 0.10711657518373574, 'B_7_3': -0.6444929969320488, 'B_7_4': -0.2401673559493046, 'B_7_5': 0.6794268327588959, 'B_7_6': 0.2855417716594435, 'B_7_7': 0.2984342809809246, 'B_7_8': 0.7263065416632086, 'B_7_9': 0.7800666537757592, 'B_8_0': 0.831189292123284, 'B_8_1': -0.8021985973587509, 'B_8_2': 0.08736948388693988, 'B_8_3': 0.4762088520122707, 'B_8_4': 0.5087151533514873, 'B_8_5': 0.21129807990060767, 'B_8_6': -0.6597610763363251, 'B_8_7': 0.5888926968044954, 'B_8_8': 0.43582044035959244, 'B_8_9': -0.8194511723329081, 'B_9_0': -0.07165654714473502, 'B_9_1': -0.645524748946829, 'B_9_2': -0.3217954494883156, 'B_9_3': -0.9541994374327428, 'B_9_4': -0.1973468605466987, 'B_9_5': 0.09875630388929668, 'B_9_6': 0.236773533797632, 'B_9_7': 0.3237108833548459, 'B_9_8': 0.6658567999428708, 'B_9_9': -0.5633834435940546, 'B_10_0': -0.7514459915718334, 'B_10_1': 0.6960436003578927, 'B_10_2': -0.6799066265125533, 'B_10_3': 0.9400249610456666, 'B_10_4': 0.6782256942093012, 'B_10_5': 0.0017356796614063086, 'B_10_6': 0.95321442134561, 'B_10_7': -0.7971940763661517, 'B_10_8': -0.38992979287329876, 'B_10_9': 0.2736843417804107}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].

```

    Built F_[u_9].
    Built F_[u_10].
    Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 13:59:26,355] Trial 57 finished with value: 211.55700061051985 and
parameters: {'B_0_0': -0.2969864004231061, 'B_0_1': 0.8481776529219678, 'B_0_2':
-0.8600603454079212, 'B_0_3': 0.09783542748070984, 'B_0_4': 0.17991520944232664,
'B_0_5': 0.30240168507890375, 'B_0_6': -0.4837808778030074, 'B_0_7':
-0.26815847351066835, 'B_0_8': 0.6804874487366557, 'B_0_9': -0.6040133174390836,
'B_1_0': 0.6471983374032568, 'B_1_1': 0.12405773734742986, 'B_1_2':
0.5191312721642827, 'B_1_3': 0.4229425852016737, 'B_1_4': -0.5791272391379396,
'B_1_5': 0.1362492181390682, 'B_1_6': 0.728091842554083, 'B_1_7':
-0.13440985113217985, 'B_1_8': 0.9238497250421671, 'B_1_9': -0.0946141017985982,
'B_2_0': -0.7072844393975883, 'B_2_1': -0.21891933344620695, 'B_2_2':
0.8943707646532635, 'B_2_3': -0.35695271830708863, 'B_2_4': 0.7288343565367845,
'B_2_5': 0.7548763497301418, 'B_2_6': -0.2966306818227171, 'B_2_7':
-0.8251498370597484, 'B_2_8': 0.4446246601344601, 'B_2_9': -0.3732056479523077,
'B_3_0': 0.5601887562299934, 'B_3_1': -0.11354340044434241, 'B_3_2':
-0.02917185016160928, 'B_3_3': 0.550260625362103, 'B_3_4': -0.2329206044979768,
'B_3_5': -0.11260785626062919, 'B_3_6': -0.24514287183614814, 'B_3_7':
-0.2626657797509857, 'B_3_8': 0.051374222264889635, 'B_3_9': 0.6114229001122784,
'B_4_0': -0.2736987423933196, 'B_4_1': 0.048024346664718554, 'B_4_2':
0.3380401043763497, 'B_4_3': -0.45650989631876304, 'B_4_4': -0.4405048462932627,
'B_4_5': 0.8581579915064406, 'B_4_6': 0.4425984551955555, 'B_4_7':
0.15282961892215163, 'B_4_8': 0.2924857077528118, 'B_4_9': -0.9319871026552317,
'B_5_0': 0.23384387181215982, 'B_5_1': 0.018827568983919685, 'B_5_2':
0.09058436596328256, 'B_5_3': -0.2783408228749822, 'B_5_4': -0.828681682528375,
'B_5_5': -0.6071883174481367, 'B_5_6': 0.8924745321610901, 'B_5_7':
-0.1807748667035508, 'B_5_8': -0.06596735145488897, 'B_5_9':
-0.8856346160901908, 'B_6_0': -0.9156188655029184, 'B_6_1': 0.43001926515258027,
'B_6_2': 0.07542743495980031, 'B_6_3': 0.017237848748208107, 'B_6_4':
-0.508613429763116, 'B_6_5': 0.4846064486272461, 'B_6_6': -0.6568915733651861,
'B_6_7': -0.6139292075511804, 'B_6_8': -0.07778990073041459, 'B_6_9':
-0.42316587301835706, 'B_7_0': -0.5000232971583702, 'B_7_1': 0.2875705362155967,
'B_7_2': 0.29892405553271817, 'B_7_3': 0.6425437982038598, 'B_7_4':
-0.16293703739350773, 'B_7_5': -0.8452915197682225, 'B_7_6':
-0.44814055785551377, 'B_7_7': 0.9504108171296974, 'B_7_8': 0.1905919789607502,
'B_7_9': 0.4152794279060201, 'B_8_0': -0.9754314973416792, 'B_8_1':
0.03604734972139134, 'B_8_2': -0.9156932580870298, 'B_8_3': 0.13707581994106688,
'B_8_4': 0.8331262822794292, 'B_8_5': 0.9274698783908075, 'B_8_6':
0.08029429727006976, 'B_8_7': -0.08275448560061238, 'B_8_8': 0.8811338466120182,
'B_8_9': 0.3437474673534564, 'B_9_0': -0.8614994265139493, 'B_9_1':
-0.06786335621835482, 'B_9_2': -0.5143148634406782, 'B_9_3':
-0.059787443531610474, 'B_9_4': -0.08817816742740381, 'B_9_5':
0.16264670957318916, 'B_9_6': -0.28359442910848254, 'B_9_7':
-0.6647445100772624, 'B_9_8': 0.007364480488113445, 'B_9_9': 0.6538017110046088,

```

'B_10_0': 0.3247442115738526, 'B_10_1': -0.3200316254680726, 'B_10_2': 0.8623355850625374, 'B_10_3': 0.8161709910733808, 'B_10_4': -0.5754901692194845, 'B_10_5': -0.9119531814616813, 'B_10_6': 0.4777912425345092, 'B_10_7': 0.6486683174713235, 'B_10_8': 0.9263718405836834, 'B_10_9': 0.7718760602948551}.

Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 13:59:43,377] Trial 58 finished with value: 155.76341772418237 and parameters: {'B_0_0': 0.1990588410487013, 'B_0_1': 0.059503631640513976,

'B_0_2': -0.9348747185000791, 'B_0_3': -0.5016202352473399, 'B_0_4': -0.0799627915688057, 'B_0_5': 0.25443772216546434, 'B_0_6': 0.1133586299704532, 'B_0_7': 0.4804192595697472, 'B_0_8': -0.43606147684365915, 'B_0_9': 0.1286061631518971, 'B_1_0': -0.8154783580862316, 'B_1_1': -0.1709422633052502, 'B_1_2': 0.4953184492501099, 'B_1_3': 0.38293368978950704, 'B_1_4': 0.9714726196009145, 'B_1_5': -0.4397755705442431, 'B_1_6': -0.003585187635521292, 'B_1_7': -0.3375297807274865, 'B_1_8': -0.11760583416283399, 'B_1_9': 0.7067612571413884, 'B_2_0': 0.3647339688732034, 'B_2_1': 0.7572335588682118, 'B_2_2': -0.5870480055028648, 'B_2_3': 0.7392550133260472, 'B_2_4': -0.8716706971267988, 'B_2_5': 0.7694265752655618, 'B_2_6': -0.3384339166677587, 'B_2_7': -0.36711648584014833, 'B_2_8': 0.023935381863354532, 'B_2_9': 0.8725665716919988, 'B_3_0': -0.8953928979308173, 'B_3_1': -0.2925826693599243, 'B_3_2': 0.7060125375930046, 'B_3_3': -0.20581213467777582, 'B_3_4': -0.5956693427578073, 'B_3_5': -0.14890752096380466, 'B_3_6': -0.22660604704649479, 'B_3_7': 0.8225240789040729, 'B_3_8': 0.5186300639713322, 'B_3_9': 0.010128800373132574, 'B_4_0': 0.9669737221087107, 'B_4_1': -0.0687504880736719, 'B_4_2': -0.874689094975855, 'B_4_3': -0.47069684280632995, 'B_4_4': -0.12658044916576106, 'B_4_5': -0.6793328041906822, 'B_4_6': 0.8861789501216051, 'B_4_7': 0.5283608176791907, 'B_4_8': -0.4587913431530102, 'B_4_9': 0.1797929763654733, 'B_5_0': 0.4384693310769525, 'B_5_1': 0.87373227923546, 'B_5_2': 0.755637991367313, 'B_5_3': 0.030080607753663058, 'B_5_4': 0.5438121871901749, 'B_5_5': 0.12304661257861826, 'B_5_6': 0.10435370229016949, 'B_5_7': -0.7628109224903801, 'B_5_8': -0.7402474371280189, 'B_5_9': 0.058476557744465696, 'B_6_0': -0.45765414094156687, 'B_6_1': -0.9483469331282721, 'B_6_2': 0.44122162372242024, 'B_6_3': -0.9256765385534733,

```

'B_6_4': 0.18864814538602803, 'B_6_5': 0.07760371237208252, 'B_6_6':
0.9282530873440529, 'B_6_7': -0.03125798409166958, 'B_6_8': -0.8640691559719612,
'B_6_9': 0.1023421762567942, 'B_7_0': 0.6753704178198807, 'B_7_1':
0.983761582877251, 'B_7_2': -0.1379205160956365, 'B_7_3': -0.35580848985644986,
'B_7_4': 0.8051735279649579, 'B_7_5': -0.6113405844292796, 'B_7_6':
0.2285492517651948, 'B_7_7': -0.1866854401217235, 'B_7_8': -0.9086569173880006,
'B_7_9': -0.3023688208747184, 'B_8_0': -0.4648475254447043, 'B_8_1':
0.3684604848081039, 'B_8_2': 0.8886051854299979, 'B_8_3': -0.4592665689902973,
'B_8_4': 0.6167025595203299, 'B_8_5': -0.8803875624619492, 'B_8_6':
0.07547959987883002, 'B_8_7': -0.8503108738544087, 'B_8_8': 0.8593853224386658,
'B_8_9': 0.5273755355036327, 'B_9_0': 0.31274082418509286, 'B_9_1':
0.6413148356168352, 'B_9_2': 0.29208919522892307, 'B_9_3': 0.9285099064050966,
'B_9_4': 0.42363673677224756, 'B_9_5': 0.8408240469728485, 'B_9_6':
0.033921146181063344, 'B_9_7': -0.7582124622959119, 'B_9_8':
0.055791888417573876, 'B_9_9': -0.7387918004350753, 'B_10_0':
0.18313708876928203, 'B_10_1': -0.03466374015522733, 'B_10_2':
0.2426992877089078, 'B_10_3': -0.9701069775114335, 'B_10_4':
-0.6042158265223792, 'B_10_5': -0.01716574481398081, 'B_10_6':
0.891997051946487, 'B_10_7': -0.23045410009678635, 'B_10_8':
-0.1497433884138034, 'B_10_9': 0.4250982562331118}. Best is trial 36 with value:
95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:00:01,783] Trial 59 finished with value: 200.2887021214345 and
parameters: {'B_0_0': 0.20550965942651378, 'B_0_1': -0.3981262228053002,
'B_0_2': -0.17658353677649297, 'B_0_3': -0.5589543982626797, 'B_0_4':
0.922722489077813, 'B_0_5': -0.7313439245164277, 'B_0_6': -0.2537631665584895,
'B_0_7': -0.9388388350043778, 'B_0_8': -0.5508766059715782, 'B_0_9':
0.5553328539927067, 'B_1_0': -0.5916730400680187, 'B_1_1': -0.831474462992648,
'B_1_2': -0.05058270390784392, 'B_1_3': -0.9702609152256534, 'B_1_4':
-0.8221630142358602, 'B_1_5': 0.6090992217346907, 'B_1_6': 0.5375172228220206,
'B_1_7': 0.8219748644016942, 'B_1_8': 0.701940563103284, 'B_1_9':
0.3626876152148397, 'B_2_0': -0.8863809425446876, 'B_2_1': 0.560040263613582,
'B_2_2': 0.6841114054643096, 'B_2_3': 0.5921826486284698, 'B_2_4':
0.01678848990989157, 'B_2_5': -0.5288543594740263, 'B_2_6': 0.6837533518636789,

```

'B_2_7': 0.4497823632854743, 'B_2_8': 0.951907125289926, 'B_2_9':
 -0.6817286564432528, 'B_3_0': -0.543184125556931, 'B_3_1': -0.5706396632346247,
 'B_3_2': 0.5792287398902467, 'B_3_3': -0.619923553028766, 'B_3_4':
 -0.8580135369498539, 'B_3_5': 0.9871503970395448, 'B_3_6': -0.9408619213386047,
 'B_3_7': 0.663783239114554, 'B_3_8': -0.2665736537297074, 'B_3_9':
 0.6763986103204742, 'B_4_0': 0.1081485725980671, 'B_4_1': 0.7952298343829929,
 'B_4_2': 0.8438621131009607, 'B_4_3': -0.9719856010640635, 'B_4_4':
 -0.8406838915342792, 'B_4_5': -0.4942897154497062, 'B_4_6': 0.3352445243997333,
 'B_4_7': 0.7687789490927819, 'B_4_8': -0.07931493643795373, 'B_4_9':
 0.2118067702230455, 'B_5_0': -0.616770291464789, 'B_5_1': 0.603834331085737,
 'B_5_2': 0.9253096604798701, 'B_5_3': -0.45817411282909704, 'B_5_4':
 -0.010244615830918291, 'B_5_5': -0.8194901011973799, 'B_5_6':
 0.11709057979667747, 'B_5_7': 0.005511214856427471, 'B_5_8':
 -0.7040157613729463, 'B_5_9': -0.11470889802621143, 'B_6_0':
 -0.32847278886860476, 'B_6_1': 0.043225133104275004, 'B_6_2':
 0.16959558414902243, 'B_6_3': -0.9731787543577252, 'B_6_4': 0.5167918462401195,
 'B_6_5': 0.4217682978862467, 'B_6_6': -0.264752955311086, 'B_6_7':
 -0.8851868475737399, 'B_6_8': 0.5047664287743998, 'B_6_9': 0.9012468177467023,
 'B_7_0': -0.5144998398764119, 'B_7_1': -0.34942126480066493, 'B_7_2':
 -0.6260504788260064, 'B_7_3': 0.15233352422278257, 'B_7_4': -0.6834140385980392,
 'B_7_5': -0.2847359919464687, 'B_7_6': 0.3900857675898406, 'B_7_7':
 0.11902964635850988, 'B_7_8': -0.22480412263068816, 'B_7_9':
 -0.15790578842106706, 'B_8_0': 0.12297954759399432, 'B_8_1':
 -0.9783878387834206, 'B_8_2': -0.8345504604618524, 'B_8_3': 0.754890974170763,
 'B_8_4': -0.8357996015992443, 'B_8_5': 0.40947112049680023, 'B_8_6':
 0.8316755649359242, 'B_8_7': -0.4540464634825724, 'B_8_8': -0.2564269546221618,
 'B_8_9': 0.13646438357253832, 'B_9_0': 0.7557837598843977, 'B_9_1':
 0.874973062775632, 'B_9_2': 0.47433523126031085, 'B_9_3': -0.8184928735077058,
 'B_9_4': 0.3349252279590045, 'B_9_5': -0.1775986306921291, 'B_9_6':
 -0.05801202100769809, 'B_9_7': -0.8926861871165495, 'B_9_8':
 -0.6116969640942108, 'B_9_9': 0.5752169429166225, 'B_10_0':
 -0.09116151056523769, 'B_10_1': -0.3927087615703253, 'B_10_2':
 -0.22964223354938285, 'B_10_3': 0.7017110363848469, 'B_10_4':
 0.49301805446181457, 'B_10_5': -0.4508155135814561, 'B_10_6':
 -0.7573463563795009, 'B_10_7': -0.49104352114473804, 'B_10_8':
 0.8330977730883726, 'B_10_9': 0.1615458989425156}. Best is trial 36 with value:
 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].

```

Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:00:17,780] Trial 60 finished with value: 172.23306997567175 and
parameters: {'B_0_0': 0.06629081331467002, 'B_0_1': 0.19085932739107392,
'B_0_2': -0.2652640338218998, 'B_0_3': 0.6080719002326354, 'B_0_4':
0.04610677361449467, 'B_0_5': -0.7588391525177849, 'B_0_6': -0.7344510528384554,
'B_0_7': 0.07298111007518426, 'B_0_8': 0.8644090317802615, 'B_0_9':
-0.37946279908112646, 'B_1_0': 0.7316915143412521, 'B_1_1': 0.522909261426181,
'B_1_2': -0.49933136304432446, 'B_1_3': -0.91804531833384, 'B_1_4':
-0.7883424150180822, 'B_1_5': 0.4524179890179316, 'B_1_6': -0.5946539186795196,
'B_1_7': -0.2548835773993203, 'B_1_8': 0.003364348692721175, 'B_1_9':
0.44642697353132244, 'B_2_0': -0.7658021779469317, 'B_2_1': -0.1853945175506263,
'B_2_2': -0.029555639897463126, 'B_2_3': 0.0808375866636486, 'B_2_4':
0.630354069962529, 'B_2_5': -0.6500771738144305, 'B_2_6': 0.6095062746167121,
'B_2_7': 0.38189928683937135, 'B_2_8': 0.11062421211393958, 'B_2_9':
0.3950651833970924, 'B_3_0': -0.1742048768757204, 'B_3_1': -0.7159727797831134,
'B_3_2': 0.32821517572542724, 'B_3_3': 0.04885116723753202, 'B_3_4':
0.27596210507469543, 'B_3_5': -0.3207624135635898, 'B_3_6': 0.805595823132875,
'B_3_7': 0.17299233977964223, 'B_3_8': -0.8209030811040698, 'B_3_9':
-0.09938809840523122, 'B_4_0': 0.507584988392396, 'B_4_1': 0.5823864322298993,
'B_4_2': -0.010697835527757604, 'B_4_3': 0.7816001902459762, 'B_4_4':
-0.4600164943552908, 'B_4_5': -0.17701457946823218, 'B_4_6': 0.7892218829265607,
'B_4_7': 0.7845882986137263, 'B_4_8': 0.3677112822728681, 'B_4_9':
0.7107814833019959, 'B_5_0': -0.8142289314266593, 'B_5_1': 0.7226900528913971,
'B_5_2': -0.35517258503055105, 'B_5_3': -0.6369036053921318, 'B_5_4':
-0.5160477240427482, 'B_5_5': -0.1624223541325036, 'B_5_6': -0.4926398308831549,
'B_5_7': -0.35527617968733194, 'B_5_8': -0.7227425975781487, 'B_5_9':
-0.4035700041931829, 'B_6_0': 0.14312195279316242, 'B_6_1':
-0.11512965797969343, 'B_6_2': -0.30349685275289584, 'B_6_3':
0.48797521513708575, 'B_6_4': 0.7791497433587695, 'B_6_5': 0.5084426613467461,
'B_6_6': -0.3326386098069952, 'B_6_7': 0.48399549231835626, 'B_6_8':
-0.6890792387013787, 'B_6_9': -0.7319703835792439, 'B_7_0':
-0.30565846666840535, 'B_7_1': 0.6092506907072328, 'B_7_2': -0.5022304704602423,
'B_7_3': -0.8647582503806144, 'B_7_4': -0.729215120291417, 'B_7_5':
0.7103150376885687, 'B_7_6': 0.26489090379962943, 'B_7_7': -0.4886616965274462,
'B_7_8': 0.10948573414979901, 'B_7_9': 0.054607652402193674, 'B_8_0':
0.3077632373125092, 'B_8_1': -0.5282886422451997, 'B_8_2': 0.7215179146449073,
'B_8_3': -0.3774502470815746, 'B_8_4': 0.384380674872949, 'B_8_5':
-0.9580732001830146, 'B_8_6': -0.2521388828095885, 'B_8_7':
-0.29361456044155954, 'B_8_8': 0.08609916859436861, 'B_8_9':
-0.8459822016811509, 'B_9_0': 0.5744601199776944, 'B_9_1': -0.5587940125002143,
'B_9_2': 0.35053487346638024, 'B_9_3': 0.6003600040233046, 'B_9_4':
0.12251637335819598, 'B_9_5': 0.47694874930953124, 'B_9_6': 0.320483418185711,
'B_9_7': -0.28605100163902497, 'B_9_8': 0.6115399072864593, 'B_9_9':
-0.40649128762665465, 'B_10_0': 0.141001577148004, 'B_10_1': 0.4858875235002482,
'B_10_2': -0.23966729857123492, 'B_10_3': -0.39052013691749843, 'B_10_4':

```

-0.080494899857811, 'B_10_5': 0.6295677188113242, 'B_10_6': 0.8512450910284162, 'B_10_7': 0.08926515746258246, 'B_10_8': 0.31970308654083057, 'B_10_9': 0.7984817720330499}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].
Built F__[u_10].
Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:00:33,109] Trial 61 finished with value: 157.5514556601053 and parameters: {'B_0_0': -0.794877234202374, 'B_0_1': -0.5304844158249933, 'B_0_2': 0.41362201237813334, 'B_0_3': 0.6410140345863533, 'B_0_4': 0.7069174760945443, 'B_0_5': -0.1680877898272335, 'B_0_6': -0.8131069934621037, 'B_0_7': -0.12319267494675257, 'B_0_8': -0.5331455113900335, 'B_0_9': -0.4134310191066841, 'B_1_0': 0.5390897625382616, 'B_1_1': -0.8404563696687439, 'B_1_2': 0.6486108106117903, 'B_1_3': -0.1432152100681452, 'B_1_4': 0.1902065835791782, 'B_1_5': -0.8124885821533556, 'B_1_6': -0.32202902485258633, 'B_1_7': -0.8123921803506053, 'B_1_8': 0.913856571304597, 'B_1_9': 0.6083780656659759, 'B_2_0': -0.07992714637420872, 'B_2_1': -0.5472669543191312, 'B_2_2': 0.8582411083947072, 'B_2_3': -0.30769874046059587, 'B_2_4': -0.6712508252589979, 'B_2_5': 0.12506548913883075, 'B_2_6': 0.8786727582827358, 'B_2_7': -0.9824160248093613, 'B_2_8': -0.5185128103570824, 'B_2_9': 0.868126328012607, 'B_3_0': -0.6512497856228192, 'B_3_1': -0.7657779657531567, 'B_3_2': 0.13897068728361295, 'B_3_3': 0.7164507701237632, 'B_3_4': -0.04532256096602627, 'B_3_5': 0.979237228235359, 'B_3_6': 0.3547599650852258, 'B_3_7': -0.8433316837233135, 'B_3_8': 0.9758076207303112, 'B_3_9': -0.18236941117522298, 'B_4_0': -0.42094298298594457, 'B_4_1': 0.11460078348421354, 'B_4_2': -0.3874709681946886, 'B_4_3': -0.8188141359020984, 'B_4_4': -0.9871718807032301, 'B_4_5': -0.09982496121788065, 'B_4_6': 0.6669895760086526, 'B_4_7': 0.14318406754755153, 'B_4_8': 0.9884304519080129, 'B_4_9': -0.422043641618544, 'B_5_0': -0.4346989717750904, 'B_5_1': -0.6736960687367904, 'B_5_2': -0.6651540952346363, 'B_5_3': -0.4253305399001054, 'B_5_4': 0.5798232896415838, 'B_5_5': 0.8033984454601668, 'B_5_6': 0.940672596021209, 'B_5_7': -0.5300664848890237, 'B_5_8': 0.7549691403159016, 'B_5_9': -0.3400549232308726, 'B_6_0': -0.7665757218565689, 'B_6_1': 0.732649352529928, 'B_6_2': 0.2694891565943647, 'B_6_3': -0.6638328257594978, 'B_6_4': -0.13893411012425494, 'B_6_5': -0.42122976414340607, 'B_6_6': 0.46885324604619005, 'B_6_7': -0.09223586336394929, 'B_6_8': 0.9687742913106423, 'B_6_9': -0.8816049351260017, 'B_7_0': -0.2870697730502316, 'B_7_1':


```

-0.03583766912224473, 'B_7_2': 0.0062044384016731735, 'B_7_3':
-0.3644470716764392, 'B_7_4': 0.45389569321318457, 'B_7_5': 0.7419992055081699,
'B_7_6': 0.8686642179161688, 'B_7_7': 0.6040848860630619, 'B_7_8':
0.6941487906670667, 'B_7_9': 0.10007969522759108, 'B_8_0': 0.6452586883250928,
'B_8_1': -0.38906225988705456, 'B_8_2': -0.20101205102375552, 'B_8_3':
-0.36251223263928223, 'B_8_4': -0.06457307581310767, 'B_8_5':
-0.42455235984749995, 'B_8_6': 0.9139906556638047, 'B_8_7': -0.3705676365487771,
'B_8_8': -0.7958718145403392, 'B_8_9': 0.05309798877981864, 'B_9_0':
-0.8769438246785508, 'B_9_1': -0.7711245307090105, 'B_9_2': -0.5556090093914323,
'B_9_3': -0.07058650523335941, 'B_9_4': -0.13216315402812473, 'B_9_5':
0.5933898867655265, 'B_9_6': 0.4741598146905446, 'B_9_7': -0.1458897778984114,
'B_9_8': -0.3292656690002528, 'B_9_9': -0.36339180755823275, 'B_10_0':
0.5425102728421378, 'B_10_1': -0.8868966356818859, 'B_10_2':
-0.9646131339690005, 'B_10_3': 0.5851778960147631, 'B_10_4':
-0.045250175541682425, 'B_10_5': -0.7308553456639195, 'B_10_6':
-0.5235794495028875, 'B_10_7': 0.8563578271328174, 'B_10_8': 0.7118158917372979,
'B_10_9': 0.41537439944346555}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:00:49,647] Trial 62 finished with value: 132.7034461789364 and
parameters: {'B_0_0': -0.035194043932093355, 'B_0_1': 0.0402407798589437,
'B_0_2': 0.940984179862238, 'B_0_3': 0.9222254788908886, 'B_0_4':
-0.5637672893440482, 'B_0_5': 0.6159870412845299, 'B_0_6': -0.5994595209893068,
'B_0_7': -0.5671957937897762, 'B_0_8': 0.3486642076458619, 'B_0_9':
-0.18809782479665416, 'B_1_0': 0.85445792607091, 'B_1_1': -0.8137607249059025,
'B_1_2': 0.022820768802796154, 'B_1_3': 0.8286726302773801, 'B_1_4':
0.8001450477494201, 'B_1_5': -0.6800721574368324, 'B_1_6': -0.2640573538090162,
'B_1_7': 0.23586037209245858, 'B_1_8': -0.6743981281526237, 'B_1_9':
0.9266839006505627, 'B_2_0': 0.8212036708807675, 'B_2_1': -0.659438645511751,
'B_2_2': -0.7501121867032489, 'B_2_3': -0.8208912040396347, 'B_2_4':
0.8106984547925296, 'B_2_5': -0.6066248566049877, 'B_2_6': -0.1657558414821776,
'B_2_7': -0.10127429600903404, 'B_2_8': 0.19604186754286368, 'B_2_9':
-0.8479640193800986, 'B_3_0': -0.5410179375469784, 'B_3_1': 0.8543509660994735,
'B_3_2': 0.8086520123917675, 'B_3_3': 0.5569554818632705, 'B_3_4':
-0.9515855336759335, 'B_3_5': -0.3425265983777178, 'B_3_6': 0.25335829597895754,

```

'B_3_7': -0.4451233806885181, 'B_3_8': 0.8109816562064791, 'B_3_9': 0.265101134428235, 'B_4_0': -0.8476246997134377, 'B_4_1': 0.8098310658946921, 'B_4_2': 0.9063779831889207, 'B_4_3': -0.8361294115843836, 'B_4_4': 0.06407326502540345, 'B_4_5': -0.3893387212159425, 'B_4_6': -0.21217862856682335, 'B_4_7': -0.6342905884484231, 'B_4_8': 0.15458276198457677, 'B_4_9': 0.9067991194328033, 'B_5_0': 0.006083748537785927, 'B_5_1': -0.8258291288389273, 'B_5_2': 0.23760264028526867, 'B_5_3': -0.31635847098651104, 'B_5_4': 0.7047318753907696, 'B_5_5': -0.12933726633338893, 'B_5_6': -0.6081775742943537, 'B_5_7': -0.611610867071533, 'B_5_8': 0.5321530778278976, 'B_5_9': -0.4641365553514274, 'B_6_0': 0.5291221204352614, 'B_6_1': -0.5823431246821332, 'B_6_2': -0.35306098997140734, 'B_6_3': -0.0683357945693952, 'B_6_4': 0.8775277823930323, 'B_6_5': 0.8548282210125728, 'B_6_6': -0.5803936866638701, 'B_6_7': -0.2807262983387159, 'B_6_8': -0.6941892108615308, 'B_6_9': 0.01977045704072422, 'B_7_0': -0.07804654962456015, 'B_7_1': 0.7935051216524502, 'B_7_2': 0.8514106480263237, 'B_7_3': -0.2305365658055617, 'B_7_4': -0.016358872822400894, 'B_7_5': 0.7845477733093436, 'B_7_6': 0.6072814593352216, 'B_7_7': -0.9827605035375945, 'B_7_8': -0.35628206291650666, 'B_7_9': -0.15488700096647068, 'B_8_0': -0.5351617583072013, 'B_8_1': -0.5948482005260413, 'B_8_2': -0.9151866769405542, 'B_8_3': 0.2838720556381258, 'B_8_4': -0.470229624843332, 'B_8_5': -0.886352586706, 'B_8_6': -0.5326609651544341, 'B_8_7': 0.5977896266155305, 'B_8_8': 0.6959485796601623, 'B_8_9': -0.03352883451726074, 'B_9_0': -0.16618007259080136, 'B_9_1': 0.9461503131335278, 'B_9_2': -0.5928819740156216, 'B_9_3': -0.5635886199478752, 'B_9_4': -0.41210298946792845, 'B_9_5': -0.6479405098034472, 'B_9_6': -0.2388561155557105, 'B_9_7': 0.8134517694397125, 'B_9_8': 0.22838872444457703, 'B_9_9': -0.17434583374771795, 'B_10_0': 0.8601333732510053, 'B_10_1': -0.8940408196978717, 'B_10_2': -0.5736531677543144, 'B_10_3': 0.13045943810148275, 'B_10_4': 0.22879419392249467, 'B_10_5': -0.911676137006026, 'B_10_6': -0.19198325934387828, 'B_10_7': 0.295324850365128, 'B_10_8': 0.3921796818787351, 'B_10_9': 0.27351700698221326}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$\langle F_{[u_i]}, F_{[u_j]} \rangle$].

Built matrix B = inverse(A).

[I 2025-04-29 14:01:06,899] Trial 63 finished with value: 183.18015314664794 and parameters: {'B_0_0': -0.3552848157653694, 'B_0_1': -0.8048301552761392,

'B_0_2': -0.14431735608971852, 'B_0_3': -0.7350662140112502, 'B_0_4':
 0.8862948438709208, 'B_0_5': -0.04916168527519604, 'B_0_6':
 -0.36182683477340216, 'B_0_7': 0.5272820608154221, 'B_0_8':
 -0.004061786687117541, 'B_0_9': -0.3761326324553489, 'B_1_0':
 0.6176961299337873, 'B_1_1': 0.7269612589822214, 'B_1_2': 0.3425561692515804,
 'B_1_3': -0.8970104436328188, 'B_1_4': 0.13131930272973347, 'B_1_5':
 0.5213239257546176, 'B_1_6': -0.7197240360745073, 'B_1_7': -0.5372319122584854,
 'B_1_8': 0.7573403080349101, 'B_1_9': -0.9501589907309134, 'B_2_0':
 0.5003997328954324, 'B_2_1': -0.05192888082144265, 'B_2_2':
 -0.10539112549757845, 'B_2_3': -0.9759488656601234, 'B_2_4': 0.6324084973136557,
 'B_2_5': -0.2775466967395984, 'B_2_6': 0.08071594997507492, 'B_2_7':
 0.3510553589446057, 'B_2_8': 0.4012065613470732, 'B_2_9': 0.902840516475623,
 'B_3_0': 0.06205310740848602, 'B_3_1': -0.19855528907063336, 'B_3_2':
 0.8983331804535817, 'B_3_3': 0.2813782653505319, 'B_3_4': -0.39966229457241986,
 'B_3_5': -0.739600794100929, 'B_3_6': 0.6782579129010589, 'B_3_7':
 -0.27288042646757993, 'B_3_8': 0.3177300483015737, 'B_3_9':
 -0.057783489655521714, 'B_4_0': -0.596119036005824, 'B_4_1':
 0.20133948506852084, 'B_4_2': -0.869847523477649, 'B_4_3': -0.17158192945503936,
 'B_4_4': 0.09229758053406867, 'B_4_5': -0.12662699124521348, 'B_4_6':
 0.49237090699696195, 'B_4_7': 0.6492914099997187, 'B_4_8': 0.5370612865856925,
 'B_4_9': -0.9131503633326318, 'B_5_0': -0.605625034516975, 'B_5_1':
 0.07248286834227313, 'B_5_2': 0.8609476400787264, 'B_5_3': 0.6632144146031913,
 'B_5_4': 0.6782183265365673, 'B_5_5': -0.3896477763558437, 'B_5_6':
 -0.13969316378416896, 'B_5_7': -0.058002224314443906, 'B_5_8':
 -0.6784042649671773, 'B_5_9': -0.9284851021159765, 'B_6_0': 0.8848677409318233,
 'B_6_1': -0.5133890803862566, 'B_6_2': -0.51490117062721, 'B_6_3':
 -0.8249094735378562, 'B_6_4': 0.8484453488486816, 'B_6_5': 0.15787948758074855,
 'B_6_6': 0.3307329771082006, 'B_6_7': 0.814922236780844, 'B_6_8':
 -0.3056720655764368, 'B_6_9': 0.2405844760534066, 'B_7_0': -0.9131770796368028,
 'B_7_1': -0.6201903941023639, 'B_7_2': 0.6558666940780862, 'B_7_3':
 0.5282013455499048, 'B_7_4': -0.2957274709148492, 'B_7_5': -0.24391495700705934,
 'B_7_6': 0.06643777843608722, 'B_7_7': -0.9895228107234336, 'B_7_8':
 -0.5123362146447048, 'B_7_9': -0.5777010189557354, 'B_8_0':
 -0.49404376543546835, 'B_8_1': 0.6056935682162139, 'B_8_2': -0.4362457640380425,
 'B_8_3': -0.6107877823407066, 'B_8_4': 0.008441652370202446, 'B_8_5':
 0.7239771720180093, 'B_8_6': -0.5121275864017503, 'B_8_7': -0.8344982392702586,
 'B_8_8': -0.28431862301766386, 'B_8_9': 0.4903506126386561, 'B_9_0':
 0.10036727849569327, 'B_9_1': 0.6272508717572078, 'B_9_2': 0.9099924522662832,
 'B_9_3': 0.5794717039682018, 'B_9_4': 0.20553009611063167, 'B_9_5':
 -0.013183655656286652, 'B_9_6': -0.910663209030432, 'B_9_7': 0.9589575180518017,
 'B_9_8': -0.23070737730141577, 'B_9_9': -0.6885881364446037, 'B_10_0':
 0.33008353368498433, 'B_10_1': -0.4271091875909905, 'B_10_2':
 0.1698409764434743, 'B_10_3': 0.17180045753229067, 'B_10_4':
 -0.10745495461231902, 'B_10_5': 0.6655148587791844, 'B_10_6':
 -0.4653652181202228, 'B_10_7': -0.1017074685968864, 'B_10_8':
 0.07463041461012354, 'B_10_9': 0.3239769471192533}. Best is trial 36 with value:
 95.85134184163296.

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```

[I 2025-04-29 14:01:25,007] Trial 64 finished with value: 138.85110762162034 and
parameters: {'B_0_0': 0.5622642640010671, 'B_0_1': -0.208281579901053, 'B_0_2':
0.28917294931126003, 'B_0_3': -0.6917903129243321, 'B_0_4': 0.8566432390741077,
'B_0_5': -0.231543563286413, 'B_0_6': 0.45445241551334803, 'B_0_7':
-0.9497246578732883, 'B_0_8': -0.5837294570484821, 'B_0_9':
-0.025576562239315592, 'B_1_0': 0.7489398861100462, 'B_1_1':
0.053965196902340695, 'B_1_2': 0.8511435549418804, 'B_1_3':
-0.03208214476606319, 'B_1_4': -0.018910788462819128, 'B_1_5':
-0.7529013587689468, 'B_1_6': 0.5992593634213952, 'B_1_7': -0.5010084833152162,
'B_1_8': 0.7912327289571839, 'B_1_9': 0.8134319562541843, 'B_2_0':
0.8777540215108164, 'B_2_1': 0.040056368424118194, 'B_2_2': 0.3506456938716158,
'B_2_3': 0.5514614425220938, 'B_2_4': -0.7305910550720781, 'B_2_5':
0.1998952460950445, 'B_2_6': -0.9220901554194428, 'B_2_7':
-0.004819154890322808, 'B_2_8': -0.6834114903919497, 'B_2_9':
-0.6278126065950722, 'B_3_0': -0.26720581794864895, 'B_3_1':
-0.13377440990132405, 'B_3_2': 0.24388140057373064, 'B_3_3': -0.842631731491739,
'B_3_4': 0.007309687250140362, 'B_3_5': -0.6284022420988598, 'B_3_6':
0.14647084656284903, 'B_3_7': 0.27206113227714335, 'B_3_8': 0.1122843279746506,
'B_3_9': 0.47182511074704925, 'B_4_0': -0.3254936964030154, 'B_4_1':
0.5249110514335991, 'B_4_2': -0.13312117821777658, 'B_4_3':
-0.03131469332125836, 'B_4_4': 0.29324221197692135, 'B_4_5':
-0.9536503996815365, 'B_4_6': 0.30490855841209163, 'B_4_7': 0.23479859516278845,
'B_4_8': -0.3765027088285058, 'B_4_9': -0.8201273158990108, 'B_5_0':
-0.8099478091721786, 'B_5_1': 0.8379830190307527, 'B_5_2': 0.3747730560271576,
'B_5_3': -0.738336824733934, 'B_5_4': -0.28368812476836547, 'B_5_5':
0.18311632792516086, 'B_5_6': 0.6644319760794843, 'B_5_7': -0.9035513611482431,
'B_5_8': -0.5214663996893825, 'B_5_9': 0.7717030007848524, 'B_6_0':
-0.9718522106360433, 'B_6_1': -0.5912297764681029, 'B_6_2': 0.9088678684264726,
'B_6_3': 0.2245356561138987, 'B_6_4': -0.4718906940577372, 'B_6_5':
-0.8573576330962217, 'B_6_6': -0.9341811459145001, 'B_6_7': -0.505404742318884,
'B_6_8': 0.7553419227860427, 'B_6_9': -0.16649096449093126, 'B_7_0':
0.43626007094866615, 'B_7_1': 0.8282043700523163, 'B_7_2': -0.1820798860264231,
'B_7_3': -0.6112708053459538, 'B_7_4': 0.03389350309670047, 'B_7_5':
0.8080057003785117, 'B_7_6': 0.6026613399926473, 'B_7_7': -0.8596569101409555,

```

```
'B_7_8': 0.38129414415361196, 'B_7_9': 0.14235709446872846, 'B_8_0':
0.8067770891659225, 'B_8_1': -0.7474867946364114, 'B_8_2': 0.06739150414522,
'B_8_3': 0.7513361331535469, 'B_8_4': -0.17815290225134883, 'B_8_5':
-0.03565006288892325, 'B_8_6': -0.8504878582194226, 'B_8_7':
-0.4392755919111566, 'B_8_8': -0.3474768346454843, 'B_8_9': 0.8813424485148762,
'B_9_0': -0.43031862612850513, 'B_9_1': -0.07747990416608908, 'B_9_2':
0.339182758657274, 'B_9_3': -0.6688769226324027, 'B_9_4': -0.026844073459088458,
'B_9_5': 0.4723072981622971, 'B_9_6': 0.7263733217935908, 'B_9_7':
0.24999336383160498, 'B_9_8': 0.33472596032755675, 'B_9_9': -0.7399587242825534,
'B_10_0': 0.8826374645477673, 'B_10_1': -0.7275033265807249, 'B_10_2':
0.5438524156199607, 'B_10_3': -0.4656636828789288, 'B_10_4': 0.8585144245284781,
'B_10_5': 0.4355198879515696, 'B_10_6': 0.5025511516971706, 'B_10_7':
-0.556262837341431, 'B_10_8': -0.1127808462623816, 'B_10_9':
0.23327012528678703}. Best is trial 36 with value: 95.85134184163296.
```

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```
[I 2025-04-29 14:01:41,318] Trial 65 finished with value: 105.33935643475834 and
parameters: {'B_0_0': 0.5345477640490699, 'B_0_1': -0.7841604356934686, 'B_0_2':
0.9154833265799662, 'B_0_3': 0.772636927698021, 'B_0_4': -0.45470979081874296,
'B_0_5': 0.6390652812040174, 'B_0_6': 0.03628136761885048, 'B_0_7':
-0.07562316650720313, 'B_0_8': -0.3674204704934173, 'B_0_9':
0.16031780153055308, 'B_1_0': 0.4551804530531236, 'B_1_1': -0.17635284817373864,
'B_1_2': 0.2558540021564184, 'B_1_3': 0.3946395838875014, 'B_1_4':
0.41802119769026613, 'B_1_5': -0.8871071257002631, 'B_1_6': 0.7123006552564326,
'B_1_7': -0.08265843676520512, 'B_1_8': -0.4704161582414954, 'B_1_9':
-0.5113062751902304, 'B_2_0': -0.03363205656897805, 'B_2_1': 0.7659264481485393,
'B_2_2': -0.2408717882683591, 'B_2_3': 0.4944569786314157, 'B_2_4':
-0.7982989126453072, 'B_2_5': -0.047407729740897686, 'B_2_6':
0.9735385579776166, 'B_2_7': 0.5354668495288273, 'B_2_8': -0.5731535928875863,
'B_2_9': 0.0028541179218631862, 'B_3_0': -0.5274229679758382, 'B_3_1':
-0.17906936402023166, 'B_3_2': 0.5545663669952303, 'B_3_3': 0.5370251342749992,
'B_3_4': -0.5383612517532774, 'B_3_5': 0.8360021699466231, 'B_3_6':
0.18328500187435703, 'B_3_7': -0.23774990389521122, 'B_3_8':
0.0038324499461040995, 'B_3_9': 0.20804108180673264, 'B_4_0':
0.11858325305465511, 'B_4_1': -0.026918029638367202, 'B_4_2':
```

0.009766952097012171, 'B_4_3': -0.6407044173527854, 'B_4_4': 0.6192111505232651, 'B_4_5': 0.08268543348321034, 'B_4_6': -0.7304618738246879, 'B_4_7': 0.5269910611339004, 'B_4_8': 0.8562390771871546, 'B_4_9': 0.8856676604891849, 'B_5_0': -0.259344889825878, 'B_5_1': -0.721676590349707, 'B_5_2': 0.03643126073346892, 'B_5_3': -0.767033725888543, 'B_5_4': 0.8592950886549822, 'B_5_5': -0.6265123378368109, 'B_5_6': -0.48952359729669515, 'B_5_7': -0.1848596767677585, 'B_5_8': -0.23133435225293097, 'B_5_9': 0.5834325269478202, 'B_6_0': -0.08649269530909454, 'B_6_1': -0.651878845193393, 'B_6_2': -0.4609244069375733, 'B_6_3': 0.01804450267129365, 'B_6_4': -0.6980412434962529, 'B_6_5': 0.22107823777910784, 'B_6_6': 0.13017966697126893, 'B_6_7': 0.4251105864368703, 'B_6_8': 0.866655138451148, 'B_6_9': 0.8162640309802882, 'B_7_0': 0.8832297237762861, 'B_7_1': -0.5054631178930722, 'B_7_2': -0.7295006928540779, 'B_7_3': -0.391492133017032, 'B_7_4': 0.7822686420773028, 'B_7_5': 0.9346658797545067, 'B_7_6': 0.7114634031195426, 'B_7_7': -0.7721649121674636, 'B_7_8': 0.8859074361583905, 'B_7_9': -0.1517511058454437, 'B_8_0': -0.9715656242905397, 'B_8_1': -0.11406039331857665, 'B_8_2': -0.8987830440425741, 'B_8_3': 0.08687828389901275, 'B_8_4': -0.31966302567463833, 'B_8_5': -0.052598831483783415, 'B_8_6': 0.22407731427137167, 'B_8_7': 0.7302261697192842, 'B_8_8': 0.32056512141441873, 'B_8_9': -0.13058324041320224, 'B_9_0': 0.6324671502673229, 'B_9_1': 0.8030526323948017, 'B_9_2': -0.7528752664642495, 'B_9_3': 0.063227295554527, 'B_9_4': -0.4266663197099606, 'B_9_5': -0.8968188282036441, 'B_9_6': 0.3337224563256911, 'B_9_7': 0.7182861393692668, 'B_9_8': -0.6674527441347597, 'B_9_9': 0.618579042095323, 'B_10_0': 0.6997235524031444, 'B_10_1': 0.8203940764796374, 'B_10_2': -0.46473552360961623, 'B_10_3': 0.7998270690759808, 'B_10_4': -0.3550877456218642, 'B_10_5': 0.10362926664189565, 'B_10_6': -0.8899952077520814, 'B_10_7': -0.9031154450298876, 'B_10_8': 0.4363389612097617, 'B_10_9': -0.10992964360397373}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:01:58,502] Trial 66 finished with value: 180.61330295011408 and parameters: {'B_0_0': -0.4825410967689492, 'B_0_1': -0.6129812778706778, 'B_0_2': 0.33650791190125173, 'B_0_3': -0.9892049428298271, 'B_0_4': 0.6056645150097713, 'B_0_5': -0.8330857798627409, 'B_0_6': -0.6108505624819235,

'B_0_7': -0.22525193730780257, 'B_0_8': -0.029979695362114867, 'B_0_9':
-0.2871096706883952, 'B_1_0': -0.3549899133310981, 'B_1_1': 0.9185537039049132,
'B_1_2': 0.28032776886192057, 'B_1_3': -0.9436027483725713, 'B_1_4':
-0.13825170128025988, 'B_1_5': 0.742517225493061, 'B_1_6': -0.3931167509068775,
'B_1_7': -0.8027784373467113, 'B_1_8': 0.5116018477480373, 'B_1_9':
0.20990738723344093, 'B_2_0': 0.1392187654964776, 'B_2_1': -0.8839184119773323,
'B_2_2': -0.7118119606048756, 'B_2_3': 0.24264780450196533, 'B_2_4':
-0.5785087939115051, 'B_2_5': 0.008058757901880154, 'B_2_6': 0.088865738338199,
'B_2_7': -0.42865576326353994, 'B_2_8': 0.8878818079721558, 'B_2_9':
0.6780789315344389, 'B_3_0': 0.440299545112941, 'B_3_1': -0.83557214450937,
'B_3_2': -0.7798625617681415, 'B_3_3': -0.014292935187387723, 'B_3_4':
-0.6717875686018484, 'B_3_5': 0.9469656177883587, 'B_3_6': -0.6761299366257293,
'B_3_7': -0.024813685781331474, 'B_3_8': -0.26904030745381924, 'B_3_9':
0.2759851144791139, 'B_4_0': -0.7345079103989387, 'B_4_1': -0.13797347247606606,
'B_4_2': -0.44608403504185556, 'B_4_3': 0.7677480188499424, 'B_4_4':
-0.9096402899733037, 'B_4_5': 0.8191804381327894, 'B_4_6': 0.6593283943349408,
'B_4_7': -0.4008099814471854, 'B_4_8': 0.5727954962606672, 'B_4_9':
-0.9281971505292799, 'B_5_0': 0.9243937895809193, 'B_5_1': 0.05265798632778296,
'B_5_2': -0.38950356839041833, 'B_5_3': -0.27208523906253934, 'B_5_4':
-0.8585174001482152, 'B_5_5': 0.9689120378032376, 'B_5_6': -0.912201948601629,
'B_5_7': 0.3067161038327608, 'B_5_8': -0.1133828631412126, 'B_5_9':
0.8888541120136559, 'B_6_0': -0.5029205141925456, 'B_6_1': 0.06353191139726233,
'B_6_2': -0.03493470700614909, 'B_6_3': 0.40167555921920073, 'B_6_4':
-0.8553397407342074, 'B_6_5': -0.6062975225537575, 'B_6_6': -0.5651616562161573,
'B_6_7': 0.4715384945865455, 'B_6_8': -0.6155053812730547, 'B_6_9':
0.14551369250413382, 'B_7_0': -0.5638172354081117, 'B_7_1': 0.7160174011662772,
'B_7_2': 0.8755271998894842, 'B_7_3': 0.03661609038563607, 'B_7_4':
-0.292428409265827, 'B_7_5': -0.5265656054653517, 'B_7_6': -0.8914336169843733,
'B_7_7': -0.4583698491289592, 'B_7_8': 0.30538684429172935, 'B_7_9':
0.06656152399223747, 'B_8_0': 0.8792017309962363, 'B_8_1': -0.20537102026643284,
'B_8_2': 0.3838494178102212, 'B_8_3': -0.0211409599925969, 'B_8_4':
0.02490090752993912, 'B_8_5': -0.4598068570838474, 'B_8_6':
-0.08919746185284061, 'B_8_7': -0.05784570591483962, 'B_8_8':
0.4091521930628892, 'B_8_9': 0.8724671983885055, 'B_9_0': -0.3379438705846415,
'B_9_1': -0.9064430722090291, 'B_9_2': -0.007335651508733387, 'B_9_3':
0.08803343765535376, 'B_9_4': -0.6564877087588078, 'B_9_5': 0.5157199204266121,
'B_9_6': 0.9601596347141534, 'B_9_7': 0.8826339396767526, 'B_9_8':
-0.5997300431291687, 'B_9_9': 0.7432192150035062, 'B_10_0': 0.40489740009513575,
'B_10_1': -0.09138380366231513, 'B_10_2': 0.08040976876296346, 'B_10_3':
0.6922083464652671, 'B_10_4': -0.03933281951602108, 'B_10_5':
-0.2510494091718488, 'B_10_6': -0.013369963808548158, 'B_10_7':
0.1566770119893972, 'B_10_8': -0.6156262455188692, 'B_10_9':
0.7937973059286416}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].

```

Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```

[I 2025-04-29 14:02:14,450] Trial 67 finished with value: 110.54252945002796 and
parameters: {'B_0_0': 0.3448901586826354, 'B_0_1': 0.263872456328228, 'B_0_2':
-0.7929974721623365, 'B_0_3': 0.20626439758173176, 'B_0_4': -0.7504978887677851,
'B_0_5': -0.8341786031131099, 'B_0_6': 0.05407557337193869, 'B_0_7':
-0.19248464258793996, 'B_0_8': 0.17584433490932194, 'B_0_9':
-0.8133401072466676, 'B_1_0': -0.1366535356242832, 'B_1_1':
-0.6439223042105873, 'B_1_2': 0.9644581701297508, 'B_1_3': 0.12107742062164517,
'B_1_4': 0.3711249846629172, 'B_1_5': -0.3456718042484366, 'B_1_6':
-0.8024169487368986, 'B_1_7': 0.9422243083114026, 'B_1_8': -0.4232071099412973,
'B_1_9': -0.15085979335632305, 'B_2_0': 0.2967305649710985, 'B_2_1':
-0.9410993505814996, 'B_2_2': 0.6143130784242281, 'B_2_3': 0.07242481625666808,
'B_2_4': -0.4391299696037322, 'B_2_5': -0.8680161891209108, 'B_2_6':
-0.668253137360606, 'B_2_7': 0.7821670405298051, 'B_2_8': 0.3683279633930643,
'B_2_9': -0.4951945911589581, 'B_3_0': -0.8601104854088415, 'B_3_1':
0.37741831852554, 'B_3_2': 0.14336729741617504, 'B_3_3': -0.18905134157269465,
'B_3_4': -0.27004933060367364, 'B_3_5': -0.5607077336603656, 'B_3_6':
0.33583028794428493, 'B_3_7': 0.9319226950318635, 'B_3_8': -0.6420838132430848,
'B_3_9': 0.19229090763069667, 'B_4_0': -0.15011358962296717, 'B_4_1':
-0.10555633094556349, 'B_4_2': -0.4867883853469275, 'B_4_3': 0.6437670538096099,
'B_4_4': -0.02919390378846254, 'B_4_5': 0.08594358735256291, 'B_4_6':
-0.386573392661295, 'B_4_7': -0.6300192379794395, 'B_4_8': -0.8198974313584154,
'B_4_9': 0.42395589903133346, 'B_5_0': -0.6014561222580459, 'B_5_1':
0.37634920129381944, 'B_5_2': -0.2170546368270313, 'B_5_3':
-0.16676589404049014, 'B_5_4': -0.061028557736311995, 'B_5_5':
-0.2619371539074645, 'B_5_6': -0.7907724450992717, 'B_5_7':
-0.37692901619436536, 'B_5_8': 0.32419403417399817, 'B_5_9': 0.3282531149083645,
'B_6_0': -0.4232758716648112, 'B_6_1': -0.07899679941577786, 'B_6_2':
-0.6069744885363637, 'B_6_3': -0.5734632785709252, 'B_6_4': -0.9443667649549173,
'B_6_5': 0.7855581688173343, 'B_6_6': -0.5728008031516857, 'B_6_7':
-0.5394829236658026, 'B_6_8': -0.436389163362639, 'B_6_9': -0.7377320180430506,
'B_7_0': 0.2757405086990983, 'B_7_1': -0.8412169527083786, 'B_7_2':
-0.7242791958738264, 'B_7_3': 0.28661831864838394, 'B_7_4':
-0.31747197144674333, 'B_7_5': 0.41173669686835423, 'B_7_6': 0.1137009944468812,
'B_7_7': -0.4830780773382042, 'B_7_8': 0.29863934196886954, 'B_7_9':
-0.7597781130200962, 'B_8_0': 0.07209255849846019, 'B_8_1':
-0.12444675201083144, 'B_8_2': -0.45961030602146724, 'B_8_3':
0.20592989552421015, 'B_8_4': 0.7352405128337594, 'B_8_5': 0.9878313055308918,
'B_8_6': 0.2133702821119805, 'B_8_7': -0.18868236882824307, 'B_8_8':

```



```

-0.5440899415719208, 'B_8_9': -0.4809497429976788, 'B_9_0': -0.040679143263464,
'B_9_1': -0.2627901514641957, 'B_9_2': 0.8193527547973896, 'B_9_3':
-0.26077497404169514, 'B_9_4': -0.2505797095320377, 'B_9_5':
-0.23060166538072557, 'B_9_6': 0.6665080165279631, 'B_9_7': 0.2985608752768656,
'B_9_8': 0.22605432363140743, 'B_9_9': -0.32839681567995294, 'B_10_0':
0.6294502701282301, 'B_10_1': -0.9140395545157117, 'B_10_2': 0.7341784012294641,
'B_10_3': 0.025244801591980615, 'B_10_4': -0.2014645349704426, 'B_10_5':
-0.005608784178982362, 'B_10_6': 0.3314827504569855, 'B_10_7':
-0.9647244554945632, 'B_10_8': 0.7356726757873304, 'B_10_9':
-0.10545573988344226}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:02:34,280] Trial 68 finished with value: 140.59936325735626 and
parameters: {'B_0_0': -0.31092629099384517, 'B_0_1': -0.7162737800445669,
'B_0_2': -0.11985855877028462, 'B_0_3': -0.007280912856929889, 'B_0_4':
-0.19077954490291393, 'B_0_5': 0.35374698577725616, 'B_0_6':
-0.15662487305673245, 'B_0_7': 0.9136068041362151, 'B_0_8': 0.19656948529071183,
'B_0_9': 0.6088063125907797, 'B_1_0': -0.7734753118432318, 'B_1_1':
-0.2033632399671067, 'B_1_2': -0.3230415511346376, 'B_1_3': 0.13757512851045184,
'B_1_4': 0.7150047697216226, 'B_1_5': -0.8348903288312268, 'B_1_6':
-0.8463018998714726, 'B_1_7': 0.6936290621520458, 'B_1_8': -0.5685568296440523,
'B_1_9': 0.3341314438106311, 'B_2_0': 0.08489870207439432, 'B_2_1':
0.6591293178767144, 'B_2_2': -0.742323958523921, 'B_2_3': -0.7330297278692297,
'B_2_4': 0.11250035335104891, 'B_2_5': 0.8838297336495744, 'B_2_6':
-0.1683044384259066, 'B_2_7': -0.342654641495704, 'B_2_8': 0.21124485993181397,
'B_2_9': 0.8522882290791616, 'B_3_0': 0.6671171090323549, 'B_3_1':
0.9810575492517755, 'B_3_2': -0.9453142052592933, 'B_3_3': -0.7665408734687308,
'B_3_4': 0.21674617146591157, 'B_3_5': 0.6347413498810224, 'B_3_6':
-0.42521620223260226, 'B_3_7': -0.5163620456922469, 'B_3_8': 0.5682120827747776,
'B_3_9': -0.8783639293706015, 'B_4_0': -0.25862235778648146, 'B_4_1':
0.011502260732812797, 'B_4_2': 0.11949037521364048, 'B_4_3': 0.4526580104496982,
'B_4_4': -0.2874054676072967, 'B_4_5': 0.22311500245572224, 'B_4_6':
0.16904333712609354, 'B_4_7': 0.09427987929441928, 'B_4_8': 0.27815692042158013,
'B_4_9': -0.4762906412852089, 'B_5_0': -0.8223954679024346, 'B_5_1':
-0.753842089361308, 'B_5_2': -0.5453268823452759, 'B_5_3':

```

```

0.00039191233511481904, 'B_5_4': 0.6740766460811198, 'B_5_5':
0.3985608173350006, 'B_5_6': 0.5723003963200533, 'B_5_7': -0.29506826462633906,
'B_5_8': -0.1887855915358707, 'B_5_9': -0.21050986573858665, 'B_6_0':
-0.610091094879087, 'B_6_1': -0.8687910168757274, 'B_6_2': 0.18001386379790896,
'B_6_3': 0.9686490331727604, 'B_6_4': -0.6596771841984796, 'B_6_5':
-0.8975640565994379, 'B_6_6': 0.5534727300113873, 'B_6_7': -0.4898797067642461,
'B_6_8': 0.6526947345987981, 'B_6_9': 0.4037946007283404, 'B_7_0':
0.8824336033605931, 'B_7_1': -0.18032407320178945, 'B_7_2': 0.32070028211225554,
'B_7_3': -0.06958753398972406, 'B_7_4': 0.25856562067091615, 'B_7_5':
-0.09979416610468006, 'B_7_6': -0.6165081520803318, 'B_7_7':
0.015636883686694247, 'B_7_8': 0.05659671884613582, 'B_7_9': 0.6149956195551189,
'B_8_0': 0.4663309974712908, 'B_8_1': 0.3243074154040593, 'B_8_2':
-0.7654935508059064, 'B_8_3': 0.4368202555141558, 'B_8_4':
0.0006042095819392967, 'B_8_5': 0.7803170814901952, 'B_8_6': 0.8313444495784748,
'B_8_7': -0.8807899506084648, 'B_8_8': 0.6348702895018679, 'B_8_9':
-0.9153591669631121, 'B_9_0': -0.2630041411238552, 'B_9_1': 0.08375395172973676,
'B_9_2': -0.6967604187182876, 'B_9_3': 0.4877013488867721, 'B_9_4':
-0.9734576857391656, 'B_9_5': -0.6561738841006071, 'B_9_6': 0.1599777231098818,
'B_9_7': 0.42448737283383897, 'B_9_8': 0.824953343333954, 'B_9_9':
0.40688337083055526, 'B_10_0': 0.34012295510463453, 'B_10_1':
-0.20920453022568464, 'B_10_2': -0.9874568374569332, 'B_10_3':
-0.7520846677034685, 'B_10_4': 0.06841799861695752, 'B_10_5':
0.34675717397195105, 'B_10_6': -0.16345239788500432, 'B_10_7':
0.4567444428262126, 'B_10_8': 0.04328734475771401, 'B_10_9':
-0.8431311327232618}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:02:51,710] Trial 69 finished with value: 152.01119324838265 and
parameters: {'B_0_0': -0.6098078230834614, 'B_0_1': 0.673196482023007, 'B_0_2':
0.4453272383163902, 'B_0_3': 0.7816287342847401, 'B_0_4': -0.7572366467405456,
'B_0_5': 0.9706406185378464, 'B_0_6': -0.5753747063791315, 'B_0_7':
-0.9666758449524488, 'B_0_8': 0.9848974193260371, 'B_0_9': -0.43347370681867603,
'B_1_0': 0.33267851597486864, 'B_1_1': -0.6401690179309156, 'B_1_2':
0.15885249375430055, 'B_1_3': 0.7999892731800273, 'B_1_4': -0.3560412884916916,
'B_1_5': 0.33710287256783134, 'B_1_6': -0.6179508941791353, 'B_1_7':

```

0.247094889463076, 'B_1_8': -0.32382830340132984, 'B_1_9': -0.5718076479958557,
 'B_2_0': -0.36989953960680244, 'B_2_1': -0.6358946843545572, 'B_2_2':
 0.7957938935190079, 'B_2_3': -0.20356996509208491, 'B_2_4': -0.5043469761507526,
 'B_2_5': -0.10215247478845291, 'B_2_6': -0.3553383897231388, 'B_2_7':
 0.4839501225503855, 'B_2_8': 0.933015425533279, 'B_2_9': -0.12269335973865925,
 'B_3_0': 0.7148461319450359, 'B_3_1': 0.13040888564804654, 'B_3_2':
 -0.6786733831134177, 'B_3_3': 0.3774330293210564, 'B_3_4': 0.026600304309123857,
 'B_3_5': 0.29820091603916477, 'B_3_6': 0.7715810768918967, 'B_3_7':
 -0.11793955010102875, 'B_3_8': 0.7943316939703642, 'B_3_9': 0.49632464827767664,
 'B_4_0': -0.022562246761590132, 'B_4_1': 0.20437982860276804, 'B_4_2':
 -0.3453493576123573, 'B_4_3': 0.957522653482773, 'B_4_4': 0.331355302839784,
 'B_4_5': -0.9031832320566728, 'B_4_6': -0.34328862661894266, 'B_4_7':
 0.2441625788429338, 'B_4_8': -0.8784601099575052, 'B_4_9': -0.9427093011880974,
 'B_5_0': -0.09895200573495089, 'B_5_1': -0.33127344420851335, 'B_5_2':
 -0.12216875609433153, 'B_5_3': -0.4698672464375898, 'B_5_4':
 -0.6638588610528977, 'B_5_5': 0.37450384682106863, 'B_5_6':
 0.028850810210244937, 'B_5_7': -0.5936946617262537, 'B_5_8':
 -0.3572129360052033, 'B_5_9': 0.1817796571733854, 'B_6_0': -0.8469858904204939,
 'B_6_1': -0.6980430416529645, 'B_6_2': -0.8449737038673676, 'B_6_3':
 0.6183883187739554, 'B_6_4': 0.4997702817597782, 'B_6_5': 0.501559605120929,
 'B_6_6': 0.4283241800825355, 'B_6_7': 0.3353831388663411, 'B_6_8':
 -0.045556007551213584, 'B_6_9': 0.28171076374408766, 'B_7_0':
 -0.9140441215198102, 'B_7_1': 0.24049955686043356, 'B_7_2': 0.5501978569630168,
 'B_7_3': 0.0007095535610925907, 'B_7_4': -0.2821236097140709, 'B_7_5':
 -0.4406950650940613, 'B_7_6': 0.2827480593209233, 'B_7_7': 0.3817463732872193,
 'B_7_8': 0.7345158477278342, 'B_7_9': -0.3121785167272384, 'B_8_0':
 0.5341491686234594, 'B_8_1': 0.12148453189488917, 'B_8_2': 0.7580162515425966,
 'B_8_3': 0.05506797442532907, 'B_8_4': 0.03490706022797507, 'B_8_5':
 0.15419662125299993, 'B_8_6': 0.4936444031820202, 'B_8_7': 0.9200871480803525,
 'B_8_8': -0.844344558652784, 'B_8_9': -0.884199487265786, 'B_9_0':
 0.8663496680665301, 'B_9_1': 0.059153738526048105, 'B_9_2': 0.22750796884253344,
 'B_9_3': -0.03480126884243684, 'B_9_4': -0.8045582438372005, 'B_9_5':
 0.5308169692689597, 'B_9_6': 0.4895047972754021, 'B_9_7': -0.37282742344423225,
 'B_9_8': -0.3836462944477672, 'B_9_9': 0.6689259341929266, 'B_10_0':
 -0.8467326296721733, 'B_10_1': 0.7545881937460004, 'B_10_2':
 -0.48180985400608944, 'B_10_3': 0.1841098005558779, 'B_10_4': 0.86326847442352,
 'B_10_5': 0.3526825530425599, 'B_10_6': -0.16942897131526757, 'B_10_7':
 0.375277733632722, 'B_10_8': -0.30892032009358705, 'B_10_9':
 -0.44102833562861143}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].

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    Built F_[u_9].
    Built F_[u_10].
    Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:03:07,982] Trial 70 finished with value: 245.27009659385044 and
parameters: {'B_0_0': 0.9834328903106657, 'B_0_1': 0.4965646577359797, 'B_0_2':
0.286545555215999, 'B_0_3': -0.28652650642574884, 'B_0_4': -0.8926615582730046,
'B_0_5': -0.9285073854503919, 'B_0_6': 0.10727888578862155, 'B_0_7':
0.5188247876573211, 'B_0_8': 0.07401079707883307, 'B_0_9': 0.6747544424394467,
'B_1_0': -0.9238154644754212, 'B_1_1': -0.2638308886766386, 'B_1_2':
-0.5206006254751274, 'B_1_3': -0.10439952304889832, 'B_1_4':
-0.8466947414756375, 'B_1_5': 0.8991806503107236, 'B_1_6': -0.5479543058826654,
'B_1_7': 0.8361693954574978, 'B_1_8': -0.14673155302737007, 'B_1_9':
0.9613377838453316, 'B_2_0': 0.1070355992512173, 'B_2_1': 0.8530490710680987,
'B_2_2': -0.9794090023854387, 'B_2_3': 0.4915128362381822, 'B_2_4':
0.7980402931429804, 'B_2_5': -0.7133532773816709, 'B_2_6': -0.7996708135703401,
'B_2_7': -0.149346237771808, 'B_2_8': 0.8586842341432115, 'B_2_9':
0.0839253891770686, 'B_3_0': 0.0543637981690297, 'B_3_1': -0.7105702857673427,
'B_3_2': -0.9884367680354268, 'B_3_3': 0.6140773796760133, 'B_3_4':
-0.8166023652872483, 'B_3_5': 0.26502264109594154, 'B_3_6': 0.8209405799400791,
'B_3_7': -0.5768692119463066, 'B_3_8': 0.32724021574908346, 'B_3_9':
-0.8341768181191312, 'B_4_0': 0.20543518479847367, 'B_4_1': -0.6654151925126586,
'B_4_2': 0.46718804982005246, 'B_4_3': -0.39356203630806175, 'B_4_4':
-0.07248580256680048, 'B_4_5': 0.19687247321469092, 'B_4_6':
-0.6508786468808456, 'B_4_7': 0.47844288086288933, 'B_4_8':
-0.39077212553910723, 'B_4_9': 0.32149628262725183, 'B_5_0':
-0.6765995596152401, 'B_5_1': 0.3724729652285734, 'B_5_2': -0.7361876158352858,
'B_5_3': -0.14779413738543246, 'B_5_4': -0.5005319739970239, 'B_5_5':
0.49178299386780155, 'B_5_6': -0.4711706910763884, 'B_5_7': 0.7558002666628845,
'B_5_8': 0.7758040185850934, 'B_5_9': -0.601046195858884, 'B_6_0':
0.22541497111529396, 'B_6_1': -0.8163190693944706, 'B_6_2': -0.2147565427887742,
'B_6_3': 0.4223770244125191, 'B_6_4': 0.3156580287628341, 'B_6_5':
0.24690732542508376, 'B_6_6': -0.43056908994836174, 'B_6_7': -0.241510600119884,
'B_6_8': -0.7861666334934334, 'B_6_9': 0.43366518445853, 'B_7_0':
0.007699817646721563, 'B_7_1': 0.4133559520443961, 'B_7_2':
-0.49709182090219195, 'B_7_3': -0.34761230730256887, 'B_7_4':
-0.32825363222497295, 'B_7_5': -0.9390300287005119, 'B_7_6':
-0.3147252117953654, 'B_7_7': 0.6005430555657063, 'B_7_8': 0.7866510215844822,
'B_7_9': 0.03607226495913629, 'B_8_0': -0.08810664346594388, 'B_8_1':
-0.9289672549752469, 'B_8_2': 0.7508245884190576, 'B_8_3': -0.964239777416772,
'B_8_4': 0.8806051588162194, 'B_8_5': -0.021351827597580475, 'B_8_6':
-0.3129697027220413, 'B_8_7': 0.839952500988467, 'B_8_8': -0.9417058336420084,
'B_8_9': 0.16122129184079248, 'B_9_0': 0.09688639850868852, 'B_9_1':
-0.5557456056014635, 'B_9_2': -0.14080288592658308, 'B_9_3': 0.301482731768665,
'B_9_4': 0.44971976982621653, 'B_9_5': -0.988580619760972, 'B_9_6':
0.8649697915802104, 'B_9_7': -0.6642397626622314, 'B_9_8': -0.5275107402397288,

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'B_9_9': -0.46832257752618045, 'B_10_0': -0.9151161296269701, 'B_10_1':
-0.7463567952911567, 'B_10_2': 0.9629728474294408, 'B_10_3':
-0.36587085999297364, 'B_10_4': -0.7617038411286823, 'B_10_5':
-0.7781573028742458, 'B_10_6': 0.43453073598654424, 'B_10_7': 0.702045954784629,
'B_10_8': -0.4812344469732297, 'B_10_9': -0.9485991655762347}. Best is trial 36
with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:03:25,967] Trial 71 finished with value: 140.74207798134668 and
parameters: {'B_0_0': -0.5482817634781956, 'B_0_1': -0.8780989322224204,
'B_0_2': -0.7852780254725178, 'B_0_3': -0.13561392270166805, 'B_0_4':
-0.5347616256939675, 'B_0_5': 0.03157283948033829, 'B_0_6':
-0.48808236855076303, 'B_0_7': 0.8722638561838363, 'B_0_8': 0.8037690564940037,
'B_0_9': 0.13227608792072276, 'B_1_0': -0.28824049586497535, 'B_1_1':
-0.23532917357122363, 'B_1_2': -0.17453247937389604, 'B_1_3':
0.45060705782718924, 'B_1_4': -0.09575478887270095, 'B_1_5': 0.7063252940816769,
'B_1_6': 0.8793076761421219, 'B_1_7': -0.324826616309711, 'B_1_8':
0.12017507312384668, 'B_1_9': -0.8017880570715225, 'B_2_0': -0.5930712854226092,
'B_2_1': -0.17133778191388294, 'B_2_2': -0.5817186225144779, 'B_2_3':
0.7455897717616609, 'B_2_4': -0.6765932130029858, 'B_2_5': -0.08779157332487797,
'B_2_6': 0.20928449611055377, 'B_2_7': 0.37989053815847185, 'B_2_8':
-0.23152264794151667, 'B_2_9': 0.7308161560749624, 'B_3_0': 0.1352719191342926,
'B_3_1': -0.11977459471384744, 'B_3_2': -0.8568635445769675, 'B_3_3':
-0.06991485247857332, 'B_3_4': 0.6350569648151458, 'B_3_5': 0.22679121607820463,
'B_3_6': 0.6576816583836542, 'B_3_7': -0.011856892213374826, 'B_3_8':
-0.7554458239129259, 'B_3_9': -0.9629931052449555, 'B_4_0':
-0.38962529749718944, 'B_4_1': 0.3920211335656292, 'B_4_2': 0.5747146098045537,
'B_4_3': -0.4044093874517477, 'B_4_4': -0.008605236651170745, 'B_4_5':
-0.15618456922051838, 'B_4_6': -0.17716488698417288, 'B_4_7':
-0.5319295478637021, 'B_4_8': -0.7199852120416724, 'B_4_9': -0.6005566547021981,
'B_5_0': -0.8804418734165748, 'B_5_1': -0.5977011311893174, 'B_5_2':
-0.9717822227017486, 'B_5_3': 0.251971633259237, 'B_5_4': -0.36357669337238785,
'B_5_5': 0.936826528625279, 'B_5_6': -0.6064402394629302, 'B_5_7':
-0.305259430787004, 'B_5_8': -0.2814745058897642, 'B_5_9': -0.9732660377963281,
'B_6_0': 0.6716317470181288, 'B_6_1': -0.926650968914237, 'B_6_2':

0.871556072763898, 'B_6_3': 0.7763072151653689, 'B_6_4': 0.07033339385020776,
 'B_6_5': 0.08631923832435828, 'B_6_6': 0.30448028745318956, 'B_6_7':
 -0.22364097636892377, 'B_6_8': 0.2915001211277539, 'B_6_9': -0.901358590396796,
 'B_7_0': -0.8829856978475414, 'B_7_1': -0.08468376915330211, 'B_7_2':
 0.0982521561899492, 'B_7_3': 0.7678789278649216, 'B_7_4': -0.7827719448623653,
 'B_7_5': 0.638842895843901, 'B_7_6': -0.5741062721617022, 'B_7_7':
 -0.20971437595013687, 'B_7_8': -0.3631271790417224, 'B_7_9': -0.19769255972651,
 'B_8_0': -0.6920189138162807, 'B_8_1': 0.007068747319038837, 'B_8_2':
 0.5972523278661073, 'B_8_3': -0.11576265944978403, 'B_8_4':
 -0.018585119721565846, 'B_8_5': 0.6905598188570021, 'B_8_6': 0.407995066983301,
 'B_8_7': -0.018900271130826263, 'B_8_8': -0.4797626165854243, 'B_8_9':
 0.7621549587530339, 'B_9_0': 0.15250039534730564, 'B_9_1': -0.5521277319808271,
 'B_9_2': -0.45714288977673945, 'B_9_3': 0.6002035160937669, 'B_9_4':
 -0.9025917870668382, 'B_9_5': 0.9743425303170636, 'B_9_6': -0.7319036741647122,
 'B_9_7': 0.586215680285936, 'B_9_8': -0.20834487279206793, 'B_9_9':
 -0.015411957090815664, 'B_10_0': 0.7536881433798641, 'B_10_1':
 -0.1724358446133578, 'B_10_2': 0.07423763227300428, 'B_10_3':
 -0.8923866684298435, 'B_10_4': -0.9853520017424428, 'B_10_5':
 -0.5052962514565826, 'B_10_6': 0.3575545239791593, 'B_10_7':
 -0.7498955111489093, 'B_10_8': -0.7647513215669921, 'B_10_9':
 -0.1762979960889115}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [⟨F_[u_i], F_[u_j]⟩].

Built matrix B = inverse(A).

[I 2025-04-29 14:03:44,871] Trial 72 finished with value: 211.46008500843328 and
 parameters: {'B_0_0': 0.29271733644428855, 'B_0_1': 0.4199025446509024, 'B_0_2':
 -0.6366676652852179, 'B_0_3': -0.832415468203356, 'B_0_4': 0.8719861272121774,
 'B_0_5': -0.4738805625270386, 'B_0_6': 0.6996260652652733, 'B_0_7':
 0.7197130719163802, 'B_0_8': 0.539594731689838, 'B_0_9': -0.6557094588279113,
 'B_1_0': 0.6371668893233167, 'B_1_1': 0.5331161072983417, 'B_1_2':
 0.19151152041030994, 'B_1_3': -0.32233997036547357, 'B_1_4':
 -0.47023296355224264, 'B_1_5': -0.7486593600808893, 'B_1_6':
 0.01931899246372537, 'B_1_7': 0.168816533509496, 'B_1_8': -0.27850879877874135,
 'B_1_9': -0.048789045991880275, 'B_2_0': -0.270241374408852, 'B_2_1':
 0.27919199349498847, 'B_2_2': 0.7944932314371953, 'B_2_3': -0.6654378811470492,
 'B_2_4': -0.23481055639181558, 'B_2_5': -0.7110801641772946, 'B_2_6':

0.24738866540750726, 'B_2_7': 0.45316360878741957, 'B_2_8': 0.8551872105429625, 'B_2_9': -0.4378335860032895, 'B_3_0': 0.8914181876135518, 'B_3_1': -0.8647399097950639, 'B_3_2': 0.2039674494618846, 'B_3_3': -0.8768772157002066, 'B_3_4': 0.7796728917686508, 'B_3_5': 0.05902193299870717, 'B_3_6': 0.3353097345103362, 'B_3_7': 0.656516373302696, 'B_3_8': -0.6923849674817418, 'B_3_9': -0.913588707362495, 'B_4_0': 0.7501846689074472, 'B_4_1': -0.3664322096313136, 'B_4_2': -0.7346217623042128, 'B_4_3': -0.062446750567568654, 'B_4_4': 0.1946460298781172, 'B_4_5': -0.9286782052234299, 'B_4_6': -0.44570659665096635, 'B_4_7': -0.9250562489317459, 'B_4_8': -0.15773592939291203, 'B_4_9': 0.7822980575511638, 'B_5_0': 0.955651774757267, 'B_5_1': 0.2713541559371824, 'B_5_2': -0.6699573663065082, 'B_5_3': 0.8035817921986255, 'B_5_4': 0.03217068993611982, 'B_5_5': -0.6855763620362222, 'B_5_6': 0.8350188055142895, 'B_5_7': 0.509788720431744, 'B_5_8': 0.45877643999184015, 'B_5_9': 0.41417779351292117, 'B_6_0': -0.5933271542290441, 'B_6_1': -0.5722067504190984, 'B_6_2': 0.37282013688596494, 'B_6_3': -0.5549489547814141, 'B_6_4': -0.1214300403299633, 'B_6_5': 0.5583636146760038, 'B_6_6': 0.5622259858337251, 'B_6_7': -0.0925416903987979, 'B_6_8': 0.5785360777849853, 'B_6_9': 0.6704711564721524, 'B_7_0': 0.3064559875428352, 'B_7_1': -0.5377168124293281, 'B_7_2': -0.4474566748366159, 'B_7_3': -0.36661451229070396, 'B_7_4': -0.758727866686775, 'B_7_5': -0.44914395166750865, 'B_7_6': -0.5751519358611864, 'B_7_7': -0.08844907537896773, 'B_7_8': 0.8087063447870846, 'B_7_9': -0.783128972827163, 'B_8_0': 0.4361304894158433, 'B_8_1': 0.37082034873799286, 'B_8_2': -0.8004066866465518, 'B_8_3': 0.8366933617134842, 'B_8_4': 0.13557496027050275, 'B_8_5': -0.26982346815134517, 'B_8_6': 0.5079463953207073, 'B_8_7': -0.4804163923647645, 'B_8_8': 0.38314947735171434, 'B_8_9': -0.9113718090069505, 'B_9_0': 0.6720279144663586, 'B_9_1': -0.10262172289778937, 'B_9_2': 0.2706519704287247, 'B_9_3': -0.5519491167827664, 'B_9_4': 0.35060595106372694, 'B_9_5': 0.5390357462686877, 'B_9_6': -0.05303788996889047, 'B_9_7': 0.23309930964235281, 'B_9_8': -0.08647467083165084, 'B_9_9': 0.7803009111305341, 'B_10_0': -0.00421902258164808, 'B_10_1': 0.6767812438770613, 'B_10_2': 0.5966770753750923, 'B_10_3': -0.17720045373025406, 'B_10_4': -0.5344327061221994, 'B_10_5': -0.18595737753165342, 'B_10_6': 0.6459104040349359, 'B_10_7': -0.539287579077603, 'B_10_8': 0.048383964798343726, 'B_10_9': 0.60607156096228}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.
 Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].
 Built matrix B = inverse(A).
 [I 2025-04-29 14:03:59,749] Trial 73 finished with value: 176.11998493264758 and
 parameters: {'B_0_0': -0.29011459924580996, 'B_0_1': 0.2815750812653408,
 'B_0_2': -0.5036494833833473, 'B_0_3': -0.7736266981710433, 'B_0_4':
 0.5828807656838555, 'B_0_5': -0.7940421256507605, 'B_0_6': -0.4526648377424225,
 'B_0_7': 0.7031166555549315, 'B_0_8': -0.3039908115842389, 'B_0_9':
 0.9366297920380391, 'B_1_0': 0.9498739828522422, 'B_1_1': -0.4588134795019577,
 'B_1_2': -0.9088631199452168, 'B_1_3': 0.4060253027266507, 'B_1_4':
 -0.0970467247562794, 'B_1_5': -0.790509368960039, 'B_1_6': 0.4108951815765247,
 'B_1_7': -0.9282736799583959, 'B_1_8': 0.7329894471746394, 'B_1_9':
 -0.7154679923556586, 'B_2_0': -0.7379224800814845, 'B_2_1': -0.6421688500222027,
 'B_2_2': 0.9113273898142917, 'B_2_3': 0.3824753333026152, 'B_2_4':
 0.5967156402180007, 'B_2_5': -0.202682728249149, 'B_2_6': -0.324100799540188,
 'B_2_7': -0.5257216665693235, 'B_2_8': -0.03831690394143028, 'B_2_9':
 0.1026208923976537, 'B_3_0': 0.2609141904324557, 'B_3_1': -0.48090536208258483,
 'B_3_2': 0.8883350801879215, 'B_3_3': -0.6538781735498711, 'B_3_4':
 0.9219932860374964, 'B_3_5': -0.9605848774351425, 'B_3_6': -0.03737113020264571,
 'B_3_7': -0.32993168052920563, 'B_3_8': -0.10073564041810346, 'B_3_9':
 0.1399158663819624, 'B_4_0': -0.12753346713483615, 'B_4_1':
 -0.16860323640284203, 'B_4_2': 0.460792362518347, 'B_4_3': 0.3680350246712001,
 'B_4_4': 0.6598771384309787, 'B_4_5': -0.4950812171457988, 'B_4_6':
 0.21686644931304588, 'B_4_7': 0.1686856750778507, 'B_4_8': -0.25791913447207626,
 'B_4_9': -0.02760606703269175, 'B_5_0': -0.3687113562546571, 'B_5_1':
 -0.7091480927183693, 'B_5_2': -0.566673602887789, 'B_5_3': -0.7810829801256218,
 'B_5_4': 0.36247775887644584, 'B_5_5': 0.31824023835612003, 'B_5_6':
 0.8922006157281481, 'B_5_7': 0.4547632590215378, 'B_5_8': -0.8869492392628597,
 'B_5_9': 0.7608054443793293, 'B_6_0': 0.718096822632537, 'B_6_1':
 -0.11222576487005664, 'B_6_2': 0.1700088400452846, 'B_6_3': 0.9426350982955873,
 'B_6_4': 0.06396264041936917, 'B_6_5': 0.8025372201846577, 'B_6_6':
 -0.8255832595358308, 'B_6_7': -0.6117172999568079, 'B_6_8': -0.1173052119861907,
 'B_6_9': 0.1678184805078633, 'B_7_0': 0.6261311466883444, 'B_7_1':
 0.9667589879835099, 'B_7_2': -0.2200740369812495, 'B_7_3': -0.8961170404426319,
 'B_7_4': 0.5963195765247431, 'B_7_5': 0.20667382378396004, 'B_7_6':
 -0.5968851394409607, 'B_7_7': -0.010239655889588084, 'B_7_8':
 0.4797742861567742, 'B_7_9': 0.5354153383959146, 'B_8_0':
 -0.0007555442424512782, 'B_8_1': -0.38567523034485895, 'B_8_2':
 0.9380103124362171, 'B_8_3': 0.32345796969385576, 'B_8_4': -0.18168971169432857,
 'B_8_5': -0.4346828160747882, 'B_8_6': 0.45758390634705304, 'B_8_7':
 -0.51047677772434, 'B_8_8': -0.6500142218320974, 'B_8_9': 0.059071100144899136,
 'B_9_0': -0.4158249625645991, 'B_9_1': 0.2552321869487577, 'B_9_2':
 -0.7564814237490177, 'B_9_3': 0.3067833870432968, 'B_9_4': -0.42907287456454957,
 'B_9_5': -0.9895189898196648, 'B_9_6': -0.7386666058051308, 'B_9_7':
 0.8585584481134863, 'B_9_8': 0.7273621854484085, 'B_9_9': -0.527454116568251,
 'B_10_0': 0.5511763710000726, 'B_10_1': 0.21497175448446892, 'B_10_2':
 0.9323913286707415, 'B_10_3': -0.25026079412256297, 'B_10_4':
 0.35929201278399914, 'B_10_5': -0.8641992696006412, 'B_10_6':

-0.20543025330351627, 'B_10_7': 0.9183589691525973, 'B_10_8':
0.07718213141123864, 'B_10_9': -0.4199985833970077}. Best is trial 36 with
value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:04:16,880] Trial 74 finished with value: 196.83573966939025 and
parameters: {'B_0_0': -0.7518634930053585, 'B_0_1': 0.40298611355943015,
'B_0_2': 0.286814965990859, 'B_0_3': -0.2496013470645263, 'B_0_4':
0.892863876694352, 'B_0_5': -0.9460384302840563, 'B_0_6': -0.3886267737689849,
'B_0_7': 0.8337175828731711, 'B_0_8': -0.35454783212750807, 'B_0_9':
-0.49053938896676264, 'B_1_0': 0.06775641166386959, 'B_1_1': 0.641127464842665,
'B_1_2': -0.7857694134608451, 'B_1_3': 0.8771810050985576, 'B_1_4':
0.5842443047387047, 'B_1_5': -0.6311244940079271, 'B_1_6': 0.4197757631723622,
'B_1_7': -0.22778172917763684, 'B_1_8': -0.19345653740888902, 'B_1_9':
0.5609143830289554, 'B_2_0': 0.9163139198018497, 'B_2_1': -0.739221371762556,
'B_2_2': -0.15715213097993763, 'B_2_3': -0.037735615993079064, 'B_2_4':
-0.24490260242910478, 'B_2_5': -0.2708855350418744, 'B_2_6':
-0.8865080382280388, 'B_2_7': 0.36736646871399414, 'B_2_8':
-0.12629545410231013, 'B_2_9': -0.604308270765025, 'B_3_0': 0.6257463345580954,
'B_3_1': 0.5990702813402233, 'B_3_2': -0.026944576383990282, 'B_3_3':
-0.3990676909274341, 'B_3_4': 0.2734235970769756, 'B_3_5': -0.41945877116074193,
'B_3_6': -0.5227042979590502, 'B_3_7': 0.4564190149721268, 'B_3_8':
-0.9101560253525176, 'B_3_9': -0.802358427351504, 'B_4_0': -0.9765594154332923,
'B_4_1': -0.6871724907786563, 'B_4_2': -0.8533997827532902, 'B_4_3':
0.9816886882042388, 'B_4_4': -0.4181562594909026, 'B_4_5': -0.6944493349148904,
'B_4_6': -0.5128274091680824, 'B_4_7': 0.8635286316494666, 'B_4_8':
0.039917241588303565, 'B_4_9': -0.26766016931044334, 'B_5_0':
0.7393150866602543, 'B_5_1': -0.20791926431291574, 'B_5_2': -0.3266830314837623,
'B_5_3': -0.5795806053515236, 'B_5_4': -0.2419622886593652, 'B_5_5':
-0.3629649750864865, 'B_5_6': 0.3510618394103193, 'B_5_7': -0.4166262447519603,
'B_5_8': 0.15285549791667163, 'B_5_9': -0.859533513236694, 'B_6_0':
-0.6112474697409016, 'B_6_1': -0.18333756220952946, 'B_6_2': 0.865159522739217,
'B_6_3': -0.24304106828292305, 'B_6_4': 0.3248667599756716, 'B_6_5':
-0.48492729960563163, 'B_6_6': -0.7384735195730012, 'B_6_7':
-0.8740975318415007, 'B_6_8': -0.3852812728853523, 'B_6_9': -0.6778311669421222,

```

'B_7_0': -0.6454593129280741, 'B_7_1': 0.5163034306061611, 'B_7_2':
-0.958640619529149, 'B_7_3': 0.11446404895616014, 'B_7_4': -0.23866646642493972,
'B_7_5': 0.012968999818923521, 'B_7_6': 0.12440386968491368, 'B_7_7':
-0.3695319649745892, 'B_7_8': -0.4056844568351151, 'B_7_9': 0.4621356616831134,
'B_8_0': 0.7351284364798911, 'B_8_1': -0.6086883677321148, 'B_8_2':
-0.7967148541404051, 'B_8_3': 0.7876356966562112, 'B_8_4': -0.29720698197815065,
'B_8_5': 0.10160843351237192, 'B_8_6': -0.2781045738503931, 'B_8_7':
-0.7409113740348183, 'B_8_8': 0.17687665699650146, 'B_8_9': 0.04977888650386664,
'B_9_0': -0.5880727528014296, 'B_9_1': -0.09529929158848982, 'B_9_2':
0.5879560579509788, 'B_9_3': 0.17904368415362493, 'B_9_4': -0.7981036924073501,
'B_9_5': 0.09508256451368746, 'B_9_6': 0.40142183259338204, 'B_9_7':
0.5795527434458878, 'B_9_8': 0.25565280569584825, 'B_9_9': -0.6877368342092482,
'B_10_0': 0.41614306784851807, 'B_10_1': 0.3511368552414025, 'B_10_2':
-0.9416347717775462, 'B_10_3': -0.17473961817131856, 'B_10_4':
-0.3307585110145008, 'B_10_5': 0.33543588722221385, 'B_10_6':
-0.562051091624377, 'B_10_7': 0.17992640748903188, 'B_10_8': 0.8181478043920611,
'B_10_9': 0.5896486969730432}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:04:33,551] Trial 75 finished with value: 150.17595133871356 and
parameters: {'B_0_0': 0.83448631770873, 'B_0_1': 0.9474418145267263, 'B_0_2':
-0.9688731485277977, 'B_0_3': 0.2430277272274055, 'B_0_4': -0.17257799243736527,
'B_0_5': 0.712511890863541, 'B_0_6': -0.49140689002191923, 'B_0_7':
0.9524383884478507, 'B_0_8': 0.22514177969661953, 'B_0_9': 0.5812040564130854,
'B_1_0': -0.8205128854109187, 'B_1_1': -0.6267533881774472, 'B_1_2':
0.5116392726077224, 'B_1_3': -0.9464666863621393, 'B_1_4': 0.62741841206969,
'B_1_5': -0.9623809429900153, 'B_1_6': 0.31109530134132113, 'B_1_7':
-0.2671140379410244, 'B_1_8': 0.22075980838401432, 'B_1_9': 0.38814218583844506,
'B_2_0': -0.3118990660111177, 'B_2_1': -0.7762343527405456, 'B_2_2':
-0.852101181858434, 'B_2_3': 0.17609332467384475, 'B_2_4': 0.040897600393540845,
'B_2_5': 0.47955799805984234, 'B_2_6': 0.01927630980779549, 'B_2_7':
-0.15355733573885932, 'B_2_8': -0.20178738709226296, 'B_2_9':
-0.9376351328566184, 'B_3_0': -0.32559674064061883, 'B_3_1': 0.9425199118984986,
'B_3_2': 0.7396400409296358, 'B_3_3': -0.337831427272249, 'B_3_4':
0.17383490203141694, 'B_3_5': 0.687647556160278, 'B_3_6': -0.11627681323486405,

```

'B_3_7': -0.9544434297747011, 'B_3_8': -0.8424692855990926, 'B_3_9': 0.020577588270439096, 'B_4_0': 0.929489963838239, 'B_4_1': 0.38255489359335826, 'B_4_2': -0.8495859409516323, 'B_4_3': -0.6150654659321118, 'B_4_4': 0.537219823569363, 'B_4_5': 0.6760507247133647, 'B_4_6': 0.23303528065340995, 'B_4_7': 0.2859578758670369, 'B_4_8': -0.9338918482408984, 'B_4_9': 0.6504899139713869, 'B_5_0': 0.5707849121782427, 'B_5_1': -0.47711041467537263, 'B_5_2': 0.4131799055317038, 'B_5_3': -0.5645878792785037, 'B_5_4': -0.9101462811180043, 'B_5_5': -0.5917496456710529, 'B_5_6': -0.636314824026487, 'B_5_7': -0.3212589664650025, 'B_5_8': -0.161469995592593, 'B_5_9': 0.01690700516546162, 'B_6_0': -0.04602732649351027, 'B_6_1': -0.454064280188727, 'B_6_2': -0.9322420762233325, 'B_6_3': 0.20810121202955845, 'B_6_4': -0.20642523690223835, 'B_6_5': -0.3986291283726515, 'B_6_6': 0.7265432648044143, 'B_6_7': 0.4885517494403564, 'B_6_8': 0.7083757857774355, 'B_6_9': -0.009934066469327263, 'B_7_0': -0.7307561207889806, 'B_7_1': 0.38851844352011944, 'B_7_2': -0.581973915724393, 'B_7_3': -0.1163652551984008, 'B_7_4': -0.8612984378812008, 'B_7_5': 0.6441106367975051, 'B_7_6': -0.7406375307259488, 'B_7_7': 0.7981330639576332, 'B_7_8': 0.721176592477883, 'B_7_9': -0.07012612641029281, 'B_8_0': 0.4838490445219015, 'B_8_1': 0.2365012461599434, 'B_8_2': 0.6000897718287348, 'B_8_3': -0.13502590044389418, 'B_8_4': 0.5659007988038129, 'B_8_5': -0.07433550221100704, 'B_8_6': 0.8731122599287318, 'B_8_7': -0.026983240366207806, 'B_8_8': -0.07401384157121549, 'B_8_9': 0.5644422858957552, 'B_9_0': 0.636723829686459, 'B_9_1': 0.10788844635591821, 'B_9_2': -0.6340003292797507, 'B_9_3': -0.06383870270399261, 'B_9_4': 0.9267835557611632, 'B_9_5': 0.5009126622175866, 'B_9_6': 0.6161019556806817, 'B_9_7': -0.9250055123379045, 'B_9_8': -0.06363826398436345, 'B_9_9': 0.29030984464195964, 'B_10_0': 0.6534647135835878, 'B_10_1': -0.8924381659061125, 'B_10_2': 0.6978289935709088, 'B_10_3': 0.0958338547489852, 'B_10_4': -0.5993183382594014, 'B_10_5': -0.361834671429594, 'B_10_6': -0.43177948753190964, 'B_10_7': 0.061100887891182065, 'B_10_8': -0.887662353775959, 'B_10_9': -0.6760984061701514}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$\langle F_{[u_i]}, F_{[u_j]} \rangle$].

Built matrix B = inverse(A).

[I 2025-04-29 14:04:48,606] Trial 76 finished with value: 110.32261350577446 and parameters: {'B_0_0': 0.6191322258840493, 'B_0_1': -0.7274440983213141, 'B_0_2':

-0.5278753864729087, 'B_0_3': -0.09232620090376853, 'B_0_4':
 0.03389935846439429, 'B_0_5': 0.521948749636034, 'B_0_6': -0.10428179520094816,
 'B_0_7': 0.5739319496503668, 'B_0_8': 0.23432568795000575, 'B_0_9':
 -0.7660625159408315, 'B_1_0': -0.6804637519821797, 'B_1_1':
 -0.12467322360945376, 'B_1_2': -0.2041590016837016, 'B_1_3':
 0.48221765388184323, 'B_1_4': 0.15985953679875675, 'B_1_5': -0.2214964996155464,
 'B_1_6': -0.6518946387682365, 'B_1_7': -0.5408625373955196, 'B_1_8':
 0.14949201656873679, 'B_1_9': -0.15159116629841096, 'B_2_0':
 0.02955646754413066, 'B_2_1': -0.6520586781015774, 'B_2_2': 0.3225543796422403,
 'B_2_3': 0.6980267068948147, 'B_2_4': -0.4792797765619369, 'B_2_5':
 -0.016801832852983423, 'B_2_6': 0.9415441114613856, 'B_2_7':
 -0.33018135370955926, 'B_2_8': 0.18819176294904705, 'B_2_9':
 -0.8114125136035162, 'B_3_0': 0.2099097049092975, 'B_3_1': -0.8125290888568565,
 'B_3_2': 0.9092861619411676, 'B_3_3': 0.4270120511031765, 'B_3_4':
 0.5705563805086986, 'B_3_5': 0.4041266343278409, 'B_3_6': 0.17766349143140214,
 'B_3_7': -0.09917269345069935, 'B_3_8': 0.7095792919269035, 'B_3_9':
 0.9045346217352648, 'B_4_0': -0.0019083776499190241, 'B_4_1':
 0.8544393973274187, 'B_4_2': 0.1649991617772646, 'B_4_3': 0.0760006900549377,
 'B_4_4': -0.09012418874839379, 'B_4_5': -0.5756102321703627, 'B_4_6':
 -0.9826350715802428, 'B_4_7': -0.5118176708249986, 'B_4_8':
 0.019251879775058933, 'B_4_9': -0.23767929452568992, 'B_5_0':
 -0.10199924855478504, 'B_5_1': 0.9516834747368181, 'B_5_2': -0.2964607060747314,
 'B_5_3': -0.09467778006698635, 'B_5_4': -0.7256392083214173, 'B_5_5':
 -0.8950436897747645, 'B_5_6': 0.5453108388931394, 'B_5_7': 0.1399526866425176,
 'B_5_8': 0.516195515966912, 'B_5_9': -0.6932716582046278, 'B_6_0':
 0.14222096694172803, 'B_6_1': -0.4877879008040764, 'B_6_2': -0.5384376025379606,
 'B_6_3': -0.9671784921750892, 'B_6_4': -0.7319253673332377, 'B_6_5':
 -0.37764913923060317, 'B_6_6': -0.7585800540299407, 'B_6_7': 0.9619997120136563,
 'B_6_8': -0.3789073559225776, 'B_6_9': -0.7319585855293902, 'B_7_0':
 -0.5958170206006437, 'B_7_1': 0.1736523921161235, 'B_7_2': -0.8631938227477187,
 'B_7_3': 0.7176195776794907, 'B_7_4': 0.9137693648895517, 'B_7_5':
 -0.5014401070785859, 'B_7_6': -0.0988113681900985, 'B_7_7': -0.9461737869615958,
 'B_7_8': 0.2786062449598985, 'B_7_9': -0.17042640505587403, 'B_8_0':
 -0.45528423711891974, 'B_8_1': -0.5804625209791152, 'B_8_2': 0.3049656937643457,
 'B_8_3': -0.16589027855601735, 'B_8_4': -0.6342746077207742, 'B_8_5':
 0.37443598134391154, 'B_8_6': 0.12884312041265344, 'B_8_7': 0.7101706687604024,
 'B_8_8': -0.7573977202245441, 'B_8_9': 0.5656198718500232, 'B_9_0':
 0.3435632378981728, 'B_9_1': -0.7622783450803663, 'B_9_2': 0.38574504514600694,
 'B_9_3': -0.11573636801793574, 'B_9_4': 0.8810698754005692, 'B_9_5':
 -0.7096095257765593, 'B_9_6': 0.2828101138151613, 'B_9_7': -0.23367950998336706,
 'B_9_8': 0.904493047769938, 'B_9_9': -0.33664970732525257, 'B_10_0':
 0.9151281007937644, 'B_10_1': 0.7949419851794832, 'B_10_2':
 -0.45464119408490045, 'B_10_3': 0.6010127181163267, 'B_10_4':
 0.9812963195709419, 'B_10_5': -0.48972725570086095, 'B_10_6':
 -0.12518120473982708, 'B_10_7': 0.03995785840218424, 'B_10_8':
 -0.4060363511083487, 'B_10_9': -0.959775617500861}. Best is trial 36 with value:
 95.85134184163296.

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```
[I 2025-04-29 14:05:05,260] Trial 77 finished with value: 152.4164855686094 and
parameters: {'B_0_0': -0.5541440964888972, 'B_0_1': -0.4317649690033183,
'B_0_2': -0.11167259138998942, 'B_0_3': -0.6512170927414642, 'B_0_4':
0.7390436262495117, 'B_0_5': 0.4057240428080455, 'B_0_6': -0.8213030544318692,
'B_0_7': 0.2971830532570643, 'B_0_8': 0.3042398232831285, 'B_0_9':
0.9033179238143829, 'B_1_0': -0.9840186747148008, 'B_1_1': -0.49909374938450174,
'B_1_2': 0.8470271291631246, 'B_1_3': 0.8556251524886005, 'B_1_4':
0.436761110353862, 'B_1_5': 0.5262113756741291, 'B_1_6': -0.8088637432992987,
'B_1_7': 0.0274741528609026, 'B_1_8': -0.5314656772477813, 'B_1_9':
-0.6311882736450378, 'B_2_0': 0.6144005322641983, 'B_2_1': -0.7226244893725161,
'B_2_2': -0.062069758747157255, 'B_2_3': 0.8683849143803455, 'B_2_4':
-0.6220525299806218, 'B_2_5': 0.9464269790017046, 'B_2_6': 0.978690211806897,
'B_2_7': 0.11012196627115767, 'B_2_8': -0.6531173744492274, 'B_2_9':
-0.3641123218770266, 'B_3_0': -0.14950419130046044, 'B_3_1':
-0.37753803577071854, 'B_3_2': -0.43434155184449874, 'B_3_3': 0.935014824248388,
'B_3_4': -0.6863376417168237, 'B_3_5': 0.4619892592593873, 'B_3_6':
-0.9891876269104337, 'B_3_7': 0.19044168541670614, 'B_3_8': 0.8235671478290236,
'B_3_9': -0.4198904936954263, 'B_4_0': -0.6226585689897159, 'B_4_1':
0.3003259938592855, 'B_4_2': 0.4209507864508184, 'B_4_3': 0.3723004275249231,
'B_4_4': -0.3577525775576794, 'B_4_5': -0.4906392437245249, 'B_4_6':
0.9027192239248476, 'B_4_7': 0.8381330263744604, 'B_4_8': -0.08451311241221582,
'B_4_9': 0.6854638549856593, 'B_5_0': -0.09439951428409388, 'B_5_1':
-0.14970974421642946, 'B_5_2': -0.6796093637533966, 'B_5_3':
-0.1457160081458938, 'B_5_4': 0.07408553842761778, 'B_5_5': -0.9454414316912008,
'B_5_6': -0.11065624975269528, 'B_5_7': 0.5570853124224966, 'B_5_8':
0.35649119162105425, 'B_5_9': 0.26761083795059126, 'B_6_0': 0.8109197483056609,
'B_6_1': -0.9542966205460446, 'B_6_2': 0.11295422066062355, 'B_6_3':
0.56895071985122, 'B_6_4': 0.08184405774850911, 'B_6_5': -0.3486868058908492,
'B_6_6': -0.7202238811178838, 'B_6_7': 0.7673748821960589, 'B_6_8':
-0.05789046307254608, 'B_6_9': 0.7486680900316887, 'B_7_0': 0.16998720241767318,
'B_7_1': 0.3449003908977015, 'B_7_2': 0.6024370856693311, 'B_7_3':
0.3151279783082661, 'B_7_4': -0.2018514154557034, 'B_7_5': 0.026994100178729052,
'B_7_6': 0.6000301388582421, 'B_7_7': -0.2501724131814913, 'B_7_8':
-0.2215195155983758, 'B_7_9': -0.8005335702051577, 'B_8_0': 0.8208771817390605,
```

'B_8_1': -0.8097480964080254, 'B_8_2': 0.29345417134814067, 'B_8_3':
-0.17187493197046055, 'B_8_4': -0.1868736072942241, 'B_8_5':
0.35551263641391406, 'B_8_6': -0.8571544093928402, 'B_8_7': 0.1855374502438405,
'B_8_8': 0.5386229362252624, 'B_8_9': 0.6498612606154566, 'B_9_0':
-0.6523498290003249, 'B_9_1': 0.7868897191547839, 'B_9_2': 0.3091692618579356,
'B_9_3': 0.756072505624952, 'B_9_4': -0.029144180121807417, 'B_9_5':
0.08322602329245998, 'B_9_6': 0.8296957885167688, 'B_9_7': 0.7918208958422501,
'B_9_8': -0.10264147072937602, 'B_9_9': -0.4466629893613502, 'B_10_0':
-0.21152253956275702, 'B_10_1': -0.2222266295110955, 'B_10_2':
-0.42200358753406797, 'B_10_3': -0.45821238342072623, 'B_10_4':
0.7149046142298445, 'B_10_5': -0.8516733970443698, 'B_10_6': 0.3516701405571965,
'B_10_7': -0.6478162981162244, 'B_10_8': -0.3078783310797548, 'B_10_9':
-0.868144792413023}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [<F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:05:22,441] Trial 78 finished with value: 184.68679838960796 and
parameters: {'B_0_0': -0.45545219902618517, 'B_0_1': 0.5314737208121394,
'B_0_2': 0.7623303426035721, 'B_0_3': -0.5513762392333132, 'B_0_4':
-0.07217527826733472, 'B_0_5': -0.0587583850843314, 'B_0_6': 0.7396791054042737,
'B_0_7': 0.5203421246940947, 'B_0_8': 0.025892892368738707, 'B_0_9':
0.33807945693411123, 'B_1_0': 0.6877630067405462, 'B_1_1': -0.971112871029421,
'B_1_2': -0.1922777385209038, 'B_1_3': 0.8063414712086157, 'B_1_4':
0.47581980107102373, 'B_1_5': 0.0848259211814677, 'B_1_6': 0.058696163529564194,
'B_1_7': -0.3014152067432132, 'B_1_8': 0.04063065661019882, 'B_1_9':
0.3076587349376374, 'B_2_0': -0.1616411839109897, 'B_2_1': 0.42402322985650454,
'B_2_2': 0.7152398099755639, 'B_2_3': -0.6837240005939749, 'B_2_4':
-0.09593689482149959, 'B_2_5': -0.18212021925553945, 'B_2_6':
0.3711803120184616, 'B_2_7': -0.4952624187892999, 'B_2_8': 0.27769681160550697,
'B_2_9': 0.7474286986283463, 'B_3_0': -0.16209943240862712, 'B_3_1':
0.8138110859588557, 'B_3_2': 0.9764642030882873, 'B_3_3': -0.7799377411030549,
'B_3_4': -0.5285546877899798, 'B_3_5': 0.8917519079345446, 'B_3_6':
0.9074888192331458, 'B_3_7': -0.7965395947342182, 'B_3_8': -0.9603029692532694,
'B_3_9': -0.6710219539522839, 'B_4_0': 0.9821053110968516, 'B_4_1':
-0.63360075911084, 'B_4_2': 0.00504426130960578, 'B_4_3': 0.6898670983324118,
'B_4_4': -0.9496048108373515, 'B_4_5': 0.20688818487174987, 'B_4_6':

-0.4414896112381438, 'B_4_7': 0.7174231345506685, 'B_4_8': -0.42874692445712326,
 'B_4_9': 0.3697763261581797, 'B_5_0': 0.344954872245846, 'B_5_1':
 -0.5514427732579128, 'B_5_2': -0.9210975842606668, 'B_5_3':
 -0.17968584390491338, 'B_5_4': -0.41739755910757803, 'B_5_5':
 0.6117759442351431, 'B_5_6': 0.5947210931415836, 'B_5_7': -0.08509257465860154,
 'B_5_8': 0.21089420613997345, 'B_5_9': 0.20643384165411183, 'B_6_0':
 0.6477731338866943, 'B_6_1': -0.14849522003606896, 'B_6_2': -0.444615574014792,
 'B_6_3': -0.46523086208171605, 'B_6_4': 0.07166071297876431, 'B_6_5':
 -0.10068655859021913, 'B_6_6': -0.11856087344982635, 'B_6_7':
 -0.8095480686023797, 'B_6_8': 0.43283540656658737, 'B_6_9': -0.5668013577958169,
 'B_7_0': -0.6000318121867316, 'B_7_1': -0.172738525850761, 'B_7_2':
 -0.5389988835967532, 'B_7_3': 0.1660393692698705, 'B_7_4': -0.2545025881177464,
 'B_7_5': -0.10115478926627564, 'B_7_6': 0.07359525677597722, 'B_7_7':
 -0.5869996654860263, 'B_7_8': -0.681611786726251, 'B_7_9': 0.7532353676897374,
 'B_8_0': -0.1312270683937864, 'B_8_1': -0.1740546529275725, 'B_8_2':
 -0.9194879424597738, 'B_8_3': -0.004006920876961417, 'B_8_4':
 0.10725604878978023, 'B_8_5': 0.11650492449023964, 'B_8_6': 0.4099217152593111,
 'B_8_7': -0.6385859667177967, 'B_8_8': -0.49059819422337475, 'B_8_9':
 0.6152463211562171, 'B_9_0': 0.8853978213499201, 'B_9_1': -0.7208531602127051,
 'B_9_2': -0.0046538820796679925, 'B_9_3': -0.565931019575618, 'B_9_4':
 -0.8710551266571348, 'B_9_5': -0.3455033905102519, 'B_9_6': -0.7054985021352588,
 'B_9_7': 0.4156674853998077, 'B_9_8': 0.06677631906945991, 'B_9_9':
 -0.7052077474391845, 'B_10_0': -0.9717016549570815, 'B_10_1':
 0.6101509570679411, 'B_10_2': -0.6411350757947899, 'B_10_3': 0.2103089515971166,
 'B_10_4': -0.3228946121080437, 'B_10_5': 0.16926215786377052, 'B_10_6':
 -0.3650058185749985, 'B_10_7': -0.8178596698293368, 'B_10_8':
 -0.5485589830556479, 'B_10_9': -0.6176440383714303}. Best is trial 36 with
 value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:05:39,626] Trial 79 finished with value: 185.70317289132356 and
 parameters: {'B_0_0': -0.14252168216950523, 'B_0_1': 0.31340189118379147,
 'B_0_2': -0.4676639716062121, 'B_0_3': -0.9661015281047765, 'B_0_4':
 0.0374959812442186, 'B_0_5': 0.20940378893076472, 'B_0_6': 0.9164835781235832,
 'B_0_7': 0.17715219012462868, 'B_0_8': -0.05707833243127913, 'B_0_9':

0.14533521927745352, 'B_1_0': 0.8472432471268134, 'B_1_1': 0.6501887282912595, 'B_1_2': -0.3505021079396602, 'B_1_3': -0.5202363599588682, 'B_1_4': -0.46998544751517424, 'B_1_5': 0.008753658636121764, 'B_1_6': 0.483217913432584, 'B_1_7': 0.006374030222264904, 'B_1_8': -0.8557631362603371, 'B_1_9': 0.4426897868711621, 'B_2_0': 0.6058208547684887, 'B_2_1': -0.4167942270629551, 'B_2_2': 0.3249689637810338, 'B_2_3': -0.14534803591230117, 'B_2_4': 0.7704563243403368, 'B_2_5': 0.6835466051582648, 'B_2_6': 0.050355356653091654, 'B_2_7': 0.379180980710917, 'B_2_8': 0.7590324080237658, 'B_2_9': -0.008480320045014533, 'B_3_0': -0.36976353383213256, 'B_3_1': 0.5022589530609709, 'B_3_2': -0.5542688155263102, 'B_3_3': -0.22070832694803677, 'B_3_4': 0.7241399294714514, 'B_3_5': 0.44435623373324384, 'B_3_6': -0.19045420960995152, 'B_3_7': -0.5648058852314524, 'B_3_8': -0.2851825861521716, 'B_3_9': -0.42747378452968265, 'B_4_0': -0.6413432098711692, 'B_4_1': 0.4817991688600003, 'B_4_2': -0.49090162403666043, 'B_4_3': -0.09074840906782766, 'B_4_4': -0.2259982952645041, 'B_4_5': 0.9545132096985491, 'B_4_6': -0.2156373565865234, 'B_4_7': -0.8316346840624255, 'B_4_8': -0.8121450099072736, 'B_4_9': 0.17969236827906832, 'B_5_0': 0.19673983244176596, 'B_5_1': -0.16085719884434313, 'B_5_2': -0.1349494892221812, 'B_5_3': -0.883213485975075, 'B_5_4': 0.03276209057073931, 'B_5_5': 0.00938416125309638, 'B_5_6': -0.6938923492572497, 'B_5_7': -0.7843980155003634, 'B_5_8': -0.7453264686130088, 'B_5_9': -0.7899640719473081, 'B_6_0': 0.2172222271627844, 'B_6_1': -0.6009643482422211, 'B_6_2': -0.49216829866882233, 'B_6_3': -0.6463271545801493, 'B_6_4': 0.906513902808036, 'B_6_5': 0.702142577040634, 'B_6_6': -0.5336202220439887, 'B_6_7': -0.34141873803419054, 'B_6_8': -0.5744121722361056, 'B_6_9': -0.016085600267569, 'B_7_0': -0.9511711457009778, 'B_7_1': 0.33930865761565543, 'B_7_2': 0.045853136638930536, 'B_7_3': 0.7893050780361845, 'B_7_4': -0.6645024850297401, 'B_7_5': 0.6021363072010406, 'B_7_6': 0.5780828086440972, 'B_7_7': -0.7381812453505638, 'B_7_8': -0.2008527361946274, 'B_7_9': -0.6530977705338565, 'B_8_0': -0.4856534090940532, 'B_8_1': -0.4687725738355222, 'B_8_2': 0.11134806017102994, 'B_8_3': -0.17540635511427827, 'B_8_4': 0.8865109519324845, 'B_8_5': -0.7397871189449299, 'B_8_6': 0.9379104270033936, 'B_8_7': -0.5333354923757574, 'B_8_8': 0.540290394036069, 'B_8_9': -0.3188395461778136, 'B_9_0': -0.7066988370737184, 'B_9_1': 0.708620586787555, 'B_9_2': -0.37845541951180284, 'B_9_3': 0.3145337003691475, 'B_9_4': 0.13155849647152063, 'B_9_5': -0.055922507406068855, 'B_9_6': -0.0082774441558926, 'B_9_7': 0.45983106990272926, 'B_9_8': -0.8162096300355877, 'B_9_9': 0.3329977106766522, 'B_10_0': 0.7837079607691859, 'B_10_1': -0.875579369900669, 'B_10_2': -0.7115063883173398, 'B_10_3': 0.42893123467002225, 'B_10_4': -0.12418233647565724, 'B_10_5': -0.190973730788897, 'B_10_6': -0.718806063602413, 'B_10_7': 0.013404680621220466, 'B_10_8': -0.7394402838311731, 'B_10_9': -0.04629149650512998}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].


```

Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```

[I 2025-04-29 14:05:57,181] Trial 80 finished with value: 129.27826737052095 and
parameters: {'B_0_0': 0.5219582919440802, 'B_0_1': -0.4786365713834162, 'B_0_2':
-0.038043387004139895, 'B_0_3': 0.9741916194057347, 'B_0_4': 0.3695678382409937,
'B_0_5': 0.6758864071842932, 'B_0_6': -0.16860173901011943, 'B_0_7':
0.5303646277891936, 'B_0_8': -0.9097918375761828, 'B_0_9': -0.746635858566355,
'B_1_0': -0.587920226018277, 'B_1_1': -0.19413518926186524, 'B_1_2':
0.6845748590850111, 'B_1_3': -0.3043932194398976, 'B_1_4': 0.3061499631606166,
'B_1_5': 0.4144423037108971, 'B_1_6': 0.17064106109491606, 'B_1_7':
-0.2826609030231303, 'B_1_8': 0.15256561659281775, 'B_1_9': -0.3123141262763025,
'B_2_0': -0.823929997667016, 'B_2_1': 0.9757518768083084, 'B_2_2':
-0.7381787418076284, 'B_2_3': -0.8467502000046064, 'B_2_4': 0.7494908834301688,
'B_2_5': -0.3923673714320677, 'B_2_6': 0.29758800866554513, 'B_2_7':
-0.338568655268779, 'B_2_8': -0.21492054382694714, 'B_2_9':
-0.47101631540674294, 'B_3_0': -0.31077286852983976, 'B_3_1':
-0.17681320683864465, 'B_3_2': 0.4364238168028389, 'B_3_3':
0.022841664584887678, 'B_3_4': 0.8782591299708142, 'B_3_5': -0.9070768201992245,
'B_3_6': 0.006075278811415785, 'B_3_7': 0.16676508188680184, 'B_3_8':
0.8319900324270624, 'B_3_9': -0.8926661232295734, 'B_4_0': 0.1455291781717276,
'B_4_1': 0.700187258485355, 'B_4_2': -0.5109830983239116, 'B_4_3':
0.745058850077396, 'B_4_4': -0.9522940496369022, 'B_4_5': 0.4602601656847849,
'B_4_6': 0.8575641779139329, 'B_4_7': 0.3112513700473232, 'B_4_8':
-0.12072209691871305, 'B_4_9': -0.9011355873840183, 'B_5_0': 0.8318253503788049,
'B_5_1': -0.05881347907068191, 'B_5_2': -0.9325864462148229, 'B_5_3':
0.3758457212309485, 'B_5_4': -0.035962237919008766, 'B_5_5':
-0.7437962714640267, 'B_5_6': -0.5657418175656861, 'B_5_7': -0.8623084526974244,
'B_5_8': 0.1471012086229757, 'B_5_9': 0.24161836891897837, 'B_6_0':
0.011369750743779772, 'B_6_1': 0.6278622233451776, 'B_6_2': 0.7147974661898668,
'B_6_3': 0.2289363054325908, 'B_6_4': -0.9647421701792875, 'B_6_5':
0.3865547183805167, 'B_6_6': -0.16236155049770384, 'B_6_7': 0.12980381678824626,
'B_6_8': 0.2951795919973912, 'B_6_9': -0.296531503632388, 'B_7_0':
0.8258550402401525, 'B_7_1': -0.7247297638354332, 'B_7_2': 0.3395933784164542,
'B_7_3': -0.9813680316859708, 'B_7_4': 0.840809188702258, 'B_7_5':
-0.7672418161885788, 'B_7_6': -0.8021292243078022, 'B_7_7':
-0.33064488655655644, 'B_7_8': -0.25033875364262037, 'B_7_9':
-0.5656858568007004, 'B_8_0': 0.16854088863570693, 'B_8_1': -0.1130300858769121,
'B_8_2': -0.9223323336986469, 'B_8_3': -0.3570889167345047, 'B_8_4':
0.6146571688748776, 'B_8_5': 0.3756090390264999, 'B_8_6': 0.44524429008816324,
'B_8_7': 0.3922715096853766, 'B_8_8': -0.37427341766159405, 'B_8_9':

```

```

-0.7926654342879838, 'B_9_0': -0.45051497561446996, 'B_9_1': 0.6569348943666153,
'B_9_2': 0.9513321796104808, 'B_9_3': -0.012182910811708525, 'B_9_4':
-0.9707932972523764, 'B_9_5': -0.7127112719742266, 'B_9_6': -0.3522804608737937,
'B_9_7': 0.15559899021404044, 'B_9_8': -0.8413145813054783, 'B_9_9':
0.8778474514455656, 'B_10_0': -0.26713432689688654, 'B_10_1':
0.8169167514911495, 'B_10_2': 0.5554331694589032, 'B_10_3': 0.5057657995808507,
'B_10_4': 0.21130476796881936, 'B_10_5': 0.2510319577626523, 'B_10_6':
0.2732464057281714, 'B_10_7': 0.25486834631065247, 'B_10_8': 0.1063818981687552,
'B_10_9': 0.9630997565961199}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:06:15,750] Trial 81 finished with value: 157.48547994503303 and
parameters: {'B_0_0': 0.45993532350287825, 'B_0_1': 0.548353849017192, 'B_0_2':
0.7374763738656749, 'B_0_3': -0.5630332698028186, 'B_0_4': -0.8240677649917796,
'B_0_5': -0.033869015534774616, 'B_0_6': 0.02787977732616076, 'B_0_7':
0.6517799243008371, 'B_0_8': -0.11130851302097533, 'B_0_9': 0.14028899057773914,
'B_1_0': 0.05360045825664739, 'B_1_1': -0.9143584881603657, 'B_1_2':
0.22812549526188963, 'B_1_3': 0.6495257058330195, 'B_1_4': -0.6781489148099085,
'B_1_5': 0.18677683317465954, 'B_1_6': 0.18534887800390054, 'B_1_7':
-0.6649137711684097, 'B_1_8': -0.7455421240131427, 'B_1_9': 0.9505028470934398,
'B_2_0': -0.21043604408698569, 'B_2_1': -0.5397049771274121, 'B_2_2':
-0.6507774019803376, 'B_2_3': 0.3318513928948055, 'B_2_4': 0.21670168870801199,
'B_2_5': 0.2842332035150865, 'B_2_6': -0.8157480655273245, 'B_2_7':
0.7791984753269052, 'B_2_8': -0.21925253431808434, 'B_2_9': -0.3695285752678913,
'B_3_0': -0.07452433444947637, 'B_3_1': -0.21766918981969607, 'B_3_2':
0.7198824662155088, 'B_3_3': 0.8499367308918264, 'B_3_4': -0.3891626314297797,
'B_3_5': 0.7205928987707104, 'B_3_6': 0.5574157964754729, 'B_3_7':
0.8529991912426285, 'B_3_8': -0.5034143818719421, 'B_3_9': 0.15237911723953856,
'B_4_0': -0.4464017222139196, 'B_4_1': 0.641707091159061, 'B_4_2':
-0.9794581026692928, 'B_4_3': 0.12834194251506426, 'B_4_4':
-0.31837720608160236, 'B_4_5': -0.3510862599859159, 'B_4_6': 0.2869145212525357,
'B_4_7': 0.6959392096417079, 'B_4_8': 0.216206874958786, 'B_4_9':
0.6992824977205105, 'B_5_0': -0.4660977184755607, 'B_5_1': 0.03861367243466263,
'B_5_2': -0.09793765264718945, 'B_5_3': -0.9669430989838694, 'B_5_4':
0.11248804341866547, 'B_5_5': 0.31981816180942624, 'B_5_6': 0.18199052074975874,

```

'B_5_7': 0.46783419959117456, 'B_5_8': 0.14125226633709276, 'B_5_9':
-0.8578344953955013, 'B_6_0': -0.7594378623122714, 'B_6_1': 0.19101966353702582,
'B_6_2': -0.44554119989219243, 'B_6_3': -0.8323026445111636, 'B_6_4':
-0.13938656797187743, 'B_6_5': 0.1075587442056376, 'B_6_6': -0.339082943067779,
'B_6_7': 0.4364587304070404, 'B_6_8': 0.2846658040404644, 'B_6_9':
-0.3310196450384896, 'B_7_0': 0.5046476072389501, 'B_7_1': 0.29441618421541027,
'B_7_2': 0.2721418498614461, 'B_7_3': 0.10578684755839474, 'B_7_4':
-0.9674484571846654, 'B_7_5': 0.3890842232842404, 'B_7_6': -0.24543725360387014,
'B_7_7': 0.876367226114138, 'B_7_8': 0.6296098467328666, 'B_7_9':
0.17363504967199828, 'B_8_0': -0.4906796759978465, 'B_8_1': -0.5866395613102202,
'B_8_2': -0.3160380213984527, 'B_8_3': -0.04639547171156333, 'B_8_4':
-0.4037902901228847, 'B_8_5': 0.029560400877115978, 'B_8_6':
-0.6625665574838919, 'B_8_7': 0.9458713693944898, 'B_8_8': -0.7763093811693121,
'B_8_9': 0.8552493965156567, 'B_9_0': 0.8188844625773359, 'B_9_1':
0.05021375400260597, 'B_9_2': 0.445322338235447, 'B_9_3': -0.12662437001427607,
'B_9_4': 0.25746846353402764, 'B_9_5': -0.31565964390618495, 'B_9_6':
-0.3937051408248182, 'B_9_7': 0.15489497661563711, 'B_9_8':
-0.003703245115718845, 'B_9_9': 0.15748201091696257, 'B_10_0':
-0.4224120490516945, 'B_10_1': -0.26342349300234935, 'B_10_2':
-0.8238820379547596, 'B_10_3': -0.8864927829010327, 'B_10_4':
-0.48815488324742784, 'B_10_5': -0.8429912185255194, 'B_10_6':
-0.7845260769032134, 'B_10_7': 0.6195982426799953, 'B_10_8':
-0.9130300799069552, 'B_10_9': 0.3147067319907906}. Best is trial 36 with value:
95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:06:34,401] Trial 82 finished with value: 199.07050986524936 and
parameters: {'B_0_0': -0.45033565734550696, 'B_0_1': 0.7356604478020903,
'B_0_2': -0.029715276536550927, 'B_0_3': -0.3849586105309051, 'B_0_4':
0.03492509266508037, 'B_0_5': -0.23372312229036774, 'B_0_6':
0.12731610804626392, 'B_0_7': -0.7632249207138305, 'B_0_8': 0.9840411955688511,
'B_0_9': -0.14739936216049931, 'B_1_0': -0.30050333337941004, 'B_1_1':
0.4162154996662044, 'B_1_2': -0.2694791751392417, 'B_1_3': 0.65988312395694,
'B_1_4': -0.09054396576635804, 'B_1_5': 0.4347256582655612, 'B_1_6':
-0.7772426314450647, 'B_1_7': 0.6853159873991213, 'B_1_8': 0.4667216138206576,

'B_1_9': -0.570800778645308, 'B_2_0': -0.7023337007033424, 'B_2_1':
 -0.36374114952326364, 'B_2_2': 0.9426197603431705, 'B_2_3':
 -0.48147348235722387, 'B_2_4': -0.08686812628566609, 'B_2_5':
 -0.9305599530796741, 'B_2_6': -0.20348949145885964, 'B_2_7':
 0.14121363055754332, 'B_2_8': 0.897186732765771, 'B_2_9': 0.033137368981327464,
 'B_3_0': 0.6318325058318217, 'B_3_1': 0.09331273228642312, 'B_3_2':
 -0.9092456733976741, 'B_3_3': -0.36628361512424745, 'B_3_4':
 0.14958802550558437, 'B_3_5': 0.8921760525436262, 'B_3_6': 0.7110781123074419,
 'B_3_7': -0.5378558839258817, 'B_3_8': -0.5541825365620563, 'B_3_9':
 -0.5916775251144766, 'B_4_0': 0.8365478146789562, 'B_4_1': 0.6982789831414069,
 'B_4_2': 0.5442691958608386, 'B_4_3': -0.21238447383602943, 'B_4_4':
 0.008673794511558253, 'B_4_5': -0.07062314530559743, 'B_4_6':
 -0.48827174703244214, 'B_4_7': 0.009955531374558513, 'B_4_8':
 -0.9296168352515461, 'B_4_9': -0.7290154604750309, 'B_5_0': 0.7654474004145138,
 'B_5_1': 0.4983259387222181, 'B_5_2': -0.15272055006300755, 'B_5_3':
 0.033957717322076864, 'B_5_4': -0.25740220579924467, 'B_5_5':
 0.47674413697813156, 'B_5_6': 0.5626906148795177, 'B_5_7': 0.5213046766022551,
 'B_5_8': 0.4407712151722101, 'B_5_9': -0.5850814093882643, 'B_6_0':
 -0.32081944110385197, 'B_6_1': -0.05970603873232361, 'B_6_2':
 -0.33502987697324194, 'B_6_3': 0.5047815963134914, 'B_6_4': -0.3378800219850897,
 'B_6_5': -0.7163004681650325, 'B_6_6': -0.19989990494435794, 'B_6_7':
 0.9502305769091131, 'B_6_8': -0.5427548023915197, 'B_6_9': 0.22207414470759357,
 'B_7_0': -0.13271336131738154, 'B_7_1': -0.8493713560543332, 'B_7_2':
 -0.024004320123842238, 'B_7_3': 0.40028537072365866, 'B_7_4':
 0.026175552139081182, 'B_7_5': 0.7367700062182305, 'B_7_6': -0.4179047070179863,
 'B_7_7': -0.8192266408181494, 'B_7_8': 0.8643999110828522, 'B_7_9':
 -0.24117122793412638, 'B_8_0': 0.15128299187819416, 'B_8_1': 0.5781661897377881,
 'B_8_2': -0.34785894552635044, 'B_8_3': 0.42800578559167013, 'B_8_4':
 -0.7218706773607004, 'B_8_5': -0.4132383805330996, 'B_8_6': -0.7034729404496906,
 'B_8_7': 0.6162365056346886, 'B_8_8': -0.549154514250965, 'B_8_9':
 0.43045903970231625, 'B_9_0': 0.8777308265463921, 'B_9_1': 0.054988973956376475,
 'B_9_2': 0.0792578006761615, 'B_9_3': -0.07715353516633217, 'B_9_4':
 -0.34473751608058734, 'B_9_5': -0.04820303876866405, 'B_9_6':
 0.2035589138881837, 'B_9_7': 0.9622633128979055, 'B_9_8': 0.866773827450084,
 'B_9_9': 0.717062857420381, 'B_10_0': 0.7704784606956194, 'B_10_1':
 -0.06365861708175491, 'B_10_2': 0.31173976781138024, 'B_10_3':
 0.7822321114137041, 'B_10_4': -0.2119574853171209, 'B_10_5': 0.9142897071240281,
 'B_10_6': -0.16674359710301767, 'B_10_7': 0.20014367747856698, 'B_10_8':
 0.41541333227929633, 'B_10_9': 0.892172185890358}. Best is trial 36 with value:
 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].

```

Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:06:52,808] Trial 83 finished with value: 149.1040145202906 and
parameters: {'B_0_0': -0.5056597773765132, 'B_0_1': 0.48733933691251163,
'B_0_2': 0.041214860565282674, 'B_0_3': -0.45825686157113066, 'B_0_4':
0.5476815427968533, 'B_0_5': 0.10340011474421629, 'B_0_6': 0.318728164886358,
'B_0_7': -0.350655884215756, 'B_0_8': -0.03707757538134204, 'B_0_9':
0.2323614169563728, 'B_1_0': -0.18827658774570888, 'B_1_1': -0.5972223711015514,
'B_1_2': 0.9140481747843019, 'B_1_3': 0.022705729284382414, 'B_1_4':
0.46316233316964706, 'B_1_5': -0.1761265567590049, 'B_1_6': -0.2868391072290817,
'B_1_7': -0.18381468903509257, 'B_1_8': -0.17979064530441335, 'B_1_9':
0.08747662956506819, 'B_2_0': 0.08607034256309909, 'B_2_1':
-0.005635444551914115, 'B_2_2': -0.7231241458634725, 'B_2_3':
-0.007770552568948319, 'B_2_4': -0.6551742771671101, 'B_2_5':
-0.740251143134385, 'B_2_6': 0.856638520335512, 'B_2_7': 0.20780102603245432,
'B_2_8': 0.31579031508667676, 'B_2_9': -0.19444853273993978, 'B_3_0':
0.809782165991431, 'B_3_1': 0.2731023693677377, 'B_3_2': 0.5329563416567031,
'B_3_3': -0.26024631132512666, 'B_3_4': 0.9108235240218685, 'B_3_5':
0.6345133545134638, 'B_3_6': 0.7008179567687964, 'B_3_7': -0.6490553504970065,
'B_3_8': -0.4534928484149854, 'B_3_9': -0.42445933262651736, 'B_4_0':
-0.4535997379711951, 'B_4_1': 0.7037081609664901, 'B_4_2': -0.9268681054869876,
'B_4_3': -0.8393220218180668, 'B_4_4': -0.44764855615279064, 'B_4_5':
-0.5942515101941015, 'B_4_6': -0.5309179645119602, 'B_4_7': 0.5899275147215339,
'B_4_8': 0.1249906559062528, 'B_4_9': -0.9785740906399234, 'B_5_0':
-0.13994092390569335, 'B_5_1': -0.18238784483628934, 'B_5_2':
-0.7643883664998038, 'B_5_3': 0.3487607414308851, 'B_5_4': 0.40015243299139236,
'B_5_5': 0.4373685737231534, 'B_5_6': 0.4523724722060123, 'B_5_7':
-0.9898954026740523, 'B_5_8': 0.89876319161651, 'B_5_9': 0.622133358768233,
'B_6_0': 0.7998837592467849, 'B_6_1': -0.9590262752912804, 'B_6_2':
0.8590404156557641, 'B_6_3': 0.6133812868024473, 'B_6_4': 0.6676939030582232,
'B_6_5': -0.8052909782505023, 'B_6_6': 0.47278049725882343, 'B_6_7':
0.9106350454913401, 'B_6_8': 0.35352373600795817, 'B_6_9': 0.1397115811567209,
'B_7_0': 0.8823467619430141, 'B_7_1': -0.11298449896581675, 'B_7_2':
0.19660126129474786, 'B_7_3': -0.04685017888115495, 'B_7_4': 0.8570355182933924,
'B_7_5': -0.8393522410042956, 'B_7_6': 0.0022170076341894163, 'B_7_7':
0.37016583394316793, 'B_7_8': 0.5779749460760291, 'B_7_9': 0.16437692193939002,
'B_8_0': 0.809657133360479, 'B_8_1': 0.875143064830066, 'B_8_2':
-0.8626479380767046, 'B_8_3': -0.19330633170807876, 'B_8_4':
-0.9435511369921142, 'B_8_5': -0.337755018398603, 'B_8_6': 0.07299907158179231,
'B_8_7': 0.8133739624096987, 'B_8_8': 0.6861775584814978, 'B_8_9':
-0.7375505411894683, 'B_9_0': -0.8057998580130266, 'B_9_1': 0.17674933073820065,
'B_9_2': 0.06359358546838978, 'B_9_3': -0.46257226996855283, 'B_9_4':
0.7909171074165937, 'B_9_5': 0.846185490733876, 'B_9_6': -0.1928293520864779,

```

'B_9_7': 0.7727990630847701, 'B_9_8': -0.22978118935931424, 'B_9_9':
0.4688973270228063, 'B_10_0': -0.047634409255506216, 'B_10_1':
-0.7472853501157777, 'B_10_2': 0.8473097184493026, 'B_10_3':
-0.09088874511247746, 'B_10_4': 0.46428433983835915, 'B_10_5':
-0.26299584742604154, 'B_10_6': -0.9526158604190553, 'B_10_7':
-0.7909498897901649, 'B_10_8': -0.0074854306099496615, 'B_10_9':
0.3761294271101976}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:07:14,501] Trial 84 finished with value: 166.35181306268902 and
parameters: {'B_0_0': -0.3272240533730274, 'B_0_1': 0.38706714917174767,
'B_0_2': 0.39329846707527283, 'B_0_3': 0.403793096509105, 'B_0_4':
-0.014806729039393396, 'B_0_5': 0.8214745772405778, 'B_0_6':
-0.021390894398381155, 'B_0_7': -0.09847491687665644, 'B_0_8':
-0.5952259783589586, 'B_0_9': -0.38084292703445366, 'B_1_0':
-0.9061777048137594, 'B_1_1': 0.264399881360017, 'B_1_2': -0.4978992270243561,
'B_1_3': -0.3877725961187791, 'B_1_4': 0.536212752895356, 'B_1_5':
-0.3421020573403213, 'B_1_6': -0.5104521338139785, 'B_1_7': -0.6552656567152626,
'B_1_8': 0.5442196363202727, 'B_1_9': -0.7331816091797467, 'B_2_0':
-0.0915675547043, 'B_2_1': 0.013721629832844862, 'B_2_2': 0.371032027117715,
'B_2_3': 0.26390129501933113, 'B_2_4': 0.6152269324314461, 'B_2_5':
-0.9013921608260484, 'B_2_6': -0.4030473649367935, 'B_2_7':
-0.43533444482474815, 'B_2_8': -0.7008865269276257, 'B_2_9':
-0.41742705736424257, 'B_3_0': -0.37666467799705117, 'B_3_1':
0.04333123461640698, 'B_3_2': 0.3241348682739871, 'B_3_3': -0.23542098304860815,
'B_3_4': 0.43455105800049476, 'B_3_5': -0.32439262961243165, 'B_3_6':
-0.41523831891677454, 'B_3_7': 0.08155235059690824, 'B_3_8':
-0.9430416065945346, 'B_3_9': -0.8320175352046792, 'B_4_0':
-0.16242030369867644, 'B_4_1': 0.8769671685556779, 'B_4_2': 0.43859332408997886,
'B_4_3': -0.8155093923888956, 'B_4_4': 0.6857096653061299, 'B_4_5':
0.4599555258041983, 'B_4_6': 0.21967412089125338, 'B_4_7': 0.634180062006171,
'B_4_8': -0.1898625014689449, 'B_4_9': -0.5514279193093716, 'B_5_0':
0.023806310435343425, 'B_5_1': 0.0967066760750781, 'B_5_2': 0.07588621657962591,
'B_5_3': 0.1288461071306084, 'B_5_4': 0.8884357381135779, 'B_5_5':
0.7964463337305381, 'B_5_6': 0.7439197323867, 'B_5_7': 0.5968052855990817,

```

'B_5_8': 0.877987652385265, 'B_5_9': -0.7949806094747816, 'B_6_0':
-0.46505594233743575, 'B_6_1': -0.00978922760392198, 'B_6_2':
0.6861382963843423, 'B_6_3': 0.01022810881501024, 'B_6_4': 0.4911087838963457,
'B_6_5': -0.7148593876071458, 'B_6_6': -0.4902206503064108, 'B_6_7':
0.4589166681745913, 'B_6_8': -0.6638266624023064, 'B_6_9': -0.29431524938324505,
'B_7_0': -0.4655648095291286, 'B_7_1': -0.29727384543949265, 'B_7_2':
0.41248161717438125, 'B_7_3': -0.26166258133725395, 'B_7_4':
0.10942137876372149, 'B_7_5': -0.7921348834322689, 'B_7_6': 0.6039350326045629,
'B_7_7': -0.25762052121671275, 'B_7_8': 0.29409763488601803, 'B_7_9':
0.4462714112904287, 'B_8_0': -0.5748713742188332, 'B_8_1': 0.7616564702076329,
'B_8_2': 0.23012401818176675, 'B_8_3': -0.14112421933206665, 'B_8_4':
-0.6289763365151253, 'B_8_5': 0.8757558570443893, 'B_8_6': 0.5050350441806177,
'B_8_7': -0.5781113657881672, 'B_8_8': -0.3213326660460355, 'B_8_9':
0.613155031014385, 'B_9_0': 0.3874417731663711, 'B_9_1': 0.5262528995771343,
'B_9_2': -0.11642181761839387, 'B_9_3': 0.42862300003149056, 'B_9_4':
-0.22474242872945083, 'B_9_5': -0.571699424462371, 'B_9_6': 0.2627005289858928,
'B_9_7': 0.9425075726856125, 'B_9_8': -0.3750100766372474, 'B_9_9':
-0.07728020741917152, 'B_10_0': 0.3445227147183101, 'B_10_1':
0.08218243860252605, 'B_10_2': 0.6799870355039175, 'B_10_3': 0.5182505588096415,
'B_10_4': -0.874571536352533, 'B_10_5': 0.9868740431690726, 'B_10_6':
0.4997568583645451, 'B_10_7': 0.7205285662623442, 'B_10_8': 0.29937146994878594,
'B_10_9': 0.7054445535325893}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:07:33,312] Trial 85 finished with value: 114.28446221938516 and
parameters: {'B_0_0': -0.7674817096411554, 'B_0_1': -0.3014790855925551,
'B_0_2': 0.7453134683368956, 'B_0_3': -0.5907226816915534, 'B_0_4':
-0.835080644811555, 'B_0_5': 0.5704755010915337, 'B_0_6': 0.6642625881737265,
'B_0_7': -0.6679486079839968, 'B_0_8': 0.7016050459097767, 'B_0_9':
0.3698002352823959, 'B_1_0': 0.03976371276883617, 'B_1_1': 0.6538953817071422,
'B_1_2': -0.1598864863484477, 'B_1_3': -0.6058919666776735, 'B_1_4':
-0.425991700310985, 'B_1_5': 0.010224756220763354, 'B_1_6': 0.06203496949288789,
'B_1_7': 0.6050500363536426, 'B_1_8': 0.5768783928316221, 'B_1_9':
0.1611057832650582, 'B_2_0': 0.5559552570636432, 'B_2_1': -0.6102885525785314,
'B_2_2': -0.8908722363552772, 'B_2_3': 0.7397949688167063, 'B_2_4':

```

```

-0.8404021023497162, 'B_2_5': -0.5015569091068853, 'B_2_6': 0.3867315202562813,
'B_2_7': -0.17929308349792794, 'B_2_8': 0.7451016604327074, 'B_2_9':
-0.40000000656870627, 'B_3_0': -0.9664947335946685, 'B_3_1':
-0.3947018316897266, 'B_3_2': 0.15458859738596664, 'B_3_3':
-0.24564201670147923, 'B_3_4': 0.06904805640233547, 'B_3_5': 0.7456642752933371,
'B_3_6': 0.8167204073761296, 'B_3_7': -0.463371351984858, 'B_3_8':
-0.07496186292559659, 'B_3_9': 0.09730515017807329, 'B_4_0': 0.8892426646387293,
'B_4_1': 0.9872622919454817, 'B_4_2': -0.5238936472937772, 'B_4_3':
0.44942014526471463, 'B_4_4': -0.24149849677827362, 'B_4_5':
-0.9739457701652978, 'B_4_6': -0.9716800055769134, 'B_4_7':
-0.07807930502673366, 'B_4_8': -0.023096872251417433, 'B_4_9':
-0.9453120007845836, 'B_5_0': 0.3701162852131188, 'B_5_1': -0.40787329058199284,
'B_5_2': 0.11858834104117988, 'B_5_3': -0.3941802351324708, 'B_5_4':
-0.6590740565587718, 'B_5_5': -0.18379599757422693, 'B_5_6':
-0.5300987377148085, 'B_5_7': 0.06890593429995273, 'B_5_8': -0.5963195100327696,
'B_5_9': 0.01664383375636147, 'B_6_0': -0.6783905405596422, 'B_6_1':
0.3385299247818543, 'B_6_2': 0.5027966151940195, 'B_6_3': -0.7714194161851311,
'B_6_4': -0.06375122416606838, 'B_6_5': -0.8496740327033503, 'B_6_6':
0.19440124217316468, 'B_6_7': -0.4664794900993021, 'B_6_8': 0.05469642454064161,
'B_6_9': 0.03457279858089657, 'B_7_0': 0.8402216054637248, 'B_7_1':
0.09718478068607128, 'B_7_2': 0.16433698984226508, 'B_7_3': -0.194726810986527,
'B_7_4': 0.8891776143019772, 'B_7_5': 0.548828100690427, 'B_7_6':
-0.4248975252003737, 'B_7_7': 0.6815349798369394, 'B_7_8': -0.1338552068335218,
'B_7_9': 0.6507609441965161, 'B_8_0': 0.639318375221134, 'B_8_1':
-0.6160370302095484, 'B_8_2': 0.39825944013948233, 'B_8_3': -0.4145269207730684,
'B_8_4': -0.4009645134145169, 'B_8_5': -0.40323080777645015, 'B_8_6':
0.4466251301686974, 'B_8_7': -0.5349444372507786, 'B_8_8': -0.9882064462127413,
'B_8_9': 0.4261689698956135, 'B_9_0': -0.651054207206452, 'B_9_1':
-0.7831248572904874, 'B_9_2': -0.519718081223761, 'B_9_3': -0.23779696580104615,
'B_9_4': -0.0477054716030626, 'B_9_5': -0.901751319388991, 'B_9_6':
0.9317351986021738, 'B_9_7': 0.4278048457707251, 'B_9_8': -0.3500757506578144,
'B_9_9': -0.3589794305749133, 'B_10_0': -0.10188575225045038, 'B_10_1':
0.24996474907080524, 'B_10_2': -0.7001813281538438, 'B_10_3':
-0.6159189687453943, 'B_10_4': 0.7116804740104308, 'B_10_5':
-0.4826152506996363, 'B_10_6': -0.23721390400914855, 'B_10_7':
0.7351182648812318, 'B_10_8': 0.3230327022116486, 'B_10_9':
-0.3134104789034571}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].

```



```

Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [F_[u_i], F_[u_j]].
Built matrix B = inverse(A).
[I 2025-04-29 14:07:50,002] Trial 86 finished with value: 104.75353605960404 and
parameters: {'B_0_0': 0.36372510053474905, 'B_0_1': -0.9510272335448009,
'B_0_2': 0.4665724668225031, 'B_0_3': -0.025708239777845865, 'B_0_4':
0.6201006516232332, 'B_0_5': 0.9001825509728072, 'B_0_6': 0.8430613279992454,
'B_0_7': 0.8103447860415747, 'B_0_8': 0.024096512749096632, 'B_0_9':
-0.2769186140049027, 'B_1_0': -0.0692300570332679, 'B_1_1': 0.3759940865604543,
'B_1_2': 0.7603155982049496, 'B_1_3': -0.5801671200597633, 'B_1_4':
0.828022453754172, 'B_1_5': 0.6240568593003804, 'B_1_6': 0.9481349100529046,
'B_1_7': 0.7528985110977404, 'B_1_8': -0.43572408721002487, 'B_1_9':
-0.7802072955387855, 'B_2_0': -0.7677626774566545, 'B_2_1': -0.4133778292150133,
'B_2_2': 0.22608377348938768, 'B_2_3': 0.6025542456724016, 'B_2_4':
0.7905156078234981, 'B_2_5': 0.24503170105108096, 'B_2_6': -0.9788500007588536,
'B_2_7': -0.6210927824933037, 'B_2_8': -0.7925421163703315, 'B_2_9':
-0.2706175383733601, 'B_3_0': 0.002674967394816874, 'B_3_1':
-0.9040735864144496, 'B_3_2': -0.3000136079246025, 'B_3_3': -0.6352242442418613,
'B_3_4': 0.2911989291925703, 'B_3_5': -0.0707407087378068, 'B_3_6':
-0.009271059070860566, 'B_3_7': -0.14317025969374708, 'B_3_8':
0.9029190267634026, 'B_3_9': -0.5648244522593048, 'B_4_0': 0.03299523449087105,
'B_4_1': 0.4673143893633651, 'B_4_2': 0.11255193800643393, 'B_4_3':
0.8494613537192661, 'B_4_4': -0.24162265583025733, 'B_4_5':
-0.07192190456910752, 'B_4_6': -0.8147258158748377, 'B_4_7':
0.21581823988281834, 'B_4_8': -0.41934030247275, 'B_4_9': -0.6725881017591737,
'B_5_0': 0.08817195995092608, 'B_5_1': 0.06397432149222304, 'B_5_2':
-0.8397741503763858, 'B_5_3': 0.392466597919056, 'B_5_4': 0.24506893748868874,
'B_5_5': -0.6157457801619355, 'B_5_6': -0.0605189729349932, 'B_5_7':
0.4362393746275488, 'B_5_8': 0.05828276627172757, 'B_5_9': -0.20693162915653096,
'B_6_0': -0.3517698529997316, 'B_6_1': 0.2185798559552261, 'B_6_2':
0.8694352226542763, 'B_6_3': 0.3400547841454171, 'B_6_4': -0.8420031108000953,
'B_6_5': -0.07784375250537512, 'B_6_6': 0.7738317803910426, 'B_6_7':
-0.802964396913848, 'B_6_8': 0.9679739100323357, 'B_6_9': 0.8891573994179343,
'B_7_0': -0.8350165281548458, 'B_7_1': 0.8581583744901187, 'B_7_2':
-0.27099747128626483, 'B_7_3': 0.6825241508102078, 'B_7_4':
-0.16160099132772243, 'B_7_5': -0.11710687684901855, 'B_7_6':
0.26580132013553426, 'B_7_7': 0.543465290166447, 'B_7_8': -0.9230974811224677,
'B_7_9': 0.18444887114372577, 'B_8_0': -0.6202686716872055, 'B_8_1':
-0.3050199546308582, 'B_8_2': -0.3010220410126714, 'B_8_3':
-0.061260526362115875, 'B_8_4': -0.3284099044851767, 'B_8_5':
-0.7774249854676778, 'B_8_6': -0.7016550977196061, 'B_8_7': -0.9237927885890266,
'B_8_8': -0.058906811049066476, 'B_8_9': 0.8998064358089173, 'B_9_0':
-0.7609630571767292, 'B_9_1': 0.8836036961461025, 'B_9_2': 0.4620557991695453,
'B_9_3': 0.272247629809643, 'B_9_4': 0.28764995554065464, 'B_9_5':
0.6469207438246385, 'B_9_6': -0.7782986685279537, 'B_9_7': 0.009413560659772613,
'B_9_8': 0.3902404537563018, 'B_9_9': 0.17190448124565605, 'B_10_0':
0.6589025606261234, 'B_10_1': -0.846902016157524, 'B_10_2': 0.3775880582083486,

```

'B_10_3': 0.06737490616174102, 'B_10_4': -0.23926148931971203, 'B_10_5':
-0.31625553517661886, 'B_10_6': 0.2392486128929019, 'B_10_7':
0.7741708417956443, 'B_10_8': -0.007571666319791781, 'B_10_9':
0.6429518648723018}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:08:07,357] Trial 87 finished with value: 213.20530672586483 and
parameters: {'B_0_0': -0.19373036394894427, 'B_0_1': -0.7872589636695104,

'B_0_2': -0.8682186418118957, 'B_0_3': -0.8883579447791183, 'B_0_4':
-0.2373027189238135, 'B_0_5': -0.32226925334503276, 'B_0_6':
0.43151530677268757, 'B_0_7': 0.4807865731380343, 'B_0_8': -0.8893091417091952,
'B_0_9': -0.5521759394797503, 'B_1_0': -0.44696153668537864, 'B_1_1':
0.09427172860276256, 'B_1_2': -0.7038598566642384, 'B_1_3': -0.7247963855879993,
'B_1_4': 0.8787020006317567, 'B_1_5': 0.6023217564496064, 'B_1_6':
-0.6323032938594828, 'B_1_7': -0.18499574034712785, 'B_1_8':
-0.2858684414700202, 'B_1_9': 0.5761880666495065, 'B_2_0': 0.020221574626681615,
'B_2_1': -0.7368575207529353, 'B_2_2': 0.47320428504936296, 'B_2_3':
-0.20416180311466048, 'B_2_4': -0.4694099896883488, 'B_2_5':
-0.2465127283989741, 'B_2_6': -0.0044420040583452725, 'B_2_7':
-0.3106312774943213, 'B_2_8': -0.5110265702683194, 'B_2_9':
-0.20096598296811896, 'B_3_0': -0.9520376367152015, 'B_3_1': 0.5395457352707016,
'B_3_2': -0.6515291107361801, 'B_3_3': -0.9030688650113382, 'B_3_4':
0.3043234119520568, 'B_3_5': 0.5423935350708711, 'B_3_6': 0.6213816805538295,
'B_3_7': -0.0128518183320121, 'B_3_8': -0.5833709943138291, 'B_3_9':
0.4697108009546569, 'B_4_0': 0.39543189276558754, 'B_4_1': -0.5023839464757169,
'B_4_2': 0.7987473566083301, 'B_4_3': -0.7833445129959986, 'B_4_4':
0.9557227143908249, 'B_4_5': 0.7132906657353799, 'B_4_6': 0.3720393180485688,
'B_4_7': 0.8687247787940691, 'B_4_8': 0.4401761132809414, 'B_4_9':
-0.6128951458805091, 'B_5_0': -0.11453084653663281, 'B_5_1': -0.848424011823362,
'B_5_2': -0.2184783038695315, 'B_5_3': 0.47837397907967305, 'B_5_4':
-0.22484216515698152, 'B_5_5': -0.3360175325730944, 'B_5_6':
0.18880433057041746, 'B_5_7': 0.736964967417812, 'B_5_8': -0.8957852787586974,
'B_5_9': -0.2267685392945239, 'B_6_0': -0.6766833312057798, 'B_6_1':
0.3868014164491018, 'B_6_2': -0.780665370346353, 'B_6_3': 0.3263193619888842,
'B_6_4': -0.4233618758201435, 'B_6_5': -0.5961447088002951, 'B_6_6':

```

0.8639780644934711, 'B_6_7': -0.9533213380430073, 'B_6_8': -0.12320100165865344,
'B_6_9': -0.9471785549256483, 'B_7_0': -0.03248576475342857, 'B_7_1':
0.002073808032948099, 'B_7_2': -0.35310830663840087, 'B_7_3':
-0.7283196805004531, 'B_7_4': -0.24475267528573885, 'B_7_5':
0.34108062106648074, 'B_7_6': 0.559143701116029, 'B_7_7': 0.060886192728521804,
'B_7_8': 0.17787210099071604, 'B_7_9': -0.5927795851731876, 'B_8_0':
-0.42108594020925405, 'B_8_1': -0.6694466920963303, 'B_8_2':
-0.49639520699086953, 'B_8_3': -0.5447241751534471, 'B_8_4':
0.03850859705465037, 'B_8_5': 0.12412373646564423, 'B_8_6': 0.8809958932265789,
'B_8_7': -0.6027701187389843, 'B_8_8': 0.280517906119762, 'B_8_9':
0.6281733525952953, 'B_9_0': 0.119278098290444, 'B_9_1': -0.48509276804787005,
'B_9_2': -0.4701505445603995, 'B_9_3': -0.9122789549194201, 'B_9_4':
-0.3852204314340063, 'B_9_5': 0.48431528442669247, 'B_9_6': -0.8959148407299214,
'B_9_7': -0.5557953406404352, 'B_9_8': 0.4083403470929283, 'B_9_9':
0.5596500943892957, 'B_10_0': -0.08846954807738172, 'B_10_1':
0.39218719018772474, 'B_10_2': -0.2218547060633138, 'B_10_3':
-0.6733717748522274, 'B_10_4': -0.20166400282464514, 'B_10_5':
0.15833628400222288, 'B_10_6': 0.8871454980687228, 'B_10_7':
-0.8538731148644189, 'B_10_8': 0.014640305813022003, 'B_10_9':
-0.3387388391427657}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:08:25,915] Trial 88 finished with value: 153.58639490109786 and
parameters: {'B_0_0': 0.016457154667855134, 'B_0_1': -0.9425286141944673,
'B_0_2': -0.3702097976908689, 'B_0_3': 0.6333389321034699, 'B_0_4':
0.10680463303218235, 'B_0_5': -0.0978466642240875, 'B_0_6':
-0.42466035800362767, 'B_0_7': -0.39987580487311536, 'B_0_8':
-0.23145916149064272, 'B_0_9': 0.6644300057298296, 'B_1_0': 0.9250821494803714,
'B_1_1': -0.1214623449262584, 'B_1_2': -0.21116137927943157, 'B_1_3':
0.6861705361233283, 'B_1_4': -0.14472746514011892, 'B_1_5': 0.10299598114538933,
'B_1_6': -0.532735504094486, 'B_1_7': 0.482135020842569, 'B_1_8':
0.06333687017614809, 'B_1_9': 0.6118884094758765, 'B_2_0': 0.5558508054567661,
'B_2_1': -0.5642905588253435, 'B_2_2': 0.42695839010436965, 'B_2_3':
0.5941077277437155, 'B_2_4': -0.30470766325085774, 'B_2_5': -0.6482339564534965,
'B_2_6': -0.9847883319524154, 'B_2_7': -0.09721076687556662, 'B_2_8':

```

```

-0.2921831915530937, 'B_2_9': 0.10872147097430962, 'B_3_0': -0.6176786366742042,
'B_3_1': 0.7121377843796208, 'B_3_2': -0.47964467305634284, 'B_3_3':
-0.5416693819264531, 'B_3_4': -0.0381861681609752, 'B_3_5':
-0.0022749729863369428, 'B_3_6': -0.5641312470055162, 'B_3_7':
-0.45171576108268674, 'B_3_8': -0.6539836264726151, 'B_3_9':
0.43255115307987024, 'B_4_0': 0.5347762818830821, 'B_4_1': -0.07458957503062047,
'B_4_2': 0.42389230491633456, 'B_4_3': 0.46715207603614206, 'B_4_4':
-0.716214455197939, 'B_4_5': 0.9230324891723827, 'B_4_6': 0.6356432552509335,
'B_4_7': -0.5851227468322695, 'B_4_8': -0.6213932292227808, 'B_4_9':
-0.7411909317524261, 'B_5_0': -0.7160104461670016, 'B_5_1': 0.4913603620768183,
'B_5_2': -0.06008457123578537, 'B_5_3': 0.560209662024937, 'B_5_4':
0.4875846908134067, 'B_5_5': 0.2071331727403387, 'B_5_6': 0.36900588102578546,
'B_5_7': -0.9536556657292993, 'B_5_8': -0.37268025932605653, 'B_5_9':
0.2728879149336323, 'B_6_0': -0.9714969920972365, 'B_6_1': -0.9247056515084651,
'B_6_2': 0.6967382052913589, 'B_6_3': 0.3643692261503022, 'B_6_4':
-0.464567928421719, 'B_6_5': 0.3721664234350439, 'B_6_6': 0.5790642048957413,
'B_6_7': -0.3825473194419995, 'B_6_8': -0.9186459925516088, 'B_6_9':
0.8569026668817143, 'B_7_0': -0.3065439201746516, 'B_7_1': -0.7791661086962813,
'B_7_2': -0.0161233946415682, 'B_7_3': 0.531566235414985, 'B_7_4':
-0.5511383170572328, 'B_7_5': -0.45640970073594056, 'B_7_6':
-0.5094243261388054, 'B_7_7': 0.9735126446067228, 'B_7_8': -0.5991406072135257,
'B_7_9': 0.14953002959292538, 'B_8_0': 0.8420239904807894, 'B_8_1':
-0.6991447936219971, 'B_8_2': 0.9322913117780465, 'B_8_3': 0.8860049241033368,
'B_8_4': 0.76145253437536, 'B_8_5': -0.5651780877665955, 'B_8_6':
0.265429010901747, 'B_8_7': 0.5165651164632554, 'B_8_8': 0.9623352978289308,
'B_8_9': -0.6327608322788866, 'B_9_0': 0.526731281048207, 'B_9_1':
-0.3191736835443163, 'B_9_2': -0.6422595981753554, 'B_9_3':
-0.15567365639656727, 'B_9_4': 0.5817951060542603, 'B_9_5':
-0.06959738866597676, 'B_9_6': 0.07429573840203596, 'B_9_7': 0.9804156321962401,
'B_9_8': -0.19976358687629647, 'B_9_9': 0.2284934943242949, 'B_10_0':
-0.2668164808015637, 'B_10_1': -0.21228442026916416, 'B_10_2':
0.38736869182190814, 'B_10_3': 0.8670162126678693, 'B_10_4': 0.0838707219747965,
'B_10_5': 0.531431859318584, 'B_10_6': -0.8133710135424116, 'B_10_7':
-0.5719070858847846, 'B_10_8': 0.004220649020674205, 'B_10_9':
-0.9503767872150259}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```

Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:08:46,440] Trial 89 finished with value: 187.137446800915 and
parameters: {'B_0_0': -0.4893471420501335, 'B_0_1': 0.03445455385343288,
'B_0_2': -0.9603768274525246, 'B_0_3': -0.4661983532724374, 'B_0_4':
0.02778102256222148, 'B_0_5': 0.1448355110028665, 'B_0_6': 0.8429883522495047,
'B_0_7': 0.2877843340362036, 'B_0_8': -0.1688231109906123, 'B_0_9':
0.8520953307234742, 'B_1_0': 0.5640683071078327, 'B_1_1': -0.5939823361967598,
'B_1_2': -0.7309321245293369, 'B_1_3': 0.34810703877061133, 'B_1_4':
0.11856866784242803, 'B_1_5': 0.5735537553096564, 'B_1_6': 0.47283399915579927,
'B_1_7': 0.24791883703787976, 'B_1_8': 0.9630366998252065, 'B_1_9':
0.06650766561860921, 'B_2_0': -0.7660506021265947, 'B_2_1': 0.649981658335657,
'B_2_2': -0.4913303425554962, 'B_2_3': 0.5490582533249582, 'B_2_4':
0.9050720500801017, 'B_2_5': 0.38965511764939564, 'B_2_6': 0.1901397186806062,
'B_2_7': -0.07843537216341967, 'B_2_8': -0.2684684653291015, 'B_2_9':
-0.2910505930757642, 'B_3_0': 0.27001795730498945, 'B_3_1': 0.4975207715367369,
'B_3_2': 0.35346379757914037, 'B_3_3': -0.5971479996367458, 'B_3_4':
-0.011905383315887419, 'B_3_5': 0.5774473429224647, 'B_3_6':
-0.048105925185668585, 'B_3_7': 0.07251409387627783, 'B_3_8': 0.536849375378228,
'B_3_9': 0.3193053064048812, 'B_4_0': -0.17949705665286264, 'B_4_1':
0.7829171681645535, 'B_4_2': -0.6681172040463124, 'B_4_3': 0.9811225682088429,
'B_4_4': -0.7546718946170503, 'B_4_5': -0.960399057213607, 'B_4_6':
0.772237424274554, 'B_4_7': -0.5049920534354069, 'B_4_8': 0.45480526234450736,
'B_4_9': 0.6158104423654973, 'B_5_0': 0.7107930542435914, 'B_5_1':
0.97335658661505, 'B_5_2': -0.40780137412080186, 'B_5_3': 0.416281817015973,
'B_5_4': -0.7615082806039601, 'B_5_5': -0.18451760311343823, 'B_5_6':
0.688983952758929, 'B_5_7': -0.2620370688029655, 'B_5_8': 0.1803688534605159,
'B_5_9': 0.007471418574775401, 'B_6_0': 0.28898403160184927, 'B_6_1':
0.936330890987636, 'B_6_2': -0.8796436575272237, 'B_6_3': 0.14644380465322682,
'B_6_4': 0.7187969829155638, 'B_6_5': -0.036862230738113566, 'B_6_6':
-0.8345161718356858, 'B_6_7': 0.11986791778520645, 'B_6_8': -0.3324936961688355,
'B_6_9': -0.6832152650308869, 'B_7_0': 0.6910883363012765, 'B_7_1':
0.0427670357224208, 'B_7_2': 0.857887296510552, 'B_7_3': 0.6059126791140768,
'B_7_4': 0.3820841516718614, 'B_7_5': -0.9800495033366824, 'B_7_6':
-0.14290955585626508, 'B_7_7': -0.598977162328976, 'B_7_8': 0.8133798638767582,
'B_7_9': -0.6443237103408066, 'B_8_0': -0.42360109771461474, 'B_8_1':
0.033336800114308396, 'B_8_2': -0.4685400343838031, 'B_8_3': -0.57446793898734,
'B_8_4': 0.19621666370344992, 'B_8_5': -0.8139195449450527, 'B_8_6':
-0.874827837732569, 'B_8_7': -0.25295431553157866, 'B_8_8': 0.4369156199255675,
'B_8_9': 0.10851738138241696, 'B_9_0': -0.15270459243577794, 'B_9_1':
0.13018809716467317, 'B_9_2': -0.26640606038651826, 'B_9_3': 0.8635697441555836,
'B_9_4': -0.8812883159966668, 'B_9_5': 0.34331512847907364, 'B_9_6':
0.774739676358946, 'B_9_7': 0.6016917232958017, 'B_9_8': -0.6255987510535355,
'B_9_9': -0.4040658290190291, 'B_10_0': 0.4297549186400673, 'B_10_1':
-0.8231442928220842, 'B_10_2': 0.6174861656685164, 'B_10_3': 0.4334357503174848,
'B_10_4': 0.4546211654859782, 'B_10_5': -0.791522086870169, 'B_10_6':
0.4311658787361372, 'B_10_7': 0.35603988793554064, 'B_10_8': 0.6552265318272641,
'B_10_9': -0.9076400066121904}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```
Built F_[u_1].  
Built F_[u_2].  
Built F_[u_3].  
Built F_[u_4].  
Built F_[u_5].  
Built F_[u_6].  
Built F_[u_7].  
Built F_[u_8].  
Built F_[u_9].  
Built F_[u_10].  
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:09:05,931] Trial 90 finished with value: 203.86619273206603 and parameters: {'B_0_0': -0.9442408764337714, 'B_0_1': -0.2514459422404497,

'B_0_2': -0.6953991979786773, 'B_0_3': 0.11560126558903949, 'B_0_4': 0.8263508249833627, 'B_0_5': -0.21487013393864884, 'B_0_6': 0.6830017469997562, 'B_0_7': -0.9895230684546243, 'B_0_8': -0.6563410059508152, 'B_0_9': -0.9297613914849104, 'B_1_0': -0.3402685825801641, 'B_1_1': 0.8082153994790755, 'B_1_2': -0.7980347952740042, 'B_1_3': 0.21069135795272476, 'B_1_4': -0.09603804054350051, 'B_1_5': 0.8681670604843086, 'B_1_6': -0.7474120977765542, 'B_1_7': -0.5701848560055867, 'B_1_8': -0.9315069021492433, 'B_1_9': -0.13080820572987994, 'B_2_0': -0.43989085886198454, 'B_2_1': 0.7324137491028353, 'B_2_2': 0.20778281013155508, 'B_2_3': 0.2258996611918893, 'B_2_4': 0.0631831798249789, 'B_2_5': -0.808136332286182, 'B_2_6': -0.10886300622379663, 'B_2_7': 0.9890202756228628, 'B_2_8': 0.6637539958751093, 'B_2_9': 0.7790489272508896, 'B_3_0': 0.6399881913080689, 'B_3_1': -0.7859774470066543, 'B_3_2': -0.5396524569516967, 'B_3_3': -0.9838191303143089, 'B_3_4': -0.38904995274133924, 'B_3_5': -0.7749573433539447, 'B_3_6': -0.053869176302865585, 'B_3_7': 0.3507124273763258, 'B_3_8': 0.7415204841753926, 'B_3_9': 0.49344998687997843, 'B_4_0': 0.8683994796414403, 'B_4_1': -0.8832813850202982, 'B_4_2': -0.965883925875248, 'B_4_3': 0.31944946539570096, 'B_4_4': -0.4170822277599525, 'B_4_5': -0.4880679881105692, 'B_4_6': -0.3348435663662873, 'B_4_7': -0.45580380537311893, 'B_4_8': 0.022644084477142723, 'B_4_9': -0.11850607449621764, 'B_5_0': 0.024608531725779592, 'B_5_1': 0.34197295077086354, 'B_5_2': -0.22348159223712638, 'B_5_3': -0.49241077945852957, 'B_5_4': -0.660922937118684, 'B_5_5': -0.2980412433254015, 'B_5_6': 0.4787768209874206, 'B_5_7': 0.5082757220207097, 'B_5_8': 0.3304726656347363, 'B_5_9': -0.5266242128923884, 'B_6_0': 0.17441076679230139, 'B_6_1': 0.13996552992215894, 'B_6_2': -0.031152927742743586, 'B_6_3': 0.09895839304005105, 'B_6_4': -0.01625454740513721, 'B_6_5': 0.12960360482673416, 'B_6_6': 0.4267017901498349, 'B_6_7': -0.7729361750811893, 'B_6_8': 0.49425898925506706, 'B_6_9': -0.7209454227473635, 'B_7_0': -0.8580765631474581, 'B_7_1': 0.9426377787185058, 'B_7_2': 0.9048437023634741, 'B_7_3': -0.8703356410176831, 'B_7_4': -0.11235818587957491, 'B_7_5': -0.621235022166446, 'B_7_6':

-0.02901710367385224, 'B_7_7': 0.6537509411391831, 'B_7_8': 0.8922113139909227, 'B_7_9': 0.26582025383404395, 'B_8_0': -0.28926677892060226, 'B_8_1': 0.20458461997630484, 'B_8_2': -0.3722374139052683, 'B_8_3': 0.8114508088474794, 'B_8_4': 0.7333929169495572, 'B_8_5': 0.11529886555372881, 'B_8_6': -0.3468981475468861, 'B_8_7': -0.40908844234570285, 'B_8_8': -0.9238323704839823, 'B_8_9': 0.6253252850300846, 'B_9_0': -0.0982412381827642, 'B_9_1': -0.3462368017674764, 'B_9_2': -0.05720149710444733, 'B_9_3': 0.6311012211002109, 'B_9_4': -0.743263868739108, 'B_9_5': 0.7081588567188182, 'B_9_6': 0.787067493514541, 'B_9_7': 0.8844816717670987, 'B_9_8': -0.20297377500319702, 'B_9_9': -0.5600620000054097, 'B_10_0': -0.25019117943593694, 'B_10_1': -0.3308340494618215, 'B_10_2': -0.6412152532443371, 'B_10_3': 0.21238800680094605, 'B_10_4': -0.04628416219291487, 'B_10_5': 0.7240879648015273, 'B_10_6': -0.9264230306455247, 'B_10_7': 0.2848584965591501, 'B_10_8': 0.520638779452891, 'B_10_9': 0.513783406992848}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].

Built F_[u_2].

Built F_[u_3].

Built F_[u_4].

Built F_[u_5].

Built F_[u_6].

Built F_[u_7].

Built F_[u_8].

Built F_[u_9].

Built F_[u_10].

Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:09:23,166] Trial 91 finished with value: 190.5583205847741 and parameters: {'B_0_0': 0.7644264548491397, 'B_0_1': 0.45348681080239817, 'B_0_2': 0.8470639220624387, 'B_0_3': -0.3313399323546349, 'B_0_4': 0.006354645168055284, 'B_0_5': -0.9621222666998764, 'B_0_6': -0.9762239771394834, 'B_0_7': -0.5145492915059906, 'B_0_8': -0.7904017496646112, 'B_0_9': -0.47478149148720783, 'B_1_0': -0.6394542768337732, 'B_1_1': -0.9335303508194764, 'B_1_2': 0.8104222098839777, 'B_1_3': -0.9737181998499947, 'B_1_4': 0.467442645787429, 'B_1_5': -0.6887471838993573, 'B_1_6': 0.8162146515592079, 'B_1_7': 0.7777359603879739, 'B_1_8': 0.30472426981112943, 'B_1_9': 0.3410236166051981, 'B_2_0': -0.9794284989000167, 'B_2_1': 0.9600457009049321, 'B_2_2': 0.7481085550801125, 'B_2_3': 0.7823103269455822, 'B_2_4': 0.3153041833607264, 'B_2_5': -0.3073218812640177, 'B_2_6': 0.779424325199152, 'B_2_7': -0.41279648226063914, 'B_2_8': 0.12416885581756176, 'B_2_9': -0.7925326747591128, 'B_3_0': 0.8354721476628453, 'B_3_1': -0.7112744936129975, 'B_3_2': -0.5447043246537936, 'B_3_3': 0.11510563309386557, 'B_3_4': 0.5229041404217034, 'B_3_5': 0.2363155857538719, 'B_3_6': -0.5303748526556908, 'B_3_7': -0.9881327757955691, 'B_3_8': 0.5091581490087296, 'B_3_9': 0.9607105053039655, 'B_4_0': 0.613627978206059, 'B_4_1':

```

-0.07879089604990164, 'B_4_2': 0.7994591775702311, 'B_4_3': -0.178501607225667,
'B_4_4': 0.049369486802065765, 'B_4_5': -0.04528306055654718, 'B_4_6':
-0.10370473212226061, 'B_4_7': 0.5448777232019333, 'B_4_8': 0.05300243639359414,
'B_4_9': -0.3629000983900722, 'B_5_0': -0.4788468280362129, 'B_5_1':
-0.9152603124044819, 'B_5_2': 0.16472642052843112, 'B_5_3': -0.5346763975059523,
'B_5_4': -0.6966948344075614, 'B_5_5': 0.30458332837880175, 'B_5_6':
0.35879433048666676, 'B_5_7': 0.4992666866701787, 'B_5_8': 0.007082254750411132,
'B_5_9': 0.7802281183036954, 'B_6_0': 0.863958651728491, 'B_6_1':
0.6085944607891289, 'B_6_2': 0.26847509437621064, 'B_6_3': -0.3886325118894465,
'B_6_4': 0.9562309448080379, 'B_6_5': -0.4472103889599621, 'B_6_6':
-0.5887392944589465, 'B_6_7': -0.11453485470256619, 'B_6_8':
-0.8774556699894709, 'B_6_9': 0.9474190400501459, 'B_7_0': 0.17509844170452116,
'B_7_1': 0.112765141495067, 'B_7_2': 0.30847663420144533, 'B_7_3':
-0.0058931774011308224, 'B_7_4': -0.9265142361528077, 'B_7_5':
0.22237635190398564, 'B_7_6': 0.42802427637373275, 'B_7_7': -0.1648368599643748,
'B_7_8': 0.509445232700962, 'B_7_9': -0.5405374273099272, 'B_8_0':
-0.5518723496889556, 'B_8_1': -0.6703943643786036, 'B_8_2':
-0.25345037561745953, 'B_8_3': 0.6870838300703854, 'B_8_4': 0.6492658830551474,
'B_8_5': -0.2320080682322625, 'B_8_6': 0.1764550972408685, 'B_8_7':
0.43721630662033295, 'B_8_8': -0.5124105862365145, 'B_8_9': -0.6570806499284008,
'B_9_0': 0.9347500665603758, 'B_9_1': 0.23276348015504467, 'B_9_2':
0.9532031292306544, 'B_9_3': -0.32124314547818633, 'B_9_4': 0.40200276594878037,
'B_9_5': -0.5492447996483494, 'B_9_6': -0.4728236608776524, 'B_9_7':
0.023566200113855462, 'B_9_8': -0.35715611149401894, 'B_9_9':
-0.6581093035956427, 'B_10_0': -0.4694962100611345, 'B_10_1':
-0.919218178957588, 'B_10_2': 0.4047587705254234, 'B_10_3':
-0.015609018969994004, 'B_10_4': -0.8033987285190393, 'B_10_5':
-0.6500458936055874, 'B_10_6': -0.7869816145369648, 'B_10_7':
0.22293376038623536, 'B_10_8': -0.8034634800472623, 'B_10_9':
0.5034790194097278}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:09:39,886] Trial 92 finished with value: 186.42080454604493 and parameters: {'B_0_0': -0.9493883797456255, 'B_0_1': -0.4962359397681336, 'B_0_2': 0.7735200182659763, 'B_0_3': 0.43269200988204704, 'B_0_4':

-0.8455227008071383, 'B_0_5': 0.6358296868806559, 'B_0_6':
 -0.013790459350394468, 'B_0_7': 0.6866303937942819, 'B_0_8':
 0.19946475341177883, 'B_0_9': 0.3600294034250602, 'B_1_0': 0.9771330895835022,
 'B_1_1': -0.529975619070444, 'B_1_2': 0.24541223802754897, 'B_1_3':
 -0.07856364768635127, 'B_1_4': -0.911885133213071, 'B_1_5': -0.9613481298243273,
 'B_1_6': -0.7539274159206398, 'B_1_7': 0.2952016462663254, 'B_1_8':
 0.1329829140685166, 'B_1_9': 0.10579053110418024, 'B_2_0': -0.926923610194045,
 'B_2_1': -0.060780067207185984, 'B_2_2': 0.8407531793282066, 'B_2_3':
 -0.20276809876491975, 'B_2_4': 0.24734649018114352, 'B_2_5': 0.5540409777338242,
 'B_2_6': 0.5621720527009375, 'B_2_7': 0.03817456314263734, 'B_2_8':
 0.708744851896626, 'B_2_9': -0.21507137513978103, 'B_3_0': 0.7696125652739008,
 'B_3_1': -0.5143502105368203, 'B_3_2': -0.4351715141586693, 'B_3_3':
 -0.8848211640859778, 'B_3_4': 0.5852520363537201, 'B_3_5': -0.16423063225924295,
 'B_3_6': -0.5363491044554298, 'B_3_7': -0.5310552503191723, 'B_3_8':
 0.7947718836338817, 'B_3_9': 0.661154868912853, 'B_4_0': 0.10599115300722306,
 'B_4_1': 0.22989938482312766, 'B_4_2': -0.050835370310423844, 'B_4_3':
 -0.21450024509995846, 'B_4_4': 0.8226442248122963, 'B_4_5': -0.7412868470298019,
 'B_4_6': -0.1614753563618686, 'B_4_7': -0.3077674221840069, 'B_4_8':
 -0.003661573902726256, 'B_4_9': -0.08388888840055342, 'B_5_0':
 0.03740127266914128, 'B_5_1': -0.7129604250905517, 'B_5_2': 0.02602017299628856,
 'B_5_3': -0.5100107485033225, 'B_5_4': -0.6250239290033813, 'B_5_5':
 -0.8616282468033414, 'B_5_6': 0.5200520663371591, 'B_5_7': -0.14366499670401123,
 'B_5_8': 0.2507544929572816, 'B_5_9': 0.4506348197433918, 'B_6_0':
 0.3550469973208359, 'B_6_1': -0.05840857620242912, 'B_6_2':
 -0.22196091267557494, 'B_6_3': -0.24559937891148587, 'B_6_4':
 0.2772966179215115, 'B_6_5': -0.010307177215090668, 'B_6_6': 0.6912818510172509,
 'B_6_7': 0.74190992896689, 'B_6_8': -0.6522904515470934, 'B_6_9':
 -0.7202573699394632, 'B_7_0': 0.258193833551279, 'B_7_1': -0.3935730188366867,
 'B_7_2': -0.9532299768451042, 'B_7_3': 0.769512485338651, 'B_7_4':
 -0.5407364808572597, 'B_7_5': 0.8499756903379019, 'B_7_6': 0.43052021273894225,
 'B_7_7': 0.520053910024739, 'B_7_8': -0.5911062383295423, 'B_7_9':
 -0.5262950372787104, 'B_8_0': 0.042110152571650294, 'B_8_1': -0.649242291753273,
 'B_8_2': -0.3852452484875961, 'B_8_3': -0.3535042936633471, 'B_8_4':
 -0.6739327321595271, 'B_8_5': -0.5837379898775817, 'B_8_6': 0.571332647176348,
 'B_8_7': -0.6589522079964086, 'B_8_8': -0.9336543641490335, 'B_8_9':
 0.2561227864052982, 'B_9_0': 0.9424465763970509, 'B_9_1': -0.33419664704621965,
 'B_9_2': -0.7550617076714204, 'B_9_3': 0.13936690422771103, 'B_9_4':
 0.8045813379693794, 'B_9_5': 0.562442324060388, 'B_9_6': 0.2425814194571776,
 'B_9_7': -0.36618813671695327, 'B_9_8': 0.0829227430869155, 'B_9_9':
 -0.8093352142175744, 'B_10_0': 0.3543119400757264, 'B_10_1':
 0.44569011741734865, 'B_10_2': 0.9207626133616393, 'B_10_3':
 -0.4744132296847582, 'B_10_4': -0.9534562902948152, 'B_10_5':
 0.09030475956575823, 'B_10_6': 0.4094451987056926, 'B_10_7': 0.5750068022483161,
 'B_10_8': -0.606396912610875, 'B_10_9': -0.47172931930669315}. Best is trial 36
 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u₁].

Built F__[u₂].

```

Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```

[I 2025-04-29 14:09:56,072] Trial 93 finished with value: 140.1757366207586 and
parameters: {'B_0_0': 0.024930032029246707, 'B_0_1': -0.3354103962593963,
'B_0_2': 0.7138127434431412, 'B_0_3': -0.051954895597333905, 'B_0_4':
0.5931657153219658, 'B_0_5': 0.15962712575506033, 'B_0_6': 0.5407184804018816,
'B_0_7': -0.3313074866699588, 'B_0_8': 0.6562049114220161, 'B_0_9':
-0.44634681520897723, 'B_1_0': 0.09127563984651577, 'B_1_1':
-0.5662659159157291, 'B_1_2': -0.048457487180137204, 'B_1_3':
-0.36001308517276776, 'B_1_4': -0.4619272073383158, 'B_1_5': 0.9246303356955141,
'B_1_6': -0.6730532711362733, 'B_1_7': 0.5689051581656426, 'B_1_8':
0.9586466197767656, 'B_1_9': -0.9505950621946458, 'B_2_0': 0.3940742798033481,
'B_2_1': -0.4892346249985515, 'B_2_2': -0.9098341078299572, 'B_2_3':
-0.19227863366587106, 'B_2_4': 0.8617312641383326, 'B_2_5': -0.9667051131533446,
'B_2_6': 0.1740136700209678, 'B_2_7': 0.0027650276187421996, 'B_2_8':
0.739300645454674, 'B_2_9': -0.8236766052491612, 'B_3_0': 0.35942153917556774,
'B_3_1': 0.3566172028978891, 'B_3_2': 0.3347784333873207, 'B_3_3':
0.8758222999713732, 'B_3_4': 0.6397133074336232, 'B_3_5': -0.9068940020378581,
'B_3_6': -0.05725595560506391, 'B_3_7': -0.2020529487213597, 'B_3_8':
-0.3759085022402119, 'B_3_9': -0.09635328419488742, 'B_4_0':
-0.22929551384656466, 'B_4_1': -0.13397887824024268, 'B_4_2':
0.9489384939051122, 'B_4_3': -0.8145067099958301, 'B_4_4': 0.8360171222806849,
'B_4_5': -0.03913130732736714, 'B_4_6': 0.2450096460890916, 'B_4_7':
0.6537622739030338, 'B_4_8': -0.276283202284474, 'B_4_9': -0.13776462198582562,
'B_5_0': -0.17915340904814636, 'B_5_1': -0.24803257537919554, 'B_5_2':
0.8425480568608519, 'B_5_3': 0.651281184562164, 'B_5_4': -0.7741800200713074,
'B_5_5': -0.7759157620953498, 'B_5_6': -0.14281836059107744, 'B_5_7':
-0.794658394069038, 'B_5_8': 0.235804328636817, 'B_5_9': -0.7054906176217668,
'B_6_0': 0.044299062654266086, 'B_6_1': -0.9757045049326388, 'B_6_2':
-0.36313923014295946, 'B_6_3': 0.35083763287185765, 'B_6_4':
-0.0316028983488601, 'B_6_5': 0.9469735822779966, 'B_6_6': 0.7407255162110973,
'B_6_7': -0.2790398574945606, 'B_6_8': -0.8999366478848962, 'B_6_9':
0.6843400742277372, 'B_7_0': 0.8142439344230494, 'B_7_1': 0.6348422389696928,
'B_7_2': -0.7366591169788655, 'B_7_3': 0.9858719661363298, 'B_7_4':
-0.47800324132550787, 'B_7_5': 0.4439257273605728, 'B_7_6': 0.8079090551447288,
'B_7_7': -0.5233395136251953, 'B_7_8': 0.8083519382605908, 'B_7_9':
-0.37722647319416336, 'B_8_0': 0.9432981803204312, 'B_8_1': -0.4331051596466603,
'B_8_2': 0.8978029330205137, 'B_8_3': -0.8839815912325036, 'B_8_4':

```

0.41435858310622975, 'B_8_5': 0.06358004499726744, 'B_8_6':
-0.04380338729699329, 'B_8_7': -0.5474812577734368, 'B_8_8': 0.4070796637574541,
'B_8_9': -0.612197206936312, 'B_9_0': 0.10462675604984106, 'B_9_1':
-0.4885741811803034, 'B_9_2': 0.902130331967085, 'B_9_3': -0.9333102362054074,
'B_9_4': -0.5945380053797515, 'B_9_5': -0.5102249993528535, 'B_9_6':
-0.17014570497972348, 'B_9_7': -0.529194121097294, 'B_9_8':
-0.017128801267805138, 'B_9_9': 0.7374817360519073, 'B_10_0':
0.41988557439015817, 'B_10_1': 0.1583247214595198, 'B_10_2':
-0.8137737075942943, 'B_10_3': 0.38410437461076574, 'B_10_4':
-0.6604997170643391, 'B_10_5': 0.5670074242041621, 'B_10_6':
-0.4948876181498967, 'B_10_7': -0.16424351792461622, 'B_10_8':
-0.2960689994105027, 'B_10_9': 0.7104515510562164}. Best is trial 36 with value:
95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:10:12,493] Trial 94 finished with value: 201.15367906256205 and
parameters: {'B_0_0': -0.7394419717474519, 'B_0_1': 0.1299768671218815, 'B_0_2':
-0.7087180133027368, 'B_0_3': 0.40549575554172956, 'B_0_4': 0.8020546930330728,
'B_0_5': -0.1294768453428382, 'B_0_6': -0.02357732467534779, 'B_0_7':
0.8462972162358693, 'B_0_8': 0.04156294952572703, 'B_0_9':
-0.024555092661869393, 'B_1_0': 0.1550916233545223, 'B_1_1':
0.37467487799864485, 'B_1_2': -0.7783709173276778, 'B_1_3': 0.6357988449282017,
'B_1_4': -0.12770004564254667, 'B_1_5': -0.8467499504029786, 'B_1_6':
0.566641685012462, 'B_1_7': -0.2436390560611813, 'B_1_8': -0.7912723597117932,
'B_1_9': 0.26488249829563615, 'B_2_0': 0.7526600797986105, 'B_2_1':
0.7677519486396929, 'B_2_2': 0.9061736801217668, 'B_2_3': 0.7280568103380423,
'B_2_4': -0.985760979695959, 'B_2_5': -0.23712249050148737, 'B_2_6':
0.6307143082128961, 'B_2_7': -0.26481904812302104, 'B_2_8': 0.20106604263181804,
'B_2_9': 0.21050755580157343, 'B_3_0': -0.26724989800313037, 'B_3_1':
0.8008583760169494, 'B_3_2': 0.006392123864512822, 'B_3_3':
-0.06972727224374842, 'B_3_4': 0.9850772164105539, 'B_3_5':
-0.012869279136219047, 'B_3_6': -0.6367714611814095, 'B_3_7':
0.025877457269607218, 'B_3_8': 0.6217531363294615, 'B_3_9': 0.5481834472585971,
'B_4_0': 0.5713918949147725, 'B_4_1': 0.4118870923158653, 'B_4_2':
-0.48664869844795733, 'B_4_3': 0.7169264899636583, 'B_4_4': 0.7790180614450049,

'B_4_5': -0.6055776606952729, 'B_4_6': -0.7316978396282401, 'B_4_7': 0.9416978378064418, 'B_4_8': 0.7292814480277285, 'B_4_9': 0.7786075714756944, 'B_5_0': 0.7650252415563024, 'B_5_1': 0.0825399115867822, 'B_5_2': 0.28987543859828335, 'B_5_3': 0.08981844722461041, 'B_5_4': -0.8886114286031485, 'B_5_5': 0.4860961089984899, 'B_5_6': 0.010557426005042947, 'B_5_7': 0.292463968084655, 'B_5_8': 0.005226813851559875, 'B_5_9': -0.016213581801537202, 'B_6_0': -0.6498575632340217, 'B_6_1': -0.43814684402052506, 'B_6_2': -0.6754716699509675, 'B_6_3': -0.6025807346618219, 'B_6_4': 0.9364952132322037, 'B_6_5': -0.6868353773710472, 'B_6_6': -0.45684455987568884, 'B_6_7': 0.17936752232239828, 'B_6_8': 0.21815358094855086, 'B_6_9': -0.6851312231072184, 'B_7_0': 0.5815605030547204, 'B_7_1': 0.5647016784673935, 'B_7_2': -0.5201109456762956, 'B_7_3': -0.4184407207025883, 'B_7_4': 0.867215973609804, 'B_7_5': 0.6448294630393665, 'B_7_6': 0.3916381254637973, 'B_7_7': -0.827178221409011, 'B_7_8': -0.16183711981290005, 'B_7_9': 0.1174177067121176, 'B_8_0': -0.6189256497921216, 'B_8_1': -0.10251136742571543, 'B_8_2': 0.2000312374276061, 'B_8_3': -0.3417941733715095, 'B_8_4': 0.5562707415219645, 'B_8_5': 0.6740901716448349, 'B_8_6': -0.3492532505205399, 'B_8_7': 0.6705411658950651, 'B_8_8': -0.7849183991160302, 'B_8_9': 0.6687036152728596, 'B_9_0': -0.6875541452981191, 'B_9_1': -0.8258042380307424, 'B_9_2': 0.28071530079320217, 'B_9_3': -0.0555640158045172, 'B_9_4': 0.8076233888174089, 'B_9_5': -0.2708853917401186, 'B_9_6': -0.4214385183934952, 'B_9_7': 0.0014551141655126942, 'B_9_8': -0.08722655217805064, 'B_9_9': -0.30803246848996724, 'B_10_0': 0.38360777142269065, 'B_10_1': 0.5139324311495383, 'B_10_2': 0.6989082586796704, 'B_10_3': 0.9362560773264788, 'B_10_4': -0.4620207395776411, 'B_10_5': 0.42786509533012373, 'B_10_6': -0.20936869371842448, 'B_10_7': 0.350622304463194, 'B_10_8': -0.04755625310854372, 'B_10_9': 0.9776707434805045}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:10:29,306] Trial 95 finished with value: 221.16625628135506 and parameters: {'B_0_0': 0.45322622886568475, 'B_0_1': 0.8959327748599626, 'B_0_2': 0.6632397079727477, 'B_0_3': 0.7977440021034747, 'B_0_4': 0.6225500104850563, 'B_0_5': -0.08086492185569549, 'B_0_6': -0.49145886237316894, 'B_0_7': -0.25676605260796603, 'B_0_8': -0.9499322593728982, 'B_0_9':

0.047506471580041065, 'B_1_0': 0.010014444535409117, 'B_1_1':
 0.04999592036436673, 'B_1_2': 0.9385818254446865, 'B_1_3': 0.8146789268405819,
 'B_1_4': -0.6698086989063862, 'B_1_5': -0.25961260731933944, 'B_1_6':
 -0.15056676466274144, 'B_1_7': -0.2761448226537374, 'B_1_8': 0.5910220079357478,
 'B_1_9': 0.04759458633782221, 'B_2_0': 0.14275763760381754, 'B_2_1':
 0.6989668467960926, 'B_2_2': 0.6057377437624101, 'B_2_3': -0.7282211848158067,
 'B_2_4': 0.423550876485596, 'B_2_5': -0.1278364647739696, 'B_2_6':
 -0.40659517193624906, 'B_2_7': -0.890962734613671, 'B_2_8':
 -0.02524587051600713, 'B_2_9': -0.2862353721654278, 'B_3_0':
 -0.4920041814407264, 'B_3_1': 0.38545002460197586, 'B_3_2': 0.5335810668848624,
 'B_3_3': -0.5053167523217389, 'B_3_4': -0.8920940753748924, 'B_3_5':
 0.6905303471191653, 'B_3_6': -0.9702797124154794, 'B_3_7': 0.8654827379645145,
 'B_3_8': 0.25268735048575763, 'B_3_9': -0.3527436623013859, 'B_4_0':
 -0.5019737390380623, 'B_4_1': -0.972168784756339, 'B_4_2': -0.5237665943184633,
 'B_4_3': -0.7172979629960605, 'B_4_4': 0.22208911703182843, 'B_4_5':
 -0.6602397345842734, 'B_4_6': 0.05668707832250619, 'B_4_7': 0.1784305162420523,
 'B_4_8': -0.3998774372731092, 'B_4_9': -0.920571400942937, 'B_5_0':
 -0.9162373163838917, 'B_5_1': -0.9378322822560431, 'B_5_2': 0.4734668409303615,
 'B_5_3': -0.3603407844676897, 'B_5_4': 0.5866743682159739, 'B_5_5':
 -0.13516900879115867, 'B_5_6': -0.09306376035667163, 'B_5_7':
 0.025507552968955904, 'B_5_8': 0.4218022836227058, 'B_5_9': -0.44846432565718,
 'B_6_0': -0.7492570479236167, 'B_6_1': 0.11099352367909376, 'B_6_2':
 -0.3425843059793763, 'B_6_3': -0.11386719893728592, 'B_6_4': 0.2909544825078647,
 'B_6_5': -0.011631032540284192, 'B_6_6': 0.23815205785966898, 'B_6_7':
 0.08751600418498318, 'B_6_8': 0.4747968164334764, 'B_6_9': 0.4077545030690075,
 'B_7_0': 0.06769598578106595, 'B_7_1': 0.35422798130221955, 'B_7_2':
 0.7396480124997733, 'B_7_3': -0.13410285017942614, 'B_7_4': -0.1727291159573886,
 'B_7_5': 0.5232063292877127, 'B_7_6': 0.1634565973431481, 'B_7_7':
 -0.8494065751592746, 'B_7_8': 0.776381791068341, 'B_7_9': -0.9290494575694725,
 'B_8_0': 0.4918247852952913, 'B_8_1': 0.5124124025545396, 'B_8_2':
 0.8971599488582223, 'B_8_3': -0.882813871363405, 'B_8_4': 0.19177658435329215,
 'B_8_5': -0.14287567015843516, 'B_8_6': -0.3353467493781068, 'B_8_7':
 -0.8111692506078354, 'B_8_8': 0.12405515694919211, 'B_8_9':
 -0.041499363164264946, 'B_9_0': -0.7829821728609719, 'B_9_1':
 0.22272922405222517, 'B_9_2': 0.4711471781008181, 'B_9_3': 0.5292110360862234,
 'B_9_4': 0.8026389127596063, 'B_9_5': -0.1792479310602212, 'B_9_6':
 0.8970840818718311, 'B_9_7': 0.7235502220637184, 'B_9_8': -0.4029268893726007,
 'B_9_9': -0.6746195860480722, 'B_10_0': -0.12792899566653348, 'B_10_1':
 -0.1942405582471861, 'B_10_2': 0.14992422455578458, 'B_10_3':
 0.16510474000894093, 'B_10_4': 0.4949742290723509, 'B_10_5':
 0.01958209744710815, 'B_10_6': -0.7824511709372842, 'B_10_7':
 0.7094168443951501, 'B_10_8': -0.37666612529332955, 'B_10_9':
 0.9331689714070521}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].

```

Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```

[I 2025-04-29 14:10:46,073] Trial 96 finished with value: 263.28842326725453 and
parameters: {'B_0_0': -0.7566369848140551, 'B_0_1': 0.9073373210203564, 'B_0_2':
-0.8097966223726164, 'B_0_3': -0.5325484468648223, 'B_0_4': -0.8977717146686477,
'B_0_5': 0.40612403034521005, 'B_0_6': 0.6935914984121476, 'B_0_7':
-0.13053925800435873, 'B_0_8': 0.8484737085840108, 'B_0_9': 0.17879216207608728,
'B_1_0': 0.257414128184116, 'B_1_1': 0.6839300193675244, 'B_1_2':
-0.40233559399735896, 'B_1_3': -0.2885538717713009, 'B_1_4': 0.8037848000053958,
'B_1_5': 0.7939077221567523, 'B_1_6': 0.9434358968777243, 'B_1_7':
0.7956398665545088, 'B_1_8': -0.7378033686539454, 'B_1_9': 0.640974086551235,
'B_2_0': -0.1785198133430037, 'B_2_1': 0.971786684772957, 'B_2_2':
-0.054883690628973913, 'B_2_3': 0.10659544335439675, 'B_2_4':
0.058954332208461135, 'B_2_5': -0.35296068088060994, 'B_2_6':
-0.841456263992022, 'B_2_7': 0.7833894361957987, 'B_2_8': 0.5992981088020479,
'B_2_9': -0.7137369744918549, 'B_3_0': 0.5480011772731443, 'B_3_1':
0.2491763987802078, 'B_3_2': -0.17989529992310782, 'B_3_3': -0.4870381247907858,
'B_3_4': 0.7122629913344203, 'B_3_5': 0.396372753402225, 'B_3_6':
0.09452502074889102, 'B_3_7': 0.21694569711895895, 'B_3_8': -0.8096889594772816,
'B_3_9': -0.051638873395839946, 'B_4_0': -0.6064734202843101, 'B_4_1':
-0.5774003155625258, 'B_4_2': 0.3193818453279962, 'B_4_3': 0.8142871806318903,
'B_4_4': -0.1905619112528738, 'B_4_5': -0.5786564394358631, 'B_4_6':
0.29451893572857535, 'B_4_7': 0.7349503517736291, 'B_4_8': 0.4916712262225318,
'B_4_9': 0.7301393164624426, 'B_5_0': 0.07237824860138264, 'B_5_1':
0.498714963850998, 'B_5_2': -0.7126721724693058, 'B_5_3': -0.30041626158088086,
'B_5_4': -0.21869265432658558, 'B_5_5': 0.02236557756714852, 'B_5_6':
0.849571193308511, 'B_5_7': 0.741107189239905, 'B_5_8': -0.7236395252674146,
'B_5_9': -0.6478764892727229, 'B_6_0': -0.28187171181395554, 'B_6_1':
-0.41432714553646566, 'B_6_2': 0.22724648724939933, 'B_6_3': 0.797496574573799,
'B_6_4': 0.9153372525117218, 'B_6_5': 0.30413384601974847, 'B_6_6':
0.6695552187741516, 'B_6_7': 0.04990035645697821, 'B_6_8': -0.5775126452171213,
'B_6_9': 0.04888305594508524, 'B_7_0': 0.10906355084654917, 'B_7_1':
0.48006597714284194, 'B_7_2': -0.5546316803794488, 'B_7_3': -0.2607454434682712,
'B_7_4': -0.26922561095515873, 'B_7_5': 0.9096903995675065, 'B_7_6':
-0.11098897855309653, 'B_7_7': 0.50508605525033, 'B_7_8': 0.4966065351213893,
'B_7_9': -0.46422251408958204, 'B_8_0': 0.14230198495602964, 'B_8_1':
0.7807197287709233, 'B_8_2': -0.04910851547524009, 'B_8_3': 0.8813326645382411,
'B_8_4': -0.5653727059381601, 'B_8_5': -0.8431649705674191, 'B_8_6':
0.7750786688266036, 'B_8_7': 0.688408771965189, 'B_8_8': 0.9753279543490125,
'B_8_9': 0.35519910646260366, 'B_9_0': 0.8646420948009561, 'B_9_1':

```

```

0.6473590696665394, 'B_9_2': 0.681964139577772, 'B_9_3': 0.1369003094994512,
'B_9_4': -0.23291809612491254, 'B_9_5': 0.8604249836356981, 'B_9_6':
-0.1344079922805289, 'B_9_7': 0.07686234715434481, 'B_9_8': -0.1066936718921283,
'B_9_9': 0.12834026164119017, 'B_10_0': -0.05839994359519218, 'B_10_1':
-0.8950242340126378, 'B_10_2': 0.7564554082952433, 'B_10_3':
-0.11010023854677586, 'B_10_4': 0.17610842404455052, 'B_10_5':
-0.7144081505391857, 'B_10_6': -0.9832920609943745, 'B_10_7':
-0.504325614585587, 'B_10_8': 0.40508034362403356, 'B_10_9':
-0.7935948303765004}. Best is trial 36 with value: 95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:11:02,405] Trial 97 finished with value: 176.0910881188633 and
parameters: {'B_0_0': 0.8095971319652064, 'B_0_1': -0.5642553126747545, 'B_0_2':
-0.5849046437022839, 'B_0_3': -0.6306142359894389, 'B_0_4': -0.9623162083166978,
'B_0_5': 0.5381748275869045, 'B_0_6': 0.742568371865612, 'B_0_7':
0.22604215118007387, 'B_0_8': -0.6173640362738726, 'B_0_9': -0.4251495660570581,
'B_1_0': -0.0476651392173727, 'B_1_1': 0.8996049842141356, 'B_1_2':
-0.5002442838008692, 'B_1_3': 0.45865888976137836, 'B_1_4':
-0.11989251788001776, 'B_1_5': -0.4002622502471568, 'B_1_6':
-0.08728644469621483, 'B_1_7': -0.8490106992683596, 'B_1_8': 0.5914009944421177,
'B_1_9': -0.508262182593725, 'B_2_0': 0.4463160039220815, 'B_2_1':
-0.8546702245325745, 'B_2_2': 0.9010951412080583, 'B_2_3': -0.6078303288967167,
'B_2_4': 0.7578293108900194, 'B_2_5': 0.6127569550385603, 'B_2_6':
0.9263626496812036, 'B_2_7': -0.04377649132741579, 'B_2_8': 0.45024664181498397,
'B_2_9': -0.2559928990651875, 'B_3_0': -0.6249508887210062, 'B_3_1':
-0.11270965279177136, 'B_3_2': -0.6618169495308593, 'B_3_3': 0.9544483531170402,
'B_3_4': -0.7994319475413816, 'B_3_5': 0.1766392124361884, 'B_3_6':
0.2503984334030307, 'B_3_7': 0.0753974948906273, 'B_3_8': 0.32112805370943054,
'B_3_9': -0.759962065476456, 'B_4_0': 0.10893667792647643, 'B_4_1':
-0.4213191091678241, 'B_4_2': 0.8715486053360921, 'B_4_3': 0.518901873810657,
'B_4_4': 0.9355786746762904, 'B_4_5': -0.9156389819144247, 'B_4_6':
-0.42543631314219676, 'B_4_7': -0.603063278456069, 'B_4_8': -0.9679139171225756,
'B_4_9': -0.5101950541122835, 'B_5_0': 0.3539764473704492, 'B_5_1':
-0.5324439341735543, 'B_5_2': 0.01782682205558417, 'B_5_3': 0.7678798676306973,
'B_5_4': 0.3836406822911689, 'B_5_5': -0.2086344415860567, 'B_5_6':

```

```

-0.2525115009540255, 'B_5_7': -0.08764308826320022, 'B_5_8':
-0.6394579467013853, 'B_5_9': 0.26532151948365645, 'B_6_0': -0.2689774181728811,
'B_6_1': 0.6976833915388465, 'B_6_2': 0.5079742669144762, 'B_6_3':
0.21881606490247685, 'B_6_4': -0.2688795991539582, 'B_6_5': -0.5743349732865008,
'B_6_6': 0.5467806909291333, 'B_6_7': -0.026909983997648657, 'B_6_8':
-0.5449949244088321, 'B_6_9': -0.6035542622207877, 'B_7_0': -0.6625062090873091,
'B_7_1': 0.548779168487078, 'B_7_2': -0.5366053731053828, 'B_7_3':
0.05275168177989098, 'B_7_4': -0.09593820527334818, 'B_7_5':
0.15762958047200204, 'B_7_6': 0.7467667027349647, 'B_7_7': 0.7281398323618267,
'B_7_8': -0.09686937778677784, 'B_7_9': -0.9595748235452478, 'B_8_0':
-0.14564126468567973, 'B_8_1': 0.21026320409979804, 'B_8_2':
0.11437422301416222, 'B_8_3': 0.29964548948440695, 'B_8_4': -0.6849300497032946,
'B_8_5': 0.2901394955858505, 'B_8_6': -0.628349794887331, 'B_8_7':
0.49603748177956297, 'B_8_8': -0.31860652573989356, 'B_8_9':
0.41559295809527796, 'B_9_0': -0.8952261141659079, 'B_9_1':
-0.18243714932999144, 'B_9_2': 0.40596835408139875, 'B_9_3':
-0.5407963776636416, 'B_9_4': 0.5846683486985007, 'B_9_5': 0.2589438715477652,
'B_9_6': 0.1733723088566006, 'B_9_7': -0.2899732686754125, 'B_9_8':
-0.034746088645387196, 'B_9_9': 0.1927516159361491, 'B_10_0':
-0.05452501444851687, 'B_10_1': 0.3926603814023091, 'B_10_2':
-0.13128007244822004, 'B_10_3': -0.5894991001507799, 'B_10_4':
-0.5664111100539264, 'B_10_5': -0.9785085376909767, 'B_10_6':
-0.3941037231938286, 'B_10_7': 0.49285692814746884, 'B_10_8':
0.5824421634984669, 'B_10_9': -0.589114266650415}. Best is trial 36 with value:
95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:11:20,271] Trial 98 finished with value: 167.93771651914204 and
parameters: {'B_0_0': 0.9089071138053872, 'B_0_1': 0.004757531090650913,
'B_0_2': 0.8256179230566352, 'B_0_3': 0.4093952580860296, 'B_0_4':
0.44041817758437785, 'B_0_5': -0.8216136298139791, 'B_0_6': -0.5544175687440773,
'B_0_7': -0.9725466300769442, 'B_0_8': 0.42793272814394023, 'B_0_9':
-0.30801218891155024, 'B_1_0': 0.5675719818311666, 'B_1_1':
-0.31890628452468794, 'B_1_2': 0.632280829551745, 'B_1_3': -0.7312174575783402,
'B_1_4': -0.6467211496201337, 'B_1_5': -0.3150644516238057, 'B_1_6':

```


-0.1811949700130988, 'B_1_7': -0.29956843453090276, 'B_1_8': 0.5220211243959754,
 'B_1_9': -0.7980362425991049, 'B_2_0': -0.29459030111579365, 'B_2_1':
 0.6677351990406095, 'B_2_2': -0.8577417870208994, 'B_2_3': 0.8758401451892877,
 'B_2_4': 0.9629073476985854, 'B_2_5': -0.8146350158263415, 'B_2_6':
 -0.4132804059336408, 'B_2_7': -0.2526466812499948, 'B_2_8': 0.5472549099323976,
 'B_2_9': 0.189935087991133, 'B_3_0': 0.2858440869752834, 'B_3_1':
 0.5098694491806419, 'B_3_2': -0.756401581114689, 'B_3_3': 0.6597547246823194,
 'B_3_4': 0.40712721227668514, 'B_3_5': 0.15313661588881322, 'B_3_6':
 0.7716759628704521, 'B_3_7': 0.39660658068993815, 'B_3_8': -0.5906340739905697,
 'B_3_9': -0.1391041454701799, 'B_4_0': -0.011795123467248092, 'B_4_1':
 -0.8806690990341968, 'B_4_2': 0.9001822764529104, 'B_4_3': -0.7354170236221806,
 'B_4_4': 0.39097330910891914, 'B_4_5': 0.012337451459942805, 'B_4_6':
 -0.9392481452288433, 'B_4_7': 0.8410000022114361, 'B_4_8': 0.6966368814614878,
 'B_4_9': 0.0480306972495943, 'B_5_0': -0.5366941744438984, 'B_5_1':
 -0.7491866764534134, 'B_5_2': -0.27048289072438936, 'B_5_3':
 -0.48425200352874076, 'B_5_4': 0.013248577992471366, 'B_5_5':
 0.9234904063243894, 'B_5_6': 0.14840791659468988, 'B_5_7': 0.7720634906623938,
 'B_5_8': 0.29874009824163883, 'B_5_9': -0.04901187735734314, 'B_6_0':
 0.7021229742120751, 'B_6_1': -0.05669114185746493, 'B_6_2': 0.7298837776105147,
 'B_6_3': -0.4126779962399847, 'B_6_4': -0.7349501139492836, 'B_6_5':
 0.7035226394386178, 'B_6_6': 0.17308202430948016, 'B_6_7': 0.7704029485329411,
 'B_6_8': 0.8668978698449188, 'B_6_9': -0.8838157870645307, 'B_7_0':
 0.0019192176805593597, 'B_7_1': -0.2475701928324321, 'B_7_2':
 0.24919962619883185, 'B_7_3': 0.5813105704606005, 'B_7_4': -0.43443137237074947,
 'B_7_5': 0.1124608010588799, 'B_7_6': 0.41328258821170105, 'B_7_7':
 -0.7238264559852466, 'B_7_8': 0.6218524737835083, 'B_7_9': 0.21009700770888662,
 'B_8_0': 0.8900841254850305, 'B_8_1': -0.3177327401212273, 'B_8_2':
 -0.7758710341723467, 'B_8_3': 0.012872378053112143, 'B_8_4': -0.820094672015311,
 'B_8_5': 0.9296625161815335, 'B_8_6': -0.9134037826507607, 'B_8_7':
 0.5019377241924698, 'B_8_8': 0.04970274832164323, 'B_8_9': -0.26013110805338846,
 'B_9_0': -0.8596325142728638, 'B_9_1': 0.24189050986374339, 'B_9_2':
 0.16873625219409139, 'B_9_3': -0.4338511193261525, 'B_9_4': 0.8122389687924181,
 'B_9_5': -0.387862185461183, 'B_9_6': -0.09954368702762983, 'B_9_7':
 0.08664966271893815, 'B_9_8': -0.7915408812109536, 'B_9_9': 0.855680204042992,
 'B_10_0': -0.24841841814693866, 'B_10_1': -0.8300571565287996, 'B_10_2':
 -0.2231634761228467, 'B_10_3': 0.19840104324330987, 'B_10_4':
 0.027032290206232545, 'B_10_5': 0.3851931085938025, 'B_10_6':
 -0.880141794705537, 'B_10_7': -0.42660959166132884, 'B_10_8':
 0.6320765421434025, 'B_10_9': -0.8318104981731055}. Best is trial 36 with value:
 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].

```

Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:11:36,787] Trial 99 finished with value: 185.1547450264969 and
parameters: {'B_0_0': 0.703862190297996, 'B_0_1': -0.8080457376873296, 'B_0_2':
0.6622193454855585, 'B_0_3': -0.2756389658641212, 'B_0_4': 0.22217394524928724,
'B_0_5': -0.6872835110169636, 'B_0_6': -0.8589859309841419, 'B_0_7':
-0.5624423560401326, 'B_0_8': -0.7368494405407828, 'B_0_9': -0.371436322594329,
'B_1_0': 0.028478828091820008, 'B_1_1': 0.17574373785871877, 'B_1_2':
0.5739388526118909, 'B_1_3': 0.30935498863175015, 'B_1_4': 0.21534148375651485,
'B_1_5': -0.1746715855781662, 'B_1_6': -0.05185929877892537, 'B_1_7':
-0.16924875780744142, 'B_1_8': 0.6774887395215385, 'B_1_9': -0.5345991039541683,
'B_2_0': -0.97867255554691, 'B_2_1': 0.2996133965079537, 'B_2_2':
-0.5891248763951138, 'B_2_3': 0.7724279698934224, 'B_2_4': 0.710911732819179,
'B_2_5': -0.8380798035821524, 'B_2_6': -0.7793509572801268, 'B_2_7':
-0.00956024735160077, 'B_2_8': 0.21626087228332214, 'B_2_9':
-0.44568149178516336, 'B_3_0': 0.4292594790647539, 'B_3_1': 0.1879337214193657,
'B_3_2': -0.34684826269786173, 'B_3_3': -0.9397644536762151, 'B_3_4':
0.4952975569710649, 'B_3_5': -0.5647877416663363, 'B_3_6': 0.6622580120560042,
'B_3_7': -0.15782528509257476, 'B_3_8': 0.15261333641089547, 'B_3_9':
0.7384384862098237, 'B_4_0': 0.7199735418215076, 'B_4_1': -0.2543912410177249,
'B_4_2': -0.5763999924246106, 'B_4_3': 0.5447892339026175, 'B_4_4':
-0.13250119686778028, 'B_4_5': -0.6097568527531719, 'B_4_6': 0.8007621777879574,
'B_4_7': -0.597753695402647, 'B_4_8': -0.9389018069813625, 'B_4_9':
-0.9498456519405084, 'B_5_0': 0.24218440577944333, 'B_5_1': 0.3201166202207102,
'B_5_2': -0.37413742398717187, 'B_5_3': -0.3563782412908145, 'B_5_4':
-0.3178020481813133, 'B_5_5': 0.3567139225825797, 'B_5_6': -0.8748950937668794,
'B_5_7': 0.8496503783288247, 'B_5_8': -0.5494083325286561, 'B_5_9':
0.9893529463793427, 'B_6_0': -0.6503850663013697, 'B_6_1': 0.5376619854785074,
'B_6_2': -0.5476431610393424, 'B_6_3': -0.8299233327463571, 'B_6_4':
0.7423679822142848, 'B_6_5': 0.710327493197966, 'B_6_6': -0.19093062294978935,
'B_6_7': 0.40074436223004506, 'B_6_8': 0.11299054671415587, 'B_6_9':
0.6988531050740225, 'B_7_0': -0.36618918074906426, 'B_7_1': 0.04687135199799064,
'B_7_2': -0.028962705014990098, 'B_7_3': -0.28926646401141975, 'B_7_4':
0.7750810278216171, 'B_7_5': -0.6615617091681798, 'B_7_6': -0.6939728937490216,
'B_7_7': 0.1522808799390476, 'B_7_8': 0.3118385753701851, 'B_7_9':
-0.604653876437361, 'B_8_0': -0.2163794643358442, 'B_8_1': 0.8841964090979126,
'B_8_2': -0.5016975030724655, 'B_8_3': 0.33335419019244017, 'B_8_4':
0.4754957995212299, 'B_8_5': -0.10270517505898413, 'B_8_6': 0.8663014537839744,
'B_8_7': 0.46858033573320523, 'B_8_8': -0.08330933187542222, 'B_8_9':
0.13728369951666597, 'B_9_0': 0.38137461798169925, 'B_9_1': 0.18253655724985207,
'B_9_2': 0.7498212159756303, 'B_9_3': -0.2833604486848057, 'B_9_4':
-0.39622623468950624, 'B_9_5': -0.5665989987993767, 'B_9_6':
-0.9151897232321016, 'B_9_7': -0.9048323505274904, 'B_9_8': 0.7327092823563799,

```

'B_9_9': 0.35321842988369645, 'B_10_0': 0.31785418256424247, 'B_10_1':
0.22699278149107438, 'B_10_2': 0.6914539064943277, 'B_10_3':
-0.03907148718205766, 'B_10_4': 0.7334134405454347, 'B_10_5':
-0.2782988481591664, 'B_10_6': 0.8448808585913994, 'B_10_7': 0.3933408205608555,
'B_10_8': -0.480789270671634, 'B_10_9': 0.12688814494347778}. Best is trial 36
with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:11:54,908] Trial 100 finished with value: 136.65946310007797
and parameters: {'B_0_0': 0.35182564957546714, 'B_0_1': -0.13595444788579933,
'B_0_2': 0.9601454969092826, 'B_0_3': 0.041017595447093144, 'B_0_4':
-0.6312470916189228, 'B_0_5': 0.7594224171026673, 'B_0_6': 0.6227806493619353,
'B_0_7': 0.6345796470368906, 'B_0_8': 0.710819609759384, 'B_0_9':
-0.2452871580909028, 'B_1_0': -0.057293430128971104, 'B_1_1':
-0.025195235214699713, 'B_1_2': -0.599014218120219, 'B_1_3':
-0.5886069202358466, 'B_1_4': 0.3638248591709649, 'B_1_5': -0.09516859360169871,
'B_1_6': -0.4361760822669791, 'B_1_7': 0.62000005112016, 'B_1_8':
0.3914729879447424, 'B_1_9': -0.2099554715635646, 'B_2_0': 0.2921408617653057,
'B_2_1': -0.9377911928765007, 'B_2_2': -0.3011526137083593, 'B_2_3':
0.060728510073532166, 'B_2_4': -0.7878738822245545, 'B_2_5':
-0.49999347746713474, 'B_2_6': -0.6355284471599453, 'B_2_7':
-0.20805546190734997, 'B_2_8': 0.05676777840570027, 'B_2_9':
-0.04348345980581411, 'B_3_0': -0.7583287186394074, 'B_3_1':
-0.9578989317507266, 'B_3_2': -0.1279752141979716, 'B_3_3': -0.6782998674416998,
'B_3_4': -0.12146993594025624, 'B_3_5': 0.7990710206406343, 'B_3_6':
-0.3383949483153814, 'B_3_7': -0.3561434499090979, 'B_3_8': 0.8878564621229361,
'B_3_9': -0.7124963069039003, 'B_4_0': 0.03913638712458999, 'B_4_1':
0.9389019753134247, 'B_4_2': -0.2568262512806336, 'B_4_3': 0.2070075154190763,
'B_4_4': -0.6414223863563979, 'B_4_5': 0.35175351289258494, 'B_4_6':
-0.8933650263998372, 'B_4_7': 0.062318087257845295, 'B_4_8':
0.21721118013113838, 'B_4_9': 0.8618603972244889, 'B_5_0': -0.307364565236368,
'B_5_1': -0.09978805684301582, 'B_5_2': 0.551405370720826, 'B_5_3':
-0.07257506903627542, 'B_5_4': 0.8365228788507079, 'B_5_5': -0.4534153083755779,
'B_5_6': -0.9874846143598242, 'B_5_7': 0.0891760047275752, 'B_5_8':
-0.9468552190278989, 'B_5_9': 0.13327167756269187, 'B_6_0':

```

-0.16567315625997933, 'B_6_1': -0.26798669284391197, 'B_6_2':
-0.10211709422555926, 'B_6_3': 0.6890561402918305, 'B_6_4': -0.5949223974015059,
'B_6_5': -0.8273695947040925, 'B_6_6': 0.10493944679896139, 'B_6_7':
0.8757146765010316, 'B_6_8': -0.48262860931380613, 'B_6_9':
-0.08117199837053253, 'B_7_0': 0.9891643747901797, 'B_7_1': -0.9646768749868069,
'B_7_2': -0.8894194874240436, 'B_7_3': 0.949942932826217, 'B_7_4':
0.7303625966364069, 'B_7_5': 0.9602236559958918, 'B_7_6': 0.7083767791363991,
'B_7_7': -0.6631749087457987, 'B_7_8': -0.9774290425452135, 'B_7_9':
0.3154576308272924, 'B_8_0': -0.7420666289072753, 'B_8_1': 0.14046006530417138,
'B_8_2': -0.6285364613256892, 'B_8_3': -0.23224937843644589, 'B_8_4':
-0.28905705448885133, 'B_8_5': -0.6951697389454843, 'B_8_6': 0.6280997627400937,
'B_8_7': 0.03206958074082042, 'B_8_8': 0.01750336875032865, 'B_8_9':
0.6888158922208051, 'B_9_0': -0.021723623220164856, 'B_9_1':
-0.6521886302970948, 'B_9_2': 0.54897695837657, 'B_9_3': 0.029534660095575045,
'B_9_4': 0.6522412571843063, 'B_9_5': -0.7958747895964325, 'B_9_6':
0.12462590927474229, 'B_9_7': -0.1671986592164646, 'B_9_8': 0.41136200500180464,
'B_9_9': -0.5928247808296705, 'B_10_0': 0.48949992701155665, 'B_10_1':
-0.67173560334652, 'B_10_2': -0.8535474642867467, 'B_10_3': -0.5887164416575342,
'B_10_4': 0.9472828804435802, 'B_10_5': -0.5882744038544385, 'B_10_6':
0.21236243811378586, 'B_10_7': -0.985079431380741, 'B_10_8':
0.21609801146114654, 'B_10_9': 0.6488770311301133}. Best is trial 36 with value:
95.85134184163296.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:12:11,542] Trial 101 finished with value: 98.35884539721778 and
parameters: {'B_0_0': 0.18994465011969067, 'B_0_1': -0.2956866694414438,
'B_0_2': 0.9179455261556249, 'B_0_3': -0.192362581320159, 'B_0_4':
-0.3061758178272759, 'B_0_5': 0.9014133321484824, 'B_0_6': 0.2179994565026994,
'B_0_7': -0.45769516994011755, 'B_0_8': 0.12915227715322258, 'B_0_9':
-0.13631272015789186, 'B_1_0': 0.24276098717337355, 'B_1_1':
-0.10170662306648423, 'B_1_2': -0.11597560735984738, 'B_1_3':
0.5862858826665257, 'B_1_4': -0.37760290926987033, 'B_1_5': 0.6930036429964777,
'B_1_6': -0.48479091965567545, 'B_1_7': 0.9893518299974537, 'B_1_8':
-0.4383670769235301, 'B_1_9': -0.2089894668068459, 'B_2_0': -0.5841939911324925,
'B_2_1': 0.5560805358049067, 'B_2_2': -0.3084252613801619, 'B_2_3':

```

0.6688440835851721, 'B_2_4': 0.5371993237753838, 'B_2_5': -0.748885251859982,
'B_2_6': -0.5568491075764934, 'B_2_7': -0.7456265682628486, 'B_2_8':
0.3419229614673631, 'B_2_9': 0.49509194874451323, 'B_3_0': -0.8296078957372188,
'B_3_1': -0.5969096346286539, 'B_3_2': 0.1800521345478452, 'B_3_3':
0.1603524038909169, 'B_3_4': -0.7327157133822501, 'B_3_5': 0.06628320974588725,
'B_3_6': 0.17340919394916218, 'B_3_7': -0.07434071411343054, 'B_3_8':
0.68562684268452, 'B_3_9': 0.09785547498803465, 'B_4_0': -0.09676028671335618,
'B_4_1': 0.6009248319766307, 'B_4_2': 0.06390019860628936, 'B_4_3':
-0.5902293452861029, 'B_4_4': -0.5265845704941674, 'B_4_5': 0.5591933441275451,
'B_4_6': -0.9663488242812937, 'B_4_7': -0.06424153527089671, 'B_4_8':
0.845775271624247, 'B_4_9': 0.5028403758045923, 'B_5_0': -0.2588067796502308,
'B_5_1': -0.01850436992699446, 'B_5_2': -0.03684653178561492, 'B_5_3':
-0.2238829752612386, 'B_5_4': -0.0920245473446174, 'B_5_5': -0.4476204503856392,
'B_5_6': -0.3788782739508789, 'B_5_7': 0.19421504823909203, 'B_5_8':
-0.5882878581670492, 'B_5_9': 0.34162919245663126, 'B_6_0':
-0.11777237783039722, 'B_6_1': 0.25357835836447096, 'B_6_2': 0.5923362815173355,
'B_6_3': 0.6919650634811667, 'B_6_4': -0.9183536699481681, 'B_6_5':
-0.7628148370282358, 'B_6_6': -0.037798496088358224, 'B_6_7':
-0.7246000505325716, 'B_6_8': 0.8042387510527909, 'B_6_9': 0.8116079226520394,
'B_7_0': -0.8204632874054458, 'B_7_1': -0.6307649336264146, 'B_7_2':
-0.8545347638761402, 'B_7_3': 0.864693758513271, 'B_7_4': 0.5342191192471724,
'B_7_5': -0.13537563195458574, 'B_7_6': 0.04102113413597425, 'B_7_7':
0.2382426876757408, 'B_7_8': -0.8143025310688401, 'B_7_9': 0.3504708096504438,
'B_8_0': -0.34878374348125535, 'B_8_1': 0.13699357890087918, 'B_8_2':
0.47518604578714296, 'B_8_3': 0.11021753441412589, 'B_8_4':
-0.45507560029523964, 'B_8_5': -0.6625050255229835, 'B_8_6':
0.15995646940847907, 'B_8_7': -0.9694725171552058, 'B_8_8': 0.18373986058997865,
'B_8_9': -0.10894871813616205, 'B_9_0': 0.4599044753566853, 'B_9_1':
-0.9606967214274366, 'B_9_2': 0.5253502584865302, 'B_9_3': 0.6955117570347591,
'B_9_4': -0.14723863813528482, 'B_9_5': -0.8807734601415634, 'B_9_6':
0.03409748092086097, 'B_9_7': -0.20305305671699606, 'B_9_8': 0.506312126977216,
'B_9_9': 0.9799152882607814, 'B_10_0': 0.07994319346183953, 'B_10_1':
-0.5429587981120045, 'B_10_2': -0.8592573489546224, 'B_10_3':
-0.825089267933191, 'B_10_4': 0.7964374969562716, 'B_10_5':
-0.09762500113355244, 'B_10_6': 0.06803978602979857, 'B_10_7':
0.8005975126346253, 'B_10_8': -0.23857986318230534, 'B_10_9':
-0.14302751831716293}. Best is trial 36 with value: 95.85134184163296.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].
Built F__[u_10].

```

Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:12:32,084] Trial 102 finished with value: 73.90742475345364 and
parameters: {'B_0_0': 0.1646905343952125, 'B_0_1': -0.3469119141819876, 'B_0_2':
-0.8727865647064216, 'B_0_3': 0.5311292188795762, 'B_0_4': -0.2990466564803879,
'B_0_5': 0.9427314253450197, 'B_0_6': 0.21339606845150458, 'B_0_7':
-0.6566399000847918, 'B_0_8': 0.1130588771229538, 'B_0_9': -0.12895970494998343,
'B_1_0': 0.14374671457327598, 'B_1_1': -0.1004437223226825, 'B_1_2':
0.07188817601743752, 'B_1_3': 0.570122859592249, 'B_1_4': -0.33792946006509283,
'B_1_5': 0.6954989754087698, 'B_1_6': -0.42977664125583304, 'B_1_7':
0.1001796843106336, 'B_1_8': -0.4022912374686786, 'B_1_9': -0.22782395504246272,
'B_2_0': -0.5301583356259296, 'B_2_1': -0.2838880777124185, 'B_2_2':
-0.4652710087396784, 'B_2_3': 0.6910849452942291, 'B_2_4': -0.40883530362551723,
'B_2_5': -0.7839379098812405, 'B_2_6': -0.5441441943134009, 'B_2_7':
-0.7410543721939358, 'B_2_8': -0.4359635950183882, 'B_2_9': 0.5420930157893576,
'B_3_0': -0.8356046051961361, 'B_3_1': -0.6081506833727635, 'B_3_2':
0.19119489190473155, 'B_3_3': -0.10792958266467736, 'B_3_4':
-0.3395636020826987, 'B_3_5': 0.03880960263241881, 'B_3_6': 0.15704107626046457,
'B_3_7': -0.09137144966771597, 'B_3_8': 0.708301140023488, 'B_3_9':
-0.5597078984311812, 'B_4_0': -0.015231088896412552, 'B_4_1':
0.5516664522085375, 'B_4_2': 0.04520884108025944, 'B_4_3': -0.579803764685582,
'B_4_4': -0.548591156767821, 'B_4_5': 0.5497009539275356, 'B_4_6':
-0.9707372410736455, 'B_4_7': -0.10168228675043572, 'B_4_8': 0.78773952123164,
'B_4_9': 0.5283178745567088, 'B_5_0': -0.25449869611305354, 'B_5_1':
-0.312387011642035, 'B_5_2': 0.297865262560292, 'B_5_3': -0.22467447833931734,
'B_5_4': -0.13496432645707024, 'B_5_5': -0.4902075405490436, 'B_5_6':
-0.35173546463273286, 'B_5_7': 0.07134233069361626, 'B_5_8':
-0.46951712764627007, 'B_5_9': 0.33663466773253103, 'B_6_0':
-0.11375533357424078, 'B_6_1': -0.7315692759685405, 'B_6_2': 0.7806808212934567,
'B_6_3': 0.6887889897668255, 'B_6_4': -0.9220513731079242, 'B_6_5':
-0.13000659637776546, 'B_6_6': 0.038720555971498954, 'B_6_7':
0.8509738947545211, 'B_6_8': 0.8200014907388334, 'B_6_9': 0.8318175018593289,
'B_7_0': -0.779333576932029, 'B_7_1': -0.6453064162809481, 'B_7_2':
-0.6988564671495892, 'B_7_3': 0.8835651843120995, 'B_7_4': 0.524524552227631,
'B_7_5': -0.31338424796022235, 'B_7_6': -0.05144668054194853, 'B_7_7':
-0.9410430563077724, 'B_7_8': -0.8132362549567976, 'B_7_9': 0.3520239842537647,
'B_8_0': -0.9101074315877875, 'B_8_1': 0.0947568281925423, 'B_8_2':
0.4841647983881299, 'B_8_3': 0.07998263782129036, 'B_8_4': -0.4360058182135264,
'B_8_5': -0.6600415717233369, 'B_8_6': 0.3193704215783295, 'B_8_7':
-0.9862969806869558, 'B_8_8': 0.19564589755506778, 'B_8_9': 0.9718491332374233,
'B_9_0': 0.4395102875696402, 'B_9_1': -0.7081675404417593, 'B_9_2':
0.5131248567136368, 'B_9_3': 0.6888221038923883, 'B_9_4': -0.14608989000887662,
'B_9_5': -0.9093643192766907, 'B_9_6': 0.0743261416126324, 'B_9_7':
0.6692677018134939, 'B_9_8': 0.4603969444590733, 'B_9_9': 0.8987287627552412,
'B_10_0': 0.12760329945765955, 'B_10_1': -0.9775430209944872, 'B_10_2':
-0.7867481191785994, 'B_10_3': -0.8367230818371215, 'B_10_4':

```

0.7944449587647588, 'B_10_5': -0.12476962013485304, 'B_10_6':
0.5782443925234957, 'B_10_7': 0.8357090283871428, 'B_10_8': -0.2233501879550217,
'B_10_9': -0.14882540096885377}. Best is trial 102 with value:
73.90742475345364.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:12:48,667] Trial 103 finished with value: 84.2271151778599 and
parameters: {'B_0_0': 0.13087307979125085, 'B_0_1': -0.27932756546231857,
'B_0_2': 0.8981694351611432, 'B_0_3': -0.1784088710141238, 'B_0_4':
-0.3312939245864033, 'B_0_5': 0.962708573701251, 'B_0_6': 0.20671589469902135,
'B_0_7': 0.7867603227751772, 'B_0_8': 0.12047184452332592, 'B_0_9':
-0.13530125103824403, 'B_1_0': 0.15891869113956794, 'B_1_1':
-0.08776001153168592, 'B_1_2': 0.08840699848470349, 'B_1_3': 0.5985891016645017,
'B_1_4': -0.37646118241431564, 'B_1_5': 0.986703248448491, 'B_1_6':
-0.49531138142487197, 'B_1_7': 0.11583655585176952, 'B_1_8':
-0.4230280590982134, 'B_1_9': -0.19972944648249227, 'B_2_0': -0.520072524862054,
'B_2_1': -0.2667400508582563, 'B_2_2': -0.3705142930987476, 'B_2_3':
0.6509439290647777, 'B_2_4': -0.4328713553512563, 'B_2_5': -0.7519978222352695,
'B_2_6': -0.5623796105563366, 'B_2_7': -0.7367421684114785, 'B_2_8':
-0.3470669122004214, 'B_2_9': 0.5038355569602421, 'B_3_0': -0.8429938477979261,
'B_3_1': -0.6413787060880456, 'B_3_2': 0.04887867651688052, 'B_3_3':
0.1303870114704087, 'B_3_4': -0.3312911399726911, 'B_3_5': 0.08416860422306757,
'B_3_6': 0.14131270541204027, 'B_3_7': -0.09052326415981629, 'B_3_8':
0.6538998686330905, 'B_3_9': 0.06482319461116859, 'B_4_0': -0.06672891095637362,
'B_4_1': 0.5934509146491646, 'B_4_2': 0.05977650572370319, 'B_4_3':
-0.5827110111514161, 'B_4_4': -0.5514237680924631, 'B_4_5': 0.5454886875424835,
'B_4_6': -0.9660918533374989, 'B_4_7': -0.10363788201955071, 'B_4_8':
0.7472899056234921, 'B_4_9': 0.5455426524060205, 'B_5_0': -0.23238685507505083,
'B_5_1': -0.29078444195712433, 'B_5_2': 0.14480036534483753, 'B_5_3':
0.8767537396508476, 'B_5_4': 0.3093366578392242, 'B_5_5': -0.4716649659089528,
'B_5_6': -0.42357730942881416, 'B_5_7': 0.1393188916818734, 'B_5_8':
-0.5766826442544075, 'B_5_9': 0.3388101509945399, 'B_6_0': -0.0818094053389069,
'B_6_1': 0.2846106271211268, 'B_6_2': 0.8025229969944702, 'B_6_3':
0.6986722458040209, 'B_6_4': -0.8914396108926977, 'B_6_5': -0.25775146191054044,
'B_6_6': -0.031073793114784976, 'B_6_7': 0.8293543239925452, 'B_6_8':

```

0.8143173337913643, 'B_6_9': 0.8483406721251022, 'B_7_0': -0.8112855144975094,
'B_7_1': -0.6685319425313434, 'B_7_2': -0.9053576897503657, 'B_7_3':
0.9050980913369056, 'B_7_4': 0.5276021874021137, 'B_7_5': -0.11388375353259111,
'B_7_6': -0.031112238853250634, 'B_7_7': 0.44526341678045145, 'B_7_8':
-0.8045594575709587, 'B_7_9': 0.3577735177851206, 'B_8_0': -0.9166835176091422,
'B_8_1': -0.04257171831418727, 'B_8_2': 0.5114651097904211, 'B_8_3':
0.0974980400380954, 'B_8_4': -0.4811128737909126, 'B_8_5': -0.6918783248110322,
'B_8_6': 0.13824392841892494, 'B_8_7': 0.1867540201646783, 'B_8_8':
-0.726489888895048, 'B_8_9': -0.12470517844758897, 'B_9_0': 0.3627413883959502,
'B_9_1': -0.9722263991982024, 'B_9_2': -0.9305674821144833, 'B_9_3':
0.6774166926285192, 'B_9_4': -0.30026462200976545, 'B_9_5': -0.9107519841852135,
'B_9_6': 0.05210664593194625, 'B_9_7': 0.6292437902086008, 'B_9_8':
0.45745425858444677, 'B_9_9': 0.9806534432362999, 'B_10_0': 0.9536006394941299,
'B_10_1': -0.9822065105571537, 'B_10_2': -0.8920302656238829, 'B_10_3':
-0.849963838477474, 'B_10_4': 0.7731063318015433, 'B_10_5':
-0.09905288257721709, 'B_10_6': 0.042829028663265434, 'B_10_7':
0.8284916669519541, 'B_10_8': -0.2256516075395666, 'B_10_9':
-0.07320915376615839}. Best is trial 102 with value: 73.90742475345364.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:13:05,244] Trial 104 finished with value: 79.02587067246036 and
parameters: {'B_0_0': 0.12131715578436444, 'B_0_1': -0.18228998503919183,
'B_0_2': 0.8130044196124975, 'B_0_3': -0.17229545697918985, 'B_0_4':
-0.2865564989395981, 'B_0_5': 0.9393617409516405, 'B_0_6': 0.21432185101888404,
'B_0_7': 0.7914450881242122, 'B_0_8': 0.2565491199366515, 'B_0_9':
-0.1417307898402445, 'B_1_0': 0.23769355905020378, 'B_1_1':
-0.08094447257175683, 'B_1_2': 0.089999880156696, 'B_1_3': 0.5410922936161869,
'B_1_4': -0.2238032026122243, 'B_1_5': 0.9698767817302565, 'B_1_6':
-0.4911546444751783, 'B_1_7': 0.966643779250299, 'B_1_8': -0.41545246283144643,
'B_1_9': -0.22028014062368434, 'B_2_0': -0.558802267256475, 'B_2_1':
-0.3068694476252818, 'B_2_2': -0.340523724543773, 'B_2_3': 0.661673479994598,
'B_2_4': -0.54272882123516, 'B_2_5': -0.7929710616579406, 'B_2_6':
-0.5664498483628302, 'B_2_7': -0.7270655107787665, 'B_2_8':
-0.35142555286686394, 'B_2_9': 0.47839965462153017, 'B_3_0':
-0.8312471708072479, 'B_3_1': -0.6346157741056585, 'B_3_2': 0.05026209263664219,

```


'B_3_3': 0.15135402219313596, 'B_3_4': -0.3314798954193968, 'B_3_5':
 0.08673219862337328, 'B_3_6': 0.15327148006099547, 'B_3_7':
 -0.09237785593476926, 'B_3_8': 0.6832896467633929, 'B_3_9':
 -0.28225795164610784, 'B_4_0': -0.060964168569644026, 'B_4_1':
 0.6224857730943577, 'B_4_2': 0.04562028148839091, 'B_4_3': -0.5895184623468896,
 'B_4_4': -0.5355476360351124, 'B_4_5': 0.5543804362512683, 'B_4_6':
 -0.8114278813983207, 'B_4_7': -0.10554384225074787, 'B_4_8': 0.8227718963240526,
 'B_4_9': 0.5022937781861914, 'B_5_0': -0.25771768650456806, 'B_5_1':
 -0.020738139825780544, 'B_5_2': 0.14390181279014208, 'B_5_3':
 -0.22189976466318104, 'B_5_4': -0.12761914519976875, 'B_5_5':
 -0.5067492335517019, 'B_5_6': -0.36903055976257754, 'B_5_7':
 0.24178805505443762, 'B_5_8': -0.4677097984513063, 'B_5_9': 0.3890243197437128,
 'B_6_0': -0.13068076189616323, 'B_6_1': -0.7862707496424912, 'B_6_2':
 0.9690715714914514, 'B_6_3': 0.6954095636597325, 'B_6_4': -0.9215706847998755,
 'B_6_5': -0.30311705681619106, 'B_6_6': -0.00677014464022728, 'B_6_7':
 0.8410163475336998, 'B_6_8': 0.8297055482540497, 'B_6_9': 0.8018668899923389,
 'B_7_0': -0.8092396598589937, 'B_7_1': -0.7807446043384522, 'B_7_2':
 -0.6969729916027572, 'B_7_3': 0.8868732429480738, 'B_7_4': 0.5406479860437429,
 'B_7_5': -0.33099986932545034, 'B_7_6': -0.056377294002136626, 'B_7_7':
 -0.9323203677950826, 'B_7_8': -0.7938060011377881, 'B_7_9': 0.36311283938204153,
 'B_8_0': -0.8841000980915441, 'B_8_1': -0.060994004462881585, 'B_8_2':
 0.46669155930602585, 'B_8_3': 0.09779062697464917, 'B_8_4':
 -0.46477111347269795, 'B_8_5': -0.641826273563988, 'B_8_6': 0.16952204757486564,
 'B_8_7': -0.9770783036889773, 'B_8_8': 0.3285808855033945, 'B_8_9':
 0.9430083077173814, 'B_9_0': 0.4107266533535712, 'B_9_1': -0.930447016027327,
 'B_9_2': -0.9725541195044862, 'B_9_3': 0.6909301378270118, 'B_9_4':
 -0.3031231043727772, 'B_9_5': -0.7820047439973661, 'B_9_6':
 0.030593030454856236, 'B_9_7': -0.19891231640357815, 'B_9_8':
 0.48317608188887423, 'B_9_9': 0.9711546212384506, 'B_10_0': 0.9644240435641727,
 'B_10_1': -0.9897626170851668, 'B_10_2': -0.9218142112464055, 'B_10_3':
 -0.7942082048529833, 'B_10_4': 0.8111805879516032, 'B_10_5':
 -0.1178264535966709, 'B_10_6': 0.06135584507832792, 'B_10_7':
 0.8401394314789726, 'B_10_8': -0.21663907286754747, 'B_10_9':
 -0.16866379144089333}. Best is trial 102 with value: 73.90742475345364.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix $B = \text{inverse}(A)$.

[I 2025-04-29 14:13:23,632] Trial 105 finished with value: 80.14069149568236 and parameters: {'B_0_0': 0.15408454318596526, 'B_0_1': -0.20068058216644058, 'B_0_2': 0.9214884300369723, 'B_0_3': -0.11490211778069821, 'B_0_4': -0.34324225714680157, 'B_0_5': 0.9332377442695704, 'B_0_6': 0.19621248897714244, 'B_0_7': 0.7778535746119813, 'B_0_8': 0.14573748909081136, 'B_0_9': -0.14388372036493624, 'B_1_0': 0.2349454576219635, 'B_1_1': -0.08601620628191368, 'B_1_2': 0.06686633131502454, 'B_1_3': 0.5659468245983547, 'B_1_4': -0.22610580345704723, 'B_1_5': 0.9719932445657142, 'B_1_6': -0.49390182595464044, 'B_1_7': 0.9898871045863402, 'B_1_8': -0.41770360047931293, 'B_1_9': -0.2846577026666963, 'B_2_0': -0.5257019843509861, 'B_2_1': -0.28401975600800566, 'B_2_2': -0.33713167913700337, 'B_2_3': 0.678910191872398, 'B_2_4': -0.5219144717437444, 'B_2_5': -0.7575697961881647, 'B_2_6': -0.5834083786357902, 'B_2_7': -0.7272929042212841, 'B_2_8': -0.34337193818436906, 'B_2_9': 0.4414911437417155, 'B_3_0': -0.8315382509937984, 'B_3_1': -0.640908522710043, 'B_3_2': 0.0431961133308042, 'B_3_3': 0.1971855507796993, 'B_3_4': -0.32896845409359476, 'B_3_5': 0.1130556503453117, 'B_3_6': 0.17319065469086542, 'B_3_7': -0.0947859875598641, 'B_3_8': 0.6672277510833199, 'B_3_9': -0.5568895884172604, 'B_4_0': -0.058613511798174506, 'B_4_1': 0.6249891756702107, 'B_4_2': 0.060669819414760046, 'B_4_3': -0.5623476342948744, 'B_4_4': -0.522684585352257, 'B_4_5': 0.5458744453837883, 'B_4_6': -0.7968673583298458, 'B_4_7': -0.10570164873349394, 'B_4_8': 0.8290739956843706, 'B_4_9': 0.5241418669137834, 'B_5_0': -0.20762937023954417, 'B_5_1': -0.31330517476106856, 'B_5_2': -0.02066318707448596, 'B_5_3': -0.23428163331951904, 'B_5_4': 0.2924030467711998, 'B_5_5': -0.5128619339260476, 'B_5_6': -0.2867868209269143, 'B_5_7': 0.22862159074099508, 'B_5_8': -0.48585801841710724, 'B_5_9': 0.3396967807112288, 'B_6_0': -0.11485203716550263, 'B_6_1': -0.7646257714629743, 'B_6_2': 0.9855011975183579, 'B_6_3': 0.7164582410507558, 'B_6_4': -0.9089972254602554, 'B_6_5': -0.11293019562396324, 'B_6_6': -0.07130723613892287, 'B_6_7': 0.8458311915353078, 'B_6_8': 0.829569490002518, 'B_6_9': 0.8049291858979801, 'B_7_0': -0.807388619444832, 'B_7_1': -0.6522519866370495, 'B_7_2': -0.6941035944415684, 'B_7_3': 0.8753609866603036, 'B_7_4': 0.5209014571495744, 'B_7_5': -0.3136951350891596, 'B_7_6': -0.0542361575313584, 'B_7_7': -0.9155974281167429, 'B_7_8': -0.7959212400825127, 'B_7_9': 0.35320212132141127, 'B_8_0': -0.9097165831062508, 'B_8_1': -0.049981115674601395, 'B_8_2': 0.4939837384544366, 'B_8_3': 0.11631734209404589, 'B_8_4': -0.46526109594016213, 'B_8_5': -0.6776164464128103, 'B_8_6': 0.1418916999280878, 'B_8_7': -0.9353669173064264, 'B_8_8': 0.34518467049836166, 'B_8_9': 0.97967122185666, 'B_9_0': 0.35374386794947893, 'B_9_1': -0.9750546763981359, 'B_9_2': -0.9898546429072426, 'B_9_3': 0.6504078875858954, 'B_9_4': -0.34026486233444214, 'B_9_5': -0.7922897634468524, 'B_9_6': 0.0861115039353341, 'B_9_7': 0.6573458791687039, 'B_9_8': 0.46636953051699936, 'B_9_9': 0.9633438395104571, 'B_10_0': 0.9658563223352479, 'B_10_1': -0.9869175623881102, 'B_10_2': -0.7595814940008545, 'B_10_3': -0.8204913482179662, 'B_10_4': 0.7867962502269016, 'B_10_5': -0.11029688908780265, 'B_10_6': 0.07354030373060093, 'B_10_7': 0.8108524015455926, 'B_10_8':

-0.23449160974485092, 'B_10_9': -0.1440953586834898}. Best is trial 102 with value: 73.90742475345364.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].
Built F__[u_10].
Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:13:41,071] Trial 106 finished with value: 82.7046130789721 and parameters: {'B_0_0': 0.15807404812793843, 'B_0_1': -0.2084019337654005, 'B_0_2': 0.9157486723916083, 'B_0_3': -0.12855954779395912, 'B_0_4': -0.34891601807056977, 'B_0_5': 0.9307528000011747, 'B_0_6': 0.24649439193705927, 'B_0_7': 0.7732182932983738, 'B_0_8': 0.15091754998232992, 'B_0_9': -0.14397024660500318, 'B_1_0': 0.19563274907102562, 'B_1_1': -0.0972387684776087, 'B_1_2': 0.07872925662765103, 'B_1_3': 0.5724776550857313, 'B_1_4': -0.24248584328855163, 'B_1_5': 0.9842376145853922, 'B_1_6': -0.5026965423483392, 'B_1_7': 0.9880741588752009, 'B_1_8': -0.421982588197597, 'B_1_9': -0.22434565923965447, 'B_2_0': -0.49765854067332355, 'B_2_1': -0.2696795161076416, 'B_2_2': -0.3201956875993712, 'B_2_3': 0.678903939855504, 'B_2_4': -0.5330662780684289, 'B_2_5': -0.764007237740269, 'B_2_6': -0.5828662197746247, 'B_2_7': -0.750159772617013, 'B_2_8': -0.3438396497061412, 'B_2_9': 0.5111713257100147, 'B_3_0': -0.8281915452078281, 'B_3_1': -0.6174026891759541, 'B_3_2': 0.022662703981830304, 'B_3_3': 0.2045008362803261, 'B_3_4': -0.34039659797497823, 'B_3_5': 0.026809255791401698, 'B_3_6': 0.1659975760772686, 'B_3_7': -0.08025066820936534, 'B_3_8': 0.6653266763518931, 'B_3_9': -0.25765762968649164, 'B_4_0': -0.0690958314335432, 'B_4_1': 0.5980252118794697, 'B_4_2': 0.07466971641380771, 'B_4_3': -0.5408807226191277, 'B_4_4': -0.5357321651469737, 'B_4_5': 0.5786694104409466, 'B_4_6': -0.8210322617609557, 'B_4_7': -0.23550413684055171, 'B_4_8': 0.786369286697391, 'B_4_9': 0.5060569491228296, 'B_5_0': -0.2391217366538439, 'B_5_1': -0.31689344633781136, 'B_5_2': 0.2898019781988908, 'B_5_3': -0.24367983115953015, 'B_5_4': 0.29387297804139484, 'B_5_5': -0.5187941935492763, 'B_5_6': -0.3633361100311281, 'B_5_7': 0.21594616440842931, 'B_5_8': -0.46552890475083264, 'B_5_9': 0.3762095048327995, 'B_6_0': -0.1136161451737778, 'B_6_1': -0.7568526199766324, 'B_6_2': 0.9506968485969032, 'B_6_3': 0.8779513027589776, 'B_6_4': -0.8878046559582228, 'B_6_5': -0.15342563814038657, 'B_6_6': 0.022470908309327478, 'B_6_7': 0.8543745022919982, 'B_6_8': 0.7936812496031049, 'B_6_9': 0.8086337586163093, 'B_7_0': -0.7587742604130419, 'B_7_1': -0.6729667510106164, 'B_7_2':

```

-0.6926797124632722, 'B_7_3': 0.8820340636459192, 'B_7_4': 0.4986526542894258,
'B_7_5': -0.2884589885035961, 'B_7_6': -0.05256186610861583, 'B_7_7':
-0.94806274921629, 'B_7_8': -0.7807828126595346, 'B_7_9': 0.35536299489024503,
'B_8_0': -0.9133818239536322, 'B_8_1': -0.05190966679840325, 'B_8_2':
0.48748224212194735, 'B_8_3': 0.1078022873709169, 'B_8_4': -0.4627032607730085,
'B_8_5': -0.6516669442406997, 'B_8_6': 0.15102473693864393, 'B_8_7':
-0.9881016091749121, 'B_8_8': 0.3483850967898516, 'B_8_9': 0.9884715479470733,
'B_9_0': 0.44449628479982456, 'B_9_1': -0.9896322550990768, 'B_9_2':
-0.9375168344850738, 'B_9_3': 0.6730145241754903, 'B_9_4': -0.3099676438228981,
'B_9_5': -0.7681521904252292, 'B_9_6': 0.11350795964310256, 'B_9_7':
-0.24037052882943208, 'B_9_8': 0.4804961028514003, 'B_9_9': 0.9885227480972713,
'B_10_0': 0.9818783065640606, 'B_10_1': -0.9870663738405288, 'B_10_2':
-0.7637050252554097, 'B_10_3': -0.8175000792618587, 'B_10_4':
0.7734947077675806, 'B_10_5': -0.11072522189198346, 'B_10_6':
0.07976489925098534, 'B_10_7': 0.8597900332512314, 'B_10_8':
-0.2089194600907605, 'B_10_9': -0.14474094500006085}. Best is trial 102 with
value: 73.90742475345364.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:13:57,783] Trial 107 finished with value: 78.91975030675466 and
parameters: {'B_0_0': 0.17059024120423474, 'B_0_1': -0.25882317570010005,
'B_0_2': 0.8957330764584449, 'B_0_3': -0.14305032083024719, 'B_0_4':
-0.3533001002412664, 'B_0_5': 0.9287734432040813, 'B_0_6': 0.23913901672515842,
'B_0_7': 0.7783593130368088, 'B_0_8': 0.12263307082581455, 'B_0_9':
-0.13957889196560755, 'B_1_0': 0.24006549992851273, 'B_1_1':
-0.08148630968184693, 'B_1_2': 0.03545318990635857, 'B_1_3': 0.5678038526747237,
'B_1_4': -0.2480071578376996, 'B_1_5': 0.64612195118482, 'B_1_6':
-0.47037728462658984, 'B_1_7': 0.985306146878672, 'B_1_8': -0.42120990574801415,
'B_1_9': -0.23185756186382955, 'B_2_0': -0.5310033677841391, 'B_2_1':
-0.2794408341630305, 'B_2_2': -0.31449265482198296, 'B_2_3': 0.6657865191336201,
'B_2_4': -0.5318641259169297, 'B_2_5': -0.7633734420134942, 'B_2_6':
-0.5521387365607564, 'B_2_7': -0.7697005068829664, 'B_2_8':
-0.38190624694846964, 'B_2_9': 0.4482225433961414, 'B_3_0': -0.8100876252417303,
'B_3_1': -0.6324393659988349, 'B_3_2': 0.05391016376072334, 'B_3_3':
0.18948035749446654, 'B_3_4': -0.3184086616749691, 'B_3_5': 0.09655146176502735,

```

'B_3_6': 0.1709514396855314, 'B_3_7': -0.06464229376301828, 'B_3_8': 0.6762309580112836, 'B_3_9': -0.26932379070052254, 'B_4_0': -0.038083593876748686, 'B_4_1': 0.5882247900648785, 'B_4_2': 0.08883465176532422, 'B_4_3': -0.5276554533492206, 'B_4_4': -0.5041500353159436, 'B_4_5': 0.5758488381583667, 'B_4_6': -0.8296431339073249, 'B_4_7': -0.1201344708853799, 'B_4_8': 0.8223064195717629, 'B_4_9': 0.5059839053834098, 'B_5_0': -0.24608207091085277, 'B_5_1': -0.2960683365322799, 'B_5_2': 0.31801314184849916, 'B_5_3': -0.23194615246372266, 'B_5_4': 0.3351433954236669, 'B_5_5': -0.5302473963452586, 'B_5_6': -0.409197326858708, 'B_5_7': 0.18844771014435122, 'B_5_8': -0.4389111111391545, 'B_5_9': 0.5181417889531037, 'B_6_0': -0.1394242795965206, 'B_6_1': -0.7738771922743364, 'B_6_2': 0.9772979860610949, 'B_6_3': 0.840689167777827, 'B_6_4': -0.9026455443665957, 'B_6_5': -0.13339237684907265, 'B_6_6': -0.047371549848526505, 'B_6_7': 0.8524690199048519, 'B_6_8': 0.8084411723145805, 'B_6_9': 0.8238302647414084, 'B_7_0': -0.7944038904614652, 'B_7_1': -0.4421337278282396, 'B_7_2': -0.8921499518932442, 'B_7_3': 0.8762931012642303, 'B_7_4': 0.5170369421289306, 'B_7_5': -0.3415273491544373, 'B_7_6': -0.0496848931668642, 'B_7_7': -0.926470835727583, 'B_7_8': -0.7831319567007234, 'B_7_9': 0.35555871547630147, 'B_8_0': -0.9063440244070681, 'B_8_1': -0.042772536190751, 'B_8_2': 0.49504485855894165, 'B_8_3': 0.0963523788493576, 'B_8_4': -0.4659253215150581, 'B_8_5': -0.6460640885126414, 'B_8_6': 0.1384349806098476, 'B_8_7': -0.9676191456006418, 'B_8_8': 0.21609759315811702, 'B_8_9': 0.9630910195297598, 'B_9_0': 0.4431591760544023, 'B_9_1': -0.9784585327409857, 'B_9_2': -0.9880289841078685, 'B_9_3': 0.68889327386496, 'B_9_4': -0.34272571259209345, 'B_9_5': -0.7596892933648065, 'B_9_6': 0.05076567346252232, 'B_9_7': -0.21303737497878938, 'B_9_8': 0.4755652883337423, 'B_9_9': 0.9813739072645254, 'B_10_0': 0.9752232029108504, 'B_10_1': -0.9797766499556113, 'B_10_2': -0.7732980772583815, 'B_10_3': -0.8172312260780169, 'B_10_4': 0.8105324768992989, 'B_10_5': -0.10235617221596124, 'B_10_6': 0.09944214914075938, 'B_10_7': 0.8382014703555327, 'B_10_8': -0.22660706916856302, 'B_10_9': -0.16154948245395323}. Best is trial 102 with value: 73.90742475345364.

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:14:14,591] Trial 108 finished with value: 84.50974780669766 and

```

parameters: {'B_0_0': 0.13833494105780986, 'B_0_1': -0.21254616029899076,
'B_0_2': 0.8809734418245821, 'B_0_3': -0.17337356940665338, 'B_0_4':
-0.339955929892523, 'B_0_5': 0.9496614645433895, 'B_0_6': 0.24809055354492682,
'B_0_7': 0.7761026697346459, 'B_0_8': 0.15707417867183746, 'B_0_9':
-0.12012310440584553, 'B_1_0': 0.23995006084045345, 'B_1_1':
-0.06980056882407082, 'B_1_2': 0.07190696172431346, 'B_1_3': 0.5793215758443641,
'B_1_4': -0.26895541384804467, 'B_1_5': 0.9819008462096724, 'B_1_6':
-0.4776645793815723, 'B_1_7': 0.9762703412690307, 'B_1_8': -0.44644115184200994,
'B_1_9': -0.23914197264028852, 'B_2_0': -0.5240316634927863, 'B_2_1':
-0.26128410641987965, 'B_2_2': -0.3188361593806977, 'B_2_3': 0.6559799691217908,
'B_2_4': -0.5035439303367579, 'B_2_5': -0.7749954966972892, 'B_2_6':
-0.5824153957966677, 'B_2_7': -0.7185816996813539, 'B_2_8':
-0.33136987839981535, 'B_2_9': 0.5284332564765056, 'B_3_0': -0.8052920396727481,
'B_3_1': -0.6509488205078581, 'B_3_2': 0.051000309424884684, 'B_3_3':
0.14990547290747916, 'B_3_4': -0.34033650056797593, 'B_3_5':
0.07369571672408151, 'B_3_6': 0.08525789715387297, 'B_3_7':
-0.22967028793333955, 'B_3_8': 0.6821996579950527, 'B_3_9': -0.2816359665874499,
'B_4_0': -0.06075080140798942, 'B_4_1': 0.596318563996691, 'B_4_2':
0.07207935786499885, 'B_4_3': -0.5553228906477274, 'B_4_4': -0.5809323270646085,
'B_4_5': 0.5685187019554434, 'B_4_6': -0.7927545753320715, 'B_4_7':
-0.12058657813467108, 'B_4_8': 0.839343636589467, 'B_4_9': 0.5186392191062152,
'B_5_0': -0.2346953157084937, 'B_5_1': -0.2852372807151231, 'B_5_2':
0.3138723661642399, 'B_5_3': -0.2500838067425038, 'B_5_4': 0.28844139062329216,
'B_5_5': -0.5265458944227532, 'B_5_6': -0.4006353639257134, 'B_5_7':
0.15566962266514517, 'B_5_8': -0.4534946593941804, 'B_5_9': 0.3584358484095811,
'B_6_0': -0.08428542681752935, 'B_6_1': -0.7709813885132991, 'B_6_2':
0.9865870762471155, 'B_6_3': 0.8473712310878494, 'B_6_4': -0.9001854906582171,
'B_6_5': -0.12720000857076652, 'B_6_6': -0.014595796445247333, 'B_6_7':
0.698776632164076, 'B_6_8': 0.7937634449418763, 'B_6_9': 0.8288788607609789,
'B_7_0': -0.8095062453941563, 'B_7_1': -0.6624869481212456, 'B_7_2':
-0.6878463121520569, 'B_7_3': 0.8689220454070208, 'B_7_4': 0.5215388110080408,
'B_7_5': -0.3166906692314732, 'B_7_6': -0.04468345434791808, 'B_7_7':
-0.9210648938467212, 'B_7_8': -0.8093869558077496, 'B_7_9': 0.36965034830830246,
'B_8_0': -0.9143031149583749, 'B_8_1': -0.040682468334852576, 'B_8_2':
0.4939808761004255, 'B_8_3': 0.12432309439944435, 'B_8_4': -0.45545575232565744,
'B_8_5': -0.6383196021378004, 'B_8_6': 0.35781850959293127, 'B_8_7':
-0.987423793432578, 'B_8_8': 0.34294178062873243, 'B_8_9': 0.9733399808253884,
'B_9_0': 0.439789344891215, 'B_9_1': -0.9811908348619791, 'B_9_2':
-0.9844918657649125, 'B_9_3': 0.705746428460712, 'B_9_4': -0.5001885348922706,
'B_9_5': -0.7970320376554362, 'B_9_6': 0.041863900659070634, 'B_9_7':
0.6688582051476125, 'B_9_8': 0.48070118343569623, 'B_9_9': 0.9753312374418207,
'B_10_0': 0.9863610293621857, 'B_10_1': -0.970613474311771, 'B_10_2':
-0.7803049900847181, 'B_10_3': -0.8236493910974806, 'B_10_4':
0.8019877339274447, 'B_10_5': -0.11147730035053094, 'B_10_6':
0.07705190315436566, 'B_10_7': 0.8245880438346538, 'B_10_8':
-0.23038572696836013, 'B_10_9': -0.15451725197025212}. Best is trial 102 with
value: 73.90742475345364.

```

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].
Built F__[u_10].
Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:14:30,306] Trial 109 finished with value: 77.0806747080115 and parameters: {'B_0_0': 0.1448461775990315, 'B_0_1': -0.21096872589607102, 'B_0_2': 0.9832434579041653, 'B_0_3': -0.15501283654623374, 'B_0_4': -0.32855529775143283, 'B_0_5': 0.9293075678500302, 'B_0_6': 0.23534408676461116, 'B_0_7': 0.7651427049047969, 'B_0_8': 0.14595024235017265, 'B_0_9': -0.13699788458251716, 'B_1_0': 0.15835468081500548, 'B_1_1': -0.08321032325441967, 'B_1_2': 0.06496193309417182, 'B_1_3': 0.5732710061246066, 'B_1_4': -0.23104241131111108, 'B_1_5': 0.9848428329056135, 'B_1_6': -0.4863156336641025, 'B_1_7': 0.9880526834806208, 'B_1_8': -0.3885405795284103, 'B_1_9': -0.2488365789272273, 'B_2_0': -0.532417618495529, 'B_2_1': -0.26721984793085946, 'B_2_2': -0.33872808868058946, 'B_2_3': 0.6586870596351254, 'B_2_4': -0.5554452743557594, 'B_2_5': -0.7592271904928407, 'B_2_6': -0.5879321572261612, 'B_2_7': -0.7351518333582258, 'B_2_8': -0.33062916227172423, 'B_2_9': 0.514588117849359, 'B_3_0': -0.7565756378255863, 'B_3_1': -0.6424016888841951, 'B_3_2': 0.040174618550877186, 'B_3_3': 0.1534653223732689, 'B_3_4': -0.4707184987240465, 'B_3_5': 0.07383075637747069, 'B_3_6': 0.10205329635608702, 'B_3_7': -0.07049032497480108, 'B_3_8': 0.6631374846545244, 'B_3_9': -0.2566450196662111, 'B_4_0': -0.05388816763122295, 'B_4_1': 0.6050441585372444, 'B_4_2': 0.07965605935911785, 'B_4_3': -0.5614252729399011, 'B_4_4': -0.5234700208012961, 'B_4_5': 0.5601029271618545, 'B_4_6': -0.7994287801146491, 'B_4_7': -0.13810953182752966, 'B_4_8': 0.7839613271029197, 'B_4_9': 0.5091686342940616, 'B_5_0': -0.22679712290357112, 'B_5_1': -0.2895320896949664, 'B_5_2': 0.311521298292047, 'B_5_3': -0.24336982298269236, 'B_5_4': 0.28424680165648214, 'B_5_5': -0.53419366181165, 'B_5_6': -0.4127707696848666, 'B_5_7': 0.2086017601488321, 'B_5_8': -0.47233163694654334, 'B_5_9': 0.36328024030150075, 'B_6_0': -0.1313788639187512, 'B_6_1': -0.7422812149764603, 'B_6_2': 0.9786400847256261, 'B_6_3': 0.8739495400190468, 'B_6_4': -0.9038619492696812, 'B_6_5': -0.13018663571232053, 'B_6_6': -0.05122000005820236, 'B_6_7': 0.7076524149356993, 'B_6_8': 0.7942789693235082, 'B_6_9': 0.8162321792930989, 'B_7_0': -0.8065554013559841, 'B_7_1': -0.6959176621976989, 'B_7_2': -0.6809701508096712, 'B_7_3': 0.8779389709245412, 'B_7_4': 0.5190925636548619, 'B_7_5': -0.2833794039188421, 'B_7_6': -0.04680305965557291, 'B_7_7': -0.9220156469831183, 'B_7_8': -0.7817025230342127, 'B_7_9': 0.37857930597873035,

'B_8_0': -0.8860974656722772, 'B_8_1': -0.03960355323580633, 'B_8_2': 0.48259597445420716, 'B_8_3': 0.08955239355636652, 'B_8_4': -0.4624894523989246, 'B_8_5': -0.6564719405075927, 'B_8_6': 0.3339178293557912, 'B_8_7': -0.9821859455868664, 'B_8_8': 0.19613603069929603, 'B_8_9': 0.980933812060848, 'B_9_0': 0.43330629048166597, 'B_9_1': -0.9597537547730127, 'B_9_2': -0.9706414174520038, 'B_9_3': 0.6818447739710329, 'B_9_4': -0.30694343521796363, 'B_9_5': -0.7948377089509859, 'B_9_6': 0.06371359051331771, 'B_9_7': -0.22201561173673157, 'B_9_8': 0.4780919869569189, 'B_9_9': 0.9687536162785932, 'B_10_0': 0.9866800598535136, 'B_10_1': -0.6367180384509886, 'B_10_2': -0.8947419152329227, 'B_10_3': -0.8170803048169892, 'B_10_4': 0.7986762414470248, 'B_10_5': -0.10959108206029189, 'B_10_6': 0.05755488099294162, 'B_10_7': 0.8185335835183143, 'B_10_8': -0.23892167756983992, 'B_10_9': -0.18826946373054398}. Best is trial 102 with value: 73.90742475345364.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:14:48,269] Trial 110 finished with value: 79.83466096666714 and parameters: {'B_0_0': 0.1389328098601491, 'B_0_1': -0.19986430909821645, 'B_0_2': 0.989201817704888, 'B_0_3': -0.17895708096565585, 'B_0_4': -0.32792755829470277, 'B_0_5': 0.9863774586163777, 'B_0_6': 0.22067726567170515, 'B_0_7': 0.7567342265504484, 'B_0_8': 0.1367542986588748, 'B_0_9': -0.14989577246382188, 'B_1_0': 0.22899190939016717, 'B_1_1': -0.05730789220216653, 'B_1_2': 0.07319245899648733, 'B_1_3': 0.5902418749651596, 'B_1_4': -0.2321594743489611, 'B_1_5': 0.9822251426124791, 'B_1_6': -0.46302560457815195, 'B_1_7': 0.9568635052806133, 'B_1_8': -0.3972880831778692, 'B_1_9': -0.22363172648810942, 'B_2_0': -0.5167431962504607, 'B_2_1': -0.25373619101746125, 'B_2_2': -0.3689295668115886, 'B_2_3': 0.6495713304801893, 'B_2_4': -0.5454272739145289, 'B_2_5': -0.7642503314450976, 'B_2_6': -0.5573961978019345, 'B_2_7': -0.728934371531377, 'B_2_8': -0.3405738074392684, 'B_2_9': 0.47857988517416294, 'B_3_0': -0.8222597666752971, 'B_3_1': -0.6429997015953313, 'B_3_2': 0.05938460890762513, 'B_3_3': 0.17876062924650224, 'B_3_4': -0.331858565526457, 'B_3_5': 0.06902413890987089, 'B_3_6': 0.08811514765910045, 'B_3_7': -0.0498606283317361, 'B_3_8': 0.6783121411500371, 'B_3_9': -0.26197775233801573, 'B_4_0': -0.0562750941020423, 'B_4_1': 0.596828051728461, 'B_4_2': 0.06565734718237272, 'B_4_3': -0.5863080676121721,

'B_4_4': -0.5295739389875577, 'B_4_5': 0.5733191958391233, 'B_4_6':
 -0.9303504663540583, 'B_4_7': -0.12479513400788321, 'B_4_8': 0.7768298966300025,
 'B_4_9': 0.5129588609767813, 'B_5_0': -0.2292954097303674, 'B_5_1':
 -0.285654541275241, 'B_5_2': 0.3183839154474618, 'B_5_3': -0.22822599490112958,
 'B_5_4': 0.2817377801087142, 'B_5_5': -0.5127557506143422, 'B_5_6':
 -0.3880350822822609, 'B_5_7': 0.2162394621059721, 'B_5_8': -0.4524638345329359,
 'B_5_9': 0.3580012613943371, 'B_6_0': -0.13340753099277394, 'B_6_1':
 -0.7570784489194216, 'B_6_2': 0.9640431678867551, 'B_6_3': 0.8522859127951794,
 'B_6_4': -0.9108548844558578, 'B_6_5': -0.27161023236544396, 'B_6_6':
 -0.04891756595186082, 'B_6_7': 0.8495470435516352, 'B_6_8': 0.8197494474987601,
 'B_6_9': 0.9863830274202758, 'B_7_0': -0.7760077397713414, 'B_7_1':
 -0.6638037556045804, 'B_7_2': -0.9102617441685662, 'B_7_3': 0.8853329007357524,
 'B_7_4': 0.5129118896444868, 'B_7_5': -0.30428822508907216, 'B_7_6':
 -0.04580781150332276, 'B_7_7': -0.9283675138518955, 'B_7_8':
 -0.7844234104104517, 'B_7_9': 0.35936333475736737, 'B_8_0': -0.9093032382518694,
 'B_8_1': -0.03665256486320795, 'B_8_2': 0.47433280558722496, 'B_8_3':
 0.10785173465405418, 'B_8_4': -0.510998233583771, 'B_8_5': -0.6391523580268259,
 'B_8_6': 0.328292098405211, 'B_8_7': -0.9896011181965343, 'B_8_8':
 0.33848170370021785, 'B_8_9': 0.9692080351603581, 'B_9_0': 0.4458843957210145,
 'B_9_1': -0.9660363691967053, 'B_9_2': -0.9893757095444725, 'B_9_3':
 0.9813313552991061, 'B_9_4': -0.31515186838976655, 'B_9_5': -0.8011349873887395,
 'B_9_6': 0.06664285171345904, 'B_9_7': -0.2253484576106916, 'B_9_8':
 0.46836986128120933, 'B_9_9': 0.9614465809607595, 'B_10_0': 0.9893895552238936,
 'B_10_1': -0.9487685730489542, 'B_10_2': -0.7697162668251386, 'B_10_3':
 -0.8199513982435536, 'B_10_4': 0.7864013844602347, 'B_10_5': -0.103580010313058,
 'B_10_6': 0.06492070281266524, 'B_10_7': 0.8463431601429204, 'B_10_8':
 -0.23280188251759754, 'B_10_9': -0.20016673752480071}. Best is trial 102 with
 value: 73.90742475345364.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:15:07,952] Trial 111 finished with value: 81.78381703800883 and
 parameters: {'B_0_0': 0.12971791983825692, 'B_0_1': -0.2069328133916874,
 'B_0_2': 0.8866808900062377, 'B_0_3': -0.168848156311849, 'B_0_4':
 -0.3339909929487066, 'B_0_5': 0.9712090778338427, 'B_0_6': 0.22969793194717048,
 'B_0_7': 0.761740502267293, 'B_0_8': 0.1333876982412556, 'B_0_9':

-0.1321489143955709, 'B_1_0': 0.21515836975889693, 'B_1_1':
 -0.07391715196014925, 'B_1_2': 0.055830760613876644, 'B_1_3':
 0.5862409959988887, 'B_1_4': -0.22825893745676312, 'B_1_5': 0.9837652248668948,
 'B_1_6': -0.5032541644187286, 'B_1_7': 0.9600362975170101, 'B_1_8':
 -0.4071777167722008, 'B_1_9': -0.23462211004844089, 'B_2_0':
 -0.5191727966856079, 'B_2_1': -0.285921934868149, 'B_2_2': -0.3651178716437006,
 'B_2_3': 0.6684260008167779, 'B_2_4': -0.543899270396245, 'B_2_5':
 -0.7478895816328217, 'B_2_6': -0.579446313347806, 'B_2_7': -0.7212770777346621,
 'B_2_8': -0.3322517037683974, 'B_2_9': 0.4902269322471181, 'B_3_0':
 -0.8246033461015315, 'B_3_1': -0.6565495694505574, 'B_3_2':
 0.053551016179316124, 'B_3_3': 0.1693058619944191, 'B_3_4': -0.330275764876423,
 'B_3_5': 0.07305098558470946, 'B_3_6': 0.07700304793924773, 'B_3_7':
 -0.055913167593034656, 'B_3_8': 0.6764380075224983, 'B_3_9':
 -0.2656367234903525, 'B_4_0': -0.058266016896710064, 'B_4_1':
 0.5970955114658738, 'B_4_2': 0.062017202770186304, 'B_4_3': -0.5459458066208563,
 'B_4_4': -0.5241390284191787, 'B_4_5': 0.5645209412910863, 'B_4_6':
 -0.9280309364344065, 'B_4_7': -0.12906026208145316, 'B_4_8': 0.7735824193678427,
 'B_4_9': 0.5102486882404863, 'B_5_0': -0.2321014352017694, 'B_5_1':
 -0.28938754761030494, 'B_5_2': 0.31497765923928894, 'B_5_3':
 -0.2339994395940735, 'B_5_4': 0.30143383913351546, 'B_5_5': -0.5069951562807593,
 'B_5_6': -0.38617033239175563, 'B_5_7': 0.21548269793895242, 'B_5_8':
 -0.46044311487541206, 'B_5_9': 0.36279778709161226, 'B_6_0':
 -0.1291745435930533, 'B_6_1': -0.7697071319229974, 'B_6_2': 0.9895076339724812,
 'B_6_3': 0.8623426097304523, 'B_6_4': -0.9025061553669437, 'B_6_5':
 -0.26620308161459494, 'B_6_6': -0.04442472878679242, 'B_6_7':
 0.8349260297213138, 'B_6_8': 0.8109313939447238, 'B_6_9': 0.9861629846083767,
 'B_7_0': -0.7635098789725252, 'B_7_1': -0.6601325740304258, 'B_7_2':
 -0.6860973325936456, 'B_7_3': 0.8954969182793082, 'B_7_4': 0.5184067090632016,
 'B_7_5': -0.32109023297768147, 'B_7_6': -0.04809820855030997, 'B_7_7':
 -0.9156368016471518, 'B_7_8': -0.771149302487704, 'B_7_9': 0.3592062496995628,
 'B_8_0': -0.875851259634763, 'B_8_1': -0.03557728545275296, 'B_8_2':
 0.47274186317446415, 'B_8_3': 0.10555096720605649, 'B_8_4': -0.4674118846529518,
 'B_8_5': -0.6613201316213608, 'B_8_6': 0.3377439005435395, 'B_8_7':
 -0.983130453138089, 'B_8_8': 0.22982123305268756, 'B_8_9': 0.9549176955765342,
 'B_9_0': 0.44616057446219953, 'B_9_1': -0.9838727850306458, 'B_9_2':
 -0.96534098944069, 'B_9_3': 0.6583548815594552, 'B_9_4': -0.2933071296472425,
 'B_9_5': -0.7973824686469284, 'B_9_6': 0.07691916932042106, 'B_9_7':
 -0.1989775630273626, 'B_9_8': 0.4806025761140887, 'B_9_9': 0.9868918828950783,
 'B_10_0': 0.9712896419364976, 'B_10_1': -0.9663631261474719, 'B_10_2':
 -0.8796709488748948, 'B_10_3': -0.8132408478999069, 'B_10_4':
 0.7977702931527706, 'B_10_5': -0.12820213100464803, 'B_10_6':
 0.06792927350687597, 'B_10_7': 0.9888559875381885, 'B_10_8':
 -0.23726653223347274, 'B_10_9': -0.16632994857039246}. Best is trial 102 with
 value: 73.90742475345364.

Building the Basis functions for the RKHS:

Built F__[u₁].

Built F__[u₂].

Built F__[u₃].

```

Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:15:27,067] Trial 112 finished with value: 77.3859626225838 and parameters: {'B_0_0': 0.13411272917338068, 'B_0_1': -0.20981961592635218, 'B_0_2': 0.875297899413604, 'B_0_3': -0.16843651604117627, 'B_0_4': -0.35720714030641937, 'B_0_5': 0.9436147672104371, 'B_0_6': 0.21826573419108836, 'B_0_7': 0.7711156776696533, 'B_0_8': 0.13322550777235567, 'B_0_9': -0.1427480261527322, 'B_1_0': 0.23929281854040493, 'B_1_1': -0.07759419919684447, 'B_1_2': 0.08068566091926584, 'B_1_3': 0.5760596208451683, 'B_1_4': -0.22646797481874198, 'B_1_5': 0.9774134684082945, 'B_1_6': -0.4604802370101002, 'B_1_7': 0.9813242507679415, 'B_1_8': -0.40302904027302955, 'B_1_9': -0.23439108109701604, 'B_2_0': -0.508520995182759, 'B_2_1': -0.2789085283513978, 'B_2_2': -0.3574184200857249, 'B_2_3': 0.6659870249606081, 'B_2_4': -0.5498481824860144, 'B_2_5': -0.7546660318664546, 'B_2_6': -0.5852427311852773, 'B_2_7': -0.7498013276862789, 'B_2_8': -0.33593742253029246, 'B_2_9': 0.4955715678443153, 'B_3_0': -0.8250974878645447, 'B_3_1': -0.6386316624709063, 'B_3_2': 0.07685846197171427, 'B_3_3': 0.1653710653381865, 'B_3_4': -0.306009030083447, 'B_3_5': 0.07546139375958773, 'B_3_6': 0.07653313803198523, 'B_3_7': -0.07492952743240755, 'B_3_8': 0.6687121584981464, 'B_3_9': -0.26622872080243015, 'B_4_0': -0.05543817244476546, 'B_4_1': 0.5950191646174448, 'B_4_2': 0.05059213240711656, 'B_4_3': -0.5772546612301087, 'B_4_4': -0.5323353332122394, 'B_4_5': 0.5705158360432018, 'B_4_6': -0.9271270267246421, 'B_4_7': -0.12522876124823093, 'B_4_8': 0.7951517842899698, 'B_4_9': 0.4863769594378837, 'B_5_0': -0.23681949254851595, 'B_5_1': -0.2875829750426848, 'B_5_2': 0.32402241285855315, 'B_5_3': -0.24836734734546698, 'B_5_4': 0.24177260855661664, 'B_5_5': -0.5294099011185691, 'B_5_6': -0.4046305102808998, 'B_5_7': 0.21763701738607666, 'B_5_8': -0.4548437874190739, 'B_5_9': 0.3649642571121324, 'B_6_0': -0.10421162389328659, 'B_6_1': -0.8077231717797136, 'B_6_2': 0.9813238719158616, 'B_6_3': 0.8628985662255078, 'B_6_4': -0.9130034194749445, 'B_6_5': -0.28299370869926854, 'B_6_6': -0.038592601021371103, 'B_6_7': 0.6873872191902421, 'B_6_8': 0.7881536120190296, 'B_6_9': 0.8271971423383928, 'B_7_0': -0.7997205025856284, 'B_7_1': -0.6715264412003046, 'B_7_2': -0.6791544444965111, 'B_7_3': 0.8784025641954262, 'B_7_4': 0.5351488622777449, 'B_7_5': -0.30806739950736906, 'B_7_6': -0.20288122223340627, 'B_7_7': -0.9210983996788042, 'B_7_8': -0.7734161597596978, 'B_7_9': 0.3538771551105812, 'B_8_0': -0.8823170640598685, 'B_8_1': -0.049666755185875516, 'B_8_2': 0.5117808623104718, 'B_8_3': 0.10998311721823466, 'B_8_4': -0.5108928595807997, 'B_8_5': -0.6530030883232867,

```

'B_8_6': 0.14358637402471758, 'B_8_7': -0.9859986998692104, 'B_8_8':
0.2141239200518545, 'B_8_9': 0.9814212455928178, 'B_9_0': 0.4550790029674546,
'B_9_1': -0.989490278511931, 'B_9_2': -0.989036356482524, 'B_9_3':
0.6737506340617987, 'B_9_4': -0.30563085331173495, 'B_9_5': -0.7854370908803278,
'B_9_6': 0.07645124763275016, 'B_9_7': -0.20642755761615852, 'B_9_8':
0.46473520453758066, 'B_9_9': 0.9691089406816693, 'B_10_0': 0.9688207430347915,
'B_10_1': -0.9881749771777543, 'B_10_2': -0.8833146295226958, 'B_10_3':
-0.8112372772359607, 'B_10_4': 0.7724857799813484, 'B_10_5':
-0.1264985978062121, 'B_10_6': 0.06491101287176966, 'B_10_7':
0.8313192014904618, 'B_10_8': -0.20616871597016298, 'B_10_9':
-0.19000436933301432}. Best is trial 102 with value: 73.90742475345364.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:15:45,973] Trial 113 finished with value: 83.01107380333177 and
parameters: {'B_0_0': 0.1304483663081559, 'B_0_1': -0.19791887353353294,
'B_0_2': 0.9889708447310899, 'B_0_3': -0.14908213312306184, 'B_0_4':
-0.3160664847653633, 'B_0_5': 0.9858487465083642, 'B_0_6': 0.253595026476187,
'B_0_7': 0.7614517091568698, 'B_0_8': 0.26556828676622884, 'B_0_9':
-0.05985130068213361, 'B_1_0': 0.15188794132725772, 'B_1_1':
-0.07118536961909433, 'B_1_2': 0.04801119187075435, 'B_1_3': 0.5658767541298071,
'B_1_4': -0.2323908383498268, 'B_1_5': 0.979115581014122, 'B_1_6':
-0.4239059063485146, 'B_1_7': 0.8924090560447527, 'B_1_8': -0.37603408355307716,
'B_1_9': -0.28315604622384133, 'B_2_0': -0.5279133269199724, 'B_2_1':
-0.25306255471262257, 'B_2_2': -0.39746588755034407, 'B_2_3':
0.6384746085471382, 'B_2_4': -0.5447156667857953, 'B_2_5': -0.7951296910275593,
'B_2_6': -0.49429650027973, 'B_2_7': -0.7065304573181984, 'B_2_8':
-0.3336294179459212, 'B_2_9': 0.5293076108111917, 'B_3_0': -0.7383878117579148,
'B_3_1': -0.6127640288194759, 'B_3_2': 0.07328450866080836, 'B_3_3':
0.21207572678932202, 'B_3_4': -0.3265958978869913, 'B_3_5': 0.08633876142347544,
'B_3_6': 0.07483315424696244, 'B_3_7': -0.0652970760129494, 'B_3_8':
0.6441281451396694, 'B_3_9': -0.2702729290706909, 'B_4_0': -0.07255922393870866,
'B_4_1': 0.5635886013608203, 'B_4_2': 0.0451635894675585, 'B_4_3':
-0.5363792786356653, 'B_4_4': -0.6298971604706036, 'B_4_5': 0.5740730925931085,
'B_4_6': -0.9179494526360855, 'B_4_7': -0.13366429265610785, 'B_4_8':
0.7747082266853719, 'B_4_9': 0.5271663616468527, 'B_5_0': -0.20457486475602335,

```

'B_5_1': -0.303015491203322, 'B_5_2': 0.34828363953234964, 'B_5_3':
-0.2327962599495816, 'B_5_4': 0.29050729875743553, 'B_5_5': -0.5177619286724408,
'B_5_6': -0.41940118669266097, 'B_5_7': 0.2029509456324112, 'B_5_8':
-0.4747725383059662, 'B_5_9': 0.5106296538955204, 'B_6_0': -0.22117386658187288,
'B_6_1': -0.7864760458513823, 'B_6_2': 0.9834069196482518, 'B_6_3':
0.8550816276362935, 'B_6_4': -0.9004013923161641, 'B_6_5': -0.2825125837765956,
'B_6_6': 0.03589552546667483, 'B_6_7': 0.6936249602768718, 'B_6_8':
0.8072374164626014, 'B_6_9': 0.6269839230773628, 'B_7_0': -0.7825319618468392,
'B_7_1': -0.6938264107215316, 'B_7_2': -0.6767835154923157, 'B_7_3':
0.8984463674841091, 'B_7_4': 0.5202050101298361, 'B_7_5': -0.31891542290338143,
'B_7_6': -0.1927028493735932, 'B_7_7': -0.9229280972289123, 'B_7_8':
-0.769685402047468, 'B_7_9': 0.24439098093409078, 'B_8_0': -0.9061834005711515,
'B_8_1': -0.05404187159104085, 'B_8_2': 0.528940159291817, 'B_8_3':
0.1609846703813422, 'B_8_4': -0.5100800089969233, 'B_8_5': -0.715770086551304,
'B_8_6': 0.011452967477116549, 'B_8_7': -0.988744382214691, 'B_8_8':
0.21261557629364525, 'B_8_9': 0.9682713217045965, 'B_9_0': 0.46329563221596076,
'B_9_1': -0.9377102815232427, 'B_9_2': -0.9563766718785192, 'B_9_3':
0.9807870016606217, 'B_9_4': -0.3174973277064199, 'B_9_5': -0.7897087159816063,
'B_9_6': 0.08056646181280283, 'B_9_7': -0.43788089663200547, 'B_9_8':
0.4482220712879029, 'B_9_9': 0.9867901281459693, 'B_10_0': 0.9869790756619983,
'B_10_1': -0.9699602304959744, 'B_10_2': -0.7582263778337043, 'B_10_3':
-0.7719444583529249, 'B_10_4': 0.7813659424740861, 'B_10_5':
-0.13661672314567064, 'B_10_6': 0.0048306239178448895, 'B_10_7':
0.9667795399780978, 'B_10_8': -0.2131140141321993, 'B_10_9':
-0.21123652797099762}. Best is trial 102 with value: 73.90742475345364.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].
Built F__[u_10].
Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:16:02,254] Trial 114 finished with value: 78.80123485726813 and
parameters: {'B_0_0': 0.11340289557144792, 'B_0_1': -0.17482964587701572,
'B_0_2': 0.975675684063785, 'B_0_3': -0.1668902905974768, 'B_0_4':
-0.35049619393218295, 'B_0_5': 0.9836515078097594, 'B_0_6': 0.23969043306580182,
'B_0_7': 0.7745644301542645, 'B_0_8': 0.27104241082134944, 'B_0_9':
-0.08934782926703601, 'B_1_0': 0.15659871604176961, 'B_1_1':
-0.0666379119853791, 'B_1_2': 0.07031211298328692, 'B_1_3': 0.5733898621083637,
'B_1_4': -0.23297340791644228, 'B_1_5': 0.9797141019893393, 'B_1_6':

-0.4431244999893995, 'B_1_7': 0.8839051958663395, 'B_1_8': -0.38972166903283295,
 'B_1_9': -0.28161646859787576, 'B_2_0': -0.527243652908716, 'B_2_1':
 -0.2763866364245415, 'B_2_2': -0.40345727269456916, 'B_2_3': 0.6429986605218794,
 'B_2_4': -0.5415992835739237, 'B_2_5': -0.7866715265821935, 'B_2_6':
 -0.4782085871558507, 'B_2_7': -0.7389076655158979, 'B_2_8':
 -0.36562434101697505, 'B_2_9': 0.5098354656536233, 'B_3_0': -0.7536345977377041,
 'B_3_1': -0.6461662898322507, 'B_3_2': 0.047931448979627866, 'B_3_3':
 0.18733266585813654, 'B_3_4': -0.31740860106937785, 'B_3_5':
 0.07793289376745523, 'B_3_6': 0.06967648226984545, 'B_3_7': -0.0702438505001403,
 'B_3_8': 0.6807185512817425, 'B_3_9': -0.2885409118419489, 'B_4_0':
 -0.05122354141346258, 'B_4_1': 0.5558401130045387, 'B_4_2': 0.06641714138984224,
 'B_4_3': -0.549304065146215, 'B_4_4': -0.6229527965430096, 'B_4_5':
 0.5767321277770541, 'B_4_6': -0.9188061952841107, 'B_4_7': -0.13038966393056622,
 'B_4_8': 0.7616134874909116, 'B_4_9': 0.5384360431342724, 'B_5_0':
 -0.325702205005838, 'B_5_1': -0.2849139988023776, 'B_5_2': 0.33038187381313716,
 'B_5_3': -0.2291126430963467, 'B_5_4': 0.31153977702442137, 'B_5_5':
 -0.5441347930729429, 'B_5_6': -0.3960796947221896, 'B_5_7': 0.20753071230993864,
 'B_5_8': -0.4660352998408534, 'B_5_9': 0.3613275820006046, 'B_6_0':
 -0.13126769380522052, 'B_6_1': -0.8190023399152713, 'B_6_2': 0.984819469589986,
 'B_6_3': 0.812325017247251, 'B_6_4': -0.8990285721997999, 'B_6_5':
 -0.2909277229633934, 'B_6_6': 0.03565015315518838, 'B_6_7': 0.6807864361090108,
 'B_6_8': 0.8062003087299376, 'B_6_9': 0.9870412066610048, 'B_7_0':
 -0.7815777777156836, 'B_7_1': -0.6713204837800008, 'B_7_2': -0.6785226498306852,
 'B_7_3': 0.9105292063743711, 'B_7_4': 0.4890958917666947, 'B_7_5':
 -0.32103730930287805, 'B_7_6': -0.18872032759904464, 'B_7_7':
 -0.9307321093784074, 'B_7_8': -0.7612635569860882, 'B_7_9': 0.25014479973427706,
 'B_8_0': -0.9094492263111512, 'B_8_1': -0.04204279992028673, 'B_8_2':
 0.537574109783366, 'B_8_3': 0.1618967309083556, 'B_8_4': -0.5266771885454444,
 'B_8_5': -0.710647168968595, 'B_8_6': -0.00507229011855273, 'B_8_7':
 -0.9626920142784985, 'B_8_8': 0.22000626497845868, 'B_8_9': 0.986645019138345,
 'B_9_0': 0.4558449978395901, 'B_9_1': -0.987004321942877, 'B_9_2':
 -0.9885423822443279, 'B_9_3': 0.8372917709390315, 'B_9_4': -0.3312710264682345,
 'B_9_5': -0.7641263357639116, 'B_9_6': 0.08275262730931898, 'B_9_7':
 -0.434626750361618, 'B_9_8': 0.4548712429145557, 'B_9_9': 0.9314038827275272,
 'B_10_0': 0.9638525274549012, 'B_10_1': -0.9824043916420516, 'B_10_2':
 -0.7844503243064496, 'B_10_3': -0.7525108432811829, 'B_10_4':
 0.8145546663555688, 'B_10_5': -0.1274105883776519, 'B_10_6':
 -0.01077923219556666, 'B_10_7': 0.849086084757923, 'B_10_8':
 -0.23954650490123724, 'B_10_9': -0.21714152696961658}. Best is trial 102 with
 value: 73.90742475345364.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].

```

Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:16:19,978] Trial 115 finished with value: 81.49152985886244 and
parameters: {'B_0_0': 0.26642622886044764, 'B_0_1': -0.13231605247673772,
'B_0_2': 0.8068461965542646, 'B_0_3': -0.11337400969921975, 'B_0_4':
-0.40467761125008955, 'B_0_5': 0.8565754615939657, 'B_0_6': 0.364221885586098,
'B_0_7': 0.7467184956011813, 'B_0_8': 0.25907829328133897, 'B_0_9':
-0.044477132922937174, 'B_1_0': 0.17367781576930488, 'B_1_1':
-0.013247664170247794, 'B_1_2': 0.07072934504366055, 'B_1_3':
0.5589455678425045, 'B_1_4': -0.22341287344328123, 'B_1_5': 0.8364928591734888,
'B_1_6': -0.4294694793923858, 'B_1_7': 0.9063574294112693, 'B_1_8':
-0.38700791324744077, 'B_1_9': -0.2877366232427743, 'B_2_0':
-0.5157907775947892, 'B_2_1': -0.2979944719943251, 'B_2_2':
-0.39626977512063793, 'B_2_3': 0.8109502742006107, 'B_2_4': -0.4218758882401523,
'B_2_5': -0.7811702757809235, 'B_2_6': -0.4686563735452892, 'B_2_7':
-0.6810171792564405, 'B_2_8': -0.37281206996439575, 'B_2_9':
0.46470714489259085, 'B_3_0': -0.7279642593434167, 'B_3_1': -0.6449196864248472,
'B_3_2': 0.0788615019131495, 'B_3_3': 0.2122976700848456, 'B_3_4':
-0.47949628733124816, 'B_3_5': 0.11358809293234105, 'B_3_6':
0.03934062587138562, 'B_3_7': 0.03521644508443214, 'B_3_8': 0.6379291335086128,
'B_3_9': -0.2591827647334213, 'B_4_0': -0.06010919442937771, 'B_4_1':
0.5441268886025225, 'B_4_2': 0.2056200521554089, 'B_4_3': -0.5351611018324754,
'B_4_4': -0.6565456279135626, 'B_4_5': 0.5932246775334976, 'B_4_6':
-0.9285222485911944, 'B_4_7': -0.2557020603205744, 'B_4_8': 0.7736968573538501,
'B_4_9': 0.5806917617671777, 'B_5_0': -0.326365873765973, 'B_5_1':
-0.34250239824408124, 'B_5_2': 0.1505350278689122, 'B_5_3': -0.2131546553809962,
'B_5_4': 0.3188942300678941, 'B_5_5': -0.5398699655020722, 'B_5_6':
-0.33776142543981247, 'B_5_7': 0.23724413601700456, 'B_5_8':
-0.46428766827356105, 'B_5_9': 0.5063570952117272, 'B_6_0':
-0.22251898994791414, 'B_6_1': -0.7067364339194423, 'B_6_2': 0.7978522500169041,
'B_6_3': 0.8771282889957824, 'B_6_4': -0.8098604693999537, 'B_6_5':
-0.2707629355706123, 'B_6_6': 0.04085069777039417, 'B_6_7': 0.8423514012893071,
'B_6_8': 0.8162712612774595, 'B_6_9': 0.9721606611332418, 'B_7_0':
-0.7499825652846888, 'B_7_1': -0.6719618769031114, 'B_7_2': -0.6306228291456638,
'B_7_3': 0.9173607596870574, 'B_7_4': 0.5008996819502322, 'B_7_5':
-0.3468307158944884, 'B_7_6': -0.17202726531581808, 'B_7_7':
-0.9208795938495665, 'B_7_8': -0.7696128513320379, 'B_7_9': 0.2675674299340662,
'B_8_0': -0.8567449995566552, 'B_8_1': -0.06683797955983395, 'B_8_2':
0.5211486808947173, 'B_8_3': 0.16292709921844134, 'B_8_4': -0.5205630166356818,
'B_8_5': -0.5291724262109598, 'B_8_6': 0.02164402243711966, 'B_8_7':
-0.9447649417774369, 'B_8_8': 0.2214881685295232, 'B_8_9': 0.9672876883470845,
'B_9_0': 0.4214531078430037, 'B_9_1': -0.8995569487061642, 'B_9_2':
-0.9301084061208603, 'B_9_3': 0.8326463936970723, 'B_9_4': -0.3072172211391904,

```

'B_9_5': -0.7529432520720248, 'B_9_6': 0.09878114527796064, 'B_9_7':
-0.4367661213208138, 'B_9_8': 0.45358768427293605, 'B_9_9': 0.9301821453785911,
'B_10_0': 0.9691942848588185, 'B_10_1': -0.9846562419070061, 'B_10_2':
-0.9256707596379519, 'B_10_3': -0.7708170729888606, 'B_10_4':
0.6509939288443714, 'B_10_5': -0.06418169814912628, 'B_10_6':
-0.0029152856294275466, 'B_10_7': 0.9732583046477246, 'B_10_8':
-0.19306191447057935, 'B_10_9': -0.25222962444180275}. Best is trial 102 with
value: 73.90742475345364.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [<F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:16:36,236] Trial 116 finished with value: 72.22876181340362 and
parameters: {'B_0_0': 0.2669910227701995, 'B_0_1': -0.11679023106268889,
'B_0_2': 0.9895261821714613, 'B_0_3': -0.0788332629957806, 'B_0_4':
-0.39106838466722016, 'B_0_5': 0.8630475989876831, 'B_0_6': 0.36766587803361617,
'B_0_7': 0.7386100984792647, 'B_0_8': 0.2590775756374013, 'B_0_9':
-0.043839106272900025, 'B_1_0': 0.29541550470678235, 'B_1_1':
0.0002504000967008535, 'B_1_2': 0.10019585554041543, 'B_1_3':
0.5421679846846852, 'B_1_4': -0.22994787051214952, 'B_1_5': 0.8520095676515229,
'B_1_6': -0.43448384075359064, 'B_1_7': 0.8940794221294247, 'B_1_8':
-0.51467049048532, 'B_1_9': -0.36653888491081976, 'B_2_0': -0.6288723730774993,
'B_2_1': -0.3150969183847812, 'B_2_2': -0.43578042716772153, 'B_2_3':
0.8241422061291943, 'B_2_4': -0.5477480344303118, 'B_2_5': -0.58159909626688,
'B_2_6': -0.4751516312079566, 'B_2_7': -0.6619532853897805, 'B_2_8':
-0.4441651237016425, 'B_2_9': 0.432719995053581, 'B_3_0': -0.7358265712041867,
'B_3_1': -0.4460041538402506, 'B_3_2': -0.08909000270193926, 'B_3_3':
0.21084072159538167, 'B_3_4': -0.45526519122899933, 'B_3_5': 0.1164819382455808,
'B_3_6': 0.04151096085646167, 'B_3_7': -0.06938420163028523, 'B_3_8':
0.5859596954577203, 'B_3_9': -0.2516163291766448, 'B_4_0': -0.04393486413315319,
'B_4_1': 0.5492149688171004, 'B_4_2': 0.2187435026879785, 'B_4_3':
-0.527830355780533, 'B_4_4': -0.6380517700981589, 'B_4_5': 0.6302066086010465,
'B_4_6': -0.8636713707099731, 'B_4_7': -0.25111095056605126, 'B_4_8':
0.7909939196910785, 'B_4_9': 0.5985770757835225, 'B_5_0': -0.3209067901744079,
'B_5_1': -0.36980931503409387, 'B_5_2': 0.3390988634565395, 'B_5_3':
-0.1287523050652758, 'B_5_4': 0.44862666287620945, 'B_5_5': -0.5403406706823127,
'B_5_6': -0.3320786508343041, 'B_5_7': 0.24735895691075444, 'B_5_8':


```

-0.47154901897355533, 'B_5_9': 0.5020435049821782, 'B_6_0':
-0.21560784955649176, 'B_6_1': -0.7066573878912633, 'B_6_2': 0.9388772153557853,
'B_6_3': 0.8863442499281852, 'B_6_4': -0.8177536176362374, 'B_6_5':
-0.17812137604373054, 'B_6_6': 0.04138367608731159, 'B_6_7': 0.6931550702949741,
'B_6_8': 0.6908041664199446, 'B_6_9': 0.9723312204639305, 'B_7_0':
-0.7573758917674752, 'B_7_1': -0.5746505880755144, 'B_7_2': -0.6174621692492684,
'B_7_3': 0.9356898214735487, 'B_7_4': 0.686957392763704, 'B_7_5':
-0.35682453164309913, 'B_7_6': -0.19106457256262976, 'B_7_7':
-0.923661066376057, 'B_7_8': -0.7475062839672368, 'B_7_9': 0.2670021044670368,
'B_8_0': -0.8482672521490309, 'B_8_1': 0.08556192749055902, 'B_8_2':
0.5444838181698947, 'B_8_3': 0.15907340696770833, 'B_8_4': -0.5234302246295717,
'B_8_5': -0.5075674240757071, 'B_8_6': 0.013218007158356165, 'B_8_7':
-0.9215201516404251, 'B_8_8': 0.22434475145562305, 'B_8_9': 0.9742193173611041,
'B_9_0': 0.426924033602023, 'B_9_1': -0.8983797539588168, 'B_9_2':
-0.9878155313365858, 'B_9_3': 0.8088809910386593, 'B_9_4': -0.3166383458370438,
'B_9_5': -0.7485546066995422, 'B_9_6': 0.0790573939192887, 'B_9_7':
-0.44412132776997015, 'B_9_8': 0.5616175700958972, 'B_9_9': 0.9046627674966328,
'B_10_0': 0.830728768352292, 'B_10_1': -0.9546290798988476, 'B_10_2':
-0.7735553553419219, 'B_10_3': -0.7415200530084933, 'B_10_4':
0.6456255983920993, 'B_10_5': -0.16237035661870047, 'B_10_6':
-0.00020135702908619796, 'B_10_7': 0.9650826364566771, 'B_10_8':
-0.18399320013068782, 'B_10_9': -0.2153392209274243}. Best is trial 116 with
value: 72.22876181340362.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:16:54,123] Trial 117 finished with value: 68.37739393874485 and
parameters: {'B_0_0': 0.22661275365337363, 'B_0_1': -0.10284355737698622,
'B_0_2': 0.82669372864905, 'B_0_3': -0.09217134445528091, 'B_0_4':
-0.3724085874922261, 'B_0_5': 0.8572777933986176, 'B_0_6': 0.28327885264760827,
'B_0_7': 0.6087414163310328, 'B_0_8': 0.2589449533942856, 'B_0_9':
0.05677495819546936, 'B_1_0': 0.3093209095823626, 'B_1_1':
-0.002898841418342221, 'B_1_2': 0.12892601425439426, 'B_1_3':
0.5316027281554444, 'B_1_4': -0.1969697703040933, 'B_1_5': 0.8712358467296352,
'B_1_6': -0.34684897699273676, 'B_1_7': 0.9157430919192395, 'B_1_8':
-0.5290166005786107, 'B_1_9': -0.38169011290340515, 'B_2_0':

```

```

-0.4005207177791399, 'B_2_1': -0.3163375245250971, 'B_2_2': -0.446780495999482,
'B_2_3': 0.8261245144730394, 'B_2_4': -0.5482629765468398, 'B_2_5':
-0.6843547159261414, 'B_2_6': -0.460028037821353, 'B_2_7': -0.7751086963885696,
'B_2_8': -0.43038033839104795, 'B_2_9': 0.4251789696163275, 'B_3_0':
-0.6969378441901133, 'B_3_1': -0.5454758992360657, 'B_3_2': -0.0802711721157184,
'B_3_3': 0.2309526518649472, 'B_3_4': -0.45327333863305086, 'B_3_5':
0.030711311473611752, 'B_3_6': 0.11228261228913963, 'B_3_7':
-0.05646771238376065, 'B_3_8': 0.5791443715899439, 'B_3_9':
-0.22757195639573113, 'B_4_0': -0.043467192641102326, 'B_4_1':
0.5065533302431814, 'B_4_2': 0.19948231311338133, 'B_4_3': -0.5138754602760641,
'B_4_4': -0.5022009381490872, 'B_4_5': 0.6235834593109113, 'B_4_6':
-0.8470100925658589, 'B_4_7': -0.2337896942930517, 'B_4_8': 0.6454404839243745,
'B_4_9': 0.5858202481329561, 'B_5_0': -0.05258551672226197, 'B_5_1':
-0.37737917027100554, 'B_5_2': 0.2611275291922101, 'B_5_3':
-0.19961178430237855, 'B_5_4': 0.339694020124519, 'B_5_5': -0.5324373808739105,
'B_5_6': -0.27501443049505, 'B_5_7': 0.2475905519330212, 'B_5_8':
-0.3237706588672626, 'B_5_9': 0.3974361637344852, 'B_6_0': -0.14385316536837459,
'B_6_1': -0.6981317119617677, 'B_6_2': 0.947536476384238, 'B_6_3':
0.7570938186202627, 'B_6_4': -0.8064173587861307, 'B_6_5': -0.18955519070866772,
'B_6_6': 0.04163862438451089, 'B_6_7': 0.8694710499356139, 'B_6_8':
0.7164715079075956, 'B_6_9': 0.9865642380000207, 'B_7_0': -0.983077460595774,
'B_7_1': -0.5854626716909108, 'B_7_2': -0.6304997353863359, 'B_7_3':
0.9409465292537778, 'B_7_4': 0.6841008027520099, 'B_7_5': -0.35985498450330083,
'B_7_6': -0.18238960998650727, 'B_7_7': -0.8948157692465916, 'B_7_8':
-0.7490507230630534, 'B_7_9': 0.5147041387869935, 'B_8_0': -0.8347284328902487,
'B_8_1': 0.08354192135590244, 'B_8_2': 0.4619552783595648, 'B_8_3':
0.06002500626871009, 'B_8_4': -0.5317200353357092, 'B_8_5': -0.5219276461949824,
'B_8_6': 0.36026764449420445, 'B_8_7': -0.9302123121545476, 'B_8_8':
0.32381272737699424, 'B_8_9': 0.9334382165764384, 'B_9_0': 0.4115575944259785,
'B_9_1': -0.8856117746470962, 'B_9_2': -0.8901577426026694, 'B_9_3':
0.81968774955744, 'B_9_4': -0.3451187167697377, 'B_9_5': -0.746642321020358,
'B_9_6': 0.12525524388994297, 'B_9_7': -0.21885939828153242, 'B_9_8':
0.575134951446814, 'B_9_9': 0.9450903838175637, 'B_10_0': 0.8368130665400279,
'B_10_1': -0.9479618821129827, 'B_10_2': -0.9330594731092849, 'B_10_3':
-0.7215541322410507, 'B_10_4': 0.6398394078412419, 'B_10_5':
-0.05039179942677996, 'B_10_6': -0.05797362338383916, 'B_10_7':
0.863108876750866, 'B_10_8': -0.16641927559099318, 'B_10_9':
0.005889755394336255}. Best is trial 117 with value: 68.37739393874485.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].

```

```

Built F_[u_10].
Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:17:12,649] Trial 118 finished with value: 70.18197763269897 and
parameters: {'B_0_0': 0.24340597463652192, 'B_0_1': -0.09830351513116214,
'B_0_2': 0.8134654485682167, 'B_0_3': -0.09601815017425197, 'B_0_4':
-0.39828796418426127, 'B_0_5': 0.8471283071345135, 'B_0_6': 0.38728318145902063,
'B_0_7': 0.6199881230770234, 'B_0_8': 0.2762907032651597, 'B_0_9':
0.06996107992760027, 'B_1_0': 0.319903278024992, 'B_1_1': -0.010874024171418474,
'B_1_2': 0.13496595246454254, 'B_1_3': 0.7491680691178373, 'B_1_4':
-0.31937776671673784, 'B_1_5': 0.8385433238571767, 'B_1_6':
-0.35543632387250373, 'B_1_7': 0.9059822078906107, 'B_1_8': -0.6423417303147392,
'B_1_9': -0.4028856234027467, 'B_2_0': -0.4073719390669822, 'B_2_1':
-0.3162621711641073, 'B_2_2': -0.44420079042976535, 'B_2_3': 0.8368686847922432,
'B_2_4': -0.42055485564531836, 'B_2_5': -0.5645404259532416, 'B_2_6':
-0.4736606602623188, 'B_2_7': -0.6673632035011802, 'B_2_8': -0.4636924515880937,
'B_2_9': 0.42504849609297896, 'B_3_0': -0.70818659985376, 'B_3_1':
-0.5472853620101126, 'B_3_2': -0.13461925366745695, 'B_3_3':
0.07137329051678984, 'B_3_4': -0.48674214430210916, 'B_3_5': 0.1275153691461065,
'B_3_6': 0.023771948839046098, 'B_3_7': 0.03983340385574183, 'B_3_8':
0.5694127332211744, 'B_3_9': -0.21581444862493693, 'B_4_0': 0.06284859398023655,
'B_4_1': 0.5021861096858748, 'B_4_2': 0.1947458960238299, 'B_4_3':
-0.4866008757249989, 'B_4_4': -0.4957211766835517, 'B_4_5': 0.6189719265432382,
'B_4_6': -0.855820876585574, 'B_4_7': -0.2614466654836204, 'B_4_8':
0.6549959022081635, 'B_4_9': 0.5908684251731545, 'B_5_0': -0.33407521975376064,
'B_5_1': -0.38917586119812725, 'B_5_2': 0.23243446701060572, 'B_5_3':
-0.12424978830967205, 'B_5_4': 0.45859269642762857, 'B_5_5':
-0.6924047939952142, 'B_5_6': -0.27456333767251895, 'B_5_7': 0.3671540693334459,
'B_5_8': -0.3352770377387774, 'B_5_9': 0.5404066023558913, 'B_6_0':
-0.15156661625187112, 'B_6_1': -0.7070960747573964, 'B_6_2': 0.9309889301727412,
'B_6_3': 0.7619642092532217, 'B_6_4': -0.8104344892304017, 'B_6_5':
-0.18385420367151734, 'B_6_6': -0.08296885341320655, 'B_6_7':
0.6452715882618382, 'B_6_8': 0.7250741717522852, 'B_6_9': 0.9837886741255518,
'B_7_0': -0.9766888219006825, 'B_7_1': -0.5739178096901753, 'B_7_2':
-0.6180372719597903, 'B_7_3': 0.9350634457367171, 'B_7_4': 0.6647193913010079,
'B_7_5': -0.379242793082068, 'B_7_6': -0.18610526461102422, 'B_7_7':
-0.8936789927047154, 'B_7_8': -0.8651417110645852, 'B_7_9': 0.4992354456732429,
'B_8_0': -0.8501337314651001, 'B_8_1': 0.09621746387941385, 'B_8_2':
0.5706702847694229, 'B_8_3': 0.0485074939126165, 'B_8_4': -0.5373242556909152,
'B_8_5': -0.5514914094890219, 'B_8_6': 0.3373796435462657, 'B_8_7':
-0.9313708573690648, 'B_8_8': 0.22484303065707784, 'B_8_9': 0.8027203569862928,
'B_9_0': 0.26342935044638405, 'B_9_1': -0.8821839418912415, 'B_9_2':
-0.8864105261210514, 'B_9_3': 0.8244706500817179, 'B_9_4': -0.4675237598995217,
'B_9_5': -0.7353543620147642, 'B_9_6': -0.05122645038604007, 'B_9_7':
-0.44028271951254105, 'B_9_8': 0.5403834544211271, 'B_9_9': 0.9205379855103603,
'B_10_0': 0.8364543340793821, 'B_10_1': -0.9416941641244915, 'B_10_2':

```

-0.9282592008025328, 'B_10_3': -0.7224833091623456, 'B_10_4':
0.6384404913110897, 'B_10_5': -0.05754793357400165, 'B_10_6':
-0.04799369650115441, 'B_10_7': 0.8841755391422009, 'B_10_8':
-0.16855250962647955, 'B_10_9': -0.25181874757989386}. Best is trial 117 with
value: 68.37739393874485.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:17:32,160] Trial 119 finished with value: 66.30887875663029 and
parameters: {'B_0_0': 0.2698443014701542, 'B_0_1': -0.10502241166581261,
'B_0_2': 0.8134913828606827, 'B_0_3': -0.06957122135981178, 'B_0_4':
-0.3865883419368485, 'B_0_5': 0.8539563056713786, 'B_0_6': 0.3619640371426069,
'B_0_7': 0.6149227638110165, 'B_0_8': 0.2574321555260409, 'B_0_9':
0.023162770148009576, 'B_1_0': 0.30097460531191694, 'B_1_1':
0.016944730976808986, 'B_1_2': 0.12702560302934798, 'B_1_3': 0.5381787643608352,
'B_1_4': -0.3336331768890229, 'B_1_5': 0.8431498052001875, 'B_1_6':
-0.3383528264353352, 'B_1_7': 0.91078893723779, 'B_1_8': -0.5068086962777998,
'B_1_9': -0.38944356185879175, 'B_2_0': -0.40911697428671767, 'B_2_1':
-0.3342796715742826, 'B_2_2': -0.46781818850466333, 'B_2_3': 0.8172235529302723,
'B_2_4': -0.40812498488540994, 'B_2_5': -0.5580507813455411, 'B_2_6':
-0.47919387210103115, 'B_2_7': -0.6734148552020183, 'B_2_8':
-0.4372569016783083, 'B_2_9': 0.4298485184752172, 'B_3_0': -0.7036334734497405,
'B_3_1': -0.5397459982001284, 'B_3_2': -0.10807400905828099, 'B_3_3':
0.07649621561053171, 'B_3_4': -0.47500485968999984, 'B_3_5':
0.13043656658903485, 'B_3_6': 0.03331879325651725, 'B_3_7':
0.043527710089277485, 'B_3_8': 0.5976262030601331, 'B_3_9':
-0.21695322517804772, 'B_4_0': 0.05227714391975473, 'B_4_1': 0.5086125900841653,
'B_4_2': 0.17908417470524138, 'B_4_3': -0.4954236381114354, 'B_4_4':
-0.6560123436297823, 'B_4_5': 0.6125316555936337, 'B_4_6': -0.8848739933370218,
'B_4_7': -0.2782827866725588, 'B_4_8': 0.654041754671326, 'B_4_9':
0.5674033764410729, 'B_5_0': -0.3019594109682048, 'B_5_1': -0.36818931936636634,
'B_5_2': 0.25153318542871944, 'B_5_3': -0.12808154572729302, 'B_5_4':
0.44555815974181423, 'B_5_5': -0.6983112072016693, 'B_5_6':
-0.27247986070039604, 'B_5_7': 0.37742476578154405, 'B_5_8':
-0.3258500462613936, 'B_5_9': 0.5151284395349766, 'B_6_0': 0.08875314592679001,
'B_6_1': -0.7187260462782211, 'B_6_2': 0.9324222501737787, 'B_6_3':

0.7859951070413336, 'B_6_4': -0.8150943701678788, 'B_6_5': -0.1888313578338373,
 'B_6_6': -0.07119894509704376, 'B_6_7': 0.6226280273542413, 'B_6_8':
 0.7137733623953024, 'B_6_9': 0.9762199886333209, 'B_7_0': -0.9858503373403411,
 'B_7_1': -0.5853842218515543, 'B_7_2': -0.6047624833187684, 'B_7_3':
 0.9564081975917388, 'B_7_4': 0.6570672681055343, 'B_7_5': -0.3763804220648461,
 'B_7_6': -0.18287909801020708, 'B_7_7': -0.8808961571977109, 'B_7_8':
 -0.8579013011394573, 'B_7_9': 0.49466057953944653, 'B_8_0': -0.796608154669808,
 'B_8_1': 0.08668738600969567, 'B_8_2': 0.6593316718354718, 'B_8_3':
 0.05865946045671927, 'B_8_4': -0.5188474662303904, 'B_8_5': -0.5198412154235463,
 'B_8_6': 0.32190847204365547, 'B_8_7': -0.9149491257388782, 'B_8_8':
 0.10566341595549675, 'B_8_9': 0.7965359292195328, 'B_9_0': 0.2440119240954058,
 'B_9_1': -0.8932376656599416, 'B_9_2': -0.8902149902635514, 'B_9_3':
 0.8186253282719969, 'B_9_4': -0.47747382729731835, 'B_9_5': -0.7410705233948421,
 'B_9_6': -0.06046912133149979, 'B_9_7': -0.41608482729246166, 'B_9_8':
 0.5503046919481479, 'B_9_9': 0.9172244201426255, 'B_10_0': 0.9205948822949714,
 'B_10_1': -0.795984861554696, 'B_10_2': -0.9174946981727129, 'B_10_3':
 -0.7297139200086868, 'B_10_4': 0.6474090211440796, 'B_10_5':
 -0.04270994622985591, 'B_10_6': -0.054576386440949254, 'B_10_7':
 0.8880106755225669, 'B_10_8': -0.168674794598198, 'B_10_9':
 0.012949021298925698}. Best is trial 119 with value: 66.30887875663029.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:17:48,467] Trial 120 finished with value: 70.52630573662614 and
 parameters: {'B_0_0': 0.2533631671699885, 'B_0_1': -0.09384377647605421,
 'B_0_2': 0.9874568033793903, 'B_0_3': 0.03913398197959683, 'B_0_4':
 -0.242993687805444, 'B_0_5': 0.7915207911340704, 'B_0_6': 0.39025529940978576,
 'B_0_7': 0.607441835706027, 'B_0_8': 0.45851291321421894, 'B_0_9':
 0.06604305458608731, 'B_1_0': 0.2947874154349671, 'B_1_1': 0.014538946781129061,
 'B_1_2': 0.1299563610969334, 'B_1_3': 0.747025012592476, 'B_1_4':
 -0.32494323266752434, 'B_1_5': 0.8406547897541982, 'B_1_6': -0.3592448545813618,
 'B_1_7': 0.9265936575286555, 'B_1_8': -0.5156569228964666, 'B_1_9':
 -0.3842134613510602, 'B_2_0': -0.38133022142901096, 'B_2_1':
 -0.1340079196749824, 'B_2_2': -0.4824049227541768, 'B_2_3': 0.8321599488837669,
 'B_2_4': -0.4000262763236794, 'B_2_5': -0.5565947647119702, 'B_2_6':
 -0.46207648622781056, 'B_2_7': -0.6339304711024265, 'B_2_8':

-0.43965188882624123, 'B_2_9': 0.28600928310540247, 'B_3_0':
 -0.6799658379941645, 'B_3_1': -0.545476799079451, 'B_3_2': -0.15941012059574655,
 'B_3_3': 0.07051196188244734, 'B_3_4': -0.457335827747146, 'B_3_5':
 0.12658231014353383, 'B_3_6': 0.12510264045848593, 'B_3_7':
 -0.16868295106527284, 'B_3_8': 0.5810709531351924, 'B_3_9':
 -0.20115009935487682, 'B_4_0': 0.06724333328649071, 'B_4_1':
 0.49983532099962713, 'B_4_2': 0.2719889226176212, 'B_4_3': -0.4856969009619504,
 'B_4_4': -0.4893889136325965, 'B_4_5': 0.5171228948158857, 'B_4_6':
 -0.8735688428935093, 'B_4_7': -0.02934981587534606, 'B_4_8': 0.6428124900803214,
 'B_4_9': 0.46364724618124853, 'B_5_0': -0.03379908728121561, 'B_5_1':
 -0.3942787816308705, 'B_5_2': 0.251227300642813, 'B_5_3': -0.1865699233806878,
 'B_5_4': 0.45551650997810056, 'B_5_5': -0.691018079745757, 'B_5_6':
 -0.23247182911507866, 'B_5_7': 0.348111963320977, 'B_5_8': -0.3192756803731712,
 'B_5_9': 0.4115361004694928, 'B_6_0': -0.030154455220057635, 'B_6_1':
 -0.8634042031731245, 'B_6_2': 0.9316716020341553, 'B_6_3': 0.7868947664822081,
 'B_6_4': -0.8104676184429231, 'B_6_5': -0.19045439469862838, 'B_6_6':
 -0.08606503701690728, 'B_6_7': 0.6666027494617134, 'B_6_8': 0.7238434976799696,
 'B_6_9': 0.5467094274311594, 'B_7_0': -0.9870885184472806, 'B_7_1':
 -0.5807353361879202, 'B_7_2': -0.7878145095197637, 'B_7_3': 0.936763193050489,
 'B_7_4': 0.6863577443080664, 'B_7_5': -0.38225843250200187, 'B_7_6':
 -0.35984566585183486, 'B_7_7': -0.7821657509724508, 'B_7_8':
 -0.8598839742551097, 'B_7_9': 0.4944933327937193, 'B_8_0': -0.7898188994937816,
 'B_8_1': 0.059870875286994016, 'B_8_2': 0.6614422362475026, 'B_8_3':
 0.05115374443194662, 'B_8_4': -0.35797766969268696, 'B_8_5':
 -0.5082947171875464, 'B_8_6': 0.3200164913987977, 'B_8_7': -0.9003394085148305,
 'B_8_8': 0.3063281098252801, 'B_8_9': 0.7956195318849301, 'B_9_0':
 0.24926664494713632, 'B_9_1': -0.8806815919010478, 'B_9_2': -0.8845975339555197,
 'B_9_3': 0.8165787650335573, 'B_9_4': -0.4765102879986295, 'B_9_5':
 -0.8428254569203247, 'B_9_6': -0.11193108828762217, 'B_9_7':
 -0.3302710669398349, 'B_9_8': 0.5517539076443726, 'B_9_9': 0.9167680059710331,
 'B_10_0': 0.8474402479596834, 'B_10_1': -0.8058952396021953, 'B_10_2':
 -0.8072195214867037, 'B_10_3': -0.7186078622210849, 'B_10_4':
 0.8989535946270129, 'B_10_5': -0.2219910072473318, 'B_10_6':
 -0.06257893262178832, 'B_10_7': 0.8788322419205751, 'B_10_8':
 -0.16191291100203256, 'B_10_9': -0.003932585233050571}. Best is trial 119 with
 value: 66.30887875663029.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.
Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].
Built matrix B = inverse(A).
[I 2025-04-29 14:18:05,597] Trial 121 finished with value: 70.70379178052808 and parameters: {'B_0_0': 0.23873079498568103, 'B_0_1': -0.06826072162039537, 'B_0_2': 0.9646017588193351, 'B_0_3': 0.04766640650426324, 'B_0_4': -0.23573840252682782, 'B_0_5': 0.8190374964157011, 'B_0_6': 0.3918825292923042, 'B_0_7': 0.6175307767895636, 'B_0_8': 0.3219619754531103, 'B_0_9': 0.07139337747044039, 'B_1_0': 0.3332326385746883, 'B_1_1': 0.020679283437328416, 'B_1_2': 0.12028725557589928, 'B_1_3': 0.7408529214458577, 'B_1_4': -0.31681363420990194, 'B_1_5': 0.8310635980714411, 'B_1_6': -0.35559577080056887, 'B_1_7': 0.9209856812012992, 'B_1_8': -0.5261853359655745, 'B_1_9': -0.37644463433863096, 'B_2_0': -0.4012534160100305, 'B_2_1': -0.13296428621216408, 'B_2_2': -0.47007676480261096, 'B_2_3': 0.8107515201414033, 'B_2_4': -0.4077546804234077, 'B_2_5': -0.5646593199768515, 'B_2_6': -0.46255632432306865, 'B_2_7': -0.6308196394999754, 'B_2_8': -0.43721804097230743, 'B_2_9': 0.29957669885494875, 'B_3_0': -0.6858085381150808, 'B_3_1': -0.541677606251292, 'B_3_2': -0.08792287070278024, 'B_3_3': 0.0579351312388644, 'B_3_4': -0.5911881909735699, 'B_3_5': 0.11970653677928828, 'B_3_6': 0.032581990050323556, 'B_3_7': 0.027472050024722944, 'B_3_8': 0.5675627128960382, 'B_3_9': -0.32903870129515245, 'B_4_0': 0.06385980649161781, 'B_4_1': 0.4995787179671709, 'B_4_2': 0.2814053432189302, 'B_4_3': -0.49985581860925155, 'B_4_4': -0.4968325906219829, 'B_4_5': 0.6232493457982855, 'B_4_6': -0.869783415249658, 'B_4_7': -0.03273733275408616, 'B_4_8': 0.6507408563985967, 'B_4_9': 0.4642005377958079, 'B_5_0': -0.3394140840038737, 'B_5_1': -0.3943529973341475, 'B_5_2': 0.2534295288903242, 'B_5_3': -0.13038582363275786, 'B_5_4': 0.46949095585766154, 'B_5_5': -0.701871456080308, 'B_5_6': -0.280049309461208, 'B_5_7': 0.34954072664510005, 'B_5_8': -0.3243929727324347, 'B_5_9': 0.5426806491174079, 'B_6_0': 0.07863525573764252, 'B_6_1': -0.843315010166401, 'B_6_2': 0.9264376761882626, 'B_6_3': 0.7615458128750899, 'B_6_4': -0.8069508565476525, 'B_6_5': -0.198689725825518, 'B_6_6': -0.07644289981861657, 'B_6_7': 0.6575754413830633, 'B_6_8': 0.7118465288637655, 'B_6_9': 0.8848588125317974, 'B_7_0': -0.9897306991225048, 'B_7_1': -0.45428500012956535, 'B_7_2': -0.7904778721165137, 'B_7_3': 0.9409419469196553, 'B_7_4': 0.6614458133016813, 'B_7_5': -0.3699452811227361, 'B_7_6': -0.360706122945862, 'B_7_7': -0.8952618832163586, 'B_7_8': -0.8880411655922407, 'B_7_9': 0.5757650892488051, 'B_8_0': -0.7962898075913419, 'B_8_1': 0.08314353896483559, 'B_8_2': 0.6575049242634898, 'B_8_3': 0.04416327279209563, 'B_8_4': -0.34827509884804186, 'B_8_5': -0.5235227682035831, 'B_8_6': 0.3292133132819764, 'B_8_7': -0.9012861588450511, 'B_8_8': 0.14062946587186287, 'B_8_9': 0.810619453276423, 'B_9_0': 0.25690939194081414, 'B_9_1': -0.8883479044040357, 'B_9_2': -0.8693835700654894, 'B_9_3': 0.7876523951286575, 'B_9_4': -0.4687695982592066, 'B_9_5': -0.8409510742326908, 'B_9_6': -0.07055314347520872, 'B_9_7': -0.342557410720506, 'B_9_8': 0.5443429934900544, 'B_9_9': 0.9203949806554457, 'B_10_0': 0.8347485530670347, 'B_10_1': -0.7835518979107943, 'B_10_2': -0.8165284463615748, 'B_10_3': -0.7284302058799599, 'B_10_4': 0.5754443894773047, 'B_10_5':

-0.1656445831509034, 'B_10_6': -0.06394757554273978, 'B_10_7':
0.8715187934600059, 'B_10_8': -0.13248945654923247, 'B_10_9':
0.01271846261017474}. Best is trial 119 with value: 66.30887875663029.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:18:23,916] Trial 122 finished with value: 74.57991098903666 and
parameters: {'B_0_0': 0.2749764489250603, 'B_0_1': -0.09888797533065934,
'B_0_2': 0.9804738862119302, 'B_0_3': 0.03762798486729799, 'B_0_4':
-0.2383051827349252, 'B_0_5': 0.7870865056604442, 'B_0_6': 0.37663907206233166,
'B_0_7': 0.6071886655795864, 'B_0_8': 0.4690936270594219, 'B_0_9':
0.02365965769758019, 'B_1_0': 0.30321190152391414, 'B_1_1':
0.010910983416233824, 'B_1_2': 0.13791237595327172, 'B_1_3': 0.7604411446109871,
'B_1_4': -0.31225350543806224, 'B_1_5': 0.7564145932873723, 'B_1_6':
-0.3146980548876714, 'B_1_7': 0.9221610394442059, 'B_1_8': -0.5167774907950469,
'B_1_9': -0.36214042742582636, 'B_2_0': -0.391587467052427, 'B_2_1':
-0.12624525587753677, 'B_2_2': -0.4643644357896174, 'B_2_3': 0.8221039298715571,
'B_2_4': -0.3191847952546752, 'B_2_5': -0.560720322571467, 'B_2_6':
-0.45979728541787085, 'B_2_7': -0.6313541661088682, 'B_2_8':
-0.4249439538934628, 'B_2_9': 0.28453176959553994, 'B_3_0': -0.6932087505571519,
'B_3_1': -0.5387878487224921, 'B_3_2': -0.08860072196041267, 'B_3_3':
0.06970126941193926, 'B_3_4': -0.46336668579332757, 'B_3_5':
-0.023825528347421868, 'B_3_6': 0.12544270848024142, 'B_3_7':
0.13579382036848142, 'B_3_8': 0.5839905123483248, 'B_3_9': -0.19794786443467147,
'B_4_0': 0.0517782111801396, 'B_4_1': 0.5051419723995599, 'B_4_2':
0.18973877967168792, 'B_4_3': -0.43331253469189224, 'B_4_4':
-0.48891052232079923, 'B_4_5': 0.6307086872065238, 'B_4_6': -0.8640072985955273,
'B_4_7': -0.021430929410915248, 'B_4_8': 0.6611418186968236, 'B_4_9':
0.5951778327910129, 'B_5_0': -0.046769270051570506, 'B_5_1':
-0.3802085220541167, 'B_5_2': 0.2415355436611616, 'B_5_3': -0.1296965663267885,
'B_5_4': 0.43803913251359183, 'B_5_5': -0.7024633544991403, 'B_5_6':
-0.22906311519579164, 'B_5_7': 0.3538980384008409, 'B_5_8': -0.3287373246732612,
'B_5_9': 0.5538549525818319, 'B_6_0': 0.080524125487975, 'B_6_1':
-0.8454550052476413, 'B_6_2': 0.916349480318184, 'B_6_3': 0.7756471438704605,
'B_6_4': -0.8168156634854681, 'B_6_5': -0.18528401137632322, 'B_6_6':
0.0777878272339912, 'B_6_7': 0.6472067999327262, 'B_6_8': 0.7392909627329193,


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'B_6_9': 0.5363659096562161, 'B_7_0': -0.9823271823488582, 'B_7_1':
-0.5774936883169195, 'B_7_2': -0.7899856258285144, 'B_7_3': 0.9450141501064635,
'B_7_4': 0.6551836893218509, 'B_7_5': -0.38434948712046063, 'B_7_6':
-0.3503261652759822, 'B_7_7': -0.7932126278545202, 'B_7_8': -0.8696556123873457,
'B_7_9': 0.5210230805798217, 'B_8_0': -0.8005867836336097, 'B_8_1':
0.09026555779231271, 'B_8_2': 0.569347317285808, 'B_8_3': 0.04012108940378452,
'B_8_4': -0.3652533321377877, 'B_8_5': -0.5267904812545109, 'B_8_6':
0.3314306306270929, 'B_8_7': -0.8951790845991588, 'B_8_8': 0.12125259112598538,
'B_8_9': 0.8000205735840393, 'B_9_0': 0.2571462983977349, 'B_9_1':
-0.8986579184067309, 'B_9_2': -0.880492989701507, 'B_9_3': 0.8067584302120021,
'B_9_4': -0.5477498872790598, 'B_9_5': -0.8429757058083754, 'B_9_6':
-0.11830901358667746, 'B_9_7': -0.3387733501248036, 'B_9_8': 0.5641153006726282,
'B_9_9': 0.9062700581788335, 'B_10_0': 0.8161684378199004, 'B_10_1':
-0.7838773055701316, 'B_10_2': -0.8116527968910541, 'B_10_3':
-0.7367220197262978, 'B_10_4': 0.6065786659568742, 'B_10_5':
-0.16898427301621982, 'B_10_6': -0.03778177883563047, 'B_10_7':
0.8723334959114528, 'B_10_8': -0.15811278996519867, 'B_10_9':
0.008378393097616411}. Best is trial 119 with value: 66.30887875663029.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:18:40,451] Trial 123 finished with value: 72.96581260305314 and
parameters: {'B_0_0': 0.25170315760424233, 'B_0_1': -0.09883934255484667,
'B_0_2': 0.8595446879494941, 'B_0_3': -0.0468280734659709, 'B_0_4':
-0.2600074197127789, 'B_0_5': 0.786514922190081, 'B_0_6': 0.37377693209081303,
'B_0_7': 0.6062998502098627, 'B_0_8': 0.32191469302983217, 'B_0_9':
0.05229866959750812, 'B_1_0': 0.301659258512183, 'B_1_1': 0.018850989741056277,
'B_1_2': 0.13766899630356227, 'B_1_3': 0.7493387874084368, 'B_1_4':
-0.3304016974857715, 'B_1_5': 0.7549343080429003, 'B_1_6': -0.3362613350121971,
'B_1_7': 0.9144659324215886, 'B_1_8': -0.6362822735885003, 'B_1_9':
-0.36219890925849413, 'B_2_0': -0.3643811683633315, 'B_2_1':
-0.13678072226933763, 'B_2_2': -0.475602520789973, 'B_2_3': 0.82304590490341,
'B_2_4': -0.39759221468845396, 'B_2_5': -0.566963427211873, 'B_2_6':
-0.5238248197099031, 'B_2_7': -0.5673965080384059, 'B_2_8': -0.4311643684446615,
'B_2_9': 0.2747407115245925, 'B_3_0': -0.7149065256644666, 'B_3_1':
-0.5402824077946301, 'B_3_2': -0.09650302701293909, 'B_3_3':

```

```

0.07035717826697567, 'B_3_4': -0.47532503919442276, 'B_3_5':
-0.02503371141835653, 'B_3_6': 0.12373746220910496, 'B_3_7':
0.12564119428352424, 'B_3_8': 0.5756069102235188, 'B_3_9': -0.20580160928375635,
'B_4_0': -0.004694986742576075, 'B_4_1': 0.48760185341264495, 'B_4_2':
0.2942968426038461, 'B_4_3': -0.49890228992736596, 'B_4_4':
-0.48744112907881976, 'B_4_5': 0.5128503150882598, 'B_4_6': -0.8557028267171414,
'B_4_7': -0.002588895117984734, 'B_4_8': 0.6432483881020112, 'B_4_9':
0.592745980834972, 'B_5_0': -0.333777327180967, 'B_5_1': -0.47118336563509533,
'B_5_2': 0.23372515035069252, 'B_5_3': -0.043272145264280915, 'B_5_4':
0.45395887073054286, 'B_5_5': -0.6879760597879626, 'B_5_6': -0.2520352361581741,
'B_5_7': 0.3862169118548874, 'B_5_8': -0.3267311451829248, 'B_5_9':
0.5559864640729909, 'B_6_0': 0.08254955973691007, 'B_6_1': -0.8460549453575372,
'B_6_2': 0.9161672517375731, 'B_6_3': 0.7793492009283757, 'B_6_4':
-0.9872571570860614, 'B_6_5': -0.20207887854869866, 'B_6_6':
0.06493600302954738, 'B_6_7': 0.6538683575019726, 'B_6_8': 0.7071457243814496,
'B_6_9': 0.8909727834227004, 'B_7_0': -0.9828552832508486, 'B_7_1':
-0.5715551578144049, 'B_7_2': -0.6084539107591647, 'B_7_3': 0.9465035169533691,
'B_7_4': 0.6860148549007561, 'B_7_5': -0.3933136852208777, 'B_7_6':
-0.37937658700981947, 'B_7_7': -0.7822719207172786, 'B_7_8':
-0.8684021534754668, 'B_7_9': 0.5020353814901779, 'B_8_0': -0.7918695292072861,
'B_8_1': 0.09461948400823604, 'B_8_2': 0.6602909253537634, 'B_8_3':
0.04651323203438335, 'B_8_4': -0.3567423540670111, 'B_8_5': -0.5179656903175647,
'B_8_6': 0.25827240365210574, 'B_8_7': -0.8854085021989566, 'B_8_8':
0.13443909336853221, 'B_8_9': 0.8072038787061349, 'B_9_0': 0.2353607690462554,
'B_9_1': -0.8896181969937209, 'B_9_2': -0.8751054707969531, 'B_9_3':
0.7975556293632591, 'B_9_4': -0.47866218112076214, 'B_9_5': -0.8427115268665168,
'B_9_6': -0.04825971708676631, 'B_9_7': -0.34366452472501896, 'B_9_8':
0.5393554830729047, 'B_9_9': 0.9252645814394442, 'B_10_0': 0.799591848225392,
'B_10_1': -0.7813934325196885, 'B_10_2': -0.8338947111955297, 'B_10_3':
-0.7214013203559816, 'B_10_4': 0.5632785489526678, 'B_10_5':
-0.16549282360171108, 'B_10_6': -0.05749248300661662, 'B_10_7':
0.8846723427787216, 'B_10_8': -0.14966821382311607, 'B_10_9':
0.016707456321800224}. Best is trial 119 with value: 66.30887875663029.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:19:00,067] Trial 124 finished with value: 74.56788671480179 and parameters: {'B_0_0': 0.2646338447579517, 'B_0_1': -0.08599761207997515, 'B_0_2': 0.9620239070755952, 'B_0_3': 0.04024997039715009, 'B_0_4': -0.24031658945766188, 'B_0_5': 0.793969103978504, 'B_0_6': 0.3837487797946185, 'B_0_7': 0.4061712976019608, 'B_0_8': 0.477956802500196, 'B_0_9': 0.0817037613129108, 'B_1_0': 0.3174697229554692, 'B_1_1': 0.035139078573799165, 'B_1_2': 0.13594278048253153, 'B_1_3': 0.7321822641054507, 'B_1_4': -0.30193222793778557, 'B_1_5': 0.7356664451800952, 'B_1_6': -0.3728017671812745, 'B_1_7': 0.8993386038257928, 'B_1_8': -0.6422902935158565, 'B_1_9': -0.38858765807434953, 'B_2_0': -0.37350000024707974, 'B_2_1': -0.1283432341396171, 'B_2_2': -0.45520688970042955, 'B_2_3': 0.9715904807884378, 'B_2_4': -0.32158073983691515, 'B_2_5': -0.5495075959925979, 'B_2_6': -0.45527696588198, 'B_2_7': -0.638073103467202, 'B_2_8': -0.4371785762634149, 'B_2_9': 0.26558739392017494, 'B_3_0': -0.6956310820488156, 'B_3_1': -0.44084148511633625, 'B_3_2': -0.10163652038788681, 'B_3_3': 0.07285320749297086, 'B_3_4': -0.45343622358660085, 'B_3_5': -0.04266508706744826, 'B_3_6': 0.04428264937582123, 'B_3_7': 0.13320594274637715, 'B_3_8': 0.47998918699082993, 'B_3_9': -0.1902712919682373, 'B_4_0': 0.19585076549086033, 'B_4_1': 0.494666492031317, 'B_4_2': 0.29328448293042975, 'B_4_3': -0.49740214383716524, 'B_4_4': -0.4824213555662775, 'B_4_5': 0.6477321279479923, 'B_4_6': -0.8598354830784014, 'B_4_7': -0.02179062911248239, 'B_4_8': 0.6335792366229368, 'B_4_9': 0.45662295801702507, 'B_5_0': -0.03653477331296981, 'B_5_1': -0.3816189192807518, 'B_5_2': 0.2552122237429474, 'B_5_3': -0.04495863044968984, 'B_5_4': 0.4742971309360989, 'B_5_5': -0.6930509856901979, 'B_5_6': -0.22889342746609811, 'B_5_7': 0.3869854527406932, 'B_5_8': -0.31832791444498265, 'B_5_9': 0.6861029343481405, 'B_6_0': 0.11261310501449796, 'B_6_1': -0.8378648727631401, 'B_6_2': 0.9133697220874913, 'B_6_3': 0.7772053263772167, 'B_6_4': -0.9857384103130569, 'B_6_5': -0.1913915449798952, 'B_6_6': 0.07719695014101911, 'B_6_7': 0.6395999120645263, 'B_6_8': 0.7245882132530939, 'B_6_9': 0.5480069701994154, 'B_7_0': -0.9796884292792728, 'B_7_1': -0.5648616999141879, 'B_7_2': -0.789613848230045, 'B_7_3': 0.9520920566125268, 'B_7_4': 0.7175483579104529, 'B_7_5': -0.382020835832432, 'B_7_6': -0.3666801999299968, 'B_7_7': -0.6721468411458887, 'B_7_8': -0.8826531583265889, 'B_7_9': 0.5007950642821045, 'B_8_0': -0.7775509474739656, 'B_8_1': 0.08375708785577538, 'B_8_2': 0.672003538082246, 'B_8_3': 0.030138728816307848, 'B_8_4': -0.36891278511493253, 'B_8_5': -0.5341098840926835, 'B_8_6': 0.2723090440783674, 'B_8_7': -0.8977933145441895, 'B_8_8': 0.1041209188279717, 'B_8_9': 0.7877393861820311, 'B_9_0': 0.2592420963166237, 'B_9_1': -0.8777134626502417, 'B_9_2': -0.8792349399232045, 'B_9_3': 0.7956809977047911, 'B_9_4': -0.4866387161591563, 'B_9_5': -0.7278852702196028, 'B_9_6': -0.06993787435587664, 'B_9_7': -0.4126978561316356, 'B_9_8': 0.5611501530694007, 'B_9_9': 0.9138191221939042, 'B_10_0': 0.8293630489530999, 'B_10_1': -0.6325638322006807, 'B_10_2': -0.8203167529848268, 'B_10_3': -0.72810164821338, 'B_10_4': 0.5735723744614059, 'B_10_5': -0.2302241816714455, 'B_10_6': -0.05561824572611899, 'B_10_7': 0.8834223947953651, 'B_10_8': -0.14815316039843776, 'B_10_9': -0.004989599046106047}. Best is trial 119 with value: 66.30887875663029.

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:19:17,771] Trial 125 finished with value: 70.4741946310292 and parameters: {'B_0_0': 0.24294076443276638, 'B_0_1': -0.09649157264154992, 'B_0_2': 0.9554940069111675, 'B_0_3': 0.04187553630751406, 'B_0_4': -0.2561033517023395, 'B_0_5': 0.7989742072888982, 'B_0_6': 0.37449773373337913, 'B_0_7': 0.39156924717871, 'B_0_8': 0.46812725037672087, 'B_0_9': 0.07089754929499428, 'B_1_0': 0.3069428356266806, 'B_1_1': 0.02023292772995658, 'B_1_2': 0.1373075987539177, 'B_1_3': 0.7659983915561704, 'B_1_4': -0.3075279625869725, 'B_1_5': 0.7446249185541238, 'B_1_6': -0.35201651119541716, 'B_1_7': 0.8995062993027843, 'B_1_8': -0.6667280800753645, 'B_1_9': -0.3863179291785325, 'B_2_0': -0.39325297753870925, 'B_2_1': -0.13159379390778128, 'B_2_2': -0.4580845686104168, 'B_2_3': 0.9713300617320785, 'B_2_4': -0.3032376008180877, 'B_2_5': -0.5658664248222453, 'B_2_6': -0.44543719885801014, 'B_2_7': -0.6440253246554205, 'B_2_8': -0.43277803500883516, 'B_2_9': 0.2795150103358959, 'B_3_0': -0.6752404005010507, 'B_3_1': -0.44044594478899335, 'B_3_2': -0.09646832610281758, 'B_3_3': 0.05023919529326676, 'B_3_4': -0.45326964203160436, 'B_3_5': 0.02540959931540418, 'B_3_6': -0.1147503108190274, 'B_3_7': 0.10803335294565207, 'B_3_8': 0.5861558457688151, 'B_3_9': -0.1768010656391204, 'B_4_0': 0.05923875837036752, 'B_4_1': 0.49955913835626253, 'B_4_2': 0.2691404266143973, 'B_4_3': -0.4369411465371752, 'B_4_4': -0.5991787188888386, 'B_4_5': 0.6329145488092043, 'B_4_6': -0.8771207184463166, 'B_4_7': -0.025516326889030744, 'B_4_8': 0.660457768463606, 'B_4_9': 0.594784764384823, 'B_5_0': -0.03173674524840453, 'B_5_1': -0.3907927557633124, 'B_5_2': 0.22430230109054924, 'B_5_3': -0.03586868720445095, 'B_5_4': 0.45196295686347226, 'B_5_5': -0.6935794916735637, 'B_5_6': -0.22904953239237508, 'B_5_7': 0.36898485355776894, 'B_5_8': -0.29994749801092396, 'B_5_9': 0.6757702444783698, 'B_6_0': 0.08638040552117067, 'B_6_1': -0.878137281049155, 'B_6_2': 0.9214209831785495, 'B_6_3': 0.7773417961367559, 'B_6_4': -0.9788447775575122, 'B_6_5': -0.1915598695820033, 'B_6_6': 0.08039498367847184, 'B_6_7': 0.6596342126436398, 'B_6_8': 0.7093840011999544, 'B_6_9': 0.8938777517697359, 'B_7_0': -0.9828750673425325, 'B_7_1': -0.5748992298079535, 'B_7_2': -0.6218331702096378, 'B_7_3': 0.944380323717769, 'B_7_4': 0.7083868607736226, 'B_7_5': -0.40461739826773657,

```
'B_7_6': -0.3827523837856955, 'B_7_7': -0.7892925513961893, 'B_7_8':
-0.8496981774966634, 'B_7_9': 0.5047576388794959, 'B_8_0': -0.7752056212606343,
'B_8_1': 0.08236452916484659, 'B_8_2': 0.6514534120889915, 'B_8_3':
0.04984171424821149, 'B_8_4': -0.3577041100206096, 'B_8_5': -0.5252581739393163,
'B_8_6': 0.26738326318180305, 'B_8_7': -0.9031026868809273, 'B_8_8':
0.13212342813506728, 'B_8_9': 0.7786948594718495, 'B_9_0': 0.25279217401423426,
'B_9_1': -0.8707181951303774, 'B_9_2': -0.8486267491400014, 'B_9_3':
0.8013464260437662, 'B_9_4': -0.47815299061904065, 'B_9_5': -0.8475650382800681,
'B_9_6': -0.08253976985201791, 'B_9_7': -0.34051034162962984, 'B_9_8':
0.5522345964758434, 'B_9_9': 0.9292966804205418, 'B_10_0': 0.8237125147956108,
'B_10_1': -0.638825949930396, 'B_10_2': -0.8288737952216763, 'B_10_3':
-0.7198114052014202, 'B_10_4': 0.5840254105743822, 'B_10_5':
-0.21463249319044064, 'B_10_6': -0.0523666599791405, 'B_10_7':
0.8884452422354424, 'B_10_8': -0.14439658536419564, 'B_10_9':
0.023645633531292457}. Best is trial 119 with value: 66.30887875663029.
```

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$\langle F_{[u_i]}, F_{[u_j]} \rangle$].

Built matrix B = inverse(A).

```
[I 2025-04-29 14:19:32,542] Trial 126 finished with value: 74.06788019235154 and
parameters: {'B_0_0': 0.260741354468142, 'B_0_1': -0.09282262917977399, 'B_0_2':
0.8551226524683588, 'B_0_3': 0.05468818190730196, 'B_0_4': -0.2466713857403351,
'B_0_5': 0.7806226414179908, 'B_0_6': 0.36465272042809704, 'B_0_7':
0.38938385740071746, 'B_0_8': 0.4558583509831693, 'B_0_9': 0.08087524617526451,
'B_1_0': 0.3194160658413915, 'B_1_1': 0.01609138463526613, 'B_1_2':
0.15646362911550082, 'B_1_3': 0.7325654728733335, 'B_1_4': -0.3132717318323562,
'B_1_5': 0.7373730995523213, 'B_1_6': -0.36635105404718576, 'B_1_7':
0.9223922901957763, 'B_1_8': -0.5134461900603976, 'B_1_9': -0.39277274149528874,
'B_2_0': -0.39007448856340665, 'B_2_1': -0.13868645728490558, 'B_2_2':
-0.45608918438244556, 'B_2_3': 0.9731952600510372, 'B_2_4':
-0.29711009379581305, 'B_2_5': -0.5609199365707229, 'B_2_6':
-0.4509311487584069, 'B_2_7': -0.6353127732980743, 'B_2_8': -0.4394377992449828,
'B_2_9': 0.264143515703832, 'B_3_0': -0.7043379452454104, 'B_3_1':
-0.44976886048284015, 'B_3_2': -0.09263507544718261, 'B_3_3':
0.05198700481215332, 'B_3_4': -0.4641594880539708, 'B_3_5':
-0.035924429717451495, 'B_3_6': 0.11466984207281943, 'B_3_7':
0.13300236838581486, 'B_3_8': 0.5733251110899464, 'B_3_9': -0.20800433436817478,
```

'B_4_0': 0.1788423588192322, 'B_4_1': 0.4861576467832374, 'B_4_2': 0.30061251255205257, 'B_4_3': -0.3569868443362568, 'B_4_4': -0.3842009805454365, 'B_4_5': 0.6447641704945057, 'B_4_6': -0.8664389811985139, 'B_4_7': -0.03979336486206664, 'B_4_8': 0.6197970578390877, 'B_4_9': 0.5965721887107899, 'B_5_0': -0.03661570201005124, 'B_5_1': -0.3823670846823498, 'B_5_2': 0.2579897650686847, 'B_5_3': -0.04163501327489415, 'B_5_4': 0.4485313123300673, 'B_5_5': -0.7174350525796264, 'B_5_6': -0.21316110901635607, 'B_5_7': 0.3883223921859269, 'B_5_8': -0.3272899164744518, 'B_5_9': 0.6890194211723146, 'B_6_0': 0.10949228722825571, 'B_6_1': -0.8722373365996874, 'B_6_2': 0.9169751614139021, 'B_6_3': 0.7530920469499386, 'B_6_4': -0.9889426418259799, 'B_6_5': -0.1829660733457572, 'B_6_6': -0.10660772258970215, 'B_6_7': 0.6202953968061751, 'B_6_8': 0.7075439760576192, 'B_6_9': 0.9049888892212572, 'B_7_0': -0.9875687381163789, 'B_7_1': -0.5657986813343281, 'B_7_2': -0.7851636152188595, 'B_7_3': 0.9482480474821784, 'B_7_4': 0.6679749250117729, 'B_7_5': -0.3953403089762301, 'B_7_6': -0.35890780404552153, 'B_7_7': -0.6710040410169649, 'B_7_8': -0.8612149468057912, 'B_7_9': 0.5023639749305451, 'B_8_0': -0.7859041953471505, 'B_8_1': 0.09268119422573366, 'B_8_2': 0.6766235797575176, 'B_8_3': 0.048255846063750565, 'B_8_4': -0.3588863901307074, 'B_8_5': -0.5042138642372571, 'B_8_6': 0.25931717007352656, 'B_8_7': -0.9058450493626506, 'B_8_8': 0.1384242199523618, 'B_8_9': 0.794854305168081, 'B_9_0': 0.22977416993995514, 'B_9_1': -0.8691009972332009, 'B_9_2': -0.8464448509699214, 'B_9_3': 0.8802332600494212, 'B_9_4': -0.4900914362142127, 'B_9_5': -0.8509252531579592, 'B_9_6': -0.09586118886579362, 'B_9_7': -0.3491247138734084, 'B_9_8': 0.5512911634735155, 'B_9_9': 0.9149845273908769, 'B_10_0': 0.8309350204040494, 'B_10_1': -0.655774343657214, 'B_10_2': -0.8260910012918913, 'B_10_3': -0.7116269216946011, 'B_10_4': 0.5678810777446547, 'B_10_5': -0.16950197022393562, 'B_10_6': -0.07059852830535011, 'B_10_7': 0.8919833814871874, 'B_10_8': -0.13734934528024742, 'B_10_9': 0.015060929968311773}. Best is trial 119 with value: 66.30887875663029.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$\langle F_{[u_i]}, F_{[u_j]} \rangle$].

Built matrix B = inverse(A).

[I 2025-04-29 14:19:47,207] Trial 127 finished with value: 69.04368616211238 and parameters: {'B_0_0': 0.24727670972291077, 'B_0_1': -0.0969727490131173, 'B_0_2': 0.8494595172467382, 'B_0_3': 0.04914395052919404, 'B_0_4':

-0.23614185387588596, 'B_0_5': 0.7781485218716799, 'B_0_6': 0.3816394821616937,
 'B_0_7': 0.38499140889949796, 'B_0_8': 0.4856842441070829, 'B_0_9':
 0.08420980924508609, 'B_1_0': 0.3106882723276912, 'B_1_1': 0.03277057390543886,
 'B_1_2': 0.14230605085301318, 'B_1_3': 0.7243219938711423, 'B_1_4':
 -0.3276178548627972, 'B_1_5': 0.7376619812298384, 'B_1_6': -0.3605776116674838,
 'B_1_7': 0.9226085663518255, 'B_1_8': -0.6290401613266667, 'B_1_9':
 -0.3906425783073769, 'B_2_0': -0.3747993962819838, 'B_2_1': -0.1477714243580574,
 'B_2_2': -0.46839270350883194, 'B_2_3': 0.9858970095043211, 'B_2_4':
 -0.30223122007757053, 'B_2_5': -0.5596935076210404, 'B_2_6':
 -0.24506114050351283, 'B_2_7': -0.6457865974938324, 'B_2_8':
 -0.44310574182922446, 'B_2_9': 0.27570501613927617, 'B_3_0':
 -0.6717632804961184, 'B_3_1': -0.4690666018113544, 'B_3_2':
 -0.13685005114138524, 'B_3_3': 0.0447885251075572, 'B_3_4': -0.4672019772617498,
 'B_3_5': -0.038333851703124866, 'B_3_6': -0.12014831531088557, 'B_3_7':
 0.12706662981050185, 'B_3_8': 0.49207520019870044, 'B_3_9':
 -0.19279031513928696, 'B_4_0': 0.1581003263729206, 'B_4_1': 0.48976754547481816,
 'B_4_2': 0.28828819476458867, 'B_4_3': -0.3568707624937605, 'B_4_4':
 -0.39573144304886027, 'B_4_5': 0.6479845670372211, 'B_4_6': -0.8770012804099004,
 'B_4_7': -0.03165107018515648, 'B_4_8': 0.6429018217908405, 'B_4_9':
 0.6250375068022197, 'B_5_0': -0.05227037387097269, 'B_5_1': -0.4750522581332458,
 'B_5_2': 0.24523495207772394, 'B_5_3': -0.039748625054240816, 'B_5_4':
 0.45972868636068254, 'B_5_5': -0.6975099461608358, 'B_5_6':
 -0.19603726660557047, 'B_5_7': 0.38894482595658725, 'B_5_8':
 -0.3075259343725123, 'B_5_9': 0.6801007476208416, 'B_6_0': 0.0592808119050946,
 'B_6_1': -0.8661116551169903, 'B_6_2': 0.9134629007522415, 'B_6_3':
 0.7582003922960339, 'B_6_4': -0.8200580124273793, 'B_6_5': -0.18113688005362064,
 'B_6_6': 0.07436497783710314, 'B_6_7': 0.6271799678044974, 'B_6_8':
 0.706407378158544, 'B_6_9': 0.5000739058316465, 'B_7_0': -0.9718998115218678,
 'B_7_1': -0.57174239842051, 'B_7_2': -0.797634247617968, 'B_7_3':
 0.9457786991122104, 'B_7_4': 0.6639116312228661, 'B_7_5': -0.3917961133540794,
 'B_7_6': -0.37019166718430074, 'B_7_7': -0.6519515213924695, 'B_7_8':
 -0.8629440678136415, 'B_7_9': 0.49717164645231365, 'B_8_0': -0.7978221856626357,
 'B_8_1': 0.0975406543237583, 'B_8_2': 0.8368405742460954, 'B_8_3':
 0.045646950124880716, 'B_8_4': -0.25631897566096146, 'B_8_5':
 -0.5113328841231333, 'B_8_6': 0.2588556286040032, 'B_8_7': -0.894983003171164,
 'B_8_8': 0.1386138115786404, 'B_8_9': 0.794921137335648, 'B_9_0':
 0.24631499569986573, 'B_9_1': -0.8722089819768826, 'B_9_2': -0.8417941696252219,
 'B_9_3': 0.8863231100986636, 'B_9_4': -0.4833456720870983, 'B_9_5':
 -0.7186242439937064, 'B_9_6': -0.08794376556864321, 'B_9_7':
 -0.34423271642883235, 'B_9_8': 0.6506335679822034, 'B_9_9': 0.8964163040444704,
 'B_10_0': 0.8290749857458724, 'B_10_1': -0.6237165375898162, 'B_10_2':
 -0.8225619060657625, 'B_10_3': -0.632854157392064, 'B_10_4': 0.5533426086605917,
 'B_10_5': -0.22103168625936126, 'B_10_6': -0.061668501467650814, 'B_10_7':
 0.896987392576158, 'B_10_8': -0.11402595598354873, 'B_10_9':
 0.023411120106718607}. Best is trial 119 with value: 66.30887875663029.

Building the Basis functions for the RKHS:

Built F_[u₁].

Built F_[u₂].

Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].
Built F__[u_10].
Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:20:01,580] Trial 128 finished with value: 72.55266961888142 and parameters: {'B_0_0': 0.24197616496729174, 'B_0_1': -0.07709515340752156, 'B_0_2': 0.8390132971316286, 'B_0_3': 0.05015500388104396, 'B_0_4': -0.2486051989962063, 'B_0_5': 0.7939198331239569, 'B_0_6': 0.39184148236774496, 'B_0_7': 0.41509536916804796, 'B_0_8': 0.46852616656152846, 'B_0_9': 0.08564639664070015, 'B_1_0': 0.31173548274167606, 'B_1_1': 0.009903879919969103, 'B_1_2': 0.12835246175947818, 'B_1_3': 0.7359241490890067, 'B_1_4': -0.32024913310657827, 'B_1_5': 0.752955223059175, 'B_1_6': -0.311433642387646, 'B_1_7': 0.9030167142695157, 'B_1_8': -0.6381790073056626, 'B_1_9': -0.3839430052334076, 'B_2_0': -0.38933757554528786, 'B_2_1': -0.11281373563949876, 'B_2_2': -0.49318589950340763, 'B_2_3': 0.979928222749694, 'B_2_4': -0.2951408038250948, 'B_2_5': -0.5653857003110848, 'B_2_6': -0.43520606095915426, 'B_2_7': -0.63197422667579, 'B_2_8': -0.43969788161193574, 'B_2_9': 0.27369449500565607, 'B_3_0': -0.6879185409138007, 'B_3_1': -0.4433036880126464, 'B_3_2': -0.08452195729844507, 'B_3_3': 0.07516949350981558, 'B_3_4': -0.5942096276245651, 'B_3_5': -0.04939736674934512, 'B_3_6': -0.11562749178430566, 'B_3_7': 0.11455119811413074, 'B_3_8': 0.4695879254804066, 'B_3_9': -0.20929669195276718, 'B_4_0': 0.18319015726435636, 'B_4_1': 0.4445826023964544, 'B_4_2': 0.2945909416464745, 'B_4_3': -0.41866344679343437, 'B_4_4': -0.38405859914535523, 'B_4_5': 0.6287982581970226, 'B_4_6': -0.8569742469213274, 'B_4_7': -0.0071185550980080985, 'B_4_8': 0.6314444166651927, 'B_4_9': 0.6199388078625652, 'B_5_0': -0.05857253552410757, 'B_5_1': -0.4696841352976464, 'B_5_2': 0.2464935404016115, 'B_5_3': -0.036208288706199215, 'B_5_4': 0.44803280641739945, 'B_5_5': -0.7010928000882888, 'B_5_6': -0.20576803694641793, 'B_5_7': 0.3886154246925352, 'B_5_8': -0.3096675435664374, 'B_5_9': 0.6351707095989266, 'B_6_0': 0.1110850629483382, 'B_6_1': -0.8725507695706728, 'B_6_2': 0.91495931076816, 'B_6_3': 0.7562791219006982, 'B_6_4': -0.8185966277573619, 'B_6_5': -0.1927793188641489, 'B_6_6': 0.08710972063331132, 'B_6_7': 0.6340761576055928, 'B_6_8': 0.6352441677995329, 'B_6_9': 0.5704622210036345, 'B_7_0': -0.9738684079154816, 'B_7_1': -0.5788903392110972, 'B_7_2': -0.7769969251798546, 'B_7_3': 0.9610036558563225, 'B_7_4': 0.6662514501404191, 'B_7_5': -0.3790824945366197, 'B_7_6': -0.37203352232909376, 'B_7_7': -0.6501258010729323, 'B_7_8': -0.8650297003063957, 'B_7_9': 0.5893012557682596, 'B_8_0': -0.7986758897814461, 'B_8_1': 0.0828630937814286, 'B_8_2': 0.665750326135222, 'B_8_3':

0.052674752372306934, 'B_8_4': -0.35828541530543495, 'B_8_5':
-0.5089089456652075, 'B_8_6': 0.2650293599011123, 'B_8_7': -0.7515111216602189,
'B_8_8': 0.12832643302437463, 'B_8_9': 0.8072620258076799, 'B_9_0':
0.24679965980522311, 'B_9_1': -0.8779378365049981, 'B_9_2': -0.840456430011155,
'B_9_3': 0.7861659161464009, 'B_9_4': -0.5142240923119359, 'B_9_5':
-0.8466592393146364, 'B_9_6': -0.08773471018892488, 'B_9_7':
-0.34791179330878236, 'B_9_8': 0.5586024259073289, 'B_9_9': 0.9087658290366724,
'B_10_0': 0.8318803835826472, 'B_10_1': -0.78274156920354, 'B_10_2':
-0.8181701176032976, 'B_10_3': -0.7211507054300133, 'B_10_4':
0.5653238666837443, 'B_10_5': -0.2199354349803562, 'B_10_6':
-0.06387687805480005, 'B_10_7': 0.9025579891887675, 'B_10_8':
-0.13871959356524713, 'B_10_9': 0.037043248540854334}. Best is trial 119 with
value: 66.30887875663029.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:20:15,404] Trial 129 finished with value: 77.80296959134823 and
parameters: {'B_0_0': 0.2472766194243589, 'B_0_1': -0.09992780105442749,
'B_0_2': 0.8449482224954251, 'B_0_3': 0.06827088139780227, 'B_0_4':
-0.2678690554665776, 'B_0_5': 0.7043682532095144, 'B_0_6': 0.5103913281059551,
'B_0_7': 0.44343385070625596, 'B_0_8': 0.5827667600800606, 'B_0_9':
0.09329638621631628, 'B_1_0': 0.3498819391286566, 'B_1_1': 0.12697917920987428,
'B_1_2': 0.14462000182202167, 'B_1_3': 0.7622349065460592, 'B_1_4':
-0.35069723627983246, 'B_1_5': 0.8395161607923741, 'B_1_6':
-0.36431416241738374, 'B_1_7': 0.8545961152963829, 'B_1_8': -0.6355559613473118,
'B_1_9': -0.44318293297601546, 'B_2_0': -0.38731902114627015, 'B_2_1':
-0.1424165417918244, 'B_2_2': -0.45448265692955275, 'B_2_3': 0.9728672157825372,
'B_2_4': -0.2694878543562687, 'B_2_5': -0.4600863367128004, 'B_2_6':
-0.5135509927811023, 'B_2_7': -0.5842402834611811, 'B_2_8':
-0.44672578903311355, 'B_2_9': 0.2672914180773245, 'B_3_0': -0.6904467355412733,
'B_3_1': -0.4300927474027877, 'B_3_2': -0.1772119918055271, 'B_3_3':
0.09281485986352628, 'B_3_4': -0.6027130456850379, 'B_3_5':
-0.05022044848501708, 'B_3_6': -0.12317649331329025, 'B_3_7':
0.12772419028092785, 'B_3_8': 0.46699319589544985, 'B_3_9':
-0.17587604905375134, 'B_4_0': 0.19173780792191025, 'B_4_1':
0.44052352346188733, 'B_4_2': 0.29115519902121273, 'B_4_3': -0.3555440447021036,

'B_4_4': -0.37330126065014263, 'B_4_5': 0.7303357593763092, 'B_4_6': -0.869356996182378, 'B_4_7': -0.02732405662802273, 'B_4_8': 0.6216451667529808, 'B_4_9': 0.4417062582052479, 'B_5_0': 0.137183446626971, 'B_5_1': -0.4579583569817592, 'B_5_2': 0.18939637993449984, 'B_5_3': -0.02754014885392417, 'B_5_4': 0.4863162777626259, 'B_5_5': -0.702539066909384, 'B_5_6': -0.19823916744477296, 'B_5_7': 0.39978578328375625, 'B_5_8': -0.2848704660092064, 'B_5_9': 0.6799078467184658, 'B_6_0': 0.09681133805232855, 'B_6_1': -0.8826006588498384, 'B_6_2': 0.7703867423438147, 'B_6_3': 0.747599810005092, 'B_6_4': -0.9834672857395916, 'B_6_5': -0.22699232218011062, 'B_6_6': -0.0998530077371591, 'B_6_7': 0.5500803989577906, 'B_6_8': 0.6551673161198519, 'B_6_9': 0.9047465308071223, 'B_7_0': -0.9897298372315205, 'B_7_1': -0.5549461623334332, 'B_7_2': -0.7583014874233773, 'B_7_3': 0.9475178569557834, 'B_7_4': 0.7265659685222827, 'B_7_5': -0.3928169130706957, 'B_7_6': -0.3705266726307614, 'B_7_7': -0.6651837163939796, 'B_7_8': -0.8650943375352279, 'B_7_9': 0.5763563762202835, 'B_8_0': -0.7888637669055318, 'B_8_1': 0.1708960939851693, 'B_8_2': 0.6694381434360345, 'B_8_3': -0.09100241641557846, 'B_8_4': -0.2544607615598943, 'B_8_5': -0.4974055387393206, 'B_8_6': 0.38654896848686354, 'B_8_7': -0.7678722664951606, 'B_8_8': 0.07551229686899583, 'B_8_9': 0.7205051251463067, 'B_9_0': 0.19132561528072214, 'B_9_1': -0.8548746343299723, 'B_9_2': -0.8275637498386873, 'B_9_3': 0.884380233334734, 'B_9_4': -0.4726677601112845, 'B_9_5': -0.7202570490765361, 'B_9_6': -0.07835398381702867, 'B_9_7': -0.47187329706036657, 'B_9_8': 0.5563205912677371, 'B_9_9': 0.8091230081125736, 'B_10_0': 0.8246493434641056, 'B_10_1': -0.6257295191405717, 'B_10_2': -0.8443379413473215, 'B_10_3': -0.6487308045881913, 'B_10_4': 0.5505444431917341, 'B_10_5': -0.22768730891808758, 'B_10_6': -0.08290768714285918, 'B_10_7': 0.9032961879494907, 'B_10_8': -0.09138915488117208, 'B_10_9': 0.0033501779354853153}. Best is trial 119 with value: 66.30887875663029.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F_[u_i], _F_[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:20:29,319] Trial 130 finished with value: 83.57240915890412 and parameters: {'B_0_0': 0.22851729191459136, 'B_0_1': -0.05811803924148245, 'B_0_2': 0.7964453075581169, 'B_0_3': -0.061451366947139485, 'B_0_4': -0.4495155417540663, 'B_0_5': 0.824002821990866, 'B_0_6': 0.38602915847461505,

'B_0_7': 0.39574506427402456, 'B_0_8': 0.34497345805822016, 'B_0_9': 0.06532701569508942, 'B_1_0': 0.29308210506032634, 'B_1_1': 0.01820076605755817, 'B_1_2': 0.18349926129565108, 'B_1_3': 0.9230652769892973, 'B_1_4': -0.31866783157490486, 'B_1_5': 0.7357244757458098, 'B_1_6': -0.23131867056643232, 'B_1_7': 0.7372837725663437, 'B_1_8': -0.6475940029441154, 'B_1_9': -0.3989117127747463, 'B_2_0': -0.41801563666126323, 'B_2_1': -0.07621561670900726, 'B_2_2': -0.5248757603694602, 'B_2_3': 0.921382412599711, 'B_2_4': -0.39944924915416424, 'B_2_5': -0.44878248443781893, 'B_2_6': -0.4446079902323944, 'B_2_7': -0.5553314969749114, 'B_2_8': -0.4672195622549134, 'B_2_9': 0.3527582842685075, 'B_3_0': -0.6992930788962968, 'B_3_1': -0.3145797876493081, 'B_3_2': -0.14671652793410506, 'B_3_3': -0.08194274225230955, 'B_3_4': -0.6128893618578966, 'B_3_5': 0.017342374315789687, 'B_3_6': -0.17787866893726978, 'B_3_7': 0.04669939388851368, 'B_3_8': 0.47358817376848666, 'B_3_9': -0.3373649157670789, 'B_4_0': 0.26969409485497137, 'B_4_1': 0.44374865688299314, 'B_4_2': 0.27839749632496813, 'B_4_3': -0.49913565689458816, 'B_4_4': -0.4011903379158101, 'B_4_5': 0.6875077305991611, 'B_4_6': -0.6685816474209061, 'B_4_7': 0.02651252581226301, 'B_4_8': 0.674723713502511, 'B_4_9': 0.6233887333166398, 'B_5_0': -0.036013214173817934, 'B_5_1': -0.4936906483066792, 'B_5_2': 0.25169679222146357, 'B_5_3': -0.06948923656674122, 'B_5_4': 0.43895254869509676, 'B_5_5': -0.7474158386523949, 'B_5_6': -0.1765245929156703, 'B_5_7': 0.4180140859346635, 'B_5_8': -0.3144019797485652, 'B_5_9': 0.6379128278274342, 'B_6_0': -0.01532886771073251, 'B_6_1': -0.9025291099453695, 'B_6_2': 0.913443524520167, 'B_6_3': 0.7519907231208207, 'B_6_4': -0.7675587129757266, 'B_6_5': -0.050447727607026344, 'B_6_6': 0.07865079763515455, 'B_6_7': 0.6112637751447704, 'B_6_8': 0.6942293421211693, 'B_6_9': 0.5546456909206116, 'B_7_0': -0.9381190328089576, 'B_7_1': -0.4567238360857629, 'B_7_2': -0.6144842214080376, 'B_7_3': 0.828298219097172, 'B_7_4': 0.6914151836260656, 'B_7_5': -0.5224613775274922, 'B_7_6': -0.38463176763947304, 'B_7_7': -0.6881663102751814, 'B_7_8': -0.8630470779926438, 'B_7_9': 0.4903193259463794, 'B_8_0': -0.7254676427138602, 'B_8_1': 0.08224028486635075, 'B_8_2': 0.8013214676271447, 'B_8_3': -0.02699654728736619, 'B_8_4': -0.2456859579556521, 'B_8_5': -0.5588375793757703, 'B_8_6': 0.25214266312255557, 'B_8_7': -0.7111647118713535, 'B_8_8': 0.00930255755532998, 'B_8_9': 0.8056399308941661, 'B_9_0': 0.19776805305601805, 'B_9_1': -0.8604529890306614, 'B_9_2': -0.811653206040845, 'B_9_3': 0.7797555382750179, 'B_9_4': -0.602222694623118, 'B_9_5': -0.929175573952733, 'B_9_6': -0.03864922308544757, 'B_9_7': -0.3830545285589135, 'B_9_8': 0.6521519084943845, 'B_9_9': 0.9154827027273205, 'B_10_0': 0.7192873131977826, 'B_10_1': -0.7864924466322618, 'B_10_2': -0.7021470068234734, 'B_10_3': -0.5543121127695327, 'B_10_4': 0.6442453251432294, 'B_10_5': -0.22194800064155484, 'B_10_6': -0.2841789921692535, 'B_10_7': 0.9225124703647977, 'B_10_8': -0.05652348773264197, 'B_10_9': 0.026372156780829548}. Best is trial 119 with value: 66.30887875663029.

Building the Basis functions for the RKHS:

Built F_[u₁].

Built F_[u₂].

```

Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```

[I 2025-04-29 14:20:43,956] Trial 131 finished with value: 68.9078163269931 and
parameters: {'B_0_0': 0.2941175852160799, 'B_0_1': -0.09855285389101562,
'B_0_2': 0.9501258238640129, 'B_0_3': 0.028158326171341272, 'B_0_4':
-0.2289163300747896, 'B_0_5': 0.7720084435446724, 'B_0_6': 0.37773376051247715,
'B_0_7': 0.576434216248347, 'B_0_8': 0.4716024405585325, 'B_0_9':
0.018878292787366233, 'B_1_0': 0.4119520232918035, 'B_1_1':
0.024051405623363286, 'B_1_2': 0.13327829031052, 'B_1_3': 0.7441176389271089,
'B_1_4': -0.3073600858217444, 'B_1_5': 0.7659581382718138, 'B_1_6':
-0.309637539607983, 'B_1_7': 0.9181733357893604, 'B_1_8': -0.5073270782725064,
'B_1_9': -0.3628065831685172, 'B_2_0': -0.3544781587151421, 'B_2_1':
-0.1332638372754362, 'B_2_2': -0.4524066456613225, 'B_2_3': 0.8275760721252586,
'B_2_4': -0.19251642893925566, 'B_2_5': -0.5613802360127031, 'B_2_6':
-0.45106594183704296, 'B_2_7': -0.6236006384906662, 'B_2_8':
-0.6065558426081471, 'B_2_9': 0.27977974955138096, 'B_3_0': -0.5914890038833494,
'B_3_1': -0.5423587376064103, 'B_3_2': -0.08413122609151066, 'B_3_3':
0.04495873508481011, 'B_3_4': -0.46857461621014557, 'B_3_5':
-0.041113065694714145, 'B_3_6': -0.09088101575248488, 'B_3_7':
0.14618091974884234, 'B_3_8': 0.5808810738014384, 'B_3_9': -0.19667613906206194,
'B_4_0': 0.06544652642191492, 'B_4_1': 0.5009616222829033, 'B_4_2':
0.20395996445689343, 'B_4_3': -0.42650111640291455, 'B_4_4':
-0.47642327581479516, 'B_4_5': 0.628262144429367, 'B_4_6': -0.855427297257923,
'B_4_7': -0.019878142497637013, 'B_4_8': 0.6728611933850268, 'B_4_9':
0.5914131879868587, 'B_5_0': 0.10845200624287454, 'B_5_1': -0.39824522145419833,
'B_5_2': 0.22633805855526645, 'B_5_3': -0.039961801724469184, 'B_5_4':
0.5371577027223091, 'B_5_5': -0.6924105312319236, 'B_5_6': -0.24651840787125692,
'B_5_7': 0.35878107254762315, 'B_5_8': -0.31997218056253357, 'B_5_9':
0.6988315253483468, 'B_6_0': 0.08688323111464105, 'B_6_1': -0.8495863262999279,
'B_6_2': 0.9156828196776595, 'B_6_3': 0.774383467500612, 'B_6_4':
-0.8157535135473148, 'B_6_5': -0.18148092683530728, 'B_6_6':
0.07833431906982667, 'B_6_7': 0.6449125241873903, 'B_6_8': 0.6161642084408937,
'B_6_9': 0.5266835175402844, 'B_7_0': -0.9868425742636373, 'B_7_1':
-0.5885184003659709, 'B_7_2': -0.7957595837212609, 'B_7_3': 0.9493588428153996,
'B_7_4': 0.6628573130030962, 'B_7_5': -0.3914669580110215, 'B_7_6':
-0.32653667746686454, 'B_7_7': -0.7897795457095141, 'B_7_8':
-0.9835759550635468, 'B_7_9': 0.5915061496463998, 'B_8_0': -0.7897397307127848,
'B_8_1': 0.07258614072289592, 'B_8_2': 0.8282867190884544, 'B_8_3':

```

0.021099495468042743, 'B_8_4': -0.3664169755199659, 'B_8_5':
-0.5294524879992046, 'B_8_6': 0.2911363389168012, 'B_8_7': -0.9005392179160003,
'B_8_8': 0.1339882258486232, 'B_8_9': 0.788008393959126, 'B_9_0':
0.27077151567105306, 'B_9_1': -0.904798677029184, 'B_9_2': -0.8735132793644697,
'B_9_3': 0.7975914642819869, 'B_9_4': -0.5654176464129081, 'B_9_5':
-0.839355468002557, 'B_9_6': -0.13666391123000804, 'B_9_7':
-0.35251549825824025, 'B_9_8': 0.5492352959631237, 'B_9_9': 0.8935722805025836,
'B_10_0': 0.824669548278836, 'B_10_1': -0.783874538147537, 'B_10_2':
-0.8090819660620865, 'B_10_3': -0.7026276926108113, 'B_10_4':
0.5401271035886378, 'B_10_5': -0.1749482115442466, 'B_10_6':
-0.06211908078630672, 'B_10_7': 0.8860984194928928, 'B_10_8':
-0.15164364166433741, 'B_10_9': -0.026706017594034025}. Best is trial 119 with
value: 66.30887875663029.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [<F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:20:58,065] Trial 132 finished with value: 77.77785301528401 and
parameters: {'B_0_0': 0.397359405993288, 'B_0_1': -0.02055886918771979, 'B_0_2':
0.8582963899720821, 'B_0_3': 0.017778103762970565, 'B_0_4':
-0.20383022208093915, 'B_0_5': 0.7711229968929406, 'B_0_6': 0.40513439211552066,
'B_0_7': 0.34922828824481034, 'B_0_8': 0.47044107769419596, 'B_0_9':
0.008513072659790219, 'B_1_0': 0.4050343238444436, 'B_1_1': 0.08494563887496637,
'B_1_2': 0.11719663878285858, 'B_1_3': 0.7277266940458889, 'B_1_4':
-0.41861037567086085, 'B_1_5': 0.762823374657187, 'B_1_6': -0.3625080099914589,
'B_1_7': 0.9166518245137193, 'B_1_8': -0.5622331055484625, 'B_1_9':
-0.34930889059144504, 'B_2_0': -0.3592884520853244, 'B_2_1':
-0.16758071637080207, 'B_2_2': -0.4821237567143047, 'B_2_3': 0.9540128846743072,
'B_2_4': -0.20018460968169224, 'B_2_5': -0.5980740747408877, 'B_2_6':
-0.23106874692805318, 'B_2_7': -0.6537991026471474, 'B_2_8':
-0.5685815233506062, 'B_2_9': 0.40510679775308195, 'B_3_0': -0.5842970127734662,
'B_3_1': -0.4424422432573844, 'B_3_2': -0.09258049536756911, 'B_3_3':
0.034342168803416014, 'B_3_4': -0.5659253227792754, 'B_3_5':
-0.20712404840960918, 'B_3_6': -0.003056128302967455, 'B_3_7':
0.11414948549055144, 'B_3_8': 0.5546934354053311, 'B_3_9': -0.20676169560048552,
'B_4_0': 0.3251208286727728, 'B_4_1': 0.49402351016135354, 'B_4_2':
0.39067660833419476, 'B_4_3': -0.3739003959921721, 'B_4_4': -0.4732850345819884,

'B_4_5': 0.6289968856533343, 'B_4_6': -0.7015503992602992, 'B_4_7': 0.10979333336144903, 'B_4_8': 0.6255663136971298, 'B_4_9': 0.5905450996672362, 'B_5_0': -0.05133709086777998, 'B_5_1': -0.3920201809759769, 'B_5_2': 0.20960596697250378, 'B_5_3': -0.05589261969663764, 'B_5_4': 0.523229834452348, 'B_5_5': -0.6687506149112745, 'B_5_6': -0.2735696335589342, 'B_5_7': 0.35876761611671093, 'B_5_8': -0.3959748694213671, 'B_5_9': 0.6882388203961078, 'B_6_0': 0.10311372279984579, 'B_6_1': -0.6927412640525543, 'B_6_2': 0.82583820536088, 'B_6_3': 0.6533718449704405, 'B_6_4': -0.8471849307653697, 'B_6_5': -0.1866802630667723, 'B_6_6': 0.11580310007315905, 'B_6_7': 0.5950187852620481, 'B_6_8': 0.6243734237158035, 'B_6_9': 0.46385267893669885, 'B_7_0': -0.9179764037690434, 'B_7_1': -0.598889534436175, 'B_7_2': -0.8000565117881113, 'B_7_3': 0.9648664856206258, 'B_7_4': 0.6647321085072827, 'B_7_5': -0.2097128201448778, 'B_7_6': -0.30851098670196364, 'B_7_7': -0.7689698666968198, 'B_7_8': -0.9880785715789125, 'B_7_9': 0.6495762943476383, 'B_8_0': -0.7753828694038359, 'B_8_1': 0.10528016505312947, 'B_8_2': 0.8196706969414096, 'B_8_3': 0.04538296471565843, 'B_8_4': -0.3412445147984434, 'B_8_5': -0.4842828292168931, 'B_8_6': 0.38439706286677366, 'B_8_7': -0.7980586292012118, 'B_8_8': 0.14141268685952105, 'B_8_9': 0.7179630350887078, 'B_9_0': 0.05365991812179563, 'B_9_1': -0.7225274068053172, 'B_9_2': -0.8520797156219291, 'B_9_3': 0.9301236744792394, 'B_9_4': -0.46502023804925957, 'B_9_5': -0.8484858792134837, 'B_9_6': -0.1496300119694743, 'B_9_7': -0.3951655355114132, 'B_9_8': 0.540417878039085, 'B_9_9': 0.8568031428111482, 'B_10_0': 0.8478322003218626, 'B_10_1': -0.6930107244787567, 'B_10_2': -0.9886310307465036, 'B_10_3': -0.6358909082477604, 'B_10_4': 0.5324414590294844, 'B_10_5': -0.05729020027798755, 'B_10_6': -0.044562575457941664, 'B_10_7': 0.9353737610692942, 'B_10_8': -0.12807968094964745, 'B_10_9': 0.074926705977081}. Best is trial 119 with value: 66.30887875663029.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:21:12,352] Trial 133 finished with value: 73.37526820819335 and parameters: {'B_0_0': 0.3120426327215289, 'B_0_1': 0.08145247523585301, 'B_0_2': 0.7095522456881126, 'B_0_3': 0.1355456341853763, 'B_0_4': -0.40269817427034466, 'B_0_5': 0.8602445030222192, 'B_0_6': 0.29186820368936883, 'B_0_7': 0.572058862182237, 'B_0_8': 0.4747065087464064, 'B_0_9': 0.08602133700401009,

'B_1_0': 0.4723233896965391, 'B_1_1': -0.022488716501294328, 'B_1_2':
 0.18543313584969312, 'B_1_3': 0.7503494168807198, 'B_1_4': -0.29281173313803577,
 'B_1_5': 0.824099669811403, 'B_1_6': -0.27723472653075587, 'B_1_7':
 0.8599617729610924, 'B_1_8': -0.6938134845215541, 'B_1_9': -0.475290171991053,
 'B_2_0': -0.6218795629765758, 'B_2_1': -0.3424861950012602, 'B_2_2':
 -0.4365958044458442, 'B_2_3': 0.9155227254078991, 'B_2_4': -0.357267823608814,
 'B_2_5': -0.5356121946317013, 'B_2_6': -0.3828142603383008, 'B_2_7':
 -0.5524398223009417, 'B_2_8': -0.6071214073061654, 'B_2_9': 0.3227357132585837,
 'B_3_0': -0.6674397189455491, 'B_3_1': -0.3335444034766548, 'B_3_2':
 -0.23082912846042242, 'B_3_3': -0.13434296086638203, 'B_3_4':
 -0.43920480755449687, 'B_3_5': 0.19247548309460732, 'B_3_6':
 0.018369529444695026, 'B_3_7': 0.21023248971180106, 'B_3_8':
 0.41125948229302545, 'B_3_9': -0.14412277830994635, 'B_4_0':
 0.16937084619175558, 'B_4_1': 0.432290784360388, 'B_4_2': 0.23118916737410966,
 'B_4_3': -0.43759377940441413, 'B_4_4': -0.5757851418868731, 'B_4_5':
 0.5054146217906967, 'B_4_6': -0.8626861068430839, 'B_4_7': -0.04554361088730481,
 'B_4_8': 0.7011641876793093, 'B_4_9': 0.6600432705651329, 'B_5_0':
 0.12035031880082703, 'B_5_1': -0.4461366184404584, 'B_5_2': 0.26680702216012575,
 'B_5_3': -0.12097471304292803, 'B_5_4': 0.4449994383144265, 'B_5_5':
 -0.736533305517034, 'B_5_6': -0.23607053395107988, 'B_5_7': 0.3321107204882392,
 'B_5_8': -0.21126298672501412, 'B_5_9': 0.7211584100629157, 'B_6_0':
 0.01604035240183655, 'B_6_1': -0.8458069644740142, 'B_6_2': 0.9032999358763264,
 'B_6_3': 0.7451958029513259, 'B_6_4': -0.6429180436035276, 'B_6_5':
 -0.2132670290809722, 'B_6_6': 0.2484681140373559, 'B_6_7': 0.5355226235454377,
 'B_6_8': 0.7320667775579245, 'B_6_9': 0.8976507052997182, 'B_7_0':
 -0.94894133480601, 'B_7_1': -0.4955848771222924, 'B_7_2': -0.6013857339142231,
 'B_7_3': 0.8201991237446926, 'B_7_4': 0.7402794397025427, 'B_7_5':
 -0.41550062475548505, 'B_7_6': -0.26267379077875896, 'B_7_7':
 -0.633786589989854, 'B_7_8': -0.9449427360069909, 'B_7_9': 0.5800195730916143,
 'B_8_0': -0.8284452320232808, 'B_8_1': 0.2833372028556883, 'B_8_2':
 0.6476440790582662, 'B_8_3': 0.014735352532451472, 'B_8_4': -0.4183195921586501,
 'B_8_5': -0.3456853034156525, 'B_8_6': 0.2916760843678667, 'B_8_7':
 -0.8722376091846417, 'B_8_8': 0.15028526780875404, 'B_8_9': 0.9128246696190746,
 'B_9_0': 0.2712888352841743, 'B_9_1': -0.8834797153346482, 'B_9_2':
 -0.8815053884423304, 'B_9_3': 0.8543780702265867, 'B_9_4': -0.5601649786957773,
 'B_9_5': -0.6180624902445145, 'B_9_6': -0.08167946202225823, 'B_9_7':
 -0.33142439848506466, 'B_9_8': 0.6338113274341044, 'B_9_9': 0.8990014303508421,
 'B_10_0': 0.917144389434254, 'B_10_1': -0.6560006725511972, 'B_10_2':
 -0.9473680738803798, 'B_10_3': -0.6998593607895718, 'B_10_4':
 0.5652981693487211, 'B_10_5': -0.19067033063783168, 'B_10_6':
 -0.07756148796767265, 'B_10_7': 0.8864702027690426, 'B_10_8':
 -0.06422672200089621, 'B_10_9': -0.03503839165736003}. Best is trial 119 with
 value: 66.30887875663029.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].

```
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```
[I 2025-04-29 14:21:26,704] Trial 134 finished with value: 63.27557267023822 and
parameters: {'B_0_0': 0.3243038847016484, 'B_0_1': 0.08574565572657387, 'B_0_2':
0.6846703574628257, 'B_0_3': 0.08358867363495745, 'B_0_4': -0.4112883203105625,
'B_0_5': 0.8701190410302024, 'B_0_6': 0.29468401246478226, 'B_0_7':
0.5833819167799018, 'B_0_8': 0.38780043540292713, 'B_0_9': 0.20007678402904383,
'B_1_0': 0.47433420686943595, 'B_1_1': -0.14559025840420683, 'B_1_2':
0.1933868587785661, 'B_1_3': 0.6925532634234322, 'B_1_4': -0.1830697072221923,
'B_1_5': 0.8293819985889673, 'B_1_6': -0.285859197344986, 'B_1_7':
0.8519369925103372, 'B_1_8': -0.6004781253218354, 'B_1_9': -0.4871659310375239,
'B_2_0': -0.6205791868022419, 'B_2_1': -0.3620441490417161, 'B_2_2':
-0.5414497205963876, 'B_2_3': 0.9032127464315927, 'B_2_4': -0.3662612325774305,
'B_2_5': -0.6178956819681858, 'B_2_6': -0.3691946226616577, 'B_2_7':
-0.4842671754174371, 'B_2_8': -0.520085657087122, 'B_2_9': 0.315809224752707,
'B_3_0': -0.569367607896657, 'B_3_1': -0.32275448990137634, 'B_3_2':
-0.22838140448122626, 'B_3_3': -0.028402810916014647, 'B_3_4':
-0.5144560116988204, 'B_3_5': 0.18229350825112484, 'B_3_6':
-0.10742536469853232, 'B_3_7': 0.23982170026034932, 'B_3_8':
0.39787945720456397, 'B_3_9': -0.03069295672600364, 'B_4_0':
0.07807211590009563, 'B_4_1': 0.3491734217768004, 'B_4_2': 0.22966335771098262,
'B_4_3': -0.321082584467052, 'B_4_4': -0.593188190832328, 'B_4_5':
0.4917742558699198, 'B_4_6': -0.755914524807435, 'B_4_7': -0.27888454930218176,
'B_4_8': 0.5734175439280079, 'B_4_9': 0.6445379205312661, 'B_5_0':
0.17440082707211824, 'B_5_1': -0.5254721356177474, 'B_5_2': 0.47970423034175325,
'B_5_3': -0.11329799138968902, 'B_5_4': 0.6350204082193145, 'B_5_5':
-0.7659234380975708, 'B_5_6': -0.3049685606964431, 'B_5_7': 0.32081204791968665,
'B_5_8': -0.24513736613708637, 'B_5_9': 0.5954643128565408, 'B_6_0':
0.014079991248429475, 'B_6_1': -0.8836637606419103, 'B_6_2': 0.8330953601659757,
'B_6_3': 0.9336784879859121, 'B_6_4': -0.6386497560131924, 'B_6_5':
-0.07712233552672401, 'B_6_6': 0.22093943030098928, 'B_6_7': 0.5120548082174027,
'B_6_8': 0.624160979944949, 'B_6_9': 0.8783196875278023, 'B_7_0':
-0.9426681355191571, 'B_7_1': -0.4783521295872878, 'B_7_2': -0.6343624787734002,
'B_7_3': 0.8076030615826344, 'B_7_4': 0.7628941183515563, 'B_7_5':
-0.4731616136890753, 'B_7_6': -0.26506276271360746, 'B_7_7':
-0.8735557584141082, 'B_7_8': -0.9548902371274374, 'B_7_9': 0.5839960602773784,
'B_8_0': -0.8260567219794912, 'B_8_1': 0.27222741450800164, 'B_8_2':
0.6397599248026717, 'B_8_3': 0.2418753758100336, 'B_8_4': -0.41570044821101587,
'B_8_5': -0.35734819377131855, 'B_8_6': 0.30516804720219665, 'B_8_7':
-0.8577682272245867, 'B_8_8': 0.057900146551581655, 'B_8_9': 0.9182036846869592,
```


'B_9_0': 0.29649882273016437, 'B_9_1': -0.8084464537162223, 'B_9_2': -0.7203462527836877, 'B_9_3': 0.8672140128779192, 'B_9_4': -0.5405414471322696, 'B_9_5': -0.8542936907260633, 'B_9_6': -0.22719746057007367, 'B_9_7': -0.33600353761734314, 'B_9_8': 0.6560878214362271, 'B_9_9': 0.7845689435807245, 'B_10_0': 0.8917481100563673, 'B_10_1': -0.4727976295786007, 'B_10_2': -0.9540932657555372, 'B_10_3': -0.6948570810806511, 'B_10_4': 0.6167460577588664, 'B_10_5': -0.17152104646886085, 'B_10_6': -0.1093862989763468, 'B_10_7': 0.8914287501343102, 'B_10_8': -0.06666007615049883, 'B_10_9': 0.11777519659037346}. Best is trial 134 with value: 63.27557267023822.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:21:40,974] Trial 135 finished with value: 65.54616488351057 and parameters: {'B_0_0': 0.31626908797590797, 'B_0_1': 0.08102954396131543, 'B_0_2': 0.7160491159025549, 'B_0_3': 0.10023459785797287, 'B_0_4': -0.4980880759784749, 'B_0_5': 0.8749217012810293, 'B_0_6': 0.296357079685644, 'B_0_7': 0.5710234128228997, 'B_0_8': 0.3857667463734862, 'B_0_9': 0.2167493517560253, 'B_1_0': 0.4584742752217299, 'B_1_1': -0.028855004990238964, 'B_1_2': 0.19834982589418765, 'B_1_3': 0.6916979639959703, 'B_1_4': -0.17534111875595054, 'B_1_5': 0.8306496395207522, 'B_1_6': -0.30901256008528283, 'B_1_7': 0.8452472124987999, 'B_1_8': -0.7126364622885412, 'B_1_9': -0.47906843722289766, 'B_2_0': -0.644383596199228, 'B_2_1': -0.3382263381018237, 'B_2_2': -0.5295684367756308, 'B_2_3': 0.9092227397778107, 'B_2_4': -0.3574074049368399, 'B_2_5': -0.619540956504866, 'B_2_6': -0.3716772627116401, 'B_2_7': -0.491561664195436, 'B_2_8': -0.6089412371413125, 'B_2_9': 0.31218089761382706, 'B_3_0': -0.5719064397361355, 'B_3_1': -0.5524064304556984, 'B_3_2': -0.1593419442697245, 'B_3_3': -0.13507481008432753, 'B_3_4': -0.5114922177671204, 'B_3_5': 0.19139303595732257, 'B_3_6': -0.09563892684504016, 'B_3_7': 0.23725941693376984, 'B_3_8': 0.3948194399888969, 'B_3_9': -0.13490646561441813, 'B_4_0': 0.1645900843500613, 'B_4_1': 0.3669141359117466, 'B_4_2': 0.22885652873375356, 'B_4_3': -0.4388033461042188, 'B_4_4': -0.5803461716769337, 'B_4_5': 0.5135678789959027, 'B_4_6': -0.7604161871072106, 'B_4_7': -0.28132214840020053, 'B_4_8': 0.6978831150454369, 'B_4_9': 0.6629930865402618, 'B_5_0': 0.14157972708729596, 'B_5_1': -0.43043650413581747, 'B_5_2':

0.37915814774363144, 'B_5_3': -0.11652166898614152, 'B_5_4': 0.5414840428764371, 'B_5_5': -0.7746625722420732, 'B_5_6': -0.3108608191035171, 'B_5_7': 0.31140084345110813, 'B_5_8': -0.21097384183358586, 'B_5_9': 0.588941118105887, 'B_6_0': 0.02669533623041058, 'B_6_1': -0.6811109516468327, 'B_6_2': 0.8374196442211084, 'B_6_3': 0.9859120225028273, 'B_6_4': -0.753748412631081, 'B_6_5': 0.03477417831007584, 'B_6_6': 0.20883225059235852, 'B_6_7': 0.5550714504924917, 'B_6_8': 0.616044629560759, 'B_6_9': 0.8877150429137908, 'B_7_0': -0.9443628695208492, 'B_7_1': -0.4735244426853012, 'B_7_2': -0.5910306371221754, 'B_7_3': 0.8103523210354014, 'B_7_4': 0.7701977244264531, 'B_7_5': -0.4603323823372152, 'B_7_6': -0.29023493144892254, 'B_7_7': -0.864045227457929, 'B_7_8': -0.9585214478405516, 'B_7_9': 0.5893177229358774, 'B_8_0': -0.8175479434645656, 'B_8_1': 0.038586013283606085, 'B_8_2': 0.6309003793158754, 'B_8_3': 0.25562659902424173, 'B_8_4': -0.4111191502804372, 'B_8_5': -0.33779919928872315, 'B_8_6': 0.30251872371196903, 'B_8_7': -0.8702175864296334, 'B_8_8': 0.06578723035155484, 'B_8_9': 0.9071197332154127, 'B_9_0': 0.27966914931140807, 'B_9_1': -0.8002344831731695, 'B_9_2': -0.7901771707662621, 'B_9_3': 0.7358773758201417, 'B_9_4': -0.5530124155714415, 'B_9_5': -0.6777092659644962, 'B_9_6': -0.24763929835551873, 'B_9_7': -0.11127413641413644, 'B_9_8': 0.6363077403087694, 'B_9_9': 0.7775556023696706, 'B_10_0': 0.9080074642490052, 'B_10_1': -0.7867277839754407, 'B_10_2': -0.9518741366694669, 'B_10_3': -0.5384331015184483, 'B_10_4': 0.6198332000930817, 'B_10_5': -0.306353425778785, 'B_10_6': 0.01601966357416333, 'B_10_7': 0.765139393399697, 'B_10_8': -0.058923744243295995, 'B_10_9': -0.036461295210943115}. Best is trial 134 with value: 63.27557267023822.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:21:55,075] Trial 136 finished with value: 62.32599854647733 and parameters: {'B_0_0': 0.3164626392150868, 'B_0_1': 0.0802620069508916, 'B_0_2': 0.6841692237852102, 'B_0_3': 0.09475243005472637, 'B_0_4': -0.49631711612977164, 'B_0_5': 0.8604079966697321, 'B_0_6': 0.32448582633800277, 'B_0_7': 0.5702054670258684, 'B_0_8': 0.37301721239442454, 'B_0_9': 0.211737852151001, 'B_1_0': 0.48794451638304603, 'B_1_1': -0.0301789311616067, 'B_1_2': 0.1983085240796461, 'B_1_3': 0.7819006453341318, 'B_1_4': -0.40053692170016997, 'B_1_5': 0.8190359385163762, 'B_1_6': -0.2986369699551671, 'B_1_7': 0.8435508858421188, 'B_1_8': -0.7147844354999645, 'B_1_9': -0.4496069254461045,

'B_2_0': -0.6423851416320546, 'B_2_1': -0.35306384414801767, 'B_2_2':
-0.5296717853396686, 'B_2_3': 0.9025345650672085, 'B_2_4': -0.3538121419860071,
'B_2_5': -0.6679334823319737, 'B_2_6': -0.23978012021988768, 'B_2_7':
-0.4987571514784955, 'B_2_8': -0.5247173801082774, 'B_2_9': 0.3146996828588703,
'B_3_0': -0.4465632524422537, 'B_3_1': -0.3137404088929066, 'B_3_2':
-0.23126628182178277, 'B_3_3': -0.013207506832949273, 'B_3_4':
-0.5560206171110359, 'B_3_5': 0.1862243679686272, 'B_3_6': -0.08767461243654018,
'B_3_7': 0.23211950383825453, 'B_3_8': 0.401597877709364, 'B_3_9':
-0.1352268800922295, 'B_4_0': 0.07585880407572482, 'B_4_1': 0.35278414171057276,
'B_4_2': 0.23038156825320866, 'B_4_3': -0.4340433698986105, 'B_4_4':
-0.6785959934726915, 'B_4_5': 0.5036916880689983, 'B_4_6': -0.7642619479249422,
'B_4_7': -0.2594973685543119, 'B_4_8': 0.7105040395176716, 'B_4_9':
0.6502014981598183, 'B_5_0': 0.13514308957134147, 'B_5_1': -0.5020537059372607,
'B_5_2': 0.5011632004273285, 'B_5_3': -0.11579113285662254, 'B_5_4':
0.6296942390602519, 'B_5_5': -0.7924106776675127, 'B_5_6': -0.30916554756327946,
'B_5_7': 0.3195875523010069, 'B_5_8': -0.20769879667455982, 'B_5_9':
0.5730873715708282, 'B_6_0': 0.02969003278401596, 'B_6_1': -0.6670583098565889,
'B_6_2': 0.8169518143721579, 'B_6_3': 0.9313709735545423, 'B_6_4':
-0.6503914771625865, 'B_6_5': -0.0766747694308187, 'B_6_6': 0.2358496975622384,
'B_6_7': 0.5232055438886651, 'B_6_8': 0.546918576658754, 'B_6_9':
0.7607337363328277, 'B_7_0': -0.9422205860597505, 'B_7_1': -0.4917839435707045,
'B_7_2': -0.6237809700143435, 'B_7_3': 0.8033730201080549, 'B_7_4':
0.7816252934642917, 'B_7_5': -0.4827885996384388, 'B_7_6': -0.27603480655311524,
'B_7_7': -0.7750026379429309, 'B_7_8': -0.9505411338016134, 'B_7_9':
0.5825186170866055, 'B_8_0': -0.8392710211161085, 'B_8_1': 0.31030336173343526,
'B_8_2': 0.6390147893825489, 'B_8_3': 0.23315152872507516, 'B_8_4':
-0.2742076189019781, 'B_8_5': -0.33052176701399305, 'B_8_6':
0.41455455757608245, 'B_8_7': -0.8457169111943286, 'B_8_8': 0.04527201022013469,
'B_8_9': 0.9162278122234192, 'B_9_0': 0.2818991129359856, 'B_9_1':
-0.8083980432716577, 'B_9_2': -0.7121521876686259, 'B_9_3': 0.7497208927571909,
'B_9_4': -0.6738119520068245, 'B_9_5': -0.6048327274027419, 'B_9_6':
-0.24053853135217132, 'B_9_7': -0.2736584182143839, 'B_9_8': 0.60544990639766,
'B_9_9': 0.7720532788987396, 'B_10_0': 0.9159867995207138, 'B_10_1':
-0.7732929403168695, 'B_10_2': -0.9520493300522515, 'B_10_3': -0.55037641926926,
'B_10_4': 0.6287710307547205, 'B_10_5': -0.3394885404453958, 'B_10_6':
-0.11146676773053639, 'B_10_7': 0.9488940482206385, 'B_10_8':
-0.07137104684570658, 'B_10_9': 0.18818652790707943}. Best is trial 136 with
value: 62.32599854647733.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].

```

    Built F_[u_10].
    Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:22:09,321] Trial 137 finished with value: 64.14894685232461 and
parameters: {'B_0_0': 0.41250339703080446, 'B_0_1': 0.0011976613306131229,
'B_0_2': 0.7825649427805957, 'B_0_3': 0.08700880380982948, 'B_0_4':
-0.49663237415335076, 'B_0_5': 0.8644397871731436, 'B_0_6': 0.486937792029428,
'B_0_7': 0.6587913174863497, 'B_0_8': 0.32826398814560753, 'B_0_9':
0.30502081441886025, 'B_1_0': 0.4058811486506544, 'B_1_1': -0.14446879294954174,
'B_1_2': 0.2860070016995537, 'B_1_3': 0.6831217033605597, 'B_1_4':
-0.38667858966312596, 'B_1_5': 0.8607582081904227, 'B_1_6': -0.3076551649258483,
'B_1_7': 0.8467111577644302, 'B_1_8': -0.7074310389838044, 'B_1_9':
-0.46176610777151134, 'B_2_0': -0.24923160097035957, 'B_2_1':
-0.2046918727889931, 'B_2_2': -0.5312211239759307, 'B_2_3': 0.8330006449180614,
'B_2_4': -0.35755983830634364, 'B_2_5': -0.6215749553831437, 'B_2_6':
-0.2438519836492563, 'B_2_7': -0.47734098802815167, 'B_2_8': -0.511890726504286,
'B_2_9': 0.40546063041852665, 'B_3_0': -0.431298753029676, 'B_3_1':
-0.5620273127116265, 'B_3_2': -0.23150350401694042, 'B_3_3':
-0.030531767430197095, 'B_3_4': -0.5195833486568978, 'B_3_5':
0.1462413231897538, 'B_3_6': -0.09949902036709235, 'B_3_7': 0.31587777262939176,
'B_3_8': 0.439770016579603, 'B_3_9': -0.003508151806832042, 'B_4_0':
0.07015556184525334, 'B_4_1': 0.3449814186429015, 'B_4_2': 0.3810599620112558,
'B_4_3': -0.30772719610813815, 'B_4_4': -0.6697435330112853, 'B_4_5':
0.5066230459480799, 'B_4_6': -0.7639304706358112, 'B_4_7': -0.2885564923260362,
'B_4_8': 0.5783111144503592, 'B_4_9': 0.8307495153265131, 'B_5_0':
0.17062343672532354, 'B_5_1': -0.5285107593851309, 'B_5_2': 0.4899674545947196,
'B_5_3': -0.1721639093090652, 'B_5_4': 0.6189633428068857, 'B_5_5':
-0.7913585419836661, 'B_5_6': -0.30213826711647923, 'B_5_7':
0.27769355304020454, 'B_5_8': -0.11423463273649999, 'B_5_9': 0.5984971113197205,
'B_6_0': 0.19830031841895668, 'B_6_1': -0.6730406422002719, 'B_6_2':
0.8331220403314009, 'B_6_3': 0.9119379077681921, 'B_6_4': -0.7565873122696181,
'B_6_5': 0.03395350079235768, 'B_6_6': 0.2735466571916398, 'B_6_7':
0.56326644566088, 'B_6_8': 0.6115656315043735, 'B_6_9': 0.7499655950739339,
'B_7_0': -0.893935178943858, 'B_7_1': -0.41748706013369297, 'B_7_2':
-0.6403975586040725, 'B_7_3': 0.7455523713830029, 'B_7_4': 0.7744520168382861,
'B_7_5': -0.4854955414088128, 'B_7_6': -0.4621441848687789, 'B_7_7':
-0.8651831693389223, 'B_7_8': -0.9530072847965756, 'B_7_9': 0.6857285609029894,
'B_8_0': -0.8362119690217361, 'B_8_1': 0.04710614208061832, 'B_8_2':
0.633610055483717, 'B_8_3': 0.26213362259767486, 'B_8_4': -0.3001747794534659,
'B_8_5': -0.5982961213994877, 'B_8_6': 0.3761685648184234, 'B_8_7':
-0.8358532668091868, 'B_8_8': 0.05080897698273095, 'B_8_9': 0.8328638687526484,
'B_9_0': 0.2954705542634679, 'B_9_1': -0.8053645125266882, 'B_9_2':
-0.715540879643801, 'B_9_3': 0.7281393830516848, 'B_9_4': -0.6797364013056445,
'B_9_5': -0.6764130067237719, 'B_9_6': -0.2473313192280511, 'B_9_7':
-0.2734583047411969, 'B_9_8': 0.6887222849335594, 'B_9_9': 0.7788460172201083,
'B_10_0': 0.8986103857298571, 'B_10_1': -0.7722801381159624, 'B_10_2':

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-0.943131479682588, 'B_10_3': -0.5390970590211326, 'B_10_4': 0.6214358345712911, 'B_10_5': -0.04694715390469448, 'B_10_6': -0.12847678902228127, 'B_10_7': 0.9506558772237641, 'B_10_8': -0.10169753911797373, 'B_10_9': 0.18136654633588475}. Best is trial 136 with value: 62.32599854647733.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:22:23,511] Trial 138 finished with value: 63.32991970489335 and parameters: {'B_0_0': 0.3979717337482112, 'B_0_1': 0.08977616059195749, 'B_0_2': 0.7855873513835077, 'B_0_3': 0.0741583451216026, 'B_0_4': -0.5135760429164419, 'B_0_5': 0.7432080798184877, 'B_0_6': 0.4949301621052795, 'B_0_7': 0.5663943990065818, 'B_0_8': 0.38376008554308627, 'B_0_9': 0.19444220058291156, 'B_1_0': 0.40410705800904734, 'B_1_1': -0.1462210789107532, 'B_1_2': 0.3148492875070382, 'B_1_3': 0.9323136869962054, 'B_1_4': -0.1689736334485934, 'B_1_5': 0.878117778625279, 'B_1_6': -0.20576569884961923, 'B_1_7': 0.8219604263771648, 'B_1_8': -0.6026913171647772, 'B_1_9': -0.43406417884468834, 'B_2_0': -0.25046091559359895, 'B_2_1': -0.22648501118483663, 'B_2_2': -0.5308531819734874, 'B_2_3': 0.8934970499722422, 'B_2_4': -0.3575935213601718, 'B_2_5': -0.6366845595144826, 'B_2_6': -0.28097041812234935, 'B_2_7': -0.5944555075488465, 'B_2_8': -0.5152382651012716, 'B_2_9': 0.39205548981606947, 'B_3_0': -0.5858761064922939, 'B_3_1': -0.2540653551258387, 'B_3_2': -0.22581458092384665, 'B_3_3': 0.0016093887335962287, 'B_3_4': -0.5110551990260024, 'B_3_5': 0.14954960640365997, 'B_3_6': -0.15235171974883865, 'B_3_7': 0.31881828046243943, 'B_3_8': 0.42587773466469836, 'B_3_9': -0.010812620469543765, 'B_4_0': 0.06979563060709601, 'B_4_1': 0.3491490198485214, 'B_4_2': 0.3742148617791638, 'B_4_3': -0.3195681102850155, 'B_4_4': -0.6986997451391629, 'B_4_5': 0.3965150250871039, 'B_4_6': -0.6082376769658363, 'B_4_7': -0.2757744890833084, 'B_4_8': 0.5489069971315071, 'B_4_9': 0.8030573480792886, 'B_5_0': 0.16505595022097946, 'B_5_1': -0.5100820053808911, 'B_5_2': 0.4943452723154668, 'B_5_3': 0.07054458463133313, 'B_5_4': 0.6322604383356462, 'B_5_5': -0.782733665000695, 'B_5_6': -0.31812414400678096, 'B_5_7': 0.27908220868671246, 'B_5_8': -0.11529979952936545, 'B_5_9': 0.598573905928217, 'B_6_0': -0.045545015045575954, 'B_6_1': -0.6708069779140785, 'B_6_2': 0.8310478355444925, 'B_6_3': 0.9256961842951333, 'B_6_4': -0.7398309253220413, 'B_6_5': 0.04997428740867594, 'B_6_6': 0.27865602392029215, 'B_6_7':

0.576822600034783, 'B_6_8': 0.6130760926225391, 'B_6_9': 0.7495677615472707,
 'B_7_0': -0.8899525175417113, 'B_7_1': -0.41549799692926137, 'B_7_2':
 -0.47982545056879267, 'B_7_3': 0.7253938154857097, 'B_7_4': 0.9818238491878399,
 'B_7_5': -0.4891621104342736, 'B_7_6': -0.4667885235715572, 'B_7_7':
 -0.8677506963128664, 'B_7_8': -0.9587713792383024, 'B_7_9': 0.7005311927752906,
 'B_8_0': -0.68175939915688, 'B_8_1': 0.03522420565959587, 'B_8_2':
 0.8558477306563816, 'B_8_3': 0.2552230230829574, 'B_8_4': -0.2810768583670075,
 'B_8_5': -0.5873213920796818, 'B_8_6': 0.4113413269031546, 'B_8_7':
 -0.8340234739038217, 'B_8_8': -0.029031638272006083, 'B_8_9':
 0.8405884489598139, 'B_9_0': 0.305793118285894, 'B_9_1': -0.8051577362187053,
 'B_9_2': -0.7191618158536744, 'B_9_3': 0.755474192404917, 'B_9_4':
 -0.6852250511281406, 'B_9_5': -0.6135775915557129, 'B_9_6':
 -0.24422962714570273, 'B_9_7': -0.10326567177566029, 'B_9_8':
 0.6921090816832021, 'B_9_9': 0.7835882984219407, 'B_10_0': 0.8975387758548977,
 'B_10_1': -0.8213832238503616, 'B_10_2': -0.9474048117699587, 'B_10_3':
 -0.5140756208996402, 'B_10_4': 0.6265005558016403, 'B_10_5':
 -0.30394781572753843, 'B_10_6': -0.12259877494033454, 'B_10_7':
 0.9469100282676638, 'B_10_8': -0.08336028285886013, 'B_10_9':
 0.2052076899568438}. Best is trial 136 with value: 62.32599854647733.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:22:38,586] Trial 139 finished with value: 78.998619206023 and
 parameters: {'B_0_0': 0.38007797493666157, 'B_0_1': 0.07440040628585097,
 'B_0_2': 0.6918868366543023, 'B_0_3': 0.10148564760621098, 'B_0_4':
 -0.5154737380650338, 'B_0_5': 0.7347401003650041, 'B_0_6': 0.4897998369236905,
 'B_0_7': 0.6735565920468086, 'B_0_8': 0.4240360806031803, 'B_0_9':
 0.29310377466503, 'B_1_0': 0.40230907299705676, 'B_1_1': 0.12896079653134707,
 'B_1_2': 0.31491297761223963, 'B_1_3': 0.941545129127239, 'B_1_4':
 -0.1744987471856685, 'B_1_5': 0.8665881261798762, 'B_1_6': -0.2471050999511048,
 'B_1_7': 0.8113637596957696, 'B_1_8': -0.7057325767585408, 'B_1_9':
 -0.48113220690699077, 'B_2_0': -0.250786995484079, 'B_2_1':
 -0.22025996283016455, 'B_2_2': -0.553822048520333, 'B_2_3': 0.91051405085727,
 'B_2_4': -0.3577100317690044, 'B_2_5': -0.6323001191213113, 'B_2_6':
 -0.25175507544910136, 'B_2_7': -0.48063062873129353, 'B_2_8':
 -0.515716721989827, 'B_2_9': 0.41079005546264025, 'B_3_0': -0.4706526303301225,

```

'B_3_1': -0.27120798348695785, 'B_3_2': -0.2599134145122552, 'B_3_3':
-0.010838153759420974, 'B_3_4': -0.5211729230310704, 'B_3_5':
0.15331068413285323, 'B_3_6': -0.15547133577069805, 'B_3_7':
0.35268791658788723, 'B_3_8': 0.4276562277731211, 'B_3_9':
-0.024249078135364168, 'B_4_0': 0.05965884947271147, 'B_4_1':
0.33536926175825477, 'B_4_2': 0.3813487701114242, 'B_4_3': -0.3077230113426716,
'B_4_4': -0.6746022662025752, 'B_4_5': 0.39576786865694225, 'B_4_6':
-0.7524418919807822, 'B_4_7': -0.28286040339357654, 'B_4_8': 0.5640942974092351,
'B_4_9': 0.7973167964916582, 'B_5_0': 0.16976687731047255, 'B_5_1':
-0.5169284535864914, 'B_5_2': 0.5172385993148073, 'B_5_3': 0.056442769552015354,
'B_5_4': 0.626083849833103, 'B_5_5': -0.7759452592380708, 'B_5_6':
-0.29757697086985246, 'B_5_7': 0.2868015254117084, 'B_5_8':
-0.10146662332660669, 'B_5_9': 0.5877847819417076, 'B_6_0': -0.0472564745709588,
'B_6_1': -0.6542978547588856, 'B_6_2': 0.8300516649827302, 'B_6_3':
0.9202940524521931, 'B_6_4': -0.7531785331476787, 'B_6_5': 0.04459249929171087,
'B_6_6': 0.13889871451634125, 'B_6_7': 0.5040080830406856, 'B_6_8':
0.5965599366445865, 'B_6_9': 0.7398451224793082, 'B_7_0': -0.8814888737280735,
'B_7_1': -0.37925866471507785, 'B_7_2': -0.4770143880063131, 'B_7_3':
0.7333405049881401, 'B_7_4': 0.621136029040681, 'B_7_5': -0.5620003484287949,
'B_7_6': -0.48868668640206936, 'B_7_7': -0.865239247357701, 'B_7_8':
-0.9172116747515326, 'B_7_9': 0.6903409244726515, 'B_8_0': -0.83359889289433,
'B_8_1': 0.02538386472083584, 'B_8_2': 0.8646682221050019, 'B_8_3':
0.2480806522364025, 'B_8_4': -0.27738231749143494, 'B_8_5': -0.6031339549216823,
'B_8_6': 0.41822975247164906, 'B_8_7': -0.8269135121026358, 'B_8_8':
-0.01787240433008469, 'B_8_9': 0.8478120164584121, 'B_9_0': 0.32388293407206215,
'B_9_1': -0.7991575236728832, 'B_9_2': -0.72154153791712, 'B_9_3':
0.7415005635620096, 'B_9_4': -0.6518679689821089, 'B_9_5': -0.6716856420871586,
'B_9_6': -0.32161247203462706, 'B_9_7': -0.12026978557401613, 'B_9_8':
0.7712900151888583, 'B_9_9': 0.7768787658015796, 'B_10_0': 0.8968867257277507,
'B_10_1': -0.49953227164967823, 'B_10_2': -0.9371551839080268, 'B_10_3':
-0.5347339698608604, 'B_10_4': 0.6308903432719656, 'B_10_5':
-0.3177595629310907, 'B_10_6': -0.12856480719809027, 'B_10_7':
0.9494387725620852, 'B_10_8': -0.08308198828954509, 'B_10_9':
0.119704736900597}. Best is trial 136 with value: 62.32599854647733.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:22:52,713] Trial 140 finished with value: 64.92745053980177 and parameters: {'B_0_0': 0.41737454185958056, 'B_0_1': 0.006459841142956987, 'B_0_2': 0.7803717725899959, 'B_0_3': -0.007536050205057737, 'B_0_4': -0.5775226505663148, 'B_0_5': 0.8702201539487147, 'B_0_6': 0.5397098355770118, 'B_0_7': 0.5703680870909433, 'B_0_8': 0.3792750455777527, 'B_0_9': 0.29599296506974954, 'B_1_0': 0.47492863526094714, 'B_1_1': -0.17027691824548288, 'B_1_2': 0.354779448221919, 'B_1_3': 0.6886303864500474, 'B_1_4': -0.40353217486689685, 'B_1_5': 0.8044952504733561, 'B_1_6': -0.21583645353704983, 'B_1_7': 0.7809465647997837, 'B_1_8': -0.5481753028470351, 'B_1_9': -0.5732576921840244, 'B_2_0': -0.6466205184062561, 'B_2_1': -0.3395958857072123, 'B_2_2': -0.5282365836687716, 'B_2_3': 0.8453365444639722, 'B_2_4': -0.16225545664027513, 'B_2_5': -0.6633357155890096, 'B_2_6': -0.35354386859846676, 'B_2_7': -0.4728178437032732, 'B_2_8': -0.5402162358111969, 'B_2_9': 0.34515262762006504, 'B_3_0': -0.42638860961791175, 'B_3_1': -0.48814616578504955, 'B_3_2': -0.2000065532213059, 'B_3_3': -0.04583117673169893, 'B_3_4': -0.6314312940425751, 'B_3_5': 0.20130202073928977, 'B_3_6': -0.08294921733733537, 'B_3_7': 0.31171935945528245, 'B_3_8': 0.5161693748352395, 'B_3_9': -0.13010607343937766, 'B_4_0': 0.09246126881447443, 'B_4_1': 0.26548248238259264, 'B_4_2': 0.15627023941524734, 'B_4_3': -0.33115475394748917, 'B_4_4': -0.7900705426214126, 'B_4_5': 0.3071363259553309, 'B_4_6': -0.5996485781118144, 'B_4_7': -0.37096512351498945, 'B_4_8': 0.5243846131384464, 'B_4_9': 0.7504187068398104, 'B_5_0': 0.19793014922345506, 'B_5_1': -0.6320516609536542, 'B_5_2': 0.4173037301382971, 'B_5_3': -0.17773483068888665, 'B_5_4': 0.6371398001940259, 'B_5_5': -0.9374221178795444, 'B_5_6': -0.16371794139130857, 'B_5_7': 0.2687957391280796, 'B_5_8': -0.13569723805219766, 'B_5_9': 0.4794411496342502, 'B_6_0': 0.23497569991978434, 'B_6_1': -0.6761939093405502, 'B_6_2': 0.757189266272686, 'B_6_3': 0.9836741315183608, 'B_6_4': -0.6342787724219121, 'B_6_5': -0.07516233870364897, 'B_6_6': 0.21818932729791013, 'B_6_7': 0.581867004444624, 'B_6_8': 0.5667816036217382, 'B_6_9': 0.7697354050746238, 'B_7_0': -0.9029010997264669, 'B_7_1': -0.4289667475285624, 'B_7_2': -0.5750231425297194, 'B_7_3': 0.7891510421632283, 'B_7_4': 0.7645637246858432, 'B_7_5': -0.47856506152575035, 'B_7_6': -0.2934379509078992, 'B_7_7': -0.8611653133700771, 'B_7_8': -0.9507148055912635, 'B_7_9': 0.7939362774885905, 'B_8_0': -0.6765789313321234, 'B_8_1': 0.26594422001033746, 'B_8_2': 0.7367776777870352, 'B_8_3': 0.2586512619835163, 'B_8_4': -0.6043261083876124, 'B_8_5': -0.45477258451189584, 'B_8_6': 0.37029021262149453, 'B_8_7': -0.6955250977315734, 'B_8_8': 0.04517584518717662, 'B_8_9': 0.7560295586943946, 'B_9_0': 0.1732814201214728, 'B_9_1': -0.7526685677930611, 'B_9_2': -0.779958177372061, 'B_9_3': 0.7499720298287841, 'B_9_4': -0.6862518952560941, 'B_9_5': -0.6313216100376928, 'B_9_6': -0.23984374778759263, 'B_9_7': -0.2714546288194761, 'B_9_8': 0.6979195707205001, 'B_9_9': 0.6791351077327501, 'B_10_0': 0.7478706601062552, 'B_10_1': -0.59068259724983, 'B_10_2': -0.9643261514349416, 'B_10_3': -0.5840108465217614, 'B_10_4': 0.6824686943694007, 'B_10_5': -0.3971663284150243, 'B_10_6': -0.2932198784151529, 'B_10_7': 0.7608991127408574, 'B_10_8': -0.03664504406865465, 'B_10_9':

0.21291568605274214}. Best is trial 136 with value: 62.32599854647733.

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:23:07,027] Trial 141 finished with value: 62.74735078499448 and parameters: {'B_0_0': 0.43303868038766347, 'B_0_1': 0.19649024818369096,

'B_0_2': 0.7861781029024525, 'B_0_3': -0.003701534075261831, 'B_0_4': -0.5719841241160507, 'B_0_5': 0.8710533301402348, 'B_0_6': 0.43744065815032895, 'B_0_7': 0.5747762704146987, 'B_0_8': 0.3847847699839067, 'B_0_9': 0.2026344875988066, 'B_1_0': 0.4963918542808438, 'B_1_1': -0.14414796111480438, 'B_1_2': 0.4471893336500869, 'B_1_3': 0.6802638999875235, 'B_1_4': -0.39165306460593413, 'B_1_5': 0.8078120108631099, 'B_1_6': -0.1961926038926923, 'B_1_7': 0.77808886556236, 'B_1_8': -0.5909343711596865, 'B_1_9': -0.44474152115046206, 'B_2_0': -0.6435816591726765, 'B_2_1': -0.3461457725279996, 'B_2_2': -0.5478445324493849, 'B_2_3': 0.8634488788868308, 'B_2_4': -0.2088401119461128, 'B_2_5': -0.6765678863029794, 'B_2_6': -0.3422345773311492, 'B_2_7': -0.5191099415386597, 'B_2_8': -0.609642894160788, 'B_2_9': 0.3604857352994994, 'B_3_0': -0.424241944337569, 'B_3_1': -0.5122467497302364, 'B_3_2': -0.1919299371943046, 'B_3_3': -0.044379440113644586, 'B_3_4': -0.6300360530880547, 'B_3_5': 0.20251177514971686, 'B_3_6': -0.059493803287702596, 'B_3_7': 0.3049984475953749, 'B_3_8': 0.381496071610253, 'B_3_9': -0.1117208816900298, 'B_4_0': 0.07159454169521519, 'B_4_1': 0.37998933124027867, 'B_4_2': 0.16229539650399458, 'B_4_3': -0.3259283492647411, 'B_4_4': -0.8047884903949477, 'B_4_5': 0.32369435346350395, 'B_4_6': -0.6242309053618685, 'B_4_7': -0.3557269075562344, 'B_4_8': 0.5074915184388625, 'B_4_9': 0.8283888869220268, 'B_5_0': 0.28560116498443733, 'B_5_1': -0.5809815439466522, 'B_5_2': 0.5836078027915802, 'B_5_3': -0.16423566993937042, 'B_5_4': 0.627411999386739, 'B_5_5': -0.9769373626644611, 'B_5_6': -0.31551039099803646, 'B_5_7': 0.3275063086845232, 'B_5_8': -0.2518878674161894, 'B_5_9': 0.48821064517891777, 'B_6_0': 0.19848680978913497, 'B_6_1': -0.629542823314496, 'B_6_2': 0.7582171321020035, 'B_6_3': 0.9667646675435531, 'B_6_4': -0.5345362882851454, 'B_6_5': 0.01281250060974709, 'B_6_6': 0.2891045056116022, 'B_6_7': 0.5666395294684075, 'B_6_8': 0.561166615319691, 'B_6_9': 0.6343248657102771, 'B_7_0': -0.9093891462849283, 'B_7_1': -0.4272404022902058, 'B_7_2': -0.5676398337588564, 'B_7_3': 0.7885733192107051, 'B_7_4': 0.9837058199548311,

```
'B_7_5': -0.4841193013593343, 'B_7_6': -0.23714369368112237, 'B_7_7':
-0.8731723498514671, 'B_7_8': -0.970304266965861, 'B_7_9': 0.7664123830510992,
'B_8_0': -0.6895135449604853, 'B_8_1': 0.16751902694034101, 'B_8_2':
0.9807785589458391, 'B_8_3': 0.22139363056452793, 'B_8_4': -0.11205290624306741,
'B_8_5': -0.4531670519110488, 'B_8_6': 0.37763884994861835, 'B_8_7':
-0.8488514087538447, 'B_8_8': 0.04968525581659536, 'B_8_9': 0.7529577808728748,
'B_9_0': 0.16878013008728154, 'B_9_1': -0.8241438303236666, 'B_9_2':
-0.7204874210580655, 'B_9_3': 0.7345430583947058, 'B_9_4': -0.6979210752004591,
'B_9_5': -0.7110040163883022, 'B_9_6': -0.2255742213223466, 'B_9_7':
-0.2717405073510151, 'B_9_8': 0.6861635784738823, 'B_9_9': 0.8250977273682515,
'B_10_0': 0.7580206600023519, 'B_10_1': -0.8242587909477235, 'B_10_2':
-0.9578933403512019, 'B_10_3': -0.568979239546701, 'B_10_4': 0.680072355894192,
'B_10_5': -0.3917285964607019, 'B_10_6': -0.29859212250827644, 'B_10_7':
0.7733417844644821, 'B_10_8': -0.03402796675156339, 'B_10_9':
0.18105599468667327}. Best is trial 136 with value: 62.32599854647733.
```

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```
[I 2025-04-29 14:23:21,360] Trial 142 finished with value: 60.4596754287619 and
parameters: {'B_0_0': 0.41719631471379115, 'B_0_1': 0.22538982048502626,
'B_0_2': 0.7961635002842358, 'B_0_3': -0.009671673932239978, 'B_0_4':
-0.5966012465073072, 'B_0_5': 0.8340902130603529, 'B_0_6': 0.537604300122936,
'B_0_7': 0.5765034677002815, 'B_0_8': 0.3691735050411238, 'B_0_9':
0.20222727413406985, 'B_1_0': 0.5013102595525208, 'B_1_1': -0.1493355093215659,
'B_1_2': 0.3987909657301586, 'B_1_3': 0.6949521717753904, 'B_1_4':
-0.4028851410266008, 'B_1_5': 0.8082549484761317, 'B_1_6': -0.21215605824613645,
'B_1_7': 0.7277981430446621, 'B_1_8': -0.5808175445814052, 'B_1_9':
-0.5691493879441503, 'B_2_0': -0.44538916198766887, 'B_2_1':
-0.3483426440581294, 'B_2_2': -0.5338939084340018, 'B_2_3': 0.8795843313381775,
'B_2_4': -0.1902429888273105, 'B_2_5': -0.6928224381374094, 'B_2_6':
-0.3623483701095994, 'B_2_7': -0.46583621644313516, 'B_2_8':
-0.6144997734176323, 'B_2_9': 0.37406844887296314, 'B_3_0': -0.4268999196886553,
'B_3_1': -0.4931184000552226, 'B_3_2': -0.20173127672942925, 'B_3_3':
-0.03317552607303255, 'B_3_4': -0.638302231965514, 'B_3_5': 0.19144454668133942,
'B_3_6': -0.08266069325529785, 'B_3_7': 0.31248794785534717, 'B_3_8':
0.37164765523163046, 'B_3_9': -0.12774061290852456, 'B_4_0':
```

0.09036945862404956, 'B_4_1': 0.37260829012984925, 'B_4_2': 0.16377828316701418, 'B_4_3': -0.3080390374069145, 'B_4_4': -0.7912748036860877, 'B_4_5': 0.352474712258169, 'B_4_6': -0.6233850265516732, 'B_4_7': -0.3695324149919509, 'B_4_8': 0.533768863406391, 'B_4_9': 0.8410265393515967, 'B_5_0': 0.3040622348222358, 'B_5_1': -0.5835053986173258, 'B_5_2': 0.4872863447568073, 'B_5_3': -0.1754186266153498, 'B_5_4': 0.6289946306040639, 'B_5_5': -0.9282488037172189, 'B_5_6': -0.14047056892185392, 'B_5_7': 0.3349107914061987, 'B_5_8': -0.24667934906870248, 'B_5_9': 0.4760379267890854, 'B_6_0': 0.034450130954316, 'B_6_1': -0.6089550295051342, 'B_6_2': 0.6524371416319231, 'B_6_3': 0.9809066837942296, 'B_6_4': -0.5323451457569877, 'B_6_5': -0.07527933741394471, 'B_6_6': 0.2722832779955919, 'B_6_7': 0.5684169537083287, 'B_6_8': 0.541602840376644, 'B_6_9': 0.665873310183323, 'B_7_0': -0.9168224520265315, 'B_7_1': -0.441221435556911, 'B_7_2': -0.4065913716209403, 'B_7_3': 0.7514048251169634, 'B_7_4': 0.9563061586161652, 'B_7_5': -0.48315837288949504, 'B_7_6': -0.2786398513428009, 'B_7_7': -0.8547423601666041, 'B_7_8': -0.9525203684651449, 'B_7_9': 0.7775197346970674, 'B_8_0': -0.6848689999918061, 'B_8_1': 0.2635124545540819, 'B_8_2': 0.8323777887713076, 'B_8_3': 0.2271774977157615, 'B_8_4': -0.22830269196814118, 'B_8_5': -0.4488192943916902, 'B_8_6': 0.5498755937898887, 'B_8_7': -0.7062619227387968, 'B_8_8': 0.03713471490200947, 'B_8_9': 0.7500832434320208, 'B_9_0': 0.17324240445152161, 'B_9_1': -0.7480366063122531, 'B_9_2': -0.7771474758517197, 'B_9_3': 0.7340097637712104, 'B_9_4': -0.6492191596303172, 'B_9_5': -0.6384037264501152, 'B_9_6': -0.24642404821066488, 'B_9_7': -0.27818526381869857, 'B_9_8': 0.6839787587127104, 'B_9_9': 0.6647331347077772, 'B_10_0': 0.7224918553341257, 'B_10_1': -0.587996382648655, 'B_10_2': -0.9775296743390158, 'B_10_3': -0.5692950538465607, 'B_10_4': 0.6886836746738318, 'B_10_5': -0.3914383338237261, 'B_10_6': -0.3070286377276603, 'B_10_7': 0.7803117694262548, 'B_10_8': -0.028726639579389243, 'B_10_9': 0.17071507187447776}. Best is trial 142 with value: 60.4596754287619.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:23:35,087] Trial 143 finished with value: 59.53391095383986 and parameters: {'B_0_0': 0.4310971112493863, 'B_0_1': 0.19771612763103835, 'B_0_2': 0.6019601882175792, 'B_0_3': -0.002818288164257235, 'B_0_4':

-0.5771992605389271, 'B_0_5': 0.8763291627565947, 'B_0_6': 0.5268423961739954,
 'B_0_7': 0.5753460067166842, 'B_0_8': 0.3914579275716875, 'B_0_9':
 0.21938267284830135, 'B_1_0': 0.48932614835283733, 'B_1_1':
 -0.15991857174111426, 'B_1_2': 0.3684293419432263, 'B_1_3': 0.6885371481371088,
 'B_1_4': -0.40640963085694426, 'B_1_5': 0.9149271777925738, 'B_1_6':
 -0.2040194295273056, 'B_1_7': 0.7545212337001828, 'B_1_8': -0.5949722207238896,
 'B_1_9': -0.5721927025706823, 'B_2_0': -0.46226307489608065, 'B_2_1':
 -0.4067873989494229, 'B_2_2': -0.5401628278452499, 'B_2_3': 0.8740574367181306,
 'B_2_4': -0.17978740821210185, 'B_2_5': -0.6853585963745499, 'B_2_6':
 -0.37098512344168116, 'B_2_7': -0.4875105922738157, 'B_2_8': -0.532591775469308,
 'B_2_9': 0.3666102366755449, 'B_3_0': -0.42521122553347945, 'B_3_1':
 -0.48119927123261963, 'B_3_2': -0.19741367799517473, 'B_3_3':
 -0.03896957591479161, 'B_3_4': -0.5333445730708659, 'B_3_5':
 0.20970463008474557, 'B_3_6': -0.08341767694011863, 'B_3_7': 0.3082140469337334,
 'B_3_8': 0.3560168643301785, 'B_3_9': -0.12932585508850053, 'B_4_0':
 0.08608284244974154, 'B_4_1': 0.2680911723161071, 'B_4_2': 0.16497536045548145,
 'B_4_3': -0.2502217240284264, 'B_4_4': -0.7916460875036644, 'B_4_5':
 0.26279927096478534, 'B_4_6': -0.6049441737259114, 'B_4_7':
 -0.36779335622487896, 'B_4_8': 0.5117949296417631, 'B_4_9': 0.8418488122374606,
 'B_5_0': 0.31281889933168494, 'B_5_1': -0.6478949744298038, 'B_5_2':
 0.5804009608424141, 'B_5_3': -0.18678178838491374, 'B_5_4': 0.6338703910513033,
 'B_5_5': -0.9288872961724752, 'B_5_6': -0.15400927443044196, 'B_5_7':
 0.30823864474432816, 'B_5_8': -0.2512896849950336, 'B_5_9': 0.4658829216424026,
 'B_6_0': 0.16678421787109482, 'B_6_1': -0.6221813391557912, 'B_6_2':
 0.6665643427969229, 'B_6_3': 0.9748320365829638, 'B_6_4': -0.5330977776432174,
 'B_6_5': -0.06291903900163953, 'B_6_6': 0.30595155761153453, 'B_6_7':
 0.5588304668484856, 'B_6_8': 0.5537566571965562, 'B_6_9': 0.6397418756288594,
 'B_7_0': -0.9014360369008323, 'B_7_1': -0.41806497874082726, 'B_7_2':
 -0.42433303085605123, 'B_7_3': 0.7233893064343595, 'B_7_4': 0.9792320974153196,
 'B_7_5': -0.4704558731551231, 'B_7_6': -0.29935004128531945, 'B_7_7':
 -0.8173219982162812, 'B_7_8': -0.9539452781888812, 'B_7_9': 0.7313642860772276,
 'B_8_0': -0.6993454106075822, 'B_8_1': 0.2900661845939228, 'B_8_2':
 0.836689490652534, 'B_8_3': 0.3370745290191435, 'B_8_4': -0.11512301346600395,
 'B_8_5': -0.29634802213117956, 'B_8_6': 0.5723135851213818, 'B_8_7':
 -0.8475763327649444, 'B_8_8': 0.05385075860973704, 'B_8_9': 0.7402746054961155,
 'B_9_0': 0.15679279490855252, 'B_9_1': -0.7566239942217374, 'B_9_2':
 -0.7867146626031953, 'B_9_3': 0.739647286001725, 'B_9_4': -0.6835264694369483,
 'B_9_5': -0.5903431024147782, 'B_9_6': -0.2504752706225048, 'B_9_7':
 -0.2972867311616121, 'B_9_8': 0.688164826587909, 'B_9_9': 0.8300584725669172,
 'B_10_0': 0.7354160009699342, 'B_10_1': -0.6015411555452169, 'B_10_2':
 -0.9745145215717858, 'B_10_3': -0.5628967756384685, 'B_10_4': 0.692435688491681,
 'B_10_5': -0.4001649498752357, 'B_10_6': -0.27911706765578037, 'B_10_7':
 0.7857751938340917, 'B_10_8': -0.037558386405723926, 'B_10_9':
 0.1881574503261135}. Best is trial 143 with value: 59.53391095383986.

Building the Basis functions for the RKHS:

Built F__[u₁].

Built F__[u₂].

Built F__[u₃].

```

Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```

[I 2025-04-29 14:23:49,057] Trial 144 finished with value: 61.09709469355234 and
parameters: {'B_0_0': 0.4204392681926363, 'B_0_1': 0.22311034747355918, 'B_0_2':
0.6155796103340316, 'B_0_3': -0.007314731735978172, 'B_0_4':
-0.5941463690224853, 'B_0_5': 0.880847589917908, 'B_0_6': 0.5331104155225243,
'B_0_7': 0.5526420705935986, 'B_0_8': 0.38563293845741425, 'B_0_9':
0.2095989299890877, 'B_1_0': 0.4913926185846133, 'B_1_1': -0.13304036759383997,
'B_1_2': 0.4076593156788393, 'B_1_3': 0.6825574828662134, 'B_1_4':
-0.4123518826105514, 'B_1_5': 0.9161844599503962, 'B_1_6': -0.22035768829085256,
'B_1_7': 0.7673340887729967, 'B_1_8': -0.5868908304701023, 'B_1_9':
-0.5765918628071736, 'B_2_0': -0.45032742332724546, 'B_2_1':
-0.4378631334663887, 'B_2_2': -0.5346008646329509, 'B_2_3': 0.8867308412005,
'B_2_4': -0.17152919979350167, 'B_2_5': -0.6770225801568053, 'B_2_6':
-0.26989300241022324, 'B_2_7': -0.4769085861022286, 'B_2_8':
-0.6101015912777195, 'B_2_9': 0.3660793246948781, 'B_3_0': -0.571105887629466,
'B_3_1': -0.4987257544979752, 'B_3_2': -0.21583918773748545, 'B_3_3':
-0.03846497515310846, 'B_3_4': -0.6511049232030497, 'B_3_5':
0.20402981289446298, 'B_3_6': -0.07785235711328391, 'B_3_7': 0.3081359226511935,
'B_3_8': 0.3861228597122498, 'B_3_9': -0.013980958175103739, 'B_4_0':
0.1462363618192211, 'B_4_1': 0.27076620453394884, 'B_4_2': 0.157804882070631,
'B_4_3': -0.2643897520615265, 'B_4_4': -0.8042038514850872, 'B_4_5':
0.3135660512032832, 'B_4_6': -0.612423887548158, 'B_4_7': -0.361177373645115,
'B_4_8': 0.49940262059195756, 'B_4_9': 0.8424359783173025, 'B_5_0':
0.28938652481980864, 'B_5_1': -0.6440903338510857, 'B_5_2': 0.5986692479397877,
'B_5_3': -0.15454529043439777, 'B_5_4': 0.6233029853716019, 'B_5_5':
-0.9674651481813288, 'B_5_6': -0.16890491464544946, 'B_5_7':
0.31164350794316037, 'B_5_8': -0.24490643750162758, 'B_5_9':
0.47139197697856017, 'B_6_0': 0.18498667277188818, 'B_6_1': -0.6221467336258452,
'B_6_2': 0.6580534630002495, 'B_6_3': 0.9862092727670335, 'B_6_4':
-0.5656834608535168, 'B_6_5': 0.02426331589635044, 'B_6_6': 0.28330196556798426,
'B_6_7': 0.5794006771126787, 'B_6_8': 0.54092387793569, 'B_6_9':
0.6086564496972161, 'B_7_0': -0.9064695472272543, 'B_7_1': -0.41088214356640634,
'B_7_2': -0.4136048747878618, 'B_7_3': 0.7154428067902363, 'B_7_4':
0.9808339478115128, 'B_7_5': -0.4742714390030775, 'B_7_6': -0.29462817705729777,
'B_7_7': -0.8324849964461621, 'B_7_8': -0.9545215560661935, 'B_7_9':
0.8062321784824845, 'B_8_0': -0.681051230491384, 'B_8_1': 0.2757551744708606,
'B_8_2': 0.9896329592934465, 'B_8_3': 0.3410369841056795, 'B_8_4':
-0.09168123470878523, 'B_8_5': -0.31946680951844425, 'B_8_6':

```

0.5720939962603231, 'B_8_7': -0.6995116696219064, 'B_8_8': 0.052133163587781056, 'B_8_9': 0.7476353866827743, 'B_9_0': 0.16315475948991162, 'B_9_1': -0.7464473641251864, 'B_9_2': -0.7734262258650695, 'B_9_3': 0.7477163315132989, 'B_9_4': -0.6686578896477487, 'B_9_5': -0.6114669206265714, 'B_9_6': -0.2477004932263129, 'B_9_7': -0.2720002435232355, 'B_9_8': 0.6878973518449468, 'B_9_9': 0.6601585193171208, 'B_10_0': 0.741824206884513, 'B_10_1': -0.5751422850119075, 'B_10_2': -0.9677214370274424, 'B_10_3': -0.5184344306478516, 'B_10_4': 0.673502959977893, 'B_10_5': -0.408471209536455, 'B_10_6': -0.28953907558161196, 'B_10_7': 0.7754107117825388, 'B_10_8': -0.03388454961461343, 'B_10_9': 0.20423311496838006}. Best is trial 143 with value: 59.53391095383986.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:24:03,341] Trial 145 finished with value: 62.57654606365661 and parameters: {'B_0_0': 0.4331773790441031, 'B_0_1': 0.19388698748504066, 'B_0_2': 0.585279621489409, 'B_0_3': 0.14185263948211885, 'B_0_4': -0.5957807751252513, 'B_0_5': 0.8805842217766398, 'B_0_6': 0.5764940235533137, 'B_0_7': 0.5623343424512194, 'B_0_8': 0.37736431040569646, 'B_0_9': 0.21161313964888262, 'B_1_0': 0.4950337245448062, 'B_1_1': -0.15092390460527907, 'B_1_2': 0.3917501748815056, 'B_1_3': 0.8463341634986685, 'B_1_4': -0.3991103292933148, 'B_1_5': 0.9190544453004305, 'B_1_6': -0.21293513615895943, 'B_1_7': 0.7601275832657894, 'B_1_8': -0.59199141058999, 'B_1_9': -0.5859140026924052, 'B_2_0': -0.667441469927359, 'B_2_1': -0.4526368049722295, 'B_2_2': -0.5291975937461663, 'B_2_3': 0.8708237121107323, 'B_2_4': -0.17054778963197292, 'B_2_5': -0.6893539243916075, 'B_2_6': -0.2679525227128396, 'B_2_7': -0.4634147567375841, 'B_2_8': -0.5850931102908286, 'B_2_9': 0.35073644561573797, 'B_3_0': -0.42360262432846996, 'B_3_1': -0.488698634163566, 'B_3_2': -0.21097191124547918, 'B_3_3': -0.044520357695384635, 'B_3_4': -0.5503928711494501, 'B_3_5': 0.18704413016966825, 'B_3_6': -0.07455017169052361, 'B_3_7': 0.31526806205026464, 'B_3_8': 0.37173799287610354, 'B_3_9': -0.021212949191727733, 'B_4_0': 0.24044407174281646, 'B_4_1': 0.363304119234994, 'B_4_2': 0.15001220303155263, 'B_4_3': -0.2587558762392231, 'B_4_4': -0.7919074936228392, 'B_4_5': 0.25430841500348017, 'B_4_6': -0.608724437086701, 'B_4_7': -0.34568762752541854, 'B_4_8': 0.44672955401251874, 'B_4_9': 0.8357677358356852, 'B_5_0':

0.30866563006927744, 'B_5_1': -0.5858959323127215, 'B_5_2': 0.6062333775330887, 'B_5_3': -0.16884692313121574, 'B_5_4': 0.6157071167471475, 'B_5_5': -0.9812689816143577, 'B_5_6': -0.1746810912632574, 'B_5_7': 0.30826358833481626, 'B_5_8': -0.2306595422722153, 'B_5_9': 0.46841267600416675, 'B_6_0': 0.2372037855640462, 'B_6_1': -0.6198284448591828, 'B_6_2': 0.6767825995746197, 'B_6_3': 0.9865884852188462, 'B_6_4': -0.5423609360463468, 'B_6_5': -0.0774628794122241, 'B_6_6': 0.28829949448499, 'B_6_7': 0.55906237048538, 'B_6_8': 0.543977743472389, 'B_6_9': 0.6362510895324894, 'B_7_0': -0.9066910075568764, 'B_7_1': -0.4232048475243624, 'B_7_2': -0.39323261949149757, 'B_7_3': 0.7717736948127906, 'B_7_4': 0.9759995061492664, 'B_7_5': -0.4796864554801755, 'B_7_6': -0.29322989142639105, 'B_7_7': -0.8388536008282088, 'B_7_8': -0.952977435386294, 'B_7_9': 0.8334577291319166, 'B_8_0': -0.6775341522319496, 'B_8_1': 0.2580550208213801, 'B_8_2': 0.9757566061762518, 'B_8_3': 0.31643075423179773, 'B_8_4': -0.10397621867268793, 'B_8_5': -0.30506487411627176, 'B_8_6': 0.5705980361175966, 'B_8_7': -0.7103744519357204, 'B_8_8': 0.042838228612840236, 'B_8_9': 0.7198920168355704, 'B_9_0': 0.17195281734638013, 'B_9_1': -0.7614430012840936, 'B_9_2': -0.7155246112054743, 'B_9_3': 0.7342004109247475, 'B_9_4': -0.6804164125196341, 'B_9_5': -0.598730700063854, 'B_9_6': -0.23084176207061835, 'B_9_7': -0.2639956384358285, 'B_9_8': 0.6871094079331637, 'B_9_9': 0.6852518343890305, 'B_10_0': 0.7208992751570642, 'B_10_1': -0.5942698543786253, 'B_10_2': -0.9883300866160645, 'B_10_3': -0.5182257841139287, 'B_10_4': 0.6928545933344317, 'B_10_5': -0.4089319304544453, 'B_10_6': -0.28905173255873867, 'B_10_7': 0.7652565578882706, 'B_10_8': -0.03842638273494261, 'B_10_9': 0.19922099578570812}. Best is trial 143 with value: 59.53391095383986.

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:24:17,330] Trial 146 finished with value: 61.01327381912604 and parameters: {'B_0_0': 0.4161630169030905, 'B_0_1': 0.19695714590237295, 'B_0_2': 0.5926948521361532, 'B_0_3': 0.14927363858801418, 'B_0_4': -0.5939060324916944, 'B_0_5': 0.8824148621751126, 'B_0_6': 0.5604440712503391, 'B_0_7': 0.5496735546356748, 'B_0_8': 0.3784239582175139, 'B_0_9': 0.215834655616883, 'B_1_0': 0.4998514129014174, 'B_1_1': -0.14743494319388004, 'B_1_2': 0.40622307327617013, 'B_1_3': 0.6835248696917866, 'B_1_4': -0.39851918838301925,

'B_1_5': 0.9258839434175661, 'B_1_6': -0.20432188633091602, 'B_1_7':
 0.6529978580265972, 'B_1_8': -0.5910554897925777, 'B_1_9': -0.5921602134582038,
 'B_2_0': -0.4678701325671801, 'B_2_1': -0.42119596787094277, 'B_2_2':
 -0.5320087951048832, 'B_2_3': 0.8842463858339359, 'B_2_4': -0.15623904502229308,
 'B_2_5': -0.6785130929881942, 'B_2_6': -0.2787476734679291, 'B_2_7':
 -0.46505264769277765, 'B_2_8': -0.598543377298511, 'B_2_9': 0.3634133136588968,
 'B_3_0': -0.4315246420287716, 'B_3_1': -0.3484666337075337, 'B_3_2':
 -0.20826172284398142, 'B_3_3': -0.039079628738294986, 'B_3_4':
 -0.6283195624243423, 'B_3_5': 0.2171666629106611, 'B_3_6': -0.0691618078891183,
 'B_3_7': 0.3142778203608189, 'B_3_8': 0.3825110260974863, 'B_3_9':
 -0.02557658798821436, 'B_4_0': 0.2404809901600674, 'B_4_1': 0.29057166966681,
 'B_4_2': 0.1640873131082754, 'B_4_3': -0.27292835191931286, 'B_4_4':
 -0.7979837829959593, 'B_4_5': 0.27815023668839356, 'B_4_6': -0.612464089705742,
 'B_4_7': -0.41241014643374685, 'B_4_8': 0.43659490608248785, 'B_4_9':
 0.8426102417918941, 'B_5_0': 0.2908886698732634, 'B_5_1': -0.6470618954652961,
 'B_5_2': 0.5893481273684029, 'B_5_3': -0.17789010731088573, 'B_5_4':
 0.6330139656334581, 'B_5_5': -0.9772473711356203, 'B_5_6': -0.15885593449927765,
 'B_5_7': 0.2977393191060855, 'B_5_8': -0.2486622155511967, 'B_5_9':
 0.46529169557569977, 'B_6_0': 0.23692123666256576, 'B_6_1': -0.6214687792739534,
 'B_6_2': 0.6591763423649799, 'B_6_3': 0.9754527269700067, 'B_6_4':
 -0.5347691645212133, 'B_6_5': 0.01731009892961849, 'B_6_6': 0.27245543999589383,
 'B_6_7': 0.5726996061464157, 'B_6_8': 0.5492179624363931, 'B_6_9':
 0.6129685364750828, 'B_7_0': -0.8663660332458432, 'B_7_1': -0.40699816269683375,
 'B_7_2': -0.39966301944763954, 'B_7_3': 0.710100203483509, 'B_7_4':
 0.9858171690074148, 'B_7_5': -0.4901261378913079, 'B_7_6': -0.2915337157957705,
 'B_7_7': -0.8328285484166618, 'B_7_8': -0.9554504055895756, 'B_7_9':
 0.8294388035044501, 'B_8_0': -0.6617441641437988, 'B_8_1': 0.2733233214701172,
 'B_8_2': 0.9723552193284483, 'B_8_3': 0.3495191249399583, 'B_8_4':
 -0.08791862321273938, 'B_8_5': -0.3054309682070648, 'B_8_6': 0.5545828969614919,
 'B_8_7': -0.7230999829870567, 'B_8_8': 0.04367804459466926, 'B_8_9':
 0.6923598474762915, 'B_9_0': 0.15148654348515891, 'B_9_1': -0.7478789044493221,
 'B_9_2': -0.7905714661710986, 'B_9_3': 0.7381916297288317, 'B_9_4':
 -0.6968325186918595, 'B_9_5': -0.6042739145217827, 'B_9_6':
 -0.23594866540182047, 'B_9_7': -0.28607274794561566, 'B_9_8':
 0.6935066278987438, 'B_9_9': 0.6818451913511906, 'B_10_0': 0.7416153879099977,
 'B_10_1': -0.6036885759311472, 'B_10_2': -0.9845993583788389, 'B_10_3':
 -0.5128628747473774, 'B_10_4': 0.6826107097895113, 'B_10_5': -0.417543113092969,
 'B_10_6': -0.27614222126447396, 'B_10_7': 0.7663513095029792, 'B_10_8':
 -0.015263039777039304, 'B_10_9': 0.194952583276849}. Best is trial 143 with
 value: 59.53391095383986.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].


```

    Built F_[u_8].
    Built F_[u_9].
    Built F_[u_10].
    Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:24:31,582] Trial 147 finished with value: 62.440850053141034
and parameters: {'B_0_0': 0.4295100891799306, 'B_0_1': 0.19280037534539807,
'B_0_2': 0.6444840625834142, 'B_0_3': 0.28514788640419664, 'B_0_4':
-0.5950978181671852, 'B_0_5': 0.89301421846004, 'B_0_6': 0.5629787312470715,
'B_0_7': 0.5481403516824483, 'B_0_8': 0.38531845271854115, 'B_0_9':
0.3035747160240551, 'B_1_0': 0.4894209875373551, 'B_1_1': -0.1513884452827949,
'B_1_2': 0.40224836845012296, 'B_1_3': 0.6837437320528006, 'B_1_4':
-0.5057053966108618, 'B_1_5': 0.9199625151208406, 'B_1_6': -0.21301448241753462,
'B_1_7': 0.6458461202352734, 'B_1_8': -0.5860808099073759, 'B_1_9':
-0.6807152810994336, 'B_2_0': -0.6620870088192733, 'B_2_1': -0.4499880309910517,
'B_2_2': -0.5345994572723181, 'B_2_3': 0.8824945584763413, 'B_2_4':
-0.17673513174829467, 'B_2_5': -0.6685462192898174, 'B_2_6':
-0.3561546210400723, 'B_2_7': -0.48260770425488025, 'B_2_8':
-0.6124396143177582, 'B_2_9': 0.35815376658751485, 'B_3_0':
-0.42101418454775974, 'B_3_1': -0.35320218507914414, 'B_3_2':
-0.1959129909348993, 'B_3_3': -0.04097166851217028, 'B_3_4':
-0.7352187661702956, 'B_3_5': 0.1967659122457615, 'B_3_6': -0.06766897673797022,
'B_3_7': 0.3127044292866906, 'B_3_8': 0.385649754762156, 'B_3_9':
0.005860147596012801, 'B_4_0': 0.3285114374269177, 'B_4_1': 0.26901585754015744,
'B_4_2': 0.14461469334998311, 'B_4_3': -0.24448881580125864, 'B_4_4':
-0.7983142658519673, 'B_4_5': 0.27827419692761657, 'B_4_6': -0.6030452993737376,
'B_4_7': -0.36450480170614163, 'B_4_8': 0.42359507134382635, 'B_4_9':
0.8397094100721857, 'B_5_0': 0.29505881453381305, 'B_5_1': -0.6041004530267851,
'B_5_2': 0.5937187660242981, 'B_5_3': -0.16659913032874213, 'B_5_4':
0.6263384247912124, 'B_5_5': -0.9858787021624846, 'B_5_6': -0.04221445371413057,
'B_5_7': 0.2935470079090481, 'B_5_8': -0.2491859376580124, 'B_5_9':
0.4613018907399618, 'B_6_0': 0.23561916509469558, 'B_6_1': -0.6203516926295105,
'B_6_2': 0.6624100179990097, 'B_6_3': 0.9891417315513444, 'B_6_4':
-0.5288296624635501, 'B_6_5': 0.04124028352416767, 'B_6_6': 0.28417454760587507,
'B_6_7': 0.5820545513129439, 'B_6_8': 0.5276594632433711, 'B_6_9':
0.654334985903723, 'B_7_0': -0.8558582717867702, 'B_7_1': -0.4074175530953102,
'B_7_2': -0.4044767204917319, 'B_7_3': 0.7103662507942471, 'B_7_4':
0.9843650755118366, 'B_7_5': -0.4767055706347322, 'B_7_6': -0.289088187260376,
'B_7_7': -0.8404434805347163, 'B_7_8': -0.9740212745103825, 'B_7_9':
0.8582079318015919, 'B_8_0': -0.6596575451972324, 'B_8_1': 0.2722346489854126,
'B_8_2': 0.9622367069073502, 'B_8_3': 0.3594203487539632, 'B_8_4':
-0.12210147048362285, 'B_8_5': -0.3041917269516581, 'B_8_6': 0.5551228302371047,
'B_8_7': -0.7107945495505761, 'B_8_8': 0.03928796302775947, 'B_8_9':
0.7455723298700858, 'B_9_0': 0.16053399030898324, 'B_9_1': -0.7564058874728237,
'B_9_2': -0.7923788381150116, 'B_9_3': 0.7530893363904266, 'B_9_4':
-0.688716718105131, 'B_9_5': -0.601846995244879, 'B_9_6': -0.23520540597325948,

```

'B_9_7': -0.2694795818596031, 'B_9_8': 0.7073446008942913, 'B_9_9': 0.6850954070308115, 'B_10_0': 0.7761151004244236, 'B_10_1': -0.5964166375234004, 'B_10_2': -0.9716703566165077, 'B_10_3': -0.5142964350621213, 'B_10_4': 0.687087895869956, 'B_10_5': -0.38456036542144134, 'B_10_6': -0.28580784031105083, 'B_10_7': 0.7787830091425532, 'B_10_8': -0.03371424007361296, 'B_10_9': 0.17139624243727752}. Best is trial 143 with value: 59.53391095383986.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:24:45,992] Trial 148 finished with value: 65.64484921791357 and parameters: {'B_0_0': 0.4285905650598457, 'B_0_1': 0.18664986996417304, 'B_0_2': 0.6083943341052219, 'B_0_3': 0.2967626635778554, 'B_0_4': -0.5652499973999956, 'B_0_5': 0.8897523292275149, 'B_0_6': 0.5616193792580173, 'B_0_7': 0.4601683666515799, 'B_0_8': 0.38393392740195137, 'B_0_9': 0.2059968120181542, 'B_1_0': 0.4994771579633771, 'B_1_1': -0.16453104915781733, 'B_1_2': 0.4429940825911013, 'B_1_3': 0.6809678221205743, 'B_1_4': -0.5023907642552786, 'B_1_5': 0.8934838770898874, 'B_1_6': -0.11737930729787853, 'B_1_7': 0.753443333410777, 'B_1_8': -0.5977178251727036, 'B_1_9': -0.6775382401841838, 'B_2_0': -0.6637782370808971, 'B_2_1': -0.4406313080407188, 'B_2_2': -0.6169650594393424, 'B_2_3': 0.9168663784908678, 'B_2_4': -0.14604517241265497, 'B_2_5': -0.6987070989546271, 'B_2_6': -0.3694164823979541, 'B_2_7': -0.46754727072802793, 'B_2_8': -0.5492964230077482, 'B_2_9': 0.3782275229746741, 'B_3_0': -0.4344601371123059, 'B_3_1': -0.2613472303882099, 'B_3_2': -0.2121106305942294, 'B_3_3': -0.039434137453382784, 'B_3_4': -0.6387981955111048, 'B_3_5': 0.21448755600440575, 'B_3_6': -0.2304259600886542, 'B_3_7': 0.317799952312478, 'B_3_8': 0.38402368233569406, 'B_3_9': -0.010206460295172266, 'B_4_0': 0.3188970288045231, 'B_4_1': 0.2613271989570383, 'B_4_2': 0.12787820150995322, 'B_4_3': -0.2521877623338467, 'B_4_4': -0.7979677504869698, 'B_4_5': 0.2713051494763976, 'B_4_6': -0.6034743946479, 'B_4_7': -0.3662804720060007, 'B_4_8': 0.41736229263490376, 'B_4_9': 0.8426594196498876, 'B_5_0': 0.30020218985124764, 'B_5_1': -0.591811455462526, 'B_5_2': 0.580341839492653, 'B_5_3': -0.17312385371337272, 'B_5_4': 0.624458859568759, 'B_5_5': -0.9690679228539618, 'B_5_6': -0.04631504625106779, 'B_5_7': 0.2649769563831922, 'B_5_8': -0.24941353838927655, 'B_5_9': 0.4542876179263928, 'B_6_0': 0.19665425049496135, 'B_6_1': -0.6104988577443013,

'B_6_2': 0.6463261683236662, 'B_6_3': 0.9714091836190466, 'B_6_4':
-0.5313340784647673, 'B_6_5': 0.014998464555590382, 'B_6_6':
0.27470820876116386, 'B_6_7': 0.5677723165282839, 'B_6_8': 0.5345730783464399,
'B_6_9': 0.6474943931070685, 'B_7_0': -0.8680524529174019, 'B_7_1':
-0.41295776809229207, 'B_7_2': -0.42213942154525136, 'B_7_3':
0.7067309057881053, 'B_7_4': 0.9891478354406664, 'B_7_5': -0.4806551685712364,
'B_7_6': -0.24656034536203747, 'B_7_7': -0.8404822010964895, 'B_7_8':
-0.9664395954582063, 'B_7_9': 0.8251961045382148, 'B_8_0': -0.6817011063224198,
'B_8_1': 0.27289185217313683, 'B_8_2': 0.9872924929001837, 'B_8_3':
0.35557238339308783, 'B_8_4': -0.1164010187398015, 'B_8_5': -0.3052861219063766,
'B_8_6': 0.5911094065664487, 'B_8_7': -0.6957267638785707, 'B_8_8':
-0.13619202257471358, 'B_8_9': 0.7438150788194483, 'B_9_0': 0.16325177018159928,
'B_9_1': -0.7610946442167955, 'B_9_2': -0.7860425255739596, 'B_9_3':
0.7303233298424517, 'B_9_4': -0.6954883850940207, 'B_9_5': -0.6159012546275277,
'B_9_6': -0.2414195345963698, 'B_9_7': -0.09395253560554012, 'B_9_8':
0.6964592645596508, 'B_9_9': 0.6836757010004065, 'B_10_0': 0.7400177283518177,
'B_10_1': -0.48190156377383636, 'B_10_2': -0.980869576304119, 'B_10_3':
-0.5117589637921394, 'B_10_4': 0.673551607255054, 'B_10_5':
-0.40524215618342563, 'B_10_6': -0.3082187814740517, 'B_10_7':
0.6558776909856681, 'B_10_8': -0.021737857006819794, 'B_10_9':
0.198120205744993}. Best is trial 143 with value: 59.53391095383986.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [<F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:25:00,456] Trial 149 finished with value: 62.91982781029871 and
parameters: {'B_0_0': 0.42620961490245646, 'B_0_1': 0.21736091523807502,
'B_0_2': 0.5942397928473209, 'B_0_3': 0.31737796599396184, 'B_0_4':
-0.5952232015951124, 'B_0_5': 0.8890387265616102, 'B_0_6': 0.5452749472191513,
'B_0_7': 0.4659923625047683, 'B_0_8': 0.3858275893074876, 'B_0_9':
0.21825787659672916, 'B_1_0': 0.49518263748756225, 'B_1_1':
-0.15388092257499994, 'B_1_2': 0.42708571499906245, 'B_1_3': 0.6899025685518313,
'B_1_4': -0.500974584868011, 'B_1_5': 0.9252687348597205, 'B_1_6':
-0.11391069162262575, 'B_1_7': 0.6629897527148693, 'B_1_8': -0.5806679353087347,
'B_1_9': -0.6805859387219376, 'B_2_0': -0.6687972906330228, 'B_2_1':
-0.44227862316613353, 'B_2_2': -0.6160449987594935, 'B_2_3': 0.8954657699704799,
'B_2_4': -0.04614311778570279, 'B_2_5': -0.671918280323435, 'B_2_6':

-0.27480805619921217, 'B_2_7': -0.47223049005852874, 'B_2_8':
 -0.5377277273656647, 'B_2_9': 0.36115995213573715, 'B_3_0': -0.4353709509476201,
 'B_3_1': -0.3605649134620593, 'B_3_2': -0.20678885843409703, 'B_3_3':
 -0.044054188465857416, 'B_3_4': -0.636915185030684, 'B_3_5':
 0.20860646915135397, 'B_3_6': -0.19086424617087028, 'B_3_7': 0.3099067044226751,
 'B_3_8': 0.37889107506965475, 'B_3_9': -0.01652505328857279, 'B_4_0':
 0.33080574885049746, 'B_4_1': 0.2551807960761676, 'B_4_2': 0.1370809265858464,
 'B_4_3': -0.2567689158766603, 'B_4_4': -0.7954494512736587, 'B_4_5':
 0.2803646602507767, 'B_4_6': -0.5609548274781392, 'B_4_7': -0.3752601073521553,
 'B_4_8': 0.4993577584012498, 'B_4_9': 0.8259001937176655, 'B_5_0':
 0.2988884060712494, 'B_5_1': -0.6471994534586789, 'B_5_2': 0.5921773258457748,
 'B_5_3': -0.16457242387479126, 'B_5_4': 0.6282367058342114, 'B_5_5':
 -0.9748685848453843, 'B_5_6': -0.040327313056415084, 'B_5_7':
 0.29420347943696834, 'B_5_8': -0.1263158729297178, 'B_5_9': 0.4616447719077048,
 'B_6_0': 0.2390259916394516, 'B_6_1': -0.6212099292246975, 'B_6_2':
 0.6517397515742156, 'B_6_3': 0.9885005968009798, 'B_6_4': -0.5532933051599391,
 'B_6_5': 0.0248646129402484, 'B_6_6': 0.2796292954979019, 'B_6_7':
 0.5686482187951217, 'B_6_8': 0.5213781231952009, 'B_6_9': 0.623442551175611,
 'B_7_0': -0.8987193944402148, 'B_7_1': -0.33401226327779965, 'B_7_2':
 -0.38430166166612467, 'B_7_3': 0.7295299120207158, 'B_7_4': 0.9862717351654301,
 'B_7_5': -0.5667294872411186, 'B_7_6': -0.28710060637505197, 'B_7_7':
 -0.8439654723458256, 'B_7_8': -0.9465217652269179, 'B_7_9': 0.8455695726561995,
 'B_8_0': -0.6930029188953375, 'B_8_1': 0.27207672644931447, 'B_8_2':
 0.9804657275869139, 'B_8_3': 0.3583922571937453, 'B_8_4': -0.09728703555084828,
 'B_8_5': -0.3081690650563949, 'B_8_6': 0.585909817958464, 'B_8_7':
 -0.7082066715179938, 'B_8_8': -0.12156742114672632, 'B_8_9': 0.6987100423218318,
 'B_9_0': 0.16730588067865035, 'B_9_1': -0.6734537525861161, 'B_9_2':
 -0.6998837183126736, 'B_9_3': 0.7240944906108855, 'B_9_4': -0.6915918251892242,
 'B_9_5': -0.6003269716954958, 'B_9_6': -0.23225724412654827, 'B_9_7':
 -0.08619484522793705, 'B_9_8': 0.700988114070581, 'B_9_9': 0.6733495386365351,
 'B_10_0': 0.7233387282447648, 'B_10_1': -0.48885851176353734, 'B_10_2':
 -0.9756187313373672, 'B_10_3': -0.5185004408868424, 'B_10_4':
 0.6915164702746067, 'B_10_5': -0.40759255946973866, 'B_10_6':
 -0.29600826458179147, 'B_10_7': 0.6817935474454213, 'B_10_8':
 -0.027261338666793873, 'B_10_9': 0.1879481712822178}. Best is trial 143 with
 value: 59.53391095383986.

Building the Basis functions for the RKHS:

Built F__[u₁].
 Built F__[u₂].
 Built F__[u₃].
 Built F__[u₄].
 Built F__[u₅].
 Built F__[u₆].
 Built F__[u₇].
 Built F__[u₈].
 Built F__[u₉].
 Built F__[u₁₀].
 Built F__[u₁₁].

Established the refined x grid with 1001 points.
 Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].
 Built matrix B = inverse(A).
 [I 2025-04-29 14:25:15,115] Trial 150 finished with value: 60.01812065467868 and
 parameters: {'B_0_0': 0.44109979307156183, 'B_0_1': 0.19561081871934766,
 'B_0_2': 0.5975647919163912, 'B_0_3': 0.23549346272051894, 'B_0_4':
 -0.5684160883021744, 'B_0_5': 0.8958897381675389, 'B_0_6': 0.5520979463987931,
 'B_0_7': 0.46973571163593053, 'B_0_8': 0.3805398948606301, 'B_0_9':
 0.21198381639995895, 'B_1_0': 0.4926485014362573, 'B_1_1': -0.14234910986762972,
 'B_1_2': 0.41012725715527615, 'B_1_3': 0.6745015503315497, 'B_1_4':
 -0.40664097594366677, 'B_1_5': 0.9228165884043937, 'B_1_6':
 -0.20238372124823065, 'B_1_7': 0.6481798894937221, 'B_1_8': -0.5909928646198892,
 'B_1_9': -0.666505265196937, 'B_2_0': -0.6626238554846344, 'B_2_1':
 -0.4545818700689123, 'B_2_2': -0.604159106836555, 'B_2_3': 0.8826895387385854,
 'B_2_4': -0.14076799270660673, 'B_2_5': -0.6640774836449201, 'B_2_6':
 -0.3621948660254496, 'B_2_7': -0.4629761236922132, 'B_2_8': -0.7423082709099439,
 'B_2_9': 0.37100179840382475, 'B_3_0': -0.41823679369722716, 'B_3_1':
 -0.3591490177442057, 'B_3_2': -0.20960561656774432, 'B_3_3':
 -0.043037287006998194, 'B_3_4': -0.6459723234560425, 'B_3_5':
 0.36061340407681974, 'B_3_6': -0.2245858823861459, 'B_3_7': 0.3182481786526116,
 'B_3_8': 0.3753318089370084, 'B_3_9': -0.017001855075416676, 'B_4_0':
 0.23410581084569812, 'B_4_1': 0.25890205446437786, 'B_4_2':
 -0.03011130524211869, 'B_4_3': -0.25930429138248584, 'B_4_4':
 -0.7683797811157852, 'B_4_5': 0.2894294895350552, 'B_4_6': -0.6097189643442354,
 'B_4_7': -0.376287537474507, 'B_4_8': 0.42746020841127813, 'B_4_9':
 0.8309960688187967, 'B_5_0': 0.30455481431138753, 'B_5_1': -0.5972649035027613,
 'B_5_2': 0.5850849142597805, 'B_5_3': -0.16545700723466866, 'B_5_4':
 0.7787507671610576, 'B_5_5': -0.9766399493133666, 'B_5_6': -0.05250953973233602,
 'B_5_7': 0.30395705372533893, 'B_5_8': -0.11785640940840172, 'B_5_9':
 0.4641956684700754, 'B_6_0': 0.25341473615202154, 'B_6_1': -0.6191585140220673,
 'B_6_2': 0.638773627636966, 'B_6_3': 0.9706485590492263, 'B_6_4':
 -0.5305207302384594, 'B_6_5': 0.02913773686004599, 'B_6_6': 0.284537163922181,
 'B_6_7': 0.5740616436978293, 'B_6_8': 0.5315873791394488, 'B_6_9':
 0.6400543112468362, 'B_7_0': -0.8611782742933378, 'B_7_1': -0.3336769575035244,
 'B_7_2': -0.40717057480545255, 'B_7_3': 0.7140662461077497, 'B_7_4':
 0.9862633866437539, 'B_7_5': -0.47823656285600824, 'B_7_6':
 -0.27973610896066436, 'B_7_7': -0.8429525916679281, 'B_7_8': -0.940125624419097,
 'B_7_9': 0.8501835935695765, 'B_8_0': -0.6540167726269476, 'B_8_1':
 0.27915540893118024, 'B_8_2': 0.986395087889283, 'B_8_3': 0.32842121838123445,
 'B_8_4': -0.07418208149045766, 'B_8_5': -0.3023212032390372, 'B_8_6':
 0.564591938795157, 'B_8_7': -0.708037984020446, 'B_8_8': -0.14343559925589192,
 'B_8_9': 0.7252219559409143, 'B_9_0': 0.16229081910568596, 'B_9_1':
 -0.6698946459308032, 'B_9_2': -0.7837234922285343, 'B_9_3': 0.7301447265917647,
 'B_9_4': -0.6961917596199215, 'B_9_5': -0.5878721176095237, 'B_9_6':
 -0.24025150562906325, 'B_9_7': -0.2654725014668132, 'B_9_8': 0.7080867060669113,
 'B_9_9': 0.6781324613911102, 'B_10_0': 0.7455166688109761, 'B_10_1':
 -0.4741095244249035, 'B_10_2': -0.9877102951193198, 'B_10_3':
 -0.4089888835276332, 'B_10_4': 0.6842623609637406, 'B_10_5':

```
-0.40359890509925916, 'B_10_6': -0.3129929976574487, 'B_10_7':
0.7653431932972503, 'B_10_8': -0.026504379830759508, 'B_10_9':
0.2202785575261365}. Best is trial 143 with value: 59.53391095383986.
```

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```
[I 2025-04-29 14:25:29,237] Trial 151 finished with value: 62.898347152558884
and parameters: {'B_0_0': 0.4301987312083521, 'B_0_1': 0.21403210917070023,
'B_0_2': 0.5931295533275498, 'B_0_3': 0.2719402741090595, 'B_0_4':
-0.5932716307663873, 'B_0_5': 0.891853690923213, 'B_0_6': 0.5612831613115673,
'B_0_7': 0.4619610749310271, 'B_0_8': 0.37898576902788683, 'B_0_9':
0.21392048271431385, 'B_1_0': 0.4909712484748201, 'B_1_1': -0.1510160070765334,
'B_1_2': 0.40837607298901624, 'B_1_3': 0.6811102851980737, 'B_1_4':
-0.508641712045713, 'B_1_5': 0.9205600259130817, 'B_1_6': -0.1222621656488185,
'B_1_7': 0.6643548746981752, 'B_1_8': -0.5925882177627884, 'B_1_9':
-0.675968205486235, 'B_2_0': -0.6702092183641651, 'B_2_1': -0.43995941537845906,
'B_2_2': -0.5379177655018563, 'B_2_3': 0.9013603675839571, 'B_2_4':
-0.1438064798735172, 'B_2_5': -0.6791747473157388, 'B_2_6': -0.2814398004589236,
'B_2_7': -0.46518736831247726, 'B_2_8': -0.7392154636290331, 'B_2_9':
0.36962312436539996, 'B_3_0': -0.41725041680878727, 'B_3_1':
-0.35507126657164617, 'B_3_2': -0.20528908302041296, 'B_3_3':
-0.046987875401003826, 'B_3_4': -0.6479347043239273, 'B_3_5':
0.3327627725208122, 'B_3_6': -0.2693939283001865, 'B_3_7': 0.3139241509756344,
'B_3_8': 0.3630381554709693, 'B_3_9': -0.01302049995763423, 'B_4_0':
0.24641571090020248, 'B_4_1': 0.2635837946160409, 'B_4_2': 0.15427851942674414,
'B_4_3': -0.2733475161562096, 'B_4_4': -0.8008493354849198, 'B_4_5':
0.28289268860462746, 'B_4_6': -0.5764584188489573, 'B_4_7': -0.3899296603429836,
'B_4_8': 0.4122197258937314, 'B_4_9': 0.8331343550861907, 'B_5_0':
0.28336796687732807, 'B_5_1': -0.6609854388489899, 'B_5_2': 0.6165775028763312,
'B_5_3': -0.16544519293949192, 'B_5_4': 0.7837888256354297, 'B_5_5':
-0.9782741411419503, 'B_5_6': -0.04079385824713652, 'B_5_7':
0.29484718844808566, 'B_5_8': -0.13542909207869552, 'B_5_9': 0.4671259582281512,
'B_6_0': 0.2432426679144383, 'B_6_1': -0.5497620425127694, 'B_6_2':
0.658710757328482, 'B_6_3': 0.9799533550736174, 'B_6_4': -0.5506258755644431,
'B_6_5': 0.025966291500329863, 'B_6_6': 0.28650251570103896, 'B_6_7':
0.5771097457594571, 'B_6_8': 0.5334173399847334, 'B_6_9': 0.6451573727583928,
```

'B_7_0': -0.8658150403443887, 'B_7_1': -0.32970051982522663, 'B_7_2':
-0.4107305631477924, 'B_7_3': 0.7017451878170545, 'B_7_4': 0.9859128190551008,
'B_7_5': -0.475324987235449, 'B_7_6': -0.26761609907610895, 'B_7_7':
-0.8288665738180473, 'B_7_8': -0.9535016522735235, 'B_7_9': 0.8464064996066085,
'B_8_0': -0.657764018229637, 'B_8_1': 0.26922431522632234, 'B_8_2':
0.9732241416614676, 'B_8_3': 0.34944544124744, 'B_8_4': -0.13350426981186764,
'B_8_5': -0.3024997580033713, 'B_8_6': 0.5599466584852882, 'B_8_7':
-0.7037423667681241, 'B_8_8': -0.1372986690264183, 'B_8_9': 0.7136547231195468,
'B_9_0': 0.1756562329750197, 'B_9_1': -0.6751234566069588, 'B_9_2':
-0.7885114657640212, 'B_9_3': 0.734119088628426, 'B_9_4': -0.6837549785557759,
'B_9_5': -0.6038023016399184, 'B_9_6': -0.2480923531838015, 'B_9_7':
-0.09691546594222802, 'B_9_8': 0.6993980850023158, 'B_9_9': 0.67901952400994,
'B_10_0': 0.7380034425357542, 'B_10_1': -0.47101849881304003, 'B_10_2':
-0.9886989116717122, 'B_10_3': -0.5278536420015041, 'B_10_4':
0.6827569836297448, 'B_10_5': -0.4175809299985884, 'B_10_6':
-0.31725543678188933, 'B_10_7': 0.6695041773916857, 'B_10_8':
-0.031405620970044276, 'B_10_9': 0.2034274155848041}. Best is trial 143 with
value: 59.53391095383986.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [<F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:25:43,385] Trial 152 finished with value: 57.68140852100222 and
parameters: {'B_0_0': 0.4982244639010941, 'B_0_1': 0.22463080015130724, 'B_0_2':
0.5399588734597648, 'B_0_3': 0.22678675076322338, 'B_0_4': -0.6114429725104985,
'B_0_5': 0.8831791418428536, 'B_0_6': 0.5840918813485786, 'B_0_7':
0.557299682713413, 'B_0_8': 0.37237488784428985, 'B_0_9': 0.299727142740192,
'B_1_0': 0.460920735212762, 'B_1_1': -0.14706134826863762, 'B_1_2':
0.3700387875771168, 'B_1_3': 0.6252292058725718, 'B_1_4': -0.40339953284137214,
'B_1_5': 0.9360781154412218, 'B_1_6': -0.20263396368477204, 'B_1_7':
0.6389290502294352, 'B_1_8': -0.8145828022215758, 'B_1_9': -0.5754348868076123,
'B_2_0': -0.675636019920226, 'B_2_1': -0.4720167727520454, 'B_2_2':
-0.5217121184023276, 'B_2_3': 0.8818680845089498, 'B_2_4':
-0.052790611466082674, 'B_2_5': -0.6626125179643751, 'B_2_6':
-0.2851752599561814, 'B_2_7': -0.505836247183069, 'B_2_8': -0.7298625782018479,
'B_2_9': 0.37174064545194335, 'B_3_0': -0.4341491761629455, 'B_3_1':
-0.3437272761838533, 'B_3_2': -0.2870514428202006, 'B_3_3':

-0.04956612978653208, 'B_3_4': -0.761792903717595, 'B_3_5': 0.35533003653282086, 'B_3_6': -0.28071838959048745, 'B_3_7': 0.24463003400702982, 'B_3_8': 0.28562262385395293, 'B_3_9': -0.07115200171626586, 'B_4_0': 0.2330910883357617, 'B_4_1': 0.3259137260388729, 'B_4_2': 0.14299315348229524, 'B_4_3': -0.28697984027068946, 'B_4_4': -0.8012915634231526, 'B_4_5': 0.2780395560267224, 'B_4_6': -0.6047370457171866, 'B_4_7': -0.45258854411907473, 'B_4_8': 0.49757115564378945, 'B_4_9': 0.9173751679374427, 'B_5_0': 0.2824696426026022, 'B_5_1': -0.6449659720726385, 'B_5_2': 0.6081796842747876, 'B_5_3': -0.09132878375136771, 'B_5_4': 0.7829075173103093, 'B_5_5': -0.9199831115144818, 'B_5_6': -0.13146567877436888, 'B_5_7': 0.2922159870983127, 'B_5_8': -0.12560974152561896, 'B_5_9': 0.465340138393002, 'B_6_0': 0.22293797288387976, 'B_6_1': -0.5318522175192546, 'B_6_2': 0.5926928950111376, 'B_6_3': 0.9896912559971204, 'B_6_4': -0.5555291212614639, 'B_6_5': 0.15916881273507166, 'B_6_6': 0.339921218048589, 'B_6_7': 0.5177564629042328, 'B_6_8': 0.41966182263106844, 'B_6_9': 0.6159708487080798, 'B_7_0': -0.9171157620780808, 'B_7_1': -0.33211208382131335, 'B_7_2': -0.37889920243614506, 'B_7_3': 0.7492619660945243, 'B_7_4': 0.9421180588544447, 'B_7_5': -0.5622551592201276, 'B_7_6': -0.3003145823627072, 'B_7_7': -0.7260396417407033, 'B_7_8': -0.9435332730907557, 'B_7_9': 0.8946709112705235, 'B_8_0': -0.6360784847532349, 'B_8_1': 0.3612730416637466, 'B_8_2': 0.9689814441983842, 'B_8_3': 0.27925942859941993, 'B_8_4': -0.09261023652894587, 'B_8_5': -0.20382862859129525, 'B_8_6': 0.5498500183925631, 'B_8_7': -0.6447951399987008, 'B_8_8': 0.04261071688142973, 'B_8_9': 0.6923403091651109, 'B_9_0': 0.051807716510639756, 'B_9_1': -0.6665003793173638, 'B_9_2': -0.722082230592477, 'B_9_3': 0.7325222577042807, 'B_9_4': -0.6951346878288828, 'B_9_5': -0.5967415441530682, 'B_9_6': -0.32303586200594747, 'B_9_7': -0.2749722026988166, 'B_9_8': 0.7374001377138228, 'B_9_9': 0.5096104686269958, 'B_10_0': 0.6061010094528017, 'B_10_1': -0.4450911576729124, 'B_10_2': -0.9879278670657397, 'B_10_3': -0.4277586306288834, 'B_10_4': 0.6861834141704644, 'B_10_5': -0.35774866178636144, 'B_10_6': -0.33351900605824214, 'B_10_7': 0.5588947197422492, 'B_10_8': 0.04163715463550817, 'B_10_9': 0.18068129272103142}. Best is trial 152 with value: 57.68140852100222.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:25:57,422] Trial 153 finished with value: 57.61808695399101 and parameters: {'B_0_0': 0.4912980349030566, 'B_0_1': 0.23218929568420993, 'B_0_2': 0.5694140926736888, 'B_0_3': 0.24833909619167244, 'B_0_4': -0.6067093944442503, 'B_0_5': 0.8939160214730858, 'B_0_6': 0.5797742221989455, 'B_0_7': 0.4858495833578533, 'B_0_8': 0.36747211306353733, 'B_0_9': 0.2992129807302228, 'B_1_0': 0.5906279851903464, 'B_1_1': -0.16385971014932868, 'B_1_2': 0.383128104786644, 'B_1_3': 0.6385278425048388, 'B_1_4': -0.39958954465527624, 'B_1_5': 0.9270240198170473, 'B_1_6': -0.20540700016660626, 'B_1_7': 0.6504738609434724, 'B_1_8': -0.5805885167325011, 'B_1_9': -0.5854855155480407, 'B_2_0': -0.7108633860703384, 'B_2_1': -0.47222093162086437, 'B_2_2': -0.5877226219186611, 'B_2_3': 0.8798438511439415, 'B_2_4': -0.056139574095826844, 'B_2_5': -0.6568418086970322, 'B_2_6': -0.20052328514853932, 'B_2_7': -0.5322847641731469, 'B_2_8': -0.65316516528388, 'B_2_9': 0.36864744187479365, 'B_3_0': -0.42181452103616585, 'B_3_1': -0.3568527299902216, 'B_3_2': -0.2884822303125472, 'B_3_3': -0.029398949368659832, 'B_3_4': -0.734344799006484, 'B_3_5': 0.3509540273986875, 'B_3_6': -0.20002295048578583, 'B_3_7': 0.3273908939094269, 'B_3_8': 0.2904165399109956, 'B_3_9': 0.045163860286471735, 'B_4_0': 0.23077873409986477, 'B_4_1': 0.31497734968486657, 'B_4_2': 0.15073615351574354, 'B_4_3': -0.2782426498221191, 'B_4_4': -0.887428385768643, 'B_4_5': 0.289355670939216, 'B_4_6': -0.5764374743232548, 'B_4_7': -0.4405097669177154, 'B_4_8': 0.49419067579710496, 'B_4_9': 0.964288181134114, 'B_5_0': 0.2911668903247107, 'B_5_1': -0.6487529179005412, 'B_5_2': 0.6184277707391123, 'B_5_3': -0.08073798591733508, 'B_5_4': 0.7792115758489044, 'B_5_5': -0.9287449067522707, 'B_5_6': -0.1339729657950598, 'B_5_7': 0.30343492381227993, 'B_5_8': -0.12462862596001668, 'B_5_9': 0.4764637499084571, 'B_6_0': 0.2613876526897567, 'B_6_1': -0.5465978537543116, 'B_6_2': 0.606679355538967, 'B_6_3': 0.9288886037391864, 'B_6_4': -0.5348279175288805, 'B_6_5': 0.16343715968045255, 'B_6_6': 0.35548590991155826, 'B_6_7': 0.5078116668376692, 'B_6_8': 0.4266881832247021, 'B_6_9': 0.6455647615613874, 'B_7_0': -0.8502040592315402, 'B_7_1': -0.3318668181766361, 'B_7_2': -0.36858952953564617, 'B_7_3': 0.7430585566381741, 'B_7_4': 0.943122240816742, 'B_7_5': -0.5638061475865973, 'B_7_6': -0.2462580174360205, 'B_7_7': -0.7351367596781673, 'B_7_8': -0.9248536511589324, 'B_7_9': 0.854992084179269, 'B_8_0': -0.6530907756720766, 'B_8_1': 0.35944559092529504, 'B_8_2': 0.9745888930294065, 'B_8_3': 0.5080555320295969, 'B_8_4': -0.05204135443019813, 'B_8_5': -0.22253555953362375, 'B_8_6': 0.5542597627470939, 'B_8_7': -0.6385226725687728, 'B_8_8': 0.03046602193525297, 'B_8_9': 0.6783327469963623, 'B_9_0': 0.060158033339826805, 'B_9_1': -0.6745286649490541, 'B_9_2': -0.696575859673988, 'B_9_3': 0.6152229012090962, 'B_9_4': -0.7004412764557821, 'B_9_5': -0.5897801248873198, 'B_9_6': -0.3155434779636032, 'B_9_7': -0.2614098690239144, 'B_9_8': 0.752475699357109, 'B_9_9': 0.6781379222771753, 'B_10_0': 0.6092460388067498, 'B_10_1': -0.4457041486126704, 'B_10_2': -0.982299562867579, 'B_10_3': -0.3290947249724163, 'B_10_4': 0.7059635393949079, 'B_10_5': -0.36859647406682683, 'B_10_6': -0.3313133266397938, 'B_10_7': 0.6959552506754079, 'B_10_8': 0.03771978703489301, 'B_10_9': 0.19588259248779097}. Best is trial 153 with value: 57.61808695399101.

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```

[I 2025-04-29 14:26:12,609] Trial 154 finished with value: 64.29593491534656 and
parameters: {'B_0_0': 0.507888723364937, 'B_0_1': 0.22593040387434954, 'B_0_2':
0.5270235426974724, 'B_0_3': 0.24069554551442196, 'B_0_4': -0.6089586285856342,
'B_0_5': 0.9008491853338862, 'B_0_6': 0.5979195327693718, 'B_0_7':
0.49071455857810453, 'B_0_8': 0.5423844386213985, 'B_0_9': 0.32143402844618135,
'B_1_0': 0.5360276837682388, 'B_1_1': -0.23412068762053123, 'B_1_2':
0.39870576163133975, 'B_1_3': 0.6385365612045152, 'B_1_4': -0.5009129782599644,
'B_1_5': 0.9075433315778036, 'B_1_6': -0.19473391545223206, 'B_1_7':
0.6453155666431724, 'B_1_8': -0.5925328434834255, 'B_1_9': -0.6650395768521561,
'B_2_0': -0.7218449612787557, 'B_2_1': -0.4528019429085731, 'B_2_2':
-0.658796801185586, 'B_2_3': 0.8770369235123473, 'B_2_4': -0.07139727631185763,
'B_2_5': -0.6783958064140839, 'B_2_6': -0.2910273387480599, 'B_2_7':
-0.5194075137605009, 'B_2_8': -0.7577739359063567, 'B_2_9': 0.3806525172609451,
'B_3_0': -0.46839025806735374, 'B_3_1': -0.3300254330694301, 'B_3_2':
-0.28660362372547904, 'B_3_3': -0.003928982172218411, 'B_3_4':
-0.7652263744814481, 'B_3_5': 0.34007264560672235, 'B_3_6': -0.2845828360954802,
'B_3_7': 0.35031395670789217, 'B_3_8': 0.27962789413661454, 'B_3_9':
0.05018951174940521, 'B_4_0': 0.23181201296305404, 'B_4_1': 0.3176090993226387,
'B_4_2': 0.13564445918150506, 'B_4_3': -0.2673990781251667, 'B_4_4':
-0.8777768130527334, 'B_4_5': 0.36156961835797347, 'B_4_6': -0.5419080672020633,
'B_4_7': -0.42554774278297725, 'B_4_8': 0.48371329424771464, 'B_4_9':
0.9528000733507143, 'B_5_0': 0.3016280704749059, 'B_5_1': -0.6627869637666614,
'B_5_2': 0.6134969329342302, 'B_5_3': -0.07859871474938837, 'B_5_4':
0.9840101125369624, 'B_5_5': -0.9803229443488114, 'B_5_6': 0.06384055210922271,
'B_5_7': 0.30873477423946005, 'B_5_8': -0.07839126269090167, 'B_5_9':
0.44901448016117806, 'B_6_0': 0.2616770429863307, 'B_6_1': -0.5312842141485098,
'B_6_2': 0.6068011767625057, 'B_6_3': 0.9288554063449307, 'B_6_4':
-0.5253220380487778, 'B_6_5': 0.16488817174542048, 'B_6_6': 0.33857667017273846,
'B_6_7': 0.5124153529948848, 'B_6_8': 0.40313234074580806, 'B_6_9':
0.611945937984044, 'B_7_0': -0.8573619376685231, 'B_7_1': -0.3363444679329163,
'B_7_2': -0.38140158523322115, 'B_7_3': 0.6400771978980182, 'B_7_4':
0.9418491066184705, 'B_7_5': -0.5729489757554166, 'B_7_6': -0.4532201848116963,
'B_7_7': -0.7408434283134588, 'B_7_8': -0.9218733179594463, 'B_7_9':
0.8810483273754127, 'B_8_0': -0.6289304412727956, 'B_8_1': 0.37970747452727416,

```

'B_8_2': 0.9818286734245851, 'B_8_3': 0.5738781890565507, 'B_8_4':
-0.05901285041966135, 'B_8_5': -0.2083468715659263, 'B_8_6': 0.5275924080755493,
'B_8_7': -0.6473485941631073, 'B_8_8': -0.20923281240035624, 'B_8_9':
0.6885685765939145, 'B_9_0': 0.0686156231241124, 'B_9_1': -0.6721258232419192,
'B_9_2': -0.7145623731393624, 'B_9_3': 0.6290486898294204, 'B_9_4':
-0.7339668452903734, 'B_9_5': -0.5910409573022585, 'B_9_6':
-0.32460773081164707, 'B_9_7': -0.2701812713729804, 'B_9_8': 0.7691495397266964,
'B_9_9': 0.5319787922881318, 'B_10_0': 0.6190103804871205, 'B_10_1':
-0.4483243916485919, 'B_10_2': -0.9833979049629628, 'B_10_3':
-0.3962438437784676, 'B_10_4': 0.7202550082789639, 'B_10_5':
-0.36343637909181387, 'B_10_6': -0.34323671621808255, 'B_10_7':
0.5828287290092222, 'B_10_8': 0.04931936398854344, 'B_10_9':
0.17683828546910751}. Best is trial 153 with value: 57.61808695399101.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [<F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:26:27,135] Trial 155 finished with value: 87.49116291386038 and
parameters: {'B_0_0': 0.453625734022283, 'B_0_1': 0.1703402120348301, 'B_0_2':
0.5545438469082432, 'B_0_3': 0.19082125538229086, 'B_0_4': -0.6815917355612262,
'B_0_5': 0.7286117722137877, 'B_0_6': 0.6347853850365427, 'B_0_7':
0.5398999916135708, 'B_0_8': 0.41143854037176314, 'B_0_9': 0.25378703817235676,
'B_1_0': 0.594459803225442, 'B_1_1': -0.13614296024268113, 'B_1_2':
0.453596557704923, 'B_1_3': 0.7995946295609366, 'B_1_4': -0.5680259156571873,
'B_1_5': 0.9358720211938666, 'B_1_6': -0.1016342767596204, 'B_1_7':
0.7027180758284094, 'B_1_8': -0.6001726629743493, 'B_1_9': -0.6039582418874581,
'B_2_0': -0.5910860374939015, 'B_2_1': -0.5038561673562589, 'B_2_2':
-0.6056838966762624, 'B_2_3': 0.8862909667223424, 'B_2_4':
-0.060491168471093204, 'B_2_5': -0.7056288401891856, 'B_2_6':
-0.20628724674800977, 'B_2_7': -0.38876305615288786, 'B_2_8':
-0.6523194565369838, 'B_2_9': 0.5852202584724812, 'B_3_0': -0.403011132300154,
'B_3_1': -0.36301270271289615, 'B_3_2': -0.3198165709469878, 'B_3_3':
-0.09373656182133189, 'B_3_4': -0.7355572965232171, 'B_3_5': 0.372860514320806,
'B_3_6': -0.18754852495252045, 'B_3_7': 0.43676759964652334, 'B_3_8':
0.3639009014764539, 'B_3_9': -0.04553510030757511, 'B_4_0': 0.3543573855527493,
'B_4_1': 0.2555412821360793, 'B_4_2': -0.023791502157748834, 'B_4_3':
-0.1143704578291655, 'B_4_4': -0.8337032169736129, 'B_4_5': 0.2561023263910963,

'B_4_6': -0.6830435235828886, 'B_4_7': -0.48031052880163494, 'B_4_8': 0.39558326958575535, 'B_4_9': 0.8358316791811029, 'B_5_0': 0.27565635703152225, 'B_5_1': -0.5818295357322069, 'B_5_2': 0.5000428868051772, 'B_5_3': -0.16274392593227618, 'B_5_4': 0.7830284222701753, 'B_5_5': -0.9188647780916236, 'B_5_6': -0.03746228522602025, 'B_5_7': 0.3153283098074016, 'B_5_8': -0.1680433992504981, 'B_5_9': 0.47760342002042605, 'B_6_0': 0.39388578046489514, 'B_6_1': -0.5600072044586559, 'B_6_2': 0.6722970093098581, 'B_6_3': 0.9327390211451124, 'B_6_4': -0.5256897729794747, 'B_6_5': 0.13342556173582318, 'B_6_6': 0.28427259017244405, 'B_6_7': 0.521373589507722, 'B_6_8': 0.5266892995594309, 'B_6_9': 0.6569882177330587, 'B_7_0': -0.7053726438499665, 'B_7_1': -0.29116042995200386, 'B_7_2': -0.33631238741797154, 'B_7_3': 0.7406504550254055, 'B_7_4': 0.9470547475602102, 'B_7_5': -0.5702512930268708, 'B_7_6': -0.25626867087574023, 'B_7_7': -0.722732860824461, 'B_7_8': -0.9264297478093255, 'B_7_9': 0.9698901797919564, 'B_8_0': -0.649264705999424, 'B_8_1': 0.31948850969198683, 'B_8_2': 0.9481329894102144, 'B_8_3': 0.2963765218568977, 'B_8_4': -0.11984255674691326, 'B_8_5': -0.2727389600863516, 'B_8_6': 0.6640115562392063, 'B_8_7': -0.7833163802959339, 'B_8_8': -0.036999467145808054, 'B_8_9': 0.6730434378347061, 'B_9_0': 0.03242917257634932, 'B_9_1': -0.6174199959045185, 'B_9_2': -0.6941790206188376, 'B_9_3': 0.622945767604286, 'B_9_4': -0.6427911182271641, 'B_9_5': -0.4762254236332063, 'B_9_6': -0.3669765456717856, 'B_9_7': -0.15927447726144117, 'B_9_8': 0.8014167057400721, 'B_9_9': 0.7024882877129356, 'B_10_0': 0.694503938727216, 'B_10_1': -0.5204894922030251, 'B_10_2': -0.8872710657232561, 'B_10_3': -0.33063842015802625, 'B_10_4': 0.6831638981531504, 'B_10_5': -0.5304498881737605, 'B_10_6': -0.34114664762310504, 'B_10_7': 0.6845598745310832, 'B_10_8': 0.010798620805742919, 'B_10_9': 0.3148043920427822}. Best is trial 153 with value: 57.61808695399101.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_F[u_i], _F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:26:41,381] Trial 156 finished with value: 64.77304660119266 and parameters: {'B_0_0': 0.5817487988104475, 'B_0_1': 0.2389860730574188, 'B_0_2': 0.6016161832954043, 'B_0_3': 0.3139122688677371, 'B_0_4': -0.6030357946266925, 'B_0_5': 0.8983456128465195, 'B_0_6': 0.587058056301555, 'B_0_7': 0.4515213946589416, 'B_0_8': 0.35314002730106675, 'B_0_9': 0.4077197684675177,

'B_1_0': 0.4910755571462704, 'B_1_1': -0.20212758454431884, 'B_1_2':
 0.383235540660404, 'B_1_3': 0.850902180102794, 'B_1_4': -0.42401746136598933,
 'B_1_5': 0.9123565000338603, 'B_1_6': -0.14737945946659198, 'B_1_7':
 0.5747066696214123, 'B_1_8': -0.7908132463009929, 'B_1_9': -0.547351708727282,
 'B_2_0': -0.7307576531012314, 'B_2_1': -0.39272781115717903, 'B_2_2':
 -0.5556314809141176, 'B_2_3': 0.778322795437518, 'B_2_4': -0.22976856139355734,
 'B_2_5': -0.6334247263973315, 'B_2_6': -0.26762665418725184, 'B_2_7':
 -0.4522891273982691, 'B_2_8': -0.7006305026996864, 'B_2_9': 0.2234186229904003,
 'B_3_0': -0.5119915695992164, 'B_3_1': -0.36669165084183264, 'B_3_2':
 -0.363741813734792, 'B_3_3': -0.04327622461104713, 'B_3_4': -0.6971665731135863,
 'B_3_5': 0.26856748494217647, 'B_3_6': -0.15592826399386134, 'B_3_7':
 0.2636033821682846, 'B_3_8': 0.2841704653862653, 'B_3_9': 0.015401535934983128,
 'B_4_0': 0.25774889913960775, 'B_4_1': 0.30881481574696684, 'B_4_2':
 0.1163804995269282, 'B_4_3': -0.30358219128745767, 'B_4_4': -0.73928647970318,
 'B_4_5': 0.29273393573307777, 'B_4_6': -0.6280991920434271, 'B_4_7':
 -0.42082074852777074, 'B_4_8': 0.48669899429704644, 'B_4_9': 0.8883026921249727,
 'B_5_0': 0.21638085085122694, 'B_5_1': -0.6433366067473516, 'B_5_2':
 0.6876739117757262, 'B_5_3': -0.08556482720662416, 'B_5_4': 0.8012929708194646,
 'B_5_5': -0.9071961038697722, 'B_5_6': -0.12892618430756694, 'B_5_7':
 0.4296311017209392, 'B_5_8': -0.11188527066421655, 'B_5_9': 0.43717041306133425,
 'B_6_0': 0.16370019924419194, 'B_6_1': -0.6333710248187887, 'B_6_2':
 0.5725713071601074, 'B_6_3': 0.9159624316571994, 'B_6_4': -0.5733563769678911,
 'B_6_5': -0.025432260272697016, 'B_6_6': 0.35544925215845774, 'B_6_7':
 0.38793349516572584, 'B_6_8': 0.45545432031779143, 'B_6_9': 0.5950948564761455,
 'B_7_0': -0.9075228559719387, 'B_7_1': -0.36377702345852775, 'B_7_2':
 -0.4218670099889248, 'B_7_3': 0.669115785668469, 'B_7_4': 0.9564664688246348,
 'B_7_5': -0.6712759030222417, 'B_7_6': -0.24473938849563057, 'B_7_7':
 -0.8220356075643973, 'B_7_8': -0.9854535159882893, 'B_7_9': 0.7436698880131288,
 'B_8_0': -0.7152890431667734, 'B_8_1': 0.32579178321442004, 'B_8_2':
 0.95447809981917, 'B_8_3': 0.516272437767728, 'B_8_4': -0.035056131540836005,
 'B_8_5': -0.28326224641782943, 'B_8_6': 0.6097524481249018, 'B_8_7':
 -0.7340094456071863, 'B_8_8': -0.12841441494771216, 'B_8_9': 0.5298483704871855,
 'B_9_0': 0.12996569944339514, 'B_9_1': -0.683185674168609, 'B_9_2':
 -0.6690664139810988, 'B_9_3': 0.7238816685625103, 'B_9_4': -0.7952373609340366,
 'B_9_5': -0.4404031733821837, 'B_9_6': -0.30648420676769983, 'B_9_7':
 -0.2743495939104963, 'B_9_8': 0.7523761692080724, 'B_9_9': 0.6488562635809021,
 'B_10_0': 0.6570445337798885, 'B_10_1': -0.3909749662625784, 'B_10_2':
 -0.978396826290866, 'B_10_3': -0.42914264098324, 'B_10_4': 0.7373802391016873,
 'B_10_5': -0.4363044375073314, 'B_10_6': -0.24571782947331505, 'B_10_7':
 0.580285299755418, 'B_10_8': 0.07152103915262499, 'B_10_9':
 0.14871544710074214}. Best is trial 153 with value: 57.61808695399101.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].

```

Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:26:55,670] Trial 157 finished with value: 66.07765054649485 and
parameters: {'B_0_0': 0.4164842448279731, 'B_0_1': 0.29239718759924205, 'B_0_2':
0.4935789643877446, 'B_0_3': 0.2281487139723648, 'B_0_4': -0.672064137808138,
'B_0_5': 0.8305526317638647, 'B_0_6': 0.5078238854675329, 'B_0_7':
0.6528689588023144, 'B_0_8': 0.3189781863372485, 'B_0_9': 0.2638376508126067,
'B_1_0': 0.6696510561937932, 'B_1_1': -0.1464375943721467, 'B_1_2':
0.44758480932621053, 'B_1_3': 0.6364853026070001, 'B_1_4': -0.39070423887971284,
'B_1_5': 0.9192465843419637, 'B_1_6': -0.19926221883442582, 'B_1_7':
0.6409435359456691, 'B_1_8': -0.8240383206571791, 'B_1_9': -0.6006477745727823,
'B_2_0': -0.6986124248403799, 'B_2_1': -0.46655998356082184, 'B_2_2':
-0.6854355616435984, 'B_2_3': 0.880142986343901, 'B_2_4': -0.017891381085468866,
'B_2_5': -0.664715653561907, 'B_2_6': -0.19642047530173506, 'B_2_7':
-0.4338405651052552, 'B_2_8': -0.75151930882031, 'B_2_9': 0.3526006809822813,
'B_3_0': -0.3740117840777136, 'B_3_1': -0.2635204930141363, 'B_3_2':
-0.224293567143707, 'B_3_3': -0.11609026181559813, 'B_3_4': -0.7813041433641738,
'B_3_5': 0.3102400422973658, 'B_3_6': -0.3265451648823016, 'B_3_7':
0.49060723410953216, 'B_3_8': 0.4303660123909289, 'B_3_9': -0.09066670558379628,
'B_4_0': 0.29000569637459966, 'B_4_1': 0.16627040677690458, 'B_4_2':
0.16139377930529408, 'B_4_3': -0.23816393702987068, 'B_4_4':
-0.7657910185473561, 'B_4_5': 0.1584206313073952, 'B_4_6': -0.5546116038872551,
'B_4_7': -0.34005664249284767, 'B_4_8': 0.5323693855046507, 'B_4_9':
0.98215426679785, 'B_5_0': 0.4519233940080848, 'B_5_1': -0.685523094054652,
'B_5_2': 0.5861661671752011, 'B_5_3': 0.002516041688689241, 'B_5_4':
0.7276344016847638, 'B_5_5': -0.9846987268651263, 'B_5_6': -0.08322870213146435,
'B_5_7': 0.2734484401830819, 'B_5_8': -0.1894894591897712, 'B_5_9':
0.4697795862972311, 'B_6_0': 0.26114923862838985, 'B_6_1': -0.37954276847477764,
'B_6_2': 0.6360618913275411, 'B_6_3': 0.9507136085103869, 'B_6_4':
-0.37694094481955814, 'B_6_5': 0.06471394664965971, 'B_6_6': 0.3986873176702956,
'B_6_7': 0.45348072075187096, 'B_6_8': 0.3928455416747233, 'B_6_9':
0.7076793050184019, 'B_7_0': -0.8536228600001865, 'B_7_1': -0.23606454311608205,
'B_7_2': -0.3801417409729677, 'B_7_3': 0.6101128518763462, 'B_7_4':
0.9077209208737785, 'B_7_5': -0.5411023128070925, 'B_7_6': -0.28815920388335414,
'B_7_7': -0.7415734571957804, 'B_7_8': -0.9131479532505675, 'B_7_9':
0.9209473806199145, 'B_8_0': -0.5961475998020276, 'B_8_1': 0.42537949373112804,
'B_8_2': 0.8546820664810799, 'B_8_3': 0.34856320243214245, 'B_8_4':
-0.10256616383244538, 'B_8_5': -0.3717901323544994, 'B_8_6': 0.5327537637358795,
'B_8_7': -0.6178007453200688, 'B_8_8': 0.009076007082392312, 'B_8_9':
0.7118726109638482, 'B_9_0': 0.08340779217314001, 'B_9_1': -0.7387454949155341,
'B_9_2': -0.6617778741642963, 'B_9_3': 0.560670997786561, 'B_9_4':
-0.6215544134275649, 'B_9_5': -0.5393053152752386, 'B_9_6': -0.1864458756105306,

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'B_9_7': -0.15025333108650618, 'B_9_8': 0.8162368290402103, 'B_9_9':
0.748464972509813, 'B_10_0': 0.5896314798830661, 'B_10_1': -0.5785937467310955,
'B_10_2': -0.9017890808359251, 'B_10_3': -0.5061616664541915, 'B_10_4':
0.6891949564477832, 'B_10_5': -0.3577933794313289, 'B_10_6': -0.391486802606223,
'B_10_7': 0.6729826561052464, 'B_10_8': 0.16104435865291752, 'B_10_9':
0.10507716333156665}. Best is trial 153 with value: 57.61808695399101.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].
Built F__[u_10].
Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:27:10,378] Trial 158 finished with value: 79.20941741441543 and
parameters: {'B_0_0': 0.49967062749212726, 'B_0_1': 0.2296254146543175, 'B_0_2':
0.6340534105484603, 'B_0_3': 0.26716721565468693, 'B_0_4': -0.538361677823695,
'B_0_5': 0.9013138675607548, 'B_0_6': 0.46019441567954716, 'B_0_7':
0.5072793051156513, 'B_0_8': 0.4138362377911019, 'B_0_9': 0.31760855727484744,
'B_1_0': 0.5201715985376563, 'B_1_1': -0.27114971359073536, 'B_1_2':
0.4757674004447947, 'B_1_3': 0.9865254629325337, 'B_1_4': -0.45537992410329603,
'B_1_5': 0.9302065493105582, 'B_1_6': -0.14525631635525602, 'B_1_7':
0.7290049388301902, 'B_1_8': -0.7299887277518697, 'B_1_9': -0.6985529446135348,
'B_2_0': -0.6023103056952076, 'B_2_1': -0.4102913014220026, 'B_2_2':
-0.5824257547593572, 'B_2_3': 0.9373613221469974, 'B_2_4': -0.11887790597528322,
'B_2_5': -0.7155443528136095, 'B_2_6': -0.31945797037056184, 'B_2_7':
-0.5376476507147026, 'B_2_8': -0.6232567466604841, 'B_2_9': 0.1587534504006779,
'B_3_0': -0.4373435403102545, 'B_3_1': -0.38748878627021216, 'B_3_2':
-0.3838043075907185, 'B_3_3': -0.16962558608475112, 'B_3_4':
-0.5517365629175693, 'B_3_5': 0.3632750320864163, 'B_3_6': -0.22525060735715288,
'B_3_7': 0.4040878790258751, 'B_3_8': 0.35495735956801194, 'B_3_9':
0.07152831796323386, 'B_4_0': 0.2370065590022509, 'B_4_1': 0.3613753527606682,
'B_4_2': 0.011037514952922857, 'B_4_3': -0.1272270141397851, 'B_4_4':
-0.7077868498811719, 'B_4_5': 0.3442224623305241, 'B_4_6': -0.5789194230479011,
'B_4_7': -0.3949653330696636, 'B_4_8': 0.45604649017243415, 'B_4_9':
0.8609264870721143, 'B_5_0': 0.33108688182537044, 'B_5_1': -0.7909099861502201,
'B_5_2': 0.6504041025014519, 'B_5_3': -0.15987627704501206, 'B_5_4':
0.7051774616049469, 'B_5_5': -0.851797758827159, 'B_5_6': -0.15805490860242621,
'B_5_7': 0.3073440779126477, 'B_5_8': -0.04935501354426822, 'B_5_9':
0.6050274951328141, 'B_6_0': 0.1946878049386239, 'B_6_1': -0.6096023397895811,
'B_6_2': 0.5336170612178285, 'B_6_3': 0.9759630615486773, 'B_6_4':

```

-0.5660004858347027, 'B_6_5': 0.022491985494162398, 'B_6_6': 0.273030477288713,
'B_6_7': 0.48543071319301323, 'B_6_8': 0.5351757023973361, 'B_6_9':
0.6321220535080652, 'B_7_0': -0.8978046306014279, 'B_7_1': -0.32342183441494193,
'B_7_2': -0.4614091044304567, 'B_7_3': 0.7565427113228992, 'B_7_4':
0.9740981341568089, 'B_7_5': -0.5960004406070001, 'B_7_6': -0.414134813528645,
'B_7_7': -0.9749757091617284, 'B_7_8': -0.9207617991716825, 'B_7_9':
0.8485025574686123, 'B_8_0': -0.7093920308576965, 'B_8_1': 0.23844166542102144,
'B_8_2': 0.9111997662403595, 'B_8_3': 0.4028861283000956, 'B_8_4':
-0.161594523916783, 'B_8_5': -0.1938658117637324, 'B_8_6': 0.5547615163095573,
'B_8_7': -0.6723394448035758, 'B_8_8': -0.05328483010141756, 'B_8_9':
0.6606630317305514, 'B_9_0': 0.1882509507864783, 'B_9_1': -0.6028446090171562,
'B_9_2': -0.7351869458717786, 'B_9_3': 0.7495953057987528, 'B_9_4':
-0.7617905597467568, 'B_9_5': -0.5884305950408979, 'B_9_6':
-0.27765672999135693, 'B_9_7': -0.050453722235401406, 'B_9_8':
0.6802084746412591, 'B_9_9': 0.590740903105154, 'B_10_0': 0.71462467239098,
'B_10_1': -0.4720294891842886, 'B_10_2': -0.9475143980162033, 'B_10_3':
-0.5709785575385921, 'B_10_4': 0.7313460104850332, 'B_10_5': -0.586250545295452,
'B_10_6': -0.2626494474833506, 'B_10_7': 0.7347619517042608, 'B_10_8':
0.11018171987945446, 'B_10_9': 0.23022297347754261}. Best is trial 153 with
value: 57.61808695399101.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:27:24,494] Trial 159 finished with value: 67.4364086275295 and
parameters: {'B_0_0': 0.47310224259648065, 'B_0_1': 0.15342327544906065,
'B_0_2': 0.5675321728481131, 'B_0_3': 0.3493028542996437, 'B_0_4':
-0.6295365147119043, 'B_0_5': 0.829534102114973, 'B_0_6': 0.5626222744509447,
'B_0_7': 0.5477693446294286, 'B_0_8': 0.3758183968961665, 'B_0_9':
0.17654862230499385, 'B_1_0': 0.5800670504872875, 'B_1_1': -0.3681981434082924,
'B_1_2': 0.3718305397892411, 'B_1_3': 0.6308622350826556, 'B_1_4':
-0.608945350382851, 'B_1_5': 0.9371460178953201, 'B_1_6': -0.254785565893422,
'B_1_7': 0.6533796798716388, 'B_1_8': -0.5793718298429491, 'B_1_9':
-0.7549948249558976, 'B_2_0': -0.6822513767416588, 'B_2_1': -0.5279749911383639,
'B_2_2': -0.6184327921589456, 'B_2_3': 0.8691756585853845, 'B_2_4':
-0.21618450229912808, 'B_2_5': -0.6593089032462833, 'B_2_6':
-0.12831187486751827, 'B_2_7': -0.5149029151813281, 'B_2_8':

```



```

-0.5825722747644707, 'B_2_9': 0.34140700078304387, 'B_3_0': -0.3481724822972666,
'B_3_1': -0.2379047116262158, 'B_3_2': -0.2603353914075955, 'B_3_3':
-0.020780863705286173, 'B_3_4': -0.6565987460116199, 'B_3_5':
0.2749604903744556, 'B_3_6': -0.27215844977516424, 'B_3_7': 0.29726279621903645,
'B_3_8': 0.2213302537575686, 'B_3_9': -0.02178707328791915, 'B_4_0':
0.46166652419750925, 'B_4_1': 0.10457448279797821, 'B_4_2':
-0.08783214698389608, 'B_4_3': -0.2847354364501159, 'B_4_4': -0.931753969479902,
'B_4_5': 0.24898833533394463, 'B_4_6': -0.6330041314140151, 'B_4_7':
-0.45054394434973044, 'B_4_8': 0.308519514662949, 'B_4_9': 0.9102948232525754,
'B_5_0': 0.4014742271088726, 'B_5_1': -0.5635757069726094, 'B_5_2':
0.7338536390016038, 'B_5_3': -0.09746534044452455, 'B_5_4': 0.5925198881670108,
'B_5_5': -0.9429734289116549, 'B_5_6': -0.03589366547774166, 'B_5_7':
0.1641099942961399, 'B_5_8': -0.14112411619580728, 'B_5_9': 0.42207760054044013,
'B_6_0': 0.43392872367017626, 'B_6_1': -0.5708477100618394, 'B_6_2':
0.675873051568586, 'B_6_3': 0.9038588475185545, 'B_6_4': -0.6224859647868327,
'B_6_5': -0.059479547038211844, 'B_6_6': 0.3085658045601388, 'B_6_7':
0.5813757578546536, 'B_6_8': 0.5737747013812873, 'B_6_9': 0.7299532789252267,
'B_7_0': -0.8526732190053685, 'B_7_1': -0.40591859035650063, 'B_7_2':
-0.22639935795194913, 'B_7_3': 0.7018639032702184, 'B_7_4': 0.8269343380570073,
'B_7_5': -0.504848428911367, 'B_7_6': -0.2296987379878915, 'B_7_7':
-0.8128658474892115, 'B_7_8': -0.9401070232431834, 'B_7_9': 0.8969990445660837,
'B_8_0': -0.666017854600542, 'B_8_1': 0.4046322364759847, 'B_8_2':
0.9483908790531466, 'B_8_3': 0.32452104769462853, 'B_8_4': -0.21230113587299626,
'B_8_5': -0.24351027869153752, 'B_8_6': 0.47982015177517934, 'B_8_7':
-0.7860967358746691, 'B_8_8': -0.1036661545760155, 'B_8_9': 0.7388177668547133,
'B_9_0': 0.011539925575951626, 'B_9_1': -0.6840049884433964, 'B_9_2':
-0.6226473419477716, 'B_9_3': 0.721183641537217, 'B_9_4': -0.6957758875254552,
'B_9_5': -0.5287853661818699, 'B_9_6': -0.2108387833726854, 'B_9_7':
-0.2852242018445769, 'B_9_8': 0.7184500560998435, 'B_9_9': 0.5260162613380972,
'B_10_0': 0.7777068767270486, 'B_10_1': -0.5373355913718194, 'B_10_2':
-0.9861301622556778, 'B_10_3': -0.31557773127459443, 'B_10_4':
0.845305442649251, 'B_10_5': -0.45887152706621226, 'B_10_6':
-0.4381991468117248, 'B_10_7': 0.7889168789180678, 'B_10_8': 0.0270848169012963,
'B_10_9': 0.17867246716908325}. Best is trial 153 with value: 57.61808695399101.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:27:38,887] Trial 160 finished with value: 66.86244893912142 and parameters: {'B_0_0': 0.6070467293316096, 'B_0_1': 0.1112229501876155, 'B_0_2': 0.6796066229025901, 'B_0_3': 0.15934116996674708, 'B_0_4': -0.48311323644411847, 'B_0_5': 0.906051187699129, 'B_0_6': 0.627852813660805, 'B_0_7': 0.4846800304225841, 'B_0_8': 0.5728727937399837, 'B_0_9': 0.21688487344407162, 'B_1_0': 0.43368190700992404, 'B_1_1': -0.20956859859847124, 'B_1_2': 0.2851458350825992, 'B_1_3': 0.8782997217065491, 'B_1_4': -0.5078755354251545, 'B_1_5': 0.8902871003873664, 'B_1_6': -0.08237221337947283, 'B_1_7': 0.5288492858015577, 'B_1_8': -0.6840206848500229, 'B_1_9': -0.6413577913864663, 'B_2_0': -0.7805863106607738, 'B_2_1': -0.48227387328888327, 'B_2_2': -0.5251047996292267, 'B_2_3': 0.7728539429675045, 'B_2_4': -0.16352473733990125, 'B_2_5': -0.6177978175641801, 'B_2_6': -0.0584962549975378, 'B_2_7': -0.3952817112093873, 'B_2_8': -0.890443237162146, 'B_2_9': 0.3813676580017668, 'B_3_0': -0.28090808631524433, 'B_3_1': -0.33705862983089063, 'B_3_2': -0.2779934947822362, 'B_3_3': -0.0713967326984792, 'B_3_4': -0.7475653344284281, 'B_3_5': 0.23455172210528222, 'B_3_6': -0.03534958713108635, 'B_3_7': 0.3685004773496399, 'B_3_8': 0.2828991135202407, 'B_3_9': -0.06390380273424756, 'B_4_0': 0.3626388893032555, 'B_4_1': 0.2896763853007081, 'B_4_2': 0.15624263699383778, 'B_4_3': -0.21310786564423695, 'B_4_4': -0.9021760215980636, 'B_4_5': 0.390925478731989, 'B_4_6': -0.7059727084279083, 'B_4_7': -0.3170449504856967, 'B_4_8': 0.3608135368185853, 'B_4_9': 0.9392683587214508, 'B_5_0': 0.26749812606231227, 'B_5_1': -0.6087828106530142, 'B_5_2': 0.5479079379718118, 'B_5_3': -0.3237197140854162, 'B_5_4': 0.756228678424533, 'B_5_5': -0.8970409376340072, 'B_5_6': -0.10673150079081686, 'B_5_7': 0.44301278794689536, 'B_5_8': -0.25007858246765885, 'B_5_9': 0.5634013299460923, 'B_6_0': 0.25439346940307106, 'B_6_1': -0.5281461649179651, 'B_6_2': 0.45722830893863114, 'B_6_3': 0.9885980177592554, 'B_6_4': -0.6988878082804387, 'B_6_5': 0.1677651034785825, 'B_6_6': 0.357546657868654, 'B_6_7': 0.5351011959026934, 'B_6_8': 0.495921933526518, 'B_6_9': 0.679704484654572, 'B_7_0': -0.9317979561032819, 'B_7_1': -0.32121046429016054, 'B_7_2': -0.29994945551368046, 'B_7_3': 0.5662946040967174, 'B_7_4': 0.9208892933321704, 'B_7_5': -0.5191090939943733, 'B_7_6': -0.3190850367606958, 'B_7_7': -0.8368405886977248, 'B_7_8': -0.9588538312032338, 'B_7_9': 0.8052736214376394, 'B_8_0': -0.5568767794554511, 'B_8_1': 0.31114747112864105, 'B_8_2': 0.9875407713087161, 'B_8_3': 0.2150361053213589, 'B_8_4': -0.08833904094659542, 'B_8_5': -0.3089914310074934, 'B_8_6': 0.7202213826013517, 'B_8_7': -0.7217369796103628, 'B_8_8': 0.03777463072852401, 'B_8_9': 0.8555863908968413, 'B_9_0': 0.12944864819994722, 'B_9_1': -0.8143421930206225, 'B_9_2': -0.5616215451288913, 'B_9_3': 0.6478799878317766, 'B_9_4': -0.6726098280179429, 'B_9_5': -0.5929725483496208, 'B_9_6': -0.3694447009647185, 'B_9_7': -0.26040652631009004, 'B_9_8': 0.7379673624424847, 'B_9_9': 0.624262413241978, 'B_10_0': 0.7548703591300215, 'B_10_1': -0.44110059162726906, 'B_10_2': -0.8769114587151454, 'B_10_3': -0.4179876854655461, 'B_10_4': 0.6883673110745081, 'B_10_5': -0.3505594663572944, 'B_10_6': -0.32017411547679414, 'B_10_7': 0.7036000230895819, 'B_10_8': 0.22014923744299333, 'B_10_9': 0.2800384298957278}. Best is trial 153 with value: 57.61808695399101.

Building the Basis functions for the RKHS:

```
Built F_[u_1].  
Built F_[u_2].  
Built F_[u_3].  
Built F_[u_4].  
Built F_[u_5].  
Built F_[u_6].  
Built F_[u_7].  
Built F_[u_8].  
Built F_[u_9].  
Built F_[u_10].  
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:27:53,070] Trial 161 finished with value: 68.28531235132574 and parameters: {'B_0_0': 0.5396292327847363, 'B_0_1': 0.22537264327385287, 'B_0_2': 0.5344478084544846, 'B_0_3': 0.21299933778271482, 'B_0_4': -0.6036748430917526, 'B_0_5': 0.8963983124046763, 'B_0_6': 0.5948846855536222, 'B_0_7': 0.5015425693510346, 'B_0_8': 0.554665997363122, 'B_0_9': 0.32076637555797155, 'B_1_0': 0.5500511175048384, 'B_1_1': -0.2351462673104533, 'B_1_2': 0.41303939224939656, 'B_1_3': 0.661642683036203, 'B_1_4': -0.4974234653669058, 'B_1_5': 0.8855606835559879, 'B_1_6': -0.20138267631086817, 'B_1_7': 0.6566867520663676, 'B_1_8': -0.5956997825729857, 'B_1_9': -0.6699953296993943, 'B_2_0': -0.46419293967437786, 'B_2_1': -0.44536297699774174, 'B_2_2': -0.6625499169823184, 'B_2_3': 0.8896598794373679, 'B_2_4': -0.06429757362962712, 'B_2_5': -0.6783779035701921, 'B_2_6': -0.3159262541609498, 'B_2_7': -0.5332247757288431, 'B_2_8': -0.7634263113124854, 'B_2_9': 0.381929712389073, 'B_3_0': -0.4642461823102696, 'B_3_1': -0.30073602297958585, 'B_3_2': -0.2885590179791011, 'B_3_3': -0.0023740622047650943, 'B_3_4': -0.8391949827077904, 'B_3_5': 0.3249122063961909, 'B_3_6': -0.38351833793099854, 'B_3_7': 0.3325234492648939, 'B_3_8': 0.31750568009614766, 'B_3_9': 0.041072234043388715, 'B_4_0': 0.23097439607716333, 'B_4_1': 0.3244695870643056, 'B_4_2': 0.12634670491291938, 'B_4_3': -0.2663643406623521, 'B_4_4': -0.8635393213909859, 'B_4_5': 0.3582413415439633, 'B_4_6': -0.5284770086235182, 'B_4_7': -0.4343407908774096, 'B_4_8': 0.49321742095832016, 'B_4_9': 0.8039778141368968, 'B_5_0': 0.3140646176123764, 'B_5_1': -0.6915914669856322, 'B_5_2': 0.5931992759640448, 'B_5_3': -0.09225020898098446, 'B_5_4': 0.9887912022817176, 'B_5_5': -0.9875446472466569, 'B_5_6': 0.03942068315893987, 'B_5_7': 0.30799701207420566, 'B_5_8': -0.06251437237140842, 'B_5_9': 0.45202608831223445, 'B_6_0': 0.28969809043892875, 'B_6_1': -0.48434640055915423, 'B_6_2': 0.6097317224607794, 'B_6_3': 0.9306014576969827, 'B_6_4': -0.5485016001950032, 'B_6_5': 0.18325054935379329, 'B_6_6': 0.336375445680533, 'B_6_7': 0.506757015102823, 'B_6_8': 0.3917762525188858, 'B_6_9': 0.6275543799050908, 'B_7_0': -0.8465229395245392, 'B_7_1': -0.33572936971315337, 'B_7_2': -0.37820982318662316, 'B_7_3': 0.6663097825687759, 'B_7_4': 0.9397153866056572, 'B_7_5': -0.7001799691097689, 'B_7_6': -0.41400411438210527, 'B_7_7': -0.7385926686855242, 'B_7_8':

```

-0.987700010286098, 'B_7_9': 0.8823664997011684, 'B_8_0': -0.6132778003572064,
'B_8_1': 0.37504330843953765, 'B_8_2': 0.9835979566028363, 'B_8_3':
0.37335389618812653, 'B_8_4': -0.036427406667399714, 'B_8_5':
-0.19847236997503676, 'B_8_6': 0.5254367016160145, 'B_8_7': -0.6457506368764158,
'B_8_8': -0.18782301664302203, 'B_8_9': 0.6989188629642242, 'B_9_0':
0.06735513744289556, 'B_9_1': -0.6764522752340709, 'B_9_2': -0.7166592662707971,
'B_9_3': 0.7666749252353526, 'B_9_4': -0.7402734940187131, 'B_9_5':
-0.5856913048804409, 'B_9_6': -0.3350023615560883, 'B_9_7':
-0.29810585422502095, 'B_9_8': 0.7701311276137437, 'B_9_9': 0.5068067866961609,
'B_10_0': 0.6883977236035639, 'B_10_1': -0.43057222374208454, 'B_10_2':
-0.9859005788057116, 'B_10_3': -0.399829596639367, 'B_10_4': 0.7257977640336984,
'B_10_5': -0.3772846283769638, 'B_10_6': -0.3570870563695285, 'B_10_7':
0.5743462077293906, 'B_10_8': 0.04383637938638699, 'B_10_9':
0.1791193213799478}. Best is trial 153 with value: 57.61808695399101.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:28:07,608] Trial 162 finished with value: 75.69609528432075 and
parameters: {'B_0_0': 0.5060617974204928, 'B_0_1': 0.2141448716554821, 'B_0_2':
0.5230035810314441, 'B_0_3': 0.2736843422238381, 'B_0_4': -0.6439341666758285,
'B_0_5': 0.9116351743829798, 'B_0_6': 0.5015204012299269, 'B_0_7':
0.690430481997242, 'B_0_8': 0.5196825171362093, 'B_0_9': 0.36839773452548197,
'B_1_0': 0.532715019557855, 'B_1_1': -0.30968348803208817, 'B_1_2':
0.4191108477131652, 'B_1_3': 0.7965261415181073, 'B_1_4': -0.5463608596409013,
'B_1_5': 0.9212734029782551, 'B_1_6': -0.16708682098194425, 'B_1_7':
0.6863113113768191, 'B_1_8': -0.5787000485552588, 'B_1_9': -0.5714778272741426,
'B_2_0': -0.7102054170988794, 'B_2_1': -0.3691215704673993, 'B_2_2':
-0.6545737330032352, 'B_2_3': 0.8681274621054483, 'B_2_4': -0.0872850868840897,
'B_2_5': -0.8537974483300118, 'B_2_6': -0.30953170654050827, 'B_2_7':
-0.5196484204772684, 'B_2_8': -0.6581460293633081, 'B_2_9': 0.321117045411302,
'B_3_0': -0.48169816447958763, 'B_3_1': -0.06212972070483175, 'B_3_2':
-0.20919239435087902, 'B_3_3': 0.007379495402001501, 'B_3_4':
-0.705736035932193, 'B_3_5': 0.3455469659885164, 'B_3_6': -0.26569221894174816,
'B_3_7': 0.24899679029648208, 'B_3_8': 0.2829837910381676, 'B_3_9':
0.02895196807923021, 'B_4_0': 0.31060995033970723, 'B_4_1': 0.40386199184316846,
'B_4_2': 0.23951327819928642, 'B_4_3': -0.33183008966997485, 'B_4_4':

```

```

-0.8120762909171715, 'B_4_5': 0.31273920433648605, 'B_4_6': -0.5653726307403049,
'B_4_7': -0.3502357943974661, 'B_4_8': 0.5770087735612921, 'B_4_9':
0.9584354294170931, 'B_5_0': 0.2792624991027429, 'B_5_1': -0.6440312504599639,
'B_5_2': 0.6230494664255574, 'B_5_3': 0.09437218319739601, 'B_5_4':
0.6627849893532658, 'B_5_5': -0.9588168705851388, 'B_5_6': 0.09845505319559596,
'B_5_7': 0.2855886243976326, 'B_5_8': -0.0838755772237994, 'B_5_9':
0.47994193532790547, 'B_6_0': 0.35747819323804336, 'B_6_1': -0.6242737044988772,
'B_6_2': 0.7257835655162721, 'B_6_3': 0.9326384748522353, 'B_6_4':
-0.498904150812226, 'B_6_5': 0.09887915259164465, 'B_6_6': 0.25226567053753896,
'B_6_7': 0.3669626324191211, 'B_6_8': 0.44387017063881534, 'B_6_9':
0.5936467462159825, 'B_7_0': -0.880891003157774, 'B_7_1': -0.3719193752736114,
'B_7_2': -0.3983151704161991, 'B_7_3': 0.6301252533228637, 'B_7_4':
0.9869101762932371, 'B_7_5': -0.6122741115936016, 'B_7_6': -0.45362203704531956,
'B_7_7': -0.7583874097821168, 'B_7_8': -0.9218975470791841, 'B_7_9':
0.745851803891222, 'B_8_0': -0.6280985952519789, 'B_8_1': 0.1655308645458711,
'B_8_2': 0.8741863411916779, 'B_8_3': 0.5431673651317835, 'B_8_4':
-0.06266011157731571, 'B_8_5': -0.223573843224983, 'B_8_6': 0.6348437246435839,
'B_8_7': -0.575633758720902, 'B_8_8': -0.23019841107798034, 'B_8_9':
0.6740805029168873, 'B_9_0': 0.15408688546849766, 'B_9_1': -0.6571000127692933,
'B_9_2': -0.7352625325857302, 'B_9_3': 0.6436286426832004, 'B_9_4':
-0.7210823573603706, 'B_9_5': -0.6609027366908724, 'B_9_6': -0.2858318453864811,
'B_9_7': -0.26004229374836374, 'B_9_8': 0.6685827470609713, 'B_9_9':
0.645733654542072, 'B_10_0': 0.636755362550032, 'B_10_1': -0.5807536974959309,
'B_10_2': -0.9175638741508788, 'B_10_3': -0.48566762665819463, 'B_10_4':
0.7362191704049154, 'B_10_5': -0.4329194135299154, 'B_10_6':
-0.2138753923391108, 'B_10_7': 0.6133151418631363, 'B_10_8':
-0.027532904574208983, 'B_10_9': 0.342954438012256}. Best is trial 153 with
value: 57.61808695399101.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:28:23,186] Trial 163 finished with value: 62.873538678868066
and parameters: {'B_0_0': 0.3752637134839302, 'B_0_1': 0.25161255601079324,
'B_0_2': 0.6368212006611416, 'B_0_3': 0.1503286991267296, 'B_0_4':
-0.7075590397359465, 'B_0_5': 0.83882700448981, 'B_0_6': 0.44586366854413295,
'B_0_7': 0.4840214678243714, 'B_0_8': 0.41907434068365224, 'B_0_9':

```

0.46158689366689787, 'B_1_0': 0.44237348483122974, 'B_1_1':
 -0.13689962616091036, 'B_1_2': 0.3334716799639791, 'B_1_3': 0.6332960644955402,
 'B_1_4': -0.4334571656482908, 'B_1_5': 0.9340762326057978, 'B_1_6':
 -0.21350135953129676, 'B_1_7': 0.7218551569594502, 'B_1_8': -0.469988717187529,
 'B_1_9': -0.6525263921264443, 'B_2_0': -0.8481605410701275, 'B_2_1':
 -0.43152621176277867, 'B_2_2': -0.5624278528025051, 'B_2_3': 0.9396579847431588,
 'B_2_4': -0.13009606859265632, 'B_2_5': -0.7124574247820272, 'B_2_6':
 -0.2639964590239326, 'B_2_7': -0.5843566720600315, 'B_2_8': -0.7389823874283084,
 'B_2_9': 0.3619435469749185, 'B_3_0': -0.5306230430520167, 'B_3_1':
 -0.39900370533276613, 'B_3_2': -0.39186573773950695, 'B_3_3':
 -0.05289233217712587, 'B_3_4': -0.7590529564760966, 'B_3_5': 0.4257481218828842,
 'B_3_6': -0.20851280852111426, 'B_3_7': 0.29250765744691143, 'B_3_8':
 0.4404169293365056, 'B_3_9': 0.12164485411273562, 'B_4_0': 0.2278288417912713,
 'B_4_1': 0.24766141273801348, 'B_4_2': 0.11728134778544982, 'B_4_3':
 -0.25042677909879146, 'B_4_4': -0.888245376733061, 'B_4_5': 0.3680961497049232,
 'B_4_6': -0.6169601268174406, 'B_4_7': -0.40138444431083925, 'B_4_8':
 0.4737775028199598, 'B_4_9': 0.8805360427120361, 'B_5_0': 0.23244499633908483,
 'B_5_1': -0.6636415426665293, 'B_5_2': 0.4953673257629637, 'B_5_3':
 -0.1813867939793625, 'B_5_4': 0.8312838908992974, 'B_5_5': -0.8721677738417738,
 'B_5_6': -0.010773266731877219, 'B_5_7': 0.3361897704750603, 'B_5_8':
 -0.19248029984923876, 'B_5_9': 0.5324223392129401, 'B_6_0': 0.25927033902519525,
 'B_6_1': -0.5367399349249076, 'B_6_2': 0.6096146925508288, 'B_6_3':
 0.9462266467046505, 'B_6_4': -0.6026033169170015, 'B_6_5':
 -0.0031292591392107945, 'B_6_6': 0.30948090154546914, 'B_6_7':
 0.5191568758992142, 'B_6_8': 0.5334900575621915, 'B_6_9': 0.6061211136574253,
 'B_7_0': -0.8488330416957195, 'B_7_1': -0.26640629210518396, 'B_7_2':
 -0.4966750747525883, 'B_7_3': 0.731723853526525, 'B_7_4': 0.8803276951043251,
 'B_7_5': -0.5581604845841304, 'B_7_6': -0.27180820797105454, 'B_7_7':
 -0.7050861889412542, 'B_7_8': -0.9056692847107034, 'B_7_9': 0.937657423502162,
 'B_8_0': -0.736023038106798, 'B_8_1': 0.3400299375673848, 'B_8_2':
 0.7753957433981291, 'B_8_3': 0.599530433264911, 'B_8_4': -0.1317022016052624,
 'B_8_5': -0.3870400894994023, 'B_8_6': 0.5592140452226453, 'B_8_7':
 -0.6739689668170272, 'B_8_8': -0.027836027325442325, 'B_8_9':
 0.6305016076444123, 'B_9_0': 0.20375103377033063, 'B_9_1': -0.7197172377075748,
 'B_9_2': -0.708015863892924, 'B_9_3': 0.6129251724697755, 'B_9_4':
 -0.7621488829318382, 'B_9_5': -0.5024502532681205, 'B_9_6': -0.4504212520061629,
 'B_9_7': -0.18184949053212135, 'B_9_8': 0.8307288352057185, 'B_9_9':
 0.713561651900659, 'B_10_0': 0.6000288185638616, 'B_10_1': -0.46800128297841387,
 'B_10_2': -0.987944903805063, 'B_10_3': -0.3737696000703886, 'B_10_4':
 0.6115426452331165, 'B_10_5': -0.40618030281557205, 'B_10_6': -0.35903321902353,
 'B_10_7': 0.5511052084166647, 'B_10_8': 0.05752233153564471, 'B_10_9':
 0.2503282223907763}. Best is trial 153 with value: 57.61808695399101.

Building the Basis functions for the RKHS:

Built F_[u₁].
 Built F_[u₂].
 Built F_[u₃].
 Built F_[u₄].
 Built F_[u₅].

```

Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:28:37,413] Trial 164 finished with value: 62.79429668167844 and
parameters: {'B_0_0': 0.3753089574735283, 'B_0_1': 0.28474305249819065, 'B_0_2':
0.6423335411539165, 'B_0_3': 0.12945434718711404, 'B_0_4': -0.6946774464187356,
'B_0_5': 0.8333835284306889, 'B_0_6': 0.4503749213938644, 'B_0_7':
0.3401784982751697, 'B_0_8': 0.33923173050434013, 'B_0_9': 0.17486291051786712,
'B_1_0': 0.3810218398109946, 'B_1_1': -0.15115901824067382, 'B_1_2':
0.3205150856911241, 'B_1_3': 0.6868987762073485, 'B_1_4': -0.39381488151671734,
'B_1_5': 0.8017140293067077, 'B_1_6': -0.23917443841525002, 'B_1_7':
0.7191933811253614, 'B_1_8': -0.4802698313554331, 'B_1_9': -0.6170638222634095,
'B_2_0': -0.8068344273507763, 'B_2_1': -0.4045920898038128, 'B_2_2':
-0.5784335388544228, 'B_2_3': 0.937255808848118, 'B_2_4': -0.13561856234766267,
'B_2_5': -0.7155640031720627, 'B_2_6': -0.17855299974204714, 'B_2_7':
-0.436452721192223, 'B_2_8': -0.49759476836402, 'B_2_9': 0.22162947919547116,
'B_3_0': -0.5482017371464895, 'B_3_1': -0.3995967259551923, 'B_3_2':
-0.3930583501232092, 'B_3_3': -0.09849470098398797, 'B_3_4':
-0.6552264148822925, 'B_3_5': 0.5030593395314376, 'B_3_6': -0.16070187656129392,
'B_3_7': 0.5420612377283494, 'B_3_8': 0.37088283589771104, 'B_3_9':
0.10052465360854416, 'B_4_0': 0.13631555480742852, 'B_4_1': 0.2344456645975088,
'B_4_2': 0.11465552271472271, 'B_4_3': -0.17813494091494547, 'B_4_4':
-0.8278824255730624, 'B_4_5': 0.415188318777735, 'B_4_6': -0.6563563303147304,
'B_4_7': -0.48645675377550085, 'B_4_8': 0.42765467433616855, 'B_4_9':
0.879678919349695, 'B_5_0': 0.2391726032589262, 'B_5_1': -0.5377687447594454,
'B_5_2': 0.4697427854593916, 'B_5_3': -0.17908413645603632, 'B_5_4':
0.8228297385037013, 'B_5_5': -0.8265427590977289, 'B_5_6':
-0.016654350155155317, 'B_5_7': 0.32882642419178437, 'B_5_8':
-0.19257221318335668, 'B_5_9': 0.5904283315514041, 'B_6_0': 0.20836355525953482,
'B_6_1': -0.5449602986692765, 'B_6_2': 0.7044757805352851, 'B_6_3':
0.9079068579090701, 'B_6_4': -0.6000330254516336, 'B_6_5': 0.013320577055481918,
'B_6_6': 0.4008623016372308, 'B_6_7': 0.5817121026721215, 'B_6_8':
0.5370587715088844, 'B_6_9': 0.6726049065778652, 'B_7_0': -0.9201742007732683,
'B_7_1': -0.2606325435903043, 'B_7_2': -0.49449337349090416, 'B_7_3':
0.7371652360616447, 'B_7_4': 0.8759983448135547, 'B_7_5': -0.4443649762820546,
'B_7_6': -0.543664451312525, 'B_7_7': -0.8169683473483381, 'B_7_8':
-0.9031573815787146, 'B_7_9': 0.9292806132370925, 'B_8_0': -0.7422226645630812,
'B_8_1': 0.4646556079866607, 'B_8_2': 0.9148018698462884, 'B_8_3':
0.43608164825486506, 'B_8_4': -0.13382636524139518, 'B_8_5':
-0.38674059589426546, 'B_8_6': 0.5764263944084098, 'B_8_7': -0.8200269510636305,
'B_8_8': -0.014211113783415795, 'B_8_9': 0.6483436343723217, 'B_9_0':
0.20271563963115125, 'B_9_1': -0.711109631021359, 'B_9_2': -0.6823944178549736,

```

'B_9_3': 0.726179113373645, 'B_9_4': -0.8226592380355294, 'B_9_5':
-0.4927548409106129, 'B_9_6': -0.1993987582373687, 'B_9_7':
-0.16651969096464672, 'B_9_8': 0.8647787623176026, 'B_9_9': 0.7170934005066603,
'B_10_0': 0.5690409940141815, 'B_10_1': -0.36780368802304675, 'B_10_2':
-0.9436735599688939, 'B_10_3': -0.534983706057531, 'B_10_4': 0.6875614301938693,
'B_10_5': -0.40560780495593124, 'B_10_6': -0.27726219367312643, 'B_10_7':
0.6467852867232037, 'B_10_8': -0.07072849105220164, 'B_10_9':
0.24648000723690716}. Best is trial 153 with value: 57.61808695399101.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:28:51,737] Trial 165 finished with value: 70.55811561329051 and
parameters: {'B_0_0': 0.657226266305122, 'B_0_1': 0.2711648177263838, 'B_0_2':
0.640261477365417, 'B_0_3': 0.14667063663849098, 'B_0_4': -0.6970880048888166,
'B_0_5': 0.8293117358834845, 'B_0_6': 0.4397497582136512, 'B_0_7':
0.3322603045790148, 'B_0_8': 0.4173528134332569, 'B_0_9': 0.4476235662281002,
'B_1_0': 0.3681967611146873, 'B_1_1': -0.176384765977036, 'B_1_2':
0.3260779673125273, 'B_1_3': 0.8314766680756127, 'B_1_4': -0.4342296259419795,
'B_1_5': 0.9472008272051046, 'B_1_6': -0.22534259490363787, 'B_1_7':
0.5788578361511226, 'B_1_8': -0.5519681939475973, 'B_1_9': -0.7189126257781093,
'B_2_0': -0.7977501974263489, 'B_2_1': -0.38927424645786013, 'B_2_2':
-0.5747641047803762, 'B_2_3': 0.9378464901030211, 'B_2_4': -0.12499684232088432,
'B_2_5': -0.7280817937082201, 'B_2_6': -0.18455023589092073, 'B_2_7':
-0.4342695029341549, 'B_2_8': -0.7985455902357609, 'B_2_9': 0.2210548784890345,
'B_3_0': -0.5431119618962591, 'B_3_1': -0.39940505069185983, 'B_3_2':
-0.3306591364258419, 'B_3_3': -0.09598085565603996, 'B_3_4':
-0.6582081864474364, 'B_3_5': 0.42877608771778164, 'B_3_6':
-0.21408011183290798, 'B_3_7': 0.27125893740301965, 'B_3_8':
0.37298999275582245, 'B_3_9': 0.10926401923695052, 'B_4_0': 0.13302296857364862,
'B_4_1': 0.2274828053418882, 'B_4_2': 0.17029230236745746, 'B_4_3':
-0.1690173538659898, 'B_4_4': -0.819708415437743, 'B_4_5': 0.43527559200344756,
'B_4_6': -0.6215351395925273, 'B_4_7': -0.4884334383288949, 'B_4_8':
0.41591281569256633, 'B_4_9': 0.8869507904390632, 'B_5_0': 0.2370246360666428,
'B_5_1': -0.5990955687471421, 'B_5_2': 0.4438203266390347, 'B_5_3':
-0.2739541826564436, 'B_5_4': 0.8163000355656739, 'B_5_5': -0.860457035779799,
'B_5_6': -0.06367552532515328, 'B_5_7': 0.3329901023399188, 'B_5_8':


```

-0.22396721841182612, 'B_5_9': 0.5258350205690823, 'B_6_0': 0.1469491300138261,
'B_6_1': -0.5546459765508941, 'B_6_2': 0.6562406108640353, 'B_6_3':
0.9882560207446452, 'B_6_4': -0.5962012455551795, 'B_6_5':
-0.007352953655450523, 'B_6_6': 0.4061736526295201, 'B_6_7': 0.4423044292272845,
'B_6_8': 0.5563709874676577, 'B_6_9': 0.65831954396019, 'B_7_0':
-0.9233147427409942, 'B_7_1': -0.2787193177895266, 'B_7_2':
-0.49493467054751455, 'B_7_3': 0.7843416374575175, 'B_7_4': 0.8604645192635921,
'B_7_5': -0.43702337108972444, 'B_7_6': -0.23170217959994135, 'B_7_7':
-0.7069956683766071, 'B_7_8': -0.9054714592080477, 'B_7_9': 0.9453242066461169,
'B_8_0': -0.711768131295261, 'B_8_1': 0.45483632947395114, 'B_8_2':
0.7727845290831954, 'B_8_3': 0.438643814032769, 'B_8_4': -0.13946789504474394,
'B_8_5': -0.38257640261318454, 'B_8_6': 0.657893075974098, 'B_8_7':
-0.7505759076625775, 'B_8_8': -0.008659569630930286, 'B_8_9':
0.6000018799900091, 'B_9_0': 0.200448257117314, 'B_9_1': -0.7195231836979495,
'B_9_2': -0.768848961032312, 'B_9_3': 0.6014168891770069, 'B_9_4':
-0.8220265294534164, 'B_9_5': -0.44475673009616934, 'B_9_6':
-0.19212384266192217, 'B_9_7': -0.1755631243670271, 'B_9_8': 0.8922802004818976,
'B_9_9': 0.720509268904611, 'B_10_0': 0.514495664974742, 'B_10_1':
-0.3551363534871649, 'B_10_2': -0.8711470652477703, 'B_10_3':
-0.6043329660851903, 'B_10_4': 0.6716327711227504, 'B_10_5':
-0.4808643164190473, 'B_10_6': -0.26422607662522146, 'B_10_7':
0.5358161956974018, 'B_10_8': -0.009237943695200826, 'B_10_9':
0.2381839444117795}. Best is trial 153 with value: 57.61808695399101.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$\langle F_{[u_i]}, F_{[u_j]} \rangle$].

Built matrix B = inverse(A).

[I 2025-04-29 14:29:06,290] Trial 166 finished with value: 62.269029969588814

and parameters: {'B_0_0': 0.3419140206357686, 'B_0_1': 0.318417802792371,

'B_0_2': 0.590638152790506, 'B_0_3': 0.18717550226628502, 'B_0_4':

-0.5779867619723081, 'B_0_5': 0.6994276565708981, 'B_0_6': 0.5382605620679317,

'B_0_7': 0.4374143373977186, 'B_0_8': 0.38079652585598983, 'B_0_9':

0.1745737344401569, 'B_1_0': 0.4447855514695623, 'B_1_1': -0.13930466254370907,

'B_1_2': 0.3487401372677857, 'B_1_3': 0.6186775008907532, 'B_1_4':

-0.47352338502395935, 'B_1_5': 0.8825115091750161, 'B_1_6':

-0.12728769659430067, 'B_1_7': 0.7381545570199477, 'B_1_8': -0.4688419154830832,

'B_1_9': -0.610435080652047, 'B_2_0': -0.6681276275783141, 'B_2_1':

```

-0.5086725679245037, 'B_2_2': -0.5641715379899657, 'B_2_3': 0.9264242717694565,
'B_2_4': -0.03503574253156898, 'B_2_5': -0.6961219823670136, 'B_2_6':
-0.1567109461038375, 'B_2_7': -0.45472818784444213, 'B_2_8':
-0.7232844440172983, 'B_2_9': 0.34488107436826676, 'B_3_0': -0.3957913534377613,
'B_3_1': -0.3521641371975874, 'B_3_2': -0.4012940952112394, 'B_3_3':
-0.13371709551381425, 'B_3_4': -0.5755463039885308, 'B_3_5':
0.46761103725961506, 'B_3_6': -0.15890334043801033, 'B_3_7': 0.5582773112964753,
'B_3_8': 0.35051466099698325, 'B_3_9': -0.08696272734292004, 'B_4_0':
0.2513061697973848, 'B_4_1': 0.18796422212606106, 'B_4_2':
-0.021768549652449648, 'B_4_3': -0.2313393273096972, 'B_4_4':
-0.7397971894769779, 'B_4_5': 0.22515344797420403, 'B_4_6': -0.6712262999255451,
'B_4_7': -0.534033047958512, 'B_4_8': 0.44818671816377387, 'B_4_9':
0.7614802987407501, 'B_5_0': 0.3639701742981424, 'B_5_1': -0.5591665526831623,
'B_5_2': 0.5361910837484423, 'B_5_3': -0.1885243225771762, 'B_5_4':
0.9202928918418005, 'B_5_5': -0.9167280114397609, 'B_5_6':
-0.012083589522677546, 'B_5_7': 0.4190865915283911, 'B_5_8':
-0.1838344921284313, 'B_5_9': 0.4077676271687969, 'B_6_0': 0.22591273793591926,
'B_6_1': -0.5925096366090212, 'B_6_2': 0.5571028824060635, 'B_6_3':
0.8953587869985914, 'B_6_4': -0.4875052756931846, 'B_6_5': -0.0902349355876261,
'B_6_6': 0.18098649615303908, 'B_6_7': 0.5854095111589974, 'B_6_8':
0.5020382267159672, 'B_6_9': 0.7038372075320256, 'B_7_0': -0.8441954620275607,
'B_7_1': -0.2046886264222774, 'B_7_2': -0.330144530475052, 'B_7_3':
0.7044243730896583, 'B_7_4': 0.8823389665092795, 'B_7_5': -0.5458325735700131,
'B_7_6': -0.2883088435355202, 'B_7_7': -0.8152827323696785, 'B_7_8':
-0.989837184207726, 'B_7_9': 0.8516451673675044, 'B_8_0': -0.7369048126134172,
'B_8_1': 0.3435973368009971, 'B_8_2': 0.9073952319106956, 'B_8_3':
0.3016318040582934, 'B_8_4': -0.2068692959643627, 'B_8_5': -0.32005781730704663,
'B_8_6': 0.5901248346932537, 'B_8_7': -0.6852365312501777, 'B_8_8':
-0.03477419860473069, 'B_8_9': 0.6519995658790267, 'B_9_0': 0.14725093422857366,
'B_9_1': -0.7621154229368842, 'B_9_2': -0.6074746816899126, 'B_9_3':
0.7150246190293352, 'B_9_4': -0.7643667874321195, 'B_9_5': -0.5024680462707654,
'B_9_6': -0.4625319204956856, 'B_9_7': -0.12573984130618374, 'B_9_8':
0.9510434488577312, 'B_9_9': 0.822829205549769, 'B_10_0': 0.5644011751378323,
'B_10_1': -0.5039586918652875, 'B_10_2': -0.9096975996654904, 'B_10_3':
-0.4534323569751205, 'B_10_4': 0.6104828004928295, 'B_10_5':
-0.5611677583208874, 'B_10_6': -0.31079153462132136, 'B_10_7':
0.6288509046028329, 'B_10_8': -0.07764053052144484, 'B_10_9':
0.2815084820609568}. Best is trial 153 with value: 57.61808695399101.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].

```

```

    Built F_[u_10].
    Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:29:20,692] Trial 167 finished with value: 61.234510173358686
and parameters: {'B_0_0': 0.331461089924362, 'B_0_1': 0.3567511219321497,
'B_0_2': 0.5974413107563071, 'B_0_3': 0.18106928931801203, 'B_0_4':
-0.5743736303204559, 'B_0_5': 0.8298405982839389, 'B_0_6': 0.5421786402229898,
'B_0_7': 0.4434239613824297, 'B_0_8': 0.35297396546109583, 'B_0_9':
0.2445914109841783, 'B_1_0': 0.611635027786872, 'B_1_1': -0.2159809716010096,
'B_1_2': 0.36891342107109515, 'B_1_3': 0.6329276848051815, 'B_1_4':
-0.4756253126438153, 'B_1_5': 0.801198089068958, 'B_1_6': -0.13216836740472976,
'B_1_7': 0.7086155849783763, 'B_1_8': -0.4687556430894489, 'B_1_9':
-0.6067482997041054, 'B_2_0': -0.8800416547538247, 'B_2_1': -0.4989305995730479,
'B_2_2': -0.5902057252217405, 'B_2_3': 0.9343420739868944, 'B_2_4':
-0.03824855826583315, 'B_2_5': -0.7099025208540259, 'B_2_6':
-0.14461237170030422, 'B_2_7': -0.37359994578399597, 'B_2_8':
-0.7123809762996784, 'B_2_9': 0.3317471284124406, 'B_3_0': -0.4008960837872255,
'B_3_1': -0.35102195269051234, 'B_3_2': -0.44627767643655686, 'B_3_3':
-0.14438436816322103, 'B_3_4': -0.5684124816375984, 'B_3_5': 0.5331219511186123,
'B_3_6': -0.19541540956411535, 'B_3_7': 0.5861741948680022, 'B_3_8':
0.22927645443487493, 'B_3_9': -0.09468091134989143, 'B_4_0':
0.25685419714527424, 'B_4_1': 0.23976426903405523, 'B_4_2': 0.10691916356316379,
'B_4_3': -0.23266658860873937, 'B_4_4': -0.7531930567897631, 'B_4_5':
0.24517728955213638, 'B_4_6': -0.6573750732233588, 'B_4_7': -0.5565174096344467,
'B_4_8': 0.44172893905182137, 'B_4_9': 0.7681952302176646, 'B_5_0':
0.3598595799979668, 'B_5_1': -0.557993839668572, 'B_5_2': 0.7780435695857575,
'B_5_3': -0.19342440983553735, 'B_5_4': 0.8936203475085249, 'B_5_5':
-0.8948079243218724, 'B_5_6': -0.00042701766202306486, 'B_5_7':
0.9714469988936116, 'B_5_8': -0.17369098871043503, 'B_5_9': 0.4142353821654015,
'B_6_0': 0.3223562214118377, 'B_6_1': -0.5919150017438707, 'B_6_2':
0.5570587879404988, 'B_6_3': 0.8239146263566839, 'B_6_4': -0.41051163979120653,
'B_6_5': -0.09117957510329486, 'B_6_6': 0.49761672226086867, 'B_6_7':
0.4889616440871357, 'B_6_8': 0.5161972504355048, 'B_6_9': 0.5823755794048827,
'B_7_0': -0.8421490679450468, 'B_7_1': -0.20679673249671562, 'B_7_2':
-0.33551713343011813, 'B_7_3': 0.7044181159642294, 'B_7_4': 0.8841006037895389,
'B_7_5': -0.6322424365520432, 'B_7_6': -0.28425745125843793, 'B_7_7':
-0.8320451485159636, 'B_7_8': -0.9006192114244713, 'B_7_9': 0.8338254910639464,
'B_8_0': -0.7415598859981141, 'B_8_1': 0.3332490513127832, 'B_8_2':
0.9182067189752755, 'B_8_3': 0.6189701575933634, 'B_8_4': -0.2154220422695191,
'B_8_5': -0.3062130687605375, 'B_8_6': 0.602965965943728, 'B_8_7':
-0.7038492909757762, 'B_8_8': -0.10373658911422343, 'B_8_9': 0.6558859049126678,
'B_9_0': 0.12289156920051356, 'B_9_1': -0.7540135415809941, 'B_9_2':
-0.5962730625596988, 'B_9_3': 0.8550323001722789, 'B_9_4': -0.7628677925551901,
'B_9_5': -0.4973744311446288, 'B_9_6': -0.4969315874744556, 'B_9_7':
-0.14779532242967566, 'B_9_8': 0.96355388707994, 'B_9_9': 0.8283728681716224,
'B_10_0': 0.5966346255309677, 'B_10_1': -0.49910489438159483, 'B_10_2':

```

-0.9052056411264037, 'B_10_3': -0.4686897131810274, 'B_10_4':
0.6962496404660244, 'B_10_5': -0.4069787760788876, 'B_10_6':
-0.3725030971549086, 'B_10_7': 0.6431162288845395, 'B_10_8': 0.0794051797965499,
'B_10_9': 0.2839349640503307}. Best is trial 153 with value: 57.61808695399101.
Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:29:34,689] Trial 168 finished with value: 63.35252690136754 and
parameters: {'B_0_0': 0.4640444856588349, 'B_0_1': 0.35130196145970594, 'B_0_2':
0.592723067629575, 'B_0_3': 0.1972830855107219, 'B_0_4': -0.7521056903029826,
'B_0_5': 0.7189959356200707, 'B_0_6': 0.547525855538386, 'B_0_7':
0.45079730605516566, 'B_0_8': 0.3070615699800771, 'B_0_9': 0.259040058300374,
'B_1_0': 0.6181560140110387, 'B_1_1': -0.22888750054463577, 'B_1_2':
0.36330855103257637, 'B_1_3': 0.6243958468803901, 'B_1_4': -0.4735086529095663,
'B_1_5': 0.7992273955627547, 'B_1_6': -0.1395790147160103, 'B_1_7':
0.7466527078478744, 'B_1_8': -0.4723738187552628, 'B_1_9': -0.6091176454292445,
'B_2_0': -0.8633955048201958, 'B_2_1': -0.5114799593147926, 'B_2_2':
-0.6233172928362518, 'B_2_3': 0.9304548312600713, 'B_2_4': 0.016601168195138627,
'B_2_5': -0.7012580654833729, 'B_2_6': -0.1409703187456878, 'B_2_7':
-0.3814526432521437, 'B_2_8': -0.7197048096357477, 'B_2_9': 0.34815116255296225,
'B_3_0': -0.3969704419433865, 'B_3_1': -0.406744265591625, 'B_3_2':
-0.39941451675012385, 'B_3_3': -0.15755377722079694, 'B_3_4':
-0.5688122969512875, 'B_3_5': 0.4960935592723545, 'B_3_6': -0.17742748249188386,
'B_3_7': 0.5136641804771289, 'B_3_8': 0.21604119259208265, 'B_3_9':
-0.10114041907822954, 'B_4_0': 0.3613895903354362, 'B_4_1': 0.1909357243115945,
'B_4_2': -0.024362684867845236, 'B_4_3': -0.22786476973576866, 'B_4_4':
-0.7758880942470813, 'B_4_5': 0.3202242439627739, 'B_4_6': -0.6667948944454509,
'B_4_7': -0.5316350553706831, 'B_4_8': 0.4452075504621143, 'B_4_9':
0.7678681816403401, 'B_5_0': 0.4895606731135013, 'B_5_1': -0.5669255276535676,
'B_5_2': 0.789236625194847, 'B_5_3': -0.19734985889638135, 'B_5_4':
0.8713404308042028, 'B_5_5': -0.9157361793257216, 'B_5_6':
-0.027564830577611527, 'B_5_7': 0.47955191689043686, 'B_5_8':
-0.17554314545123928, 'B_5_9': 0.42456312726145035, 'B_6_0':
0.32840485183622997, 'B_6_1': -0.47585393096954876, 'B_6_2': 0.5616705465103976,
'B_6_3': 0.8264340215885001, 'B_6_4': -0.40841258169129524, 'B_6_5':
-0.03443304896130262, 'B_6_6': 0.49943447932967966, 'B_6_7': 0.4676927668681397,

'B_6_8': 0.5152034433907161, 'B_6_9': 0.5806417362358386, 'B_7_0':
-0.841384962076182, 'B_7_1': -0.21110381968022732, 'B_7_2': -0.3387696488881796,
'B_7_3': 0.6918831364383363, 'B_7_4': 0.8965637322227624, 'B_7_5':
-0.558611528178784, 'B_7_6': -0.12688945066019805, 'B_7_7': -0.8213851247753992,
'B_7_8': -0.9048121221797458, 'B_7_9': 0.8371821060226127, 'B_8_0':
-0.7343791756715626, 'B_8_1': 0.34500842472825755, 'B_8_2': 0.8985512189179264,
'B_8_3': 0.6195031818989433, 'B_8_4': -0.20757528268996311, 'B_8_5':
-0.32383419705393635, 'B_8_6': 0.5821248143009954, 'B_8_7': -0.6944307541286531,
'B_8_8': -0.1510244248656067, 'B_8_9': 0.6457818456454845, 'B_9_0':
0.11902342302310631, 'B_9_1': -0.632228327673783, 'B_9_2': -0.5972138427053468,
'B_9_3': 0.37590369622399844, 'B_9_4': -0.7808760897060358, 'B_9_5':
-0.4891900083811039, 'B_9_6': -0.46367000143001974, 'B_9_7':
-0.16290029760628483, 'B_9_8': 0.8426132799367607, 'B_9_9': 0.8292443352759648,
'B_10_0': 0.5703988378120365, 'B_10_1': -0.5111846175594384, 'B_10_2':
-0.9020119750924013, 'B_10_3': -0.3683211304577066, 'B_10_4':
0.5090291788071757, 'B_10_5': -0.41080165910550975, 'B_10_6':
-0.3811694497828488, 'B_10_7': 0.6403664753379296, 'B_10_8':
0.10678303397132935, 'B_10_9': 0.2914137866421811}. Best is trial 153 with
value: 57.61808695399101.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].
Built F__[u_10].
Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [_{<F}[u_i], F[u_j]].

Built matrix B = inverse(A).

[I 2025-04-29 14:29:48,982] Trial 169 finished with value: 72.27642783006893 and
parameters: {'B_0_0': 0.36379963552131905, 'B_0_1': 0.3245003906375792, 'B_0_2':
0.4950533180266734, 'B_0_3': 0.34654515104062966, 'B_0_4': -0.5790277848127606,
'B_0_5': 0.8312931887984747, 'B_0_6': 0.6535977138586899, 'B_0_7':
0.5302908457531459, 'B_0_8': 0.34459072503072313, 'B_0_9': 0.14964908192508244,
'B_1_0': 0.4458325828627002, 'B_1_1': -0.28883477877391456, 'B_1_2':
0.4866056257735402, 'B_1_3': 0.6581909336517812, 'B_1_4': -0.5838340607987338,
'B_1_5': 0.795564257261678, 'B_1_6': -0.10312002349133816, 'B_1_7':
0.7122919332339666, 'B_1_8': -0.45873076911352434, 'B_1_9': -0.6438136161965342,
'B_2_0': -0.9113791249911601, 'B_2_1': -0.5688759814811424, 'B_2_2':
-0.7857475914021536, 'B_2_3': 0.9394379974176464, 'B_2_4':
-0.030414496447536174, 'B_2_5': -0.8325786270482388, 'B_2_6':
-0.21880980077771195, 'B_2_7': -0.4259348984496669, 'B_2_8':
-0.6798828813423688, 'B_2_9': 0.35384576681707974, 'B_3_0': -0.3499802146197048,

```

'B_3_1': -0.4903185214039245, 'B_3_2': -0.4382380911666265, 'B_3_3':
-0.19698199882070966, 'B_3_4': -0.70580422889229, 'B_3_5': 0.50069990039684,
'B_3_6': -0.05431468865377191, 'B_3_7': 0.5622360526569562, 'B_3_8':
0.32756392360751857, 'B_3_9': -0.06163692548686283, 'B_4_0': 0.2741448767987851,
'B_4_1': 0.25174114301762035, 'B_4_2': 0.10241382363543056, 'B_4_3':
-0.19517600662746676, 'B_4_4': -0.7423860957766066, 'B_4_5':
0.11430320599715468, 'B_4_6': -0.5107989124294404, 'B_4_7':
-0.39383250895082694, 'B_4_8': 0.40071184274233196, 'B_4_9': 0.8624463142110834,
'B_5_0': 0.39608121352368203, 'B_5_1': -0.650437275337178, 'B_5_2':
0.5503142884738311, 'B_5_3': -0.2825833835456692, 'B_5_4': 0.9070331448846634,
'B_5_5': -0.8829437209537959, 'B_5_6': -0.0013478334890007163, 'B_5_7':
0.16716801820784913, 'B_5_8': -0.19439214934743815, 'B_5_9': 0.4885764147508206,
'B_6_0': 0.227178997321457, 'B_6_1': -0.59543607379041, 'B_6_2':
0.7029032356714059, 'B_6_3': 0.8958128429142054, 'B_6_4': -0.48840242594936256,
'B_6_5': 0.11324612791970078, 'B_6_6': 0.37526311975800153, 'B_6_7':
0.7417059176621839, 'B_6_8': 0.45252101634453573, 'B_6_9': 0.4877582989000313,
'B_7_0': -0.8344393519727906, 'B_7_1': -0.2600078431712191, 'B_7_2':
-0.4224842351317893, 'B_7_3': 0.7596428874724628, 'B_7_4': 0.8769198378987929,
'B_7_5': -0.6502420714284539, 'B_7_6': -0.3164386422931298, 'B_7_7':
-0.8158496949492373, 'B_7_8': -0.9884412281440376, 'B_7_9': 0.7807150811153156,
'B_8_0': -0.6583520973036839, 'B_8_1': 0.2972604350759628, 'B_8_2':
0.9251811190525057, 'B_8_3': 0.5022246699142319, 'B_8_4': 0.07707427774551712,
'B_8_5': -0.4134697252977905, 'B_8_6': 0.6965261915150012, 'B_8_7':
-0.6635019018414845, 'B_8_8': -0.0847737527374546, 'B_8_9': 0.5797511736360105,
'B_9_0': -0.00866095593408897, 'B_9_1': -0.5296719088452484, 'B_9_2':
-0.6653692722017193, 'B_9_3': 0.7142921587003823, 'B_9_4': -0.8562114834561555,
'B_9_5': -0.5480660645012352, 'B_9_6': -0.4652415991131411, 'B_9_7':
-0.06317961329184171, 'B_9_8': 0.9213018023755467, 'B_9_9': 0.6958855158614345,
'B_10_0': 0.47710332941100303, 'B_10_1': -0.4015537335724309, 'B_10_2':
-0.9044248710901356, 'B_10_3': -0.45066414073231315, 'B_10_4':
0.7525510843715181, 'B_10_5': -0.5680435706091262, 'B_10_6':
-0.32372932089793743, 'B_10_7': 0.6262700526120729, 'B_10_8':
0.07053107968921446, 'B_10_9': 0.3611670757625287}. Best is trial 153 with
value: 57.61808695399101.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix $B = \text{inverse}(A)$.

[I 2025-04-29 14:30:03,552] Trial 170 finished with value: 64.67347661843587 and parameters: {'B_0_0': 0.36586774046434123, 'B_0_1': 0.17621137521808183, 'B_0_2': 0.5739976451664073, 'B_0_3': 0.13702678535981472, 'B_0_4': -0.6624729757684652, 'B_0_5': 0.9520096094581223, 'B_0_6': 0.5314906168758916, 'B_0_7': 0.43154441752163997, 'B_0_8': 0.6254763134569408, 'B_0_9': 0.2392863792488099, 'B_1_0': 0.5956065802589063, 'B_1_1': -0.18330591816046643, 'B_1_2': 0.4214444231506514, 'B_1_3': 0.6192619190309583, 'B_1_4': -0.43341386925280057, 'B_1_5': 0.9408157578074846, 'B_1_6': -0.05648870890009097, 'B_1_7': 0.6020219751418591, 'B_1_8': -0.4778740117667749, 'B_1_9': -0.5456919022632267, 'B_2_0': -0.8145841277521985, 'B_2_1': -0.4250611406361236, 'B_2_2': -0.6910495664196894, 'B_2_3': 0.8654528757175222, 'B_2_4': -0.11509002602591212, 'B_2_5': -0.7182617098994865, 'B_2_6': -0.08803506003278277, 'B_2_7': -0.29779766828700227, 'B_2_8': -0.8627002961907317, 'B_2_9': 0.22917739751798846, 'B_3_0': -0.5184486638621277, 'B_3_1': -0.36261125340701617, 'B_3_2': -0.49066856921687546, 'B_3_3': -0.11757996261820097, 'B_3_4': -0.6314534111127079, 'B_3_5': 0.5625458088861609, 'B_3_6': -0.34985780440317693, 'B_3_7': 0.5990070989372318, 'B_3_8': 0.3431311039929781, 'B_3_9': 0.11586070462285099, 'B_4_0': 0.41846106965192653, 'B_4_1': 0.13876424923565156, 'B_4_2': 0.0260036714795472, 'B_4_3': -0.2651570451253081, 'B_4_4': -0.9787090075146626, 'B_4_5': 0.2319473491538692, 'B_4_6': -0.6585175201503906, 'B_4_7': -0.5621227062869097, 'B_4_8': 0.33712415878894614, 'B_4_9': 0.9057116125515398, 'B_5_0': 0.3615265353803726, 'B_5_1': -0.6208208657581294, 'B_5_2': 0.6654392840842582, 'B_5_3': -0.16544608703539107, 'B_5_4': 0.8501164907725722, 'B_5_5': -0.8242942780876404, 'B_5_6': -0.1230543562112939, 'B_5_7': 0.6279567151892806, 'B_5_8': -0.2697448430703464, 'B_5_9': 0.4110317381754876, 'B_6_0': 0.42217150427508254, 'B_6_1': -0.5343177176882901, 'B_6_2': 0.4650639469249236, 'B_6_3': 0.9522475044284838, 'B_6_4': -0.5789749748311378, 'B_6_5': -0.08856590226277967, 'B_6_6': 0.18368870680168645, 'B_6_7': 0.5407375469245289, 'B_6_8': 0.4818888936675643, 'B_6_9': 0.6861287101334947, 'B_7_0': -0.7329261946836196, 'B_7_1': -0.15234952187458134, 'B_7_2': -0.2822706321732697, 'B_7_3': 0.6583648363525602, 'B_7_4': 0.81586927290962, 'B_7_5': -0.6107103061767891, 'B_7_6': -0.27988353250157605, 'B_7_7': -0.7532449142494084, 'B_7_8': -0.8306375103861134, 'B_7_9': 0.926864788035414, 'B_8_0': -0.7426894175763763, 'B_8_1': 0.22815370953363234, 'B_8_2': 0.9295534072781577, 'B_8_3': 0.9826252205449886, 'B_8_4': -0.14239119292416125, 'B_8_5': -0.2643078990129493, 'B_8_6': 0.5572907087345788, 'B_8_7': -0.6127231098228476, 'B_8_8': -0.06369763924410747, 'B_8_9': 0.526961921083583, 'B_9_0': 0.15581794615653724, 'B_9_1': -0.7666199164171, 'B_9_2': -0.8057946422774156, 'B_9_3': 0.6176386479626292, 'B_9_4': -0.830290086749318, 'B_9_5': -0.50903913205331, 'B_9_6': -0.5584810117656387, 'B_9_7': -0.13793806177599038, 'B_9_8': 0.9555313822942565, 'B_9_9': 0.6029980530093999, 'B_10_0': 0.600821502545221, 'B_10_1': -0.5940528696791708, 'B_10_2': -0.8650379423160222, 'B_10_3': -0.47788589219803035, 'B_10_4': 0.6921262456511975, 'B_10_5': -0.5002526512371951, 'B_10_6': -0.42811393535311637, 'B_10_7': 0.5204805643015931, 'B_10_8': 0.012584396371337264, 'B_10_9': 0.2382705972149852}. Best is trial 153 with value: 57.61808695399101.

Building the Basis functions for the RKHS:

```
Built F_[u_1].  
Built F_[u_2].  
Built F_[u_3].  
Built F_[u_4].  
Built F_[u_5].  
Built F_[u_6].  
Built F_[u_7].  
Built F_[u_8].  
Built F_[u_9].  
Built F_[u_10].  
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:30:18,179] Trial 171 finished with value: 67.45523510939854 and parameters: {'B_0_0': 0.3341976117139682, 'B_0_1': 0.2619832856429142, 'B_0_2': 0.6608759605581652, 'B_0_3': 0.18265483947602157, 'B_0_4': -0.7063264259410211, 'B_0_5': 0.8639810409034574, 'B_0_6': 0.5785347155379951, 'B_0_7': 0.4736763724005496, 'B_0_8': 0.36133047230323745, 'B_0_9': 0.17966660223606284, 'B_1_0': 0.504432627703672, 'B_1_1': -0.12550240858869458, 'B_1_2': 0.37914930684473785, 'B_1_3': 0.7016451854620374, 'B_1_4': -0.3821395707639557, 'B_1_5': 0.8901150388035254, 'B_1_6': -0.2689076069088061, 'B_1_7': 0.7586583646294989, 'B_1_8': -0.54933358180573, 'B_1_9': -0.5946501819841861, 'B_2_0': -0.670692854042074, 'B_2_1': -0.4829168776720086, 'B_2_2': -0.5686815740693079, 'B_2_3': 0.7845396797970668, 'B_2_4': -0.1758164616989189, 'B_2_5': -0.6736588918949253, 'B_2_6': -0.1587694181059084, 'B_2_7': -0.4606464799071971, 'B_2_8': -0.5861107272608614, 'B_2_9': 0.3240836826192315, 'B_3_0': -0.40323772926685575, 'B_3_1': -0.3028309136825904, 'B_3_2': -0.41967676868937287, 'B_3_3': -0.07230923401662664, 'B_3_4': -0.6695041442715075, 'B_3_5': 0.41430766226977334, 'B_3_6': -0.19641507405322636, 'B_3_7': 0.3822684026290013, 'B_3_8': 0.3952633220897128, 'B_3_9': -0.08161978204066915, 'B_4_0': 0.2202561304798565, 'B_4_1': 0.27187130453638464, 'B_4_2': -0.15124916507408853, 'B_4_3': -0.14605710436184807, 'B_4_4': -0.8524699245720022, 'B_4_5': 0.18375698164685073, 'B_4_6': -0.5776359132191599, 'B_4_7': -0.4931528372869589, 'B_4_8': 0.49068540639529995, 'B_4_9': 0.7387244227761476, 'B_5_0': 0.3344133687566362, 'B_5_1': -0.550510146890873, 'B_5_2': 0.6994717397762713, 'B_5_3': -0.1456135945141969, 'B_5_4': 0.7366617073569365, 'B_5_5': -0.9407059422880418, 'B_5_6': 0.05131254462543995, 'B_5_7': 0.42761714699192854, 'B_5_8': -0.23321062148694197, 'B_5_9': 0.5417447877169969, 'B_6_0': 0.30831268081896523, 'B_6_1': -0.6168562144261853, 'B_6_2': 0.6281335429012905, 'B_6_3': 0.9541799512958522, 'B_6_4': -0.43644342008759707, 'B_6_5': -0.001042852307276857, 'B_6_6': 0.31842705093915463, 'B_6_7': 0.5010798401683191, 'B_6_8': 0.5486506360385665, 'B_6_9': 0.6374928528169377, 'B_7_0': -0.9267801356237414, 'B_7_1': -0.4952450755304664, 'B_7_2': -0.4452383314341247, 'B_7_3': 0.705630680249739, 'B_7_4': 0.9242934962397691, 'B_7_5': -0.5403345325930413, 'B_7_6': -0.22670953996085386, 'B_7_7':


```

-0.8024133891606896, 'B_7_8': -0.8989221511689167, 'B_7_9': 0.8679149130949512,
'B_8_0': -0.5707798977967863, 'B_8_1': 0.2597289427324696, 'B_8_2':
0.9562545748497014, 'B_8_3': 0.29801870354317334, 'B_8_4': -0.09590258702682605,
'B_8_5': -0.3617022723125628, 'B_8_6': 0.612484211332504, 'B_8_7':
-0.771996657294288, 'B_8_8': 0.02681683712217906, 'B_8_9': 0.7369817666678941,
'B_9_0': 0.20091805233488422, 'B_9_1': -0.7191104859603403, 'B_9_2':
-0.6811864271537837, 'B_9_3': 0.85739796509022, 'B_9_4': -0.6282700376954578,
'B_9_5': -0.4069137637638397, 'B_9_6': -0.40479707853677027, 'B_9_7':
-0.237860134596581, 'B_9_8': 0.9599386932296481, 'B_9_9': 0.7341427780520052,
'B_10_0': 0.6706131883775369, 'B_10_1': -0.4922291680795407, 'B_10_2':
-0.9895393760945314, 'B_10_3': -0.4296162309370556, 'B_10_4':
0.6060513528717157, 'B_10_5': -0.44679465656486106, 'B_10_6':
-0.2868398859705446, 'B_10_7': 0.6861601156613308, 'B_10_8':
-0.03884920782987204, 'B_10_9': 0.14250550629130515}. Best is trial 153 with
value: 57.61808695399101.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$\langle F_{[u_i]}, F_{[u_j]} \rangle$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:30:32,294] Trial 172 finished with value: 82.53329936941118 and
parameters: {'B_0_0': 0.4414820390720494, 'B_0_1': 0.140828173095502, 'B_0_2':
0.6290323659720101, 'B_0_3': 0.2952571333435496, 'B_0_4': -0.5581721277409339,
'B_0_5': 0.67811949377053, 'B_0_6': 0.6017319257884743, 'B_0_7':
0.541016124547991, 'B_0_8': 0.425450100885884, 'B_0_9': 0.27511036413097784,
'B_1_0': 0.4711513058488167, 'B_1_1': -0.25505759162669844, 'B_1_2':
0.5532635325840993, 'B_1_3': 0.7901809928217969, 'B_1_4': -0.5362273129375856,
'B_1_5': 0.7968166021192645, 'B_1_6': -0.16997164696347428, 'B_1_7':
0.6866360844693268, 'B_1_8': -0.5583327155404257, 'B_1_9': -0.6950110837381214,
'B_2_0': -0.7475068269361964, 'B_2_1': -0.5345562172519022, 'B_2_2':
-0.5969358405927214, 'B_2_3': 0.9042199458074106, 'B_2_4': -0.04475867023852098,
'B_2_5': -0.7080356031017643, 'B_2_6': -0.2741490396298106, 'B_2_7':
-0.3600898585076964, 'B_2_8': -0.6428506304493234, 'B_2_9': 0.3218919950326954,
'B_3_0': -0.4911164323953572, 'B_3_1': -0.35072721267745893, 'B_3_2':
-0.31364808289581186, 'B_3_3': -0.04515759640291062, 'B_3_4':
-0.7824176945928523, 'B_3_5': 0.5287402766973409, 'B_3_6': -0.24585114466855004,
'B_3_7': 0.733367353589384, 'B_3_8': 0.24578814172498206, 'B_3_9':
-0.03638253518002457, 'B_4_0': 0.26004686752561507, 'B_4_1':

```

```

0.08206344582928946, 'B_4_2': 0.10381149894628888, 'B_4_3':
-0.23564914169744014, 'B_4_4': -0.9118811772293276, 'B_4_5': 0.283370555420824,
'B_4_6': -0.674877402847573, 'B_4_7': -0.4625933200773658, 'B_4_8':
0.4557099806382921, 'B_4_9': 0.8195664851150783, 'B_5_0': 0.23971423015580193,
'B_5_1': -0.6989336935918785, 'B_5_2': 0.5250169110503612, 'B_5_3':
-0.19243524420913605, 'B_5_4': 0.9247775795167447, 'B_5_5': -0.881399265009197,
'B_5_6': -0.07929159054681975, 'B_5_7': 0.3323712780585397, 'B_5_8':
-0.14364281999412704, 'B_5_9': 0.5081990822628992, 'B_6_0': 0.16201060261043065,
'B_6_1': -0.5797445474618212, 'B_6_2': 0.5781208823677311, 'B_6_3':
0.9010776586943355, 'B_6_4': -0.6634777396061391, 'B_6_5': -0.0779229339295269,
'B_6_6': 0.21873523553148796, 'B_6_7': 0.5971566165413565, 'B_6_8':
0.5107359259881485, 'B_6_9': 0.6993729580229021, 'B_7_0': -0.9280878028004833,
'B_7_1': -0.3554198710693597, 'B_7_2': -0.33257166179816283, 'B_7_3':
0.7918749147526067, 'B_7_4': 0.9477875679542547, 'B_7_5': -0.44835063672552045,
'B_7_6': -0.2728373123909418, 'B_7_7': -0.7014680032260856, 'B_7_8':
-0.9474204013666018, 'B_7_9': 0.8135701185980282, 'B_8_0': -0.7009263662461781,
'B_8_1': 0.47155189503425055, 'B_8_2': 0.7987259128340456, 'B_8_3':
0.37222080468581487, 'B_8_4': -0.2094273589862575, 'B_8_5':
-0.30436385667718036, 'B_8_6': 0.4737050738793961, 'B_8_7': -0.7133104875085491,
'B_8_8': -0.014044699886626261, 'B_8_9': 0.6280970156581509, 'B_9_0':
0.0790499520097238, 'B_9_1': -0.7516479219531097, 'B_9_2': -0.636998284582788,
'B_9_3': 0.9144077697211004, 'B_9_4': -0.7573181411204046, 'B_9_5':
-0.5448726381519382, 'B_9_6': -0.28226014315359194, 'B_9_7':
-0.1800723203434188, 'B_9_8': 0.8764120550597216, 'B_9_9': 0.8236529593866166,
'B_10_0': 0.723700567038679, 'B_10_1': -0.4575655527063519, 'B_10_2':
-0.941578098210576, 'B_10_3': -0.3010868327845121, 'B_10_4': 0.6928741355250643,
'B_10_5': -0.4023586028903823, 'B_10_6': -0.22821303435137105, 'B_10_7':
0.7339744036919285, 'B_10_8': -0.061762833336290054, 'B_10_9':
0.08834077504724873}. Best is trial 153 with value: 57.61808695399101.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:30:46,797] Trial 173 finished with value: 66.7868749462815 and
parameters: {'B_0_0': 0.4846413328326358, 'B_0_1': 0.18874447238229805, 'B_0_2':
0.6042119319032002, 'B_0_3': 0.24428429964605392, 'B_0_4': -0.6093073109017141,
'B_0_5': 0.8830936183529698, 'B_0_6': 0.443844877448364, 'B_0_7':

```

0.4964396487828493, 'B_0_8': 0.30122707161817935, 'B_0_9': 0.130772787549652,
'B_1_0': 0.5495625270251342, 'B_1_1': -0.13045334985746526, 'B_1_2':
0.46408135762154057, 'B_1_3': 0.6914047007540883, 'B_1_4': -0.46655043290517023,
'B_1_5': 0.9385182232072533, 'B_1_6': -0.24631876723420934, 'B_1_7':
0.7225526934073875, 'B_1_8': -0.6745856017939722, 'B_1_9': -0.5064888204870274,
'B_2_0': -0.6233560512836649, 'B_2_1': -0.38799763132266424, 'B_2_2':
-0.5654287395054284, 'B_2_3': 0.9358499446787213, 'B_2_4': -0.2377000069978607,
'B_2_5': -0.6174194044429768, 'B_2_6': -0.3503332187618135, 'B_2_7':
-0.5719559320948705, 'B_2_8': -0.7289187925676361, 'B_2_9': 0.4524862320031262,
'B_3_0': -0.5405771783446108, 'B_3_1': -0.40690420043022957, 'B_3_2':
-0.362354858851475, 'B_3_3': -0.06354717371938666, 'B_3_4': -0.5617602836921994,
'B_3_5': 0.26414873389641974, 'B_3_6': -0.14338163234835072, 'B_3_7':
0.6184144261376476, 'B_3_8': 0.3778759965363694, 'B_3_9': 0.06513573331861588,
'B_4_0': 0.31924686697481847, 'B_4_1': 0.17359584523614768, 'B_4_2':
-0.06549694572079784, 'B_4_3': -0.386487638334824, 'B_4_4': -0.7870596235500159,
'B_4_5': 0.25309384678067204, 'B_4_6': -0.7049774429028621, 'B_4_7':
-0.4019564150019492, 'B_4_8': 0.37432097194255975, 'B_4_9': 0.7700789579249963,
'B_5_0': 0.44206867983151993, 'B_5_1': -0.7949740499279394, 'B_5_2':
0.5808886425561955, 'B_5_3': -0.11359997389802587, 'B_5_4': 0.7787749485095048,
'B_5_5': -0.9241105071073811, 'B_5_6': 0.0007771775730310449, 'B_5_7':
0.2615705276667933, 'B_5_8': -0.2587773397249254, 'B_5_9': 0.45765954915957485,
'B_6_0': 0.26654927162804254, 'B_6_1': -0.5190613217061719, 'B_6_2':
0.7436698403636637, 'B_6_3': 0.9871644359748333, 'B_6_4': -0.6082218972821711,
'B_6_5': 0.07893894199497169, 'B_6_6': 0.4486132340508896, 'B_6_7':
0.41050320204958485, 'B_6_8': 0.5748836998056743, 'B_6_9': 0.600998693503473,
'B_7_0': -0.8747760725915321, 'B_7_1': -0.3065349380004542, 'B_7_2':
-0.5173903570084512, 'B_7_3': 0.5823720416381426, 'B_7_4': 0.8979722359419746,
'B_7_5': -0.5183870877985403, 'B_7_6': -0.3272799971864464, 'B_7_7':
-0.5557943723836789, 'B_7_8': -0.9019601987399573, 'B_7_9': 0.7639322510999451,
'B_8_0': -0.7376384817589507, 'B_8_1': 0.34302589683149515, 'B_8_2':
0.8897265631136385, 'B_8_3': 0.4643533860377663, 'B_8_4': -0.15306357025489736,
'B_8_5': -0.39809828388632573, 'B_8_6': 0.5737175704703837, 'B_8_7':
-0.8011676388139449, 'B_8_8': -0.10246658352107382, 'B_8_9': 0.6930838128995255,
'B_9_0': 0.1394970331734477, 'B_9_1': -0.7028568665366273, 'B_9_2':
-0.7556856084477517, 'B_9_3': 0.6999620944606683, 'B_9_4': -0.7187928710418545,
'B_9_5': -0.6433235126103455, 'B_9_6': -0.16522531077730862, 'B_9_7':
-0.30184327763101676, 'B_9_8': 0.8085573178448193, 'B_9_9': 0.6692411832197483,
'B_10_0': 0.5171606048281479, 'B_10_1': -0.29272077200538305, 'B_10_2':
-0.9198055727134266, 'B_10_3': -0.5633703765642616, 'B_10_4':
0.6620027308335502, 'B_10_5': -0.5407145178209505, 'B_10_6':
-0.36363116950723434, 'B_10_7': 0.7815220091721138, 'B_10_8':
0.19524735462038706, 'B_10_9': 0.2772565156519737}. Best is trial 153 with
value: 57.61808695399101.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].

```

Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```

[I 2025-04-29 14:31:01,142] Trial 174 finished with value: 55.46637859185029 and
parameters: {'B_0_0': 0.5608377190870901, 'B_0_1': 0.3691204870771957, 'B_0_2':
0.689059353945652, 'B_0_3': 0.11853810367696477, 'B_0_4': -0.45392486414364275,
'B_0_5': 0.8231701849077839, 'B_0_6': 0.5334812679277712, 'B_0_7':
0.4339271821906765, 'B_0_8': 0.40095812400359093, 'B_0_9': 0.21699242722296813,
'B_1_0': 0.5037739916309686, 'B_1_1': -0.19859862795310868, 'B_1_2':
0.3343339481897833, 'B_1_3': 0.6151097483834207, 'B_1_4': -0.4098771467177668,
'B_1_5': 0.8864357501322666, 'B_1_6': -0.1188414425093795, 'B_1_7':
0.5310274002628917, 'B_1_8': -0.6113077068462368, 'B_1_9': -0.5376705100508947,
'B_2_0': -0.8467110989401998, 'B_2_1': -0.4573034810037547, 'B_2_2':
-0.6250972909421758, 'B_2_3': 0.8651755012581981, 'B_2_4': -0.1368477521264802,
'B_2_5': -0.6784227166982032, 'B_2_6': -0.24186291826621517, 'B_2_7':
-0.5024932196621796, 'B_2_8': -0.6869034621576818, 'B_2_9': 0.2288501344648558,
'B_3_0': -0.43397742659544863, 'B_3_1': -0.2936819122718262, 'B_3_2':
-0.260432431560655, 'B_3_3': -0.1479233496073371, 'B_3_4': -0.725650190996076,
'B_3_5': 0.4616852109148943, 'B_3_6': -0.07366442729457157, 'B_3_7':
0.4257264539030465, 'B_3_8': 0.1782006012195811, 'B_3_9': -0.13427585451077662,
'B_4_0': 0.14285438271705986, 'B_4_1': 0.28991414336198473, 'B_4_2':
0.16952285893107524, 'B_4_3': -0.192862652594385, 'B_4_4': -0.8373276752013409,
'B_4_5': 0.32747064450870955, 'B_4_6': -0.6346603724100461, 'B_4_7':
-0.32469700128270595, 'B_4_8': 0.5152176968801417, 'B_4_9': 0.9291137369312503,
'B_5_0': 0.29725522225218665, 'B_5_1': -0.6013751162947403, 'B_5_2':
0.45176057349500465, 'B_5_3': -0.19650551524650256, 'B_5_4': 0.6914681516044425,
'B_5_5': -0.9885026211054312, 'B_5_6': -0.12266919695156078, 'B_5_7':
0.9888628029866736, 'B_5_8': -0.1795183471875904, 'B_5_9': 0.5569192951592477,
'B_6_0': 0.22119247761053815, 'B_6_1': -0.6432107156665827, 'B_6_2':
0.5295613668199292, 'B_6_3': 0.9546012777357761, 'B_6_4': -0.472046937977992,
'B_6_5': -0.05731795638699545, 'B_6_6': 0.16874733332663305, 'B_6_7':
0.5329034019306728, 'B_6_8': 0.43372126418244256, 'B_6_9': 0.6619953579369797,
'B_7_0': -0.8434739748971654, 'B_7_1': -0.23627407026640038, 'B_7_2':
-0.2496236883879152, 'B_7_3': 0.8351352300294271, 'B_7_4': 0.9864175111624608,
'B_7_5': -0.5874617458018225, 'B_7_6': -0.2738871581400583, 'B_7_7':
-0.8390837860710847, 'B_7_8': -0.9396382464392812, 'B_7_9': 0.9638787793104947,
'B_8_0': -0.635350966839773, 'B_8_1': 0.3962504057553418, 'B_8_2':
0.9895298314525842, 'B_8_3': 0.649324755544547, 'B_8_4': -0.038371261947561776,
'B_8_5': -0.35256274498855394, 'B_8_6': 0.6195723688161705, 'B_8_7':
-0.681537670157586, 'B_8_8': 0.0796840559106727, 'B_8_9': 0.7448809843304165,
'B_9_0': 0.022596461209072674, 'B_9_1': -0.634482815917382, 'B_9_2':

```

```

-0.7882394131861187, 'B_9_3': 0.5272940150800831, 'B_9_4': -0.5993577953794988,
'B_9_5': -0.49746247871232574, 'B_9_6': -0.6318624598463249, 'B_9_7':
-0.02137080177286349, 'B_9_8': 0.986994978937746, 'B_9_9': 0.7534390867315218,
'B_10_0': 0.6245572364461374, 'B_10_1': -0.5223541927629842, 'B_10_2':
-0.9898932264865556, 'B_10_3': -0.23508161808295294, 'B_10_4':
0.6036844681820756, 'B_10_5': -0.3339808505471673, 'B_10_6':
-0.48001611581896064, 'B_10_7': 0.7124140780999487, 'B_10_8':
0.039376252057709435, 'B_10_9': 0.1332970983425773}. Best is trial 174 with
value: 55.46637859185029.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:31:15,692] Trial 175 finished with value: 58.53586527187915 and
parameters: {'B_0_0': 0.5485608954022696, 'B_0_1': 0.3053870334253306, 'B_0_2':
0.5591217422920448, 'B_0_3': 0.1262758367197785, 'B_0_4': -0.4489851638241278,
'B_0_5': 0.8264376479241814, 'B_0_6': 0.5379887178393299, 'B_0_7':
0.3563093618075779, 'B_0_8': 0.4337147532562644, 'B_0_9': 0.23400050075914505,
'B_1_0': 0.5189310978683389, 'B_1_1': -0.205514559549569, 'B_1_2':
0.34138375210926, 'B_1_3': 0.4982664440814738, 'B_1_4': -0.39296162883106933,
'B_1_5': 0.8947150551566528, 'B_1_6': -0.14227015038818053, 'B_1_7':
0.5338356436367216, 'B_1_8': -0.7525076471635948, 'B_1_9': -0.530381119633351,
'B_2_0': -0.8862936629168378, 'B_2_1': -0.46653420742153573, 'B_2_2':
-0.6324005884138929, 'B_2_3': 0.858216749934796, 'B_2_4': -0.14995538814825604,
'B_2_5': -0.6744315719998742, 'B_2_6': -0.23162613941884652, 'B_2_7':
-0.5075581782875489, 'B_2_8': -0.6766785661684559, 'B_2_9': 0.15955172339762566,
'B_3_0': -0.3002879807740255, 'B_3_1': -0.4052003630704162, 'B_3_2':
-0.1861700875409321, 'B_3_3': -0.14759963408180396, 'B_3_4':
-0.7263597933530478, 'B_3_5': 0.3852951486814522, 'B_3_6':
-0.021550067234890107, 'B_3_7': 0.41796327665707345, 'B_3_8':
0.32437198109949883, 'B_3_9': -0.11811890525313629, 'B_4_0':
0.13376036128491045, 'B_4_1': 0.23239138641936474, 'B_4_2': 0.15863097043130514,
'B_4_3': -0.19331156441441158, 'B_4_4': -0.8365089359659874, 'B_4_5':
0.32677411938246614, 'B_4_6': -0.6351373803363769, 'B_4_7': -0.6710358829196679,
'B_4_8': 0.5271204623790228, 'B_4_9': 0.8747973164680655, 'B_5_0':
0.30575448813751666, 'B_5_1': -0.6659845156180795, 'B_5_2': 0.6237435131725916,
'B_5_3': -0.18902779471393916, 'B_5_4': 0.7065046297567423, 'B_5_5':

```

-0.9859903948611831, 'B_5_6': -0.13971498795704007, 'B_5_7': 0.4233151214569315, 'B_5_8': -0.17000443096074325, 'B_5_9': 0.39582049905916905, 'B_6_0': 0.2175515466536278, 'B_6_1': -0.6353779233826623, 'B_6_2': 0.5277127310241023, 'B_6_3': 0.8257126862339426, 'B_6_4': -0.5001346122208754, 'B_6_5': -0.03683183284865381, 'B_6_6': 0.16416177157000586, 'B_6_7': 0.5924367051335271, 'B_6_8': 0.3640162546923835, 'B_6_9': 0.6555120628109781, 'B_7_0': -0.8308259994834258, 'B_7_1': -0.2180458206174152, 'B_7_2': -0.3707312978291956, 'B_7_3': 0.7304253575672008, 'B_7_4': 0.986177537576347, 'B_7_5': -0.6454281148090385, 'B_7_6': -0.1471487699329924, 'B_7_7': -0.7730295319319417, 'B_7_8': -0.891856896550513, 'B_7_9': 0.988120003762795, 'B_8_0': -0.529697998218599, 'B_8_1': 0.3983639985255307, 'B_8_2': 0.9268054067749953, 'B_8_3': 0.6388510624247847, 'B_8_4': -0.027664991200432254, 'B_8_5': -0.2665664167321545, 'B_8_6': 0.7503243947495899, 'B_8_7': -0.679032039892523, 'B_8_8': 0.08631166370535162, 'B_8_9': 0.7394525287008464, 'B_9_0': 0.013463424442595795, 'B_9_1': -0.5740365195663495, 'B_9_2': -0.7847917114330141, 'B_9_3': 0.5669743375834871, 'B_9_4': -0.7727467281684757, 'B_9_5': -0.49488419883311224, 'B_9_6': -0.5159213764301627, 'B_9_7': -0.026561608976472, 'B_9_8': 0.9749014041116421, 'B_9_9': 0.752262506004205, 'B_10_0': 0.6230921192001769, 'B_10_1': -0.5446633131818615, 'B_10_2': -0.9897580746305301, 'B_10_3': -0.22555147376493614, 'B_10_4': 0.7508020360023752, 'B_10_5': -0.3373130711475371, 'B_10_6': -0.4826159753332669, 'B_10_7': 0.6563131618405286, 'B_10_8': 0.03214697783093211, 'B_10_9': 0.24752396776758814}. Best is trial 174 with value: 55.46637859185029.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:31:30,724] Trial 176 finished with value: 64.57612451271352 and parameters: {'B_0_0': 0.5658103548941704, 'B_0_1': 0.3620759682982449, 'B_0_2': 0.5445685104353741, 'B_0_3': 0.13648795475231346, 'B_0_4': -0.5533904810522012, 'B_0_5': 0.8240235077735447, 'B_0_6': 0.4592329136677964, 'B_0_7': 0.23935150374544334, 'B_0_8': 0.41917875211524835, 'B_0_9': 0.38263821096861783, 'B_1_0': 0.5729412775662105, 'B_1_1': -0.19834655549783334, 'B_1_2': 0.33580931822721066, 'B_1_3': 0.5007290836054906, 'B_1_4': -0.4141432684999354, 'B_1_5': 0.8735675490816769, 'B_1_6': -0.17258524957855614, 'B_1_7': 0.5211495273294114, 'B_1_8': -0.8199246341568555, 'B_1_9': -0.5300263031351972,

'B_2_0': -0.8622200647275947, 'B_2_1': -0.46249009470802405, 'B_2_2':
 -0.5113401966408995, 'B_2_3': 0.8582411919111864, 'B_2_4': -0.14130469901399392,
 'B_2_5': -0.5035327471998847, 'B_2_6': -0.22548474071224123, 'B_2_7':
 -0.5224753801687403, 'B_2_8': -0.6880487705629164, 'B_2_9': 0.14515579440220588,
 'B_3_0': -0.22245474107764165, 'B_3_1': -0.4898950655068336, 'B_3_2':
 -0.26583744126904196, 'B_3_3': -0.158892507147436, 'B_3_4': -0.8208670814973392,
 'B_3_5': 0.45293730958773815, 'B_3_6': -0.01223264604034624, 'B_3_7':
 0.4675118014129892, 'B_3_8': 0.16899003202205262, 'B_3_9': -0.11928661503529667,
 'B_4_0': 0.12535720309363965, 'B_4_1': 0.2247617123758572, 'B_4_2':
 0.18997819399357962, 'B_4_3': -1.6862834212483246e-05, 'B_4_4':
 -0.8846023314634828, 'B_4_5': 0.34971140189764494, 'B_4_6': -0.637965435522762,
 'B_4_7': -0.6840691406678945, 'B_4_8': 0.4285310228860741, 'B_4_9':
 0.9841928427004765, 'B_5_0': 0.37736582826391685, 'B_5_1': -0.5975374572838383,
 'B_5_2': 0.6470196553217283, 'B_5_3': -0.1932796557850751, 'B_5_4':
 0.6973917102496651, 'B_5_5': -0.9414943204730974, 'B_5_6': -0.13935013333787016,
 'B_5_7': 0.9059941017812795, 'B_5_8': -0.17512125942562873, 'B_5_9':
 0.3049397813491508, 'B_6_0': 0.185493098761206, 'B_6_1': -0.6435204784618493,
 'B_6_2': 0.5274945612761865, 'B_6_3': 0.884628863451455, 'B_6_4':
 -0.3734406225675736, 'B_6_5': -0.04014502610030195, 'B_6_6':
 0.16267753026949344, 'B_6_7': 0.47461227709704606, 'B_6_8': 0.361001865569486,
 'B_6_9': 0.6576691703076413, 'B_7_0': -0.8385924960955, 'B_7_1':
 -0.2492810498056654, 'B_7_2': -0.22676120017964296, 'B_7_3': 0.8303272190527651,
 'B_7_4': 0.8630478988974077, 'B_7_5': -0.7083210327809664, 'B_7_6':
 -0.21917864232957018, 'B_7_7': -0.7855764654925814, 'B_7_8':
 -0.8336757441037844, 'B_7_9': 0.9822678585163216, 'B_8_0': -0.6089946425523785,
 'B_8_1': 0.40611043354663057, 'B_8_2': 0.9238508226230009, 'B_8_3':
 0.637666973400075, 'B_8_4': -0.016028926299759196, 'B_8_5': -0.1599307915560391,
 'B_8_6': 0.6729205736208053, 'B_8_7': -0.5703331913765826, 'B_8_8':
 0.08278766214896031, 'B_8_9': 0.7419202642550284, 'B_9_0':
 0.0010063213651589076, 'B_9_1': -0.6329838286649109, 'B_9_2':
 -0.798651764671112, 'B_9_3': 0.53396742406544, 'B_9_4': -0.9189125104934389,
 'B_9_5': -0.43011930197261217, 'B_9_6': -0.5329634408062823, 'B_9_7':
 -9.259488834661989e-05, 'B_9_8': 0.9002494283396442, 'B_9_9':
 0.7393103711314114, 'B_10_0': 0.6172659504016154, 'B_10_1': -0.5554527454586885,
 'B_10_2': -0.9879953752690162, 'B_10_3': -0.3755414446075581, 'B_10_4':
 0.7603011812197279, 'B_10_5': -0.3376925495513437, 'B_10_6':
 -0.5080987606032027, 'B_10_7': 0.6070459870055472, 'B_10_8':
 0.03544338234141165, 'B_10_9': 0.2527638786317962}. Best is trial 174 with
 value: 55.46637859185029.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].

```

    Built F_[u_9].
    Built F_[u_10].
    Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:31:45,303] Trial 177 finished with value: 66.41829010330821 and
parameters: {'B_0_0': 0.5403368897347, 'B_0_1': 0.2984384296670973, 'B_0_2':
0.47282920255579797, 'B_0_3': 0.17664494178188642, 'B_0_4': -0.4544749993865462,
'B_0_5': 0.7371440529794882, 'B_0_6': 0.6469814568661201, 'B_0_7':
0.3497274348624044, 'B_0_8': 0.5042888203874706, 'B_0_9': 0.4911927597778829,
'B_1_0': 0.6922747793444273, 'B_1_1': -0.2158588495685576, 'B_1_2':
0.36333919289967953, 'B_1_3': 0.6128034215518825, 'B_1_4': -0.3705931226552992,
'B_1_5': 0.8943991255384666, 'B_1_6': 0.018272192310983842, 'B_1_7':
0.5597470643159996, 'B_1_8': -0.48105281766877545, 'B_1_9': -0.6220534588916832,
'B_2_0': -0.9268111688294761, 'B_2_1': -0.503521599401516, 'B_2_2':
-0.6844425788289721, 'B_2_3': 0.7862515239747649, 'B_2_4': -0.19073972503830966,
'B_2_5': -0.7313587207992371, 'B_2_6': -0.17379733367936612, 'B_2_7':
-0.40699087989780675, 'B_2_8': -0.7722892801233995, 'B_2_9':
0.23587539197606203, 'B_3_0': -0.306621672184051, 'B_3_1': -0.4075519299715365,
'B_3_2': -0.46395965154884805, 'B_3_3': -0.12081278680398483, 'B_3_4':
-0.7236974377252983, 'B_3_5': 0.40186629522872447, 'B_3_6':
-0.07071829060730907, 'B_3_7': 0.41719103592087825, 'B_3_8': 0.2985215032670074,
'B_3_9': -0.14852764348674577, 'B_4_0': 0.15719098568695522, 'B_4_1':
0.30687733556537583, 'B_4_2': 0.157955311164981, 'B_4_3': -0.18140361169824087,
'B_4_4': -0.8399158531394668, 'B_4_5': 0.1720957114133563, 'B_4_6':
-0.48144106465973247, 'B_4_7': -0.5489721816238872, 'B_4_8':
0.44535160020145415, 'B_4_9': 0.9281874223096149, 'B_5_0': 0.34231257098462975,
'B_5_1': -0.7559131815484319, 'B_5_2': 0.9857839688296363, 'B_5_3':
-0.2914428869488051, 'B_5_4': 0.8054348121200032, 'B_5_5': -0.989857874631521,
'B_5_6': -0.17155565434371767, 'B_5_7': 0.9158875674423453, 'B_5_8':
-0.20610354531643071, 'B_5_9': 0.401060780328488, 'B_6_0': 0.3671545703292099,
'B_6_1': -0.5648196358426716, 'B_6_2': 0.5951173412634136, 'B_6_3':
0.826706740691577, 'B_6_4': -0.48744160642231904, 'B_6_5':
-0.0047833714011979445, 'B_6_6': 0.31024019042927997, 'B_6_7':
0.5933886222650873, 'B_6_8': 0.42882843038812135, 'B_6_9': 0.7239669652716145,
'B_7_0': -0.7019535918610795, 'B_7_1': -0.2143006909161616, 'B_7_2':
-0.2981844925889623, 'B_7_3': 0.7657457445615783, 'B_7_4': 0.9374665293624133,
'B_7_5': -0.6525716141019989, 'B_7_6': -0.14493677828083412, 'B_7_7':
-0.706894041047404, 'B_7_8': -0.9888510672537301, 'B_7_9': 0.9091569672712504,
'B_8_0': -0.6395704841049463, 'B_8_1': 0.36127384385266825, 'B_8_2':
0.8764369972884369, 'B_8_3': 0.6255239775551076, 'B_8_4': -0.055095663973512746,
'B_8_5': -0.2505727172535439, 'B_8_6': 0.7328933097301236, 'B_8_7':
-0.644295971532138, 'B_8_8': 0.003550416220430702, 'B_8_9': 0.6550736950513844,
'B_9_0': 0.043865692011906376, 'B_9_1': -0.5850603884180461, 'B_9_2':
-0.5289677515945749, 'B_9_3': 0.4593661676427532, 'B_9_4': -0.7786384522496183,
'B_9_5': -0.49634733116981417, 'B_9_6': -0.6718796511099754, 'B_9_7':
0.043350116028654295, 'B_9_8': 0.9828072388750362, 'B_9_9': 0.6336096583644913,

```


'B_10_0': 0.5625154465234512, 'B_10_1': -0.5389686864076215, 'B_10_2': -0.915102529969094, 'B_10_3': -0.45997953116565116, 'B_10_4': 0.8334490255057523, 'B_10_5': -0.2953888554252418, 'B_10_6': -0.41912379945529976, 'B_10_7': 0.5362716190184099, 'B_10_8': 0.12893840447866461, 'B_10_9': 0.31417102737910974}. Best is trial 174 with value: 55.46637859185029.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:31:59,468] Trial 178 finished with value: 63.32626702545103 and parameters: {'B_0_0': 0.37559541480286157, 'B_0_1': 0.42156511857551204, 'B_0_2': 0.4305054871004558, 'B_0_3': 0.11977136780633069, 'B_0_4': -0.6520275916989298, 'B_0_5': 0.9321357659040025, 'B_0_6': 0.5208981810128835, 'B_0_7': 0.4457831708879833, 'B_0_8': 0.3408920074493319, 'B_0_9': 0.15754272965218216, 'B_1_0': 0.44030926094472983, 'B_1_1': -0.2792233311961948, 'B_1_2': 0.2829298250210841, 'B_1_3': 0.6359128533474246, 'B_1_4': -0.4570720469510646, 'B_1_5': 0.7952070843222386, 'B_1_6': -0.13408939420370897, 'B_1_7': 0.7916645675563923, 'B_1_8': -0.6668164060468188, 'B_1_9': -0.5643319342574684, 'B_2_0': -0.8426598885050798, 'B_2_1': -0.41281673961758714, 'B_2_2': -0.631731326426054, 'B_2_3': 0.9877905893251316, 'B_2_4': -0.10196925143312903, 'B_2_5': -0.6552608362431462, 'B_2_6': -0.33510382490634544, 'B_2_7': -0.5729460584277968, 'B_2_8': -0.8056050059185843, 'B_2_9': 0.20258148410642163, 'B_3_0': -0.369404163252239, 'B_3_1': -0.29462569927208365, 'B_3_2': -0.18289161558742906, 'B_3_3': -0.25892601708029694, 'B_3_4': -0.882187220429448, 'B_3_5': 0.47086748097246744, 'B_3_6': -0.030489240023759627, 'B_3_7': 0.5197952289893435, 'B_3_8': 0.24193090596405767, 'B_3_9': -0.0942605752432101, 'B_4_0': 0.20828944092359145, 'B_4_1': 0.39092579896972984, 'B_4_2': 0.10806327169673643, 'B_4_3': -0.20539530129713873, 'B_4_4': -0.7224312826144841, 'B_4_5': 0.21182202499846087, 'B_4_6': -0.6096844045333523, 'B_4_7': -0.7579221470911471, 'B_4_8': 0.38755105036981313, 'B_4_9': 0.8719005481617133, 'B_5_0': 0.26298348432594965, 'B_5_1': -0.677884136029436, 'B_5_2': 0.44963520249194455, 'B_5_3': -0.33516820872327324, 'B_5_4': 0.7498299260080653, 'B_5_5': -0.8340551272928936, 'B_5_6': -0.107714012091401, 'B_5_7': 0.9772453218490511, 'B_5_8': -0.157299154416217, 'B_5_9': 0.5514236468840652, 'B_6_0': 0.31591201293236915, 'B_6_1': -0.4628508304335026, 'B_6_2': 0.6981728824527605,

'B_6_3': 0.9510977001181262, 'B_6_4': -0.45636562552339943, 'B_6_5':
-0.10404228147905424, 'B_6_6': 0.37194876762285606, 'B_6_7': 0.5381569984082616,
'B_6_8': 0.4698602730511219, 'B_6_9': 0.6710789471478459, 'B_7_0':
-0.8307212368817043, 'B_7_1': -0.19310101101454413, 'B_7_2':
-0.3525488124119208, 'B_7_3': 0.68302079380942, 'B_7_4': 0.8999072853834922,
'B_7_5': -0.5948812381263948, 'B_7_6': -0.32360367810343726, 'B_7_7':
-0.7651256099209585, 'B_7_8': -0.9021269809009724, 'B_7_9': 0.937536325001932,
'B_8_0': -0.5249368495701261, 'B_8_1': 0.19848008328047656, 'B_8_2':
0.9472557611418648, 'B_8_3': 0.5686058122391588, 'B_8_4': -0.00369706030051848,
'B_8_5': -0.435976324228084, 'B_8_6': 0.6291753230912602, 'B_8_7':
-0.7548221099048489, 'B_8_8': -0.05215057563110078, 'B_8_9': 0.5997959315998966,
'B_9_0': 0.09190657842527925, 'B_9_1': -0.7393857932366249, 'B_9_2':
-0.7709434710760504, 'B_9_3': 0.5810310463259369, 'B_9_4': -0.590508355954745,
'B_9_5': -0.5687544388124198, 'B_9_6': -0.5864165754753932, 'B_9_7':
-0.12604885336565214, 'B_9_8': 0.9323508145841102, 'B_9_9': 0.749912806853299,
'B_10_0': 0.664234141904275, 'B_10_1': -0.6921636295709541, 'B_10_2':
-0.8655926313075926, 'B_10_3': -0.21994122903371482, 'B_10_4':
0.7167865656210773, 'B_10_5': -0.45579891629225844, 'B_10_6':
-0.3478861034841134, 'B_10_7': 0.7343525781082467, 'B_10_8':
0.09024126346024575, 'B_10_9': 0.22655622154643823}. Best is trial 174 with
value: 55.46637859185029.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [<F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

[I 2025-04-29 14:32:13,987] Trial 179 finished with value: 66.47662254093642 and
parameters: {'B_0_0': 0.485292030220849, 'B_0_1': 0.26667037126965326, 'B_0_2':
0.6505987612475157, 'B_0_3': 0.22618602537683252, 'B_0_4': -0.7774629197690237,
'B_0_5': 0.8243412992414514, 'B_0_6': 0.5653283153726708, 'B_0_7':
0.31749958621573243, 'B_0_8': 0.4358146314824441, 'B_0_9': 0.2604510192587728,
'B_1_0': 0.5059632423233338, 'B_1_1': -0.3898655840827141, 'B_1_2':
0.5167107704596738, 'B_1_3': 0.6570295472540542, 'B_1_4': -0.4095328267376244,
'B_1_5': 0.949797877253483, 'B_1_6': -0.2353949655038609, 'B_1_7':
0.6282310767972306, 'B_1_8': -0.7556846091120438, 'B_1_9': -0.6436333045237386,
'B_2_0': -0.9068573716306109, 'B_2_1': -0.5791368552425791, 'B_2_2':
-0.5804954160816169, 'B_2_3': 0.7471722536276045, 'B_2_4': -0.2547591416929829,
'B_2_5': -0.8182950323105909, 'B_2_6': -0.21906909760519788, 'B_2_7':

-0.43545569753892155, 'B_2_8': -0.7244958624221242, 'B_2_9':
 0.18440763195231746, 'B_3_0': -0.4582948700878667, 'B_3_1':
 -0.20952428962981612, 'B_3_2': -0.311991357994795, 'B_3_3': -0.2108217668255467,
 'B_3_4': -0.6870673923648557, 'B_3_5': 0.6008799097142257, 'B_3_6':
 -0.1501869897306986, 'B_3_7': 0.4222873892590517, 'B_3_8': 0.1898586075115758,
 'B_3_9': 0.01738444332635445, 'B_4_0': 0.23202573661629547, 'B_4_1':
 0.19247389036885035, 'B_4_2': -0.004882388911727417, 'B_4_3':
 -0.09932415706538472, 'B_4_4': -0.8647951619772727, 'B_4_5':
 0.31510309067391246, 'B_4_6': -0.7193675989080989, 'B_4_7': -0.3448462662514209,
 'B_4_8': 0.5259338938329297, 'B_4_9': 0.9205506552878908, 'B_5_0':
 0.20259937361855457, 'B_5_1': -0.7321202772221378, 'B_5_2': 0.5466915982869839,
 'B_5_3': -0.19895969814030257, 'B_5_4': 0.6832297213856657, 'B_5_5':
 -0.8938944836284692, 'B_5_6': -0.15186559243105183, 'B_5_7': 0.9500015468679037,
 'B_5_8': -0.21297353340122227, 'B_5_9': 0.50852930706969, 'B_6_0':
 0.21495976843167341, 'B_6_1': -0.5883185848658118, 'B_6_2': 0.5600101010972187,
 'B_6_3': 0.8401555801629665, 'B_6_4': -0.537733991137878, 'B_6_5':
 -0.04460713129085292, 'B_6_6': 0.2372738782963553, 'B_6_7': 0.599787107107772,
 'B_6_8': 0.37036091270949817, 'B_6_9': 0.5744066109660703, 'B_7_0':
 -0.869551397139755, 'B_7_1': -0.11606349545106812, 'B_7_2': -0.4475525247665366,
 'B_7_3': 0.6334275406961709, 'B_7_4': 0.8316671363512119, 'B_7_5':
 -0.43659562853918465, 'B_7_6': -0.22384348709510865, 'B_7_7':
 -0.8117123320399287, 'B_7_8': -0.8309531404300655, 'B_7_9': 0.9862102244419203,
 'B_8_0': -0.5845962032508172, 'B_8_1': 0.5448058018611206, 'B_8_2':
 0.9064376868127741, 'B_8_3': 0.6970714495097898, 'B_8_4': -0.1402663790740521,
 'B_8_5': -0.29017058408926705, 'B_8_6': 0.7939004429105516, 'B_8_7':
 -0.6817485933776174, 'B_8_8': 0.07899424995999238, 'B_8_9': 0.7554152708885744,
 'B_9_0': 0.03291020482675825, 'B_9_1': -0.5935931033558994, 'B_9_2':
 -0.8174170597625384, 'B_9_3': 0.5476169166768603, 'B_9_4': -0.6406645500958321,
 'B_9_5': -0.46837260137900044, 'B_9_6': -0.4385880873619068, 'B_9_7':
 -0.046527063364148705, 'B_9_8': 0.848083108774952, 'B_9_9': 0.7060090371772327,
 'B_10_0': 0.4682996230845997, 'B_10_1': -0.6013213513623713, 'B_10_2':
 -0.9439260034838339, 'B_10_3': -0.28557448378277583, 'B_10_4':
 0.6035251710956889, 'B_10_5': -0.37071509028988503, 'B_10_6':
 -0.4725243496160156, 'B_10_7': 0.6392731565967622, 'B_10_8':
 0.004538183512206709, 'B_10_9': 0.1575041671713175}. Best is trial 174 with
 value: 55.46637859185029.

Building the Basis functions for the RKHS:

Built F_[u_1].
 Built F_[u_2].
 Built F_[u_3].
 Built F_[u_4].
 Built F_[u_5].
 Built F_[u_6].
 Built F_[u_7].
 Built F_[u_8].
 Built F_[u_9].
 Built F_[u_10].
 Built F_[u_11].

Established the refined x grid with 1001 points.
 Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].
 Built matrix B = inverse(A).
 [I 2025-04-29 14:32:28,174] Trial 180 finished with value: 56.26293916020178 and
 parameters: {'B_0_0': 0.65213627386122, 'B_0_1': 0.31462535005788345, 'B_0_2':
 0.7275910756565347, 'B_0_3': -0.011965832429150534, 'B_0_4':
 -0.5273051695718652, 'B_0_5': 0.7540052111372693, 'B_0_6': 0.5981257495704259,
 'B_0_7': 0.4247823796682268, 'B_0_8': 0.35288755535467325, 'B_0_9':
 0.3387285073273231, 'B_1_0': 0.5429742812374835, 'B_1_1': -0.1865558757958669,
 'B_1_2': 0.3915778111869891, 'B_1_3': 0.6088204856903771, 'B_1_4':
 -0.3877274770505127, 'B_1_5': 0.865774502096443, 'B_1_6': -0.19199346678231663,
 'B_1_7': 0.46262388117450504, 'B_1_8': -0.7652018687317852, 'B_1_9':
 -0.592658311682299, 'B_2_0': -0.7793072018657967, 'B_2_1': -0.5288956842860667,
 'B_2_2': -0.6455273736015852, 'B_2_3': 0.9449662606296466, 'B_2_4':
 -0.147133500928256, 'B_2_5': -0.7121994868813579, 'B_2_6': -0.05748500407194944,
 'B_2_7': -0.5084708453170534, 'B_2_8': -0.6254571860028587, 'B_2_9':
 0.4030689703280771, 'B_3_0': -0.40722723570586383, 'B_3_1':
 -0.40951556469056966, 'B_3_2': -0.38612333326351445, 'B_3_3':
 -0.10523418520798543, 'B_3_4': -0.7580787217365466, 'B_3_5':
 0.37401747440540434, 'B_3_6': -0.3060443281418831, 'B_3_7': 0.5648742011217925,
 'B_3_8': 0.10993119143433727, 'B_3_9': -0.11117945320995148, 'B_4_0':
 0.1325827186411769, 'B_4_1': 0.28873669945096164, 'B_4_2': 0.1645119592333097,
 'B_4_3': -0.13846047815593482, 'B_4_4': -0.763499461054117, 'B_4_5':
 0.4076025999210584, 'B_4_6': -0.649328707014029, 'B_4_7': -0.44290245165707004,
 'B_4_8': 0.510691013310387, 'B_4_9': 0.8564313484435665, 'B_5_0':
 0.30548754177874476, 'B_5_1': -0.5683222566525263, 'B_5_2': 0.7563145028605003,
 'B_5_3': -0.27792725680758934, 'B_5_4': 0.8580444473959002, 'B_5_5':
 -0.8656888649629297, 'B_5_6': -0.09804475658270079, 'B_5_7': 0.4311680884184194,
 'B_5_8': -0.18340013191131174, 'B_5_9': 0.3993564908949274, 'B_6_0':
 0.14473827307721776, 'B_6_1': -0.5012119427638728, 'B_6_2': 0.48091933536744347,
 'B_6_3': 0.8939224953965592, 'B_6_4': -0.5061392896626055, 'B_6_5':
 0.1299570881460804, 'B_6_6': 0.5011871034658176, 'B_6_7': 0.4345802698091202,
 'B_6_8': 0.42979209517308814, 'B_6_9': 0.5194758823222332, 'B_7_0':
 -0.8235529882254171, 'B_7_1': -0.28264872457670687, 'B_7_2':
 -0.11468959039518536, 'B_7_3': 0.7176087569404419, 'B_7_4': 0.9521257795234538,
 'B_7_5': -0.621472773015876, 'B_7_6': -0.144201944286952, 'B_7_7':
 -0.7589682859511244, 'B_7_8': -0.8936501731220253, 'B_7_9': 0.8749220189902981,
 'B_8_0': -0.649542414553431, 'B_8_1': 0.44505076318299963, 'B_8_2':
 0.8313654568408945, 'B_8_3': 0.5847650119307239, 'B_8_4': -0.07596729666852603,
 'B_8_5': -0.3387464588656005, 'B_8_6': 0.621884757551098, 'B_8_7':
 -0.607761650857761, 'B_8_8': 0.022441908268080686, 'B_8_9': 0.6411842064275153,
 'B_9_0': -0.056037533992310734, 'B_9_1': -0.778060337217746, 'B_9_2':
 -0.6195670681711783, 'B_9_3': 0.6630924156937917, 'B_9_4': -0.7277432888936217,
 'B_9_5': -0.5152953569980396, 'B_9_6': -0.6449999932303305, 'B_9_7':
 -0.2271362011138235, 'B_9_8': 0.6108167721989826, 'B_9_9': 0.5831469121276758,
 'B_10_0': 0.6188163821866443, 'B_10_1': -0.5479918225924117, 'B_10_2':
 -0.9053594222102402, 'B_10_3': -0.22308843863336253, 'B_10_4':
 0.6608815715719382, 'B_10_5': -0.34077735333323234, 'B_10_6':

-0.4447957367136869, 'B_10_7': 0.7067163232685123, 'B_10_8': 0.0578400973946776, 'B_10_9': 0.0715489183555091}. Best is trial 174 with value: 55.46637859185029.

Building the Basis functions for the RKHS:

Built F__[u_1].
Built F__[u_2].
Built F__[u_3].
Built F__[u_4].
Built F__[u_5].
Built F__[u_6].
Built F__[u_7].
Built F__[u_8].
Built F__[u_9].
Built F__[u_10].
Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:32:42,744] Trial 181 finished with value: 62.094558817500726
and parameters: {'B_0_0': 0.6648592361241321, 'B_0_1': 0.30805467139858717, 'B_0_2': 0.7308389624742181, 'B_0_3': -0.0011714630461077125, 'B_0_4': -0.5439595151183421, 'B_0_5': 0.7612846994033039, 'B_0_6': 0.6141565177350319, 'B_0_7': 0.3771957801172758, 'B_0_8': 0.3527714749761777, 'B_0_9': 0.293158023516949, 'B_1_0': 0.6350955477568636, 'B_1_1': -0.11423114658472558, 'B_1_2': 0.3920999101165009, 'B_1_3': 0.6554883685812755, 'B_1_4': -0.3726028171388708, 'B_1_5': 0.8652179310357788, 'B_1_6': -0.1905105813301098, 'B_1_7': 0.6006842901409772, 'B_1_8': -0.8736668306160189, 'B_1_9': -0.5958816635741682, 'B_2_0': -0.7800298322247556, 'B_2_1': -0.5110515859823995, 'B_2_2': -0.6370256152665719, 'B_2_3': 0.9364011349061959, 'B_2_4': -0.14438127519949662, 'B_2_5': -0.6969679780349226, 'B_2_6': -0.13304989695878233, 'B_2_7': -0.5006565912703548, 'B_2_8': -0.6343413291927478, 'B_2_9': 0.39959373722267416, 'B_3_0': -0.4185346397685094, 'B_3_1': -0.4803827297799741, 'B_3_2': -0.37607480968605717, 'B_3_3': -0.149791977547744, 'B_3_4': -0.9862562188930717, 'B_3_5': 0.3772926292153002, 'B_3_6': -0.3024268263896191, 'B_3_7': 0.5749891582590261, 'B_3_8': 0.3328998317362983, 'B_3_9': -0.07707061254582617, 'B_4_0': 0.12184380127440472, 'B_4_1': 0.27538394262462845, 'B_4_2': 0.1714486265216527, 'B_4_3': -0.1493924556711712, 'B_4_4': -0.7582742678042305, 'B_4_5': 0.38602630144435135, 'B_4_6': -0.6643961516035629, 'B_4_7': -0.43841800559962146, 'B_4_8': 0.47417289627921994, 'B_4_9': 0.8648192466880749, 'B_5_0': 0.316729698582763, 'B_5_1': -0.5645500500190364, 'B_5_2': 0.7873277949971035, 'B_5_3': -0.2659036075221339, 'B_5_4': 0.945007309763474, 'B_5_5': -0.8467606066315164, 'B_5_6': -0.09121796761221618, 'B_5_7': 0.45745978786079594, 'B_5_8': -0.16313058894655477, 'B_5_9': 0.39954042695018377, 'B_6_0': 0.2736781298760548, 'B_6_1': -0.5015881986638793, 'B_6_2': 0.4972155242409116, 'B_6_3': 0.8915701525688048, 'B_6_4': -0.48265352573122283, 'B_6_5': 0.1281658104525797, 'B_6_6': 0.5092071147525065, 'B_6_7': 0.443594721100513, 'B_6_8': 0.32232929072981753, 'B_6_9': 0.5266830774689589, 'B_7_0': -0.8309788018844351, 'B_7_1': -0.2777577262005261, 'B_7_2': -0.25016081380457555, 'B_7_3':

```

0.7106604410899569, 'B_7_4': 0.9894764644344296, 'B_7_5': -0.6449376666910824,
'B_7_6': -0.15212202540229294, 'B_7_7': -0.7599554620535194, 'B_7_8':
-0.9127467264048608, 'B_7_9': 0.8753033764448245, 'B_8_0': -0.6512154920294368,
'B_8_1': 0.449039700961438, 'B_8_2': 0.8541234945479679, 'B_8_3':
0.4129950713335411, 'B_8_4': -0.0751790337666223, 'B_8_5': -0.34044577127194114,
'B_8_6': 0.5092488690587315, 'B_8_7': -0.619615795124969, 'B_8_8':
0.024106021892777177, 'B_8_9': 0.7131778341190557, 'B_9_0': 0.11869472920473503,
'B_9_1': -0.7698632147009372, 'B_9_2': -0.6225404175689838, 'B_9_3':
0.6546603687135026, 'B_9_4': -0.7377354548001174, 'B_9_5': -0.5067628919970623,
'B_9_6': -0.6476663333045725, 'B_9_7': -0.987016468746516, 'B_9_8':
0.613756013041154, 'B_9_9': 0.5776987165623807, 'B_10_0': 0.5465521379601195,
'B_10_1': -0.5417939495885364, 'B_10_2': -0.9046050422134516, 'B_10_3':
-0.24466909279568583, 'B_10_4': 0.663376291652132, 'B_10_5':
-0.34434210709980523, 'B_10_6': -0.45364088654443685, 'B_10_7':
0.7105200088050085, 'B_10_8': 0.05415845000055952, 'B_10_9':
0.07717276713578447}. Best is trial 174 with value: 55.46637859185029.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:32:57,281] Trial 182 finished with value: 58.48223097071771 and
parameters: {'B_0_0': 0.6162024585484348, 'B_0_1': 0.31521183404967135, 'B_0_2':
0.7301904073601684, 'B_0_3': -0.006256753944283863, 'B_0_4':
-0.5280105218267673, 'B_0_5': 0.7497642291069149, 'B_0_6': 0.6176105356059739,
'B_0_7': 0.3600813812442254, 'B_0_8': 0.31070924314358156, 'B_0_9':
0.3616878382401118, 'B_1_0': 0.6470978413231036, 'B_1_1': -0.11051382134717225,
'B_1_2': 0.33827549550129277, 'B_1_3': 0.4915545831498534, 'B_1_4':
-0.37970901076612434, 'B_1_5': 0.8630507161320774, 'B_1_6':
-0.19050147073776805, 'B_1_7': 0.6024887513852625, 'B_1_8': -0.8390495174206041,
'B_1_9': -0.5799803936575776, 'B_2_0': -0.8119679563577692, 'B_2_1':
-0.5399637797928384, 'B_2_2': -0.6505280520928862, 'B_2_3': 0.9425732410362505,
'B_2_4': -0.20258426818802083, 'B_2_5': -0.7186795080585394, 'B_2_6':
-0.021519518500019014, 'B_2_7': -0.5068992602468504, 'B_2_8':
-0.6344087960388617, 'B_2_9': 0.4232008472757176, 'B_3_0': -0.3908803804807321,
'B_3_1': -0.46906858631207593, 'B_3_2': -0.3921836630500876, 'B_3_3':
-0.10456279680334643, 'B_3_4': -0.9855878938640705, 'B_3_5':
0.39381058402051106, 'B_3_6': -0.059819367016568654, 'B_3_7':

```

```

0.6376179087532516, 'B_3_8': 0.3277488275942803, 'B_3_9': -0.06634819978738224,
'B_4_0': 0.11971347361913326, 'B_4_1': 0.2813943697648573, 'B_4_2':
0.19426719794799877, 'B_4_3': -0.05853905030815268, 'B_4_4':
-0.7615915120436791, 'B_4_5': 0.3824632808772661, 'B_4_6': -0.6940141775245231,
'B_4_7': -0.4534925551308658, 'B_4_8': 0.5243788047437259, 'B_4_9':
0.8829937964113427, 'B_5_0': 0.3217885538510578, 'B_5_1': -0.5626438680143043,
'B_5_2': 0.7749116283768779, 'B_5_3': -0.26692498324186553, 'B_5_4':
0.942645511969918, 'B_5_5': -0.8455580294654576, 'B_5_6': -0.11048349879376956,
'B_5_7': 0.5684659689898084, 'B_5_8': -0.26606181217354047, 'B_5_9':
0.4017322407889139, 'B_6_0': 0.1314131454539078, 'B_6_1': -0.49974923869681254,
'B_6_2': 0.48401988148342445, 'B_6_3': 0.8902541478784204, 'B_6_4':
-0.5049922681175141, 'B_6_5': 0.1324948448992955, 'B_6_6': 0.6114448503931094,
'B_6_7': 0.4390544988273643, 'B_6_8': 0.4339375811602893, 'B_6_9':
0.512469506742642, 'B_7_0': -0.8406670091484542, 'B_7_1': -0.2773734061456363,
'B_7_2': -0.27297921169188666, 'B_7_3': 0.7580193613188732, 'B_7_4':
0.9431146011830095, 'B_7_5': -0.6297451824908669, 'B_7_6': -0.15014198667599898,
'B_7_7': -0.760102013693531, 'B_7_8': -0.9002438305079089, 'B_7_9':
0.8899744924760239, 'B_8_0': -0.7461890517373718, 'B_8_1': 0.4543526272562848,
'B_8_2': 0.831083323734824, 'B_8_3': 0.6511519847853522, 'B_8_4':
0.06545195493025692, 'B_8_5': -0.33784468605970674, 'B_8_6': 0.4960805564764741,
'B_8_7': -0.5848292847321669, 'B_8_8': 0.02669086769736885, 'B_8_9':
0.645361523001693, 'B_9_0': -0.036980747015209314, 'B_9_1': -0.7807804679508612,
'B_9_2': -0.6219605454782389, 'B_9_3': 0.6660915351523735, 'B_9_4':
-0.7364793109345352, 'B_9_5': -0.5174242071548047, 'B_9_6': -0.6089209637796376,
'B_9_7': -0.22606748866382242, 'B_9_8': 0.9337581016188723, 'B_9_9':
0.5584587763863162, 'B_10_0': 0.5485871628267434, 'B_10_1': -0.5550717460174684,
'B_10_2': -0.9021871210357424, 'B_10_3': -0.26309561866385706, 'B_10_4':
0.7497718474830545, 'B_10_5': -0.27893304660445184, 'B_10_6':
-0.4732270943183744, 'B_10_7': 0.7549912228840778, 'B_10_8': 0.0592510148049817,
'B_10_9': 0.06949758139640526}. Best is trial 174 with value: 55.46637859185029.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:33:11,469] Trial 183 finished with value: 61.31004718266563 and parameters: {'B_0_0': 0.5549282309323562, 'B_0_1': 0.3778927629617559, 'B_0_2': 0.7373220528904872, 'B_0_3': -0.0056998815815201415, 'B_0_4':

-0.5318054495207184, 'B_0_5': 0.687320660209328, 'B_0_6': 0.609422979416361,
 'B_0_7': 0.36912654307128406, 'B_0_8': 0.29983045677323594, 'B_0_9':
 0.3446191104328731, 'B_1_0': 0.6510069281068864, 'B_1_1': -0.1947327741837842,
 'B_1_2': 0.23912770400465425, 'B_1_3': 0.6140885301007162, 'B_1_4':
 -0.36506817216911985, 'B_1_5': 0.8530314465234369, 'B_1_6': -0.194547762775948,
 'B_1_7': 0.4230409488284814, 'B_1_8': -0.8771215608270546, 'B_1_9':
 -0.52461543620702, 'B_2_0': -0.8050468015865274, 'B_2_1': -0.5325301015258329,
 'B_2_2': -0.6158135623149978, 'B_2_3': 0.8580434343622216, 'B_2_4':
 0.004060246563408004, 'B_2_5': -0.7239138928241876, 'B_2_6':
 -0.008054184612362597, 'B_2_7': -0.5059766457852763, 'B_2_8':
 -0.6313390829594048, 'B_2_9': 0.4227587792858398, 'B_3_0': -0.33468715798262305,
 'B_3_1': -0.4828780021940753, 'B_3_2': -0.3491406197285136, 'B_3_3':
 -0.14393087839367782, 'B_3_4': -0.9844283308620517, 'B_3_5': 0.3833387319661194,
 'B_3_6': -0.0648900773396157, 'B_3_7': 0.5914179347751203, 'B_3_8':
 0.11359196487393114, 'B_3_9': -0.06832232440648901, 'B_4_0':
 0.12859193317290818, 'B_4_1': 0.29733374297615084, 'B_4_2': 0.2317034838244187,
 'B_4_3': -0.14321902751675616, 'B_4_4': -0.7363760256964919, 'B_4_5':
 0.4085381202598304, 'B_4_6': -0.6637023053274536, 'B_4_7': -0.4497813854902156,
 'B_4_8': 0.5195739792047521, 'B_4_9': 0.9275225270484552, 'B_5_0':
 0.3656290966164825, 'B_5_1': -0.5590427818761119, 'B_5_2': 0.7773453436939701,
 'B_5_3': -0.258146913464909, 'B_5_4': 0.9307995617073906, 'B_5_5':
 -0.8400445101207907, 'B_5_6': -0.08812749755630221, 'B_5_7':
 0.44495494021859916, 'B_5_8': -0.26019864876413357, 'B_5_9':
 0.40037151365675583, 'B_6_0': 0.16049815422352548, 'B_6_1': -0.3162542095196181,
 'B_6_2': 0.4894723779385613, 'B_6_3': 0.8865586863320579, 'B_6_4':
 -0.5073222213729344, 'B_6_5': 0.11296209224206276, 'B_6_6': 0.599319265370278,
 'B_6_7': 0.3046281132223594, 'B_6_8': 0.41728313674660605, 'B_6_9':
 0.5146555464823699, 'B_7_0': -0.7876459054281993, 'B_7_1': -0.3801181605143269,
 'B_7_2': -0.12399122487178277, 'B_7_3': 0.8376572881781581, 'B_7_4':
 0.9413565396873925, 'B_7_5': -0.6992517298789946, 'B_7_6': -0.15051946186675744,
 'B_7_7': -0.7655217461879967, 'B_7_8': -0.8842945527685184, 'B_7_9':
 0.8808950868650646, 'B_8_0': -0.5272896163578034, 'B_8_1': 0.4170119330160632,
 'B_8_2': 0.8306803774119915, 'B_8_3': 0.4616415476261008, 'B_8_4':
 0.0877039254275669, 'B_8_5': -0.33337151636596035, 'B_8_6': 0.5180199377125806,
 'B_8_7': -0.5897199298570031, 'B_8_8': 0.031450915897506045, 'B_8_9':
 0.6607644110681656, 'B_9_0': -0.0738182922787326, 'B_9_1': -0.7781623939235512,
 'B_9_2': -0.6200941227905535, 'B_9_3': 0.6453220476193355, 'B_9_4':
 -0.7235275812232985, 'B_9_5': -0.5301439265223079, 'B_9_6': -0.6434503306225091,
 'B_9_7': -0.2322401008097977, 'B_9_8': 0.9763908173550141, 'B_9_9':
 0.5014849149935965, 'B_10_0': 0.5406617627191503, 'B_10_1': -0.547280518021175,
 'B_10_2': -0.9017602378694005, 'B_10_3': -0.23316260812255155, 'B_10_4':
 0.7473781572796622, 'B_10_5': -0.3332433869142705, 'B_10_6':
 -0.45122874214520864, 'B_10_7': 0.7668066426541651, 'B_10_8':
 0.14587244093563656, 'B_10_9': 0.0659110560262385}. Best is trial 174 with
 value: 55.46637859185029.

Building the Basis functions for the RKHS:

Built F_[u₁].

Built F_[u₂].


```

Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```

[I 2025-04-29 14:33:26,139] Trial 184 finished with value: 64.05420954283403 and
parameters: {'B_0_0': 0.6304420496566635, 'B_0_1': 0.3833259518629137, 'B_0_2':
0.7414817726094138, 'B_0_3': 0.008926776480182155, 'B_0_4': -0.5344428832709038,
'B_0_5': 0.690017590960211, 'B_0_6': 0.7023981302409253, 'B_0_7':
0.4205899952162934, 'B_0_8': 0.3237628735939236, 'B_0_9': 0.3376413500710772,
'B_1_0': 0.656915628095755, 'B_1_1': -0.19545546459157984, 'B_1_2':
0.2532965352020961, 'B_1_3': 0.6075970821040115, 'B_1_4': -0.355091670217422,
'B_1_5': 0.8667448112990714, 'B_1_6': -0.17537595324785263, 'B_1_7':
0.41673588297474634, 'B_1_8': -0.8877142730669233, 'B_1_9': -0.5322826605049168,
'B_2_0': -0.727088536998774, 'B_2_1': -0.6830147205256578, 'B_2_2':
-0.6985212574529902, 'B_2_3': 0.8549457762247948, 'B_2_4': 0.15116660399269605,
'B_2_5': -0.609700673819565, 'B_2_6': -0.03257358477602619, 'B_2_7':
-0.5174335081606739, 'B_2_8': -0.63467110847306, 'B_2_9': 0.4501967280284969,
'B_3_0': -0.33871355979266904, 'B_3_1': -0.5042676890352729, 'B_3_2':
-0.3282311575261601, 'B_3_3': -0.1849847969533335, 'B_3_4': -0.982311645683417,
'B_3_5': 0.38555908977164915, 'B_3_6': -0.05997002157623953, 'B_3_7':
0.6248688418134585, 'B_3_8': 0.3162142772076195, 'B_3_9': -0.07475578198149418,
'B_4_0': 0.11960442026941857, 'B_4_1': 0.3702159644097148, 'B_4_2':
0.23325946312512447, 'B_4_3': -0.1381156076150936, 'B_4_4': -0.7501232160652158,
'B_4_5': 0.4242811039117356, 'B_4_6': -0.6881471739655982, 'B_4_7':
-0.45024385333491684, 'B_4_8': 0.5271536708096923, 'B_4_9': 0.940020146492881,
'B_5_0': 0.42206180368557433, 'B_5_1': -0.5778531248579571, 'B_5_2':
0.7700758111246526, 'B_5_3': -0.27760584321154186, 'B_5_4': 0.9048052236773141,
'B_5_5': -0.9402108101410347, 'B_5_6': -0.08393715654593942, 'B_5_7':
0.4511056818675532, 'B_5_8': -0.27051485508273787, 'B_5_9': 0.39159940528341913,
'B_6_0': 0.16116066685380645, 'B_6_1': -0.6610234497465424, 'B_6_2':
0.4848756918613646, 'B_6_3': 0.8876379674438499, 'B_6_4': -0.49989667230654183,
'B_6_5': 0.2101439391549003, 'B_6_6': 0.6362559996246259, 'B_6_7':
0.3539470175883747, 'B_6_8': 0.34378321390161976, 'B_6_9': 0.49777115243706277,
'B_7_0': -0.7271299820868968, 'B_7_1': -0.3813866674254026, 'B_7_2':
-0.16307714443321158, 'B_7_3': 0.783907423879387, 'B_7_4': 0.9381315978631672,
'B_7_5': -0.7416192471381303, 'B_7_6': -0.11406333630908079, 'B_7_7':
-0.7692128034946604, 'B_7_8': -0.8359298753935959, 'B_7_9': 0.8678924949184705,
'B_8_0': -0.5470493828646137, 'B_8_1': 0.4105292156444147, 'B_8_2':
0.8260079218915892, 'B_8_3': 0.6613907605434058, 'B_8_4': 0.045949072315407354,
'B_8_5': -0.33407506178429264, 'B_8_6': 0.5039066241557775, 'B_8_7':

```

```

-0.6098801329199443, 'B_8_8': 0.0230374124109659, 'B_8_9': 0.760491903534619,
'B_9_0': -0.035337392026788095, 'B_9_1': -0.7714817381784459, 'B_9_2':
-0.5897182189268582, 'B_9_3': 0.6641656993131393, 'B_9_4': -0.7213930575598392,
'B_9_5': -0.5552410222628061, 'B_9_6': -0.501646763614723, 'B_9_7':
-0.24226555627877228, 'B_9_8': 0.6169496533473876, 'B_9_9': 0.4711269947569633,
'B_10_0': 0.6328386271540194, 'B_10_1': -0.563405491541189, 'B_10_2':
-0.8756968745673743, 'B_10_3': -0.22459651610109838, 'B_10_4':
0.7476457882847578, 'B_10_5': -0.32822876826957925, 'B_10_6':
-0.5350176335087026, 'B_10_7': 0.7852211573240611, 'B_10_8': 0.1657298907496361,
'B_10_9': 0.08532473932731693}. Best is trial 174 with value: 55.46637859185029.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:33:40,626] Trial 185 finished with value: 62.569153183297146
and parameters: {'B_0_0': 0.6808104600939927, 'B_0_1': 0.32245742533316635,
'B_0_2': 0.704184092039613, 'B_0_3': -0.011662599537606386, 'B_0_4':
-0.4685265761877738, 'B_0_5': 0.7508699871451708, 'B_0_6': 0.635955899638006,
'B_0_7': 0.38493175427719484, 'B_0_8': 0.29419292074057213, 'B_0_9':
0.34063120684170134, 'B_1_0': 0.7202028173487147, 'B_1_1': -0.04604997034332037,
'B_1_2': 0.3861995628287843, 'B_1_3': 0.5043294236044791, 'B_1_4':
-0.3685429739215284, 'B_1_5': 0.8638396329184288, 'B_1_6': -0.05727182895617179,
'B_1_7': 0.4593896350460498, 'B_1_8': -0.9795785750686528, 'B_1_9':
-0.5746242723828583, 'B_2_0': -0.7665782011725014, 'B_2_1': -0.6096711448668602,
'B_2_2': -0.6613753964726362, 'B_2_3': 0.7974251623092671, 'B_2_4':
-0.19950646112424242, 'B_2_5': -0.6531420983404754, 'B_2_6':
-0.06481940673559189, 'B_2_7': -0.5055289096447657, 'B_2_8':
-0.6784838353514474, 'B_2_9': 0.40652340962645644, 'B_3_0': -0.3814800414055156,
'B_3_1': -0.4707617835708442, 'B_3_2': -0.2687564194309191, 'B_3_3':
-0.16003483036652033, 'B_3_4': -0.8684490959503169, 'B_3_5': 0.2964761060534189,
'B_3_6': -0.36803082993136627, 'B_3_7': 0.5924345105345197, 'B_3_8':
0.14527108580651393, 'B_3_9': -0.15075737415072663, 'B_4_0':
0.02033192754441935, 'B_4_1': 0.2943436632572602, 'B_4_2': 0.18666621925973165,
'B_4_3': -0.07741811397285134, 'B_4_4': -0.7486664941825465, 'B_4_5':
0.23312707913758576, 'B_4_6': -0.7243759383072745, 'B_4_7': -0.6625749929912786,
'B_4_8': 0.5145305906582747, 'B_4_9': 0.7907446100295632, 'B_5_0':
0.36618427162586104, 'B_5_1': -0.6122819743042653, 'B_5_2': 0.7188044917311098,

```

'B_5_3': -0.3098559672135639, 'B_5_4': 0.9468189577322915, 'B_5_5': -0.9043621607185581, 'B_5_6': -0.12025482574810832, 'B_5_7': 0.47890852956581986, 'B_5_8': -0.260204874559918, 'B_5_9': 0.29629245523010217, 'B_6_0': 0.14403660410258923, 'B_6_1': -0.34301509315657364, 'B_6_2': 0.5318102643826995, 'B_6_3': 0.8278512746041035, 'B_6_4': -0.4406956731723332, 'B_6_5': 0.1213668268474949, 'B_6_6': 0.4992943448317221, 'B_6_7': 0.41560864640340817, 'B_6_8': 0.3070495832487331, 'B_6_9': 0.4554500536781423, 'B_7_0': -0.819434902430875, 'B_7_1': -0.5156016241559988, 'B_7_2': -0.10289159312436298, 'B_7_3': 0.8410035390572429, 'B_7_4': 0.9360903479859739, 'B_7_5': -0.6897530789755066, 'B_7_6': -0.15507390746084065, 'B_7_7': -0.6119925775356448, 'B_7_8': -0.8894651994222198, 'B_7_9': 0.7788974005674211, 'B_8_0': -0.5983414956939401, 'B_8_1': 0.4361503810897389, 'B_8_2': 0.862394322460743, 'B_8_3': 0.48392781030424725, 'B_8_4': 0.09254797723262058, 'B_8_5': -0.23968102043905226, 'B_8_6': 0.45094001856996413, 'B_8_7': -0.49721666074935245, 'B_8_8': 0.07595937058257925, 'B_8_9': 0.5559816553730393, 'B_9_0': -0.10507048420301329, 'B_9_1': -0.7830020365608154, 'B_9_2': -0.5503100038780007, 'B_9_3': 0.6596873673258754, 'B_9_4': -0.6126597521228455, 'B_9_5': -0.5427608451940276, 'B_9_6': -0.6537506030121114, 'B_9_7': -0.30638137290360434, 'B_9_8': 0.9851052622106128, 'B_9_9': 0.5685218117506737, 'B_10_0': 0.5232822410218116, 'B_10_1': -0.5386133037996914, 'B_10_2': -0.8863232661409346, 'B_10_3': -0.26569421987463065, 'B_10_4': 0.7549463343389504, 'B_10_5': -0.34105868430929315, 'B_10_6': -0.4495205246294916, 'B_10_7': 0.721730126681405, 'B_10_8': 0.09540318229596845, 'B_10_9': 0.05820001605831762}. Best is trial 174 with value: 55.46637859185029.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].
 Built F__[u_8].
 Built F__[u_9].
 Built F__[u_10].
 Built F__[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:33:54,880] Trial 186 finished with value: 65.96866035151936 and parameters: {'B_0_0': 0.6782648785271606, 'B_0_1': 0.4547335285026374, 'B_0_2': 0.7072635393751875, 'B_0_3': -0.02422674904445817, 'B_0_4': -0.4711557300628585, 'B_0_5': 0.7552869447974885, 'B_0_6': 0.6201101130716301, 'B_0_7': 0.36643675639337636, 'B_0_8': 0.3019293247819482, 'B_0_9': 0.4004797325950977, 'B_1_0': 0.7131426056982743, 'B_1_1': -0.04498344097247012, 'B_1_2': 0.3882137489584468, 'B_1_3': 0.5309412613644438, 'B_1_4': -0.36261153416366165, 'B_1_5': 0.8744181838813806, 'B_1_6': -0.06353981936414593, 'B_1_7': 0.3974990285190357, 'B_1_8': -0.9727944850493315, 'B_1_9': -0.572302304231106,

'B_2_0': -0.7555533267075483, 'B_2_1': -0.5909470932997525, 'B_2_2':
-0.6474291780913513, 'B_2_3': 0.9889926236670228, 'B_2_4': 0.018075369904123784,
'B_2_5': -0.6470295910084647, 'B_2_6': -0.06580395153003826, 'B_2_7':
-0.5519968456266902, 'B_2_8': -0.6629868355294511, 'B_2_9': 0.4043570889252227,
'B_3_0': -0.2963726533384459, 'B_3_1': -0.46899649276834865, 'B_3_2':
-0.34905859016774343, 'B_3_3': -0.13806010463916016, 'B_3_4':
-0.9643645594112166, 'B_3_5': 0.3020874561813312, 'B_3_6': -0.2991525692984499,
'B_3_7': 0.5829725878062716, 'B_3_8': 0.001989524172979054, 'B_3_9':
-0.14337254772292518, 'B_4_0': 0.016971221904582973, 'B_4_1':
0.29676764177340154, 'B_4_2': 0.20074346931177214, 'B_4_3':
-0.04379854338233469, 'B_4_4': -0.7536806158565226, 'B_4_5':
0.23069536310214844, 'B_4_6': -0.7287095532819302, 'B_4_7': -0.6727035322048452,
'B_4_8': 0.5927648793618686, 'B_4_9': 0.7731124218705261, 'B_5_0':
0.4690117703514707, 'B_5_1': -0.6204740671039375, 'B_5_2': 0.7170101282173762,
'B_5_3': -0.36227434393700214, 'B_5_4': 0.9375408148526116, 'B_5_5':
-0.9102421686712006, 'B_5_6': -0.17278178692978516, 'B_5_7': 0.5485532042781192,
'B_5_8': -0.029631186754783448, 'B_5_9': 0.41358793771514746, 'B_6_0':
0.13150817782811322, 'B_6_1': -0.3899551394455965, 'B_6_2': 0.40923128939695314,
'B_6_3': 0.8246816049615027, 'B_6_4': -0.4433060220991011, 'B_6_5':
0.1335474245427457, 'B_6_6': 0.6059181694422661, 'B_6_7': 0.43804946110875326,
'B_6_8': 0.30874881334835097, 'B_6_9': 0.45662457326464656, 'B_7_0':
-0.8025339824999674, 'B_7_1': -0.2889582374514096, 'B_7_2':
-0.12547122269791267, 'B_7_3': 0.8344579591209923, 'B_7_4': 0.9276930757515022,
'B_7_5': -0.8031189642216986, 'B_7_6': -0.14795749273270653, 'B_7_7':
-0.6019825616133063, 'B_7_8': -0.8847786837675046, 'B_7_9': 0.8095299320816094,
'B_8_0': -0.5275835111896706, 'B_8_1': 0.5006089277363248, 'B_8_2':
0.848750382664215, 'B_8_3': 0.4798879118131071, 'B_8_4': 0.14986196389193737,
'B_8_5': -0.2473880470921029, 'B_8_6': 0.4621396794466584, 'B_8_7':
-0.5692491664290853, 'B_8_8': 0.08345014164652408, 'B_8_9': 0.577846891498065,
'B_9_0': -0.12075675937409557, 'B_9_1': -0.776865520564163, 'B_9_2':
-0.6060890565137259, 'B_9_3': 0.4990575503207425, 'B_9_4': -0.6087810889225816,
'B_9_5': -0.5256139498141384, 'B_9_6': -0.6563510079384925, 'B_9_7':
-0.3048441898192189, 'B_9_8': 0.9786609136241876, 'B_9_9': 0.4951778503933549,
'B_10_0': 0.530887619934365, 'B_10_1': -0.5269384203951479, 'B_10_2':
-0.9031339487474055, 'B_10_3': -0.14557981184961655, 'B_10_4':
0.8968896625750338, 'B_10_5': -0.2596197042176115, 'B_10_6':
-0.4633622407264524, 'B_10_7': 0.7225334631272813, 'B_10_8': 0.2421120994042306,
'B_10_9': 0.06221040251200178}. Best is trial 174 with value: 55.46637859185029.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].

```

    Built F_[u_10].
    Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:34:09,401] Trial 187 finished with value: 73.18340341699671 and
parameters: {'B_0_0': 0.760244549590345, 'B_0_1': 0.32352184203779, 'B_0_2':
0.6856988511914193, 'B_0_3': -0.03814049129258165, 'B_0_4': -0.5156813009491044,
'B_0_5': 0.59881368953805, 'B_0_6': 0.6728489897223833, 'B_0_7':
0.2999973810049206, 'B_0_8': 0.2187173782223301, 'B_0_9': 0.3535583392770839,
'B_1_0': 0.7539343876435189, 'B_1_1': -0.24101067436652274, 'B_1_2':
0.3817098806264928, 'B_1_3': 0.48054924640271235, 'B_1_4': -0.4615421186144975,
'B_1_5': 0.8507361759580746, 'B_1_6': -0.07961822275736011, 'B_1_7':
0.4465939542225253, 'B_1_8': -0.9346146519560599, 'B_1_9': -0.5148018847421288,
'B_2_0': -0.8862178215262869, 'B_2_1': -0.537843507195033, 'B_2_2':
-0.7750137126556917, 'B_2_3': 0.8065067524075542, 'B_2_4': -0.08409611875756276,
'B_2_5': -0.7332952440189009, 'B_2_6': 0.049040761650330805, 'B_2_7':
-0.4939513035617023, 'B_2_8': -0.6843863588274726, 'B_2_9': 0.42753395173173175,
'B_3_0': -0.2547872739233986, 'B_3_1': -0.4698317606701974, 'B_3_2':
-0.2666466702333895, 'B_3_3': -0.2940919860656242, 'B_3_4': -0.9297921674548861,
'B_3_5': 0.38109241467118904, 'B_3_6': -0.33359866052804465, 'B_3_7':
0.6983433806292921, 'B_3_8': 0.13073764125436999, 'B_3_9': -0.1541110427906995,
'B_4_0': 0.18665288796173818, 'B_4_1': 0.2931949657384917, 'B_4_2':
0.24049725509721803, 'B_4_3': -0.06600771865464405, 'B_4_4':
-0.6961518908568232, 'B_4_5': 0.1256309842583361, 'B_4_6': -0.6678760910079381,
'B_4_7': -0.7271533256559077, 'B_4_8': 0.5343606028913502, 'B_4_9':
0.9890091942294625, 'B_5_0': 0.37609447551657066, 'B_5_1': -0.5541150622443477,
'B_5_2': 0.806766332260288, 'B_5_3': -0.31656151731595106, 'B_5_4':
0.9599523510508976, 'B_5_5': -0.869975085433704, 'B_5_6': -0.11345988262044746,
'B_5_7': 0.4854753762444829, 'B_5_8': -0.23723147319875174, 'B_5_9':
0.3042069997621886, 'B_6_0': 0.15456689147621555, 'B_6_1': -0.47898656330133466,
'B_6_2': 0.43414865212041903, 'B_6_3': 0.8647462644949646, 'B_6_4':
-0.4007823512506295, 'B_6_5': 0.092048841780832, 'B_6_6': 0.5080692446921466,
'B_6_7': 0.3262148459246746, 'B_6_8': 0.25774977163064183, 'B_6_9':
0.5015864195633699, 'B_7_0': -0.7532762183723641, 'B_7_1': -0.06626077373218456,
'B_7_2': -0.09228471632435353, 'B_7_3': 0.8418805062248372, 'B_7_4':
0.945023779032479, 'B_7_5': -0.6925564511449811, 'B_7_6': -0.09152568522136025,
'B_7_7': -0.7380846566129753, 'B_7_8': -0.9315733749671642, 'B_7_9':
0.8877228618371386, 'B_8_0': -0.5912639367979697, 'B_8_1': 0.6181288500599895,
'B_8_2': 0.8024308159759487, 'B_8_3': 0.5537964326463584, 'B_8_4':
0.08395838899415933, 'B_8_5': -0.1779612406910785, 'B_8_6': 0.509422586026281,
'B_8_7': -0.6260014902701223, 'B_8_8': 0.08870593094048701, 'B_8_9':
0.5503066883213693, 'B_9_0': -0.026225117570951834, 'B_9_1':
-0.8345383109503168, 'B_9_2': -0.4062964192397384, 'B_9_3': 0.5781622097383088,
'B_9_4': -0.7444742918194333, 'B_9_5': -0.36514014183596066, 'B_9_6':
-0.6144785691224413, 'B_9_7': -0.23243367612070479, 'B_9_8': 0.9897204558033262,
'B_9_9': 0.5652538198619736, 'B_10_0': 0.46205474383668443, 'B_10_1':
-0.6052836048587804, 'B_10_2': -0.9029761807146408, 'B_10_3':

```

-0.20338786607827775, 'B_10_4': 0.7494310805202016, 'B_10_5':
-0.284841425246703, 'B_10_6': -0.5780633566985917, 'B_10_7': 0.7079873036867751,
'B_10_8': 0.14147650242057364, 'B_10_9': 0.10994946614096487}. Best is trial 174
with value: 55.46637859185029.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:34:23,721] Trial 188 finished with value: 61.602828474001875

and parameters: {'B_0_0': 0.6440590539526245, 'B_0_1': 0.31932669812942466,

'B_0_2': 0.7251835747471778, 'B_0_3': 0.09684929038984563, 'B_0_4':

-0.44339478385514286, 'B_0_5': 0.6600471470979583, 'B_0_6': 0.6100040049813399,

'B_0_7': 0.26717891191880566, 'B_0_8': 0.3556607494633418, 'B_0_9':

0.2947737149251183, 'B_1_0': 0.620948953544551, 'B_1_1': -0.10083777314563658,

'B_1_2': 0.29181141659904547, 'B_1_3': 0.5028005340782015, 'B_1_4':

-0.42493208479205535, 'B_1_5': 0.8971895936484188, 'B_1_6': -0.1590146381460143,

'B_1_7': 0.5069645091039607, 'B_1_8': -0.862999154173588, 'B_1_9':

-0.5975314433445559, 'B_2_0': -0.7725422973883154, 'B_2_1': -0.6216563396318237,

'B_2_2': -0.7231805100562299, 'B_2_3': 0.9469383570058238, 'B_2_4':

-0.01838763831676249, 'B_2_5': -0.8778671306762157, 'B_2_6':

-0.02960334162545615, 'B_2_7': -0.34688531787032295, 'B_2_8':

-0.6396303871372264, 'B_2_9': 0.45806090432213425, 'B_3_0': -0.3768347946233475,

'B_3_1': -0.42626754997162014, 'B_3_2': -0.2742742767583764, 'B_3_3':

-0.15787548396576945, 'B_3_4': -0.9290126159023784, 'B_3_5': 0.3551483726116603,

'B_3_6': -0.384422642328801, 'B_3_7': 0.47579579601526323, 'B_3_8':

0.17348265922413897, 'B_3_9': -0.05711395230702558, 'B_4_0':

0.10314889513625777, 'B_4_1': 0.31906697376602305, 'B_4_2': 0.1947087691588711,

'B_4_3': -0.13888031690386549, 'B_4_4': -0.7437442756328655, 'B_4_5':

0.38098135741166955, 'B_4_6': -0.7129772739684928, 'B_4_7': -0.5689671417965629,

'B_4_8': 0.474612467031949, 'B_4_9': 0.9177050960201452, 'B_5_0':

0.32691732962712755, 'B_5_1': -0.5034901788657394, 'B_5_2': 0.8872537213453633,

'B_5_3': -0.25918042298313065, 'B_5_4': 0.8784661587899802, 'B_5_5':

-0.8440100779322517, 'B_5_6': -0.07074293431796853, 'B_5_7': 0.6071421516679494,

'B_5_8': -0.3622393592322486, 'B_5_9': 0.38758145754702933, 'B_6_0':

0.2876153001381167, 'B_6_1': -0.31381220211932215, 'B_6_2': 0.5205843103761774,

'B_6_3': 0.8913275797554433, 'B_6_4': -0.47858422577735105, 'B_6_5':

0.2539062229996791, 'B_6_6': 0.526273564834612, 'B_6_7': 0.2406132785188334,

```

'B_6_8': 0.4193797493116558, 'B_6_9': 0.5400576027792996, 'B_7_0':
-0.8124917063997459, 'B_7_1': -0.5207241609035426, 'B_7_2':
-0.24559627089180885, 'B_7_3': 0.6714442534078354, 'B_7_4': 0.9161078451368058,
'B_7_5': -0.6151210882167342, 'B_7_6': -0.15701634851782498, 'B_7_7':
-0.7687212942577356, 'B_7_8': -0.8291281651737241, 'B_7_9': 0.8171309133895214,
'B_8_0': -0.6387587736856069, 'B_8_1': 0.4379207198710474, 'B_8_2':
0.8732043786131783, 'B_8_3': 0.4984927638340583, 'B_8_4': 0.11567868878723744,
'B_8_5': -0.2628682811678421, 'B_8_6': 0.6260818708371406, 'B_8_7':
-0.5066689615308404, 'B_8_8': 0.022420119041398824, 'B_8_9': 0.4752688008443594,
'B_9_0': -0.0963129426348673, 'B_9_1': -0.7630541595353816, 'B_9_2':
-0.5155306403439539, 'B_9_3': 0.6478965782742936, 'B_9_4': -0.6348305917166968,
'B_9_5': -0.44180745738868954, 'B_9_6': -0.6325043228140118, 'B_9_7':
-0.20892313053720926, 'B_9_8': 0.9236859560912756, 'B_9_9': 0.5703983586938854,
'B_10_0': 0.5371247611679117, 'B_10_1': -0.553940047266454, 'B_10_2':
-0.8531509774149874, 'B_10_3': -0.25528078074624394, 'B_10_4':
0.8526926481004927, 'B_10_5': -0.3235716844356644, 'B_10_6':
-0.44495027220147015, 'B_10_7': 0.752978300656478, 'B_10_8':
0.08466137822625548, 'B_10_9': 0.12999649160603954}. Best is trial 174 with
value: 55.46637859185029.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:34:38,522] Trial 189 finished with value: 62.86347287261261 and
parameters: {'B_0_0': 0.6414380448028901, 'B_0_1': 0.3682214980366341, 'B_0_2':
0.7433757475631626, 'B_0_3': -0.001986987756965742, 'B_0_4':
-0.4361171208714472, 'B_0_5': 0.6679239257933771, 'B_0_6': 0.7846563332557923,
'B_0_7': 0.36761666527553605, 'B_0_8': 0.285409482522058, 'B_0_9':
0.2918342761015248, 'B_1_0': 0.624528117208248, 'B_1_1': -0.0996680254821968,
'B_1_2': 0.24263671822209765, 'B_1_3': 0.5088907577952549, 'B_1_4':
-0.2792802341679921, 'B_1_5': 0.8495953321215769, 'B_1_6':
-0.017192185687422784, 'B_1_7': 0.4930964318311004, 'B_1_8':
-0.8643871161197382, 'B_1_9': -0.7446164308422504, 'B_2_0': -0.9620230705312031,
'B_2_1': -0.6233171830005275, 'B_2_2': -0.7177449745982335, 'B_2_3':
0.9480037664630169, 'B_2_4': -0.0026424517484940624, 'B_2_5':
-0.816334021082682, 'B_2_6': -0.020084331740429017, 'B_2_7':
-0.33725852401050327, 'B_2_8': -0.6305326125136188, 'B_2_9': 0.467024839478667,

```

```

'B_3_0': -0.36527875397128506, 'B_3_1': -0.4292636981908202, 'B_3_2':
-0.2948284334344413, 'B_3_3': -0.17125880236895574, 'B_3_4':
-0.9114950016315368, 'B_3_5': 0.3621719960348547, 'B_3_6': -0.45855976283672567,
'B_3_7': 0.4849850504482628, 'B_3_8': 0.08995611899879702, 'B_3_9':
-0.07541546743941448, 'B_4_0': 0.1436106944448948, 'B_4_1': 0.20124488609487096,
'B_4_2': 0.192619470602414, 'B_4_3': -0.08727661888735695, 'B_4_4':
-0.7262847906219273, 'B_4_5': 0.38094111378365736, 'B_4_6': -0.7218444913416455,
'B_4_7': -0.5683379903247493, 'B_4_8': 0.5614847181313868, 'B_4_9':
0.9387902882942607, 'B_5_0': 0.513268642977786, 'B_5_1': -0.5070456709736323,
'B_5_2': 0.8912465040333162, 'B_5_3': -0.3947926904552299, 'B_5_4':
0.8804594561133346, 'B_5_5': -0.8302519999366279, 'B_5_6': -0.07252904995057442,
'B_5_7': 0.6090660334944259, 'B_5_8': -0.3649911831263845, 'B_5_9':
0.37863678500033476, 'B_6_0': 0.2903458874122852, 'B_6_1': -0.3369289038206841,
'B_6_2': 0.5068252907235143, 'B_6_3': 0.8298651588231294, 'B_6_4':
-0.46684510767496545, 'B_6_5': 0.30381714967499457, 'B_6_6': 0.5917752495620303,
'B_6_7': 0.2779825240855287, 'B_6_8': 0.4187664063511276, 'B_6_9':
0.5389715215407703, 'B_7_0': -0.8037889337248524, 'B_7_1': -0.5090597553937576,
'B_7_2': -0.2595110282725658, 'B_7_3': 0.6747127920502415, 'B_7_4':
0.9095150873581455, 'B_7_5': -0.6361282711115422, 'B_7_6': -0.14454076171546518,
'B_7_7': -0.7605663308472954, 'B_7_8': -0.7306566176012191, 'B_7_9':
0.7291548195413511, 'B_8_0': -0.6346127814507013, 'B_8_1': 0.43792131740058116,
'B_8_2': 0.8620285893967169, 'B_8_3': 0.7215779210572504, 'B_8_4':
0.13958881108572788, 'B_8_5': -0.12193858366499472, 'B_8_6': 0.628706084129201,
'B_8_7': -0.541070089829263, 'B_8_8': 0.016816059771748347, 'B_8_9':
0.6681257645960583, 'B_9_0': -0.18777939947252617, 'B_9_1': -0.7815343381998883,
'B_9_2': -0.5814655988694652, 'B_9_3': 0.668740296160028, 'B_9_4':
-0.6362235103326754, 'B_9_5': -0.4255346815087234, 'B_9_6': -0.6191025889954345,
'B_9_7': -0.8521593490618583, 'B_9_8': 0.9315199307006929, 'B_9_9':
0.5692871779442426, 'B_10_0': 0.5616495649154235, 'B_10_1': -0.5498722839642831,
'B_10_2': -0.8571408823917757, 'B_10_3': -0.26776790447237575, 'B_10_4':
0.8672928679903823, 'B_10_5': -0.3415449515734759, 'B_10_6':
-0.4498644949671334, 'B_10_7': 0.740901545260338, 'B_10_8': 0.07006360453776062,
'B_10_9': 0.1408118987115566}. Best is trial 174 with value: 55.46637859185029.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:34:53,079] Trial 190 finished with value: 77.45678817216148 and parameters: {'B_0_0': 0.5866623936255745, 'B_0_1': 0.3188418765235512, 'B_0_2': 0.7147646398494399, 'B_0_3': -0.03209036056807189, 'B_0_4': -0.4832160099523878, 'B_0_5': 0.6467301786901207, 'B_0_6': 0.7155581763790883, 'B_0_7': 0.26989450146338845, 'B_0_8': 0.293019954055913, 'B_0_9': 0.3478344244922487, 'B_1_0': 0.7826492643223822, 'B_1_1': -0.11286972747413657, 'B_1_2': 0.2920467007644775, 'B_1_3': 0.43744195225244054, 'B_1_4': -0.43899441940793776, 'B_1_5': 0.8976679525002355, 'B_1_6': -0.13625464558478764, 'B_1_7': 0.36710417344493096, 'B_1_8': -0.8223450994400552, 'B_1_9': -0.5446796062817172, 'B_2_0': -0.7884970154534058, 'B_2_1': -0.6781443073218585, 'B_2_2': -0.8044720353386502, 'B_2_3': 0.9502426332545444, 'B_2_4': 0.08439698261621517, 'B_2_5': -0.6081949429460574, 'B_2_6': -0.11723903697400404, 'B_2_7': -0.4081374664069619, 'B_2_8': -0.6870671221206479, 'B_2_9': 0.5656943793940377, 'B_3_0': -0.3247419349821174, 'B_3_1': -0.2990607957668811, 'B_3_2': -0.5120207171664942, 'B_3_3': -0.20819921339283343, 'B_3_4': -0.9892628033582099, 'B_3_5': 0.297676103911327, 'B_3_6': -0.3154077982812914, 'B_3_7': 0.5475202524935632, 'B_3_8': 0.03944030030934373, 'B_3_9': -0.05657230625379572, 'B_4_0': 0.10978466231019515, 'B_4_1': 0.32272052623663094, 'B_4_2': 0.2501340995966964, 'B_4_3': 0.0548454888942336, 'B_4_4': -0.7604666041770516, 'B_4_5': 0.3305117006210216, 'B_4_6': -0.6906740381024757, 'B_4_7': -0.5293443985030061, 'B_4_8': 0.4782132209300673, 'B_4_9': 0.7254161447028674, 'B_5_0': 0.33987723440970724, 'B_5_1': -0.6226777023308689, 'B_5_2': 0.8322097786508595, 'B_5_3': -0.2601309027888039, 'B_5_4': 0.9548026122821023, 'B_5_5': -0.9064968111910867, 'B_5_6': -0.11887764051671343, 'B_5_7': 0.6813770903504637, 'B_5_8': -0.38308932428542414, 'B_5_9': 0.324879345986067, 'B_6_0': 0.3312765528567122, 'B_6_1': -0.2620902126947314, 'B_6_2': 0.5233570364100315, 'B_6_3': 0.8824715676068186, 'B_6_4': -0.3436519657560902, 'B_6_5': 0.25012289916479025, 'B_6_6': 0.5246715464014033, 'B_6_7': 0.41313042454960713, 'B_6_8': 0.33215885887204655, 'B_6_9': 0.3515309280878056, 'B_7_0': -0.7765295947765921, 'B_7_1': -0.14991946191686087, 'B_7_2': -0.07667230174405623, 'B_7_3': 0.6081399642500214, 'B_7_4': 0.8341140716647321, 'B_7_5': -0.6318009849733099, 'B_7_6': -0.16837601319655687, 'B_7_7': -0.699449713253093, 'B_7_8': -0.8300955318830067, 'B_7_9': 0.7882049498592172, 'B_8_0': -0.5869574988992378, 'B_8_1': 0.38039147308569915, 'B_8_2': 0.7229862944268146, 'B_8_3': 0.5139004136985688, 'B_8_4': 0.09440811327634377, 'B_8_5': -0.23434358070440087, 'B_8_6': 0.6955128167825251, 'B_8_7': -0.48943677939653, 'B_8_8': -0.04813899164344201, 'B_8_9': 0.5011136082124642, 'B_9_0': -0.07069486219708104, 'B_9_1': -0.6382801547155771, 'B_9_2': -0.5390380796625729, 'B_9_3': 0.6362556947683852, 'B_9_4': -0.7803438805317924, 'B_9_5': -0.4624661142880389, 'B_9_6': -0.6567920213932125, 'B_9_7': -0.21065983156827933, 'B_9_8': 0.9378545573054347, 'B_9_9': 0.5996729326957416, 'B_10_0': 0.5021542072930364, 'B_10_1': -0.5287512650912254, 'B_10_2': -0.8640390876376461, 'B_10_3': -0.3281732962437583, 'B_10_4': 0.8493368555283357, 'B_10_5': -0.27878772259957524, 'B_10_6': -0.401596803270979, 'B_10_7': 0.8006649518504404, 'B_10_8': 0.10381918925099742, 'B_10_9': 0.09151012462489308}. Best is trial 174 with value: 55.46637859185029. Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

```
Built matrix A = [<F_[u_i], F_[u_j]>].
```

```
Built matrix B = inverse(A).
```

```
[I 2025-04-29 14:35:07,466] Trial 191 finished with value: 59.66090426553775 and
parameters: {'B_0_0': 0.6933441336375377, 'B_0_1': 0.43545926805867663, 'B_0_2':
0.5490847486810341, 'B_0_3': 0.093224815658398, 'B_0_4': -0.5391742358918729,
'B_0_5': 0.7495335252637191, 'B_0_6': 0.600400818714238, 'B_0_7':
0.3692021125189203, 'B_0_8': 0.3531558359111984, 'B_0_9': 0.2697498076038398,
'B_1_0': 0.6032719605939806, 'B_1_1': -0.18213905383605458, 'B_1_2':
0.3420854767760376, 'B_1_3': 0.4746665512892627, 'B_1_4': -0.36993670316097876,
'B_1_5': 0.9011599121604317, 'B_1_6': -0.17308144096411657, 'B_1_7':
0.5386280042371164, 'B_1_8': -0.9846802993666481, 'B_1_9': -0.5768460599529723,
'B_2_0': -0.829496585911005, 'B_2_1': -0.5014307577597746, 'B_2_2':
-0.6317707694872577, 'B_2_3': 0.8608981274208951, 'B_2_4': -0.17940568860942763,
'B_2_5': -0.8936755362740954, 'B_2_6': 0.019514556718852077, 'B_2_7':
-0.492809377998906, 'B_2_8': -0.5778624409300979, 'B_2_9': 0.4108900580412029,
'B_3_0': -0.3925504351926869, 'B_3_1': -0.4752441819598655, 'B_3_2':
-0.44817835212459833, 'B_3_3': -0.1250636272164589, 'B_3_4':
-0.8681928723506607, 'B_3_5': 0.4516176421630084, 'B_3_6': -0.36155724477883167,
'B_3_7': 0.6552183799887088, 'B_3_8': 0.16044913390724105, 'B_3_9':
-0.11449966992978781, 'B_4_0': 0.19150510576271543, 'B_4_1': 0.2946699231940621,
'B_4_2': 0.17807288831763485, 'B_4_3': -0.13017752239148805, 'B_4_4':
-0.691276091919531, 'B_4_5': 0.2596754459523026, 'B_4_6': -0.647542077443536,
'B_4_7': -0.6298586353348167, 'B_4_8': 0.4663438089538394, 'B_4_9':
0.9022874342520811, 'B_5_0': 0.3263328140932922, 'B_5_1': -0.5607571224072112,
'B_5_2': 0.9190749771885149, 'B_5_3': -0.31533215214168286, 'B_5_4':
0.8692166475377963, 'B_5_5': -0.9395651517499046, 'B_5_6': -0.18055633543784333,
'B_5_7': 0.5633598738233657, 'B_5_8': -0.1728908424624993, 'B_5_9':
0.4274750572485341, 'B_6_0': 0.13414988262416408, 'B_6_1': -0.4099341860143644,
'B_6_2': 0.5372598635853855, 'B_6_3': 0.9064063847477689, 'B_6_4':
-0.4300604925386029, 'B_6_5': 0.14466473074736286, 'B_6_6': 0.4764466916616244,
'B_6_7': 0.38275480092670866, 'B_6_8': 0.4158517147348467, 'B_6_9':
0.5610702589170948, 'B_7_0': -0.8222451737115778, 'B_7_1': -0.4628068250227633,
'B_7_2': -0.2194892927888971, 'B_7_3': 0.7586810255964492, 'B_7_4':
0.9533111907535042, 'B_7_5': -0.7325888812646337, 'B_7_6': -0.19653793884303775,
'B_7_7': -0.794769316734457, 'B_7_8': -0.8923342557359704, 'B_7_9':
0.8296664224738197, 'B_8_0': -0.6534159967556661, 'B_8_1': 0.4377967553535683,
```

'B_8_2': 0.8896502932278549, 'B_8_3': 0.4122637806964048, 'B_8_4':
-0.006377286353452985, 'B_8_5': -0.271073359010567, 'B_8_6': 0.512875083627508,
'B_8_7': -0.5989156997144396, 'B_8_8': 0.03814525650230568, 'B_8_9':
0.7140070214131646, 'B_9_0': -0.12532162228363317, 'B_9_1': -0.7496095101077157,
'B_9_2': -0.4912013363637383, 'B_9_3': 0.660497132877579, 'B_9_4':
-0.6527849305777991, 'B_9_5': -0.5520641468987204, 'B_9_6': -0.7018148475216989,
'B_9_7': -0.24105387986838062, 'B_9_8': 0.9136605676953066, 'B_9_9':
0.5483825256016738, 'B_10_0': 0.6794800437761552, 'B_10_1': -0.5781206757061498,
'B_10_2': -0.9194840722496702, 'B_10_3': -0.23879678228964385, 'B_10_4':
0.7602726375318593, 'B_10_5': -0.3484828156408588, 'B_10_6':
-0.5183758601918732, 'B_10_7': 0.7516837941702067, 'B_10_8':
0.026873142752156867, 'B_10_9': 0.06496115053079159}. Best is trial 174 with
value: 55.46637859185029.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$\langle F_{[u_i]}, F_{[u_j]} \rangle$].

Built matrix B = inverse(A).

[I 2025-04-29 14:35:21,841] Trial 192 finished with value: 60.94879565292185 and
parameters: {'B_0_0': 0.7371056704344305, 'B_0_1': 0.42319001521607047, 'B_0_2':
0.6880894164783219, 'B_0_3': 0.06966488712481649, 'B_0_4': -0.5333513152799033,
'B_0_5': 0.7046272783475944, 'B_0_6': 0.6197866040001168, 'B_0_7':
0.42198352694228924, 'B_0_8': 0.3432854261499521, 'B_0_9': 0.2880988878971929,
'B_1_0': 0.6500957293569579, 'B_1_1': -0.19683393399403637, 'B_1_2':
0.3481070504777185, 'B_1_3': 0.4787847563153279, 'B_1_4': -0.365914884952443,
'B_1_5': 0.8773986443321542, 'B_1_6': -0.15828773364098503, 'B_1_7':
0.5058798889103105, 'B_1_8': -0.986171743810598, 'B_1_9': -0.6000884023181063,
'B_2_0': -0.8103799606422514, 'B_2_1': -0.5483381455020045, 'B_2_2':
-0.6588683456196776, 'B_2_3': 0.8511303375267418, 'B_2_4': -0.26910629187324175,
'B_2_5': -0.9439580414894135, 'B_2_6': 0.08568195683847839, 'B_2_7':
-0.5063187817442549, 'B_2_8': -0.5689869920463403, 'B_2_9': 0.42066026528769335,
'B_3_0': -0.3905884116811762, 'B_3_1': -0.4339641426531888, 'B_3_2':
-0.4691227054823223, 'B_3_3': -0.13769192274769237, 'B_3_4':
-0.8648224237829316, 'B_3_5': 0.4327465624088871, 'B_3_6': -0.467847262675203,
'B_3_7': 0.6218936034338951, 'B_3_8': 0.1852412731101028, 'B_3_9':
-0.0934155254490291, 'B_4_0': 0.16606809922006993, 'B_4_1': 0.28372994453646283,
'B_4_2': 0.21237673564276816, 'B_4_3': -0.12525544258634538, 'B_4_4':
-0.7059058770910057, 'B_4_5': 0.4425174960450873, 'B_4_6': -0.6571642752845124,

```

'B_4_7': -0.6331375695206348, 'B_4_8': 0.5275930313993918, 'B_4_9':
0.8999296246658228, 'B_5_0': 0.4107033854978818, 'B_5_1': -0.5063506210234919,
'B_5_2': 0.9278221543352188, 'B_5_3': -0.42692318132552964, 'B_5_4':
0.8935223957115911, 'B_5_5': -0.8573694630114342, 'B_5_6': -0.13063172322083286,
'B_5_7': 0.5625902866544643, 'B_5_8': -0.16390133249363353, 'B_5_9':
0.433580475684437, 'B_6_0': 0.13409793392533823, 'B_6_1': -0.3181447058620495,
'B_6_2': 0.3587470651355805, 'B_6_3': 0.9057365238280847, 'B_6_4':
-0.42235633235385, 'B_6_5': 0.18518267209175984, 'B_6_6': 0.479580809652147,
'B_6_7': 0.2835612566946209, 'B_6_8': 0.3848964635296287, 'B_6_9':
0.4343165395038156, 'B_7_0': -0.6768498738499636, 'B_7_1': -0.38303170412536136,
'B_7_2': -0.2256039766785913, 'B_7_3': 0.7095581213475316, 'B_7_4':
0.9534233101712579, 'B_7_5': -0.7546196267580917, 'B_7_6': -0.09981139090804747,
'B_7_7': -0.7807143968873702, 'B_7_8': -0.8910534778816226, 'B_7_9':
0.8930692004954013, 'B_8_0': -0.6434930638541031, 'B_8_1': 0.48818203756519807,
'B_8_2': 0.8838176031726663, 'B_8_3': 0.41291175116913315, 'B_8_4':
0.05921358862243417, 'B_8_5': -0.2691899373967491, 'B_8_6': 0.5294766521482983,
'B_8_7': -0.4979117097187901, 'B_8_8': 0.013324724423241544, 'B_8_9':
0.613446878945908, 'B_9_0': -0.10209367709163131, 'B_9_1': -0.7319178730436737,
'B_9_2': -0.5076324012065869, 'B_9_3': 0.660896926125535, 'B_9_4':
-0.5884589956153387, 'B_9_5': -0.5491259532846561, 'B_9_6': -0.7040379028077841,
'B_9_7': -0.19762152346341896, 'B_9_8': 0.9034333202651934, 'B_9_9':
0.547508240754157, 'B_10_0': 0.6382659382808361, 'B_10_1': -0.6769499847502857,
'B_10_2': -0.9224773271913541, 'B_10_3': -0.12221635381909488, 'B_10_4':
0.8290392229558626, 'B_10_5': -0.3398784221606486, 'B_10_6':
-0.6340734914068665, 'B_10_7': 0.70555056419333, 'B_10_8': 0.02088509413128209,
'B_10_9': 0.0538813132637184}. Best is trial 174 with value: 55.46637859185029.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:35:36,640] Trial 193 finished with value: 63.80179102600272 and
parameters: {'B_0_0': 0.7532639036980916, 'B_0_1': 0.43613707977114513, 'B_0_2':
0.6716659430599744, 'B_0_3': 0.09644947965092522, 'B_0_4': -0.5334942132029028,
'B_0_5': 0.70043400310867, 'B_0_6': 0.6132551873523779, 'B_0_7':
0.4180551203298039, 'B_0_8': 0.34159908215233364, 'B_0_9': 0.2863656910406879,
'B_1_0': 0.6559289451058209, 'B_1_1': -0.1929955823002574, 'B_1_2':
0.3469300978073431, 'B_1_3': 0.43248884895391737, 'B_1_4': -0.48030449179117474,

```

'B_1_5': 0.9464173851530113, 'B_1_6': -0.1641006933642138, 'B_1_7':
 0.5345780589245849, 'B_1_8': -0.942129535772838, 'B_1_9': -0.6082256216596413,
 'B_2_0': -0.8277800328753909, 'B_2_1': -0.5461083134723204, 'B_2_2':
 -0.6188111632636153, 'B_2_3': 0.8475696860247441, 'B_2_4': -0.25378339439867853,
 'B_2_5': -0.9830722118493977, 'B_2_6': 0.13488710272896406, 'B_2_7':
 -0.5468932697581287, 'B_2_8': -0.5647837099919127, 'B_2_9': 0.4570047438650813,
 'B_3_0': -0.39375739782395586, 'B_3_1': -0.42351063877747436, 'B_3_2':
 -0.46628445660296525, 'B_3_3': -0.10992497934910497, 'B_3_4':
 -0.9504256945305132, 'B_3_5': 0.447378762871991, 'B_3_6': -0.4353456043867559,
 'B_3_7': 0.6606691169750813, 'B_3_8': 0.11173490729084101, 'B_3_9':
 -0.11232996262556544, 'B_4_0': 0.1857124409683177, 'B_4_1': 0.14536543609830288,
 'B_4_2': 0.22103235637677798, 'B_4_3': -0.14731572864924855, 'B_4_4':
 -0.6842727763435714, 'B_4_5': 0.44596125479748866, 'B_4_6': -0.6514438675725572,
 'B_4_7': -0.6351182117314912, 'B_4_8': 0.47342565708452766, 'B_4_9':
 0.9046723395077839, 'B_5_0': 0.4123931911240887, 'B_5_1': -0.5107155820327853,
 'B_5_2': 0.9106295453338722, 'B_5_3': -0.438521890581424, 'B_5_4':
 0.8688937468810859, 'B_5_5': -0.8527565353229136, 'B_5_6': -0.07951317041246833,
 'B_5_7': 0.5872414109500196, 'B_5_8': -0.15810535938654477, 'B_5_9':
 0.4261413398895198, 'B_6_0': 0.18336004327991093, 'B_6_1': -0.4264663592576684,
 'B_6_2': 0.3436112449175118, 'B_6_3': 0.9072648494085324, 'B_6_4':
 -0.49316533829172415, 'B_6_5': 0.2214620813642647, 'B_6_6': 0.5515363934187753,
 'B_6_7': 0.24750497878174904, 'B_6_8': 0.39407126429777994, 'B_6_9':
 0.5231681955266009, 'B_7_0': -0.670751470921592, 'B_7_1': -0.3836701905944558,
 'B_7_2': -0.18421310891731857, 'B_7_3': 0.6559683187528189, 'B_7_4':
 0.9557606392058544, 'B_7_5': -0.7712582357393979, 'B_7_6': -0.11014038412050112,
 'B_7_7': -0.7738184859181824, 'B_7_8': -0.8828420221102765, 'B_7_9':
 0.8742571065239167, 'B_8_0': -0.6463454972850132, 'B_8_1': 0.5297732407244173,
 'B_8_2': 0.900181893130018, 'B_8_3': 0.412684926823758, 'B_8_4':
 -0.0001749529495641694, 'B_8_5': -0.3402638957493367, 'B_8_6':
 0.5273336772993834, 'B_8_7': -0.6003819932687664, 'B_8_8': 0.027242345272216053,
 'B_8_9': 0.6258457116055196, 'B_9_0': -0.0629251492026751, 'B_9_1':
 -0.7243128453271803, 'B_9_2': -0.4821561353389387, 'B_9_3': 0.5955989761235556,
 'B_9_4': -0.7312638718133799, 'B_9_5': -0.6326311496593621, 'B_9_6':
 -0.6957641707048601, 'B_9_7': -0.613636939986772, 'B_9_8': 0.8937230737601217,
 'B_9_9': 0.5415717250272523, 'B_10_0': 0.6414377073015899, 'B_10_1':
 -0.6708333888985214, 'B_10_2': -0.9140227774231752, 'B_10_3':
 -0.12291392897603733, 'B_10_4': 0.9318310602735898, 'B_10_5':
 -0.31950712559432953, 'B_10_6': -0.609840863953642, 'B_10_7':
 0.8015999736442236, 'B_10_8': 0.028406704341328484, 'B_10_9':
 0.1379900333676225}. Best is trial 174 with value: 55.46637859185029.

Building the Basis functions for the RKHS:

Built F__[u_1].
 Built F__[u_2].
 Built F__[u_3].
 Built F__[u_4].
 Built F__[u_5].
 Built F__[u_6].
 Built F__[u_7].

```

Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
Established the refined x grid with 1001 points.
Built matrix A = [<F_[u_i], F_[u_j]>].
Built matrix B = inverse(A).
[I 2025-04-29 14:35:51,193] Trial 194 finished with value: 63.86436430388649 and
parameters: {'B_0_0': 0.70496416519818, 'B_0_1': 0.388041350964329, 'B_0_2':
0.5533028695814689, 'B_0_3': 0.08665599253118127, 'B_0_4': -0.535552848962845,
'B_0_5': 0.7318347743985073, 'B_0_6': 0.6532840726113436, 'B_0_7':
0.1905586836896594, 'B_0_8': 0.3533829565747458, 'B_0_9': 0.25365097278721815,
'B_1_0': 0.6076444839631027, 'B_1_1': -0.25664099176968386, 'B_1_2':
0.2608036903610772, 'B_1_3': 0.6094895315055837, 'B_1_4': -0.4312392202997158,
'B_1_5': 0.8334560786123075, 'B_1_6': -0.18745997799865935, 'B_1_7':
0.6068221968600809, 'B_1_8': -0.8929328924617985, 'B_1_9': -0.5221691803541482,
'B_2_0': -0.8823625518319956, 'B_2_1': -0.5005493875609122, 'B_2_2':
-0.7154505879130908, 'B_2_3': 0.9039856727713904, 'B_2_4': -0.09255604687020078,
'B_2_5': -0.9682863041072824, 'B_2_6': 0.08792398901763616, 'B_2_7':
-0.37310987528337425, 'B_2_8': -0.6220005793081966, 'B_2_9': 0.4201483146521344,
'B_3_0': -0.4816107830018577, 'B_3_1': -0.38117592155752794, 'B_3_2':
-0.6029750959984004, 'B_3_3': -0.2322119687134488, 'B_3_4': -0.8197090661623007,
'B_3_5': 0.4791213687766837, 'B_3_6': -0.40758460563875937, 'B_3_7':
0.6373049276406372, 'B_3_8': 0.19170346611805816, 'B_3_9': -0.09483947644919384,
'B_4_0': 0.14622175019049297, 'B_4_1': 0.278618004367422, 'B_4_2':
0.33917652717896385, 'B_4_3': -0.12354190513011731, 'B_4_4':
-0.7162401680872693, 'B_4_5': 0.3938925695063047, 'B_4_6': -0.5286765959884976,
'B_4_7': -0.5952660763125432, 'B_4_8': 0.5531569448086353, 'B_4_9':
0.8661148951197822, 'B_5_0': 0.3251300861186146, 'B_5_1': -0.5494964173192451,
'B_5_2': 0.9421672268908456, 'B_5_3': -0.3470417928376228, 'B_5_4':
0.9014263088903358, 'B_5_5': -0.8030088696640781, 'B_5_6': -0.14177042429892642,
'B_5_7': 0.5454870833280145, 'B_5_8': -0.08170048414949853, 'B_5_9':
0.39038943537385606, 'B_6_0': 0.2908818834267125, 'B_6_1': -0.3990011966512911,
'B_6_2': 0.43341584765718605, 'B_6_3': 0.8674535606968993, 'B_6_4':
-0.42106547940126343, 'B_6_5': 0.18087441625135398, 'B_6_6':
0.42781619422659867, 'B_6_7': 0.2696862668462025, 'B_6_8': 0.44933628135877757,
'B_6_9': 0.565509491816701, 'B_7_0': -0.7101040122858175, 'B_7_1':
-0.46808730631788825, 'B_7_2': -0.20609836326974343, 'B_7_3':
0.7061351147107563, 'B_7_4': 0.8995994952587623, 'B_7_5': -0.7423185645887831,
'B_7_6': -0.19457531482734752, 'B_7_7': -0.796541265428276, 'B_7_8':
-0.9258550125424283, 'B_7_9': 0.908972180766225, 'B_8_0': -0.6975988815691898,
'B_8_1': 0.4756443453361607, 'B_8_2': 0.8110606263710973, 'B_8_3':
0.43611710215130495, 'B_8_4': 0.0445347770656491, 'B_8_5': -0.26561300423698286,
'B_8_6': 0.6096352897992673, 'B_8_7': -0.5803593291107612, 'B_8_8':
0.001680371413580399, 'B_8_9': 0.6877035114912893, 'B_9_0':
-0.13459241876070616, 'B_9_1': -0.6819992071726274, 'B_9_2':
-0.42275447409984346, 'B_9_3': 0.6810635542654339, 'B_9_4': -0.6607296407073074,
'B_9_5': -0.5616434050157723, 'B_9_6': -0.572345411434947, 'B_9_7':

```

```
-0.9890861201273394, 'B_9_8': 0.9406673012092528, 'B_9_9': 0.6221576360519293,
'B_10_0': 0.6850311881862101, 'B_10_1': -0.7077488449538146, 'B_10_2':
-0.9327965899835337, 'B_10_3': -0.2404429683484672, 'B_10_4':
0.8293047308473797, 'B_10_5': -0.36728481132518265, 'B_10_6':
-0.6671460468104085, 'B_10_7': 0.6924685405210492, 'B_10_8': 0.0268302666398198,
'B_10_9': 0.059805263276922996}. Best is trial 174 with value:
55.46637859185029.
```

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```
[I 2025-04-29 14:36:05,727] Trial 195 finished with value: 54.28004122812132 and
parameters: {'B_0_0': 0.6014514607562307, 'B_0_1': 0.5735267971646951, 'B_0_2':
0.7577753644449686, 'B_0_3': 0.17838432293651585, 'B_0_4': -0.5052041761399297,
'B_0_5': 0.7004926302713071, 'B_0_6': 0.5287147032837481, 'B_0_7':
0.32414022562922773, 'B_0_8': 0.44334356457907276, 'B_0_9': 0.4235941193837888,
'B_1_0': 0.5633255094831823, 'B_1_1': -0.2073753641420864, 'B_1_2':
0.3108958580453045, 'B_1_3': 0.4806708020049012, 'B_1_4': -0.3595718312981263,
'B_1_5': 0.8954605488194237, 'B_1_6': -0.275063090788184, 'B_1_7':
0.5507783648758756, 'B_1_8': -0.8414038592913677, 'B_1_9': -0.6213914664372135,
'B_2_0': -0.8226917230267179, 'B_2_1': -0.48918397667370106, 'B_2_2':
-0.6417633831384946, 'B_2_3': 0.9029950021739956, 'B_2_4':
-0.024116832075911313, 'B_2_5': -0.9117044601584461, 'B_2_6':
-0.010019856663255222, 'B_2_7': -0.4925270615596452, 'B_2_8':
-0.5774568482167913, 'B_2_9': 0.306326257054636, 'B_3_0': -0.34652571227601553,
'B_3_1': -0.43398290612229695, 'B_3_2': -0.4331846111166192, 'B_3_3':
-0.12755020712175713, 'B_3_4': -0.8318423355001994, 'B_3_5':
0.42417477809041354, 'B_3_6': -0.37898675088082195, 'B_3_7':
0.46191383348667736, 'B_3_8': 0.22774745927110007, 'B_3_9': -0.0616870950491558,
'B_4_0': 0.29002463373034526, 'B_4_1': 0.21758452084490185, 'B_4_2':
0.2517373133306647, 'B_4_3': -0.20269439965916397, 'B_4_4': -0.8426190284553837,
'B_4_5': 0.34581971585037863, 'B_4_6': -0.5827838365041906, 'B_4_7':
-0.43870602124746993, 'B_4_8': 0.3395972619516492, 'B_4_9': 0.9210130999591085,
'B_5_0': 0.3891890627593494, 'B_5_1': -0.5013991416130413, 'B_5_2':
0.8721910510124169, 'B_5_3': -0.27254058083022514, 'B_5_4': 0.8440727748963478,
'B_5_5': -0.9362810054724999, 'B_5_6': -0.1932762602376178, 'B_5_7':
0.9874218990790898, 'B_5_8': -0.16658154569033307, 'B_5_9': 0.4257150039888762,
```

```

'B_6_0': 0.13167471733215752, 'B_6_1': -0.3067550064498173, 'B_6_2':
0.48283741216854653, 'B_6_3': 0.9282345047605515, 'B_6_4': -0.2973341914381555,
'B_6_5': 0.28274017567144316, 'B_6_6': 0.636175241971961, 'B_6_7':
0.29293182838344023, 'B_6_8': 0.42415613502490257, 'B_6_9': 0.4353070131409812,
'B_7_0': -0.8229798752408009, 'B_7_1': -0.21742736182128108, 'B_7_2':
-0.261565598734661, 'B_7_3': 0.7570440238811458, 'B_7_4': 0.8589819399252865,
'B_7_5': -0.7132100345791336, 'B_7_6': -0.20859462119199176, 'B_7_7':
-0.7314426342828411, 'B_7_8': -0.8336592016231347, 'B_7_9': 0.834185665139375,
'B_8_0': -0.6457858651074311, 'B_8_1': 0.3960518525150276, 'B_8_2':
0.9241392958258403, 'B_8_3': 0.5758198283231161, 'B_8_4': 0.1899683004294745,
'B_8_5': -0.3506810818208961, 'B_8_6': 0.49139392682578054, 'B_8_7':
-0.6378914603385553, 'B_8_8': -0.07634121341565378, 'B_8_9': 0.7082599388579243,
'B_9_0': 0.007740729781086059, 'B_9_1': -0.7370900176818378, 'B_9_2':
-0.6275252085950989, 'B_9_3': 0.5493325786504435, 'B_9_4': -0.5851081067721684,
'B_9_5': -0.5133453990415077, 'B_9_6': -0.7191419308534533, 'B_9_7':
-0.2018250758097277, 'B_9_8': 0.9135057522378175, 'B_9_9': 0.45925671799243684,
'B_10_0': 0.5965845082192951, 'B_10_1': -0.6177496704364677, 'B_10_2':
-0.9421537805650833, 'B_10_3': -0.19977444095766553, 'B_10_4':
0.6472722282120762, 'B_10_5': -0.29115144828945094, 'B_10_6':
-0.5118823888557921, 'B_10_7': 0.7515823152900177, 'B_10_8':
0.18383201116787773, 'B_10_9': 0.11072137082128672}. Best is trial 195 with
value: 54.28004122812132.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:36:19,634] Trial 196 finished with value: 63.663084762016375
and parameters: {'B_0_0': 0.604410786855454, 'B_0_1': 0.5768733628267584,
'B_0_2': 0.5024039980136367, 'B_0_3': 0.19854599637555964, 'B_0_4':
-0.43315484361634093, 'B_0_5': 0.6309158647706965, 'B_0_6': 0.5135218747216168,
'B_0_7': 0.41517892323934014, 'B_0_8': 0.4419322481280793, 'B_0_9':
0.44069330452246236, 'B_1_0': 0.5722167373382923, 'B_1_1': -0.32983481475088783,
'B_1_2': 0.30586050080219973, 'B_1_3': 0.4604182382076777, 'B_1_4':
-0.3625642034205156, 'B_1_5': 0.8924645557612326, 'B_1_6': -0.2657521667233414,
'B_1_7': 0.49637448599343453, 'B_1_8': -0.8495358405897313, 'B_1_9':
-0.611930128838758, 'B_2_0': -0.8223016589780583, 'B_2_1': -0.526868298829969,
'B_2_2': -0.6924233415642713, 'B_2_3': 0.9512263948242454, 'B_2_4':

```



```

-0.02366141198595359, 'B_2_5': -0.8516308952569337, 'B_2_6':
0.015685308917978015, 'B_2_7': -0.5799035641785824, 'B_2_8': -0.570878530970782,
'B_2_9': 0.3164398342159157, 'B_3_0': -0.34072724916880237, 'B_3_1':
-0.444055984718123, 'B_3_2': -0.42992580453202567, 'B_3_3': -0.1344511811749828,
'B_3_4': -0.9314955531480069, 'B_3_5': 0.5348282239136718, 'B_3_6':
-0.37913511269298433, 'B_3_7': 0.44789385314870045, 'B_3_8':
0.23267221220305803, 'B_3_9': -0.11637019941812465, 'B_4_0':
0.10907444304289599, 'B_4_1': 0.23121740477804087, 'B_4_2': 0.25466229342562463,
'B_4_3': -0.016832369688863008, 'B_4_4': -0.8437808620695225, 'B_4_5':
0.4290467347619229, 'B_4_6': -0.5827310242197818, 'B_4_7': -0.4515182019815568,
'B_4_8': 0.2655397358308257, 'B_4_9': 0.9108699949864073, 'B_5_0':
0.3929681072822193, 'B_5_1': -0.5133802003280957, 'B_5_2': 0.8669560170279402,
'B_5_3': -0.2547208689235404, 'B_5_4': 0.8592364541851574, 'B_5_5':
-0.9380758787425623, 'B_5_6': -0.1853563243264028, 'B_5_7': 0.9769726886845118,
'B_5_8': -0.16540614690522792, 'B_5_9': 0.22822342083185107, 'B_6_0':
0.13493353804996056, 'B_6_1': -0.3512574499671808, 'B_6_2': 0.47575209328030343,
'B_6_3': 0.9119869334880155, 'B_6_4': -0.37033545308499083, 'B_6_5':
0.2924307562098477, 'B_6_6': 0.6212366983457924, 'B_6_7': 0.2965201641833971,
'B_6_8': 0.41939825318246715, 'B_6_9': 0.41662941540244847, 'B_7_0':
-0.7793321755240716, 'B_7_1': -0.225188633314136, 'B_7_2': -0.23834399123207112,
'B_7_3': 0.7367269526069987, 'B_7_4': 0.8580433950333098, 'B_7_5':
-0.8921482259832, 'B_7_6': -0.19635988953544692, 'B_7_7': -0.729134236309005,
'B_7_8': -0.8372906376462428, 'B_7_9': 0.9874121513309196, 'B_8_0':
-0.6219782897442877, 'B_8_1': 0.38997503519164245, 'B_8_2': 0.7691857649101517,
'B_8_3': 0.5835929648296525, 'B_8_4': 0.2751302931675093, 'B_8_5':
-0.3516660392874609, 'B_8_6': 0.487626760831, 'B_8_7': -0.6333898672968951,
'B_8_8': -0.07310294116389093, 'B_8_9': 0.46443836777146974, 'B_9_0':
-0.05289509299323439, 'B_9_1': -0.8226139132763849, 'B_9_2':
-0.5086005315514125, 'B_9_3': 0.5161223119421818, 'B_9_4': -0.5790895982413583,
'B_9_5': -0.5200466939832591, 'B_9_6': -0.7747376592713484, 'B_9_7':
-0.1960578085388156, 'B_9_8': 0.9208868058025602, 'B_9_9': 0.4575873692007047,
'B_10_0': 0.5983901305188094, 'B_10_1': -0.6343401873156818, 'B_10_2':
-0.8559455847727866, 'B_10_3': -0.18633970882772471, 'B_10_4':
0.6481189907774267, 'B_10_5': -0.262891409449601, 'B_10_6': -0.5101555183510457,
'B_10_7': 0.6566253310077349, 'B_10_8': 0.17514706909426617, 'B_10_9':
0.10615713238248825}. Best is trial 195 with value: 54.28004122812132.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.
 Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].
 Built matrix B = inverse(A).
 [I 2025-04-29 14:36:34,105] Trial 197 finished with value: 59.77987623459607 and parameters: {'B_0_0': 0.5696418318240855, 'B_0_1': 0.4613047527297802, 'B_0_2': 0.6795539849783326, 'B_0_3': 0.10326473270153338, 'B_0_4': -0.4944347990200065, 'B_0_5': 0.69740178366043, 'B_0_6': 0.6008240182963913, 'B_0_7': 0.242744551106005, 'B_0_8': 0.22837113782543925, 'B_0_9': 0.4083762738285869, 'B_1_0': 0.6429626378089506, 'B_1_1': -0.19188294598198508, 'B_1_2': 0.34385051717873893, 'B_1_3': 0.5314718869100058, 'B_1_4': -0.2733628076821604, 'B_1_5': 0.8197036612111153, 'B_1_6': -0.2687744275420388, 'B_1_7': 0.5568966239177752, 'B_1_8': -0.7883218921866855, 'B_1_9': -0.5428111147860645, 'B_2_0': -0.8673799278262384, 'B_2_1': -0.5620898659229603, 'B_2_2': -0.6395928765416113, 'B_2_3': 0.9099663991079281, 'B_2_4': -0.0016165667410204254, 'B_2_5': -0.9043134947762271, 'B_2_6': -0.0055412455627108365, 'B_2_7': -0.44076986186102435, 'B_2_8': -0.6603106563697394, 'B_2_9': 0.4016411782109275, 'B_3_0': -0.30066595762111625, 'B_3_1': -0.41418060826071995, 'B_3_2': -0.37324743182946585, 'B_3_3': -0.08940194519023834, 'B_3_4': -0.8821339462427903, 'B_3_5': 0.425265976221391, 'B_3_6': -0.4704973411783565, 'B_3_7': 0.5105040745391194, 'B_3_8': 0.18903490831698033, 'B_3_9': -0.06311102276723807, 'B_4_0': 0.18576208513823728, 'B_4_1': 0.18086976531695356, 'B_4_2': 0.2142321181160443, 'B_4_3': -0.17525528049358272, 'B_4_4': -0.9286088315778933, 'B_4_5': 0.3530519035935195, 'B_4_6': -0.6520171613026297, 'B_4_7': -0.5153632975450076, 'B_4_8': 0.4970391114254606, 'B_4_9': 0.9463316242773355, 'B_5_0': 0.5626267852017286, 'B_5_1': -0.4803357854618584, 'B_5_2': 0.7846995447484366, 'B_5_3': -0.2788522064775907, 'B_5_4': 0.9236772448458598, 'B_5_5': -0.8532095748044712, 'B_5_6': -0.20362932615149376, 'B_5_7': 0.6596052438054341, 'B_5_8': -0.1169670269187087, 'B_5_9': 0.39444933927092213, 'B_6_0': 0.04831063849079739, 'B_6_1': -0.3096357049799101, 'B_6_2': 0.48596080038608624, 'B_6_3': 0.9381077154161742, 'B_6_4': -0.48766365706737136, 'B_6_5': 0.18911586701317826, 'B_6_6': 0.4834444948641696, 'B_6_7': 0.18896207742936372, 'B_6_8': 0.37195461432450444, 'B_6_9': 0.4933918842313716, 'B_7_0': -0.8158937191876994, 'B_7_1': -0.1775023798957767, 'B_7_2': -0.26149418051414236, 'B_7_3': 0.7570478214296739, 'B_7_4': 0.9090180076123352, 'B_7_5': -0.6990646342068141, 'B_7_6': -0.09538628741321567, 'B_7_7': -0.7495918762126801, 'B_7_8': -0.8741003262265422, 'B_7_9': 0.8237875899012943, 'B_8_0': -0.5582029613839792, 'B_8_1': 0.5752242049940902, 'B_8_2': 0.9255445513415936, 'B_8_3': 0.6879786506293621, 'B_8_4': 0.2227604801371762, 'B_8_5': -0.20766733530003845, 'B_8_6': 0.4981322717346895, 'B_8_7': -0.5500793763477666, 'B_8_8': -0.03487344119783922, 'B_8_9': 0.606785270631128, 'B_9_0': 0.0248621932736333, 'B_9_1': -0.7029446293030529, 'B_9_2': -0.6184122657164253, 'B_9_3': 0.5568952802356656, 'B_9_4': -0.6492728482248346, 'B_9_5': -0.4604708568838772, 'B_9_6': -0.5977067722263484, 'B_9_7': -0.13114014379236122, 'B_9_8': 0.8772911333267481, 'B_9_9': 0.41648570489880976, 'B_10_0': 0.5468522916353188, 'B_10_1': -0.5763139397673276, 'B_10_2': -0.9141108093881449, 'B_10_3': -0.14314102270338488, 'B_10_4': 0.7647611227719289, 'B_10_5': -0.3038775265470799, 'B_10_6':

-0.48317908291505035, 'B_10_7': 0.7492921681150387, 'B_10_8':
0.14507270604310293, 'B_10_9': 0.06659908532479716}. Best is trial 195 with
value: 54.28004122812132.

Building the Basis functions for the RKHS:

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

[I 2025-04-29 14:36:48,526] Trial 198 finished with value: 64.01926581159125 and
parameters: {'B_0_0': 0.5510040924606987, 'B_0_1': 0.4844566400694616, 'B_0_2':
0.7697167086340683, 'B_0_3': 0.16999716796684217, 'B_0_4': -0.5503763737698388,
'B_0_5': 0.7014250272680543, 'B_0_6': 0.6691869394774835, 'B_0_7':
0.21999272833661604, 'B_0_8': 0.43472319166359374, 'B_0_9': 0.38353740810177356,
'B_1_0': 0.6468373613648447, 'B_1_1': -0.22096284563738797, 'B_1_2':
0.351231216893757, 'B_1_3': 0.5290342877247555, 'B_1_4': -0.36675029866362835,
'B_1_5': 0.7757949217095065, 'B_1_6': -0.11958942427915964, 'B_1_7':
0.5665546328120856, 'B_1_8': -0.9099717396422429, 'B_1_9': -0.5544938956975173,
'B_2_0': -0.9378130835685593, 'B_2_1': -0.5720093726306612, 'B_2_2':
-0.7468196788892542, 'B_2_3': 0.847374239052725, 'B_2_4': 0.006261351066878772,
'B_2_5': -0.9005579603712068, 'B_2_6': 0.12046694265324016, 'B_2_7':
-0.41472391165187383, 'B_2_8': -0.6499847360243809, 'B_2_9':
0.47352686548695444, 'B_3_0': -0.30913306806623814, 'B_3_1':
-0.5015984646629656, 'B_3_2': -0.36588700632697807, 'B_3_3':
-0.09953809941998101, 'B_3_4': -0.8841970176945226, 'B_3_5': 0.4224400318555213,
'B_3_6': -0.5025813750190307, 'B_3_7': 0.701551552872306, 'B_3_8':
0.18063802862785444, 'B_3_9': -0.061473183018939365, 'B_4_0':
0.18247431067594028, 'B_4_1': 0.18085515438426392, 'B_4_2': 0.19934589943719316,
'B_4_3': -0.18192308838217233, 'B_4_4': -0.928965998215538, 'B_4_5':
0.3539963864092763, 'B_4_6': -0.6635948549509232, 'B_4_7': -0.5593220725270052,
'B_4_8': 0.5111560248823411, 'B_4_9': 0.9587941139187859, 'B_5_0':
0.4633829009052348, 'B_5_1': -0.4711546234984965, 'B_5_2': 0.7717229939044428,
'B_5_3': -0.2798648646893852, 'B_5_4': 0.9095838925251165, 'B_5_5':
-0.8601118048688672, 'B_5_6': -0.20346245404227326, 'B_5_7': 0.8632259066803026,
'B_5_8': -0.051514826732362864, 'B_5_9': 0.42288887121067187, 'B_6_0':
0.19295571932211678, 'B_6_1': -0.26982227980981865, 'B_6_2': 0.3642236060426252,
'B_6_3': 0.8748625261012895, 'B_6_4': -0.2933079463326834, 'B_6_5':
0.19160836877385656, 'B_6_6': 0.47401733130334867, 'B_6_7': 0.21358639968838694,
'B_6_8': 0.3727233939790469, 'B_6_9': 0.4630809327084649, 'B_7_0':

```

-0.7405554667539036, 'B_7_1': -0.20220818008408312, 'B_7_2':
-0.26738108103765806, 'B_7_3': 0.7481318002112367, 'B_7_4': 0.9152527488558153,
'B_7_5': -0.7279750153229171, 'B_7_6': -0.09542054216424362, 'B_7_7':
-0.7047931006509842, 'B_7_8': -0.8195895963822887, 'B_7_9': 0.8097475242585116,
'B_8_0': -0.5045383737414212, 'B_8_1': 0.5719136521147787, 'B_8_2':
0.9243090205612106, 'B_8_3': 0.6588546173062154, 'B_8_4': 0.17264321315420994,
'B_8_5': -0.20187065149302263, 'B_8_6': 0.6468675363980735, 'B_8_7':
-0.46102397940416157, 'B_8_8': -0.03795419832265601, 'B_8_9':
0.6076554986209352, 'B_9_0': 0.0070533161374946585, 'B_9_1':
-0.5310008083276826, 'B_9_2': -0.6303731543871498, 'B_9_3': 0.5640388796481706,
'B_9_4': -0.6449819655857809, 'B_9_5': -0.39941225427242527, 'B_9_6':
-0.7015893851404169, 'B_9_7': -0.028171032759937722, 'B_9_8':
0.8795824606867695, 'B_9_9': 0.4099488071970951, 'B_10_0': 0.5707310172489589,
'B_10_1': -0.5746399608730719, 'B_10_2': -0.925020739854744, 'B_10_3':
-0.08624483027848401, 'B_10_4': 0.7651117561406086, 'B_10_5':
-0.31045813310117865, 'B_10_6': -0.5459210237683058, 'B_10_7':
0.7482073753194354, 'B_10_8': 0.2651413428060927, 'B_10_9': 0.075625019763787}.
Best is trial 195 with value: 54.28004122812132.

```

Building the Basis functions for the RKHS:

```

Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].

```

Established the refined x grid with 1001 points.

Built matrix A = [$F_{[u_i]}$, $F_{[u_j]}$].

Built matrix B = inverse(A).

```

[I 2025-04-29 14:37:02,879] Trial 199 finished with value: 56.71116877723013 and
parameters: {'B_0_0': 0.6367398156710441, 'B_0_1': 0.4683243154586166, 'B_0_2':
0.743429586212445, 'B_0_3': 0.07269537051332828, 'B_0_4': -0.636791999178035,
'B_0_5': 0.6153753699191127, 'B_0_6': 0.5945267914543898, 'B_0_7':
0.3081273405594004, 'B_0_8': 0.22121946766499104, 'B_0_9': 0.5230664413409316,
'B_1_0': 0.5935583274649338, 'B_1_1': -0.1944385007774049, 'B_1_2':
0.30286149720695144, 'B_1_3': 0.3435220013910445, 'B_1_4': -0.28197223851687647,
'B_1_5': 0.8721523072262695, 'B_1_6': -0.2534860565900485, 'B_1_7':
0.5457966404108986, 'B_1_8': -0.7855399511005225, 'B_1_9': -0.5047909204349131,
'B_2_0': -0.8681435960819753, 'B_2_1': -0.4903211024062979, 'B_2_2':
-0.6329494599697755, 'B_2_3': 0.9878885677559858, 'B_2_4': 0.05236632798672581,
'B_2_5': -0.9039625588103573, 'B_2_6': -0.012981471529530938, 'B_2_7':
-0.4605486540039883, 'B_2_8': -0.691503835870633, 'B_2_9': 0.406311894170944,
'B_3_0': -0.19073971728329908, 'B_3_1': -0.4209175530338671, 'B_3_2':
-0.5068947056212378, 'B_3_3': -0.1922954462013559, 'B_3_4': -0.8492578650447229,

```

'B_3_5': 0.39115034987761016, 'B_3_6': -0.4831042485217351, 'B_3_7':
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 -0.050221767590287555, 'B_4_0': 0.277298168029651, 'B_4_1': 0.04466514435466379,
 'B_4_2': 0.18122124206391912, 'B_4_3': -0.14528080899596513, 'B_4_4':
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 'B_4_7': -0.5093077861554215, 'B_4_8': 0.34983567545036615, 'B_4_9':
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 'B_5_2': 0.8366657851979318, 'B_5_3': -0.40427946599795817, 'B_5_4':
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 'B_5_7': 0.6281242885868392, 'B_5_8': -0.09939311017752486, 'B_5_9':
 0.38068622196638624, 'B_6_0': 0.060623028924433345, 'B_6_1':
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 'B_6_4': -0.4769889028190915, 'B_6_5': 0.3429780325030949, 'B_6_6':
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 'B_6_9': 0.5186299058833672, 'B_7_0': -0.8187661568415785, 'B_7_1':
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 'B_7_4': 0.9549411950014229, 'B_7_5': -0.8104605620117479, 'B_7_6':
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 -0.8843268278973736, 'B_7_9': 0.8845048840608636, 'B_8_0': -0.5594492824070907,
 'B_8_1': 0.5085591118769357, 'B_8_2': 0.8844242249185911, 'B_8_3':
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 'B_8_6': 0.5196069944232747, 'B_8_7': -0.5329110013601546, 'B_8_8':
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 -0.5682255976062464, 'B_9_3': 0.49208418002484955, 'B_9_4': -0.5970426327475735,
 'B_9_5': -0.4652765737086035, 'B_9_6': -0.721527962104569, 'B_9_7':
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 'B_10_0': 0.6314128996772091, 'B_10_1': -0.5024764286035803, 'B_10_2':
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 'B_10_7': 0.7141769858810355, 'B_10_8': 0.14593108451593018, 'B_10_9':
 0.13047758436718174}. Best is trial 195 with value: 54.28004122812132.

Best vertical scaling factors:

B_0_0 = 0.6015
 B_0_1 = 0.5735
 B_0_2 = 0.7578
 B_0_3 = 0.1784
 B_0_4 = -0.5052
 B_0_5 = 0.7005
 B_0_6 = 0.5287
 B_0_7 = 0.3241
 B_0_8 = 0.4433
 B_0_9 = 0.4236
 B_1_0 = 0.5633
 B_1_1 = -0.2074
 B_1_2 = 0.3109
 B_1_3 = 0.4807
 B_1_4 = -0.3596

B_1_5 = 0.8955
B_1_6 = -0.2751
B_1_7 = 0.5508
B_1_8 = -0.8414
B_1_9 = -0.6214
B_2_0 = -0.8227
B_2_1 = -0.4892
B_2_2 = -0.6418
B_2_3 = 0.9030
B_2_4 = -0.0241
B_2_5 = -0.9117
B_2_6 = -0.0100
B_2_7 = -0.4925
B_2_8 = -0.5775
B_2_9 = 0.3063
B_3_0 = -0.3465
B_3_1 = -0.4340
B_3_2 = -0.4332
B_3_3 = -0.1276
B_3_4 = -0.8318
B_3_5 = 0.4242
B_3_6 = -0.3790
B_3_7 = 0.4619
B_3_8 = 0.2277
B_3_9 = -0.0617
B_4_0 = 0.2900
B_4_1 = 0.2176
B_4_2 = 0.2517
B_4_3 = -0.2027
B_4_4 = -0.8426
B_4_5 = 0.3458
B_4_6 = -0.5828
B_4_7 = -0.4387
B_4_8 = 0.3396
B_4_9 = 0.9210
B_5_0 = 0.3892
B_5_1 = -0.5014
B_5_2 = 0.8722
B_5_3 = -0.2725
B_5_4 = 0.8441
B_5_5 = -0.9363
B_5_6 = -0.1933
B_5_7 = 0.9874
B_5_8 = -0.1666
B_5_9 = 0.4257
B_6_0 = 0.1317
B_6_1 = -0.3068
B_6_2 = 0.4828

B_6_3 = 0.9282
B_6_4 = -0.2973
B_6_5 = 0.2827
B_6_6 = 0.6362
B_6_7 = 0.2929
B_6_8 = 0.4242
B_6_9 = 0.4353
B_7_0 = -0.8230
B_7_1 = -0.2174
B_7_2 = -0.2616
B_7_3 = 0.7570
B_7_4 = 0.8590
B_7_5 = -0.7132
B_7_6 = -0.2086
B_7_7 = -0.7314
B_7_8 = -0.8337
B_7_9 = 0.8342
B_8_0 = -0.6458
B_8_1 = 0.3961
B_8_2 = 0.9241
B_8_3 = 0.5758
B_8_4 = 0.1900
B_8_5 = -0.3507
B_8_6 = 0.4914
B_8_7 = -0.6379
B_8_8 = -0.0763
B_8_9 = 0.7083
B_9_0 = 0.0077
B_9_1 = -0.7371
B_9_2 = -0.6275
B_9_3 = 0.5493
B_9_4 = -0.5851
B_9_5 = -0.5133
B_9_6 = -0.7191
B_9_7 = -0.2018
B_9_8 = 0.9135
B_9_9 = 0.4593
B_10_0 = 0.5966
B_10_1 = -0.6177
B_10_2 = -0.9422
B_10_3 = -0.1998
B_10_4 = 0.6473
B_10_5 = -0.2912
B_10_6 = -0.5119
B_10_7 = 0.7516
B_10_8 = 0.1838
B_10_9 = 0.1107
Best MSE: 54.28004122812132

```
[44]: s_values = np.zeros((11, 10), dtype=float)
      for i in range(11):
          for j in range(10):
              s_values[i, j] = study.best_params[f"B_{i}_{j}"]

      print("Best vertical scaling factors:")
      print(np.round(s_values, decimals=2))

      rkhs = RKHS(data_series=data_series, m=11, fractal_depth=2, s_values=s_values)
      kernel = fractal_kernel(rkhs)
      svr_model = SVR(kernel=kernel, C = 10)
      svr_model.fit(X, y)
      y_pred = svr_model.predict(X)
```

Best vertical scaling factors:

```
[[ 0.6   0.57  0.76  0.18 -0.51  0.7   0.53  0.32  0.44  0.42]
 [ 0.56 -0.21  0.31  0.48 -0.36  0.9  -0.28  0.55 -0.84 -0.62]
 [-0.82 -0.49 -0.64  0.9  -0.02 -0.91 -0.01 -0.49 -0.58  0.31]
 [-0.35 -0.43 -0.43 -0.13 -0.83  0.42 -0.38  0.46  0.23 -0.06]
 [ 0.29  0.22  0.25 -0.2  -0.84  0.35 -0.58 -0.44  0.34  0.92]
 [ 0.39 -0.5   0.87 -0.27  0.84 -0.94 -0.19  0.99 -0.17  0.43]
 [ 0.13 -0.31  0.48  0.93 -0.3   0.28  0.64  0.29  0.42  0.44]
 [-0.82 -0.22 -0.26  0.76  0.86 -0.71 -0.21 -0.73 -0.83  0.83]
 [-0.65  0.4   0.92  0.58  0.19 -0.35  0.49 -0.64 -0.08  0.71]
 [ 0.01 -0.74 -0.63  0.55 -0.59 -0.51 -0.72 -0.2   0.91  0.46]
 [ 0.6  -0.62 -0.94 -0.2   0.65 -0.29 -0.51  0.75  0.18  0.11]]
```

Building the Basis functions for the RKHS:

```
Built F_[u_1].
Built F_[u_2].
Built F_[u_3].
Built F_[u_4].
Built F_[u_5].
Built F_[u_6].
Built F_[u_7].
Built F_[u_8].
Built F_[u_9].
Built F_[u_10].
Built F_[u_11].
```

Established the refined x grid with 1001 points.

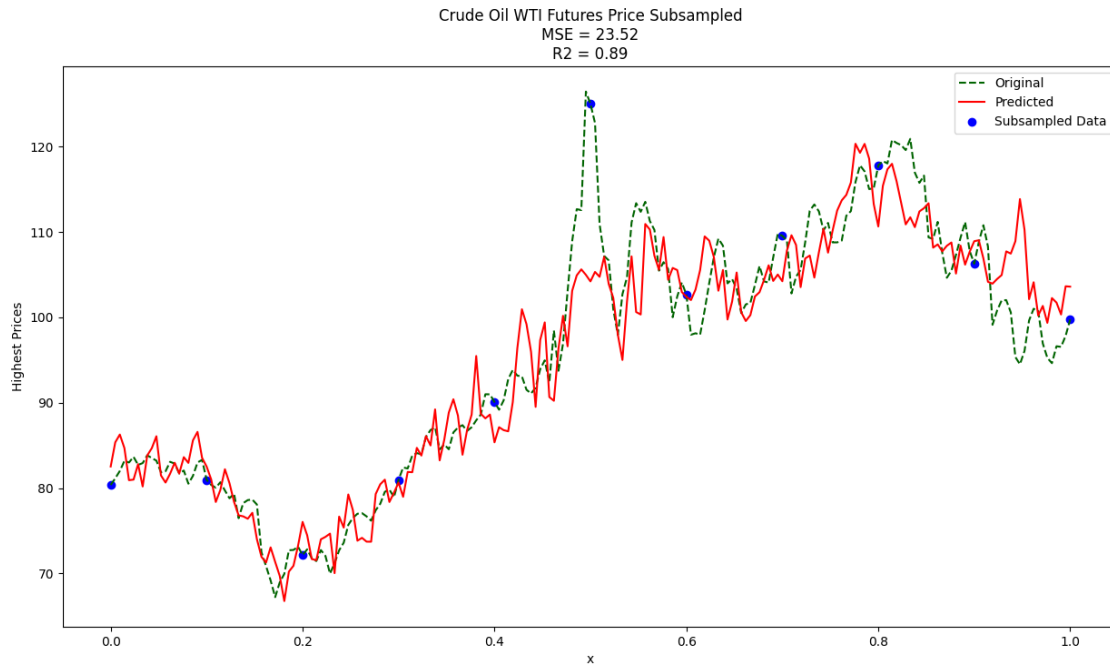
Built matrix A = [F_[u_i], F_[u_j]>].

Built matrix B = inverse(A).

```
[45]: plt.figure(figsize=(15, 8))
      plt.plot(X[:,0], y, label='Original', color='darkgreen', linestyle='--')
      plt.plot(X[:,0], y_pred, label='Predicted', color='red')
      plt.scatter(data_series[:,0], data_series[:, 1], color='blue',
                  label='Subsampled Data', marker='o')
```



```
plt.xlabel('x')
plt.ylabel('Highest Prices')
plt.title('Crude Oil WTI Futures Price Subsampled\n'
          f'MSE = {mean_squared_error(y, y_pred):.2f}\n'
          f'R2 = {r2_score(y, y_pred):.2f}')
plt.legend()
plt.show()
```



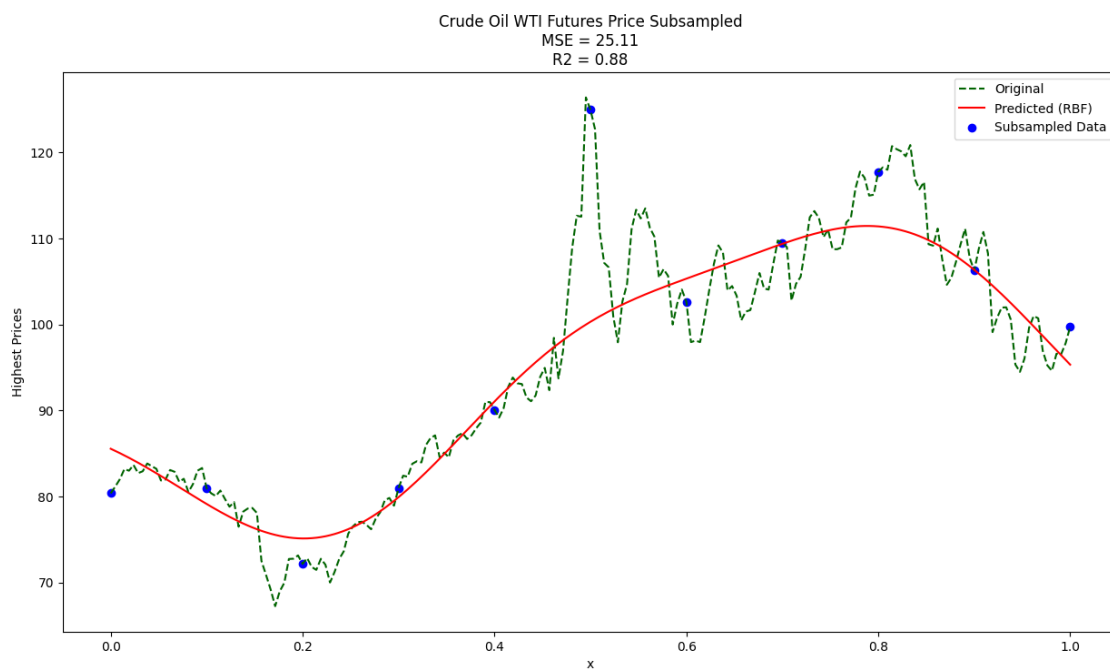
1.5 With RBF

```
[52]: rbf_svr = SVR(kernel='rbf', C = 10)
rbf_svr.fit(X, y)
y_pred_rbf = rbf_svr.predict(X)
```

```
[53]: plt.figure(figsize=(15, 8))
plt.plot(X[:,0], y, label='Original', color='darkgreen', linestyle='--')
plt.plot(X[:,0], y_pred_rbf, label='Predicted (RBF)', color='red')
plt.scatter(data_series[:,0], data_series[:, 1], color='blue',
            label='Subsampled Data', marker='o')

plt.xlabel('x')
plt.ylabel('Highest Prices')
plt.title('Crude Oil WTI Futures Price Subsampled\n'
          f'MSE = {mean_squared_error(y, y_pred_rbf):.2f}\n')
```

```
f'R2 = {r2_score(y, y_pred_rbf):.2f}'  
plt.legend()  
plt.show()
```



```
[ ]:
```